
RC WORLD

Master Development Plan

INVESTOR PROSPECTUS • OPERATIONS MANUAL • ENGINEERING HANDBOOK

Thirteen volumes covering the executive master plan, market research, the four operating divisions (Motorsport, Construction, Aviation, Marine), the engineering & workshop manual, the China procurement handbook, customer experience & loyalty, finance, architecture & park design, the franchise manual, and the RC WORLD OS technology platform.

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RC WORLD — Master Development Plan: Front Matter

RC WORLD — Master Development Plan · Volume 0 (Front Matter) of 13 volumes **Revision 1.0 — July 2026** · Status: Living document — bump revision on material change

Purpose of this volume. This volume is the reader's entry point to the entire Master Development Plan. It carries the title page and document-control record, tells each class of reader — investor, lender, architect, operator, technician, franchisee — which volumes to read and in what order, defines the living-document maintenance workflow that keeps thirteen separately maintained volumes consistent, presents the master table of contents with a synopsis of every volume, and fixes the glossary of canonical terms used identically throughout the plan. Nothing in this volume introduces new facts; where a number appears here, its source of truth is `style-guide.md` and the volume cited.

Intended readers. Everyone who opens any part of the Master Development Plan, before they open it.

Chapters

1. Title Page
2. Document Control & Revision History
3. How to Use This Master Plan
4. Living-Document Maintenance Workflow
5. Master Table of Contents
6. Glossary of Canonical Terms
7. Volume Summary & Cross-References

1. Title Page

RC WORLD MASTER DEVELOPMENT PLAN

Investor Prospectus • Operations Manual • Engineering Handbook

A complete plan for the design, financing, construction, and operation of a large multi-zone radio-controlled vehicle theme park — a **miniaturized industrial complex** spanning motor-sport racing, open-pit mining and construction simulation, commercial agriculture, aviation, and marine operations, supported by in-house telemetry hardware (the **RCW Node**), a custom enterprise platform (**RC WORLD OS**), and a central engineering facility (**The Works**).

Thirteen volumes plus this front matter. Approximately 440–490 finished pages.

Internal project codename: Omni-Zone (legacy; reference only). Prepared: **July 2026**. Currency: USD. Units: metric first, imperial in parentheses. Funding ask: **\$2.6 M**, staged tranches (Volume 1, Chapter 6; Volume 10).

Confidential. Prepared for investors, lenders, design professionals, staff, and prospective franchisees of RC WORLD. Distribution restricted; see document control below.

2. Document Control & Revision History

2.1 Control record

Field	Value
Document title	RC WORLD — Master Development Plan
Document set	13 volumes + front matter, one markdown file per volume in volumes/
Canonical style & numbers	style-guide.md (single source of canonical facts)
Source material	source-material/original-notes-digest.md (founder's three planning PDFs + brief)
Build	python3 build/assemble.py → single combined document (.md , .docx , .pdf if pandoc present)
Owner	Founder / Managing Director, RC WORLD
Review unit	One volume = one file = one independent review and revision unit
Classification	Confidential — investor and internal distribution only

2.2 Revision history

Rev	Date	Scope	Author	Notes
1.0	July 2026	Initial complete issue of all volumes	Founder + drafting team	Baseline for Phase 1 financing round
—	—	(next entries added per-volume; see Chapter 4)	—	—

Each volume carries its own revision line in its header block. The revision of the set is the table above; the revision of a volume is its header. A combined build lists both.

3. How to Use This Master Plan

This plan serves five end uses at once: investor presentations, financing applications, architect/contractor guidance, employee training and daily operations, and future franchising. No reader needs all ~450 pages; each needs a defined path through them.

3.1 Investors (equity)

Read **Volume 1** in full — it is the whole business in one volume, including the \$2.6 M ask, tranche structure, phase gates, and returns summary. Then go straight to **Volume 10 (Finance)** for the full model, sensitivities, and scenario analysis, and **Volume 2 (Market Research)** for the demand case and competitive landscape. Volumes 3–6 are your diligence layer: each division volume shows exactly what the money builds and how it earns. Volume 13 explains the technology moat.

3.2 Lenders and grant bodies

Start with Volume 1, Chapters 6–9 (ask, phases, financial overview, risk summary), then **Volume 10** for debt-service coverage and break-even (canonically operating Month 16–19 monthly operating break-even, where operating Month 1 = the opening month), then **Volume 11 (Architecture & Park Design)** for the fixed-asset base and **Volume 1, Chapter 9** plus the per-volume risk registers for the risk picture. The staged-tranche structure in Volume 1, Chapter 6 is written to map onto milestone-based drawdowns.

3.3 Architects, engineers, and contractors

Volume 11 is your primary document: site plan, zone-by-zone civil specification, utilities, drainage, netting and pond engineering, and CAD-style layout concepts. Read it alongside the division volumes for functional requirements — Volume 3 (track construction), Volume 4 (mining circuit and haul roads, maximum 15° incline; agriculture field grid), Volume 5 (netted airfield), Volume 6 (pond complex) — and Volume 13 for structured cabling, Wi-Fi mesh, and the charging bunker's electrical load.

3.4 Operations staff and managers

Your induction sequence: Volume 1, Chapters 1–5 (what RC WORLD is), then **Volume 9 (Customer Experience & Loyalty)** for the customer journey, RC WORLD License tiers, and service standards, then the division volume for your assignment, then the SOPs, checklists, and emergency procedures in your division volume and **Volume 7**. Managers add Volume 10 (budget ownership, KPI definitions) and Volume 13 (RC WORLD OS role-based interfaces).

3.5 Technicians (Artisans)

Volume 7 (Engineering & Workshop Manual) is your handbook: maintenance schedules, rebuild procedures, failure catalogues, trade hacks, custom-build instruction, and the preventive-maintenance matrices. Pair it with **Volume 8 (Procurement Handbook)** for parts sourcing and the high-turnover spares list, the maintenance chapters of your division volume, and Volume 13 for logging work in RC WORLD OS (`maintenance_logs`, predictive-maintenance flags).

3.6 Franchisees

Volume 12 (Franchise Manual) is the contract-adjacent document: what the RC WORLD franchise includes, localization method, fees, and support model. Volumes 1, 9, and 10 give you the business logic; Volume 8 gives you the supply chain you inherit; Volume 13 explains the multi-site architecture of RC WORLD OS that makes a franchise operable from day one.

Investor Note. The plan is deliberately modular. A financing application, a contractor tender pack, and a staff handbook can each be assembled from whole volumes without editing — which is also why canonical numbers live in one place and are never restated with local variations.

4. Living-Document Maintenance Workflow

The plan is a living document. Suppliers change models, prices drift, and phase experience will correct assumptions. The workflow that keeps thirteen volumes coherent:

1. **One volume = one review unit.** Any volume may be revised independently. On material change, bump that volume's revision line (e.g. 1.0 → 1.1) and add one row to its change note; the set-level table in Chapter 2 records set-wide reissues only.
2. **Canonical numbers change in one place first.** Pricing anchors, fleet counts, capex, phase definitions, and headline financials live in `style-guide.md`. To change one: update the style guide, then sweep every volume that cites it, then bump affected revisions. Never patch a canonical number in a single volume.
3. **Supersession is explicit.** Where a volume extends or overrides a decision recorded in the source-material digest, it must say so in the text ("this supersedes..."). Silent divergence is a defect.
4. **Rebuild after every sweep.** Run `build/assemble.py` to regenerate the combined document and verify per-volume word counts; the combined build is the only artifact distributed externally.
5. **Ownership.** Each volume has a natural owner (division lead, finance lead, CTO for Volume 13); the founder/MD owns the style guide and arbitrates conflicts.

5. Master Table of Contents

#	Volume	Words (target)
0	Front Matter (this volume)	2,500
1	Executive Master Plan	14,000
2	Market Research & Competitive Landscape	15,000
3	RC Motorsport Division	16,000
4	RC Construction Division	14,000
5	Aviation Division	12,000
6	Marine Division	12,000
7	Engineering & Workshop Manual	20,000
8	Procurement Handbook	18,000
9	Customer Experience & Loyalty	12,000
10	Finance	15,000
11	Architecture & Park Design	18,000
12	Franchise Manual	13,000
13	IT, IoT & ERP Systems (RC WORLD OS)	16,000

Volume 1 — Executive Master Plan. The whole business in one volume: the miniaturized-industrial-complex thesis, mission and values, the revenue engines and the high-utilization business model with worked vehicle-hour economics, the park at a glance, the \$2.6 M investment opportunity with staged tranches and use of funds, the three development phases with gating criteria, the headline five-year financial trajectory, a top-10 risk summary, and the five-year expansion roadmap. It is the only volume that summarizes every other volume, and the first document any external reader receives.

Volume 2 — Market Research & Competitive Landscape. Sizes the global RC industry (cited as ranges per the style guide), profiles the Chinese, Japanese, European, and US ecosystems, and delivers the commissioned special study on Chinese diorama and miniature-equipment manufacturers who replicate real work environments at scale. It then analyzes the experience economy RC WORLD actually competes in, every adjacent venue format, customer segmentation with a TAM/SAM/SOM model reconciled to Year-1 revenue, and market risks.

Volume 3 — RC Motorsport Division. The complete specification for Tracks A (asphalt/drift), B (baja/rally), and C (crawler trail): fleet classes and counts within the 64-car Phase 1 baseline, the standardized-chassis doctrine with lap-time parity within 2-3% and personality-through-tuning, track construction, race formats and championships, timing and telemetry integration, marshaling, maintenance, and the division's KPI dashboard.

Volume 4 — RC Construction Division. Build-out and operating specification for the Mining Zone and Agriculture Zone: machine catalogue, the electromechanical-rental versus hydraulic-premium doctrine, fleet-balancing mathematics behind the 1:3 excavator-to-dump-truck ratio, haul-road and pit design (maximum 15° incline), the field grid and implement library, earthmoving competitions, maintenance, and safety.

Volume 5 — Aviation Division. The Phase 2 netted airfield: 30 m runway, helipads, high-tensile enclosure engineering, the ~20-aircraft fleet (fixed-wing trainers through FPV), flight-line procedures, instruction programs, airspace and regulatory compliance, and weather doctrine.

Volume 6 — Marine Division. The Phase 2 pond complex: pond engineering and water management, the ~24-vessel fleet from scale tugs to fast electrics and sailing one-designs, retrieval systems, event formats, and winterization.

Volume 7 — Engineering & Workshop Manual. The Artisans' handbook for The Works: workshop layout and tooling, maintenance schedules and preventive-maintenance matrices, FMEA-driven failure catalogues, rebuild procedures, battery doctrine and charging-bunker operation, RCW Node installation and service, trade hacks, best-practice skills, and complete instruction on building custom RC cars.

Volume 8 — Procurement Handbook. The purchasing manual: supplier landscape and vetting, direct-from-China wholesale channels, the full fleet purchasing matrix with cost bands, the high-turnover spares inventory, batteries and chargers, quality-assurance inspection protocols, landed-cost calculation, and reorder automation through RC WORLD OS.

Volume 9 — Customer Experience & Loyalty. The customer journey end to end: Toolbox Talk induction, Shift booking and queueing, the tiered **RC WORLD License** driver-progression program, badges and skill mastery, **Gears** loyalty currency, memberships (Apprentice/

Operator/Foreman), season passes, parties and corporate events, and service-recovery standards.

Volume 10 — Finance. The source of truth for all financial figures: full five-year P&L, cash flow and balance sheet, capex schedules per phase, unit economics, sensitivity and scenario analysis, break-even (operating Month 16-19), funding structure for the \$2.6 M ask, investor returns, and local-adaptation methodology.

Volume 11 — Architecture & Park Design. The physical park on 4.8 ha (~12 acres): master site plan and CAD-style layout concepts, zone-by-zone civil and landscape specification, buildings (entry pavilion, The Works, charging bunker, restaurant, indoor arena), utilities and drainage, circulation and sightlines, accessibility, and the phased construction sequence.

Volume 12 — Franchise Manual. The replication playbook: what the franchise package contains, site-selection criteria, localization of pricing and fleet, fee structure, training and opening support, brand standards, and the multi-site operating model — activated by the Phase 3 franchise pilot.

Volume 13 — IT, IoT & ERP Systems (RC WORLD OS). The technology volume: RCW Node hardware (ESP32-C3, Micro-Node and Heavy-Node), kill-switch logic, Wi-Fi mesh, and the full ERP covering finance, HR/payroll, payments, bookings, fleet management, live telemetry, POS/retail, F&B, inventory, CRM, events, access control, reporting/BI, and franchise multi-site support — built on Supabase (PostgreSQL) + Kotlin services, with Kotlin Multiplatform / Compose Multiplatform client apps on **both Android and iOS** and role-based interfaces for each user class.

6. Glossary of Canonical Terms

Term	Definition
RC WORLD	The park brand (all caps in headings; “RC World” acceptable in prose). Successor to internal codename Omni-Zone.
Omni-Zone	Legacy internal project codename from the original blueprint; reference only.
Division	Top-level operating unit: Motorsport Division; Construction Division (incl. Mining & Agriculture zones); Aviation Division; Marine Division.
The Works	The central engineering and workshop facility; home base of the Artisans.
Artisan	Customer-facing title for RC WORLD technicians (“technician” acceptable in technical prose).
Shift	The billing unit: one 20-minute operating block, decoupled from battery life (~30% buffer returned).
Casual Shift	One-block session; vehicle returned with charge buffer intact. Anchor price \$15 standard / \$22 premium classes.
Operator Shift	Two or more blocks with a mandatory pit stop for an Artisan battery swap at 20 minutes. \$26 / \$38.
Top-Up Session	Instant in-app Shift extension offered at T-5 minutes when no one is queued for that vehicle class.

Term	Definition
RCW Node	In-house telemetry PCBA (ESP32-C3) inline between receiver and ESC; monitors voltage and GPS, executes remote kill. Micro-Node ~25×25 mm (racing); Heavy-Node ~40×30 mm (construction).
RC WORLD OS	The custom ERP running the entire business — finance, HR/payroll, payments, bookings, fleet, live telemetry, and all other functions — with role-based interfaces per user class. Supabase + Kotlin backend; Kotlin Multiplatform apps on Android and iOS. Volume 13.
Toolbox Talk	The digital induction module completed in-app before first booking; doubles as the signed legal waiver.
RC WORLD License	The tiered driver-license progression program (levels, badges, equipment endorsements). Volume 9.
Gears	The loyalty currency: points earned through visits, performance, and program milestones.
Tow-Truck Retrieval Protocol	Breakdown gameplay: customers never walk onto live tracks; they pilot a 1/10 winch-equipped recovery crawler to retrieve their dead vehicle, then receive a fresh one.
Charging bunker	Fire-isolated cinderblock/sandbag charging room; all LiPo charging occurs here on multi-port smart balance chargers.
Battery doctrine	2S/3S LiPo, XT60 standard (Deans legacy acceptable), 3:1 battery-to-vehicle ratio, 3.4–4.2 V/cell operating window.
1:3 ratio	Canonical excavator-to-dump-truck fleet ratio in the Mining Zone.
Rental vs premium fleet	Rental construction machines are electromechanical (lead-screw); hydraulic machines (Kabolite class) are reserved for premium supervised experiences and display.
Phase 1 / 2 / 3	“Core Park” (project Months 0–12); “Full Park” (project Months 13–30); “Destination & Beyond” (project Months 31–60). Build phasing uses project months (park opens at project Month 12); financials and division opening schedules use operating months (M1 = opening month).

7. Volume Summary & Cross-References

This front matter established the document set, its control and maintenance rules, the reading path for each audience, the master table of contents, and the canonical vocabulary. Begin with **Volume 1** for the business as a whole; consult `style-guide.md` whenever a number or name is in question — it, not any individual volume, is the source of canonical facts, with **Volume 10** the source of truth for financial detail and **Volume 13** for the technology stack. The reading paths in Chapter 3 are the fastest route to any specific answer.

Volume 1 — Executive Master Plan

RC WORLD — Master Development Plan · Volume 1 of 13 Revision 1.0 — July 2026 ·
Status: Living document — bump revision on material change

Purpose of this volume. This volume is the whole business in one document. It states the concept and the thesis behind it, the mission and values that govern every operating decision, the business model and the unit economics that make a fleet of small machines a real company, the park itself at a glance, the \$2.6 M investment opportunity with its staged tranches and use of funds, the three development phases with their gating criteria, the headline financial trajectory, the ten risks that matter most, and the five-year expansion roadmap. Every chapter here is a summary with a pointer: the detail lives in Volumes 2-13, and where a number appears in this volume it is the canonical headline figure whose source of truth is the volume cited. An investor should be able to read this volume alone and understand exactly what is being built, what it costs, what it earns, and what could go wrong.

Intended readers. Investors and lenders (primary); the founder and senior management as the master reference; architects, senior hires, and prospective franchisees as orientation before their specialist volumes.

Chapters

1. Executive Summary
2. Vision
3. Mission & Values
4. Business Model
5. The Park at a Glance
6. Investment Opportunity
7. Development Phases
8. Financial Overview
9. Risk Summary
10. Expansion Roadmap

1. Executive Summary

1.1 The concept in one paragraph

RC WORLD is a purpose-built outdoor theme park where the vehicles are radio-controlled and the customers are the operators. On 4.8 hectares (~12 acres) of leased peri-urban land,

the park recreates a working industrial world at miniature scale: a motorsport complex with three engineered tracks, an open-pit mine where customers load real aggregate into articulated dump trucks, a commercial farm with tractors and implements, and — from Phase 2 — a netted airfield and a marine pond complex. Roughly 150 powered vehicles at opening, every one fitted with an in-house telemetry node, are rented in 20-minute blocks called **Shifts**, managed end-to-end by a custom enterprise platform (**RC WORLD OS**), and kept in service by an on-site engineering workshop (**The Works**) staffed by technicians the customers know as **Artisans**. It is not a hobby shop with a track behind it and it is not an arcade; it is a **miniaturized industrial complex** run as a location-based entertainment business, and to our knowledge no directly comparable venue exists at this scope anywhere in the world (Volume 2, Chapter 6).

1.2 The market

RC WORLD sits at the intersection of two markets and monetizes the gap between them. The first is the radio-control equipment industry: industry analysts (2025–2026) place the global RC car market between **\$1.5 B and \$6.2 B** depending on scope, growing at 5.8–7.5% annually, with hobby radio control overall at roughly **\$3.2–7.0 B**; the toy-grade segment is shrinking in share while serviceable, tunable hobby-grade equipment grows. The second is location-based entertainment — the experience economy of karting halls, trampoline parks, family entertainment centers, and competitive-socializing venues, into which consumers direct a rising share of discretionary spend (Volume 2, Chapter 5). The gap: the RC industry sells millions of capable machines a year into a world with almost no purpose-built public venues to use them, while the experience-economy industry keeps recycling the same dozen formats. RC WORLD occupies that white space with an experience nobody else offers: families try RC without buying anything; hobbyists get engineered terrain they cannot build at home; corporate groups get a team event (“run a mine together”) that is genuinely novel.

Demand is modeled bottom-up, not from market share. Volume 2, Chapter 7 builds a catchment model for a mid-size international city — resident families within a 45–60-minute drive, the local hobbyist community, schools, and the corporate-events market — and shows that the canonical Year-1 revenue of $\approx \$1.28$ M requires a capture rate of the addressable catchment that sits comfortably below what established family-entertainment formats routinely achieve in comparable cities. The model needs no assumption that RC WORLD converts the general public into hobbyists; it needs a fraction of the people who already pay for karting, trampolines, and mini-golf to try the substantially more novel option once, and the retention machinery (Section 4.4) to do its work from there.

1.3 The model

The business runs on a high-utilization thesis: commercial-grade, modifiable RC equipment is now so cheap and repairable that a \$75 wholesale buggy earning \$15 per 20-minute Shift repays its capital cost in a weekend and everything after maintenance is margin. Nine revenue engines stack on that base — Shift rentals, memberships, corporate events, birthday parties, education programs, F&B, retail, repair services, and sponsorship — with franchising as a tenth from Phase 3. Three proprietary systems protect the margin: the **RCW Node** telemetry board (under \$15 per unit, built in-house) that meters usage, enforces geofences, and kill-switches any vehicle before a customer can damage a battery; **RC WORLD OS**, the Kotlin/Supabase ERP that runs bookings, fleet, payments, payroll, and live telemetry with role-based interfaces on Android and iOS; and a gamification layer — driver licenses, badges,

leaderboards, the **Gears** loyalty currency, and the **Tow-Truck Retrieval Protocol** that converts breakdowns into gameplay — engineered to turn first visits into repeat visits (Chapter 4; Volume 9).

1.4 The ask and the plan

RC WORLD seeks **\$2.6 M** — Phase 1 capital expenditure of \$1,850 K (pre-opening costs and a 10% contingency capitalized inside it) plus working capital, operating funding through break-even, corporate costs, and a board-controlled reserve — released in three milestone-gated tranches: \$1.2 M at close, \$850 K about five months before opening, \$550 K at opening (Chapter 6). Phase 1 (“Core Park”, Months 0–12) opens the Motorsport and Construction Divisions, The Works, and the entry pavilion. Phase 2 (“Full Park”, Months 13–30, capex ≈\$1.15 M) adds the Aviation and Marine Divisions, a full restaurant, and grandstand viewing, funded substantially from operating cash flow. Phase 3 (“Destination & Beyond”, Months 31–60, capex ≈\$0.9 M) adds an indoor all-weather arena, the RC Academy, a corporate event centre, night racing, and the franchise pilot. Five-year total capex ≈**\$3.9 M** (Volume 10).

1.5 The returns

Base case (Volume 10 is the source of truth): Year 1 revenue ≈**\$1.28 M**, growing to ≈\$2.6 M in Year 3 and ≈\$3.4 M in Year 5, with steady-state EBITDA margins of **24–28% from Year 3**. Monthly operating break-even is reached in **operating Month 16–19** (operating Month 1 = the opening month); full payback of invested capital arrives in **Year 4–5**. The modeled base-case investor outcome is ≈**1.6–2.0× MOIC** over a five-year hold — a high-single to low-double-digit IRR — with the ambitious case at ≈2.2–2.9× and the franchise and second-site options as further upside carried at zero (Chapter 6.4). Exit routes include trade sale to a location-based-entertainment operator, sale of the franchisor entity, secondary sale to growth capital, and structured founder buyback.

Investor Note. The plan’s conservatism is structural, not rhetorical. Revenue is modeled from bottom-up Shift capacity and catchment capture, not top-down market share; the fleet doctrine deliberately excludes fragile premium hardware from rental duty; the ask covers Phase 1 fully so that Phases 2–3 are options exercised with evidence, not obligations. The single largest risk is the Year-1 demand ramp, and Chapter 9 addresses it first.

1.6 The plan at a glance

Dimension	Headline	Detail
Site	4.8 ha (~12 acres), leased, peri-urban, mid-size international city	Volume 11
Divisions	Motorsport; Construction (Mining & Agriculture); Aviation (Ph 2); Marine (Ph 2)	Volumes 3–6
Fleet at opening	~150 powered assets, all telemetry-equipped	Section 5.3; Volume 8
Billing unit	20-minute Shift; Casual \$15/\$22, Operator \$26/\$38	Section 4.2; Volume 9
Funding ask	\$2.6 M, three milestone-gated tranches (\$1.2 M / \$850 K / \$550 K)	Chapter 6

Dimension	Headline	Detail
Capex	Ph 1 ≈\$1.85 M; Ph 2 ≈\$1.15 M; Ph 3 ≈\$0.9 M; 5-yr ≈\$3.9 M	Volume 10
Revenue	≈\$1.28 M (Y1) → ≈\$2.6 M (Y3) → ≈\$3.4 M (Y5)	Chapter 8; Volume 10
Margin	24–28% EBITDA from Year 3	Volume 10
Break-even / payback	operating Month 16–19 / Year 4–5 (base: Month ~53)	Volume 10
Investor outcome	base ≈1.6–2.0× MOIC; ambitious ≈2.2–2.9×	Chapter 6; Volume 10, \$7.5
Technology	RCW Node (<\$15/unit) + RC WORLD OS (Supabase + Kotlin, Android & iOS)	Volume 13
Expansion	Arena, Academy, night racing (Ph 3); franchise pilot; second site option	Chapter 10; Volume 12

1.7 Why now, why this team

Three curves crossed in the mid-2020s and made this business possible for the first time. Hobby-grade RC entry pricing collapsed (a repairable brushless 1/14 car now wholesales for \$60–90); commodity IoT hardware (ESP32-class microcontrollers, sub-\$5 GPS modules) made per-vehicle telemetry cost under \$15 instead of hundreds; and the experience economy normalized paying \$15–60 for a premium hour of operated equipment. The founder’s preparation — three technical planning documents distilled in this plan’s source material — covers fleet engineering, battery logistics, telemetry hardware design, and ERP architecture at a depth unusual for a pre-revenue venue business, and the operating doctrine throughout this plan (1:3 excavator-to-truck ratio, 15° haul-road limit, 3:1 battery rotation, electromechanical rental fleet) comes from that engineering-first analysis rather than from post-hoc rationalization.

2. Vision

2.1 The miniaturized industrial complex

Every RC venue that has ever failed made the same category error: it treated RC as a toy and built a sandbox. A sandbox — a pile of dirt, a rented excavator, no goal — exhausts its novelty in a single visit, and single-visit businesses die of marketing cost. RC WORLD’s founding thesis inverts this. The park is a **miniaturized industrial complex**: every zone is a working system with inputs, outputs, jobs, and measurable production. The Mining Zone is not a sandpit; it is a simulated open-pit operation with graded haul roads (maximum 15° incline, an engineering limit that prevents motor burnout), a central processing hopper that weighs every load, and a live tonnage leaderboard. The Agriculture Zone is a grid-based commercial farm — pre-tilled fields, crop rows, irrigation trenches, a barn — where tractors plow, disc, and haul with tandem trailers. The tracks are engineered racing circuits with professional timing, standardized fleets, and championship structures. From Phase 2 the airfield and pond extend the same logic to air and water.

The commercial consequence of the thesis is retention. A working system generates goals, and goals generate return visits. The customer who moved 38 kg of aggregate in their first Shift knows exactly what they want to do next time: move 50. The driver who qualified 9th wants 5th. The License holder one badge short of an excavator endorsement books a return visit to earn it. RC WORLD is designed so that every departure gate conversation is about next time — and the entire gamification architecture (Volume 9) exists to make that mechanical rather than accidental.

2.2 Why RC WORLD transcends the hobby-sandbox model

Four structural differences separate this park from every hobby-adjacent venue that preceded it:

1. **Systems, not stations.** Zones are interdependent production circuits, not isolated activity stations. Excavators need trucks; trucks need haul roads; the hopper needs both. This creates cooperative multiplayer gameplay no single-machine rental can offer — and it is why the canonical fleet arithmetic (1:3 excavator-to-truck ratio) is an experience-design decision as much as a logistics one.
2. **Instrumentation, not supervision.** Every vehicle carries an RCW Node. Telemetry meters the Shift, scores the driving, enforces the geofence, protects the battery, and feeds the leaderboard. The park knows where every asset is, what state its battery is in, and who is driving it, in real time. Competitors would need to build this; we manufacture it for under \$15 per vehicle.
3. **Progression, not admission.** The RC WORLD License program means customers hold status with the park — levels, endorsements, badges, Gears balances — that appreciates with every visit and is worthless anywhere else. The venue business becomes a membership business (Volume 9).
4. **An engineering company underneath.** The Works is not a repair corner; it is a production workshop with maintenance matrices, FMEA-driven spares inventory, and rebuild procedures (Volume 7) that keep fleet availability at the $\geq 92\%$ operating target (Volume 10 KPI dashboard) at hobby-grade capital cost. The moat is not any single machine — it is the operating system, in both senses, that keeps 150 machines earning.

The contrast, side by side:

Dimension	Hobby-sandbox model	RC WORLD model
Experience unit	Free play with a machine	A job inside a working system, scored
Repeat-visit driver	Novelty (decaying)	Goals, status, progression (compounding)
Fleet knowledge	Staff memory	Per-asset telemetry, runtime hours, maintenance logs
Battery management	Charged when someone remembers	3:1 rotation, bunkered, telemetry-enforced window
Breakdown	Staff walks out, apologizes	Tow-Truck Retrieval Protocol — gameplay
Revenue model	Ticket per session	Nine stacked engines on a recurring-membership base

Dimension	Hobby-sandbox model	RC WORLD model
Replicability	Depends on the founder being present	Documented to franchise depth (this plan)

2.3 The experience promise

To the customer, the promise is simple and specific: **you will operate real machines doing real work, and you will get measurably better at it.** A first-time visitor completes the Toolbox Talk induction on their phone, scans a QR code on a transmitter, and within minutes is loading a dump truck or lapping Track B. When their machine dies mid-field, they are not embarrassed by a staff walk-of-shame retrieval; they are handed the controls of a winch-equipped recovery crawler and sent to rescue it themselves — the Tow-Truck Retrieval Protocol, friction converted into the most-photographed moment in the park. When their Shift ends, the app shows them their tonnage, their lap consistency, their smoothness score, and precisely which License badge they are closest to. The promise to parents is that this is screen-time inverted: thumbs on real controls, consequences in real physics. The promise to hobbyists is terrain, timing infrastructure, and fleet quality they cannot assemble privately. The promise to companies is the most novel two-hour team event in the city.

The promise has a fairness clause that matters more than any single feature: **the machines are equal; the drivers are not.** Racing fleets are tuned to lap-time parity within 2–3%, so a customer never loses to a better car — only to a better driver, which is a loss that motivates a return rather than a refund request. The same principle runs through the whole park: the leaderboards measure skill the telemetry can verify, the License certifies what the holder can actually do, and nothing in the progression system can be bought instead of earned. Venues that let spending substitute for skill train their best customers to leave; RC WORLD's economics depend on training them to come back.

Investor Note. “Experience promise” is not brochure language; it is the retention mechanism the revenue model depends on. Volume 10's base case assumes repeat visitation consistent with mid-tier family-entertainment benchmarks — not the higher rates the gamification layer is designed to achieve. Everything Volume 9 delivers above benchmark is upside.

2.4 A Saturday at RC WORLD

The clearest way to test the vision is to walk it. It is 09:40 on a Saturday in Year 2. At the entry pavilion a family of four checks in; both children completed the Toolbox Talk in the car, and the app has already suggested a Family Bundle — four drivers, two Shifts each. The eight-year-old is on a Mining Zone excavator by 10:05, guided by an Artisan through her first three-channel load cycle; her father runs a dump truck in the same circuit, hauling her spoil to the hopper, because the 1:3 fleet ratio means there is always a truck seat open next to an active excavator. Across the park, a six-driver drift heat is forming on Track A — five regulars and a first-timer, all on identical MST-based chassis within 2–3% lap parity, so the regulars' advantage is skill, visible and earnable. On Track C a crawler stalls in the water hazard; its driver is already at the recovery station drawing the winch crawler, and a small crowd films the rescue. In The Works' viewing window two Artisans have a buggy on the bench, swapping a spur gear between Shifts; a customer watches, then buys the same gear set at retail for the car he owns at home. At 11:55 the app pings a Track B driver: no queue for his class — Top-Up for \$9? He taps yes without lowering the transmitter. By noon, the F&B kiosk queue is the longest line in the park, which is the plan working: dwell time is the second

product. Nothing in this scene requires a staff member to make a decision that RC WORLD OS has not already framed — and every element of it (the ratio, the parity, the protocol, the top-up, the sightline from the bench to the till) appears somewhere in this plan as an engineered choice.

2.5 The long view

The five-year vision is a **category**, not a site. Phase 1 proves the unit economics of one park; Phase 3's franchise pilot and the multi-site architecture already built into RC WORLD OS (Volume 13) convert one park into a replicable format. The end-state, described in Chapter 10, is RC WORLD as the definitive global brand for operated-miniature experiences — parks, an academy curriculum, a license program that transfers between sites, and telemetry hardware that would cost any imitator years to replicate.

3. Mission & Values

3.1 Mission statement

To build the world's first true RC theme park — a working miniature industrial world where anyone can operate, compete, learn, and progress — run to engineering standards, priced for families, and designed to be replicated.

The mission is bounded as deliberately as it is stated. RC WORLD is not a hobby shop with attractions attached (retail exists to serve the experience, never the reverse); it is not a technology company that happens to own a park (the RCW Node and RC WORLD OS exist to run Shifts, and their platform value is an option, not the plan); and it is not a theme park in the costumed-entertainment sense (the immersion is industrial authenticity, and the moment the park fakes what the machines do, the customers who matter most — the ones who know real machines — will know).

3.2 Values

The five values below are operating constraints, not wall posters. Each carries a “this means / this forbids” test that appears again in staff manuals (Volumes 7 and 9) and the franchise standards (Volume 12).

Safety first, without theatre. The park's safety architecture is engineered, not performed: LiPo charging happens only in the fire-isolated charging bunker within the 3.4–4.2 V/cell window; customers never set foot on a live track or field — the Tow-Truck Retrieval Protocol exists precisely so that the safety rule is more fun than the violation; telemetry kill-switches fire automatically on under-voltage or geofence breach; every zone has marshaling sightlines designed in (Volume 11). This forbids: any process that relies on customers reading signs, and any charging outside the bunker, ever.

Engineering excellence. Every fleet decision is made on serviceability and duty-cycle math, not on spec-sheet glamour. The rental construction fleet is electromechanical (lead-screw) because leak-prone hydraulics belong in the premium supervised tier; racing chassis are standardized with lap-time parity within 2–3% because fair racing is an engineering outcome; batteries rotate 3:1 because that is what the duty cycle requires. This forbids: buying any asset without a spares plan, and “personality” achieved through unequal hardware rather than tuning.

Immersion. The industrial fiction is maintained everywhere the customer looks: Shifts, not rentals; Artisans, not staff; pit stops, not battery swaps; a Toolbox Talk, not a waiver form. Immersion is the cheapest premium the park sells — it costs vocabulary and costume and returns pricing power. This forbids: breaking the fiction to save a sentence of explanation.

Accessibility. A seven-year-old, a wheelchair user, and a forty-year veteran hobbyist must all have a genuine version of the experience. Entry pricing (\$15 Casual Shift) sits at cinema-ticket level; control stations are designed for seated and standing operators; machine classes are tiered so novices succeed in minutes while experts chase ceilings for years. This forbids: pricing or designing the park as a hobbyist clubhouse.

Skill progression. The park teaches, measures, and certifies. Every customer has a License trajectory; every Shift produces skill telemetry; the RC Academy (Phase 3) formalizes the curriculum. Progression is the bridge from entertainment revenue to education revenue and from customer to community. This forbids: any experience design in which a tenth visit feels the same as the first.

3.3 Organization and people

The values only exist if the organization is shaped to enforce them. The Phase 1 structure is deliberately flat: a **General Manager** accountable for the P&L; a **Works Lead** (chief Artisan) accountable for fleet availability and the maintenance program; two **Division Leads** (Motorsport; Construction) accountable for their zones' revenue, safety, and customer experience; a part-time **Finance/Administration** function that leans heavily on RC WORLD OS automation; and the front line — Artisans, marshals, and front-of-house, most cross-trained across at least two roles. Phase 2 adds Aviation and Marine leads plus an F&B manager; Phase 3 adds an Academy lead and a franchise-support function. The founder holds strategy, brand, product (RC WORLD OS roadmap), and this document.

Three people-principles carry more weight than the org chart. First, **hire wrenchers who like people, not greeters who tolerate machines** — the Artisan role is the park's brand in human form, and the hiring bar (Volume 7, Chapter 1) tests mechanical aptitude and hospitality together. Second, **the induction is the plan**: every hire's training curriculum is drawn from the volumes named in their role's reading path (Volume 0, Chapter 3), which keeps the documentation honest — if a volume cannot train a new hire, it gets revised. Third, **progression mirrors the customer's**: staff hold internal License tiers and equipment endorsements in the same system customers use, which builds credibility on the floor and gives the franchise system (Volume 12) a portable, verifiable staff-certification standard from day one.

3.4 Operating doctrine

Beneath the values sits a short list of non-negotiable doctrine, inherited from the founding engineering analysis and binding on every volume: the 1:3 excavator-to-dump-truck ratio; the 15° haul-road maximum incline; the 3:1 battery-to-vehicle ratio inside the 3.4–4.2 V/cell window with bunkered charging; the 20-minute Shift decoupled from battery life with ~30% buffer returned; telemetry kill-switch on under-voltage or geofence breach; no customer on a live field, ever; electromechanical rental fleet with hydraulics reserved for premium supervised experiences; standardized racing chassis with parity within 2–3%. Any proposal that contradicts doctrine must amend the style guide first — which is to say, must convince the founder and be swept through every volume.

4. Business Model

4.1 The high-utilization thesis

The business model rests on one arithmetic fact: **hobby-grade RC equipment has become so cheap, durable, and repairable that its rental payback period is measured in days, not years.** A WLtoys-class racing buggy wholesales at \$60–90. It earns \$15 per 20-minute Casual Shift. Eight paid Shifts — one busy Saturday afternoon — repay the vehicle. A Huina-class excavator at \$350–420 wholesale repays in under thirty premium-class Shifts. Compare any other equipment-rental business: karts repay in months, boats in seasons, aircraft in years. The corollary is that **utilization, not asset cost, is the entire game**, and every proprietary system in the park exists to raise utilization or lower the cost of sustaining it:

- **RC WORLD OS** raises utilization by eliminating queue dead-time (live queue roster, T-5-minute Top-Up offers when no one waits), by dynamic scheduling of fleet across zones, and by making booking frictionless on Android and iOS.
- **The RCW Node** lowers sustaining cost by preventing the two great killers of RC fleets — over-discharged LiPo cells and lost/abused vehicles — via automated under-voltage response (push alert, throttle limited to 20%, order back to pit, kill-switch at the 3.4 V/cell floor) and geofencing.
- **The Works** lowers sustaining cost by converting maintenance from emergency to schedule: predictive flags from logged runtime hours, a spares inventory shaped by failure data, and Artisans trained to rebuild rather than replace (Volume 7).
- **The gamification layer** raises utilization on the demand side by manufacturing return visits (Section 4.4).

4.2 Nine revenue engines

#	Engine	What it is	Anchor pricing (Year 1)	Character
1	Shift rentals	The core: 20-minute operating blocks across all divisions	Casual \$15 / \$22 premium; Operator Shift \$26 / \$38; Day Pass \$59; Family bundle \$89	Largest engine (~55–60% of Year-1 revenue); weekend-peaked
2	Memberships	Monthly programs with included Shifts, discounts, priority booking	Apprentice \$29 / Operator \$59 / Foreman \$99 per month	Recurring revenue; smooths seasonality; grows with License program
3	Corporate events	Exclusive-use team events, mine-operations challenges, race series	From \$1,400 (2 h, 20 pax)	High margin; weekday utilization filler
4	Birthday parties	Packaged parties with dedicated Artisan host	From \$349 (10 children)	Volume driver; feeds family memberships
5	Education	School programs, holiday camps, STEM curriculum; RC Academy from Phase 3	Program-priced	Weekday daytime utilization; grant-eligible

#	Engine	What it is	Anchor pricing (Year 1)	Character
6	F&B	Kiosk (Phase 1) → full restaurant (Phase 2)	Venue-typical	Margin capture on dwell time
7	Retail	Curated RC starter equipment, parts, branded merchandise	Keystone-plus on merch	Converts inspired visitors into hobbyists
8	Repairs & services	The Works services customer-owned equipment	Bench rate + parts	Monetizes workshop slack; builds hobbyist loyalty
9	Sponsorship	Zone naming, fleet livery, event series sponsorship	Negotiated	Pure margin; grows with audience (Chapter 10.5)

Franchising (Volume 12) becomes the tenth engine when the Phase 3 pilot validates the package: fees and royalties on replicated sites, with RC WORLD OS multi-site support already architected (Volume 13).

The engines are designed to feed each other rather than merely coexist, and the interlocks are worth naming because they are where the model's compounding lives. Parties recruit first-time visitors at negative acquisition cost; first-time visitors who start a License become membership prospects; members generate the weekday base-load that makes corporate-event capacity schedulable; corporate guests return as family customers; the retail counter converts inspired visitors into equipment owners, who become The Works' repair customers and the hobbyist community's next members; and every leaderboard season and event weekend produces the social-media material that fills the next season's parties. The F&B and retail lines are deliberately modeled at modest capture rates — they are dwell monetization, and dwell is manufactured by everything else on the list.

4.3 Unit economics of one vehicle-hour

The atomic unit of the business is one vehicle operating for one hour — three 20-minute Shifts. The table below shows the standard-class racing case at Year-1 anchor prices; Volume 10 carries the full class-by-class model and the canonical figures.

Line	Basis	Per vehicle-hour
Revenue potential	3 Casual Shifts × \$15	\$45.00
Battery amortization	~\$18-25 LiPo pack, ~200-300 cycles, 3 packs/hr	≈ \$0.25
Maintenance & parts	Fleet-average consumption of tires, gears, arms, servos (Volume 7 failure data)	≈ \$1.80-3.00
Vehicle depreciation	\$60-90 wholesale plus hardening, over a 1,200-Shift service life with two rebuilds (Volume 3)	≈ \$0.35-0.50
Charging energy	3 × ~25 Wh at commercial tariff	< \$0.05
Direct labor share	Artisan + marshal time allocated across concurrent vehicles	≈ \$3.50-5.00
Direct margin	at full utilization	≈ \$36-39 (80-87%)

The construction classes run the same arithmetic with different shapes. A standard rental excavator (\$350–420 wholesale, \$22 premium Casual Shift) earns \$66 per vehicle-hour potential against even lower drivetrain wear — lead-screw machines working soil at walking pace shed almost no consumables — but adds a share of the Mining Zone’s supervision labor. The single hydraulic showcase machine sits outside rental arithmetic entirely: it earns through supervised premium experiences and display value, and its economics are evaluated as marketing spend with revenue offset, not as fleet (Volume 4, Chapter 3).

Two honest caveats govern the reading of this table. First, no fleet runs at full utilization: park-wide effective utilization of theoretical Shift capacity is modeled in single digits for Year 1 (weekend peaks of 40–60% on popular classes, long weekday troughs), which is why fixed costs — rent, salaries, insurance, marketing — dominate the P&L and why break-even arrives in operating Month 16–19 rather than Month 3. Second, premium classes invert some lines: a \$22 premium Shift on a hydraulic showcase machine carries higher supervision labor and maintenance, which is exactly why doctrine confines hydraulics to supervised premium duty. The strategic point survives both caveats: **every incremental Shift above the fixed-cost floor is ~80% margin**, so the entire management system — pricing, gamification, events, memberships, marketing — is aimed at filling troughs.

Investor Note. The model’s leverage is visible in one sentence: the Year-1 revenue line averages ≈\$3,500 per calendar day — the equivalent of roughly 230 Casual Shifts at the standard anchor price, before any contribution from events, memberships, F&B, or retail is counted — against a theoretical Phase 1 Shift capacity more than an order of magnitude higher. The base case does not require the park to be busy; it requires it to be found. Sensitivities on capture rate and per-capita spend are in Volume 10, Chapter 9.

4.4 Gamified mechanics as retention drivers

Location-based entertainment lives or dies on repeat visitation, and RC WORLD’s retention machinery is built into the product rather than bolted on as a coupon program:

- **The RC WORLD License** (Volume 9) — a tiered driver-license progression with levels earned through performance and repeat visits, and **equipment endorsements** (excavator, drift, FPV, crawler...) that unlock machine classes. Status loss-aversion is the strongest retention force in gamified systems; the License makes every lapsed month feel like surrendered progress.
- **Badges and skill telemetry.** The RCW Node scores what customers actually do — smoothness, lap consistency, clean loads, winch recoveries — so badges certify real skill, not attendance. Real certification is shareable, and shareable status is free marketing.
- **Gears**, the loyalty currency, earned on spend, performance, and milestones, redeemable against Shifts, upgrades, and retail — designed with expiry pressure and member multipliers.
- **Leaderboards and seasons.** Every zone has a live leaderboard (tonnage, lap time, harvest score); seasons reset competition and give lapsed customers a fresh start, which re-acquisition marketing exploits.
- **The Tow-Truck Retrieval Protocol** — the signature conversion of friction into gameplay. A breakdown triggers an 85 dB locator buzzer; the customer hands in their transmitter and pilots a 1/10 winch-equipped recovery crawler to rescue their own vehicle, then receives a fresh machine. The park’s worst moment becomes its most-shared one.

- **F1-style power management.** Telemetry reads voltage sag in real time; aggressive throttle drains the allocation faster and smooth drivers are rewarded — a skill layer that deepens with every visit and directly protects battery life. The mechanics that retain customers are the same mechanics that protect assets.

Because every mechanic runs through RC WORLD OS, retention is measured, not asserted: the system attributes each return visit to the mechanics that plausibly drove it (an expiring Gears balance, a near-complete badge, a season reset, a league fixture) and reports the 90-day second-visit rate and License progression velocity on the management dashboard (Section 8.5). The gamification layer is a tunable system with a feedback loop — under-performing mechanics get redesigned mid-season, not post-mortemed annually. Volume 9 specifies the rules; Volume 13 the instrumentation.

4.5 Pricing architecture

The price ladder is designed around three principles. First, **the entry price is an impulse price**: \$15 for a Casual Shift sits at cinema-ticket level, below the psychological threshold at which a family debates the purchase, and low enough that trying RC WORLD requires no commitment to the hobby. Second, **every rung up the ladder buys more immersion, not just more time**: the Operator Shift (\$26/\$38) adds the pit-stop battery swap — theatre that costs the park ninety seconds of Artisan time and is remembered as the highlight; premium classes (\$22 Casual) buy hydraulics, FPV, or big crawlers under closer supervision; the Day Pass (\$59) and Family Bundle (\$89) convert single-activity visitors into full-park explorers, which is where cross-division discovery (and second-visit intent) is manufactured. Third, **the ladder tops out in recurring revenue**: memberships (Apprentice \$29, Operator \$59, Foreman \$99 per month) bundle included Shifts, booking priority, Gears multipliers, and License acceleration, priced so that the break-even visit frequency is two to three visits a month — below what an engaged License-chaser actually does. Local price adaptation is a Volume 10 sensitivity exercise and a Volume 12 franchise-localization procedure; the ratios between rungs, not the absolute dollars, are the protected design.

Investor Note. The pricing anchors are Year-1 figures chosen for penetration, not for margin maximization. Volume 10 models a Year-2 price review with modest uplift on premium classes and events once the demand curve is observed. No revenue upside from that review is in the base case.

4.6 Demand shaping and seasonality

An outdoor venue's enemy is the empty Tuesday, and the model addresses troughs structurally rather than hoping marketing fills them. Weekday daytime belongs to **education** (school programs, from Phase 1 informally and Phase 3 via the Academy) and **corporate events** (from \$1,400 for two hours, booked precisely because the park is exclusive-feeling when quiet). Weekday evenings grow into **league nights** and member sessions (night racing from Phase 3 extends this year-round). Weekends run the full retail engine, with the live queue roster and Top-Up mechanics harvesting every idle vehicle-minute. Weather resilience improves by phase — the Construction Division is the most rain-tolerant anchor (earthmoving in light rain is more atmospheric, and the machines are rated for it within limits set in Volume 4), Phase 2's restaurant monetizes shelter, and Phase 3's arena removes the ceiling entirely. RC WORLD OS automates weather-triggered rebooking so a rained-out Shift becomes a banked credit plus a bounce-back offer, not a refund.

4.7 Cost structure in brief

The Year-1 cost structure (detail: Volume 10, Chapter 5) is dominated by people ($\approx 35\text{--}40\%$ of revenue at maturity: roughly 18–19 FTE — 18.5 in the Volume 10 payroll model — plus casuals: General Manager, division leads, Artisans, marshals, front-of-house), then lease and utilities, then marketing (front-loaded in Years 1–2), then fleet maintenance and parts (remarkably small — single-digit percent of revenue — because the fleet doctrine was designed backwards from this line), then insurance, payments, and platform costs. The capex the customer never sees — civil works, netting, ponds, buildings — is what the funding ask actually buys; the machines themselves are $\sim 4\%$ of Phase 1 capex (\$80 K for the entire ~ 150 -asset fleet, batteries and spares included) — a consumable in the fullest sense.

Labor deserves the extra sentence because it is the largest line and the most commonly mis-managed in venue businesses. The staffing model (Volume 9 for front-of-house, Volume 7 for The Works) is built on cross-training: every marshal completes Artisan basic training, every Artisan can run a division's customer operation, and RC WORLD OS's role-based interfaces mean a staff member switching roles switches apps, not systems. Because telemetry supervises what human eyes once had to — battery states, geofences, session clocks — the park's staff-to-concurrent-customer ratio is materially better than attraction-industry norms, and the staff the park does employ spend their time on the two things software cannot do: hospitality and wrenching.

4.8 Anatomy of one strong Saturday

To make the model concrete, the table below decomposes a strong (not record) Year-2 Saturday — the kind the base case needs roughly forty of per year, supported by far quieter weekdays. Figures are illustrative composites consistent with the Volume 10 model, not additional canon.

Revenue stream	Volume	Revenue
Casual Shifts (standard, \$15)	210	\$3,150
Casual Shifts (premium, \$22)	45	\$990
Operator Shifts (\$26/\$38 blended)	60	\$1,780
Top-Up Sessions	35	$\approx \$400$
Day Passes (\$59) & Family Bundles (\$89)	22 + 18	\$2,900
Birthday parties (from \$349)	3	$\approx \$1,150$
F&B	~ 600 visitors $\times$ modest capture	$\approx \$2,100$
Retail & repairs	—	$\approx \$700$
Day total	≈ 600 visitors	$\approx \\$13,200$

The instructive lines are the small ones. Top-Ups are revenue that did not exist until the T-5-minute queue query made idle minutes sellable. The parties line looks minor but seeds the next cohort of members — party guests are first-time visitors acquired at negative marketing cost. And the whole table rides on perhaps 350–400 vehicle-hours of fleet time, a fraction of what the ~ 150 -asset fleet could physically supply: the ceiling above a strong Saturday is high, which is precisely the utilization headroom the thesis monetizes as the brand compounds.

5. The Park at a Glance

5.1 Site

4.8 hectares (~12 acres) of leased peri-urban land in a mid-size international city: flat with 2–4% natural grade (useful fall for drainage and haul-road design), road frontage for visibility, municipal power and water. Volume 11 carries the site-selection criteria, master plan drawings, and civil specification; Volume 10's sensitivity analysis covers local adaptation of the financial model, and the same method localizes franchise sites (Volume 12).

5.2 Divisions and zones

Division	Zones / assets	Phase	Signature experience
Motorsport	Track A (asphalt/drift), Track B (baja/rally), Track C (crawler trail) + Crawler Park	1	Timed racing on standardized fleets, parity within 2–3%; drift and rally classes; championships
Construction	Mining Zone (open-pit circuit, haul roads $\leq 15^\circ$, processing hopper); Agriculture Zone (field grid, barn, implements)	1	Load, haul, and dump real aggregate; live tonnage leaderboard; farm operations with tandem trailers
Aviation	Netted airfield: 30 m paved runway, elevated helipads, high-tensile flyaway netting	2	Fixed-wing, heli, and FPV flight inside a fully enclosed volume
Marine	Pond complex	2	Scale vessels, fast electrics, sailing one-designs
The Works	Central engineering & workshop facility, pit lane, charging bunker	1	Visible engineering: customers watch Artisans rebuild the fleet
Commercial	Entry pavilion, F&B kiosk → restaurant (Ph 2), retail, grandstand (Ph 2), events lawn (Ph 2), indoor arena + RC Academy + corporate centre (Ph 3)	1–3	Everything that monetizes dwell

5.3 Headline fleet (Phase 1 baseline, ~150 powered assets)

Category	Count	Composition
Motorsport cars	64	24 touring/GT (1/14 entry + 1/10 standard), 12 drift (MST-based), 12 buggy/rally (WLtoys 144010-class, upgraded), 8 short-course, 8 formula/drag special-event
Construction & mining	30	6 excavators (Huina 1580 class), 18 dump trucks (the 1:3 ratio), 3 wheel loaders, 2 dozers, 1 premium hydraulic showcase machine (Kabolite class)
Agriculture	12	Double E E351-class 1/16 tractors + implement library

Category	Count	Composition
Crawler park	16	MN99S / WPL C24 class + 4 premium 1/10 TRX-4 class
Recovery fleet	4	1/10 winch-equipped recovery crawlers (Tow-Truck Retrieval Protocol)
Utility / marshal	balance	Marshal and course vehicles

Phase 2 adds ~20 aircraft and ~24 vessels. Batteries follow doctrine: 3:1 per vehicle minimum, 2S/3S LiPo, XT60 standard. Full specifications: Volumes 3–6; purchasing detail: Volume 8.

The physical arrangement (master plan drawings: Volume 11) follows three layout rules. The entry pavilion sits on the road frontage with The Works directly behind it, its workshop window facing the main circulation spine — engineering as spectacle is a deliberate merchandising decision. The two Phase 1 anchor zones flank the spine so that every visitor walks past both regardless of what they booked; the Phase 2 pond and airfield occupy the site’s quieter rear half, where netting height and water works have room. And the 2–4% natural grade is used, not fought: haul roads climb with it, drainage follows it, and the grandstand (Phase 2) sits on the high side where sightlines come free.

5.4 The customer journey in brief

Before arrival: book in the app (Android/iOS), complete the **Toolbox Talk** digital induction — an interactive module on battery limits, collision liability, and track rules that doubles as the signed waiver. **On arrival:** check in at the entry pavilion; the app wallet holds Shifts, Gears, and License status. **The Shift:** scan the QR on an assigned transmitter to bind the vehicle; drive from a designated operator station with sightlines engineered per zone; live dashboard shows battery, time, and score. **Mid-Shift events:** at T-5 minutes, if no one queues for the class, a Top-Up button offers instant extension; a breakdown triggers the Tow-Truck Retrieval Protocol; an Operator Shift gets its pit-stop battery swap with full “refueling” theatre. **After:** results, badge progress, and License advancement land in the app; F&B and retail capture the dwell; the departure prompt is always next time. Full journey mapping, service standards, and failure-mode recovery: Volume 9.

5.5 What the customer never sees

Behind the fiction: the charging bunker cycling hundreds of packs a day inside the 3.4–4.2 V/cell window; the Wi-Fi mesh (outdoor enterprise APs) that RCW Nodes hand off between without dropping telemetry; RC WORLD OS pushing under-voltage alerts and predictive-maintenance flags to Artisan tablets; the spares wall at The Works stocked to the failure-rate data in Volume 7; and the nightly reconciliation in which every asset’s runtime hours, revenue, and maintenance state agree with the ledger — because the ERP that runs the leaderboard also runs the payroll (Volume 13).

5.6 Who comes, and what each division earns

The demand model (Volume 2, Chapter 7) segments the catchment into five customer groups, each with a primary division affinity and a distinct economic signature:

Segment	Primary affinity	Economic signature
Families with children 6–14	Construction, Crawler Park	Bundles and parties; highest F&B capture; membership conversion via children's License progression
Adult hobbyists & enthusiasts	Motorsport, Aviation (Ph 2)	Highest visit frequency; membership and league revenue; retail and repair customers; community anchors
Young adults / social groups	Drift, FPV (Ph 2), night racing (Ph 3)	Evening and weekend peaks; social-media reach; event ticket buyers
Corporate & organized groups	Construction (team formats), full-park events	Weekday utilization; highest per-booking value; repeat annual bookings
Schools & education	Agriculture, Academy (Ph 3)	Weekday daytime; prepaid programs; long-cycle but highly durable revenue

Division revenue contributions shift by phase: in Year 1, Motorsport and Construction carry effectively all Shift revenue in roughly comparable shares — Motorsport on throughput, Construction on premium pricing and dwell; by Year 3, Aviation and Marine add breadth (each modeled to cover direct costs quickly but neither assumed to match the anchors), while F&B, events, and memberships grow their combined share of total revenue materially. The full mix model, by month and by division, is Volume 10, Chapter 4 — and the honest summary is that the two Phase 1 anchor divisions must succeed for the park to succeed; everything after them is diversification, not rescue.

5.7 Operating calendar

The park operates year-round with a season-shaped schedule: extended hours and full staffing across weekends, school holidays, and the peak season; a leaner weekday pattern built around bookings (school programs, corporate events, member sessions) rather than walk-in traffic; and one scheduled weekday closure per week in the off-season for deep maintenance — the day The Works catches up on the preventive-maintenance matrix without competing with revenue (Volume 7, Chapter 5). Evening operation arrives with Phase 3's lighting; until then, seasonal daylight sets the closing bell. Weather closures follow published thresholds per division (wind limits for aviation, standing-water limits for tracks, lightning protocol park-wide — division volumes, Safety chapters), with RC WORLD OS handling automated re-booking. The events calendar — league nights, championships, harvest festivals, corporate seasons — is planned annually and is a first-class revenue plan, not an afterthought: Volume 9, Chapter 7 carries the template calendar.

5.8 Safety architecture at a glance

Safety at RC WORLD is layered so that no single failure — human or technical — can produce a serious outcome. Layer one is **design**: customers operate from stations physically separated from live fields; spectator areas are barriered; the airfield is fully netted; the charging bunker is fire-isolated construction. Layer two is **telemetry**: every vehicle's RCW Node enforces geofences and the battery window automatically, and any marshal or Artisan can kill any vehicle from the app. Layer three is **procedure**: the Toolbox Talk gates first use; marshals hold defined per-zone authority; SOPs and emergency procedures (fire, injury, severe weather, lost child) are drilled per the schedules in each division volume. Layer four

is **culture**: the incident and near-miss log in RC WORLD OS is blame-free and reviewed weekly, because a park that hides its near-misses schedules its accidents. The insurance program sits behind all four layers, specified in Volume 10.

6. Investment Opportunity

6.1 The ask

RC WORLD seeks **\$2.6 M** to fund Phase 1 in full and carry the business to sustained monthly break-even without a further raise. The structure below follows Volume 10, Chapter 3 exactly; final instruments (equity, convertible, or blended with asset-backed debt) are settled in term sheets with Volume 10 as the modeling basis.

Component	Amount	Notes
Phase 1 capital expenditure	\$1,850 K	Section 6.2 and Volume 10, Chapter 2. Pre-opening ($\approx$ \$162 K) and 10% contingency ($\approx$ \$168 K) are capitalized inside this figure
Working capital at opening	\$130 K	Opening stock, consumables float, deposits, till/wallet float
Operating funding through break-even	\$350 K	Funds the operating trough to break-even (operating Month 16–19) plus sustaining capex, with margin for a slow ramp
Corporate, legal & transaction costs	\$90 K	Entity setup, raise legals, audit/tax setup, pre-opening corporate overhead
Unallocated reserve	\$180 K	Board-controlled; released only against Volume 10's downside triggers
Total	\$2.6 M	Three tranches (Section 6.3). The four non-capex lines sum to the \$750 K opening cash position in Volume 10's monthly model

Phases 2 and 3 ($\approx$ \$1.15 M and $\approx$ \$0.9 M) are funded from operating cash flow, with an optional \$600 K equipment-and-fit-out facility for Phase 2 (Volume 10, §3.5). The \$2.6 M ask therefore buys the whole plan's option value, not just Phase 1: no further mandatory capital calls are modeled in the base case.

6.2 Use of funds (Phase 1 capex, $\approx$ \$1.85 M)

Category	Amount	%	What it buys
Site works & earthworks	\$455 K	24.6%	Bulk earthworks, Mining Zone pit and haul roads ($\leq 15^\circ$), Agriculture grid, drainage, roads/parking, fencing, survey and design

Category	Amount	%	What it buys
Track & operating surfaces	\$172 K	9.3%	Track A asphalt; Track B dirt, jumps and berms; Track C stone/bridges/water; pit lane surfaces
Buildings	\$418 K	22.6%	Entry pavilion (reception, briefing room, F&B kiosk, retail), The Works, fire-rated charging bunker
Fleet acquisition	\$80 K	4.3%	All ~150 powered assets landed, 450 batteries (3:1), chargers, transmitters, ~90-day spares
RCW Node telemetry program	\$25 K	1.4%	Production run and fleet-wide fitment
Wi-Fi mesh & IT infrastructure	\$87 K	4.7%	Outdoor mesh, timing systems, POS hardware, comms backbone
RC WORLD OS v1 development	\$165 K	8.9%	Bookings, wallet, telemetry, fleet management, POS, finance core
FF&E, workshop equipment & theming	\$118 K	6.4%	Workshop tooling, furniture, zone theming
Pre-opening (capitalized)	\$162 K	8.8%	Hiring and training, launch marketing, licences
Contingency @ 10.0%	\$168 K	9.1%	Drawn only against Volume 11's change-control process
Total Phase 1	\$1,850 K	100%	Derived line-by-line in Volume 10, Chapter 2

Two allocation facts deserve an investor's attention. First, well over half of capex — earthworks, surfaces, and buildings, ≈\$1,045 K — is built environment: 10-20-year assets with residual value largely independent of the operating concept. Second, the entire powered fleet is **4.3%**: \$80 K buys all ~150 vehicles including their 450 batteries, radios, and ninety days of spares — less than half the cost of the software that manages them. That inversion is the whole point: the moat is the built environment, the telemetry, and the operating system; the machines customers crash are a consumable replaced from operating cash flow (Volume 8), and no plausible fleet-cost shock can move the capital plan by even two percent. The capital's risk profile is closer to a small leisure-property development than to a technology venture.

Equally important is what the money does not buy. It buys no land (leased, per canon — the capital works the operating business, not a property position). It funds no speculative technology: the RCW Node is a completed design on commodity components, and the RC WORLD OS line completes a specified v1 whose end-to-end beta must be demonstrated before the second tranche releases. It carries no Phase 2 or 3 commitment — those are funded from operations and gated on evidence. And it pays no founder liquidity: the raise is entirely primary capital deployed into the asset.

6.3 Staged tranches

Capital is drawn in three milestone-gated tranches per Volume 10, §3.2, so investors never fund the next stage of risk before the previous one has been retired. Gates are objective and verifiable in a site visit plus an RC WORLD OS data pull; Volume 10 carries the full gate definitions and remediation ladder.

Tranche	Amount	Released at	Milestone gate (all must be true — Volume 10, §3.2)	Risk retired
T1	\$1.2 M	At close (project Month 0, ~12 months before opening)	Lease executed; planning permits granted; civil tender within budget; GM contracted	Site & permitting
T2	\$850 K	≈ operating Month –5 (~project Month 7)	Buildings weather-tight; Tracks A/B/C substantially complete; ≥80% of fleet landed and commissioned; RC WORLD OS beta end-to-end (booking → telemetry → kill-switch demo); opening date fixed	Construction & technology
T3	\$550 K	Opening month (operating M1)	Soft opening complete; safety sign-off (Volume 7 audit); first 1,000 paid Shifts sold; insurance in force	Launch

Drawdown mechanics are conventional and specified in the subscription documents: gate evidence is compiled by management, independently verified (quantity surveyor for construction; OS data exports for operations), and put to the board; a failed gate pauses the tranche and triggers Volume 10's pre-agreed remediation ladder (descope, re-tender, or — worst case — orderly wind-down with land improvements as residual value). Undrawn tranches sit in escrow-equivalent instruments, not in the operating account.

Investor Note. The gates are deliberately harsher than convention: T3 releases only at opening, after the soft opening is complete and the first 1,000 paid Shifts have been sold — demand evidence, not just construction completion. If presales disappoint, T3 funds the corrective actions with investors' eyes open rather than being consumed by momentum.

6.4 Investor returns summary

Volume 10 carries the full model; the headline shape at the base case is:

- **Revenue:** ≈\$1.28 M (Y1) → ≈\$2.6 M (Y3) → ≈\$3.4 M (Y5), driven by phase openings, membership growth, and events.
- **Profitability:** steady-state EBITDA margin 24–28% from Year 3, implying ≈\$620–730 K EBITDA in Year 3 and ≈\$820–950 K in Year 5.
- **Break-even and payback:** monthly operating break-even operating Month 16–19; full payback Year 4–5 (base model: Month ~53 on Phase 1 capital).

- **Modeled investor outcome:** at the exit multiples used in Volume 10 (\$7.5), the base case supports $\approx 1.6\text{--}2.0\times$ **MOIC** over a five-year hold — a high-single to low-double-digit IRR. The **ambitious case reaches $\approx 2.2\text{--}2.9\times$** , stated explicitly as upside, not as the plan. The franchise system (Volume 12), second site (Chapter 10), the RC WORLD OS platform, and Year-4+ dividend capacity are further upside carried at zero.
- **Downside architecture:** the tranche gates cap construction risk; the asset base is dominated by long-life civil works; the fleet — the asset class customers actually consume — is 4.3% of capex; and Volume 10's conservative case still returns $\approx 0.9\text{--}1.1\times$, because the assets, cash, and a functioning business retain value.

A stylized worked example makes the multiple concrete (illustrative; Volume 10, \$7.5 carries the full analysis):

Line	Base-case illustration
Year-5 EBITDA (base)	$\approx$ \$930 K (27.4% of $\approx$ \$3.4 M)
Exit multiple (single-site LBE/FEC transactions, industry ranges 2024–2026)	$\approx$ 6.5–8.5 $\times$ EBITDA
Equity value at exit (incl. closing cash, facility repaid)	$\approx$ \$6.3–8.1 M
Investor share (65%, Structure A cap table — Volume 10, \$3.4)	$\approx$ \$4.1–5.3 M
Base-case outcome on \$2.6 M	$\approx$ 1.6–2.0$\times$ MOIC; $\approx$8–12% IRR
Ambitious case (explicit upside, not the base)	$\approx$ 2.2–2.9 $\times$ MOIC; $\approx$ 14–19% IRR
Excluded upside	franchise royalties, second site, platform and Academy licensing, Year-4+ dividends — all carried at zero

The returns architecture in plain speech: the base case is an infrastructure-flavored return — real assets, self-funding expansion, capped follow-on exposure, dividend capacity from Year 4 — whose risk-reward shape, not its midpoint, is the attraction: the conservative case still returns most of the capital, while the ambitious case and the franchise option carry the venture-like upside. Investors seeking more than a $\sim 20\%$ IRR should read Volume 12 and price the franchise program into their thesis; the plan refuses to sell it as the base case.

6.5 Exit options

1. **Trade sale** to a location-based-entertainment or attractions operator seeking a differentiated format — the most probable route; the park sells as a proven unit with a replication manual (this document).
2. **Sale of the franchisor entity** once the Phase 3 pilot validates the package: recurring royalty streams command higher multiples than single-site EBITDA.
3. **Secondary sale** to growth capital funding multi-site rollout, allowing early investors to exit at the format-proof point.
4. **Structured founder buyback** from free cash flow — the model's dividend capacity from Year 4 (post-facility amortization) makes a negotiated multi-year buyback feasible if no external exit is preferred.
5. **Hold for yield** — at steady state the park is a cash-yielding asset; investors preferring distributions over exit are accommodated in the shareholder agreement.

6.6 Governance and investor protections

The investment package includes: a board with investor representation proportional to the round; the tranche-gate reviews of Section 6.3 conducted as formal board decisions with independently verified gate evidence (quantity surveyor sign-off on construction; presales and soft-opening data pulled directly from RC WORLD OS); monthly reporting on the KPI set of Volume 10, §10.4, generated by the ERP's reporting layer rather than assembled by hand; standard reserved matters (budget approval, capex beyond plan, senior hires, related-party transactions, any change to canonical doctrine with financial effect); and information rights extending to read-only RC WORLD OS dashboard access — offered deliberately: a business built on telemetry should be diligenced and monitored through it (Volume 13).

7. Development Phases

The park is built in three phases over sixty months. Each phase is a complete, self-supporting park state — no phase depends on the next to justify itself — and each transition is governed by explicit gating criteria reviewed by the board. Detailed schedules and construction sequencing: Volume 11, Chapter 9; capex line detail: Volume 10, Chapter 2. Month numbering follows the canonical two-frame convention: **build phasing in project months** (the park opens at project Month 12); **financials and division opening schedules in operating months** (operating Month 1 = the opening month). Each table below labels its frame.

7.1 Phase 1 — “Core Park” (project Months 0-12, capex ≈\$1.85 M)

Scope. The two anchor divisions and everything needed to run them: the Motorsport Division (Tracks A, B, C and the Crawler Park), the Construction Division (Mining Zone and Agriculture Zone), The Works, the pit lane, the charging bunker, and the entry pavilion with F&B kiosk and small retail counter. Technology: park-wide Wi-Fi mesh, RCW Node fleet fitment, and **RC WORLD OS v1** (bookings, wallet, telemetry, fleet management, POS, finance core).

Why this scope. Motorsport is the volume engine — racing is legible to every customer and generates the highest Shift throughput per square meter. Construction is the signature — the experience no competitor offers and the park's marketing identity. Together they cover the full demographic (families, hobbyists, corporate) with the lowest-capex, lowest-regulatory-friction divisions; aviation (netting capex, airspace compliance) and marine (pond civil works) are sequenced later deliberately.

Milestones.

Project month	Milestone
0-1	Financial close (T1: \$1.2 M, gates per Section 6.3); civil contract mobilized
2-6	Earthworks and civil construction; buildings up; fleet and spares POs placed and received; RCW Node production batch
~7	T2 gate review (\$850 K, ≈ operating Month −5, gates per Section 6.3)
7-10	Zone fit-out; track surfacing; hopper and timing systems; OS v1 hardened in production; staff hiring and training (Toolbox Talk content, SOPs from Volumes 3, 4, 7)

Project month	Milestone
10-11	Soft opening (≥ 2 weekends, invited + limited public); membership and Shift presales push
12	T3 gate review at opening (\$550 K, gates per Section 6.3); grand opening — operating Month 1 begins

Phase 1 exit criteria (the Phase 2 gate; must be met before Phase 2 capital is committed): revenue $\geq 80\%$ of the Volume 10 plan for three consecutive months; 90-day second-visit rate $\geq 20\%$ (gate floor); fleet availability $\geq 90\%$ (gate floor; operating target $\geq 92\%$); membership base ≥ 400 ; safety record clean of notifiable incidents.

The twelve-month Phase 1 schedule is aggressive but achievable: the construction scope is civil works and simple single-story buildings; the fleet is bought, not developed; technology invention risk is retired early (the RCW Node is a finished design on commodity parts, and RC WORLD OS must demonstrate its end-to-end beta at the T2 gate); and the long-lead items — freight, netting (Phase 2 anyway), and permits — are either outside Phase 1 or resolved before close (permits are a T1 condition). The real exposure is weather during earthworks and contractor performance, both priced into the contingency and the fixed-price contract structure.

7.2 Phase 2 — “Full Park” (project Months 13-30, capex $\approx$ \$1.15 M)

Scope. The Aviation Division (netted airfield: 30 m paved runway, elevated helipads, full high-tensile enclosure; ~ 20 aircraft), the Marine Division (pond complex; ~ 24 vessels), the full restaurant replacing kiosk-only F&B, expanded retail, grandstand viewing on the motorsport complex, and the events lawn. RC WORLD OS v2 adds the modules the new divisions need (flight-line management, vessel telemetry adaptations) plus the CRM and events depth (Volume 13 roadmap).

Why this scope. Phase 2 converts a strong regional attraction into a full-day destination. Aviation and marine widen the experience portfolio into categories with almost no public-access competition; the restaurant and grandstand convert longer dwell into F&B revenue; the events lawn unlocks the corporate and festival formats that fill weekdays. Phase 2 is also the phase that most improves weather resilience of revenue mix — marine and F&B behave differently from racing across seasons.

Milestones.

Operating month (M1 = opening)	Milestone
M2-M12	Phase 2 design, permitting, and procurement run alongside Year-1 trading; no construction commitment before the gate
$\approx$ M12-13	Phase 2 gate review on Year-1 actuals (criteria in §7.1); optional \$600 K facility arranged (Volume 10, §3.5); pond and airfield civil works commence (segregated access, zero disruption to operating zones — Volume 11)
M13-19	Netting erection; runway paving; pond lining and filtration; restaurant fit-out; fleets commissioned; flight-line and pond SOPs live; staff cross-training
M16	Aviation staged opening begins: simulator lab (Volume 5 staging plan)

Operating month (M1 = opening)	Milestone
M19–20	Aviation fixed-wing program opens; restaurant opens
M20–24	Marine Division opens progressively (Volume 6); grandstand and events lawn complete
M23–24	Aviation full program (FPV, heli, drone racing) live; first full-park event season

Phase 2 exit criteria: aviation and marine divisions each covering their direct operating costs within two quarters of opening; park-wide revenue tracking the Volume 10 Year-3 trajectory; restaurant gross margin at industry-normal levels; no material erosion of Phase 1 division performance (cannibalization check).

Investor Note. Phase 2’s cash-flow funding is a plan, not a covenant. If Phase 1 outperforms, the board may accelerate Phase 2 with a small facility against the completed asset base rather than waiting for retained earnings; if Phase 1 underperforms, Phase 2 is deferred at near-zero sunk cost, because no Phase 2 commitment is contracted before the gate. The sequencing is designed so that the decision to spend the second million is always made with a year of real trading data in hand.

7.3 Phase 3 — “Destination & Beyond” (project Months 31–60, capex ≈\$0.9 M)

Scope. The indoor all-weather arena (year-round revenue floor, small-scale racing and crawler courses under roof); the **RC Academy** classrooms (structured curriculum, school programs, holiday camps — the education engine at full power); the corporate event centre; night-racing lighting on Tracks A and B; and the **franchise pilot** — the first RC WORLD site operated under license, using Volume 12 and the multi-site architecture of RC WORLD OS.

Why this scope. Phase 3 attacks the two structural constraints of an outdoor single-site business — weather and geography. The arena and lighting extend the operating calendar into evenings and winters; the Academy converts the License program into accredited education revenue; the franchise pilot converts the operating manual you are reading into a licensable product.

Milestones and exit criteria (project months): arena and Academy open by ≈Month 40; night racing live by ≈Month 36; franchise pilot site selected by ≈Month 40, opened by ≈Month 54 against the Volume 12 package. Each Phase 3 element must clear a ≥25% ROIC hurdle on incremental EBITDA at the operating Month-30 review (Volume 10, §10.2). Phase 3 “exit” is the five-year review: cumulative performance per Volume 10, a validated franchise unit, and a board decision on the expansion paths in Chapter 10.

Investor Note. The phase structure is an option ladder, not a treadmill. Each gate is a genuine decision point at which the board can accelerate, hold, or re-scope with current evidence. The base-case financials assume all three phases proceed on schedule; the downside case in Volume 10 shows Phase 1 standing alone as a viable, cash-generating business — smaller, but solvent.

7.4 Phase gates at a glance

Gate	Timing	Capital at stake	Decision evidence (summary — full definitions in Volume 10)	Decision maker
T1 → T2	≈ operating Month –5 (~project Month 7)	\$850 K	T2 gate per Section 6.3 (Volume 10, §3.2)	Board, on QS-verified data
T2 → T3	Opening month (operating M1 / project Month 12)	\$550 K	T3 gate per Section 6.3 (Volume 10, §3.2)	Board
Phase 1 → 2	≈ operating Month 12–13	≈ \$1.15 M (cash flow + optional \$600 K facility)	Revenue ≥80% of the Volume 10 plan for three consecutive months; 90-day second-visit rate ≥20% (gate floor); fleet availability ≥90% (gate floor); operating target ≥92%; membership base ≥400; safety record clean of notifiable incidents	Board
Phase 2 → 3	Operating Month-30 review	≈ \$0.9 M (cash flow)	New divisions covering direct costs; Year-3 trajectory on track; no anchor-division erosion; ≥25% ROIC hurdle per element (Volume 10, §10.2)	Board

Two construction principles apply across all phases and are binding on Volume 11's contractor packages: **no phase's construction may degrade an operating zone's revenue** (segregated construction access, dust and noise controls, and scheduling around peak trading are contractual requirements, not requests); and **every structure is designed for its next phase** (the entry pavilion's services are sized for the Phase 2 restaurant; conduit runs anticipate Phase 3 lighting; the pond's edge design anticipates the grandstand sightline). Retrofitting is the most expensive construction there is, and the master plan exists to avoid buying anything twice.

7.5 The pre-opening critical path

Within Phase 1 (all months here are project months), four workstreams must converge in Months 10–12, and the schedule risk lives in their interaction rather than in any one of them. **Civil and buildings** is the longest pole and the least compressible; it is contracted fixed-price with milestone payments aligned to the tranche gates. **Fleet and spares** is the shortest but has ocean freight in it: purchase orders per Volume 8 are placed by Month 2 so that vehicles arrive in Month 4–5, leaving time for RCW Node fitment, QA, and the 3-deep spares wall before staff training needs working machines — ≥80% of the fleet landed is itself a T2 condition. **Technology** must demonstrate the RC WORLD OS end-to-end beta at the T2 gate, roughly five months before opening, so the final stretch is hardening, mesh commissioning, and load testing — integration, not invention. **People** starts hiring at Month

5, trains against the volumes from Month 7, and runs the soft opening as its graduation exercise. The soft opening and presales push are the deliberate collision point where all four streams meet under low stakes, generating the evidence the T3 gate requires — soft opening complete, safety sign-off, and the first 1,000 paid Shifts sold. Grand opening follows only when the punch list is closed — an opening delayed two weeks costs less than a launch that teaches the catchment the park isn't ready.

8. Financial Overview

This chapter carries only the canonical headline figures. **Volume 10 is the source of truth** for every financial number in this plan; nothing here may be cited against it, and any apparent conflict is resolved in Volume 10's favor.

8.1 Five-year headline trajectory (base case)

Year	Revenue	EBITDA margin	Commentary
Y1	≈ \$1.28 M	negative → approaching break-even	First operating year; marketing-heavy; monthly break-even reached operating Month 16–19 (i.e., during Y2)
Y2	between Y1 and Y3 on the ramp	low single-digit → low-teens	First full year of Phase 1 at maturity; Phase 2 construction in background; membership base compounding
Y3	≈ \$2.6 M	≈ 24–28% (steady state reached)	Full Park operating (aviation, marine, restaurant, events lawn); events and membership engines mature
Y4	between Y3 and Y5	24–28%	Arena and Academy revenue begins; night racing extends the calendar
Y5	≈ \$3.4 M	24–28%	Destination state; franchise pilot revenue begins (royalties modeled at zero in base case)

A note on how these figures were built, because method matters more than the numbers. The model is assembled bottom-up in Volume 10: Shift capacity per zone per operating day, shaped by the calendar in Section 5.7; capture and repeat-visit rates benchmarked against family-entertainment norms rather than the plan's own gamification ambitions; per-visitor F&B and retail capture at modest venue-typical rates; membership growth as a conversion funnel from visitation; and events revenue built from a bookings calendar, not a percentage of the total. Where a judgment call had two defensible answers, the model takes the lower. The interim years (Y2, Y4) are shown as ranges on the ramp deliberately — false-precision midpoints for transition years invite anchoring on figures that phase timing will inevitably move; Volume 10 carries the monthly model.

8.2 Break-even and payback

Monthly operating break-even — the month from which monthly EBITDA is positive and stays positive — is modeled at **operating Month 16-19** (base model: Month 16). The ask carries the business through the top of that range with margin: the four non-capex components of Section 6.1 place **\$750 K of cash at opening**, and the monthly model never breaches the **\$250 K minimum-cash policy floor** (Volume 10, §6.4). Full payback occurs in **Year 4-5** — the base model crosses at **Month ~53** on Phase 1 capital; the ambitious case pulls payback into Year 4. The sensitivity drivers, in order of impact: Year-1 demand capture, average revenue per visitor, and the Phase 2 opening date; Volume 10, Chapter 9 quantifies each.

8.3 What makes the margin

Steady-state EBITDA of 24-28% is a structural outcome of decisions documented elsewhere in this plan, and diligence should test those decisions rather than the percentage: the near-zero marginal cost of an incremental Shift (Section 4.3); a maintenance line engineered down by the electromechanical fleet doctrine and the spares-and-rebuild program (Volume 7); telemetry that prevents the battery and asset losses that quietly bleed rental fleets; labor productivity from RC WORLD OS automation (one Artisan supervises multiple concurrent Shifts because the software watches every battery); and a revenue mix that shifts toward high-margin memberships, events, and sponsorship as the park matures.

8.4 Funding recap

\$2.6 M ask; three tranches (\$1.2 M / \$850 K / \$550 K) gated per Section 6.3; Phases 2-3 funded from operating cash flow, with Phase 2 backed by an optional \$600 K equipment-and-fit-out facility analyzed, but not required, in Volume 10, §3.5; five-year total capex ≈\$3.9 M.

8.5 The numbers management actually runs on

The P&L is a lagging indicator; the park is managed on leading indicators that RC WORLD OS computes continuously and reports monthly to the board (definitions and targets: Volume 10, §10.4):

KPI	Why it leads the P&L
Paid Shifts per operating day (by division and class)	The direct revenue driver; the trough-filling scoreboard
Effective fleet utilization (paid Shifts ÷ available Shift capacity)	Measures whether demand shaping (Section 4.6) is working
Fleet availability %	Gate floor ≥90%; operating target ≥92% Y1, ≥94% Y3 (Volume 10 KPI dashboard). Below target, revenue is lost to the workshop; triggers Volume 7 escalation
Average revenue per visitor (Shift + F&B + retail)	The dwell-monetization health check
Member count, churn, and included-Shift redemption rate	The recurring-revenue base and its true cost
90-day second-visit rate; License progression velocity	The retention machinery's output. Operating target ≥22% Y1 rising to ≥28% Y3; Phase 2 gate floor ≥20%
Battery packs retired per 1,000 Shifts; maintenance cost per Shift	The doctrine-compliance meters; drift here predicts margin erosion quarters early
Marketing cost per first visit	Acquisition efficiency; front-loaded spend must show declining unit cost through Year 2

Investor Note. Every figure in this table is computed from primary system data — bookings, telemetry, POS, and maintenance logs in the same database — not assembled from spreadsheets. Diligence can and should test the model’s assumptions against these definitions, because post-investment reporting will use exactly them.

8.6 Sensitivity headlines

Volume 10, Chapter 9 carries the full sensitivity and scenario analysis; the shape of it belongs here. The model is **most sensitive to Year-1 demand capture**: a ramp 25% below base pushes break-even toward the far end of the operating Month 16–19 band and consumes much of the operating funding and reserve (which is what they exist for), while the ambitious case pulls Phase-1 payback to Month ~44 (Year 4). It is **moderately sensitive to average revenue per visitor** (the F&B and retail capture assumptions are deliberately modest, so the risk is asymmetric to the upside) and to the **Phase 2 opening date** (a six-month slip defers, but does not destroy, the Year-3 revenue step). It is **notably insensitive** to the variables outsiders usually worry about: fleet cost inflation (the fleet is 4.3% of capex and wholesale prices have been deflating for a decade), battery costs, and energy prices. The downside scenario — Phase 1 standing alone, Phases 2–3 never built — models as a smaller but solvent business, which most distinguishes this plan from attraction developments whose economics only work at full build-out.

9. Risk Summary

9.1 Scope and method

The full risk registers — division-level operational risks, construction risks, market risks, technology FMEAs — live in the specialist volumes (Volume 2 Chapter 9; Volumes 3–6 per division; Volume 7 for engineering FMEA; Volume 10 for financial sensitivities; Volume 11 for construction; Volume 13 for technology and data). The table below is the board-level top ten: the risks that could materially impair the investment case, with owner and mitigation. Scoring: likelihood × impact on a 1–5 scale, assessed July 2026, reviewed quarterly.

One methodological note: mitigation and insurance are different instruments for different layers. Mitigation reduces likelihood or impact through design and procedure and is the primary defense for every risk below; insurance transfers the residual financial tail (program specified in Volume 10) and is never accepted as a substitute for a design fix. A risk “mitigated” only by an insurance policy would fail this plan’s standard.

9.2 Top-10 risk register

#	Risk	L	I	Score	Mitigation	Owner
1	Year-1 demand ramp slower than modeled	3	5	15	\$750 K non-capex funding at opening incl. \$180 K board-controlled reserve (Vol 10, \$3.1); T3 releases only against presales evidence; front-loaded, measurable marketing (Vol 2, Vol 9); pricing flexibility; cost-flex ladder (Vol 10, \$6.5)	Founder/GM
2	Construction cost overrun or delay (Phase 1)	3	4	12	Fixed-price/GMP civil contract at T1; 10% contingency capitalized inside Phase 1 capex; staged tranches; standard buildings, no bespoke structures in Phase 1 (Vol 11)	Project Director
3	Serious safety incident (LiPo fire; spectator injury)	2	5	10	Bunkered charging is doctrine; 3.4–4.2 V/cell telemetry enforcement with automatic kill; customers never on live fields (Tow-Truck Retrieval Protocol); netted airfield; marshaling and emergency SOPs in every division volume; insurance program (Vol 10)	GM / Safety Officer

#	Risk	L	I	Score	Mitigation	Owner
4	Fleet availability below the $\geq 90\%$ gate floor (maintenance overwhelm; operating target $\geq 92\%$)	3	3	9	3-deep high-turnover spares inventory (Vol 8); preventive-maintenance matrices and predictive flags from runtime telemetry (Vol 7); electromechanical rental doctrine excludes leak-prone hydraulics; Artisan rebuild training	Works Lead
5	Novelty decay — repeat visitation below model	2	4	8	The entire gamification architecture (License, badges, Gears, seasons, leaderboards — Vol 9); events calendar; membership economics; Phase 2/3 refresh cadence	GM / Marketing
6	Supply-chain disruption (China wholesale channel)	2	4	8	Multi-supplier substitution matrix per asset class (Vol 8); 3-deep spares buffer; standardized batteries/connectors make fleets cross-compatible; no single-source dependencies in doctrine	Procurement Lead

#	Risk	L	I	Score	Mitigation	Owner
7	RC WORLD OS delivery risk (v1 late or unstable)	2	4	8	End-to-end beta is a T2 gate condition five months before opening; Supabase + Kotlin stack is proven commodity infrastructure; manual-operations fallbacks documented (Vol 13); scope discipline — v1 ships bookings/telemetry/POS core only	CTO
8	Weather dependence of outdoor revenue	4	2	8	Season-shaped staffing and marketing; construction division is the most weather-tolerant anchor; Phase 2 restaurant and Phase 3 indoor arena progressively de-risk the calendar; weather-triggered re-booking automation in OS	GM
9	Regulatory/permitting friction (planning, aviation netting, noise)	2	4	8	Planning permits granted is a T1 closing condition — no capital draws before permits; fully netted airfield designed to model-aircraft containment standards (Vol 5); electric-only fleet keeps noise profile low; early authority engagement documented in Vol 11	Project Director

#	Risk	L	I	Score	Mitigation	Owner
10	Key-person concentration (founder)	3	3	9	This Master Development Plan is itself the mitigation — the operating knowledge is documented to franchise-manual depth; GM and division leads hired in Phase 1; board oversight via tranche gates	Board

Investor Note. Note what is absent from the top ten: competitive response. Volume 2’s analysis concludes that the format’s defensibility rests on the combination of civil works, proprietary telemetry, the ERP, and this documentation — a package a copycat needs years and comparable capital to assemble, by which time the License-holding customer base and the brand are entrenched. Competition appears in the Volume 2 register, ranked below the risks above.

9.3 Governance and accepted risks

Risk governance is continuous, not annual: the register above is reviewed quarterly by the board; each specialist register is reviewed monthly by its owning manager; and the safety-relevant subset (risks 3, 8, 9) is standing business at every management meeting, with incident and near-miss data flowing from RC WORLD OS’s logging into the review. A risk whose score rises two consecutive quarters is escalated to the board with a revised mitigation plan regardless of its absolute rank.

Finally, the risks the plan consciously accepts: the concept is unproven at this scope anywhere in the world — the mitigations reduce the cost of being wrong, but no benchmark park exists, and the presales-gated T3 is the plan’s answer; the gamified-retention thesis could underdeliver, in which case the park falls back to benchmark repeat-visit economics (which the base case already assumes); and a leased site means long-term lease-renewal exposure, accepted in exchange for keeping \$1M+ of land capital out of the ask and managed through option periods negotiated at signing (Volume 11, Chapter 2).

10. Expansion Roadmap

10.1 The five-year growth logic

Growth follows a strict sequence: **prove the unit, extend the calendar, then replicate.** Years 1–2 prove Phase 1 unit economics. Years 2–3 extend the revenue calendar (Phase 2 divisions, restaurant, events). Years 3–5 attack weather and geography (arena, night racing) and convert the format into products that scale beyond one site: the Academy curriculum, the franchise package, and the technology platform. Nothing in the roadmap requires capital beyond the ≈\$3.9 M five-year total; everything in Sections 10.2–10.4 is expansion option, exercised only on evidence.

10.2 Growth within the site

- **Indoor all-weather arena (Phase 3, canonical).** Small-scale racing and crawler courses under roof; converts winter and rain days from zero-revenue to base-load; doubles as the corporate and party venue in bad weather.
- **RC Academy (Phase 3, canonical).** Classroom-based structured curriculum: school STEM programs, holiday camps, adult engineering courses built on Volume 7's custom-build instruction. Education is the park's highest-defensibility revenue — timetabled, pre-paid, weekday, and grant-eligible — and the Academy formalizes the License program's top tiers into certification.
- **Night racing (Phase 3, canonical).** Lighting on Tracks A and B opens evening sessions, league nights, and a summer race-series product aimed at the adult hobbyist segment.
- **Event franchise formats.** Recurring owned events — endurance races, mining production championships, drift nights, harvest festivals — each a marketing beacon and a high-margin revenue day (event and operations chapters of Volumes 3-6; format calendar in Volume 9).

10.3 Growth beyond the site

- **Second site (Years 4-5 option).** A company-owned second park in a comparable city, using Volume 11's design package and Volume 8's supply chain at meaningfully lower cost and risk than site one — the construction drawings, supplier relationships, OS platform, and trained-manager pipeline all exist, and the second site's pre-opening marketing starts with a License-holding diaspora rather than a cold audience. Site two also answers the strategic question franchising raises: whether replication economics are better captured as owned EBITDA or as royalties. The plan's position is to run both experiments — one owned site, one franchised pilot — and let five-year evidence decide the mix. Decision gate: site-one performance at or above the Volume 10 base case for four consecutive quarters.
- **Franchise pilot (Phase 3, canonical) and system (Year 5+).** The pilot validates Volume 12's package: site selection, localization method, fee structure, training, and brand standards, on the multi-site RC WORLD OS architecture (Volume 13) that gives the franchisor live telemetry-level visibility into every franchised park — an oversight capability no conventional venue franchisor has. Success converts RC WORLD's economics from single-site EBITDA to royalty streams.
- **Compact format (“RC WORLD Arena”, exploratory).** An indoor-only urban format derived from the Phase 3 arena learnings — one to two divisions, small footprint, mall-compatible. Deliberately unmodeled; noted here so the franchise architecture keeps it possible.

10.4 Platform and product options

The technology built for the park has standalone option value, carried at zero in the model: the **RCW Node** as a product for RC clubs, schools, and competing venues (sold or licensed with the fleet-management layer of RC WORLD OS); the **Academy curriculum** as licensable STEM content; and the **License program** as a portable credential across the future site network — the loyalty asset that makes each new site open with an installed base.

10.5 Strategic partnerships and sponsorship

The sponsorship strategy (detailed in Volume 9, with the partner-market context in Volume 2) targets four partner classes, each with a natural asset to sponsor: **RC manufacturers and distributors** (fleet supply partnerships, liveried vehicle classes, “powered by” zone naming — suppliers gain the world’s most visible product demonstration venue); **construction, mining, and agricultural equipment brands** (zone naming for the Mining and Agriculture Zones — miniature versions of their real-world machines are marketing they cannot buy elsewhere); **regional consumer brands** (event-series title sponsorship, grandstand and lawn naming from Phase 2); and **education partners** (schools, STEM bodies, vocational institutions channeling Academy demand). Sponsorship is modeled conservatively in Volume 10 and treated as margin upside; the strategic value — credibility, fleet cost offsets, and marketing reach — exceeds the revenue line.

Two disciplines govern the pipeline. First, **sponsorship follows audience**: serious approaches to zone-naming and title partners begin after the park can show verified attendance and social reach from a full season — sponsors bought early are sponsors bought cheap. Second, **no sponsorship may bend doctrine**: a fleet-supply partnership that required non-standard batteries, a non-parity vehicle class, or hydraulics in rental duty would fail the doctrine test regardless of its value. Supplier partnerships are structured instead around what RC WORLD genuinely offers manufacturers — the highest-duty-cycle public demonstration of their products in the world, and telemetry-backed durability data no factory test rig can replicate (a data asset Volume 8 develops into negotiating leverage).

10.6 Roadmap timeline

Period	Expansion action	Funding source	Decision gate
Build year (project Mo 0–12)	Build and open Core Park (opening = operating M1)	T1–T3 tranches	Section 6.3 gates
Operating Y1 (M1–12)	Core Park trading ramp; Phase 2 design, permits, procurement; first full events calendar	Operating revenue	Phase 2 gate (≈M12–13, \$7.1 criteria)
Operating Y2 (M13–24)	Phase 2 build; aviation staged opening M16–24; restaurant M19–20; marine M20–24; grandstand and events lawn	Cash flow + optional \$600 K facility (Vol 10, \$3.5)	Phase 2 milestones (7.2)
Operating Y3 (M25–36)	Full Park at maturity; night racing; arena and Academy build; franchise pilot site selection	Operating cash flow	Month-30 review; ≥25% ROIC per element
Operating Y4 (M37–48)	Arena + Academy at full operation; franchise pilot opens; second-site feasibility	Cash flow; franchisee capital	Board review vs Vol 10 base case
Operating Y5 (M49–60)	Franchise system decision; second-site decision; exit-readiness review	—	Five-year review (7.3)

10.7 The five-year horizon, stated plainly

By project Month 60, the base case delivers: one destination park at $\approx$ \$3.4 M revenue and 24-28% EBITDA margin; an operating manual proven deep enough to franchise; one franchised site open; a telemetry and ERP platform with standalone value; and a credentialed customer community whose status lives in the RC WORLD system. That is the asset the exit options in Section 6.5 price.

Volume summary & cross-references

This volume compressed the entire Master Development Plan into its investment logic: a miniaturized industrial complex on 4.8 ha, $\sim$ 150 telemetry-equipped vehicles billed in 20-minute Shifts, nine stacked revenue engines on a high-utilization thesis, a \$2.6 M staged ask buying Phase 1 outright and the option ladder to a \$3.9 M five-year build-out, headline economics of $\approx$ \$1.28 M $\rightarrow$ \$3.4 M revenue with 24-28% steady-state EBITDA margins, break-even in operating Month 16-19, a base-case investor outcome of $\approx$ 1.6-2.0 $\times$ MOIC, and expansion into franchise, education, and platform revenue.

Where to go deeper: market and competition — **Volume 2**; the divisions this volume summarized in Chapter 5 — **Volumes 3-6**; the engineering and maintenance doctrine behind the margin claims — **Volume 7**; supply chain and fleet purchasing — **Volume 8**; the customer journey, License program, and retention mechanics — **Volume 9**; every financial figure at full resolution (the source of truth) — **Volume 10**; site, buildings, and construction — **Volume 11**; the franchise package — **Volume 12**; RCW Node hardware and RC WORLD OS — **Volume 13**. Canonical numbers cited here originate in `style-guide.md` and may not be varied by any volume.

Volume 2 — Market Research & Competitive Landscape

RC WORLD — Master Development Plan Revision 1.0 — July 2026 · Status: living document — bump revision on material change

Purpose of this volume. This volume is the factual foundation of the Master Development Plan. It sizes the global radio-control industry and explains why published estimates disagree; it maps the four regional ecosystems that will supply RC WORLD's fleet and shape its culture (China, Japan, Europe, the United States); it delivers the founder-commissioned special study on Chinese diorama and miniature-environment suppliers — the supply chain that lets the park build a convincing miniaturized industrial world at defensible cost; and it analyses every competitor and precedent format that exists today, from indoor RC tracks and drone-racing arenas to full-size “dig” attractions, with real venues, real prices, and honest lessons. It closes with the customer segmentation and demand assumptions that feed the financial model in Volume 10, and a positioning and SWOT analysis. Where this volume states a market figure it states a range with a source category, per the style guide: single-point precision in a fragmented hobby market is false precision, and any single-figure market claim — ours or a competitor's — deserves suspicion.

Intended readers. Investors and lenders performing diligence on the \$2.6 M funding ask; the founder and General Manager setting pricing and marketing strategy (with Volumes 9 and 10); the procurement lead (Volume 8 consumes the supplier landscape mapped in Chapters 3–7); franchise developers adapting the demand model to new territories (Volume 12).

Chapters

1. Research Methodology & Sources
2. The Global RC Industry
3. China — the Manufacturing Engine
4. Japan — Heritage, Precision, and Circuit Culture
5. Europe — Distribution, Racing Pedigree, and Scale-Truck Culture
6. USA — Brand Power and the Bashing Economy
7. Diorama & Miniature-Environment Suppliers (Special Study)
8. Competitor & Precedent Analysis
9. Customer Segmentation & Demand Analysis
10. Market Positioning & SWOT

Closing section: Volume summary & cross-references.

1. Research Methodology & Sources

1.1 How to read this volume

Every factual claim in this volume falls into one of two classes. The first class is **verifiable public fact**: a company exists, was founded in a given year, sells a given product at a given listed price, operates a venue at a given address. These we state plainly and, where prices or programs change frequently (as attraction pricing does), we date them. The second class is **estimate**: market sizes, growth rates, market shares, willingness-to-pay. For these, no single authoritative number exists — the RC industry has no equivalent of automotive registration data — so we state ranges and attribute them to a source category rather than pretending one analyst’s figure is truth.

The reader should also understand what this volume is not. It is not a purchased syndicated market report; it is a synthesis of publicly available research, manufacturer and venue primary sources, hobby-trade press, and the founder’s own operating knowledge of the industry, assembled in mid-2026. Where the public record is thin — and for some questions, notably Chinese factory revenues and the finances of small RC venues, it is very thin — we say so rather than fabricate.

1.2 Data confidence levels

Every table and material claim in this volume carries an implicit confidence grade. Where the grade matters to a decision, we state it explicitly using this scale:

Grade	Meaning	Typical sources	Example
A — Verified fact	Directly confirmed from primary sources; stable	Company registries, venue price pages, manufacturer sites	Dig This Las Vegas sells a “Mega Dig” combo from \$595 (July 2026 pricing)
B — Corroborated estimate	Multiple independent sources agree within a reasonable band	2+ analyst firms, trade press, distributor catalogues	Global RC car market \$1.5–6.2 B (2025) depending on scope
C — Single-source or inferred	One credible source, or triangulated from adjacent data	One analyst report, one interview, one forum consensus	Market-share estimates for individual manufacturers
D — Founder judgment / assumption	No reliable external data; stated as an assumption to be tested	Operating experience, first-principles reasoning	Visit-frequency assumptions in Chapter 9

Investor Note. Treat Grade C and D items as hypotheses with a test plan, not facts. Every Grade D assumption that materially drives the financial model is flagged in Chapter 9 and carried into the sensitivity analysis in Volume 10, Chapter 8. If a number in this plan ever hardens from D to A — for example, once the park has twelve months of real visit-frequency data — the relevant volumes get a revision.

1.3 Source categories used in this volume

- **Industry analysts (2025–2026)** — commercial market-research publishers covering RC vehicles, remote-control toys, hobby goods, and family entertainment centers. Their

scopes differ wildly (toy-grade in or out; drones in or out; parts and batteries in or out), which is precisely why we quote ranges.

- **Manufacturer and distributor primary sources** — company websites, investor pages, catalogues, and trade-fair listings (e.g., Intermodellbau Dortmund exhibitor lists, China Toy Expo profiles).
- **Venue primary sources** — published price lists and booking pages of the competitor venues analysed in Chapter 8, captured June–July 2026.
- **Hobby trade press and community** — RC-specialist media, club forums (e.g., the German Modell-Baustelle and Modelltruckforum communities), and event reporting. Used for culture, practice, and precedent; never for financial figures.
- **Attractions-industry bodies** — location-based-entertainment industry reporting (IAAPA-category data as republished by analysts) for the demand-side market in Chapter 9.

1.4 The honesty rule

Three commitments govern everything that follows. First: **no fabricated precision**. If the best available answer is “somewhere between \$1.5 B and \$6.2 B depending on what you count,” that is the answer we print. Second: **no fake citations**. Companies, products, venues, and prices named in this volume are real and were verified against primary sources during drafting; estimates are labelled as estimates. Third: **negative results are reported**. Where we searched for something and found little — for instance, ready-made 1/14 scale human figures at fleet quantities (Section 7.5), or audited financials for any RC venue anywhere (Section 8.7) — the gap is itself a finding, and we record it.

2. The Global RC Industry

2.1 Market size: the honest range

Anyone who reads three market reports on radio control comes away with three different numbers, because the category has no agreed definition. Some analysts count only hobby-grade cars sold through hobby channels; others fold in toy-grade product in mass retail; others include or exclude drones, aircraft, boats, parts, batteries, and radios. The defensible statement is a range (Grade B):

- **RC cars and trucks (ground vehicles)**: industry analysts (2025–2026) place the global market between **\$1.5 B and \$6.2 B (2025)**, growing at roughly **5.8–7.5% CAGR** into the early 2030s. The low end reflects narrow hobby-grade-only scopes; the high end sweeps in toy-grade product and aftermarket. At least one narrow-scope analyst publishes a figure below \$1 B and a growth rate near 11% — an outlier that illustrates the scope problem rather than a contradiction.
- **Hobby radio control overall** (cars, aircraft, boats, radios, parts, batteries; excluding commercial drones): approximately **\$3.2–7.0 B**, growing **6–7.5%** annually (industry analysts, 2025–2026).
- **Segment structure**: analysts consistently report ground vehicles as the largest hobby-RC category, hobby-grade product at roughly **45–57% of RC-car market value** and rising, and toy-grade at roughly **30–34% and declining in share** through 2034 (Grade B on direction, C on the specific percentages).

Investor Note. RC WORLD does not need the market to be \$6 B rather than \$1.5 B. The park sells location-based entertainment, not RC products; the industry figures matter because they prove three things that hold at either end of the range — the equipment supply chain is deep, mature, and brutally price-competitive; a large global hobbyist population exists to anchor community and events; and the category is growing, not shrinking.

2.2 The toy-grade / hobby-grade bifurcation

The single most important structural trend, reported consistently across analyst firms and trade sources (2025–2026), is **bifurcation**: mass-market toy-grade RC (blister-pack, non-serviceable, sub-\$80, sold on gifting occasions) is stagnant to declining in share, while hobby-grade (serviceable, tunable, LiPo-powered, with spare-parts ecosystems) grows in both units and value. Trade-platform data published by a major Chinese e-commerce group in 2026 described the low-end segment as “collapsing under the weight of its own commoditization” while the hobby-grade segment flourishes on adult enthusiast demand.

The driver is the collapse in hobby-grade entry pricing. A genuinely fast, repairable, brush-less-optioned 1/14 buggy now retails for \$60–120 — a price point that did not exist in 2015 and which Chinese manufacturers (Chapter 3) created. This trend is the industry handing RC WORLD its fleet doctrine: the park’s entire operating model (Volume 1; Volume 8) is built on cheap, durable, repairable hobby-grade equipment bought at wholesale.

Field Note. How to read RC market reports without being fooled: check three scope questions before trusting any figure — Is toy-grade included? Are drones included? Are parts, batteries, and radios included? The \$897 M and \$6.2 B “RC car market” figures both circulating in 2025–2026 are not contradictory; they answer differently on all three questions. Any pitch deck (including ours) quoting a single unqualified number has either not read the reports or hopes you haven’t.

Three further structural observations matter more than any headline number:

1. **Fragmentation at every level.** No manufacturer holds a dominant global share. The strongest brands (Traxxas, the Horizon Hobby portfolio, Tamiya) are strong regionally and by category; analyst share estimates for the leader alone range from ~14% to ~22% depending on scope (Grade C). Hundreds of Chinese factories and a long tail of specialists share the rest. Fragmentation is a gift to a fleet buyer: genuine price competition, substitutable suppliers for nearly every asset class, no single point of supply failure.
2. **Equipment-rich, venue-poor.** The industry ships millions of vehicles a year into a world with remarkably few purpose-built places to run them. Most hobbyists drive in car parks and back gardens; club tracks are volunteer-run and often intimidating to novices; almost nothing exists for a family that wants to try RC without first buying it. That supply-demand gap between equipment and venues is the white space RC WORLD occupies (Chapter 10).
3. **Asia-Pacific leads, everywhere grows.** Analysts consistently put Asia-Pacific at the largest regional revenue share of the RC market (roughly 35–40%, Grade C) with North America and Europe close behind — meaning the concept is not hostage to one region’s taste, which matters for the franchise thesis (Volume 12).

2.3 Technology trends and what each one does for a park

Trend	What it is	Status 2025-2026	Consequence for RC WORLD
Brushless motors	Electronically commutated motors: more power, less wear than brushed	Standard above ~\$100 re-tail; entry models increasingly brushless	Racing fleet gets speed + low maintenance; brushed 540/550 retained for construction fleet torque and cheapness (Volume 7)
LiPo batteries	High-density lithium-polymer packs, 2S/3S dominant	Ubiquitous; electric has displaced nitro across mainstream RC	Enables the 20-minute Shift and 3:1 battery doctrine; imposes the charging-bunker fire discipline (style-guide canon; Volume 7)
FPV (first-person view)	Live video from the vehicle to goggles/screen	Mature in drones; spreading fast into cars and crawlers	Premium-priced experiences (\$22 Casual Shift tier); spectator screens; Aviation Division core tech (Volume 5)
App-connected / telemetry	Phones and cloud services linked to the vehicle	Mainstream in consumer drones; fragmented in cars	Validates the RCW Node + RC WORLD OS bet: nobody yet operates fleet telemetry at park scale — defensible ground (Volume 13)
Scale realism	Licensed bodies, functional hydraulics, working lights/sound, 1/14 truck ecosystems	The fastest-growing aesthetic in the hobby; licensed Volvo/JCB/Mercedes construction models mainstream	The Construction Division's entire premise; drives the diorama supply chain in Chapter 7
E-commerce distribution	Direct-from-China platforms, brand webstores, marketplace sellers	Dominant for parts; hobby shops consolidating; specialist e-tailers (AMain, rcMart, Banggood-class) fill the gap	Wholesale direct sourcing is normal, not exotic — the park's procurement model (Volume 8) rides an established channel

2.4 Distribution: from hobby shop to checkout cart

The traditional three-tier chain (factory → national distributor → local hobby shop) is eroding. Analysts and trade press (2025-2026) report steady share shift toward e-commerce: brand-direct webstores, specialist e-tailers, and direct-from-China marketplaces. Local hobby shops survive where they add community — a track, a workshop counter, race nights — which is exactly the bundle RC WORLD industrializes. Two consequences:

- **For procurement:** the park buys at or near factory prices through the same channels hobbyists already trust; the wholesale relationships specified in Volume 8 are conventional practice at unusual volume, not a special favor to be negotiated.
- **For retail strategy:** the park's shop (Volume 9) should not attempt to beat online prices on commodity parts. It wins on immediacy (broken part, fixed today), curation (the exact spares for the park's own fleet classes), and impulse/gift purchases at the exit.

2.5 Segment map: what the industry actually sells

A park buyer needs the industry mapped by operating characteristics, not just revenue. The working segmentation used throughout this plan:

Sub-segment	Typical scales	RTR retail band	Demand character	RC WORLD fleet linkage
On-road touring / GT	1/10, 1/14, 1/28	\$80-600	Stable core; club racing base	Track A touring classes (Volume 3)
Drift	1/10 RWD	\$150-700	Fast-growing on social video; Japan-led	Track A drift class (MST-based, canon)
Off-road buggy / short-course	1/16-1/8	\$60-900	Largest sub-segment; entry prices collapsed	Track B fleet (WLtoys 144010-class)
Crawlers / scale trail	1/24-1/6	\$30-1,200	Strongest growth in surface RC (analyst consensus, 2025-2026); adult-skewed	Crawler park (MN/WPL + TRX-4 class)
Construction / functional	1/14-1/16	\$60-12,500	Niche but rising with scale-realism; hydraulics at the top	Construction Division entire (Volume 4)
Aircraft (fixed-wing, heli)	various	\$80-2,000+	Mature; club-field constrained	Aviation Division Phase 2 (Volume 5)
FPV drones	2-7 inch	\$150-800	Post-boom consolidation; racing niche healthy	Aviation premium tier
Boats	1/20 scale to unlimited	\$50-1,500	Small, stable, geographically pond-bound	Marine Division Phase 2 (Volume 6)
Radios, ESCs, motors, LiPo	—	\$15-600	Consumables + upgrade economy; high margin	The Works spares & retail (Volumes 7-8)

Electric power now dominates every one of these segments — analysts put electric at roughly 80-85% of RC units sold (2025-2026, Grade B) — with nitro surviving as a competition and nostalgia niche. This matters operationally: the park's all-electric doctrine (noise, fumes, insurance, simplicity) aligns with where the industry already is, not against it.

2.6 Regional split

Analyst regional breakdowns differ in detail but agree on shape (Grade B): **Asia-Pacific** holds the largest share of RC product revenue (roughly 35-40%), driven by both manufacturing-country domestic demand and Japan's mature hobby base; **North America** is the largest single national market with the strongest premium-RTR culture; **Europe** is comparable in aggregate to North America with a stronger kit/scale culture and more fragmented distribution. Growth skews to Asia-Pacific and to emerging middle-income markets. For RC WORLD the read is simple: the concept's supply chain, culture references, and demand precedents all exist in every candidate region for the reference site and for later franchises — no region is disqualified on market-structure grounds, and Volume 12's franchise screening therefore turns on city-level variables (catchment, land, climate) rather than country-level RC penetration.

2.7 Adjacent categories that matter

Three adjacent markets bear on the plan. **Consumer drones** are an order of magnitude larger than hobby RC and culturally separate, but FPV racing (Section 8.4) supplies proven arena formats. **Static and diecast scale models** form a multi-billion-dollar collector market

that feeds the scale-realism trend and overlaps RC WORLD's retail assortment. **STEM/robotics education** is a fast-growing purchasing category for schools — the wedge for the schools segment in Chapter 9 and the RC Academy in Phase 3.

3. China — the Manufacturing Engine

3.1 The geography: two clusters and a delta

Chinese RC manufacturing concentrates in Guangdong province, in two distinguishable clusters plus a supporting delta-wide ecosystem:

- **Shantou / Chenghai district** — historically “China’s toy capital.” Toy and RC manufacturing here grew out of the plastic-toy industry: Chenghai hosts thousands of toy factories and the RC brands that graduated from them. Huina, MJX (Meijiixin), Double Eagle, and WLtoys all have Shantou-area roots. The cluster’s DNA is high-volume injection molding, aggressive cost engineering, and fast imitation — and, over the last decade, a genuine climb up the quality ladder.
- **Shenzhen / Dongguan / Pearl River Delta** — the electronics cluster. Radio systems (Flysky, Radiolink), ESCs and motors (Hobbywing, Surpass Hobby), batteries (Grepow/ Gens Ace and the direct-to-consumer LiPo houses such as CNHL), and every PCB, sensor, and connector the park’s own RCW Node needs (JLCPCB-class fabrication; Volume 13). Shenzhen’s advantage is not cheap labor but component density: any electronic subassembly can be sourced, prototyped, and revised within days.
- **Foshan and the wider delta** — precision metal work. The high-end 1/14 hydraulic and machined-metal model makers (LESU in Foshan’s Nanhai district; JDModel and Kabolite’s premium lines) sit closer to the CNC and hydraulics supply base than to the toy cluster.

3.2 The vehicle manufacturers RC WORLD will actually buy from

The table below profiles the manufacturers named in the park’s fleet doctrine (source digest and Volume 8), with confidence grades on the company facts.

Manufacturer	Base	Founded	Specialty	Price tier	Relevance to RC WORLD
WLtoys	Shantou (Chenghai)	1990s toy roots (C)	High-volume hobby-grade cars, buggies (104/124/144-series), helis, boats	Entry (\$40–120)	Track B racing fleet backbone (144010-class); cheapest credible spares ecosystem in the industry
Huina	Shantou	2010s (C); trade profiles cite ~2012	RC construction: excavators, dumpers, loaders (15xx series)	Entry-mid (\$60–450)	Rental construction fleet core: 1580 excavator, 1582/1573 dumpers, 1583 loader (canon fleet, Volume 4)

Manufacturer	Base	Founded	Specialty	Price tier	Relevance to RC WORLD
Kabolite	Shantou (Huina premium brand)	2010s (C)	Full-hydraulic metal construction machines	Premium (\$670–12,500)	Showcase tier: K970 hydraulic excavator ≈ \$10.5–12.5 k retail, 31 kg (verified 2026); premium supervised experiences only
MJX / Hyper Go	Shantou (Chenghai) → Guangdong Meijiaxin Innovation Technology	1983 as Jiaxin Toys; renamed 2016 (A)	Licensed toy RC → hobby-grade Hyper Go brushless line	Entry-mid (\$80–200)	Performance-value racing classes; example of the cluster's toy-to-hobby climb
LDRC	Guangdong (C)	2010s (C)	1/14 scale on-road, licensed-look minis	Entry (\$50–120)	Entry touring/GT fleet (canon: 24 touring cars incl. LDRC/MJX 1/14)
MN Models / WPL	Guangdong (C)	2010s (C)	Micro scale crawlers and trucks (MN99S, WPL C24/D12)	Ultra-entry (\$30–70)	Crawler-park volume fleet: “unbeatable value” class with metal-gear upgrade paths
Double Eagle (Double E)	Shantou	1988 (A)	Licensed RC construction & trucks (Volvo, JCB, Mercedes-Benz licenses); CaDA bricks; Double E Hobby premium line	Entry-mid (\$40–250)	Agriculture-zone tractors (E351 class) and licensed kid-friendly construction units; European retail links via Carson and Jamara partnerships (A)
LESU	Foshan (Nanhai)	2000s; among earliest 1/14 truck OP-parts makers in China (A, per company)	All-metal 1/14 hydraulic excavators, trucks, upgrade parts	Mid-premium (\$300–3,000+)	Premium construction/display units; deep parts catalogue; also a diorama-adjacent supplier (Chapter 7)
JDModel	Guangdong (C)	2010s (C)	1/14 hydraulic excavators, dozers, mining dumpers; metal site accessories	Premium (\$800–4,000)	Alternative premium hydraulic supplier to Kabolite/LESU; second-source insurance

Grades: (A) verified primary source; (C) inferred from trade profiles — Chinese private-company registry data is not consistently public. Wholesale price bands for the rental fleet are canon from the source digest and Volume 8.

3.3 The electronics and battery suppliers

Supplier	Base	Founded	Specialty	Relevance to RC WORLD
Hobbywing	Shenzhen	2005 (A)	Brushless ESCs and motor systems; de-facto standard in racing	Standard ESC across the racing fleet; tunable profiles create “personality” within lap-time parity doctrine
Flysky	Shenzhen (R&D) / Dongguan (production)	2006 (A)	Radio systems (Noble NB4, entry GT-class wheels); national “Little Giant” high-tech designation	Fleet-standard transmitters: cheap, rebindable, lockable; multi-protocol receivers
Radiolink	Shenzhen	2000s (C)	Value radio systems, GPS modules	Second-source radios; ATGM336H-class GPS feeds the RCW Node BOM (Volume 13)
Surpass Hobby	Guangdong (C)	2010s (C)	Value brushless/brushed motors	Consumable motor supply for high-turnover classes
Gens Ace / Tattu (Grepow)	Shenzhen/Dongguan	1998 as Grepow (C)	Mainstream quality LiPo	Primary battery supplier: consistency and documented C-ratings matter more than peak specs at fleet scale
CNHL (“China Hobby Line”)	Direct-from-China e-tail	2000s (C)	Value LiPo direct to consumer	Secondary battery source; price benchmark for Gens Ace negotiations

3.4 OEM/ODM culture: how the cluster actually works

Western buyers routinely misread the Chinese RC industry by assuming brand = factory. The operative culture is **OEM/ODM**: a factory tools a design and sells it under its own brand, under other brands, in white-box form, and in tweaked variants to whoever orders a container. The same excavator casting may appear under three names at three prices; a European distributor’s “own brand” (Chapter 5: Amewi’s AMEWI-branded construction line, Carson’s licensed models produced with Double Eagle) is frequently a Chenghai or Shantou product with different stickers, packaging, and — critically — different quality-control gates.

For RC WORLD this culture is an asset with three handles:

1. **Price discovery.** Because identical hardware ships under multiple labels, the park can benchmark true factory cost and negotiate wholesale within 10–20% of it (procurement method, Volume 8, Chapter 3).
2. **Fleet customization.** ODM factories accept paint, sticker, and spec changes at modest MOQs (typically 50–300 units for cosmetic changes, Grade C). A park ordering 24 identical touring cars is a small but real ODM customer, and park-liveried fleets are achievable from the first container.

3. **Second-sourcing.** The 1:3 excavator-to-dumper doctrine fails if dumpers go out of stock. Every fleet class in Volume 8 carries a named second source precisely because the OEM/ODM web makes near-equivalents available.

Field Note. The cluster’s weakness is documentation and revision control. A “V4” machine may quietly ship with a different ESC than last quarter’s V4. The park’s receiving SOP (Volume 8, Chapter 6) therefore teardown-inspects the first unit of every batch and logs component-level deltas in RC WORLD OS before a batch enters service.

3.5 The quality ladder, illustrated by one product family

Nothing explains the Chinese RC industry faster than the Huina/Kabolite ladder, because one Shantou company spans it end to end:

Rung	Example	Retail (2025-2026)	Construction	Who buys it
Toy	Huina 1550-class plastic excavator	\$60-90	Plastic body, plastic gears	Gift market
Rental-grade hobby	Huina 1580 V4	\$350-420 wholesale (canon)	Full metal, lead-screw electromechanical booms — no hydraulic fluid to leak	RC WORLD’s rental fleet, by doctrine
Enthusiast hydraulic	Kabolite K961/K963	≈\$1,750-1,900	Metal + real hydraulics	Serious hobbyists, German Parcours drivers
Flagship hydraulic	Kabolite K970	≈\$10,500-12,500, 31 kg, brushless, 18-channel radio	Full hydraulic, near-industrial	Collectors, commercial display — RC WORLD’s premium supervised show-case (canon)

The same badge covers a 175× price range. This is why the plan’s fleet doctrine (source digest; Volume 4) is precise about which rung serves which purpose: electromechanical machines rent to the public because Artisans can rebuild a lead-screw actuator in minutes and there is no fluid to leak into the sand; hydraulics are reserved for supervised premium experiences and display, where their realism justifies their fragility and price. A competitor who buys the wrong rung — hydraulics for rental, or toys for enthusiasts — bleeds maintenance hours or credibility respectively; this single procurement judgment is worth more than any marketing budget, and it is the kind of encoded know-how Section 10.2 counts as defensibility.

3.6 Trade channels and how the park buys

Practical sourcing routes, in ascending order of commitment: (1) **retail-adjacent e-tail** (Ali-Express/Banggood-class, brand stores) for samples and urgent spares; (2) **B2B platforms** (Alibaba, and for those with mainland purchasing capability, 1688.com at domestic-China prices) for case-quantity orders and ODM conversations; (3) **direct factory accounts** with pro-forma invoicing and freight forwarding for container-scale fleet buys; (4) **trade fairs** (Canton Fair; China Toy Expo, where Double Eagle exhibits) for relationship-building and new-product scouting. The park’s procurement SOPs (Volume 8, Chapters 3-5) formalize the ladder; the point for this volume is that every route is mature, English-transactable, and

used daily by thousands of Western buyers — China sourcing is a discipline, not an adventure.

3.7 What China means for the plan

China is not merely where RC WORLD buys machines; it is why the business model works at all. The Phase 1 fleet of ~150 powered assets is budgeted (Volume 10) at a small fraction of what the same capability would have cost in 2010, and the parts stream that keeps utilization high is a commodity market with published prices. The strategic risks — tariff exposure, quality drift, IP ambiguity, single-region concentration — are real and carried in the risk register (Volume 1, Chapter 8; Volume 8, Chapter 2), but they are managed procurement risks, not existential ones.

4. Japan — Heritage, Precision, and Circuit Culture

4.1 The manufacturers

Manufacturer	Base	Founded	Specialty	Price tier	Relevance to RC WORLD
Tamiya	Shizuoka	1946 (models); RC from 1976	Kit culture, scale realism, entry racing (TT-02), 1/14 truck kits	Mid	The template for approachable RC; 1/14 truck ecosystem underpins scale-truck culture (Chapter 5); retail/circuit lessons below
Kyosho	Tokyo (Atsugi R&D)	1963	Broad line; Mini-Z 1/28 indoor racers; premium replicas	Mid-premium	Mini-Z proves micro-scale indoor racing as a venue format (Section 8.2); Kyosho Village shows brand-owned experience retail
Yokomo	Tokyo	1970s	Drift and competition; world-standard drift chassis	Mid-premium	Drift-class reference: 2024–2026 lineup overhaul retired the YD-2 for the RD/SD/MD tiers (SD3.0 Jan 2025; belt-drive MD3.0 Oct 2025) — the park’s drift fleet spec in Volume 3 tracks this generation

Manufacturer	Base	Founded	Specialty	Price tier	Relevance to RC WORLD
Futaba	Chiba	1948 (electronics); RC radios from 1962	Premium radio systems	Premium	Marshal/staff-grade radios where reliability is safety-critical (Aviation Division, Volume 5)
Sanwa (Airtronics)	Osaka	1960s	Premium surface radios; dominant in top-level car racing	Premium	Benchmark for low-latency surface radio; premium-experience hardware
MST	Taiwan (included here as East-Asian premium)	2000s	Premium drift chassis (RMX series)	Mid-premium	Canon drift platform for the Motorsport Division (source digest; Volume 3)

A note of record: **Shunsaku Tamiya, the chairman who built Tamiya into the global model institution, died in July 2025 at age 90.** His governing insight — that the box art, the build experience, and the culture around a product matter as much as the product — is arguably the single most transferable idea from Japanese RC into RC WORLD’s customer experience design (Volume 9).

4.2 Drift culture and the Mini-Z lesson

Japan invented RC drift as a discipline: rear-wheel-drive chassis, spec low-grip tires, judged style rather than raw lap time, tight indoor tracks with polished surfaces, and a social scene closer to a skate park than a race club. The discipline exports superbly — the leading US drift venue (Section 8.2) runs Japanese P-tile flooring and Japanese-designed chassis — and it is tailor-made for a park: low speeds, high skill ceiling, spectacular to watch, gentle on equipment.

Kyosho’s **Mini-Z** (1/28 scale, launched 1999) proved a second venue-relevant thesis: shrink the car and the venue shrinks with it. A 40-foot Mini-Z track fits where no 1/10 track can, rentals run ~\$20/hour retail (Section 8.2), and the format is weatherproof. RC WORLD’s Phase 3 indoor arena (Volume 11) carries a Mini-Z/1/28 layout for exactly this reason.

4.3 What RC WORLD learns from Japanese RC retail and circuit culture

Japan is the only country where major manufacturers operate destination circuits and experience retail at scale, and the lessons are direct:

- **Tamiya’s Kakegawa Circuit** (Shizuoka): a 290 m outdoor on-road circuit — among the world’s largest — plus off-road course and rock-crawling field, run on a membership basis, **Tamiya equipment only**, with amenities up to a women-only lounge. Lessons: single-marque control simplifies marshalling and parity (RC WORLD achieves the same via a standardized rental fleet); membership converts a track into a community; comfort amenities widen the demographic.
- **Kyosho Village** (Tokyo, opened 2022) and Kyosho shop-in-shop corners with free-to-run Mini-Z tracks: retail wrapped around a try-it-now experience. Lesson: the track sells the

product; the product sells the track. RC WORLD's retail counter sits in sight of the tracks for the same reason (Volume 11, Chapter 5).

- **Circuit etiquette culture:** posted rules, staffed operation on weekends, tire and equipment standards enforced without apology. Lesson: firm technical rules read as professionalism, not unfriendliness, when the staff are helpful — the model for RC WORLD's marshal doctrine (Volume 9, Chapter 4).

Field Note. Japanese RC retail treats beginners as future experts, not walk-in revenue: free beginner tracks, staffed advice, visible progression ladders. RC WORLD's License program (Volume 9) is that ladder, industrialized.

4.4 The kit-culture lesson and its limits

Japan's RC market is distinctive in one more way: it never abandoned the **kit**. Tamiya's business remains anchored in boxes of parts — the build is half the product — and the TT-02 chassis has introduced hundreds of thousands of people to RC assembly. The transferable insight is that making deepens attachment: a customer who has assembled or tuned something returns more often and spends more. The limit of the lesson is throughput — a park cannot ask walk-in customers to build before driving. RC WORLD's translation is layered: rentals are RTR by definition; the pit-stop battery swap and Tow-Truck Protocol give casual customers a taste of the mechanical world; The Works offers watch-the-Artisan repair theatre; and the Phase 3 Academy sells the full build experience (Volume 7's custom-build curriculum) to the minority who want it. Japan proves that minority is loyal beyond its size.

One further Japanese datapoint for the record: Japanese manufacturers hold significant global share by analyst estimates (Tamiya alone variously estimated at 6-12% of RC-car revenue, Kyosho at high single digits — Grade C, scope-dependent), and their products command premium pricing on quality reputation. For fleet procurement this cashes out narrowly: the park buys Japanese where reliability is safety-critical or parts standardization pays (Futaba/Sanwa radios for staff use; MST drift platforms per canon), and Chinese everywhere the 3:1-spares doctrine absorbs failure cheaply.

5. Europe — Distribution, Racing Pedigree, and Scale-Truck Culture

5.1 The commercial landscape

Europe's RC market is a distribution market first and a manufacturing market second. The names that matter commercially:

Company	Base	Founded	Role	Relevance to RC WORLD
Carson Model Sport	Fürth, Germany	1990s (Simba Dickie Group)	Brand + distributor; distributes Tamiya in DACH; own RC line (much of it Chinese ODM incl. Double Eagle partnership)	Model of ODM-based brand building; evidence of the China-Europe pipeline the park will use

Company	Base	Founded	Role	Relevance to RC WORLD
Amewi Trade GmbH	Borchen, Germany	2004 (A)	Import wholesaler: 10,000+ SKUs, 700+ dealers across Europe; own brands (AMEWI, AMXRacing) incl. hydraulic construction models	Proof that a one-founder import business can build a pan-European RC supply network in 20 years; candidate EU-side spares buffer for a European franchise (Volume 12)
Absima	Fürth, Germany	2010s	Value-to-midrange RC brand and distributor	Benchmark for European pricing of Chinese-made hobby-grade product
LRP	Stuttgart region, Germany	1980s; insolvency filed May 2017 , brand continues under restructured ownership (A)	Racing electronics legacy brand	Cautionary tale: mid-size Western RC firms carrying distribution overhead are fragile; brands outlive companies
Schumacher Racing	UK	1980 (Cecil Schumacher) (A)	Competition off-road; invented the RC ball differential	Racing pedigree supplier; niche but durable specialist model
XRAY	Trenčín, Slovakia	2000 (Juraj Hudý) (A); HUDY parts business since 1989, Serpent supplier from 1990	Premium competition cars (“luxury RC”)	The European premium-racing benchmark; relevant to Phase 3 competitive events (Volume 3, Chapter 9)
Serpent	Heemstede, Netherlands	1980 (A)	Competition cars; design in NL, factory in China	Early example of design-West/build-China — now the industry norm

The LRP case deserves one more sentence: its 2017 Stuttgart insolvency — a company that had been Europe’s premier racing-electronics house and the importer for HPI, Team Associated, and Sanwa — shows that in RC, **brands are durable but balance sheets are not**. RC WORLD’s procurement policy of multi-sourcing and modest inventory depth (Volume 8) exists partly because supplier failure is a normal industry event.

5.2 German scale-truck culture: the deepest precedent for the Construction Division

Germany hosts the world’s most developed **Funktionsmodellbau** (functional model-building) culture: 1/14 and 1/16 trucks and construction machines built to near-industrial standards and operated on elaborate model construction sites (“Parcours”). Key elements, all verified against 2025–2026 sources:

- **Manufacturers: Wedico** (the legacy name that defined premium 1/16 German truck models; today primarily a heritage/parts marque — Grade B), **ScaleART** (Gaggenau-area manufacturer of made-in-Germany 1/14 trucks with 15-bar hydraulics, sold RTR or as kits), **Servonaut/tematik** (electronics for scale trucks), **Pistenking**, **Veroma**, and

Chinese premium imports — above all **LESU** — which now dominate the value end of the same culture.

- **Clubs and venues:** permanent club-run indoor and outdoor Parours across the German-speaking world — the **Stonebreaker Area** (Lower Saxony), Indoor-Parours Rain am Lech (Bavaria), Modellbaustelle Süddeutschland (Ehingen), RCTKA Karlsruhe, and dozens more organized through the Modell-Baustelle and Modelltruckforum communities. These are club facilities, not commercial attractions — members build the diorama, then drive on it.
- **Events: Intermodellbau Dortmund** — the world’s largest model-making fair, running since 1978; the 2026 edition (16–19 April) featured the mini-Truck-Club Recklinghausen’s 40th-anniversary display: a **345 m² model landscape** with bridges and tunnels plus an **81 m² public action Parours** issuing free “mini trucker licenses” to young visitors.

Field Note. The German clubs prove the demand for exactly what RC WORLD’s Construction Division sells — hours of absorbed, purposeful operation of miniature machines in a believable world — and simultaneously prove the gap: access today requires club membership, your own \$1,000+ machine, and volunteer labor. Nobody in Europe has industrialized it into a ticketed attraction. The Recklinghausen club’s “mini trucker license” for children is, in embryo, the RC WORLD License program.

5.3 European events and channels worth tracking

Beyond Intermodellbau, the European calendar (model fairs in Nuremberg — the Spielwarenmesse toy fair where Amewi and Carson exhibit; regional truck meets; the EOS/ETS racing series where LRP-class electronics compete) is the park’s radar for product trends and a future marketing channel for a European franchise. European distribution partners (Amewi-class wholesalers) become relevant the moment a franchise territory needs EU-warehoused spares with EU warranty law compliance (Volume 12, Chapter 6).

5.4 What the German Parours teach about diorama engineering

Because Chapter 7 will send the park shopping for miniature scenery, it is worth recording what the German club builders — the world’s most experienced miniature-worksite engineers — have converged on after decades of trial (community consensus from the Modell-Baustelle/Modelltruckforum record, Grade B):

- **Substrate is everything.** Successful Parours run compacted fine gravel and sharp sand over drainage, not loose play sand; loose sand ingests into gearboxes and bogs wheeled machines. The canon aggregate specification (loose sand to 10 mm crushed stone in graded zones) matches this learning.
- **Timber-and-steel infrastructure, scenic skin.** Bridges, quay walls, and ramps are built structurally (plywood, steel section) and then dressed to scale — the reverse (scenic material asked to bear load) fails within weeks.
- **Grades kill motors.** Club haul roads stay shallow; the canon 15° maximum incline encodes the same physics.
- **Water is the highest-maintenance feature per square meter** — but also the highest-engagement. Clubs that added water crossings and wash bays report disproportionate dwell time; the Marine Division and Track C water hazards inherit both the warning and the promise.

- **Staging beats sprawl.** The most-photographed Parcours are dense vignettes (a weigh-bridge, a fuel point, a site office with lit windows) connected by working roads — not uniform detail everywhere. RC WORLD’s theming budget (Volume 11, Chapter 7) concentrates on vignette nodes for the same reason.

The strategic conclusion repeats Section 5.2’s: this culture has perfected the product RC WORLD needs and never attempted the business. Hiring or contracting from this community — German and Dutch club builders take commissions — is a credible route to Phase 1 theming quality that no local landscaping contractor will match unaided.

6. USA — Brand Power and the Bashing Economy

6.1 The market structure

The United States is the largest single national RC market by most analyst breakdowns (Grade B) and the most brand-concentrated. Its structure in 2025–2026:

Company	Base	Founded	Specialty	Price tier	Relevance to RC WORLD
Traxxas	McKinney, Texas	1986	RTR “bash-ing” (Slash, X-Maxx, Rustler); privately held, fiercely independent	Mid-premium	The masterclass in mainstreaming RC: water-proof RTR, dealer network, customer service, motorsport marketing. Analyst share estimates 14–22% of RC cars (Grade C) — the largest single brand
Horizon Hobby	Champaign, Illinois	1985	Portfolio owner/distributor: Arrma , Axial , Losi , TLR , Element RC , Spektrum , Dynamite , plus Pro-Line (acquired 2020)	Entry-premium	The other half of the US duopoly; its 2025 troubles (below) are an industry health signal
Team Associated	Lake Forest, California	1964	Competition racing (RC10 heritage); Reedy electronics	Mid-premium	Racing pedigree; club-racing standard hardware
Pro-Line Racing	Banning, California	1982	Tires, wheels, bodies — the aftermarket standard	Mid	Proof of the accessories economy (Section 6.3); park consumables benchmarking
Redcat Racing	Phoenix, Arizona	2005	Value RTR imports	Entry	Benchmark for US retail pricing of Chinese ODM product

Two 2025 events are material context (Grade A, widely reported in hobby trade press): Horizon Hobby raised prices across brands — some models 15–20% in a single February 2025 move — citing cost pressure, and cancelled its flagship community events for 2025 (AxialFest Badlands, AxialFest Europe, Horizon RC Fest) citing economic uncertainty. Read across: even the industry’s biggest players run thin margins on hardware, and community events are the first casualty of a hardware-margin squeeze. A venue business whose revenue **is** the event does not face that trade-off — it is the trade-off, solved.

6.2 RC park precedents on US soil

The US has thousands of RC tracks but almost no RC parks in the RC WORLD sense. The closest precedents:

- **Club and commercial tracks** — hundreds of volunteer club circuits plus a thin layer of commercial operations (SDRC Raceway, Nor-Cal Hobbies — analysed with prices in Section 8.2/8.3).
- **Hobby-shop/track hybrids** — the dominant surviving retail format: a shop financing a track that feeds the shop.
- **Full-size dig attractions** — Dig This (Las Vegas) and Diggerland USA (New Jersey), which prove the construction-play demand thesis at full scale (Section 8.6).
- **Crawler parks** — informal rock-crawling courses at hobby shops and public parks; a handful of paid crawler courses attached to attractions.

The absence is the finding: **no US operator combines racing, construction, crawling, marine, and aviation in one ticketed multi-zone venue**. The format RC WORLD proposes does not exist there yet (Grade A as of July 2026 research; Section 8.1 covers the nearest global analogues).

6.3 The SEMA-style aftermarket lesson

American RC replicates the American car-culture economy in miniature: the vehicle is the entry ticket, the aftermarket is the lifestyle. Pro-Line built a durable business purely on tires, wheels, and bodies; entire YouTube genres exist around upgrading a \$300 truck with \$600 of parts. For RC WORLD this validates two revenue lines: **customization retail** (bodies, wheels, lights fitted at The Works while the customer watches — Volume 9, Chapter 6) and **build-your-own programs** (Volume 7’s custom-build tutorials as paid Academy courses in Phase 3). The park should treat the aftermarket impulse — make it mine — as a first-class monetization mechanic, not an afterthought.

6.4 Traxxas as the customer-experience benchmark

Traxxas deserves a section of its own, not as a supplier (the park’s fleet doctrine buys mostly Chinese) but as the industry’s best case study in **removing friction from a technical hobby**. Its playbook, assembled from public record: ready-to-run as the default (batteries and radio in the box, decades before rivals); waterproof electronics as standard, ending the hobby’s oldest anxiety; a self-righting monster-truck feature so novices never walk to flip a truck; plain-language support and generous service; dealer-network merchandising that looks like power tools, not toys; and marketing through full-size motorsport sponsorship rather than hobby channels. The result is analyst-estimated category leadership (Section 6.1) at premium prices.

Every element maps to a park decision: RTR philosophy → rental-first fleet; waterproofing → all-weather operating envelope specs (Volume 7); self-righting → the Tow-Truck Retrieval Protocol solving the same “dead vehicle, embarrassed customer” moment with gameplay; plain-language support → Artisan hosting culture and the Toolbox Talk; motorsport aesthetics → the park’s pit lane, grandstand, and livery standards (Volume 11). Traxxas proved that the mass market wants RC minus its frictions; RC WORLD extends the same logic one step further — RC minus ownership itself.

6.5 US market data points for the record

For calibration (Grade C, analyst estimates 2025–2026): North America is generally credited with roughly 30–35% of global RC-car revenue; US hobby distribution has consolidated around Horizon’s house channels, AMain-class e-tailers, and a shrinking independent-shop base; and US analyst breakdowns consistently show crawling/trail and bashing growing while traditional club racing participation is flat — a hint that experience-seeking (trails, jumps, social video) is outgrowing competition-seeking, which favors a park built around experiences with competition as one layer rather than the reverse.

7. Diorama & Miniature-Environment Suppliers (Special Study)

This chapter fulfils the founder’s explicit commission: research Chinese diorama suppliers and manufacturers, including makers of miniature equipment replicating real work environments. It maps where a park buys convincing miniature industrial scenery at scale, and grades the confidence of each finding.

7.1 Why this supply chain is strategic

RC WORLD’s differentiation is not vehicles — anyone can buy a Huina excavator — it is the **world**: a mining pit that reads as a mine, a farm that reads as a farm, haul roads with real signage, scaffolding, containers, barriers, and figures at coherent scale. That world is a procurement problem in three layers: (1) **miniature work-environment equipment** at 1/14–1/16 scale; (2) **built environment** — structures, buildings, large scenic elements; (3) **terrain and scenery materials** in bulk. Each layer has a different supplier landscape, and the honest finding up front is that **no single “diorama industry” exists to order from**: the park assembles its world from four adjacent industries — the 1/14 truck-accessory ecosystem, architectural model-making, model-railway scenery manufacturing, and general Chinese contract fabrication — plus in-house work by The Works.

7.2 Layer 1 — 1/14 site accessories and miniature work equipment

The 1/14 scale-truck ecosystem (Chapters 3 and 5) has spawned a real accessories industry, most of it Chinese:

Supplier / channel	Base	What they supply	Confidence & notes
LESU	Foshan	Beyond vehicles: metal buckets, quick couplers, gantry cranes, chain hoists, load tensioners, trailer hardware — functional metal site equipment at 1/14	A — catalogue verified 2026; one of the earliest 1/14 OP-parts makers in China by its own (credible) account
JDModel	Guangdong	Metal site accessories alongside machines: chain blocks, gantry cranes, ratchet binders, dozer blades, quick couplers	A — retailer catalogues (Toucan Hobby-class) verified 2026
Specialist e-tailers (Toucan Hobby, RC Construction Vehicle Store, rcMart-class)	China/HK	Aggregated 1/14 site hardware: containers, pallets, drums, tool sets, figures where available	A — active 2026; these are the discovery layer for the long tail
3D-print STL vendors (Angry Flite-class) and print-on-demand	Global/digital	STL files for 1/14 concrete barriers, site fencing, traffic cones, bollards — printed in-house at The Works	A — verified 2026; a 1/14 site barrier STL measures ~143 × 77 mm printed, and a five-piece roadwork bundle covers the basics
Alibaba/1688 contract fabricators	Guangdong/Zhejiang	Custom runs of anything above: laser-cut steel guard rails, scaffolding sections, sheet-metal containers, rotomolded aggregate hoppers	B — the channel is proven for adjacent goods; specific 1/14 site-goods vendors churn, so Volume 8 maintains a live supplier register rather than this volume freezing names

The scaffolding, barrier, cone, container question (asked directly in the founder’s brief): ready-made 1/14 traffic and site furniture exists but at hobby quantities and hobby prices — single cones and barrier packs aimed at individual diorama builders. At park scale (hundreds of barriers, meters of scaffolding, dozens of containers), the economical path is **hybrid**: buy reference samples from the hobby ecosystem, then (a) print consumable items in-house from purchased or self-drawn STLs (PETG/ASA for UV resistance), and (b) contract-fabricate durable metal items (container shells, scaffold frames, hopper structures) through Guangdong sheet-metal and laser-cutting shops at commodity metalwork rates. The Works owns this pipeline; Volume 7, Chapter 8 carries the build standards and Volume 8 the sourcing SOP.

7.3 Layer 2 — architectural model-making firms

China has a mature **sand-table / architectural model industry** serving real-estate marketing, urban planning, and museum exhibits, concentrated in Shenzhen and Guangzhou:

- **Shenzhen QZY Models Design Co., Ltd.** (founded 2013) — custom architectural and industrial scale models; materials ABS, acrylic, metal; techniques from laser cutting and CNC to electroplating; MOQ of one model; export crating and international branches. Verified 2026 (Grade A).
- **Shenzhen Lvjian (“Green Building”) Model Design** — 25 years in sand-table models: industrial plants, urban planning, multimedia/automated models. Verified 2026 (Grade A).
- **Shenzhen Haojing Model Co.** (Guangzhou-founded 2006) — architectural and landscape detail models with in-house laser and precision engraving. Verified 2026 (Grade A).

- Dozens of peer firms across the delta; the industry norm is fully custom work quoted from CAD, with 7–30 day lead times on medium pieces.

Fit for RC WORLD: these firms build display-grade miniatures — a park entrance centerpiece, the Phase 2 “master plan” souvenir model, museum-quality structures for indoor areas. They are generally **not** the right source for outdoor, driven-on, weatherproof scenery: their materials (acrylic, ABS sheet) and pricing (skilled-labor-intensive one-offs) do not suit a mining pit that takes daily hits from 5 kg trucks. The park’s outdoor built environment is instead a **construction trade job at small scale** — rendered blockwork, cast concrete, treated timber, GRP — specified in Volume 11, Chapter 7, with the model firms supplying only the crown jewels.

Investor Note. This split matters to capex credibility. Volume 11 budgets outdoor theming as landscaping/light-construction cost per square meter, not as “model making” — an order-of-magnitude cheaper and far more durable. The architectural-model line item in Phase 1 capex is deliberately small (display pieces only).

7.4 Layer 3 — scenery and terrain materials at scale

The model-railway industry solved miniature landscapes decades ago, and its supply chain now has a strong Chinese tier:

- **Chinese manufacturers/brands (Evemodel-class):** trees, painted figures, fencing, containers, building kits, benches, lamps — shipped worldwide via brand stores and marketplaces at a fraction of European brand pricing; customization accepted on items like containers (verified 2026, Grade A). Ideal for indoor dioramas, the Agriculture Zone’s crop rows and farmstead details, and spectator-side scenic dressing.
- **European reference brands (Faller, Noch, Viessmann — all Intermodellbau exhibitors):** the quality benchmark for scenic materials; used selectively where longevity or realism justifies the price.
- **Bulk terrain reality check:** at 4.8 ha, the park’s “terrain materials” are overwhelmingly **real civil materials** — graded aggregates (loose sand to 10 mm crushed stone per the canon mining spec), soils, turf, and plantings — procured as landscaping supplies, not hobby goods. Hobby-grade scatter and static grass are for enclosed dioramas and photo zones only; outdoors they are consumables that weather away.

7.5 The figures problem — an honest negative result

Convincing human figures at 1/14–1/16 anchor scale realism (a mining pit with no crew reads as a toy). Research finding (Grade A on the gap): supply is **thin and fragmented**. Model-railway figures stop at ~1/43–1/32 for painted bulk packs; 1/14 truck-scene figures exist (driver figures, a few worker poses) but at single-figure hobby prices with limited pose variety; 1/18 action-figure and garage-kit ecosystems overlap useably but off-scale. Conclusion carried into Volume 7: the park **prints and paints its own figure library** (purchased and commissioned STLs, resin/FDM, batch-painted by Artisans in workshop downtime), buying commercial figures only for hero close-up positions. This converts a supply gap into a small, ownable craft capability — and printed figures are replaceable at pennies when the inevitable excavator strike occurs.

7.6 Procurement summary for a believable world

Need	Primary source	Secondary	In-house share
Site furniture (cones, barriers, fencing)	In-house 3D print from purchased STLs	Hobby e-tailers (samples/hero items)	High
Functional site equipment (cranes, hoists, couplers)	LESU / JDModel catalogues	Alibaba fabricators	Low
Containers, scaffolding, hoppers (durable metal)	Guangdong contract sheet-metal/laser shops	LESU-class accessories	Medium (design in-house)
Display-grade models & centerpieces	Shenzhen architectural model firms (QZY-class)	European model builders	Low
Scenic dressing (trees, details, indoor dioramas)	Chinese model-railway suppliers (Evemodel-class)	Faller/Noch premium	Medium
Outdoor built environment	Local light-construction trades to Volume 11 specs	—	Design + finishing
Figures (1/14–1/16)	In-house print & paint program	Hobby figures for hero spots	High

7.7 Worked example: dressing the Mining Zone

To make the stack concrete, here is the indicative theming bill for the Phase 1 Mining Zone's scenic layer (excluding earthworks and vehicles, which belong to Volumes 11 and 4 budgets). Quantities from the Volume 11 zone plan; unit costs are Grade B-C estimates from 2026 catalogue and fabrication-quote research, to be hardened at purchase order:

Item	Qty	Source (per 7.6)	Indicative cost
Concrete-look barriers, printed PETG	200	In-house print (~\$0.60 material each + labor)	≈ \$400 + 50 print-hours
Site fencing sections, printed	120	In-house print	≈ \$250 + 30 print-hours
Traffic cones/bollards (consumable)	300	In-house print, batch	≈ \$150
20 ft-style containers, sheet steel, park-liveried	12	Guangdong contract fabrication	≈ \$60–110 each → \$900–1,300
Scaffolding assemblies around processing hopper	6	Contract laser-cut + in-house assembly	≈ \$500–800 total
Central processing hopper (functional, steel)	1	Custom fabrication to Works drawings	≈ \$1,200–2,000
Gantry crane + chain hoists (functional)	2	LESU/JDModel catalogue	≈ \$250–450 total
Signage package (scale site signs, printed + UV laminate)	80	In-house	≈ \$200
Worker figure library, painted	60	In-house print & paint	≈ \$250 + paint labor
Vignette props (site office, fuel point, weighbridge dressing)	3 nodes	Mixed: Evemodel-class details + in-house	≈ \$600–900

Item	Qty	Source (per 7.6)	Indicative cost
Hero display model (mine cross-section, entrance case)	1	Shenzhen model firm (QZY-class), quoted from CAD	≈ \$2,500-5,000

Indicative scenic-layer total: ≈ **\$7,500-11,500 plus in-house labor** for the park’s most theming-intensive zone — a rounding error against the \$1.85 M Phase 1 capex, which is exactly the point: the believable-world advantage costs little if procured through the right stack, and would cost 5-10× if naively commissioned entirely from model-making firms.

Trade Hack. Order one physical sample of every hobby-ecosystem accessory before committing to in-house printing: measuring a LESU coupler or an Evemodel container with calipers and replicating proportions costs a \$20 sample and an evening in CAD, and guarantees the park’s printed items sit at true 1/14 alongside purchased hardware. Scale coherence — everything agreeing about how big a human is — is the cheapest special effect in the park.

8. Competitor & Precedent Analysis

Method note: prices and program details below were captured from venue primary sources in June-July 2026 and are Grade A at capture; attraction pricing changes seasonally, so treat figures as anchors, not quotes. Financial performance of private venues is almost never published; where we infer economics, we say so.

8.1 RC parks & RC theme attractions worldwide

What exists. The multi-zone RC park as this plan defines it — several vehicle disciplines, rental fleet, ticketed shifts, built miniature world — does not yet exist at RC WORLD’s scale anywhere we could verify. What exists is a fast-thickening layer of smaller precedents:

- **Fuji Subaru Land “Heavy Machinery RC Space”** (Yamanashi, Japan; opened June 2026 at the park-adjacent Sylvans Cafe): guests operate RC excavators and a dump truck in a sandbox, hunting buried “treasure” stones; five vehicles (4 excavators + 1 dumper — note the near-1:4 ratio running opposite to our 1:3 doctrine, workable only because their dumper is decorative rather than a haulage bottleneck); **¥2,000/hour (~\$13-14), shareable within a group, walk-in only.** A major Japanese theme-park operator validating RC construction as paid family entertainment is the single most encouraging datapoint of this research cycle.
- **Dig IT at Thunderbird Park** (Tamborine Mountain, Australia): a kids’ mini-excavator attraction whose four zones include one literally named **“RC World”** — 1/14-scale, fully hydraulic, ~30 kg RC dozers, dumpers, excavators and loaders operated “just like on the work site”; **A\$65 (~US\$43) per person** admission across the four zones. Proves families pay premium prices for supervised scale-machine operation, bundled with full-size mini-diggers.
- **Dig Dig Land** (Singapore, opened late 2024): construction-themed play garden with kid-operated mini excavators, cranes, and an **RC Zone** rationed at 10 minutes per turn with requeue — a crude ancestor of the Shift and queue mechanics this plan formalizes.

- **RC Roastery** (Ramsgate, UK, opened January 2026): a café built around RC — drift track with scenery, separate adult and child construction sites, crawler course with a suspension bridge, gamified digging (“dig a kilo of gold, trade for a prize”), weekly leaderboard competitions. One founder, one retail unit; the RC-café format (already established in Thailand and Dubai, per the founder’s own account) is the smallest viable expression of RC WORLD’s thesis.

Business-model reading. All four run rental-fleet, time-rationed, staff-supervised models — i.e., they have independently converged on the same mechanics this plan specifies, at 1/50th the scope. None operates telemetry, licensing progression, or multi-division breadth; none is on more than ~0.2 ha of RC-dedicated ground.

Lessons for RC WORLD: 1. The demand thesis is being validated in four countries simultaneously, right now (2024–2026), by unrelated operators. 2. Every precedent is capacity-starved (walk-in queues, 10-minute rations, five-vehicle fleets) — evidence that supply, not demand, is the binding constraint. RC WORLD’s 150-asset fleet and Shift system attack exactly that constraint. 3. Window of opportunity: the format is being discovered but nobody has capitalized it properly. First-mover advantage at destination scale is available and will not remain so indefinitely (see SWOT, Chapter 10).

Investor Note. When a format appears independently in Japan, Australia, Singapore, and the UK within 24 months, at price points from \$13/hour to \$43/visit, the correct read is “category emerging, consolidation play available” — the same pattern trampoline parks showed circa 2011–2014 before chains (Sky Zone-class) industrialized them.

Per-precedent scorecard. Applying a consistent lens (business model / pricing / strengths / weaknesses / lesson) across the four verified attractions:

	Fuji Subaru Land RC Space	Dig IT (Australia)	Dig Dig Land (Singapore)	RC Roastery (UK)
Model	Add-on attraction at existing theme park café	Premium zone inside adventure park	Weekend-operated play garden	Independent RC café
Pricing	¥2,000/h shared	A\$65/person, 4 zones	Admission + rationed 10-min turns	Coffee + track/site time
Strengths	Trusted host brand; zero-friction walk-in; treasure-hunt gamification	Bundles full-size 1.7 t diggers with 1/14 RC; strong per-head yield	Family targeting; queue discipline	Community feel; gamified digging (“kilo of gold” prize); accessibility (wheel-chair-accessible site); leaderboards
Weaknesses	5 vehicles; no progression; no booking	Single-visit design; remote location	Weekend-only; tiny RC fleet	One founder’s capacity; micro footprint
Lesson for RC WORLD	Host-venue partnerships could seed franchise-lite outposts (Volume 12)	Full-size + miniature pairing is additive, not cannibalizing	Ration + requeue works but frustrates — the Shift + app queue solves it properly	Gamified excavation and weekly leaderboards belong in the core product (Volume 9)

8.2 Indoor RC tracks

Real examples and economics (US, prices verified 2026):

Venue	Format	Key prices	Model notes
SDRC Raceway (San Diego, CA)	Indoor electric off-road, clay/dirt; 7 days/week	\$20 practice day (\$12 after 7 pm); \$25 race entry; rental \$25/20 min , max 2 rentals on track, 1 session/driver/day; parties from \$200/h (2 h min)	Pro shop attached; rentals deliberately capped — rental novices disrupt owner traffic
Nor-Cal Hobbies (N. California)	Shop + track hybrid	\$25 all-day practice; rentals \$25/25 min; race \$25 first class; Platinum membership \$185/mo unlimited; kids ≤12 race free Fridays	Retail subsidizes track; explicit kid-acquisition funnel
Super-G R/C Drift Arena (Baldwin Park, CA)	8,500 sq ft dedicated drift; Japanese P-tile surface; plus Mini-Z track	\$20/day track pass; limited rentals by advance email; strong retail web-store	Destination status: partnered hotel rate (\$135/night) for traveling drifters — a niche venue generating overnight tourism
RC Street Shop track (SoCal)	Indoor Mini-Z club	\$20/day practice; \$20/h rentals (McLaren/Ferrari-bodied fleet); \$30/mo membership; VIP party area	Micro-scale minimizes footprint and cost; rentals are genuinely beginner-friendly

Structural economics (hobby-trade analyses, 2025–2026, Grade B): standalone indoor tracks rarely clear their rent; viability comes from hybrid revenue — retail (20–40% margin on consumables), parties/events, memberships (\$50–185/month observed), coaching. Indoor carpet buildouts run \$75,000–200,000+; insurance \$3,000–7,000/year for track-only operations. One published back-of-envelope for a well-run indoor track: ~\$210,000 annual revenue against ~\$150,000 costs — a living, not a business. Trade commentary also notes stronger results in Southeast Asia (Singapore, Indonesia, Philippines), where commercial RC tracks with membership-plus-events models reportedly reach several hundred thousand dollars in annual revenue (Grade C, unaudited) — consistent with this plan’s thesis that professionalized formats, not the activity itself, are the constraint in the West.

Weaknesses of the format (and RC WORLD’s answers): single-discipline monotony (→ multi-division park); hobbyist-first culture that intimidates novices (→ rental-first fleet, Toolbox Talk induction, Artisan hosting); rentals treated as a nuisance (→ rentals are the product; owners’ hours are the add-on, Volume 3, Chapter 8); revenue ceiling set by track count (→ ten+ simultaneous venues on one ticket).

8.3 Outdoor RC tracks & club circuits

The global base layer of RC infrastructure is the volunteer club circuit: mowed-and-groomed off-road tracks, oval clubs, 1/8 nitro circuits. Characteristics verified across US/UK/EU club sources: entry fees \$10–30/day where charged at all; membership dues \$50–200/year; working-bee labor; insurance via national federations; no rentals in most cases; opening hours hostage to volunteer availability.

- **Strengths:** authentic community; deep skills; incredibly low cost base; tolerant of weather damage.

- **Weaknesses:** no rental path for newcomers, inconsistent operation, no F&B, no families as a designed-for segment, and chronic land insecurity (clubs lose leases to development routinely).
- **Precedent value:** clubs prove that even unmarketed, amenity-free RC venues sustain decades of loyal use — the underlying activity has durable pull. Tamiya's Kakegawa membership circuit (Section 4.3) is the professionalized ceiling of this format, and the German Parcours clubs (Section 5.2) its construction-side equivalent.
- **Scale of the base:** no global census of RC tracks exists (honest gap), but national federation listings and community directories consistently indicate hundreds of active club circuits in the US and comparable density across Western Europe and Japan (Grade C). For any candidate park city, Volume 12's site-screening SOP includes a club audit within the catchment: existing clubs are simultaneously demand evidence, partner pipeline, and — if alienated — the loudest possible critics.
- **Lessons:** (1) partner, don't fight — club racers become RC WORLD's event competitors, coaches, and credibility (Volume 3, Chapter 9 sets club-night programming); (2) the park's groomed, staffed, always-open tracks are a service upgrade clubs cannot match; (3) offer clubs what they lack — secure venue, timing infrastructure, spectators — in exchange for what they have: skilled community.

8.4 Drone racing arenas

Two tiers exist, both real and current:

- **Professional/spectacle: Drone Racing League (DRL).** Twelve salaried pilots on identical spec drones racing LED courses through stadium venues (2025 Miami Invitational at loanDepot Park, ~10,000 expected spectators); acquired by Infinite Reality in April 2024 for a reported \$250 M. DRL is a **media property**, not a venue business — it books stadiums per event. Lesson: drone racing is spectacular content; nobody has made it a sustainable daily attraction.
- **Grassroots: MultiGP.** The world's largest FPV network — on the order of 500 chapters and 30,000+ registered pilots (2025–2026, Grade B) — racing on temporary gate courses in fields and, increasingly, indoor venues (the 2025 championship ran indoors at Tulsa's Hardesty National BMX Hall of Fame, enabled by HDZero digital video under a metal roof). Chapters charge modest annual memberships plus small race fees.

Reading for RC WORLD's Aviation Division (Volume 5): the FPV community is equipment-rich and venue-poor, exactly like surface RC. A permanent netted airfield with gates, timing, charging, and spectator screens is a service no chapter can build. But FPV rental to novices is harder than car rental (nausea, crash rates, goggle hygiene) — hence Volume 5's tiered model: buzz-wire-style tethered and simulator introductions before rental flight, and the airfield rented to MultiGP-style chapters for events. Full-scale DRL-style spectacle is a Phase 3 event-format import, not a daily operation.

8.5 RC marine clubs & model boating ponds

The marine precedent is old and instructive:

- **San Francisco Model Yacht Club / Spreckels Lake** (Golden Gate Park): club founded 1898 — the oldest continuously operating model-boating body in the western hemisphere;

the city built Spreckels Lake specifically for model boating (1902–1904) and the WPA-era clubhouse is deeded to the club. ~120–175 members, public sailing always permitted.

- **UK pond culture:** Solent RC Model Boat Club at Setley Pond (New Forest) operates on a Forestry permit with paid annual rental, insurance, and risk assessments, sharing a public pond. Colwyn Bay MBC took over management of Eirias Park’s 1935 boating lake from the council in 2016, maintaining it (including ecologist-advised dye treatments against weed growth) in exchange for use.

Business reading: marine RC has essentially **no commercial layer** — it is clubs on public water. Costs that RC WORLD must internalize are visible in the club record: weed and water-quality management, wildlife conflicts, wind exposure, retrieval of dead boats. Lessons for the Marine Division (Volume 6): purpose-built shallow ponds with engineered edges and retrieval design (the marine cousin of the Tow-Truck Protocol is Volume 6’s retrieval tug — friction converted to gameplay again); scale-sailing and steering courses rent well because boats are slow, robust, and legible to spectators; and the century-long club record proves the activity’s durability while its zero-commercialization proves the open field.

8.6 RC construction parks & “dig” attractions — including full-size analogues

The founder’s construction thesis has two proof layers.

Full-size precedents (the demand proof):

Venue	Format	Verified pricing (2026)	Reading
Dig This (Las Vegas)	Adults/teens operate real Caterpillar excavators & dozers on 5 acres; games (excavator basketball, tire stacking); breathalyzer at entry	Skid steer from ~\$195–215; Big Dig excavator/dozer ~\$249–329 (60–90 min); Aggression Session \$799; Mega Dig combo \$595 ; Full Throttle \$1,311; corporate digs to 100+ pax	People pay hundreds of dollars for 60–90 minutes of supervised machine operation with zero output — the purest willingness-to-pay signal in this volume. Named machines (“Thor”, “Brutus”) show personality merchandising
Diggerland USA (New Jersey; from the UK Diggerland chain, four UK parks)	Family construction theme park: 40+ rides/attractions built from real machinery, plus water park	Day tickets ~\$47–54 gate, dated online ~\$47; season pass ~\$151; under-36” free	A construction-themed park sustains chain economics at family price points; seasonality is managed with a water park — RC WORLD’s all-weather answer is Phase 3’s indoor arena

RC-scale precedents (the format proof): the Section 8.1 attractions (Fuji Subaru Land, Dig IT, Dig Dig Land, RC Roastery) plus the German club Parours (Section 5.2) and mobile trailer-based RC dig operators serving events. Between \$13/hour (Japan), ~\$43/visit (Australia), and \$15–22/Shift (this plan’s canon pricing), RC WORLD’s Construction Division pricing sits comfortably inside the demonstrated band, at a fraction of Dig This’s price for a recognizably similar psychological product.

Field Note. Dig This converts liability into theatre: the safety orientation, the wireless-headset instructor, the breathalyzer — all staged as part of the experience. RC WORLD’s Toolbox Talk induction should copy the stagecraft: safety ritual done well reads as professionalism and raises perceived value rather than friction.

Synthesis across all six competitor formats:

Format	Real revenue anchor	Fatal limitation	What RC WORLD takes
RC parks/attractions	\$13/h – \$43/visit	Tiny fleets, no depth	Validated demand; Shift/queue mechanics
Indoor RC tracks	\$20-25/day; \$25/20-min rentals	Single discipline; hobbyist-first	Rental price anchors; retail hybrid
Club circuits	\$10-30/day	Volunteer ceiling; no novice path	Community programming; event partners
Drone arenas	Membership + event fees	No daily-attraction model	FPV as premium tier + event content
Marine ponds	Club dues only	Never commercialized	Pond engineering lessons; open field
Full-size dig attractions	\$195-1,311/session	Capex + liability of real machines	WTP proof; safety-as-theatre; naming/gamification

8.7 What we could not find — negative results, stated honestly

For the record, and per the honesty rule: we found **no audited financial statements for any dedicated RC venue anywhere** (all are private and small); **no syndicated market report on “RC entertainment venues”** as a category (the category is too new — this volume’s estimates for it are built from the venue-level prices above); and **no verified multi-division RC park precedent** at hectare scale. Each absence is simultaneously a diligence limitation and evidence of the white space.

9. Customer Segmentation & Demand Analysis

9.1 The market RC WORLD actually competes in

The park’s revenue does not come out of the ~\$1.5-6.2 B RC product market; it comes out of **location-based entertainment (LBE) spend**. Industry analysts (2025-2026) size the global family-entertainment-center market at roughly **\$30-40 B (2025)** growing **7-14% CAGR** — another legitimately wide range driven by scope differences — with attractions-industry reporting citing over a billion annual global visits to location-based entertainment. Two structural LBE trends favor the concept: the shift toward **participatory** formats (guests doing, not watching) and **multi-revenue-stream venues** (admission + F&B + retail + events per visit). RC WORLD is both by construction.

9.2 The five segments

1. Families (primary volume segment). Parents seeking a 2-4 hour weekend activity for children ~6-15; purchase driver is the parent, experience driver is the child; the Family Bundle (4 drivers × 2 Shifts, \$89 canon) is built for them. Comparable spend per verified

precedents: A\$65 at Dig IT, ~\$50/head at Diggerland with extras, \$200+ birthday parties at US RC tracks. Demand character: weekend/holiday peaked; weather-sensitive until Phase 3's indoor arena; birthday parties (canon from \$349) are the repeat-acquisition engine — every party delivers ~10 trial users.

2. Hobbyists & enthusiasts (frequency segment). Existing RC owners plus lapsed adults (“I had a Tamiya in 1995”). They come for groomed tracks, timing systems, events, and The Works; they buy memberships (Apprentice \$29 / Operator \$59 / Foreman \$99 monthly, canon), own-equipment track passes, parts, and coaching. Reference behavior: \$185/month unlimited memberships sell at plain Californian club tracks; drift tourists book hotel nights around Super-G. Modest headcount, outsized revenue-per-head and community value: they populate leagues, teach informally, and legitimize the venue.

3. Corporate & groups (yield segment). Team-building buyers currently choosing karting, escape rooms, axe-throwing. RC WORLD's pitch is unique mechanics: excavator-and-dumper logistics challenges (the 1:3 doctrine as a literal teamwork exercise), pit-crew races, telemetry-scored leaderboards on the big screens. Canon pricing from \$1,400 (2 h, 20 pax) sits well under Dig This's corporate digs while offering larger simultaneous capacity. Weekday daytime demand — precisely when families are absent — makes this segment the utilization flywheel.

4. Schools & STEM (weekday-volume and legitimacy segment). Curriculum-linkable content is genuinely abundant: gear ratios and torque in the crawler park, energy management in LiPo doctrine, logistics arithmetic in the mining zone, embedded systems via the RCW Node. Priced as per-student programs with teacher materials (Volume 9, Chapter 7); the RC Academy (Phase 3) upgrades this to structured courses. Strategic value beyond revenue: municipal goodwill, grant eligibility, and the pipeline of future hobbyists.

5. Tourists & event audiences (opportunistic segment). Depends entirely on the site city's visitor economy; the Dig This model (a novelty attraction 20 minutes off the Strip) shows what a strong tourist market yields, but the base financial model (Volume 10) conservatively treats tourist revenue as upside, not base case — Grade D assumption flagged.

Segment summary matrix:

Segment	Primary product	Canon price anchor	Demand window	Key conversion metric (KPI, Volume 10 Ch. 9)
Families	Casual Shifts, Family Bundle, parties	\$15 Shift / \$89 bundle / \$349 party	Weekends, holidays	Party → repeat-visit conversion rate
Hobbyists	Memberships, own-equipment passes, events	\$29-99/mo	Evenings, events	Member count & monthly churn
Corporate	Bespoke events	from \$1,400	Weekday daytime	Events/month; repeat bookings
Schools/STEM	Per-student programs	priced per Volume 9 Ch. 7	Weekday termtime	Schools signed; program NPS
Tourists	Day Pass, premium Shifts	\$59 Day Pass	Seasonal	Share of visits from >90 min origin

A demographic note the whole plan should carry: every verified precedent that targeted families (Dig IT, Dig Dig Land, Diggerland, Fuji Subaru Land) reports or displays broad gender

participation, while traditional RC club tracks skew heavily male-adult. The difference is framing — “operate machines in a world” versus “compete in a technical hobby.” RC WORLD’s positioning language, imagery, and staffing (Volume 9, Chapter 2) deliberately follow the family framing, because the addressable market is roughly double.

9.3 Willingness-to-pay: triangulation

Anchors from verified 2026 pricing, cheapest to dearest, per person:

Anchor	Price point	Time	\$/hour equivalent
Fuji Subaru Land RC space	~\$13-14	60 min (shareable)	~\$13
US indoor track rental	\$25	20-25 min	~\$60-75
Dig IT (Australia)	~US\$43	multi-zone visit (~2 h)	~\$20-22
Diggerland USA day	~\$47-54	full day	~\$7-10
Karting (typical Western market, for reference)	\$25-40	10-15 min session	~\$100-160
Dig This Las Vegas	\$195-329	60-90 min	~\$130-220

RC WORLD canon pricing — Casual Shift \$15/20 min (~\$45/h), premium \$22 (~\$66/h), Day Pass \$59 — sits between the family-park floor and the karting/real-machine ceiling: materially cheaper per hour than karting or Dig This, materially richer than a passive park day. That is the correct pocket for a premium-but-family-accessible positioning, and it survives the comparison in both directions (Grade B by triangulation).

9.4 A worked TAM/SAM/SOM sketch for the reference site

The reference site is a mid-size international city (canon). Using deliberately conservative structure — every input Grade D and stress-tested in Volume 10:

- **TAM (catchment):** a metro of ~1.5–2.5 M within 60 minutes’ drive. FEC-industry norms suggest 20–35% of households consume location-based family entertainment at least annually; at 2.7 persons/household this yields roughly **450,000-900,000 people in LBE-active households**, plus the corporate, schools, and tourist layers.
- **SAM (format-relevant):** the share for whom a vehicles/machines/skill proposition beats cinema-class alternatives — boys and girls ~6–15 and their parents, adult hobbyists and lapsed hobbyists, teambuilding buyers, STEM coordinators. Assumed at 25–40% of TAM: roughly **120,000-350,000 addressable individuals**.
- **SOM (Year 1):** the canonical Year-1 revenue of ≈\$1.28 M at a blended ~\$30–35 per visitor-visit (admission + F&B + retail mixed across segments) implies roughly **37,000-43,000 visits**. Against the SAM midpoint that requires penetrating on the order of **10–20% of addressable individuals once, or proportionally fewer with repeat visits** — demanding but consistent with what a well-marketed regional FEC achieves in its opening year, and far below the throughput ceiling of the Phase 1 plant (~150 assets × Shift arithmetic, Volume 3/4 capacity tables).

The sketch’s purpose is honesty about magnitude, not precision: the plan does not require heroic penetration, and the binding constraint in Year 1 is expected to be awareness and weekend capacity, not addressable demand.

9.5 Visit-frequency assumptions feeding Volume 10

The following assumptions are **Grade D** (no operator history yet) and are stated here so Volume 10 can stress them explicitly:

Segment	Share of Year-1 visits	Assumed frequency	Basis
Families	~55%	1.5-2.5 visits/year/household	FEC-industry norms for regional attractions; party-driven repeat
Hobbyists/members	~20%	15-40 visits/year (members)	Club-membership behavior at \$50-185/mo precedents
Corporate/groups	~10%	n/a (booked events)	Karting/teambuilding market norms
Schools/STEM	~10%	1-2 programs/school/year	Field-trip cycles
Tourists/other	~5%	one-off	Conservative floor

These blend to the visits and yields behind the canonical **Year-1 revenue ≈ \$1.28 M** (Volume 10 is the source of truth for the arithmetic; this volume supplies the behavioral rationale). The plan's sensitivity analysis treats family visit frequency and member count as the two highest-leverage demand variables.

Investor Note. The honest demand risk is not “will anyone come” — Section 8.1's capacity-starved precedents answer that — but **repeat frequency after novelty**: does a family return in month 9? The design answers (License progression, rotating divisions, events calendar, leaderboards — Volume 9) exist precisely to convert novelty into habit, and the Year-1 KPI dashboard (Volume 10, Chapter 9) tracks repeat rate weekly from opening day.

10. Market Positioning & SWOT

10.1 Where RC WORLD sits

Positioning statement: **the world's first full-scale miniature industrial world — a destination park where anyone can drive, dig, haul, race, sail, and fly real machines in miniature, without owning anything.** Against each adjacent category the position is differentiated, not merely better:

- vs. **hobby tracks**: a hospitality business with hobby-grade depth, not a hobby space tolerating guests;
- vs. **FECs/arcades**: real physics and real skill progression, not redemption games;
- vs. **karting**: comparable competitive thrill at lower price, zero physical risk, from age ~6, weather-extendable, and with construction/marine/aviation breadth karting cannot copy;
- vs. **Diggerland/Dig This**: 80-90% of the psychological payload of machine operation at 10-30% of the price and a fraction of the liability, insurance, and land cost;
- vs. **staying home with a \$120 WLtoys buggy**: the world (tracks, mines, ponds), the fleet variety, the telemetry gamification, and the community — the same reason gyms coexist with home dumbbells.

10.2 Barriers to entry and defensibility

Honest assessment: **no single element of RC WORLD is patentable or secret.** Vehicles are catalogue items; tracks are earthworks. Defensibility is a stack:

1. **Operational know-how** — fleet balancing (1:3 doctrine), battery logistics at 450+ pack scale, 20-minute-Shift throughput engineering, repair-hour economics. This is exactly the know-how this Master Plan encodes, and it is genuinely hard-won (the German clubs took decades to learn their fraction of it).
2. **The telemetry/ERP layer** — RCW Node + RC WORLD OS delivers kill-switch safety, anti-abuse enforcement, queue automation, and per-asset economics no copycat gets from off-the-shelf parts quickly (Volume 13).
3. **Capital and land** — \$1.85 M Phase 1 on 4.8 ha is modest for an attraction but far beyond hobby-shop copycats; the realistic fast-follower is an FEC chain, which would need years to acquire the RC-specific operating depth.
4. **Community and license lock-in** — Gears balances, License tiers, leaderboard history, and club partnerships create switching costs no new venue inherits.
5. **Brand-of-first-mover** in an emerging category (Section 8.1's window).

What defensibility does **not** include: exclusivity of supply (anyone can buy Huina), location monopoly beyond ~60–90 minute catchment, or IP moats. Franchise speed (Volume 12) is therefore itself a defensive strategy: occupy the category name before the trampoline-park consolidation moment arrives.

10.3 SWOT

	Helpful	Harmful
Internal	Strengths — first-mover at destination scale in a validated emerging format; hobby-grade unit economics (\$60–450 fleet assets vs \$50k+ karting/real machines); multi-division breadth = repeat-visit variety + all-demographic reach; proprietary telemetry/ERP stack; safety story (nobody rides anything); founder domain expertise encoded in this plan	Weaknesses — no operating history (Grade D demand assumptions); weather exposure until Phase 3 indoor arena; single-region (China) supply concentration; novelty-dependence risk on repeat visits; staff skill bottleneck (Artisans must be trained, not hired ready-made); modest brand budget vs FEC incumbents
External	Opportunities — RC industry growth (5.8–7.5% CAGR) and hobby-grade shift feeding equipment quality down-market; LBE market growth (7–14% CAGR) and participatory-format trend; category consolidation play (franchise, Volume 12); corporate/STEM weekday demand largely unserved; media-friendliness of the format (every zone is a video set); club/community partnerships available cheaply	Threats — FEC or karting chain fast-follower after our proof-of-concept; tariff/trade disruption to Chinese supply (mitigation: Volume 8 dual-sourcing and spares depth); LiPo regulatory tightening (mitigation: bunker doctrine already exceeds norms); macro discretionary-spend downturn; land-lease insecurity (the classic club killer — mitigated by lease terms in Volume 1, Chapter 6); reputational risk from a single serious battery/injury incident

10.4 Competitor-response scenarios

Positioning is only as good as its behavior under attack. Three scenarios, with prepared responses:

1. **FEC chain adds an “RC zone” (most likely, 2-4 years post-launch).** A trampoline-park or karting operator bolts a rental RC track onto an existing venue. Expected execution: toy-grade or low-hobby fleet, no telemetry, no progression, maintenance neglect leading to dead-vehicle rates that poison the experience. Response: publicize fleet uptime and lap-parity standards; accelerate License-program lock-in; treat their zone as category marketing that sends disappointed customers toward the real thing. The German club record (Section 5.2) and US track economics (Section 8.2) both show that under-maintained RC fleets fail visibly within months — this attack defeats itself unless the attacker buys the operating discipline, which is the expensive part.
2. **A funded copycat clones the full park in another city.** Not a threat to the reference site (LBE catchments rarely overlap at 60+ minutes) but a race for the category brand and franchise position. Response: the Volume 12 franchise timetable exists precisely to win this race; documentation depth (this plan) makes RC WORLD the franchisor of record before a copycat writes its first SOP.
3. **A major manufacturer goes direct-to-experience.** Tamiya already operates circuits; a Traxxas- or Horizon-branded park is conceivable. Assessment: manufacturers are structurally biased toward single-marque venues (Takegawa’s Tamiya-only rule) that under-serve the breadth a destination needs, and their 2025 retrenchment from community events (Section 6.1) signals appetite in the opposite direction. A manufacturer partnership — branded zones, fleet supply deals, sponsorship (Volume 1, Chapter 9’s sponsorship strategy) — is the more probable and more valuable outcome; the park should court it.

10.5 Positioning risks we accept knowingly

Two final honesty items. First, the plan bets that **operation beats expectation** durably — that driving a miniature excavator stays compelling after the third visit in a way passive attractions do not; the precedents suggest yes, but only Year-1 cohort data will prove it. Second, the plan bets that **breadth is a moat rather than a focus-dilution** — that five divisions run to this plan’s standards beat one division run twice as deep. The phasing (racing + construction first, canon Phase 1) is deliberately sequenced so that if the second bet weakens, the park is still a viable two-division business before Phase 2 capital commits.

Volume summary & cross-references

This volume established the market foundation for the Master Development Plan. The global RC industry is a **\$1.5-6.2 B ground-vehicle market inside a \$3.2-7.0 B hobby-RC market** (industry analysts, 2025-2026), growing 5.8-7.5% annually, bifurcating toward hobby-grade product, fragmented across hundreds of suppliers, and structurally **equipment-rich but venue-poor**. China’s Shantou toy cluster, Shenzhen electronics cluster, and Foshan metal-model specialists supply every fleet class in the plan at wholesale, with OEM/ODM culture enabling park-liveried fleets and second-sourcing (Chapter 3 → Volume 8). Japan contributes the circuit-and-retail culture template (Takegawa, Mini-Z, drift); Europe contributes the scale-truck Funktionsmodellbau proof-of-demand and distribution lessons (LRP’s fragility,

Amewi’s wholesale model); the US contributes brand-building and aftermarket-monetization lessons (Traxxas, Pro-Line) plus the 2025 Horizon Hobby warning that hardware margins are thin — venues monetize what manufacturers cannot.

The commissioned diorama study (Chapter 7) found no single “diorama industry” but a workable four-layer supply stack: LESU/JDModel-class 1/14 site equipment, Shenzhen architectural model firms (QZY-class) for display pieces, Chinese model-railway scenery suppliers (Evemodel-class) for dressing, and Guangdong contract fabrication plus in-house 3D printing for everything at fleet quantity — with figures identified as a genuine supply gap the park fills in-house.

Competitor research (Chapter 8) verified the format emerging independently in Japan, Australia, Singapore, and the UK during 2024-2026, all capacity-starved; indoor tracks anchor rental pricing at \$25/20 min; full-size dig attractions prove willingness-to-pay to \$329/session; and no multi-zone RC park exists at hectare scale — the white space is real and time-limited. Customer analysis (Chapter 9) triangulated the canon Shift pricing into the demonstrated \$13-130/hour band and flagged repeat-frequency as the honest demand risk.

Key facts established for other volumes to reference: the \$1.5-6.2 B / 5.8-7.5% CAGR industry range and \$30-40 B LBE market range; verified competitor price anchors (Dig This \$195-1,311; Diggerland ~\$47-54; US track rentals \$25/20 min; Fuji Subaru Land ¥2,000/h; Dig IT A\$65); the manufacturer profiles and confidence grades in Chapters 3-6; the four-layer diorama sourcing stack and the figures gap; the five-segment demand model and Grade-D frequency assumptions; and the defensibility stack of Section 10.2.

Cross-references: Volume 1 (executive summary consumes this volume’s headline findings); Volume 3, Chapters 2 and 9 (racing fleet classes and club/event programming); Volume 4 (construction division builds on Sections 5.2, 7.2, 8.6); Volume 5 (aviation division consumes Section 8.4); Volume 6 (marine division consumes Section 8.5); Volume 7, Chapter 8 (in-house scenery and figure fabrication standards); Volume 8 (procurement executes the supplier landscape of Chapters 3 and 7); Volume 9 (segmentation and WTP feed experience and membership design); Volume 10 (demand assumptions of Sections 9.4-9.5 feed the financial model and its sensitivities); Volume 11, Chapter 7 (outdoor theming specs per Section 7.3); Volume 12 (franchise strategy executes the consolidation window of Sections 8.1 and 10.2); Volume 13 (telemetry/ERP defensibility per Section 10.2).

Volume 3 — RC Motorsport Division

RC WORLD — Master Development Plan · Volume 3 of 13 Revision 1.0 — July 2026 ·
Status: Living document — bump revision on material change

Purpose of this volume. This volume is the complete engineering and operating specification for the RC WORLD Motorsport Division: the Phase 1 fleet of **64 racing cars** across seven vehicle classes, the three-track Speed & Off-Road Hub (Tracks A and B in Motorsport scope; Track C belongs to the crawler program in Volume 4's adventure annex), and the race-operations business built on top of them. It codifies the division's founding doctrine — **standardized chassis, personalities created through tuning, races decided by driver skill, lap-time parity within 2-3% across every car in a class** — and turns that doctrine into working engineering: platform selection, parts commonality, fleet blueprinting and balancing procedures, ESC programming tables, suspension setup sheets, gearing mathematics, corner-weighting procedure, rental-hardening specifications, RCW Node integration, Shift turnaround workflow, and race-day operations. A competent race engineer hired into RC WORLD should be able to run this division from this volume plus the workshop practices in Volume 7.

Intended readers. The Motorsport Lead and race director; Artisans assigned to the racing fleet; the General Manager; investors and lenders assessing the largest single revenue engine of Phase 1; procurement staff executing the fleet purchase (with Volume 8); the RC WORLD OS development team implementing telemetry and timing features.

Chapters

1. Division Concept & Fleet Philosophy
2. The Class Ladder: Vehicle Classes & Fleet Composition
3. Standard Chassis Strategy & Parts Commonality
4. Performance Balancing: The Parity Engineering Method
5. ESC Programming & Powertrain Electronics
6. Suspension Tuning Guide
7. Weight Distribution & Corner Weighting
8. Gearing, Rollout & Thermal Management
9. Drift Program Deep-Dive
10. Surface-Specific Setup: Touring, Rally & Short-Course
11. Formula RC & Drag Racing Programs
12. Rental Fleet Engineering & the Shift Turnaround

- 13. Race Operations
- 14. Division KPIs & Dashboard
- 15. Volume Summary & Cross-References

1. Division Concept & Fleet Philosophy

1.1 What the Motorsport Division sells

The Motorsport Division sells the feeling of being a racing driver — not the feeling of owning a fast toy. A customer who buys a Casual Shift (\$15, one 20-minute block) steps into a complete motorsport frame: a pit lane, a scrutineered car with an asset tag and live telemetry, a professional timing loop, a leaderboard with their name on it, a setup sheet clipped to the car stand. The car is deliberately not the hero. The hero is the lap time, and the lap time must be earned.

This is why the division’s founding doctrine, inherited directly from the original RC Racing Experience Park strategy, reads as it does:

Doctrine. One or two standardized 1/10 hobby-grade chassis platforms per class, with common electronics and common batteries. Vehicle “personalities” are created through tuning — ESC profiles, gearing, suspension, weight distribution, drive layout — never through different hardware. Races are decided primarily by driver skill. Lap times across every car in a class stay within **2-3%** of each other, verified weekly.

Every clause carries operational weight. Standardized chassis means an Artisan who can rebuild one touring car can rebuild all fourteen, and the spares shelf holds one suspension-arm SKU instead of five. Common electronics and batteries means one ESC programming procedure, one connector standard (XT60, per park-wide battery doctrine), and a 3:1 battery pool shared across the class. Personalities through tuning means a customer choosing between “GT Stable” and “Touring Agile” is choosing spring rates, drag brake, and ballast placement — dialed in or reverted in fifteen minutes — not choosing between inventories. Decided by driver skill is the commercial promise the RC WORLD License progression (Volume 9) monetizes. And 2-3% parity is the falsifiable engineering claim that holds the promise together; Chapter 4 defines how it is measured and maintained.

1.2 Rental-grade durability doctrine

Hobby-grade RC cars are designed for owners who crash their own property. RC WORLD’s cars are crashed by strangers, forty times a day, into other people’s cars. The fleet therefore follows a **rental-grade durability doctrine** that intervenes at four levels:

1. **Platform level.** Select chassis with proven crash architecture: shaft-drive tub chassis over exposed belt-drive carbon plates for the standard classes; foam bumpers oversized one grade; steel driveline components where the aftermarket offers them. A rental car’s Bill of Materials is optimized for **mean time between failures and mean time to repair**, not for the last 0.2 seconds of pace. We give away laptime headroom deliberately and claw parity back with tuning (Chapter 4).

2. **Hardening level.** Every car receives a standard hardening pass before entering service (Section 12.2): metal drive gears where plastic is marginal, threadlock on all metal-to-metal fasteners, silicone-sealed electronics, reinforced body mounts, skid plates on off-road classes, and the RCW Micro-Node soldered inline with strain relief.
3. **Electronic level.** Rental ESC profiles cap punch, lock timing at zero, limit reverse, and enforce soft low-voltage cutoffs (Chapter 5). Transmitter end-point adjustment (EPA) caps steering throw and throttle percentage by customer license tier. The RCW Node adds the park-side kill-switch, geofence, and under-voltage intercept.
4. **Procedural level.** The 20-minute Shift turnaround (Section 12.5), the daily fleet-readiness checklist (Section 12.6), and the weekly parity audit (SOP-MS-002) catch degradation before customers do.

Field Note. The single most damaging failure mode in a rental race fleet is not the spectacular crash — it is the slow car that nobody notices. A car 6% off the pace poisons every race it enters: the customer driving it concludes the game is rigged, and they are right. Treat “car feels slow” reports from staff or customers with the same urgency as a broken suspension arm. The parity audit exists because slow cars hide.

1.3 Where Motorsport sits in the park

Phase 1 gives the division two of the three Speed & Off-Road Hub tracks:

- **Track A (Asphalt/Drift)** — polished concrete/fine asphalt circuit, ~180 m racing line, 3.5–4.5 m lane width, with a dedicated drift annex and a straight long enough to double as the drag strip (Section 11.5). Home surface for entry 1/14, 1/10 touring/GT, drift, Formula RC, and drag.
- **Track B (Baja/Rally)** — packed-dirt circuit, ~160 m line, controlled jumps (≤ 0.5 m landing-matched tabletops), bermed corners, watered and rolled daily. Home surface for buggy/rally and short-course.

Track C (Crawler Trail) is operationally part of the crawler/adventure program and is covered with the Construction Division’s outdoor experiences in Volume 4; its 16-crawler fleet is not part of the 64-car Motorsport count. Physical track engineering — subgrade, drainage, barriers, driver rostrums, pit lane geometry — is specified in Volume 11; this volume states only the requirements racing imposes on those designs (Sections 9.6, 10.4, 11.5, 13.2).

The division also owns the racing calendar (Section 13.6), the leaderboard and license integrations in RC WORLD OS, and the pit-lane theatre — the Artisan battery swap that turns an Operator Shift’s mid-session pause into a Formula 1 pit stop.

1.4 Commercial logic

At the pricing anchors set in the style guide (Casual Shift \$15; Operator Shift \$26; race-event entries priced per event), motorsport is the highest-throughput revenue engine of Phase 1: 64 cars against two tracks and a structured queue means the division’s ceiling is set by track slots, not fleet size. The financial model (Volume 10) carries the consolidated projections; the division-level operating targets it assumes are in Chapter 14.

Investor Note. The parity doctrine is margin protection, not engineering vanity. A parity fleet lets any car serve any customer in its class, pushing utilization toward the theoretical maximum — roughly 15% fewer cars for the same throughput than a “personality hardware” fleet where specific cars are demanded and others idle. It also collapses spares inventory: Section 3.4 holds the racing spares catalogue to ~130 SKUs for 64 cars.

2. The Class Ladder: Vehicle Classes & Fleet Composition

2.1 Design of the ladder

The class ladder serves three masters: the **customer journey** (an eight-year-old’s first Shift and a club racer’s Tuesday night must both find their class), the **RC WORLD License progression** (each rung is a license gate — Volume 9), and **fleet economics** (each class must justify its spares line and training hours). Seven classes ride on effectively four chassis platforms (Chapter 3) — that is how the ladder stays affordable.

The canonical Phase 1 motorsport fleet is **64 cars**, allocated as follows. The 24-car touring/GT allocation spans two rungs — the 1/14 entry class and the 1/10 standard class — per the park-wide fleet canon.

#	Class	Scale	Fleet	Platform (primary)	Track	Speed (rental cap / cap-ability)	Skill gate
1	Entry / Academy	1/14	10	MJX Hyper Go 14301/14210-class; LDRC entry scale	A (+ B pad-dock loop)	30 km/h / ~55 km/h	Learner (age 8+, Toolbox Talk)
2	Touring / GT	1/10	14	Shaft-drive 4WD touring (TT-02 Type-S / Carten T410 class)	A	40 km/h / ~55 km/h	Apprentice
3	Drift	1/10	12	MST RMX 2.5 S+ (RWD)	A drift annex	25 km/h / ~35 km/h	Apprentice
4	Buggy / Rally	1/14	12	WLtoys 144010-class, upgraded	B	50 km/h / ~75 km/h	Apprentice
5	Short-Course	1/10	8	2WD/4WD SCT (Slash-class; ZD Racing value alternative)	B	40 km/h / ~55 km/h	Apprentice
6	Formula RC	1/10	4	F104/TRF103-class pan car, 21.5T spec	A	45 km/h / ~55 km/h	Foreman + Motorsport Master endorsement (supervised)

#	Class	Scale	Fleet	Platform (primary)	Track	Speed (rental cap / capability)	Skill gate
7	Drag	1/10	4	Touring-derived no-prep drag conversion	A straight (40.2 m strip)	event only / 100+ km/h	Any license (staged launch); Foreman + endorsement for free-brake passes
Total			64				

Skill gates use the RC WORLD License tiers (Learner → Apprentice → Operator → Foreman → Site Manager → Legend — Volume 9), enforced twice: the OS refuses a transmitter pairing above the customer’s license tier (Section 12.1), and each tier maps to a locked **transmitter profile** — Rookie (Learner/Apprentice), Standard (Operator), Licensed (Foreman+), per Volume 9 §3.6 — that caps EPA and throttle curve regardless. Electronics stack per class is consolidated in Section 3.3; per-class detail follows.

Classes 6 and 7 (8 cars combined) are **special-event classes**: they do not run walk-up Shifts but anchor scheduled events, exhibitions, and top-tier license experiences (Chapter 11).

2.2 Class 1 — Entry / Academy (1/14, 10 cars)

Role. The first rung: walk-up customers, children 8+, birthday parties, the Learner license test bed. Everything about this class minimizes the cost of a mistake — kinetic energy at 30 km/h in a 1.4 kg car is about 49 J, roughly a tenth of a 1/10 buggy at speed.

Hardware. The MJX Hyper Go 1/14 family is the reference platform: metal chassis, 4WD, oil-filled shocks, 2845/2852-class brushless motor with 45 A independent ESC, 2S LiPo. The 14301/14302 rally-pattern cars suit Track A duty; the 14210 truck variant tolerates paddock abuse. Manufacturer speed claims reach 55 km/h on 2S; RC WORLD caps the class at ~30 km/h via throttle EPA and ESC profile (Chapter 5). LDRC’s entry-scale line is the secondary source for licensed-body variety (supplier detail in Volume 8). Wholesale cost per car including spares kit: \$90–140.

Fleet notes. Ten cars support the highest churn in the park; the class runs a 4-car hot-spare rotation because turnaround damage is concentrated here. Batteries: 2S 3000 mAh packs from the shared 2S pool, XT60 (factory connectors replaced at induction).

2.3 Class 2 — Touring / GT (1/10, 14 cars)

Role. The signature class: the “real racing car” a customer graduates into, the backbone of league racing, and the class where the parity doctrine is most visible. GT-styled polycarbonate bodies (Killerbody / Pandora RC catalogue) give the fleet its showroom face.

Hardware. Shaft-drive 4WD 1/10 touring chassis, 257–258 mm wheelbase, 190 mm width — the club-racing standard geometry. Platform selection analysis is Chapter 3; the short version: a TT-02 Type-S-class tub chassis or the factory-assembled Carten T410-class rolling chassis, fitted with a sensored 13.5T motor (Surpass Hobby Rocket 3650 class), Hobbywing QuicRun 10BL120 G2 sensored ESC, 15 kg metal-gear digital servo, and 2S 5000 mAh hard-

case LiPo (Gens Ace / CNHL). Control tire: one spec 24 mm pre-glued touring slick for the whole class (Section 4.5). Rental cap 40 km/h; race profile ~55 km/h.

Cost. \$220–320 per car complete depending on platform mix, before the hardening pass.

2.4 Class 3 — Drift (1/10, 12 cars)

Role. The style class — the park’s strongest social-media surface and the anchor of evening programming. Drift rewards throttle finesse over speed; it flatters intermediate drivers and photographs beautifully under lights.

Hardware. MST RMX 2.5 S+ RWD chassis — the current generation of the industry’s reference rear-drive drift platform (257 mm wheelbase, adjustable 6/8/10/12° caster, switchable RMX/RRX gearbox modes). Sensored 13.5T motor for smooth low-RPM torque, QuicRun 10BL120 G2, high-torque low-profile steering servo (25 kg class — drift steering loads are constant), drift-specific gyro, and spec HDPE/ABS drift rings. Full drift engineering is Chapter 9. Cost: \$260–340 per car complete.

2.5 Class 4 — Buggy / Rally (1/14, 12 cars)

Role. The adrenaline class on dirt: jumps, berms, and the fastest sensation-per-dollar in the park. This is the class most first-time adult customers pick second, after one touring Shift.

Hardware. WLtoys 144010-class 1/14 4WD buggy — 2845 4300KV brushless, 60 A ESC on current versions, metal differentials and driveshafts, oil shocks, ~75 km/h capability — upgraded per the hardening pass: metal steering knuckles and servo saver, steel main gears, upgraded body posts, factory battery replaced by pool 2S 2200–3000 mAh XT60 packs. At \$70–110 wholesale plus ~\$30 hardening, the class is cheap enough to run 12 cars with 4 sealed donor units in crates.

Fleet notes. Rally-pattern bodies over buggy chassis give Track B a visual identity distinct from short-course. Rental cap 50 km/h (dirt forgives; barriers on Track B are engineered for this energy — Volume 11). The platform’s stock metal diff cups and CVD front driveshafts are retained; hardening replaces only what fleet experience proves marginal, and shielded bearings swap to rubber-sealed at first bearing service (dirt duty).

2.6 Class 5 — Short-Course (1/10, 8 cars)

Role. The contact-tolerant “truck racing” class: bigger, softer, bouncier — the best class for group events where beginners and intermediates share a grid, because short-course bodies shrug off door-to-door contact.

Hardware. 1/10 short-course truck with full-perimeter body protection. The Slash-class 2WD architecture remains the durability reference with an unmatched spares ecosystem; ZD Racing’s SC-class trucks are the direct-from-China value alternative — Volume 8 carries the trade-off (landed cost vs. parts lead time). Spec: 3650-size sensored 17.5T motors for controllable power, QuicRun 10BL120 G2, 2S 5000 mAh pool packs. Rental cap 40 km/h.

2.7 Class 6 — Formula RC (1/10, 4 cars)

Role. Special events and top-license experiences: open-wheel pan cars that look like Formula 1, reward precision, and punish clumsiness — the graduation class. Class rules and event formats are Section 11.1–11.2.

Hardware. F104/TRF103-class pan chassis, 21.5T sensored “blinky” (zero-timing) spec, foam or spec-rubber F1 tires, 2S LiPo shorty packs. Four cars support exhibition grids and the Formula Experience (guided, instructor-supervised Shifts).

2.8 Class 7 — Drag (1/10, 4 cars)

Role. Pure spectacle in 3-second doses: two cars, a Christmas tree, a 40.2 m strip (the scale 132-foot no-prep standard), and a crowd. Drag racing anchors event nights and monetizes as head-to-head bracket entries rather than Shifts (Sections 11.4–11.6).

Hardware. Touring-derived drag conversions on the standard 1/10 electronics stack, with wheelie bars, drag-specific gearing, and (for exhibition passes only) 3S packs under Artisan control. Rental customers never free-drive drag cars; launches are staged and speed-limited by profile.

2.9 Fleet capacity arithmetic

A class’s revenue ceiling is $\text{cars on track} \times \text{Shifts per hour} \times \text{operating hours}$. With 20-minute Shifts plus 5-minute turnarounds in a 25-minute cycle (Section 12.5), each car supports up to 2.4 Shifts/hour; practical scheduling yields 2.0. Track A runs 8 concurrent touring/entry cars plus 6 in the drift annex; Track B runs 8 off-road. The binding constraint is therefore **track slots (~22 concurrent) rather than the 56 Shift-serving cars** — exactly where a rental fleet wants to be: cars queue for customers, not customers for cars, and turnaround/maintenance can breathe.

3. Standard Chassis Strategy & Parts Commonality

3.1 Selection criteria

Platform selection for a rental race fleet is a different exercise from platform selection for a racer. RC WORLD scores candidate chassis against seven weighted criteria; the weights are the doctrine in numeric form.

Criterion	Weight	What “good” looks like for a rental fleet
Parts availability & ecosystem depth	25%	Every wear part orderable from ≥ 3 suppliers; aftermarket metal upgrades exist for known weak points; parts still available 5 years after purchase
Crash architecture / MTBF	20%	Tub or protected chassis; sacrificial cheap parts break before expensive ones; no exposed belts; foam bumper standard
Mean time to repair (MTTR)	15%	Front-end rebuild < 15 min; motor/pinion access without teardown; modular gearbox
Unit cost incl. hardening	15%	Complete car $\leq$ \$350 at fleet quantities

Criterion	Weight	What “good” looks like for a rental fleet
Tunability envelope	10%	Enough adjustment (camber, toe, droop, oils, gearing) to create personalities AND to equalize slow cars back to parity
Setup consistency car-to-car	10%	Molded-tub repeatability; cars built from the same kit measure the same
Wholesale/direct sourcing viability	5%	Factory-direct or distributor pricing at 12–24 unit volume (Volume 8)

Two consequences deserve emphasis. First, **competition chassis lose**: a belt-drive carbon-plate touring car out-corners a TT-02 by seconds a lap but fails criteria 1–3 catastrophically in rental service — exposed belts eat gravel, carbon plates delaminate at customer crash rates, MTTR triples. Second, **orphan bargains lose**: a cheap chassis with no parts ecosystem becomes disposable the day its donor stock runs out; the WLtoys 144010 passes only because its ecosystem is enormous and the whole car costs less than a competition chassis’s spare drivetrain.

3.2 Recommended platforms (verified July 2026)

Touring/GT — two-platform strategy. The class runs a deliberate split:

- **Tamiya TT-02 Type-S class (8 cars).** The TT-02 remains, in 2026, the world’s default club-racing entry tub: shaft-drive 4WD, enclosed drivetrain, molded bathtub chassis, and the broadest spare-parts network in the hobby (B-parts arms #51528, A-parts uprights #51527, driveshafts #51006 are stocked by effectively every hobby retailer on earth). The Type-S adds the better suspension package out of the box. Its known rental weaknesses — kit-plastic bearings on the base model, soft friction shocks, plastic steering — are solved by a standardized \$45 hop-up pass (full bearings, oil shocks, aluminum steering set, metal motor mount) at induction.
- **Carten T410-class ARTR rolling chassis (6 cars).** The T410/T410R arrives factory-assembled with oil dampers, CVDs, full bearings and metal-gear diffs on the same 258 mm / 190 mm club geometry — a stronger race base with near-zero build labor at a moderate premium. Running both platforms in one class hedges supply risk and lets the weekly parity audit prove the doctrine works across chassis makes. 3Racing’s Sakura line is the tracked third option and standing price benchmark; not stocked in Phase 1, protecting SKU discipline.

Drift — MST RMX 2.5 S+ (single platform). MST’s RMX 2.5 family superseded the long-running RMX 2.0S and is the current RWD reference: switchable RMX/RRX gearbox modes, adjustable caster uprights, HT short rear lower arms, and MST’s unrivalled drift option-parts catalogue. The S+ kit (532205, ~\$170–220 street) is the fleet spec, built with fleet-standard electronics (Section 3.3). MST’s TCR platform is a FWD touring design, explicitly out of scope for the drift fleet.

Buggy/Rally — WLtoys 144010-class. Selected on overwhelming value: metal-drivetrain 4WD brushless at \$70–110 wholesale with a giant parts ecosystem. Run as a consumable platform: harden at induction, cannibalize retired cars, keep four sealed donor units per twelve fleet cars.

Short-course — Slash-class 2WD primary. Chosen for the deepest durability record and spares network in the category; ZD Racing SC-class held as the value alternative if landed-cost pressure demands it (decision matrix in Volume 8).

Formula — F104/TRF103-class pan car. Chosen because pan-car simplicity (two diffs' worth of drivetrain fewer than a touring car) suits a 4-car special-events micro-fleet.

Trade Hack. When qualifying any chassis for fleet duty, buy two units and give one to your roughest staff driver for a week with instructions to drive like a customer. Strip both cars side by side afterwards. The delta between the two teardowns is your real hardening list — manufacturer weak-point folklore from forums is usually one product generation out of date.

3.3 The common electronics stack

Every 1/10 car in the division, regardless of class, carries the same electronic architecture:

Component	Fleet standard	Rationale
ESC	Hobbywing QuicRun 10BL120 Sensored G2	120 A continuous; sensed smoothness at zero throttle (critical for drift and novice control); 12 programmable items incl. punch, drag brake, timing (Chapter 5); programmable by LCD program box in seconds per car
Motor	Surpass Hobby Rocket 3650 sensored: 13.5T (touring, drift), 17.5T (short-course 2WD), 21.5T (formula)	One motor family, three windings; interchangeable sensor boards and bearings
Servo	15 kg-cm metal-gear digital standard; 25 kg low-profile for drift	Two SKUs total
Receiver	Fleet transmitter system's 4-ch receiver (Section 12.1)	Model-memory binding, EPA lockout
Telemetry	RCW Micro-Node (~25×25 mm, ES-P32-C3, soldered pads, conformal coated)	Voltage, GPS, kill-switch PWM intercept, lap beacons
Battery	2S 5000 mAh hard-case LiPo, XT60 (Gens Ace / CNHL)	Shared pool, 3:1 ratio, bunker-charged

The 1/14 classes keep their factory 45–60 A ESCs (adequate and cheap to replace whole) but standardize batteries to the pool and receive the same Micro-Node.

3.4 Parts commonality matrix

The matrix below is the inventory argument for standardization, expressed as SKU counts. “Shared” means one stock line serves multiple classes.

Spares category	Touring (14)	Drift (12)	Buggy (12)	SCT (8)	Entry (10)	F/Drag (8)	SKU lines
Motors (3650 family)	13.5T	13.5T	factory	17.5T	factory	21.5T	3 + 2 factory
ESC	10BL120 G2	shared	factory 60 A	shared	factory 45 A	shared	1 + 2 factory

Spares category	Touring (14)	Drift (12)	Buggy (12)	SCT (8)	Entry (10)	F/Drag (8)	SKU lines
Servos	15 kg std	25 kg LP	15 kg mini	15 kg std	factory	15 kg std	3
Batteries	2S 5000 pool	shared	2S 2200–3000 pool	shared	shared w/ buggy	shared + 3S event	3
Bearings	5×10, 10×15 sets	largely shared	144010 set	Slash set	MJX set	shared w/ touring	5
Suspension arms	TT-02 B-parts; T410 NHA402	MST arm sets	144010 arms	Slash arms	MJX arms	pan-car links	6
Drive-shafts/CVDs	2 types	1	1	1	1	1	~6
Spur/pinion (48P std where possible)	48P range	48P	factory Mod	32P/48P	factory	48P/64P	~10
Tires (control spec)	1 slick	1 drift ring	1 knobby	1 SCT	1	2	7
Body posts/clips/hardware	shared M3 hardware system park-wide						~8

Total racing spares catalogue: **≈130 active SKUs for 64 cars**. The equivalent unstandardized fleet (seven unrelated platforms with per-class electronics) models out at 320–380 SKUs — roughly 2.7× the working capital in spares and bin locations, and an Artisan training matrix that never converges. High-turnover lines (arms, knuckles, shock shafts, pinions, tires, bearings — Volume 7’s failure catalogue) are stocked at min/max levels driven by the parts-per-100-Shifts KPI (Chapter 14).

3.5 Training leverage

Standardization compounds in labor. The Artisan certification path (Volume 7) requires a full teardown-rebuild sign-off per platform; with four core platforms instead of seven-plus, a new Artisan reaches full fleet coverage in roughly half the training hours, and any on-shift Artisan can service any car that rolls into the pit lane. That is what makes the 5-minute Shift turnaround (Section 12.5) staffable with two Artisans per track instead of a specialist per class.

3.6 Sample procurement specification

Fleet purchases are executed by Volume 8’s procedures against specifications written in this format. The touring-class example below is the template; every class has an equivalent sheet as a controlled document in RC WORLD OS.

Procurement spec MS-PS-02 — Touring/GT fleet car (extract)

Line	Specification	Qty	Target landed cost
Chassis kit	1/10 shaft-drive 4WD touring, 257-258 mm WB, 190 mm width, tub chassis, enclosed drivetrain; TT-02 Type-S class or Carten T410-class ARTR per fleet split (Section 3.2)	14 + 2 donor	\$95-165/unit
Hop-up pass	Full ball-bearing set, oil dampers (if kit lacks), aluminum steering set, steel motor mount, spare B-parts arms x2 sets/car	14 kits	\$45/car
Motor	Sensored 13.5T 3650, ROAR-legal wind, replaceable sensor board (Surpass Hobby Rocket class)	16 (2 bench spares)	\$30-45/unit
ESC	Hobbywing QuicRun 10BL120 Sensored G2 (P/N 30125002) + 2 × LCD Program Box Pro	16 + boxes	\$45-60/unit
Servo	15 kg-cm metal-gear digital, 6.0 V, standard case	16	\$15-25/unit
Body	190 mm polycarbonate GT shells, 1.0 mm (Killerbody / Pandora RC catalogue), unpainted	21 (1.5/car)	\$25-45/unit
Tires	Spec 24 mm pre-glued touring slick, single compound, single supplier lot	40 sets	\$12-18/set
Batteries	2S 5000 mAh hard-case LiPo, XT60, ≥50C (Gens Ace / CNHL)	42 (3:1)	\$25-40/unit

Acceptance terms on every spec: goods inspected at The Works within 14 days; motors bench-tested on arrival (Section 4.3), >4%-off-median units rejected as a lot-quality signal; one full spares assortment per 6 cars in the same consignment (Volume 8, spares-with-fleet rule).

4. Performance Balancing: The Parity Engineering Method

4.1 What “2-3% parity” means, precisely

For each class, on its home track, a **reference driver** (a designated staff driver, rotated to prevent drift in the reference itself) drives every car for a 5-lap flying run on control tires. For each car, take the **best 3-lap average** (discarding the out-lap and any incident-flagged lap). The class parity statistic is:

$$\text{parity spread} = (T_{\text{slowest}} - T_{\text{fastest}}) / T_{\text{fastest}} \times 100\%$$

Pass: spread ≤ 3.0%. Target: ≤ 2.0%. Action threshold: any single car > 1.5% off the class median goes to the balancing bench even if the class as a whole passes. On a 22-

second touring lap, 3% is 0.66 s — noticeable to a good driver over a race, invisible within the noise of a novice's driving, which is exactly the tolerance the customer promise requires.

Lap-time parity, not component parity, is the governing metric: components are balanced only as the means to the lap-time end. The audit is SOP-MS-002 (Section 4.8), run weekly per class and after any car's major repair.

4.2 The five balancing levers

A car's lap time decomposes into power, rolling stock, mass, gearing, and chassis condition. The balancing method attacks them in a fixed order — cheapest and most-determinative first:

1. **Control tires (Section 4.5)** — one spec tire per class, age-tracked. Tires are 40-60% of lap-time variance in a spec fleet; nothing else is worth measuring until tires are equalized.
2. **Motor blueprinting (Section 4.3)** — bench-test every motor; bin into performance groups; pair strong motors with heavy chassis.
3. **ESC profile equalization (Chapter 5)** — identical firmware parameters per class, verified by LCD program box readback, not by memory.
4. **Ballast to equal mass (Section 4.4)** — all cars in a class ballasted to the heaviest car's mass $+0/-5$ g, placed to also equalize static distribution (Chapter 7).
5. **Gearing trim (Chapter 8)** — final ± 1 pinion-tooth adjustments to close residual straight-line deltas.

Suspension settings are deliberately not a parity lever — they are the personality lever. Parity is achieved with the five levers above at a neutral baseline setup; personalities (Section 5.3) are then layered on top and verified not to break parity.

4.3 Motor blueprinting and binning

Mass-produced motors vary. Two “identical” 13.5T sensored motors can differ 3-5% in output from winding tolerance, magnet strength (zeta), bearing drag, and sensor-timing offset. The Works maintains a motor bench (detail in Volume 7, Chapter 6) and every fleet motor is characterized at induction and at every 200-Shift service:

Bench procedure (no dyno required). With a charged reference pack, a bench PSU log, and a phototach: 1. Measure **free-run RPM** at a fixed 7.60 V supply with a spec pinion-mass flywheel. Record RPM and free-run current. 2. Measure **loaded current** driving the spec test fixture (a fan-brake disc) at fixed throttle. 3. Check bearing drag by spin-down time from 10,000 RPM; reject or re-bearing below spec. 4. Record all values to `fleet_inventory` (motor sub-asset) in RC WORLD OS.

Binning. Sort the class's motors by free-run RPM into three bins — **A (top third), B (middle), C (bottom)** — and pull any motor $>4\%$ below the class median for inspection (typically worn bearings or a damaged sensor board; replace, don't shim). Bins are assigned to chassis to cancel variance: A-bin motors to the cars that weighed heaviest or tested slowest as rollers; C-bin to the freest chassis. A true motor dyno (Fleet Analyzer class) is a justified Year-2 purchase; the bench method holds parity well inside 3% without it.

4.4 Ballast equalization

Weigh every car race-ready (body, battery of median pool mass, transponder, Node). The heaviest car defines class minimum weight, rounded up to the next 5 g; every other car is ballasted up to that figure with self-adhesive 5 g/10 g plates placed per the class ballast map (Chapter 7). Mass equality does double duty: it equalizes acceleration ($a = F/m$) and removes the temptation to “fix” a slow car by stripping weight, which would silently break corner-weight balance.

4.5 The control tire rule

Doctrine. One tire specification per class. No exceptions, including staff cars used in customer races. Tires are lifed in Shifts and retired on schedule or on wear indicator, whichever first.

Tires are bought in bulk lots, each sample-tested (three cars, reference driver) before release; a lot shifting lap times >1% against the incumbent triggers a full-class re-audit on changeover day. Tire age is tracked by a paint-dot code on the wheel: touring slicks life at 60 Shifts, buggy knobbies 80, drift rings 150 (wear-driven), SCT 80. Mixed-age sets are forbidden; cars re-tire as sets.

4.6 Worked example: bringing three cars into parity

Three touring cars from the fleet, audited on Track A (reference driver, 5-lap runs, best-3 average). Baseline audit:

Car	Chassis	Best-3 avg	Δ vs. fastest	Race-ready mass	Motor free-run @7.60 V	Tire age
T-04	TT-02 S	22.14 s	—	1,512 g	31,800 RPM (A-bin)	22 Shifts
T-09	T410	22.61 s	+2.1%	1,478 g	30,450 RPM (B-bin)	41 Shifts
T-11	TT-02 S	23.05 s	+4.1%	1,530 g	29,150 RPM (C-bin, −8.3% vs. T-04)	58 Shifts

Class spread = $(23.05 - 22.14)/22.14 = 4.1\%$ — **audit fail**, with T-11 the actionable outlier and T-09 marginal.

Step 1 — Tires. T-11’s tires are at 58 of 60 Shifts — end of life. New control set fitted to T-11; T-09’s 41-Shift set retained (mid-life is the fleet norm and the parity target must hold across normal tire life). Re-run: T-11 improves to 22.71 s (tires alone recovered ~0.34 s, 1.5%).

Step 2 — Motor. T-11’s motor free-run is 8.3% below the class fastest — outside the 4% bin rule. Inspection finds rough motor bearings; bearings replaced, re-bench: 30,900 RPM (B-bin). Motors stay in place post-repair; ballast does the cancelling in the next step.

Step 3 — Mass. Heaviest car is T-11 at 1,530 g → class minimum set at 1,530 g. T-04 receives +18 g (two 5 g plates at the rear cross-brace, one 5 g plate + 3 g trim on the battery tray fore position, per the touring ballast map to hold 50/50 — Chapter 7). T-09 receives +52 g (30 g low-center battery-tray plates, 22 g split front/rear to preserve its measured 49.8/50.2 distribution). Note the deliberate effect: the fastest car gained the most mass.

Step 4 — ESC readback. Program-box readback on all three: T-09 found with punch at Level 6 versus the rental-standard Level 3 — a leftover from an event night, and the explanation for its strong baseline launches. Reset to the Touring RENTAL profile (Table 5.2); profiles now identical.

Step 5 — Gearing trim. Post-correction straight-line trap speeds (RCW Node telemetry over the main-straight sector): T-04 38.9 km/h, T-09 38.1 km/h, T-11 38.4 km/h. T-09 gets +1 pinion tooth (27T → 28T, FDR 6.16 → 5.94, +3.7% top-end, negligible on a 22 s lap's single straight but closes the trap gap; motor temp check after 10 min shows 71 °C — inside the 85 °C ceiling, accepted).

Verification audit:

Car	Best-3 avg	Δ vs. fastest
T-04	22.31 s	—
T-09	22.48 s	+0.8%
T-11	22.63 s	+1.4%

Class spread **1.4% — pass, inside target**. Bench time: ~2.5 Artisan-hours for three cars; parts cost one bearing set and one tire set, both scheduled consumables. Note that T-04 got slower (+0.17 s from ballast): parity engineering levels down as well as up, deliberately. The customer promise is a fair race, not the fastest possible car.

4.7 Continuous parity monitoring between audits

The weekly audit is the calibrated measurement; the RCW Node makes the other six days observable too. Every customer lap generates a telemetry lap record tagged with car ID, driver license tier, battery pack ID, and ambient. Raw customer lap times are useless for parity directly — driver skill variance swamps car variance — but two derived statistics are not:

- **Per-car best-decile drift.** RC WORLD OS tracks each car's 90th-percentile-best lap by Foreman-and-above drivers over a rolling 14 days, normalized to the class's same-day median. A car drifting >1% slow over a week is flagged for the bench before the weekly audit would catch it. The filter matters: good drivers find a car's real pace; novices find their own.
- **Straight-line sector consistency.** The Node's speed trace over the main-straight geofence sector is driver-insensitive at full throttle. A car whose trap-speed distribution shifts down 2%+ against class peers has a powertrain problem regardless of what its lap times say.

Both statistics render on the fleet dashboard (Chapter 14) as per-car sparklines. They do not replace the audit — telemetry cannot control for tires, traffic, or track state the way a reference driver can — but they turn the weekly snapshot into a continuous control loop, and they usually name the guilty subsystem before the car reaches the bench.

4.8 SOP-MS-002 — Weekly Fleet Parity Audit

SOP ID	SOP-MS-002
Revision	1.0 (July 2026)
Owner	Motorsport Lead

PPE	None beyond track-standard (closed shoes, hi-vis on live track)
Tools	Reference transmitter; charged reference packs (storage-rotated); timing system live; LCD Program Box Pro; scales; pit thermometer/IR gun; tablet with RC WORLD OS fleet audit form
Frequency	Weekly per class (rolling schedule, one class per week-day); additionally after any major repair, motor/ESC swap, or tire-lot changeover

1. Close the track to customers for the audit window (schedule pre-opening; ~45 min per 12-car class).
2. Verify conditions: surface dry, ambient within 10 °C of last audit (log both); sweep/blow the racing line.
3. Stage all class cars race-ready: control tires within life, pool battery at full charge, bodies on, Nodes reporting.
4. Reference driver runs each car: 1 out-lap + 5 flying laps. Timing system records automatically via transponder ID.
5. RC WORLD OS computes best-3 averages and class spread; the audit form flags cars >1.5% off median.
6. For each flagged car, execute the balancing ladder in order (tires → motor bench → ESC readback → mass → gearing) and re-run.
7. If ambient/surface shifted mid-audit (>0.3 s drift in the reference car re-baseline), re-baseline and re-run affected cars.
8. Sign off the audit in RC WORLD OS; cars failing after balancing go to maintenance status and out of the rental pool.
9. File exceptions to the Motorsport Lead same-day; two consecutive weekly failures for the same car mandates a full teardown (Volume 7).

4.9 Quarantine rules for out-of-spec cars

A car that cannot be balanced back inside 1.5% of class median at the bench enters **quarantine** — a formal state, not a shelf:

- **Status flip.** The car goes to `maintenance` in RC WORLD OS with reason code `PARITY-FAIL`, its transponder and transmitter pairings are voided, and its stand gets a red quarantine tag. A quarantined car physically leaves the ready line; it is never “kept handy just in case” — busy Saturdays are exactly when a slow car sneaks back into service.
- **Diagnosis before parts.** Quarantine work follows the balancing-ladder evidence already collected, plus a corner-weight check and a full drivetrain teardown if free-roll spin-down is >15% below class median. The rule is find the watt, don’t mask it: re-gearing a car with a dragging diff hides the fault and moves the heat somewhere else.
- **Exit criteria.** Release requires: fault identified and logged (`maintenance_logs` with parts used), bench free-roll within spec, and a solo parity re-run landing within 1.5% of the last full audit’s class median (ambient-corrected). The closing Artisan and the Motorsport Lead both sign.
- **Escalation.** Two quarantines in 60 days on the same subsystem escalates to The Works for full rebuild and a pattern review (chronic crasher assignment, bad donor batch, old-impact chassis tweak). Three unexplained quarantines retire the car to the parts-harvest

bench (Section 12.8) regardless of age — fleet math never justifies a car the engineers cannot trust.

5. ESC Programming & Powertrain Electronics

5.1 Why the ESC is the division’s most powerful tuning tool

The ESC is software-defined drivability. On the fleet-standard Hobbywing QuicRun 10BL120 Sensored G2 (~\$45–70 street, verified July 2026), twelve programmable items govern how the same motor, gearing, and chassis feel: running mode, low-voltage cutoff, punch (9 steps), drag brake force (8 steps), maximum brake force (4 steps), initial brake force, maximum reverse force, neutral range, timing (8 steps), overheat protection, BEC voltage (6.0/7.4 V switchable), and motor rotation. Three of these — punch, drag brake, and timing — do most of the personality and safety work in a rental fleet, and one of them (timing) is permanently locked at zero in rental profiles.

Parameter physics, briefly:

- **Punch** (Levels 1–9) rate-limits how fast the ESC ramps current against throttle input. Low punch converts a novice’s throttle stab into a progressive launch; high punch delivers the input verbatim. Punch is the single biggest “feel” lever in the fleet and the first line of defense against novice spin-outs and stripped spur gears.
- **Drag brake** (0–100% in 8 steps) applies braking torque at neutral throttle, simulating brushed-motor coast-down. It defines corner-entry character: touring cars run modest drag brake for stable entry; drift cars run higher values as a weight-transfer tool (Chapter 9); buggies on loose dirt run little to none so the rear doesn’t step out on lifted throttle. Drag brake generates motor heat — every step up must be checked against the temperature ceiling (Section 8.3).
- **Timing** (8 steps on the 10BL120 G2, roughly 0–30° of electronic advance) trades top-end RPM for heat, consumption, and abrupt top-of-band delivery. In rental service added timing is pure liability: **timing step 0, locked**, verified at every profile audit. Race-night profiles for licensed leagues may run modest timing (steps 1–2) under Artisan supervision only, always paired with a motor-temperature check (Section 8.3). Dynamic boost/turbo advance — the XeRun-class competition feature that ramps additional degrees with RPM and full-throttle dwell — is not present on the fleet ESC by design; “blinky” (zero-timing) running is both the formula class rule and the fleet’s default posture.
- **Cutoff voltage** is set to 3.4 V/cell fleet-wide — deliberately above the ESC’s available lower options and aligned with the park’s LiPo doctrine (3.4–4.2 V/cell). The RCW Node enforces the same threshold independently by PWM intercept, so battery protection survives an ESC misconfiguration. Two independent guards, one doctrine.

5.2 Fleet ESC profiles

Profiles are written to cars with the LCD Program Box Pro from a laminated parameter card kept at the programming bench; RC WORLD OS stores the canonical table and the per-car last-write log. **Readback, not memory, is the verification standard** (see the worked example in Section 4.6 for why).

Table 5.2 — Hobbywing QuicRun 10BL120 G2 fleet profiles

Parameter	Touring RENTAL	Touring RACE	Drift RENTAL	Drift RACE	SCT RENTAL	Buggy RENTAL (factory 60 A ESC equiv.)	Formula SPEC
Running mode	Fwd/Brake	Fwd/Brake	Fwd/Rev/Brake	Fwd/Rev/Brake	Fwd/Rev/Brake	Fwd/Rev/Brake	Fwd/Brake
Cutoff voltage	3.4 V/cell	3.4 V/cell	3.4 V/cell	3.4 V/cell	3.4 V/cell	3.4 V/cell	3.4 V/cell
Punch	Level 3	Level 6	Level 4	Level 5	Level 3	Level 3 (soft start)	Level 5
Drag brake	10%	5%	40%	60%	10%	0–5%	5%
Max brake force	50%	75%	75%	100%	50%	50%	75%
Max re-verse force	n/a	n/a	25%	25%	25%	25%	n/a
Neutral range	9%	6%	6%	6%	9%	9%	6%
Timing	step 0 (locked)	step 0–2 (Lead sign-off)	step 0 (locked)	step 0	step 0 (locked)	n/a	step 0 (“blinky” class rule)
BEC voltage	6.0 V	6.0 V	7.4 V (25 kg servo)	7.4 V	6.0 V	factory	6.0 V

Rationale notes: rental punch Level 3 gives a ~0.4 s softer 0–30 km/h ramp than race Level 6 — enough to protect drivetrains and dignity without feeling numb. Reverse is enabled on off-road and drift rentals (self-recovery reduces Tow-Truck Protocol events) but disabled on touring and formula, where reversing on a live circuit is a collision generator. The wider 9% neutral range on rentals absorbs transmitter centering noise; race profiles tighten to 6%.

5.3 Motor pairing and the “personality” overlay

Personalities are published, named setups layered on top of a parity-verified baseline — never hardware differences:

Personality (touring ex-ample)	Punch	Drag brake	Spring set	Ballast map	Character sold to customer
GT Stable	3	15%	+1 step front stiff	neutral	Planted, forgiving, easy to place
Touring Neutral (baseline)	3	10%	baseline	neutral	The reference car
Touring Agile	4	5%	+1 step rear stiff	10 g rearward	Pointier turn-in, livelier rear

Each personality must re-pass the parity audit as configured (the reference driver’s best-3 within 1.5% of class median). If a personality can’t pass, it’s a gimmick, not a setup, and it doesn’t ship.

5.4 Throttle curves at the transmitter

Punch shapes how fast the ESC delivers current; the transmitter's throttle curve shapes how much throttle a given trigger position requests. The two are tuned together, and the curve lives in the locked transmitter profiles (Section 12.1; license-tier mapping Volume 9 §3.6), which is why it belongs to fleet configuration rather than per-car setup. Fleet curves, as output % at 25/50/75/100% trigger travel:

Transmitter profile (Volume 9 §3.6)	25%	50%	75%	100%	Character
Rookie — Learner/Apprentice (expo –30%, EPA 60%)	10%	25%	42%	60%	Long, gentle lower band; a nervous full pull still lands at the class rental cap
Standard — Operator (expo –15%, EPA 80%)	16%	36%	58%	80%	Progressive but honest; the default customer feel
Licensed — Foreman+ (linear, EPA 100%)	25%	50%	75%	100%	Verbatim input inside class gearing caps
Drift all profiles (expo –20%)	14%	33%	56%	profile EPA	Widens the partial-throttle band where wheelspeed control lives

Design logic: negative expo flattens the curve through the first half of trigger travel — where novices actually drive — so small hand movements make small speed changes, while the top of the band is reachable deliberately rather than accidentally. Drift keeps moderate expo at every tier because partial-throttle modulation is the class skill. Brake-side curves stay linear on all profiles (nobody should meet a nonlinear brake in an emergency), with authority set by the ESC's max-brake-force parameter. Curve revisions are fleet-configuration changes: Motorsport Lead sign-off, OS config log entry, and a spot parity re-run.

5.5 SOP-MS-001 — ESC Profile Write & Verification

SOP ID	SOP-MS-001
Revision	1.0 (July 2026)
Owner	Motorsport Lead
PPE	None (bench task); car on stand, wheels free
Tools	Hobbywing LCD Program Box Pro; laminated profile card (current revision); charged bench pack; RC WORLD OS asset scanner
Frequency	At car induction; after any ESC replacement; at every 200-Shift service; on any parity-audit flag; before/after event-night profile swaps

1. Scan the car's asset tag; open its ESC profile record in RC WORLD OS. Confirm which profile the car should hold.

2. Power the car on stand (wheels clear of the bench). Connect the program box to the ESC's programming port.
3. **Read before writing:** page through all 12 parameters and record actuals. Any mismatch with the recorded profile is logged as a deviation (who/when investigated later — deviations are a process failure signal, not just a fix).
4. Write the target profile parameter by parameter from the laminated card. Save.
5. Power-cycle the ESC. Read back all 12 parameters and verify against the card. Two-person check for RACE→RENTAL reversions after event nights.
6. Function test on stand: throttle ramp (punch behavior), neutral (drag brake engagement audible), brake, reverse lockout as applicable.
7. Update the RC WORLD OS profile record (profile name, revision, Artisan ID, timestamp). The car may not re-enter the rental pool with an unverified profile record.

Safety Warning. Never program an ESC with the car on the ground or with the pinion engaged and wheels loaded. Programming sequences can command unexpected throttle. Stand, wheels free, hands clear — every time, including “quick” changes on race night.

6. Suspension Tuning Guide

6.1 The tuning stack and how to think about it

Suspension turns chassis motion into tire grip. For fleet purposes the tuning stack is, from largest effect to smallest: tires (fixed by the control rule) → springs → damper oil → geometry (camber, caster, toe) → droop → ride height → anti-roll bars. The division's working rules: **change one variable at a time; three laps minimum before judging; write everything on the setup sheet.** RC WORLD OS stores every car's current setup digitally; the paper sheet on the car stand is the working copy.

6.2 Springs and damper oils

Springs carry the chassis; oil controls how fast weight transfers. Fleet damper oils are silicone, specified in **cSt (centistokes)** — the unambiguous international scale — with WT equivalents noted for staff raised on them:

Silicone oil	~WT equiv.	Fleet use
200-300 cSt	~17.5-25	Buggy/SCT big-bore, rough dirt
350-450 cSt	~30-35	Buggy smooth dirt; SCT baseline (400)
500-600 cSt	~40-45	Touring baseline (500); drift front (550)
700-900 cSt	~50-60	Touring high-grip; drift rear (700); Formula side dampers

Effects, memorized by every Artisan: **stiffer springs** = less roll, quicker response, less mechanical grip on that axle, better on smooth high-grip surfaces; **softer** = the reverse. **Thicker oil** slows weight transfer — planted mid-corner, lazy in direction changes; **thinner** transfers faster — responsive but nervous. On bumpy dirt, thick oil makes the car skip; go

thinner and let the springs work. Oil thins as it heats, which is why baselines are set at operating temperature and the parity audit logs ambient.

6.3 Geometry: camber, caster, toe, droop, ride height, bars, roll centers, anti-squat

- **Camber** (top of tire leaning in = negative): compensates for body roll so the outside tire stays flat under load. Touring baseline -1.5° front / -1.5° rear on spec slicks; more negative front camber adds mid-corner steering until the contact patch shrinks on the straights. Verify with a camber gauge on a flat setup board, wheels at ride height.
- **Caster** (kingpin tilted rearward at top): more caster = more straight-line stability and more camber gain on the steered wheel = stronger mid-corner front grip, at the cost of heavier initial turn-in. Touring $4-6^\circ$; drift runs high caster (Chapter 9); buggies run higher still for jump stability.
- **Toe**: front toe-out (0 to -1°) sharpens turn-in; front toe-in calms it. **Rear toe-in is stability medicine** — touring $2.5-3^\circ$, buggy 3° , never toe-out at the rear of anything a customer drives.
- **Droop** (downtravel beyond static ride height): more droop = more weight transfer available = more grip on the opposite end under load transfer, and better bump absorption; less droop = flatter, more responsive on smooth asphalt. Measured with droop blocks and gauge; touring baseline 4 mm front / 5 mm rear.
- **Ride height**: lower = lower CG = less roll, until the chassis scrubs. Touring 5.5–6 mm on our asphalt; buggy 22–25 mm; SCT 28–32 mm. Always set with the battery installed at pool-median mass.
- **Anti-roll bars**: tune roll stiffness without touching bump behavior. Stiffer bar on one end reduces grip on that end in steady-state cornering. Touring runs 1.2 mm front / 1.3 mm rear as baseline; buggies on our Track B run none (bar-free compliance wins on dirt this loose).
- **Roll centers**: set by the inner/outer heights of the suspension links (camber-link tower position, arm-mount shims). Raising a roll center toward the CG reduces body roll on that end — quicker but edgier response; lowering it rolls the chassis more and builds grip progressively. Fleet doctrine: roll centers are a **baseline decision, not a turnaround adjustment** — link positions are recorded as tower-hole coordinates and changed only with Motorsport Lead sign-off, because a one-hole change on one car is invisible to the eye and audible in the parity audit.
- **Anti-squat / kick-up** (off-road): anti-squat is the rearward tilt of the rear inner hinge pins — more ($2-3^\circ$) resists squat under throttle for crisper on-power steering and better jump takeoffs on packed dirt, at the cost of rear traction on bumpy acceleration; less ($0-1^\circ$) plants the rear on loose surfaces. Kick-up is the front-end equivalent: more absorbs landings and square-edged bumps but slows steering response. Track B baseline: buggies 2° anti-squat, $\sim 10^\circ$ kick-up (platform fixed); SCT 3° anti-squat via pill inserts. On-road classes run effectively zero of both and tune the same behaviors with droop and oils.

6.4 Setup-change effects table

The laminated pit-lane card every Artisan carries:

Change (one step)	Turn-in	Mid-corner steering	Corner-exit stability	Straight-line	Bumps/jumps	Watch out for
Softer front springs	+	+	–	0	+	Sluggish transitions
Stiffer rear springs	+	+	–	0	–	Snap over-steer on cold tires
Thinner front oil	+	0	–	0	+	Nervous on high grip
Thicker rear oil	0	+	+ (smooth)	0	–	Kicks on square bumps
More front camber (–)	0	+	0	– (drag)	0	Inner-shoulder tire wear
More caster	– (initial)	+	+	+	+ (jumps)	Heavy steering feel
Front toe-out	+	0	0	– (wander)	0	Twitchy on the straight
More rear toe-in	–	–	+	– (drag)	0	Scrubs speed; use ≤3°
More rear droop	0	0	+ (traction)	0	+	Roll into corner-exit wobble
Lower ride height	+	+	+	+	–	Chassis scrub, bottoming
Stiffer rear bar	+	+	–	0	–	Loose on power

6.5 Baseline setup sheets

Baselines are the parity-audited neutral configurations. Personalities and race setups are recorded as deltas from these sheets.

Table 6.5a — Touring/GT baseline (Track A asphalt, spec slick)

Parameter	Front	Rear
Springs	medium (silver)	medium (silver)
Damper oil	500 cSt	500 cSt
Camber	-1.5°	-1.5°
Toe	0°	3.0° in
Caster	4°	—
Droop	4 mm	5 mm
Ride height	5.5 mm	6.0 mm
Anti-roll bar	1.2 mm	1.3 mm
Diff	gear diff 3,000 cSt	gear diff 1,000 cSt
FDR / mass / distribution	6.16 · 1,530 g · 50/50 (Ch. 7–8)	

Table 6.5b — Drift baseline (MST RMX 2.5 S+, RWD — full rationale Ch. 9)

Parameter	Front	Rear
Springs	soft	medium
Damper oil	550 cSt	700 cSt
Camber	−6°	−1°
Caster	10°	—
Toe	(Ackermann-dominated; ~0° static)	0°
Ride height	5 mm	5.5 mm
Steering angle	~65° effective	—
Gyro gain	60% rental / 45% race	—
Mass / distribution	1,450 g · 46/54	

Table 6.5c — Buggy/Rally baseline (WLtoys 144010-class, Track B packed dirt)

Parameter	Front	Rear
Springs	factory +10% (upgrade set)	factory +10%
Damper oil	300 cSt	350 cSt
Camber	−1°	−1.5°
Toe	0°	3° in
Ride height	22 mm	24 mm
Mass / distribution	1,050 g · 45/55	

Table 6.5d — Short-course baseline (2WD, Track B)

Parameter	Front	Rear
Springs	medium	medium-soft
Damper oil	400 cSt	350 cSt
Camber	−1°	−1°
Toe	0°	3° in
Ride height	28 mm	30 mm
Mass / distribution	2,050 g · 42/58	

Trade Hack. Build shock pairs on the bench to matched rebound: after filling and bleeding, compress both shocks fully and time the rebound stroke side by side. A mismatched pair (one seal dragging, one aerated) is the most common invisible setup fault in a rental fleet — it steers the car under braking and no geometry change will fix it. Rebuild shocks as axle pairs, never singly.

7. Weight Distribution & Corner Weighting

7.1 Static distribution targets

Where the mass sits determines which tires do the work. Class targets, race-ready (battery in, body on):

Class	Front/Rear target	Cross-weight tolerance	Rationale
Touring/GT	50/50 (± 1)	$\pm 1.0\%$	Neutral 4WD balance; equal tire duty on a flowing circuit
Drift (RWD)	46/54 (± 1)	$\pm 1.0\%$	Rear bias for traction under constant wheelspin; enough nose weight to keep steered-wheel authority
Buggy/Rally	45/55 (± 2)	$\pm 1.5\%$	Rear bias plants forward drive on loose dirt and stabilizes jump attitude
Short-course 2WD	42/58 (± 2)	$\pm 1.5\%$	Motor-over-axle traction; high body mass makes cross-weight drift with body damage — recheck after body swaps
Formula	46/54 (± 1)	$\pm 1.0\%$	Pan-car convention: rear traction, front handled by tire compound
Drag	30/70 launch bias (ballast forward as needed to control wheelies with the wheelie bar as backstop)	—	Launch traction is everything on a 40 m strip

7.2 Corner-weighting procedure

Four-scale corner weighting, standard bench task (15 minutes once practiced):

1. Level the setup board (bubble level, shim the board not the scales). Zero all four scales.
2. Car race-ready: pool-median battery, body, transponder, Node. Shocks settled — roll the car back and forth, then lift and set down twice.
3. Record FL, FR, RL, RR. Compute: front % = $(FL+FR)/\text{total}$; **cross-weight % = $(FR+RL)/\text{total}$** (should be 50% $\pm$ class tolerance).
4. Correct distribution by moving ballast/battery position fore-aft along the ballast map; correct cross-weight by spring pre-load collars in diagonal pairs (small turns — 0.5 mm at a time), never by bending anything.
5. Re-settle and re-measure after every change. Log final corners to the setup record.

Ballast maps. Ballast lives low and central by default: touring — battery-tray floor first, then the chassis centerline fore/aft trim positions; drift — rear of battery slot (bias) and over the front axle line (steering authority trim); buggy — under-receiver-box plate; drag — nose plate. Ballast is self-adhesive lead-free plate (5/10 g), covered with a taped label showing grams and date so audits can reconcile mass records at a glance.

Field Note. Cross-weight (“wedge”) is the setup fault customers can feel but never name — the car that turns left better than right. Any car reported as “pulls one way” goes to the corner scales before anyone touches the steering trim. Nine times out of ten it’s a tweaked shock collar or a post-crash bent arm loading one diagonal.

7.3 Worked example: ballasting a touring car to target

Car MS-T-07, corner-weighed race-ready after a front-end rebuild:

	Left	Right	Axle total
Front	FL 362 g	FR 371 g	733 g
Rear	RL 391 g	RR 382 g	773 g
Total			1,506 g

Derived numbers: front % = $733/1,506 = 48.7\%$ (target 50 ± 1 — out of spec); cross-weight = $(FR + RL)/total = (371 + 391)/1,506 = 50.6\%$ (inside the $\pm 1.0\%$ tolerance); class minimum mass = 1,530 g, so the car is also **24 g light**.

The ballast requirement solves both problems at once. At 50.0% front and 1,530 g final mass, the front axle must carry $0.500 \times 1,530 = 765$ g — a gain of 32 g while the rear gains -8 g, impossible by addition alone. So the battery first shifts one tray position forward (measured effect on this chassis: +14 g front / -14 g rear), giving 747/759. The 24 g of ballast then lands as: +18 g front (10 g + 5 g + 3 g trim at the front tray edge), +6 g rear (5 g ahead of the rear brace + 1 g lead tape), all on the centerline and split left/right so cross-weight is untouched. Re-weigh confirms 765/765 (50.0/50.0), cross 50.3%, total 1,530 g. Bench time: 12 minutes. Plate positions and grams go to the setup record — the audit trail is what lets next month’s Artisan trust the sheet instead of re-deriving it.

8. Gearing, Rollout & Thermal Management

8.1 The arithmetic

Three formulas govern everything in this chapter:

- **Final Drive Ratio:** $FDR = (\text{spur} \div \text{pinion}) \times \text{internal transmission ratio}$
- **Rollout** (distance per motor revolution): $\text{rollout} = (\pi \times \text{tire } \varnothing) \div FDR$
- **Theoretical top speed:** $v = \text{motor RPM} \times \text{rollout} \div 60$ (m/s), where loaded motor RPM $\approx Kv \times \text{sagged pack voltage} \times \sim 0.85$ drivetrain/load factor.

Worked touring example: spur 64T, pinion 27T, internal ratio 2.6 (TT-02 class) $\rightarrow FDR = (64/27) \times 2.6 = 6.16$. Tire $\varnothing$ 63 mm $\rightarrow \text{rollout} = (\pi \times 63)/6.16 = 32.1$ mm/rev. A 13.5T sensored motor at $\sim 3,100$ Kv on a sagged 7.4 V under load: $3,100 \times 7.4 \times 0.85 \approx 19,500$ RPM $\rightarrow v \approx 19,500 \times 0.0321/60 \approx 10.4$ m/s ≈ 37.5 km/h — which is why the rental cap of 40 km/h for this class is enforced primarily by gearing and only secondarily by throttle EPA. Gearing caps cannot be un-clicked by a customer.

(The Carten T410-class cars run internal ratio 2.53; their pinion spec is adjusted one tooth to land within 1% of the same rollout — parity is a rollout number, not a pinion number.)

8.2 FDR tables per class and track

Fleet gears standardize on 48P for the 1/10 classes (tooth-strength vs. adjustability sweet spot; 64P reserved for formula fine-trim; the 1/14 classes keep factory Mod-0.7).

Class / duty	Tire Ø	Internal ratio	Spur/pinion	FDR	Rollout	Est. top speed
Touring RENTAL, Track A	63 mm	2.6	64/27	6.16	32.1 mm	~38 km/h
Touring RACE, Track A	63 mm	2.6	64/31	5.37	36.9 mm	~44 km/h
Drift (all duty)	64 mm	2.6 (RMX mode)	64/24	6.93	29.0 mm	~33 km/h (wheelspeed higher)
Buggy RENTAL, Track B	90 mm	factory	factory –1 pinion	~8.5 eff.	33 mm	~50 km/h
SCT RENTAL, Track B	108 mm	2.72	86/22 (48P)	10.6	32.0 mm	~40 km/h
Formula SPEC, Track A	62 mm	direct (1.0 axle)	96/33 (64P)	2.91	66.9 mm	~48 km/h (21.5T)
Drag EXHIBITION, 40.2 m	66 mm	2.6	64/38	4.38	47.3 mm	100+ km/h (3S)

Gearing changes for heat or track layout move in single pinion teeth: one 48P tooth $\approx$ 3–4% rollout — conveniently, about one parity-spread unit, which is why gearing is the last trim lever in the balancing ladder.

Touring pinion sweep chart (64T spur, 48P, internal 2.6, tire Ø 63 mm) — the benchmark for Track A trim decisions:

Pinion	FDR	Rollout	Δ speed vs. 27T	Motor temp tendency	Duty
24T	6.93	28.6 mm	–11%	coolest	Heat-wave rental fallback; new-driver events
25T	6.66	29.7 mm	–7%	cool	Summer afternoon rental state
26T	6.40	30.9 mm	–4%	cool-normal	Shoulder-season rental
27T	6.16	32.1 mm	baseline	normal	Rental baseline (Table 8.2)
28T	5.94	33.3 mm	+4%	warm	Parity trim ceiling for rentals
30T	5.55	35.7 mm	+11%	hot — check at 10 min	League race state
31T	5.37	36.9 mm	+15%	hot — mandatory temp check	Race profile (Table 8.2), finals only

Operational rule: rentals may move within 25–28T on turnaround Artisans’ authority under the thermal rules below; anything beyond requires the Motorsport Lead, because it changes the class speed envelope rather than the trim. Equivalent sweep cards for the other classes hang at each class bench as controlled documents in RC WORLD OS.

8.3 Motor temperature management

The fleet thermal ceiling is **85 °C at the motor can**, measured by IR gun at the endbell within 60 seconds of the car stopping; 90 °C+ demags rotors and cooks sensor boards. Standing rules:

- Turnaround Artisans spot-check the hottest recent runner each cycle; any car over 85 °C is pulled, geared **down** one pinion tooth (higher FDR runs cooler), and its drag-brake setting reviewed (drag brake is hidden heat).
- Summer afternoons on Track A: pre-emptive –1 tooth across touring when ambient exceeds 32 °C, reverted in the evening — logged as a fleet-wide gearing state in RC WORLD OS so parity math stays coherent.
- ESC temperatures are watched via the ESC’s own thermal protection plus the Node’s battery-side telemetry; an ESC hitting thermal cutback in normal rental duty indicates wrong gearing or a dragging drivetrain, not a bad ESC — bench-roll the car before swapping electronics.

9. Drift Program Deep-Dive

9.1 Why drift gets its own chapter

Drift is the only class where the customer is supposed to exceed the grip limit at all times. That inverts most of Chapter 6’s tuning logic and makes the class disproportionately dependent on geometry, electronics, and surface. Done well, RWD drift is the stickiest repeat-visit product in the division — progressing from “spins instantly” to “links three corners” takes exactly the handful of visits a membership wants it to take.

9.2 RWD drift geometry

RWD versus AWD/CS. The hobby ran for years on AWD “counter-steer” (CS) chassis — 4WD drivetrains overdriven at the rear (CS ratio ≈ 1.3 –2.0, rear wheels turning 30–100% faster than the fronts) so the car breaks traction predictably while the driven front axle keeps it recoverable. CS is easier for a first-timer, and the fleet platform’s switchable gearbox could support it if ever needed. The division standardizes on **RWD with gyro** anyway: it is what modern drift competition runs, so the product reads as authentic; the skill ceiling is higher, which the license ladder monetizes; and the rental gyro (Section 9.3) gives novices the recoverability CS hardware used to provide — software solving what gearing once did. AWD/CS exists in this volume only as a documented fallback, not a stocked class.

Modern RC drift is RWD with gyro assistance, mimicking full-size technique. The MST RMX 2.5 S+ fleet spec:

- **Steering angle:** $\sim 65^\circ$ effective lock. The fronts’ job while drifting is to point along the direction of travel (countersteer) while the rear drives the car sideways; insufficient lock is the hard ceiling on drift angle. The RMX’s uprights and rack deliver this out of the box.

- **Ackermann:** near-parallel steering (low Ackermann) keeps both fronts at consistent slip when countersteered far beyond normal ranges. Fleet spec sets rack position for slight anti-Ackermann at full lock.
- **Caster & KPI:** 10° caster (adjustable 6–12°) with 6° KPI gives progressive camber gain across the huge steering sweep, keeping the front contact patch usable at full counter-steer. High caster also self-centers the steering, cooperating with the gyro rather than fighting it.
- **Weight transfer:** drift initiation is a weight-transfer event. The 46/54 static bias, soft front springs, and heavier rear oil (700 cSt) let throttle lift rotate the car and throttle application settle it — the front/rear damping split is the personality dial for how aggressively the chassis initiates.

9.3 Gyro doctrine

The gyro applies countersteer faster than any human; the tuning question is how much authority to give it.

Duty	Gyro gain	Behavior
Rental / novice	55–65%	Car self-catches almost every over-rotation; customer learns line and throttle first
Intermediate (license-gated)	45–55%	Customer supplies most counter-steer; gyro cleans up
Race / competition nights	30–45%	Judge-visible driver input; club-standard feel

Gains are set on the (locked) transmitter profile per license tier, not on the car, so any drift car serves any tier. Gyro mounting is standardized (same orientation, foam-tape isolation, loom dressed clear of the spur) because a vibrating gyro oscillates the servo and eats servo gears — a known fleet-killer in drift rentals.

9.4 Tires and surface interaction

Drift rings are the one place the control-tire rule meets materials science. Fleet spec: **HDPE rings on the polished-concrete annex** (consistent low grip, ~150-Shift life, cheap), with **ABS as the alternate compound** if the annex is ever refinished smoother — grippier, faster-wearing, more progressive on very slick sealed concrete. Compounds are never mixed in service; a compound change is a class-wide event with a parity re-audit. Surface doctrine: the annex is sealed, dust-mopped before open and between event heats (dust on polished concrete acts as ball bearings), and never treated with traction compound. Fronts may run a slightly grippier ring than rears if the class needs steering authority — the one permitted front/rear asymmetry, fixed class-wide.

9.5 Drift electronics character

Sensored 13.5T + the drift ESC profiles (Table 5.2) exist for one reason: **wheelspeed control at partial throttle is the entire skill**. High drag brake (40–60%) gives the “off-throttle rotation” that drift technique is built on; sensored commutation removes the low-RPM cogging that makes cheap sensorless systems undriveable sideways. Punch stays moderate — drift launches are irrelevant, but transient throttle response mid-drift matters.

9.6 Drift annex requirements (interface to Volume 11)

Racing imposes on the architects: $\geq 250 \text{ m}^2$ sealed polished concrete; painted clipping-point dots and outline curbs (no raised curbs — drift cars slide into them side-on); perimeter kick-rail 80 mm high; spectator rail on the long side (drift is a spectator product); lighting to 300 lux minimum for night events; power and mounting for the timing bridge and two fixed cameras (leaderboard replays).

9.7 Drift competition scoring

Drift is judged, not timed, and the quarterly competitions run the three-judge club format scaled to a rental fleet:

- **Judged criteria (100 points): line** (40) — proximity to the painted clipping points and outside zones on the published judged course; **angle** (30) — sustained drift angle without corrections; **style/commitment** (30) — initiation speed, transition fluidity, no straightening moment. A spin, stalled drift, or two wheels off course zeroes the run.
- **Format:** two qualifying runs per driver (best counts), top 16 into **tandem battles** — lead-and-chase pairs, judges scoring the chase car's ability to mirror the lead; one "one more time" rerun per battle on a split decision.
- **Fleet fairness:** competition cars are drawn by lot from the parity-verified drift fleet, race-profile flashed and readback-verified; drivers may adjust gyro gain within the competition band (30–45%) and nothing else. The battle is between drivers.
- **Scoring capture:** judges score on tablets into the RC WORLD OS events module; scores render live on the leaderboard screens with camera replay, and results feed Gears and license credit like any race (Volume 9).

10. Surface-Specific Setup: Touring, Rally & Short-Course

10.1 Touring on Track A asphalt

Track A's fine asphalt/polished concrete is a **medium-grip, low-abrasion** surface, which drives the touring baseline in Table 6.5a. Operating guidance beyond the baseline:

- **Grip evolution:** the racing line rubbers in through the day; expect 0.3–0.5 s of "free" lap-time by afternoon and a nervous rear in the first morning Shifts. The parity audit runs pre-opening precisely so cars are compared on the green track.
- **Temperature:** spec slicks peak around 35–50 °C surface temperature. Below ~ 15 °C ambient, add one step of drag brake and advise league drivers to run a heat lap; above 35 °C, watch for greasy mid-corner push and drop ride height 0.5 mm as the standard response.
- **Rain:** Track A drains (Volume 11 crossfall spec) but wet running is a program decision, not a setup decision: rental touring Shifts pause in standing water (electronics are splash-resistant, not submersible; Node conformal coating is not a hull).
- **The racing line:** Track A's reference line — painted faintly at the apexes and taught in coaching — is classic outside-inside-outside geometry: enter wide, apex late on the two corners feeding the main straight (exit speed beats entry speed wherever a straight follows), sacrifice the line only in traffic. The reference driver's telemetry lap is the line's

definition of record; coaching (Section 13.7) overlays a customer's Node trace on it corner by corner.

- **Tire management on control tires:** with every car on the identical spec slick, tire condition is the remaining tire variable — managed, not tuned: sets stay as sets (Section 4.5), wear dots checked at every turnaround, a damp-cloth wipe at re-tire only. No traction additives, league included; a caught additive is a disqualification, because one treated set breaks parity for everyone racing against it.

10.2 Rally/buggy on Track B dirt

Packed dirt watered and rolled daily is a **variable-grip, high-contamination** surface. Standing setup doctrine: chase compliance, not stiffness — thin oils (300/350 cSt), bar-free, full droop, and the 45/55 bias doing the traction work. Operationally:

- **Moisture window:** the track crew waters to “dark but not shiny.” Dry-slick afternoons lower grip 20%+; the class response is a –1 pinion (cooler motors working harder against wheelspin) and nothing else — customers adapt faster than setups on dirt.
- **Jump discipline:** rental buggies fly flat off the tabletops at cap speed by design (landing-matched profiles, Volume 11). Cars returning with bent hinge pins or tweaked arms feed the corner-scale check (Section 7.2) at turnaround.
- **Dust:** air filters don't exist on electric cars, but bearings suffer; Track B cars run on the accelerated 150-Shift bearing schedule vs. 300 for Track A (maintenance matrices in Volume 7).

10.3 Short-course on Track B

The SCT setup (Table 6.5d) accepts body roll as a feature: the class's forgiving, truck-like motion is what lets mixed grids race door-to-door. The only setup rule that is enforced rigidly is rear toe-in $\geq 3^\circ$ — a loose short-course truck at 40 km/h on a crowded grid is the division's highest collision-energy scenario.

10.4 Track requirements summary (interface to Volume 11)

Track	Racing needs (this volume)	Design authority
A — Asphalt	~180 m line, 3.5–4.5 m width, drift annex per 9.6, drag straight ≥ 55 m incl. braking area, timing loop conduit at S/F, rostrum for 10 drivers, 300 lux night lighting (Phase 3 up-rated)	Volume 11
B — Baja/Rally	~160 m line, landing-matched jumps ≤ 0.5 m, berms, watering system, timing bridge mounts, splash-safe marshal posts	Volume 11

11. Formula RC & Drag Racing Programs

11.1 Formula RC class rules

Formula is run as a strict spec class — the point is precision, not development war:

1. Chassis: fleet F104/TRF103-class pan cars only; no customer chassis in fleet events.

2. Motor: 21.5T sensored, fleet-sealed. ESC: fleet profile “Formula SPEC” (Table 5.2) — **zero timing (“blinky”)**, verified by readback before finals.
3. Tires: spec F1 rubber, fleet-issued per event; no additives.
4. Weight: class minimum per current parity sheet; ballast maps only.
5. Contact: open-wheel contact rules — any avoidable contact is a position penalty; two = black flag. (This rule is the driver-education payload of the class.)

11.2 Formula event formats

- **Formula Experience** (weekly; gated at Foreman + Motorsport Master endorsement, instructor-supervised): 30-minute guided session — briefing, 2 practice Shifts’ worth of track time, 8-minute sprint with grid and podium. Priced as a premium experience, not a Shift.
- **Formula Cup** (monthly): qualifying + two 8-minute finals, grid of 8 (4 fleet cars double-stinted across two semis), championship points into the annual calendar (Section 13.6).

11.3 Energy-allocation racing (the power-management format)

The formula program turns the park’s founding “F1-style power management” doctrine into an actual race format, built entirely on the RCW Micro-Node’s voltage telemetry and PWM intercept:

- **The allocation.** Each race entry buys an **energy budget**, expressed as a percentage of a healthy pack’s usable capacity for the race distance (e.g. 85% for a standard final). The Node streams per-car voltage to RC WORLD OS; the race module integrates voltage sag against time-at-throttle into a live consumption estimate, rendered as a shrinking energy bar beside the driver’s name on the leaderboard screens.
- **The physics is honest.** Aggressive throttle produces deeper sag and higher integrated draw; a smooth driver genuinely covers the same laps on less budget. The format is a thin scoring layer over real electrochemistry, which is why it teaches real technique.
- **The consequences.** At 100% consumed, the Node’s intercept steps the car to a 40% throttle “limp” cap for the remainder — running out of fuel on the last lap, visible and dramatic without stranding the car. Budget remaining at the flag converts to bonus points, so the strategy space is push-versus-conserve.
- **Where it runs.** Standard in Formula Cup finals and as a touring league-night variant (“Eco-GP”). Deliberately not used in walk-up Shifts — novices should learn throttle before strategy — but the same telemetry drives the smoothness statistics on every customer’s post-Shift report (Volume 9).

Field Note. Energy racing quietly solves an operations problem too: it is the only format where the slower driving style wins races, which makes it the best teaching format for customers who arrive throttle-happy and the cheapest race night on the fleet — motor temperatures and crash rates both drop measurably.

11.4 Drag racing class rules

1. Strip: **40.2 m (scale 132 ft)** measured, plus ≥ 15 m braking runoff; paired lanes 1.2 m wide; centerline no-cross rule (crossing = loss).

2. Cars: fleet drag cars only for open events; “run what you brung” exhibition nights for member-owned cars under scrutineering (wheelie bar mandatory, battery $\leq 3S$, LiPo condition inspected — park battery doctrine applies to guest packs without exception).
3. Starts: staged on the Christmas tree per Section 11.5; red light = loss.
4. Rental format: customers never free-drive; a staged launch with speed-limited profile lets any license tier take a pass safely.

11.5 The strip, staging and reaction timing

Track A's main straight doubles as the strip: 40.2 m race distance fits inside the ≥ 55 m straight with braking area (Section 10.4). Timing: staged beams at launch, mid, and trap connected to the timing system's drag module; the RCW Node's telemetry provides trap-speed corroboration and the kill-switch covers a brake failure into the runoff (geofence hard-stop at the barrier line).

Staging procedure. Cars are placed by the lane marshal (never the customer) with the nose breaking the pre-stage beam, then rolled forward to stage — both beams lit, both lanes staged, tree armed. The fleet runs a **sportsman tree** (three ambers at 0.5 s intervals, then green); leaving before green trips the red light and forfeits the run. **Reaction time** is measured from green to the launch beam clearing and published alongside elapsed time (ET) on every slip — anticipation and release is a skill requiring zero throttle talent, so the format works at every license tier. League drag runs **bracket rules** (dial-in your predicted ET; breakout = loss) so mismatched cars and drivers still produce close racing — the drag program's own version of the parity doctrine.

11.6 Why these programs exist at 4 cars each

Neither class earns walk-up Shift revenue; both earn **event nights, license aspirations, and media**. Eight special-event cars generate the imagery that sells the other 56 — Volume 10 treats them as marketing-adjacent assets with event-revenue offset, and Chapter 14's utilization KPI excludes them from the denominator.

12. Rental Fleet Engineering & the Shift Turnaround

12.1 Transmitter standardization and lockout

The transmitter is the customer's entire interface with the division, and it is fleet property, never customer property, for Shift rentals. Standards:

- **One transmitter model park-wide for surface vehicles** (4-channel 2.4 GHz surface radio with model memory, EPA, dual rate, physical menu lockout — procurement spec in Volume 8). One model means one Toolbox Talk diagram, one battery type, one spare pool.
- **Menu lockout is physical and procedural:** menu access disabled on customer units (button combination set and taped internally where supported; units audited weekly). Customers get steering wheel, throttle trigger, and nothing else that changes state.
- **Rate limiting by license tier is a transmitter profile, not a car change.** Steering EPA and throttle EPA/curve are pre-programmed per transmitter profile (Volume 9 §3.6): Rookie for Learner/Apprentice (steering 75%, throttle 60%, softened center), Standard for Operator (90/80), Licensed for Foreman+ (100/100 within class caps). The transmitter binds to the car at the counter by QR scan (RC WORLD OS pairs `asset_tag` to transmitter

ID and writes the session to the queue roster) and refuses a pairing whose profile exceeds the customer's license tier.

- **The RCW Node is the enforcement backstop.** Whatever the transmitter commands, the Micro-Node's PWM intercept enforces the park-side rules: under-voltage throttle-back at 3.4 V/cell, geofence kill at track boundaries, remote kill from the director's console, and telemetry streaming to RC WORLD OS over the Wi-Fi mesh. Transmitter EPA shapes the experience; the Node guarantees the limits. Gyro-stability aids, where a class uses them (drift gyro; optional steering assist on entry class), are configured on the car and license-gated the same way.

12.2 The hardening pass (car induction)

Every car entering the fleet receives the standard hardening pass at The Works before its first Shift. Per class it varies in detail (Chapter 2 notes); structurally it is always:

1. Full fastener pass with threadlock on metal-to-metal; nylon-insert nuts on suspension pivots.
2. Known-weak-point upgrades fitted (per platform list in Volume 7, Chapter 5: TT-02 bearing/steel-pinion/oil-shock set; 144010 metal knuckles and steel main gear; SCT servo-saver spring uprated).
3. Electronics: fleet ESC/servo installed (1/10 classes), connectors converted to XT60, wiring loomed and strain-relieved, RCW Micro-Node soldered inline with conformal coat verified, receiver bound to the fleet radio system.
4. Body prep per Section 12.3; asset tagging per Section 12.4.
5. ESC profile written and verified (SOP-MS-001), baseline setup sheet applied (Chapter 6), corner-weighted (Chapter 7), motor benched and binned (Chapter 4).
6. Induction parity run: the car must land within 1.5% of class median before its first customer.

12.3 Damage-resistant body strategy

Bodies absorb the first hit, so bodies are engineered as consumables with a service life, not decorations:

- **Material:** polycarbonate (Lexan) only — 1.0 mm standard, 1.5 mm on SCT/buggy where available. No hard-plastic display bodies on rental cars, ever.
- **Reinforcement at prep:** shoe-goo/mesh laminate on nose and wheel arches; polycarbonate doublers and rubber grommets at body-post holes; front splitters trimmed back 5 mm for curb clearance.
- **Paint:** backside-painted (scratches don't show), fleet livery with class color coding and large car numbers readable at 30 m.
- **Wheelie bars** where apt: mandatory on drag cars, fitted to SCT and buggy rentals where the platform accepts them — they save rear body sections and stop novice throttle-stab flips.
- **Life and rotation:** bodies are tagged assets in RC WORLD OS; a body retires at the third structural repair or when cracks reach a wheel arch. Two spare painted bodies per 4 cars per class sit prepped in The Works, so a body swap is a 90-second turnaround action.

12.4 Asset tagging and telemetry integration

Every car carries three identities that must always agree: the **physical asset tag** (laser-etched QR + human-readable ID, e.g. MS-T-04, on chassis and body), the **RCW Node MAC** (the telemetry identity in `fleet_inventory`), and the **timing transponder ID** (Section 13.3). RC WORLD OS is the master record: lifecycle status (seven states: available → bound → on_track → pit → maintenance → charging → retired; see Volume 13 §6.3), total Shifts run, motor/ESC/tire sub-asset histories, last parity result, last profile write. The Node makes the car self-reporting: voltage and position land in `live_telemetry`, lap crossings corroborate the timing system, and the maintenance module opens flags on runtime thresholds (Volume 7; KPI hooks in Chapter 14).

12.5 The Shift turnaround

The Shift is the park's 20-minute billing block; motorsport runs a **25-minute wheel: 20 minutes on track, 5 minutes turnaround**. The turnaround is a pit crew operation with a fixed sequence — two Artisans per track handle 8+ concurrent cars because the sequence never varies:

Checklist 12.5 — Between-Shift turnaround (per car, target ≤4 minutes)

- ☐ Receive car and transmitter; scan asset tag to close the session in RC WORLD OS
- ☐ Battery out → voltage check → to charging bunker trolley (LiPo doctrine: ~30% buffer expected on a Casual Shift; log any pack below 3.5 V/cell as an over-drain event against the session)
- ☐ 20-second visual: wheels/tires (wear dots, glue joints), arms/knuckles/links (grab-and-wiggle), body damage grade (A/B/C), loose fasteners
- ☐ IR temp spot-check on motor if the car ran hard (>85 °C = pull and re-gear, Section 8.3)
- ☐ Blow down chassis (Track B cars: bearing dust check)
- ☐ Fresh pool battery in; connector fully seated; strap secure
- ☐ Node handshake confirmed in OS (voltage reporting, geofence armed)
- ☐ Radio function check: steering left-right, throttle blip on stand, brake, kill-switch ping
- ☐ Setup sheet still matches (springs/tires haven't been "borrowed" — it happens)
- ☐ Status flip to active in OS → car to the ready line; or maintenance with a one-line fault note → car to The Works trolley

Cars flagged C-grade body or any mechanical fault leave the pool immediately; the 64-car fleet against ~22 track slots exists precisely so a pulled car never cancels a booking (Section 2.9).

Battery logistics inside the wheel. The turnaround runs the park's 3:1 battery doctrine made physical: one pack in the car, one charging in the bunker, one rested on the pit-lane ready rack. Packs are pooled per class, not per car, and each pack's ID is scanned into the session record at installation, so Node voltage telemetry is always attributable to a specific pack's history. Two hard rules bound the flow: **no pack charges anywhere except the bunker** (the pit-lane rack is staging only), and **no pack rests less than 20 minutes between charge completion and installation** (warm packs sag differently and would quietly break parity). The Operator Shift's theatrical mid-session battery swap draws from the same rack under the same rules — the theatre is real logistics performed in view. Pack retirement, internal-resistance tracking, and bunker fire procedures are Volume 7's jurisdiction.

Safety Warning. A pack returning from a session below 3.4 V/cell means two independent guards failed or were bypassed. That is never treated as a battery event only: the car's ESC cutoff and its Node are both bench-verified before the car re-enters the pool, and the session log is reviewed to establish how it happened. Chronic over-drain clustering on one car is an electronics fault; clustering on one customer is a Toolbox Talk and license-tier conversation (Volume 9).

12.6 Daily fleet-readiness

Checklist 12.6 — Motorsport daily opening (fleet portion, before first Shift)

☐

Charging bunker log reviewed: overnight storage-charge state confirmed, no swollen/hot packs, day's first rotation staged at 3:1 ratio

☐

Every ready-line car: tire life check against Shift ledger, wheel-nut torque pass, battery-strap and connector inspection

☐

One reference car per class driven 3 laps by opening Artisan — track and car sanity check; time logged (drift >0.5 s from yesterday's reference = investigate surface or car before opening)

☐

All Nodes reporting on the mesh (OS fleet map shows green for every ready-line car); any silent Node = car pulled

☐

Transmitter pool: count, battery trays swapped, tier profiles spot-audited on two random units

☐

Timing system self-test: decoder up, test transponder pass registered on both tracks

☐

Track walk: Track A swept/dust-mopped (drift annex mandatory), Track B watered and rolled, barriers seated, marshal points equipped (hooks, flags, spare body clips)

☐

Tow-Truck Retrieval crawlers: charged, winches function-checked (customers meet these on their worst moment — they must always work)

☐

Maintenance queue reviewed: yesterday's pulls triaged with The Works; today's expected returns confirmed

☐

Parity audit due today? (rolling schedule, SOP-MS-002) — if yes, completed before opening

12.7 Weekly deep-check

Checklist 12.7 — Weekly fleet deep-check (per class, alongside the parity audit)

☐

Corner-weight verification on 3 random cars per class (full fleet monthly)

☐

Shock pairs rebound-matched on any car reported “inconsistent” during the week

☐

Drivetrain free-roll test: chassis on stand, timed spin-down; outliers get bearing/diff inspection

☐

ESC profile readback on 25% of the class (full class after any event night)

☐

Transmitter EPA/profile audit across the pool (menu-lockout integrity check)

☐

Tire ledger reconciliation: paint-dot ages vs. OS Shift counts; retire due sets

☐

Body fleet review: regrade, pull third-repair bodies, restock prepped spares to 2-per-4-cars

☐

Battery pool: internal-resistance spot check on 10% of packs; retire per battery doctrine (Volume 7)

☐

Spares min/max review against parts-per-100-Shifts consumption (Chapter 14 KPI feed)

☐

Node firmware/telemetry health report reviewed with the OS team

12.8 Damage triage, retirement and parts harvesting

Damage triage classes. Every incident-returned car is triaged at the turnaround bench into one of three classes, logged against the session:

Class	Definition	Disposition	Target resolution
A — Cosmetic	Body scuffs/cracks short of a wheel arch, bent body clips, scraped bumper foam, tire scrub marks	Stays in service; body re-grade noted; swap to prepped spare body at next natural pause	Same wheel cycle
B — Serviceable	Bent steering link, popped ball cup, torn arm, stripped servo saver, suspected shock leak, any “feels wrong” report	Pulled to maintenance; pit-lane repair if <10 min from stocked spares, else to The Works trolley	Same day

Class	Definition	Disposition	Target resolution
C — Structural	Cracked chassis/tub, bent motor mount or bulkhead, drivetrain lock, water ingress, any electronics fault, post-crash geometry that fails the corner-scale check	Out of pool; work order in RC WORLD OS; parity re-run mandatory before return	≤72 h or retirement review

Class B/C events feed the MTBF KPI (Chapter 14); Class A does not. A car with three Class C events in a rolling 90 days triggers the same retirement review as a triple quarantine (Section 4.9).

Retirement criteria. A car retires when any of the following holds: (1) cumulative repair spend reaches **60% of replacement landed cost** (tracked per asset by the OS); (2) structural tub/chassis damage on a platform where the tub is the car (touring, drift); (3) triple quarantine or triple Class C review concludes “cannot be trusted”; (4) platform end-of-life — spares supply below two active sources (the WLtoys consumable doctrine reaches this by design and answers it with donor stock); or (5) fleet refresh economics — Volume 10’s depreciation assumes a 3-year/2,000-Shift service life for 1/10 classes and 18-month/1,200-Shift for the 1/14 classes.

Parts harvesting. Retired cars go to the harvest bench at The Works, never to a skip whole: electronics (ESC, servo, receiver, Node) are re-flashed and re-tagged; motors re-benched and binned or stripped for sensor boards and bearings; unworn suspension and drivetrain parts return to stock as used-serviceable (marked, never mixed with new in min/max counts); batteries join the pack-retirement stream (Volume 7); bodies and tires are binned. A harvested touring car typically returns \$60–90 of stock value, closed against the asset as a salvage credit in Volume 10’s depreciation model. Bare chassis with life left become training mules for Artisan certification builds (Volume 7).

Transmitter management and hygiene. Transmitters live on a numbered charging wall behind the counter, checked out per session with a wrist lanyard. Between sessions every unit gets a 30-second wipe-down (alcohol-free disinfectant on grips, wheel, trigger — alcohol clouds polycarbonate and perishes foam) and a drop inspection; the pool carries 20% spare units so cleaning or damage never shortens the ready line. Foam wheel covers are consumables, replaced monthly or immediately when torn. Any unit with a suspected fault (glitchy steering report, drop onto concrete) is bench-checked against a reference receiver before re-joining the wall — radio faults masquerade as car faults and waste bench hours if the pool is not policed.

13. Race Operations

13.1 Race formats

The division runs four customer-facing competition formats, tiered to the license ladder:

- **Sprint races** (daily, walk-up): 6–8 minute heats inside a standard Shift slot, grid formed from the queue, class-locked. Results post to the leaderboard and earn Gears — any Apprentice-or-higher customer’s Shift can become a race.

- **Endurance events** (monthly): 30–60 minute team races (2–3 drivers per car) with mandatory pit windows for the Artisan battery swap. Endurance sells memberships: it requires teammates.
- **Ladder leagues** (six-week seasons): weekly heats, points, promotion/relegation between Rookie/Clubman/Pro divisions per class — the retention backbone (Volume 9 ties standing to license tier and Gears multipliers).
- **Special-event formats:** Formula Cup and drag brackets (Chapter 11), drift competitions judged on line/angle/style (three-judge club format), and the seasonal championship (Section 13.6).

13.2 Race-day track configuration

Racing overlays the rental layout rather than replacing it: sprint races run the standard circuits; league and championship nights may invert Track A's direction (a cheap “new track” — barriers and rostrum sightlines are designed for bidirectional running, Volume 11) and add the chicane kit on the main straight. The drag strip build (Christmas tree, staging beams, lane dots) is a 20-minute crew task documented as a rigging checklist in the race director's binder.

13.3 Timing systems

Verified against the July 2026 market, the division runs a two-tier timing architecture:

- **Tier 1 (both tracks, permanent): MYLAPS RC4** — the industry-standard inductive-loop system: detection loop embedded at start/finish (conduit cast at construction — Volume 11), RC4 decoder per track, 0.001 s resolution, rated beyond any fleet speed. Every fleet car carries a wired RC4-class transponder powered from the receiver rail; transponder ID is a field on the asset record. MYLAPS ships no scoring software; the division runs a compatible package (LiveTime Scoring / RC Scoring Pro class) feeding RC WORLD OS leaderboards via API.
- **Tier 2 (drift annex + drag module + backup): EasyLap-class IR** — the low-cost infrared system (transponder LED in the car, receiver bridge over the line, Zround/Live-Time software). IR needs line-of-sight, so transponders mount under the body window. The drag strip's staged beams integrate here.
- **Corroboration:** RCW Node geofenced lap detection provides a third, park-owned lap record — not race-grade timing, but enough to sanity-check the leaderboard and to keep casual Shift lap counts flowing when a customer's car runs outside race sessions.

Options evaluated and rejected. UHF **RFID timing** (passive tags, reader antenna at the line — the architecture of running events and some kart tracks) was evaluated as a cheap middle tier: tags cost cents and survive crashes, but read reliability degrades at RC closing speeds with multiple cars in the window, and ~0.1 s-class resolution is marginal against a 22-second lap where podium gaps run hundredths. It stays on file for paddock asset-tracking only. Relying on the **RCW Node alone** was also rejected for race duty: GPS geofencing is meters-class and Wi-Fi hand-off jitter adds tens of milliseconds — fine for Shift lap counts, not for declaring a winner by 0.02 s. The doctrine: **the Node observes every lap; MYLAPS-class timing declares race results.**

Cost note for the financial model: the RC4-class system is the expensive tier (system-plus-transponders is a four-figure system cost and transponders run ~\$100-class each across 64

cars — Volume 8 carries the negotiated numbers); the IR tier is an order of magnitude cheaper. The split architecture buys race-grade credibility where customers see it and economy where they don't.

13.4 Marshaling

Marshals are staff (customers never walk onto live tracks — park doctrine; a dead car in a rental session triggers the Tow-Truck Retrieval Protocol instead). Race sessions run two marshal posts per track at the designed refuge points, equipped with recovery hooks (reach, don't step), spare body clips, and a radio to the race director. Rules of engagement: recover between packs, never into traffic; a car that can't be made safe in 5 seconds is lifted out and rejoins from pit lane; any barrier breach red-flags the session pending reseal.

13.5 SOP-MS-003 — Race Director: Session Operations

SOP ID	SOP-MS-003
Revision	1.0 (July 2026)
Owner	Motorsport Lead (delegable to certified Race Director on duty)
PPE	Track-standard; marshals hi-vis
Tools	Director console (RC WORLD OS race module + timing software + Node kill-switch panel), PA, flag set, two charged marshal radios
Frequency	Every competition session

1. Pre-session (T−30): confirm class car pool parity-current (last audit ≤7 days); ESC profiles per class rules verified by readback sample (2 cars minimum, all finalists before finals); timing decoder test pass; marshal posts staffed and radio-checked.
2. Drivers' briefing (T−10): format, flags, contact rules, pit-lane direction, where marshals are, what the kill-switch means. Every driver confirms Toolbox Talk induction current in OS.
3. Grid formation: OS race module assigns cars and transponder IDs to drivers; scan-pair transmitters; stage on grid.
4. Session start: timing armed → grid released per format (standing start touring; rolling for SCT mixed grids; tree for drag).
5. During session: director monitors timing feed + Node telemetry map; enforces flags (yellow = local hazard, no passing between posts; red = all stop, throttle-kill via Node panel if not obeyed within 3 s).
6. Incidents: log every red flag and every contact penalty in the OS incident register during the session, not after.
7. Session end: results confirmed in timing software → published to leaderboard → Gears/license credits post automatically. Announce podium; direct cars to parc fermé for post-race spot checks (profile readback on winners — the parity promise is audited at the podium, visibly).
8. Post-session (T+15): cars to turnaround; RACE→RENTAL profile reversions executed and two-person verified (SOP-MS-001 step 5); director's session report closed in OS.

13.6 Championship calendar and night racing

The annual calendar gives the leagues somewhere to go: four **six-week league seasons** per year per major class, punctuated by monthly endurance rounds, the Formula Cup series, quarterly drift competitions, two drag bracket nights per quarter, and a season-ending **RC WORLD Championship weekend** that doubles as the park's flagship marketing event (Volume 2). **Night racing** is a Phase 3 unlock (permanent track lighting is Phase 3 capex per park canon); until then, two "twilight league" pilots per year run under temporary event lighting to build the case and the content. Drift under lights is the priority night product.

13.7 Coaching and spectator products

Two adjacencies round out race operations. **Coaching:** thirty-minute one-on-one sessions with a certified staff driver, priced at a premium over a normal Shift and structured around telemetry the customer already generates — the coach overlays the driver's Node speed trace on the class reference lap, sets one corner as the objective, and re-measures at the end. It converts the park's data exhaust into a product and is the natural upsell for a league driver stuck mid-division. **Spectating:** sprint heats and league nights are free from the rails (spectators become customers); championship weekends and drift/drag nights ticket the Phase 2 grandstand. The director's live-timing feed mirrors to the public screens with a 10-second delay, so an incident is marshaled before it is broadcast.

Investor Note. Competition revenue is deliberately modeled as secondary to Shift revenue (entries, spectator F&B uplift, event sponsorship — Volume 10), but its strategic value is retention: league drivers pre-book the same night for six weeks, and the license ladder they climb is priced in memberships. The division's racing calendar is a subscription engine wearing a sporting costume.

14. Division KPIs & Dashboard

14.1 KPI definitions

Five numbers, defined precisely so they cannot be gamed, reviewed weekly by the Motorsport Lead and monthly at park level:

1. **Utilization per car** = billed Shifts ÷ available Shift slots per car (operating hours × 2.0 practical Shifts/hour, cars in active status). Fleet target ≥ 55% weekday, ≥ 80% weekend. Special-event cars (formula/drag) are excluded from the denominator (Section 11.6).
2. **Cost per Shift** = (division direct labor + parts consumed + battery amortization + fleet depreciation) ÷ billed Shifts. Target ≤ \$4.10 at Year-1 volumes against the \$15 Casual Shift anchor. Illustrative build-up: labor ~\$1.90, parts ~\$0.85, battery amortization ~\$0.45, fleet depreciation ~\$0.70 (landed cost + hardening over a 3-year/2,000-Shift life), consumables ~\$0.20. Full costing model in Volume 10; the division owns the levers (labor via turnaround efficiency, parts via MTBF, batteries via over-drain discipline).
3. **MTBF (Shifts)** = billed Shifts ÷ failures requiring a car pull. Class targets: touring ≥ 120, entry ≥ 90, drift ≥ 150, buggy ≥ 70, SCT ≥ 100. Buggy's lower target is priced into its consumable-platform doctrine.
4. **Parts cost per 100 Shifts** = spares consumed (at landed cost) × 100 ÷ billed Shifts, per class. Feeds spares min/max (Checklist 12.7) and the procurement forecast (Volume 8).

5. **Parity spread** (SOP-MS-002 weekly result) — the doctrine number itself, trended per class.

14.2 Mini dashboard

The RC WORLD OS motorsport dashboard renders the weekly view in exactly this shape (illustrative Year-1 steady-state values):

Class	Cars active	Utilization	Cost/Shift	MTBF (Shifts)	Parts \$/100 Shifts	Parity spread
Entry 1/14	10	78%	\$2.60	95	\$18	2.4%
Touring/GT	14	71%	\$3.90	130	\$31	1.6%
Drift	12	64%	\$3.40	160	\$22	1.9%
Buggy/Rally	12	74%	\$3.10	75	\$38	2.7%
Short-course	8	61%	\$4.30	105	\$34	2.2%
Formula/Drag	8 (event)	event-based	per event	n/a	per event	1.4%
Division	64	≥65% blended	≤\$4.10	per class	trend ↓	≤3.0% all classes

Red lines: any class parity >3.0% (immediate re-audit), utilization <45% for two consecutive weeks (pricing/queue review with the GM), MTBF trending down 20% month-over-month (engineering review with The Works), parts \$/100 Shifts up 25% (failure-mode investigation before reorder).

15. Volume Summary & Cross-References

The Motorsport Division converts a standardized 64-car fleet into the park's highest-throughput revenue engine by holding one engineering promise: any car a customer is handed is within 2–3% of the fastest car in its class, so the race is theirs to win. This volume specified the class ladder; the four-platform standardization strategy and its ~130-SKU spares consequence; the parity engineering method (control tires, motor binning, ESC readback discipline, ballast equalization, gearing trim) with a worked three-car example landing at 1.4% spread and formal quarantine rules; the full tuning apparatus — ESC profile tables, suspension baselines and effects tables, corner weighting with a worked ballast calculation, rollout mathematics and thermal rules; the drift, touring, rally and special-event programs including energy-allocation racing; the rental hardening, transmitter lockout, RCW Node enforcement, Shift turnaround and lifecycle machinery; race operations from sprint heats to the championship weekend; and the five-KPI dashboard the division is managed by.

Maintenance interface note. This volume defines what the fleet must do and how it is configured; Volume 7 defines how it is kept doing it — preventive-maintenance matrices, tear-down/rebuild procedures per platform, the motor bench build-out (Section 4.3), battery pool management, failure-mode catalogue and FMEA, and Artisan training. The hard handshake: every car pull recorded at turnaround (Checklist 12.5) opens a work order in RC WORLD OS that The Works must close against Volume 7's procedures, and no car returns to active without the closing Artisan's sign-off plus, after major work, a parity re-run under SOP-MS-002. See **Volume 7, Chapters 4–5**.

Cross-references.

- **Volume 2** — market sizing and competitor scan behind the class ladder's demand assumptions; marketing/sponsorship programs the racing calendar feeds.
 - **Volume 4** — Construction Division; Track C crawler program and the recovery-crawler fleet that executes the Tow-Truck Retrieval Protocol on motorsport tracks.
 - **Volume 7** — Engineering & Workshop Manual (see maintenance interface note above); custom-build practices behind the drag conversions.
 - **Volume 8** — Procurement Handbook: supplier negotiations and landed costs for MST, MJX/LDRC, WLtoys, Carten/Tamiya-class chassis, Hobbywing/Surpass electronics, Gens Ace/CNHL batteries, MYLAPS/EasyLap timing.
 - **Volume 9** — Customer Experience & Loyalty: the RC WORLD License tiers that gate transmitter profiles (Section 12.1), Gears earning from races, league-membership mechanics.
 - **Volume 10** — Finance: division revenue model, cost-per-Shift build-up, timing-system and fleet capex lines.
 - **Volume 11** — Architecture & Park Design: Tracks A/B construction, drift annex and drag straight geometry, barriers, rostrums, timing-loop conduits, lighting phases.
 - **Volume 12** — Franchise Manual: which parts of this volume are franchise-mandatory doctrine (parity method, safety architecture) versus locally adaptable (class mix, calendar).
-

Volume 4 — RC Construction Division

RC WORLD — Master Development Plan · Volume 4 of 13 **Revision:** 1.0 — July 2026 · **Status:** Living document — bump revision on material change

Purpose of this volume. This is the handbook of the RC WORLD Construction Division: the Mining Zone and the Agriculture Zone that open with Phase 1 as one of the park's two anchor attractions. It specifies every machine class the division operates — from the Huina 1580 rental excavators to the single Kabolite-class hydraulic flagship — with anatomy, operating envelopes, verified 2026 market pricing and duty ratings; it derives the 1:3 excavator-to-dump-truck logistics ratio with worked queueing arithmetic; it engineers the haul roads, the open-pit material recirculation loop, the agricultural field grid, and the operator stations customers stand at; it defines the competition formats that convert earthmoving into repeat revenue; and it closes with the division-specific maintenance program, preventive-maintenance matrix and FMEA table that keep forty-two machines in daily public service. It is written to be handed out chapter by chapter: the civil contractor gets Chapters 5, 6 and 10, the Artisans get Chapters 2, 8 and 12, the events team gets Chapter 11, and the investor reads all of it.

Intended readers. Investors and lenders assessing Phase 1; the General Manager and Construction Division Lead; Artisans assigned to the heavy fleet; the civil/landscape contractor building the Mining and Agriculture zones (with Volume 11); the events and marketing team designing competition formats (with Volume 9).

Chapters

1. Division Overview & Philosophy
2. Excavators
3. Bulldozers & Graders
4. Wheel Loaders
5. Dump Trucks & Haul Logistics
6. The Mining Fleet System
7. The Agriculture Zone
8. Hydraulics Deep-Dive
9. Fleet Planning & Capacity
10. Site & Track Design for Construction Play
11. Earthmoving Competitions & Skill Programs
12. Maintenance & Reliability

1. Division Overview & Philosophy

1.1 The miniaturized industrial complex

The Construction Division is the purest expression of the founding thesis recorded in the original Omni-Zone blueprint: RC WORLD is a **miniaturized industrial complex, not a hobby sandbox**. A sandbox hands a customer an excavator and a pile of sand and leaves the rest to imagination. RC WORLD hands the customer a job. The Mining Zone is a simulated open-pit operation whose production circuit actually functions: excavators loading at bench faces, articulated dump trucks hauling on engineered roads that never exceed the canonical 15° incline, a central processing hopper that weighs and swallows every load, a wheel loader keeping the haul roads dressed, and a live tonnage leaderboard that turns twenty minutes of digging into a measurable shift of production. The Agriculture Zone applies the same logic to farming: pre-tilled fields, crop rows at scale spacing, implements that hitch and unhitch, irrigation trenches, a barn-and-silo receiving point, and seasonal harvest campaigns with their own scoring.

The distinction is commercial, not cosmetic. A sandbox exhausts itself in one visit. A production system generates goals, and goals generate return visits: the customer who moved 38 kg of aggregate through the hopper on their first Shift knows exactly what they want to do next time — move 50.

1.2 Simulation-of-work as entertainment

The division's product is best described as **simulation-of-work**: the satisfaction of competent labour, compressed into a 20-minute Shift and stripped of consequence. Four properties make this an unusually strong entertainment format:

1. **Low skill floor, high skill ceiling.** A first-timer is productive within two minutes — swing, curl, dump. The ceiling (smooth multi-channel blending, spill-free truck loading, clean trench walls) holds hobbyists for years. Pilot testing consistently shows construction operators using their full Shift while racing customers often crash out early; longer dwell means more F&B capture, more spectating, more Top-Up Sessions.
2. **The widest demographic in the park.** Racing skews young and male. Earthmoving draws families with children who recognize excavators before they can read; adult hobbyists and trades-adjacent enthusiasts for whom the machines are professionally familiar; and corporate groups, because “run a mine together for two hours” is a genuinely novel team exercise (Chapter 11.4).
3. **Almost no competition.** Club racing tracks exist in most large cities; public-access RC earthmoving is close to nonexistent worldwide. In Volume 2's competitive-landscape terms, the Mining Zone opens into an effectively empty field.
4. **It monetizes patience, not adrenaline.** Construction machines draw far less current than racing machines and crash less. The division has the park's best battery economics and lowest damage rate per Shift; revenue per machine is comparable to Motorsport while maintenance cost per Shift is materially lower (Chapter 12).

1.3 The two-tier fleet doctrine: electromechanical rentals, hydraulic premium

The single most important engineering decision in this volume is inherited directly from the founder's doctrine and must never be reversed: **the rental fleet is electromechanical — motor-and-lead-screw actuation — and hydraulics are reserved for premium supervised experiences and display.**

The reasoning, developed in full in Chapters 2 and 8, compresses to four lines:

- **Capital at risk.** A rental Huina 1580 puts roughly \$500 (landed, prepped) in a stranger's hands. A Kabolite K970-class machine puts \$8,000–12,000 there. At equal mishandling probability, expected loss per Shift differs by an order of magnitude.
- **Failure asymmetry.** An abused lead-screw machine strips a \$6 gear or kills a \$1 micro-switch — a 20–45 minute Artisan fix from shelf stock. An abused hydraulic machine ingests grit, scores a pump, or cooks a seal — a multi-day, \$150–400 event.
- **Turnaround.** Rental machines must flip between customers in under five minutes. Hydraulic machines want a rod wipe, level glance and temperature check between sessions — exactly what a supervised premium format provides and a walk-up format destroys.
- **Scarcity is the product.** Because only one machine in the park is hydraulic, K970 seat time is a sellable aspiration at premium pricing (\$22 Casual / \$38 Operator, Artisan always present) rather than a maintenance liability.

Investor Note. The division's fleet asymmetry is the doctrine expressed in dollars: the single hydraulic showcase machine costs more than the other 29 mining machines combined (Chapter 9.1). The cheap machines earn volume fees and are rebuilt for \$30 in parts; the expensive machine earns premium fees under professional supervision. Reversing this — renting hydraulics to walk-ins — is how an operator turns a \$14,000 fleet line into a \$90,000 one with worse uptime.

1.4 The customer journey

A Casual Shift in the Mining Zone, end to end: the customer books in RC WORLD OS and completes the Toolbox Talk digital induction once, before first visit (Volume 9). At the zone gate an Artisan scans the booking QR, issues a transmitter bound to a specific machine, and walks the customer to a numbered operator station on the pit rim. The machine is already staged — excavator at a dig face, or dump truck in the loading queue. The 20-minute Shift clock starts at first stick input. The machine's RCW Heavy-Node streams voltage and position; the hopper's load cells credit every delivery to the customer's account. At T-5 minutes the app offers a Top-Up Session if nobody is queued for the class. At Shift end the customer sees their production stats — tonnage, loads, spillage penalty, license progress — and the machine turns around with ~30% battery buffer intact, per the park's billing doctrine. If a machine dies mid-Shift, the customer does not walk onto the pit floor — the **Tow-Truck Retrieval Protocol** converts the breakdown into gameplay (Chapter 10.5).

1.5 Scale doctrine

1/14 for earthmoving, 1/16 for tractors and dozers. 1/14 is where the Chinese construction-model industry concentrates its engineering — Huina, Kabolite, LESU and JDModel all anchor on 1/14 — which means the deepest spare-parts pools, the widest attachment ecosystems, and the best payload-to-cost ratio. Going larger (1/8 hydraulics) doubles cost for little

experiential gain; going smaller (1/24) collapses payload, presence and durability. The founder's notes are explicit and the market confirms them.

2. Excavators

2.1 The rental workhorse: Huina 1580 V4 anatomy

The **Huina 1580 (V4, 2025/2026 production)** is the canonical rental excavator — six units, fleet canon — and it is worth recording exactly why it wins the seat. It is the only machine in its price band that is genuinely all-metal: tracks, undercarriage, slew deck, boom, stick and bucket are alloy castings and pressings, while every working motion remains **entirely electromechanical**. Verified 2026-market specification:

Parameter	Huina 1580 V4
Scale / configuration	1/14 crawler excavator, 360° continuous slew (slip-ring equipped)
Channels	23 (independent proportional tracks, boom, stick, bucket, slew; lights, sound, smoke)
Dimensions	~700 mm max reach length × 180 mm width × 480 mm max boom height; track frame ~290 mm
Weight	~7–8 kg
Battery	2S 7.4 V 2,000 mAh LiPo (park fleet: XT60-converted, 3:1 pool)
Stock runtime	~40 min per charge — comfortably one 20-min Shift plus the ~30% buffer
Attachments (factory case)	Bucket, wood grapple, jackhammer (park: bucket only for rentals; grapple/hammer reserved for events)
Wholesale (direct, Volume 8 channel)	\$350–420
Western retail (observed 2026)	\$560–800 (typical dealer ~\$580; some markets to \$860+)
Duty rating	RD-1 — rental, unrestricted (duty codes: Chapter 9.1)

Lead-screw boom actuation. Open a 1580 actuator and you find the whole rental doctrine in one assembly: a brushed 380/540-class motor, a small reduction gearbox, and a **lead screw** — a threaded steel rod turning inside a bronze/steel nut pinned to the boom linkage. Motor spins, screw turns, nut travels, boom rises. Where a Kabolite meters oil, the Huina meters rotation. At each end of travel a **micro-switch** cuts motor power before the mechanical stop. Three actuators (boom, stick, bucket) plus the slew motor and two track motors complete the drivetrain; a slip ring carries power and signal through the continuously rotating turret.

Two mechanical properties define the machine's character. First, the lead screw is **self-locking** at these helix angles: bucket load cannot back-drive the screw, so the machine holds a loaded bucket mid-air indefinitely, drawing zero current — a party trick no hydraulic machine can match without a counterbalance valve. Second, motion speed is fixed by screw pitch and motor RPM, so movements are steady and deliberate rather than proportionally

fluid. For a rental audience this is a feature: the machine is predictable, and predictable machines survive strangers.

Servo and motor layout (bench card, The Works): boom actuator under the boom root cover; stick actuator inside the stick; bucket actuator on the stick head; slew motor and pinion under the turret deck driving the slew ring; two brushed track motors with steel final drives in the undercarriage; main PCB, sound module, smoke unit and battery bay in the counterweight. The park adds one RCW Heavy-Node inline between receiver and ESC, on JST XH latching connectors for two-minute swaps.

2.2 Operating envelope and digging-force expectations

Honest numbers, from pilot testing and community consensus rather than any laboratory claim:

- **Dig depth:** ~150–200 mm below track datum before geometry runs out — which sets the pit-floor bench heights in Chapter 10.
- **Bucket payload:** 150–300 g per pass in the park’s washed 0–2 mm sand; 4–6 passes fill a Huina 1582 truck to its ~4 kg working load.
- **Breakout force at the teeth:** of the order of 2–3 kgf — enough to cut a moist sand face cleanly, not enough to self-injure against the pit’s timber edging. The face material spec (Chapter 6.4) is tuned so the machine always feels strong.
- **Cycle time:** a relaxed swing-dig-swing-dump cycle runs 25–35 s for a casual operator, under 20 s for a skilled one — the numbers the 1:3 ratio derivation in Chapter 5.2 is built on.
- **Travel:** walking speed is deliberately slow (~0.1 m/s class); an excavator walks perhaps 10 m in a whole Shift. Drive current is trivial; actuator current is intermittent 2–8 s bursts. Net pack draw per Shift: 25–40% of 2,000 mAh.

Field Note. The V4 transmitter has an LCD with password lock and play-timer. Both are disabled fleet-wide — Shift timing lives in RC WORLD OS, not in the transmitter — but the 23-channel layout stays factory-standard on all six units so any transmitter binds to any excavator after an Artisan swap. Never let transmitter configs drift per machine; config drift is how a fleet quietly becomes six unique machines.

2.3 The premium tier: Kabolite ‘Master Operator’ experience

Kabolite is Huina’s premium hydraulic marque (same Shantou industrial group, different engineering universe). Two models matter to this division:

Parameter	Kabolite K336GC	Kabolite K970 (-100S / -100S Pro)
Prototype	CAT 336 GC replica	Flagship 1/14 hydraulic excavator
Scale / weight	1/16, ~9.2 kg	1/14, 31 kg , 944 × 315 × 334 mm
Drive	Brushless motor, hydraulic pump + planetary travel drives	Brushless drive + brushless pump
Radio	24-channel 2.4 GHz	18-ch FlySky Paladin PL18 EV Lite touch-screen

Parameter	Kabolite K336GC	Kabolite K970 (-100S / -100S Pro)
Hydraulics	Entry hydraulic circuit	~3.5 MPa working pressure, 6–8 MPa limit, cab pressure display, 6-way valve, quick-coupler (Pro adds powered-attachment contacts)
Battery	7.4 V 10,000 mAh (~40–45 min)	11.1 V 3S 15,000 mAh
Street price (verified July 2026)	~\$1,400–1,900 (MSRP \$1,900)	Dealer/direct \$6,500–9,000; Western retail to \$10,500–12,500 (canon range)
Park role	Phase 2 candidate: second premium machine / technician trainer	The Phase 1 premium showcase machine

The park’s single premium machine is a **K970-class excavator**, and it carries three jobs: it is the **‘Master Operator’ experience** — a supervised, license-gated, one-on-one session at the premium pad (\$22 Casual / \$38 Operator, Artisan present throughout); it is the **display anchor** that makes the whole division legible (“that’s what the real hydraulics feel like”); and it is the **top rung of the license ladder** — customers grind tonnage on the 1580 fleet partly to earn K970 seat time. The K336GC is the designated second premium machine if Master Operator occupancy sustains above 70% (growth trigger, Chapter 9.5): at ~\$1,700 it adds a second supervised seat for a fifth of the flagship’s capital, and doubles as the technician-education hydraulic trainer in Chapter 8.

For the competitive map: **LESU** (Aoue) builds 1/14 hydraulic excavators from roughly \$2,000 (compact PC30-class) to \$7,000+ (LR960-class) plus hydraulic backhoes and skid-steers; **JD-Model** occupies the \$2,500–6,000 hydraulic band with excavators and wheel loaders. Both are credible second sources for premium machines and hydraulic spares (Volume 8 carries them); neither competes with Huina 15xx economics in the rental tier — which is precisely the point of the two-tier doctrine.

2.4 Operator controls tutorial (staff teaching script)

The division teaches the **ISO excavator pattern**, matching the Huina factory layout, and the tutorial below is the standard 90-second station brief every first-timer receives. Artisans deliver it verbatim until it is muscle memory.

1. **Left stick.** Left/right = **slew** (turret swing). Up/down = **stick** (the middle arm): push away to reach out, pull back to curl in. “Push to reach, pull to bring home.”
2. **Right stick.** Up/down = **boom**: pull back to raise, push to lower. Left/right = **bucket**: left curls (fills), right dumps. “Pull up, curl left, dump right.”
3. **Tracks.** Two shoulder controls, one per track, like a tank. Both forward = drive straight. Opposite = pivot. Move the machine only when the marshal light is green.
4. **The dig cycle.** Boom down, stick out, bucket cut, curl, boom up, slew to the truck, dump low over the bed — spilled material is tonnage you don’t get paid for.
5. **The golden rule.** Slow is smooth; smooth scores. Voltage-sag telemetry rewards smooth operators with longer effective allocations (F1-style power doctrine, Volume 9); jerky sticks drain the allocation and dent the smoothness index.

Skill progression from this baseline — two-function blending, grading with the bucket back, trench walls — is packaged into the badge tasks of Chapter 11.5.

3. Bulldozers & Graders

3.1 Fleet dozers: Huina 1554/1569 class

The dozer class is deliberately light: two machines, both 1/16, both cheap, charming and expendable — the entry drug of the whole division, issued mostly to children.

Parameter	Huina 1554	Huina 1569
Scale / build	1/16, ABS body, ~350 mm	1/16, semi-metal, heavier blade gear
Functions	11-function, proportional blade lift/tilt, rear ripper	8-channel, blade + rear scarifier
Runtime	40–50 min on 7.4 V pack	~35–40 min
Street price (2026)	~\$100–130 retail; wholesale \$45–70	~\$110–150 retail; wholesale \$60–90
Duty rating	RD-1	RD-1

Park units receive an Artisan-fitted aluminium blade wear-edge and a metal-gear service pack at first refurbishment (Chapter 12); blade edges on ABS machines are the first thing the Mining Zone’s crushed-stone areas eat.

3.2 Blade control as a taught skill

Dozer work is one control dimension harder than it looks, and the division teaches it as such: **the blade is a metering device, not a shovel**. The station brief teaches three moves — carry (blade just above grade, walking material forward), cut (blade 5–10 mm into loose material, short passes), and back-blade (reverse with blade floated to dress a surface flat). Children discover within minutes that a full blade stalls the tracks; the machine itself teaches load management, which is exactly the sort of embedded lesson the simulation-of-work format sells.

3.3 Haul-road maintenance duty — dozers as track-grooming assets

The dozers are not just toys with jobs; they are **rostered grooming assets**. Between customer waves and at close, Artisans use them for spill-drift back-blading in the dozer boxes and along haul-road shoulders — work that is deliberately performed in public view, because a staff machine visibly working the site reads as authenticity and reliably triggers “what is it doing?” conversations that convert into license-program interest. The heavy grooming (crown re-dressing, berm repair) belongs to the service wheel loader (Chapter 4.3); the dozers handle the fine work between waves. This dual duty is why the PM matrix (Chapter 12.2) gives two RD-1 machines a weekly track-tension check normally reserved for higher-duty assets.

3.4 Graders — the honest gap

As of July 2026 **no mass-market grader exists in the Huina rental class**; 1/14 motor graders are boutique hydraulic builds (LESU/JDModel custom territory, typically \$3,000–5,000 landed). The division therefore carries no rentable grader, and this volume says so plainly rather than padding the catalogue. Haul-road grading is performed by the service-duty 1583 wheel loader wearing a straight-blade attachment fabricated in The Works for

roughly \$40 of aluminium (drawing in Volume 7). A display-grade hydraulic grader enters the wish list only in Phase 3, and only after the premium program saturates — it is a want, not a need.

Trade Hack. The loader’s fabricated grading blade is drilled to the same pin pattern as its bucket, so blade-for-bucket swaps take under two minutes. Any attachment The Works fabricates for the division must use the standard pin pattern — one machine, many tools, zero adapters. Volume 7 carries the pattern drawing.

4. Wheel Loaders

4.1 Huina 1583 class

Parameter	Huina 1583 (2026 model)
Scale / layout	1/14 articulated wheel loader, die-cast alloy + ABS, ~570 mm long
Channels	10; proportional bucket; 2.4 GHz, ~30 m range
Capability	Carries up to ~10 kg over the front axle; push force ~1.5 kg; weight ~4.6–5 kg
Battery / runtime	7.4 V 2,000 mAh, ~40–45 min
Street price (2026)	\$285–320 retail; wholesale \$180–220
Duty rating	RD-2 (rental, license-gated) and SD (one unit reserved for service)

Fleet count: **3** (canon). Two rentable, one permanently assigned to grooming and hopper-apron duty. The loader is gated behind the second license tier because an articulated machine that can carry 10 kg can genuinely hurt a finger — it is the strongest machine in the rental pool, and the only rental machine whose issue requires either a license tier or a staff spot-check.

4.2 The hopper-loading cycle and bucket technique

The loader’s core gameplay is the **V-cycle**, taught exactly as real operators run it: approach the stockpile square, bucket flat on the ground; drive in until the wheels just begin to slip; **crowd** (curl) the bucket while lifting slightly — curling while stationary is what fills a bucket, pushing harder is what spins tires; reverse out along one leg of the V; steer to the truck or hopper along the other leg; dump low and centred. Score-relevant technique points, which the telemetry smoothness index and marshal spot-scoring both reward:

- **Fill factor:** a properly crowded bucket carries 300–500 g of sand; a flat-pushed one carries half that and spills on articulation.
- **The articulation trap:** an articulated frame mid-turn with a raised loaded bucket is the least stable state in the whole rental fleet. The taught rule is “low and level until lined up” — bucket below axle height except during the final approach.
- **Apron discipline:** at the hopper the loader gives way to trucks (trucks are on the billing clock of other customers; the loader isn’t).

4.3 The service loader

The SD-assigned 1583 — with the fabricated straight blade and a screen bucket — is the zone’s groundskeeper: haul-road crown dressing, berm repair to the 60 mm gauge, spill reclaim on the apron, stockpile management, and re-segregation of mixed aggregates (sand back to sand faces, pea gravel back to the coarse faces) during nightly grooming (SOP-CD-004, Chapter 6.6). One machine, roughly 90 minutes of Artisan operation per day, keeps the entire circuit within spec — the cheapest site-maintenance department in the amusement industry.

5. Dump Trucks & Haul Logistics

5.1 The circulation fleet: Huina 1582 / 1573

Eighteen trucks — the largest single machine population in the park — sized by the 1:3 ratio against six excavators. Two models share the pool deliberately:

Parameter	Huina 1582	Huina 1573
Scale / layout	1/14 rigid dump truck, alloy cab, metal tipping bed and linkage	1/14 rigid dump truck, ABS-dominant, metal bed floor
Channels	10 (drive, steer, tip, lights, sound)	10
Working payload	~4 kg per load (self-tips at 4 kg; chassis rated well above)	~3 kg working; self-tips ~4 kg max
Dimensions / weight	~450 × 150 × 190 mm; ~3.6–7 kg depending on trim	~450 × 140 × 190 mm; ~3.2–3.6 kg
Battery / runtime	7.4 V (park: 2S LiPo XT60 pool), ~30–45 min	7.2 V stock (park-converted to pool standard), ~30 min
Street price (2026)	~\$200–300 retail; wholesale \$150–200	~\$165 retail; wholesale \$120–160
Duty rating	RD-1	RD-1

Fleet split: **12 × 1582 + 6 × 1573**. The 1573s are the training haulers issued to younger children and absorb the roughest treatment; the 1582s are the production fleet whose bed geometry is correctly scaled to receive a 1580 bucket pass — the bucket drops between the bed rails with ~15 mm clearance per side, which is what makes clean loading a learnable skill rather than a lottery.

5.2 The 1:3 ratio, derived

The canonical **excavator : dump truck = 1 : 3** ratio is elementary queueing arithmetic applied to the physical circuit, and every future fleet decision must survive this derivation. Measured cycle elements (pilot testing on the Chapter 10 geometry, rounded conservative):

Circuit element	Time
Excavator loads one truck (4–6 bucket passes at 25–35 s)	~2.5–3.0 min
Loaded haul to hopper (~45 m at scale speed, incl. 14° ramp)	~1.5 min

Circuit element	Time
Queue + tip at hopper + weigh	~1.0 min
Empty return	~1.5 min
Truck away-time per cycle (excluding loading)	~4.0 min

An excavator can start loading a new truck every ~3 minutes. Each truck is away ~4 minutes per cycle. Trucks required for continuous face service:

$$N = (\text{loading interval} + \text{away time}) \div \text{loading interval} = (3 + 4) \div 3 \approx \mathbf{2.3 \rightarrow \text{round to 3}}$$

Round up, because customers are supposed to be having fun, not hitting takt time: human variance, spillage stops and photo pauses all inflate away-time. With **two** trucks the excavator operator stands idle waiting for a bed — the most expensive boredom in the division, since the excavator seat is the experience most customers came for. With **four** trucks they stack at the face, which reads as congestion and invites bumper-car behaviour. Three keeps every seat busy and every queue short. Fleet-wide: 6 excavators $\times$ 3 = **18 trucks**, canon.

5.3 Worked utilization calculation — payload per Shift

The arithmetic every capacity claim in Chapter 9 hangs from. Assume one **pod** (1 excavator + 3 trucks, four customers) running a casual-pace 20-minute Shift:

- Truck cycle time = loading (3.0 min) + away (4.0 min) = **7.0 min** → each truck completes **2-3 cycles per Shift** (2.86 theoretical; call it 2.5 with start/stop losses).
- Payload per cycle $\approx$ 3.5 kg blended (1582s at ~4 kg, 1573s at ~3 kg).
- **Per truck per Shift: ~8.5-10.5 kg. Per pod per Shift: ~26-31 kg through the hopper.**
- Excavator-side check: 2.5 truckloads $\times$ 3 trucks = 7.5 loads $\times$ ~5 bucket passes = ~38 bucket cycles in 20 min — one pass every ~32 s. Consistent with the observed casual cycle time; the circuit is balanced, with the excavator as the governing resource exactly as a real pit is planned.
- Skilled-pace ceiling (competition data, Chapter 11.3): ~20 kg in 11 minutes for a two-person team — roughly double casual throughput, which is the headroom that makes timed contests meaningful.

Utilization ratio: at casual pace the excavator is loading ~85% of the Shift, each truck is moving ~90% — no seat spends more than a couple of minutes waiting. That number is the product; guard it.

5.4 Haul-road design standard

The haul roads are the division's civil signature, built like real ones at 1/14:

- **Maximum incline 15° — canonical, and engineering-derived.** Brushed 540-class drive motors hauling a loaded 1582 sustain 14-15° indefinitely; beyond that, stall-current heating climbs steeply and continuous duty burns motors. All built ramps run **14°** (rising 1.0 m over 4.0 m of run) to leave margin under the canon ceiling. Grades are constant-slope — no roller-coaster profiles that shock-load gearboxes — with 300 mm vertical curves top and bottom.

- **Width for passing.** A 1/14 hauler is ~150 mm wide. Single-lane running width = 3 vehicle widths = 450 mm; the main loop is **two-lane, 900 mm formed width**, so a bogged or dawdling truck never blocks the circuit. Ramps stay two-lane; only the final hopper-apron approach necks to marshalled single-lane.
- **Berms.** Continuous compacted windrows, **60 mm high** (~40% of truck wheel height; real mines use 50% — ours is slightly under so the recovery crawler can climb them), on every edge with a drop over 100 mm. Berms passively absorb the highest-frequency customer error — drifting off-edge — with zero damage.
- **Surface.** 40 mm of compacted ≤ 10 mm crushed stone over geotextile, crowned 2% to shed water inward to the drain. The service loader re-dresses the crown weekly. Dig faces are loose sand; roads are never sand — the full surface-grade map from loose sand to 10 mm crushed stone is Chapter 6.4.
- **Circulation.** One-way, counter-clockwise. One-way running halves child-driver conflicts and is non-negotiable; painted arrows every 2 m and a give-way line at the apron are part of the nightly grooming checklist.

5.5 Payload cycles per Shift — the customer-facing number

Marketing and the app express hauling in honest, physical numbers: a Casual Shift in a 1582 moves **~10 kg through the hopper in 2-3 deliveries**; an Operator Shift (two blocks with a pit-stop battery swap) moves ~20-22 kg in 5-6. Career tonnage accumulates from exactly these weighed deliveries (± 20 g load-cell resolution, Chapter 6.5) — no estimates, no inflation. The division's unit of progress is the kilogram, and the hopper is its notary.

Field Note. Resist the temptation to “help” small children by overfilling their trucks at the face. An overloaded 1573 on the 14° ramp is the single most common motor-heat event in the fleet. The excavator brief teaches four passes for a 1573, five for a 1582 — counting passes is load discipline made child-legible.

6. The Mining Fleet System

6.1 Open-pit simulation: how the system reads

The Mining Zone must read, at first sight and from 40 m away, as a working mine: benches, haul roads, a headwall, a hopper with a conveyor, controlled dust in the light. This chapter specifies the production system — benches, hopper, material loop, medium management; the civil dimensions and safety architecture around it are Chapter 10.

6.2 Bench levels and pit-floor rotation

The pit is cut as a real open-pit in miniature, two benches deep:

- **Upper bench** at –300 mm below site datum; **lower pit floor** at –600 mm, connected by the in-pit ramp. Bench heights match the 1580's ~150-200 mm effective dig depth so a machine on each level always has a workable face below its tracks.
- **Six dig faces (DF1-DF6)**, each 3 m wide, cut into the bench walls directly below their assigned operator stations, with a marked truck spot beside each. Faces hold a 30-40° angle in the moist-sand spec and are re-cut nightly to the template so every morning

presents fresh, diggable material with 150–250 mm of loose won material heaped at the toe — first-minute success for the day’s first customers is a designed property, not luck.

- **Pit-floor rotation.** Faces retreat as customers dig. On a ~2-week cadence the active face set rotates around the pit walls: as DF1–DF3 approach their retreat limit (500 mm from the bench crest guard line), grooming shifts new cuts to the rested wall sections and the loader backfills the worked-out faces from the stockpile. Rotation keeps every face inside its station’s sight line, evens wear on the pit geometry, and means the pit visibly changes week to week — regulars notice, and noticing is retention.

6.3 Central hopper metering

The hopper is scenography, scoring system and traffic anchor in one structure — a functional scale plant, not a prop:

- **Receiving grizzly:** 600 × 600 mm steel bar grate at apron level, 20 mm bar spacing — passes all aggregate grades, stops buckets, phones and hands — over a 0.15 m³ surge bin.
- **Weighing:** the surge bin hangs on four 50 kg load cells (HX711-class amplifiers into an ESP32 on the park mesh). Every tip is weighed to **±20 g** and credited within ~2 s to the delivering customer via the truck’s Heavy-Node identity. This is the tonnage leaderboard’s ground truth and the settlement layer for every competition in Chapter 11.
- **Metering conveyor:** a 2.5 m inclined belt (200 mm belt, geared 24 V motor, variable 0.1–0.3 m/s) lifts weighed material to the stockpile discharge at +1.4 m. Belt speed is the system’s metering valve: run slow on quiet days so the stockpile grows theatrically; run fast ahead of grooming so the surge bin is empty at close.
- **Traffic control:** a stack light (green = accepting, amber = weighing, red = locked out) doubles as apron queue control. Lockout is a captive-key system — the conveyor cannot run while the bin access panel is open (Chapter 10.4).

Build cost ≈ \$6,000–9,000 including load cells, conveyor drive and corrugated cladding, fabricated in The Works to Volume 7 drawings.

6.4 Aggregate grades — the designed medium

The dig medium is a material system, not “sand.” Three grades in mapped areas, spanning the canonical range from loose sand to 10 mm crushed stone, each selected on the tension between diggability at 1/14 breakout forces, dust, and drainage:

Material	Spec	Where	Why
Washed coarse sand	0–2 mm, <3% fines, kept slightly moist	Dig faces DF1–DF4, dozer boxes, tillage fields (Ch. 7)	Best bucket penetration and heap behaviour; washed spec kills airborne dust; holds a 35° face overnight
Pea gravel	5–8 mm rounded	DF5–DF6 (“hard rock” faces), hopper surge bed	Satisfying rattle-and-pour physics for skilled operators; free-draining; too coarse to blow as dust
Crushed stone	≤10 mm angular, compacted	Haul roads, ramps, aprons — structure, never dig medium	Interlocks under compaction into a firm running surface; the canon ceiling grade

Total initial fill $\approx 55 \text{ m}^3$ (sand 30, pea 10, crushed 15), $\approx \$2,200\text{--}3,500$ delivered in most markets — among the cheapest attraction surfaces per square metre in the industry, and (unlike track asphalt) infinitely repairable with a loader.

6.5 The material recirculation loop

The zone's material cycle is **closed and visibly honest** — nothing is trucked in or out after commissioning:

1. Customers dig sand/pea from the faces and haul it to the hopper.
2. The hopper weighs it and the conveyor lifts it onto the **conical stockpile** — the zone's growing "product" landmark.
3. When the stockpile passes its 0.5 m^3 mark, the service loader reclaims it and carries it back to the worked-out faces per the material map (sand to sand faces, pea to pea faces — the screen bucket re-segregates any mixing).
4. Nightly grooming re-cuts the replenished faces to template.

One loop, roughly 1.5 t of material in perpetual rotation, driven entirely by customer labour plus ~ 30 minutes of loader time per day. Spillage is the loop's only leak: drifts on the apron and haul loop are reclaimed nightly to their source areas so the grade map stays true.

6.6 Dust and moisture management of the medium

Moisture is a tool, and the sand faces run best at **4-6% moisture** — dark, cohesive, dust-free, holding clean vertical cuts like real ground. Controls, in order of importance:

1. **Washed aggregate spec** ($<3\%$ fines) — the structural fix; dust you never buy is dust you never breathe.
2. **Nightly misting pass** to 4-6% (dark and cohesive, no standing water), verified each morning by the opening test: a face must hold a bucket-cut without slumping. Bone-dry sand is the mark of lazy grooming — it digs worse, scores worse and coats every machine in fines.
3. **Hopper misting ring** — a hose-end ring on the surge bin, because tipping is the dustiest event in the zone.
4. **Hard rule:** the compressed-air blow-down gun lives in The Works, never in the zone. Blowing a machine down in the pit just relocates fines into the next machine's bearings.

SOP-CD-004 — Mining Zone Nightly Re-Grooming SOP ID: SOP-CD-004 · **Rev:** 1.0 · **Owner:** Construction Division Lead · **PPE:** gloves, safety glasses, dust mask during dry raking · **Tools:** SD loader + blade/screen bucket, landscape rake, misting hose, face-cut template board, torque driver · **Frequency:** nightly at close, 40-55 min, 2 staff

1. Confirm zone clear; close marshal gate; hopper stack light to red; apply conveyor captive-key lockout.
2. Sweep the pit for foreign objects (dropped items, track pins, bucket teeth); log any fleet hardware found against the day's machine roster.
3. Loader pass 1: reclaim spill drifts from apron and haul loop to their source material areas (screen bucket where grades have mixed).
4. Loader pass 2: re-dress haul-road crown; repair berms to the 60 mm gauge board; refresh one-way arrows if scuffed.

5. Re-cut all six faces to template: 30–40° face angle, 150–250 mm loose toe heap; advance the face-rotation plan if any face is at its retreat limit.
6. Reclaim stockpile above the 0.5 m³ mark back to faces per the material map.
7. Misting pass on all sand areas to 4–6% moisture.
8. Hopper: empty and brush the surge bin; verify load-cell zero (± 50 g); torque-check grizzly bolts (weekly, logged).
9. Rake operator rim and spectator terrace; empty station bins; walk the drainage sump and clear debris.
10. Log completion in RC WORLD OS grooming register with photos of DF1–DF6; lockout key stays pocketed until the morning opening checklist releases it.

7. The Agriculture Zone

7.1 Concept

The Agriculture Zone is the Mining Zone’s calmer sibling: **1,500 m² (50 m × 30 m)** modelled on a commercial tobacco/maize operation per the founder’s notes — fields, not gardens. Where mining sells production, agriculture sells **husbandry**: plowing a straight furrow, disking a field to even tilth, threading a tandem trailer between crop rows without shedding the load. It is the division’s best fit for younger children, its best photographic contrast (green rows against the pit’s ochre), and home of the park’s most distinctive seasonal event, the Harvest Campaign.

7.2 The tractor fleet: Double E E351 class

Parameter	Double E E351 (E351-003), park-spec
Scale / build	1/16, ABS body, ~375 × 170 × 210 mm, ~1.1 kg stock
Drive	Rear-wheel drive stock; park units upgraded to 4WD-motor spec where available, high-torque gearing
Radio	2.4 GHz, proportional drive/steer; horn, engine sound, lights; controls compatible powered accessories via rear data socket
Battery	Stock 4.8 V 300 mAh NiCd — replaced fleet-wide with 2S-compatible park packs and XT60 during prep (the single most important prep step; stock packs would die mid-Shift)
Street price (2026)	\$50–90 retail; park landed cost \$80–120 each including battery conversion, standard hitch, Heavy-Node fitment
Duty rating	RD-1 (tandem trailers RD-2)

Fleet count: **12** (canon). Why the E351 wins the seat: it is the only widely available 1/16 RC tractor with a powered-accessory ecosystem (tipping trailer, tedder) at a toy-grade price, and its market saturation means bodies, wheels and gearboxes are perpetually cheap. The German alternative — **Siku Control** (Fendt/John Deere/Claas, €99–172 retail, superb detail) — runs at 1:32 scale in its RC line, too small for outdoor field work and incompatible with the park’s 1/16 implement library; it is noted for the retail shop (Volume 8), not the fleet.

7.3 The implement library

Tractors are the platform; implements are the content. Every tractor is fitted at The Works with the park-standard **pin hitch** (3 mm clevis pin on a 20 mm plate) so every implement fits every tractor. The library (~20 pieces, \$15-60 each wholesale, several shop-fabricated to Volume 7 drawings):

Implement	Qty	Function	Gameplay
Single/dual-bottom plow	4	Cuts real furrows in tillage fields	Straightness scored against chalk line
Disc harrow	3	Breaks furrows to tilth	Coverage scoring (photo overlay)
Tipping trailer (single-axle)	5	Haulage to the barn	Load/deliver cycles, weighed
Tandem trailer set	2	Advanced haulage (canon: tandem trailers)	RD-2, license-gated; a jackknife is a retrieval event
Water bowser	2	Fills at the irrigation valve, wets fields	Staff grooming + campaign use
Bale/crop flatbed	2	Carries crop tokens	Harvest Campaign core
Front blade (loader-arm fit)	2	Lane dressing	Staff grooming + skill badge task
Rake/tedder (Double E powered accessory)	2	Row dressing, campaign theatre	Powered via tractor data socket

7.4 Field grid layout

The zone is a grid of **eight field cells, each 9 m × 5.5 m**, separated by 600 mm compacted access lanes, with the barn/silo complex on the east edge:

- **F1-F4 — tillage fields:** 100 mm depth of the same washed 0-2 mm sand as the pit (one aggregate supply chain, deliberate), held at 4-6% moisture, re-groomed flat nightly so morning customers cut first furrows into clean ground.
- **F5-F6 — row-crop fields:** permanent simulated crop rows at 250 mm spacing. Row materials (durability-tested against 1/16 wheel strikes): UV-stable artificial boxwood strip for “young maize”; bundled natural-look raffia in drilled timber battens for mature tobacco/maize — convincing at spectator distance, replaceable in 1 m sections. Between-row lanes are exactly 1.4 tractor widths: threading them clean is the zone’s core skill.
- **F7-F8 — harvest/haul fields:** where campaigns stage crop tokens.
- **Irrigation trenches:** 150 mm wide × 80 mm deep formed channels along the grid spines — dry by default, hose-flooded for events — crossed by four single-tractor timber culvert bridges. The trenches are working scenery: a wheel dropped into one is a taught recovery lesson, not a design flaw.
- **Barn & silo:** a 2.4 m × 1.8 m scale barn with a drive-through **weighbridge bay** (same HX711 load-cell architecture as the mining hopper, 20 kg cells) and a silo tower deliveries are credited against. Same scoring backbone as mining, gentler physics.
- **Operator line:** six stations along the south edge, built to the same rail/shade/sight-line standard as the mining rim (Chapter 10.3); the whole grid is visible from every station.

7.5 Crop-row simulation and task cards

A standard Agriculture Shift issues a tractor plus one implement chosen at booking. The marshal assigns a field cell and a **task card**: plow F2 north-south, disc F3 to full coverage, haul six trailer loads from F7 to the barn. Completion, straightness scores and delivered weights post to the customer profile exactly like mining tonnage. The zone runs 8 concurrent customers comfortably (6 field tasks + 2 haul circuits) against 12 tractors — the same ~33% rotation reserve as the pit.

7.6 Seasonal Harvest Campaign events

Twice a year (spring planting, autumn harvest; inverted for southern-hemisphere deployments) the zone runs a two-week **Harvest Campaign**: F5-F8 are dressed with crop tokens — weighted 40 mm “maize bundle” and “tobacco bale” pucks, ~200 g each, RFID-tagged — and every Shift’s deliveries accrue to a park-wide campaign total on a public silo-gauge display. Individual contributions earn campaign badges (Volume 9); the finale week-end crowns a Harvest Champion per age class. Token set cost ≈ \$400 per campaign; comparable seasonal-event benchmarks reliably show a campaign fortnight doubling zone utilization, making this the cheapest utilization lever in the volume.

Field Note. The Agriculture Zone is the park’s birthday-party engine. A tractor is the machine parents trust a six-year-old with unsupervised-feeling (it isn’t — geofenced, marshalled, 10 km/h scale speed), and eight children on eight tractors with task cards is a self-running party format. Book parties into agriculture by default; upsell the pit for the second visit.

8. Hydraulics Deep-Dive

This chapter serves two audiences: the Artisans who maintain the premium tier, and the technician-education program (Volume 7) that uses the K336GC trainer to teach fluid power. Even though only one Phase 1 machine is hydraulic, every heavy-fleet Artisan must understand the system — the K970 is the division’s most expensive asset, its most fragile revenue line, and the machine most closely examined by exactly the customers whose opinions travel furthest.

8.1 Anatomy of a 1/14 hydraulic system

A scale hydraulic machine is a real hydraulic machine with the decimal point moved. The K970-class circuit contains every element of a full-size excavator’s:

1. **Reservoir.** ~200–400 ml of hydraulic oil in the counterweight area, with filler, sight window and coarse suction strainer. Miniature tanks are proportionally smaller relative to flow than full-size tanks, so the oil works harder and runs hotter per millilitre — the root cause of half the maintenance schedule below.
2. **Pump.** A miniature **gear pump** — two hardened steel gears meshing in a machined housing — driven by a **brushless motor** with its own ESC. Gear pumps dominate the scale world: simple, flat efficiency curve, dirt-tolerant relative to piston pumps. Representative aftermarket spec (LESU-class, 2026): housings rated ~10 MPa, flow ~400 ml/min, relief mandatory well below rating. The pump motor is the “engine”: the operator

hears it load up under digging effort, which is a large part of the sensory magic customers pay premium rates for.

3. **Relief valve.** Spring-loaded bypass that caps system pressure — the component that decides whether a stalled cylinder is “realistic engine lug” or “burst hose.” Set at commissioning, verified at every fluid change (SOP-CD-003).
4. **Directional valve block.** A bank of miniature spool valves — the K970 runs a **6-way block** — each spool shifted by a micro servo. Proportionality comes from spool position: the servo meters flow, so stick finesse translates directly into cylinder speed. This servo-on-spool architecture is why hydraulic machines feel alive.
5. **Cylinders.** Honed-tube rams with NBR piston seals (often PTFE backup) and rod wipers. Bores in 1/14 run 8–16 mm: a 12 mm bore at 20 bar develops ~23 kgf of rod force — how a 31 kg model genuinely digs compacted ground.
6. **Hose and hard line.** 3–4 mm OD nylon/polyurethane tube with brass compression or push-fit fittings; braided scale-appearance hose on premium boom sets.
7. **Instrumentation.** The K970 carries an in-cab display showing pressure, oil temperature and system voltage — the single most useful diagnostic on the machine and the anchor of the weekly health check.

8.2 Operating pressures and the two real enemies

Verified factory figures for the K970-100S: **~3.5 MPa (35 bar) working pressure, 6–8 MPa limit pressure**, ~80 kg track thrust, 60 kg-class slew torque. For intuition: 35 bar is about fifteen times road-bicycle tire pressure, delivered through hoses the diameter of spaghetti. Two engineering consequences dominate:

- **Heat, not pressure, is the enemy.** With a healthy relief valve nothing bursts; instead, sustained relief bypass (an operator holding a function against its stop) converts full pump power into oil heat. A 40-minute enthusiastic session can lift tank temperature 20–25 °C over ambient; above ~60 °C viscosity collapses, internal leakage rises and seals age fast. The Master Operator supervision script therefore teaches customers to **feather off at end-of-stroke** — a genuine operator skill that also protects the machine.
- **Cleanliness is everything.** Pump gears run micron-scale clearances; one grain of the zone’s own 0–2 mm sand inside the circuit scores the pump in minutes. This is the engineering core of the premium doctrine: the K970 works the same dirt as the rental fleet, but with an Artisan present whose job is partly to keep the filler cap, quick-coupler and rod wipers clean.

8.3 Fluid selection

Scale-hydraulic OEMs (Kabolite’s own manuals included) standardize on **ISO VG 32 anti-wear hydraulic oil**, stepping to **ISO VG 46** in sustained hot climates (tank temps >55 °C). Absolute prohibitions, posted on the workshop wall: **never brake fluid** (glycol swells NBR seals), **never motor oil** (detergent packages foam and corrode), **never silicone shock oil** (wrong lubricity; wrecks pump gears). RC WORLD stocks VG 32 in 1 L bottles dyed with a trace of red tracer so any drop on a machine or the pit floor is instantly attributable.

Trade Hack. Keep a white ceramic tile in the K970’s parking bay and park the machine over it every night. A red-tinged drop on the tile at morning inspection localizes a developing leak before the reservoir level moves — the sight glass is a lagging indicator; the tile is a leading one.

8.4 Common leak points

Every hydraulic machine leaks eventually; the discipline is making leaks scheduled events rather than surprises. The leak hierarchy, in descending frequency across the scale-hydraulics community:

Leak site	Cause	Fix	Typical interval
Cylinder rod seal	Rod wiper wear, dust ingestion	Seal kit (\$8-15/cylinder), 30-min bench job	150-300 run-hours
Hose push-fit fitting	Vibration walk-out, heat cycling	Re-cut tube end, re-seat; replace tube at every 2nd service	100-200 h
Valve spool O-rings	Heat aging	Spool seal kit	300-500 h
Pump shaft seal	Normal wear	Pump rebuild/swap (\$120-180)	400-600 h
Tank filler / gauge threads	Over-torque, missing PTFE tape	Re-tape, torque to spec	Commissioning errors only

Leak doctrine: any visible external leak takes the machine out of service immediately (fleet_inventory status → maintenance). At 200-400 ml total charge, a “small” leak is a large fraction of the system, and a starved gear pump destroys itself in minutes.

8.5 Bleeding procedure and fluid change

SOP-CD-003 — Hydraulic Fluid Change & Air Bleed (Premium Fleet) SOP ID: SOP-CD-003 · **Rev:** 1.0 · **Owner:** Heavy Fleet Artisan Lead · **PPE:** nitrile gloves, safety glasses · **Tools:** dyed ISO VG 32, catch tray, extraction syringe/filler bottle, lint-free wipes, IPA, PTFE tape, torque driver, fresh leak tile · **Frequency:** 100 run-hours or 6 months, and after any contamination event

1. Run the machine 5 minutes at light load to warm the oil and suspend contaminants; park over the catch tray; power off; transmitter secured by the Artisan.
2. Photograph the in-cab gauge at idle (pressure reference for post-change comparison).
3. **Clean the filler area before opening** — the filler neck is the contamination gateway; this step is the SOP’s soul.
4. Drain/extract the reservoir into the catch tray. Inspect against the reference vial: darkening = normal aging; milkiess = water ingress (find it — check rod wipers and filler seal); glitter = metal — **stop and escalate** to pump inspection before any refill.
5. Extend all cylinders fully to push residual oil back to tank; extract again. Do not full-system flush unless contamination was found — over-servicing opens more entry points than it closes.
6. Clean the suction strainer; replace the breather element if fitted.
7. Refill to the sight-window line; run the pump at idle 60 s; then **bleed**: cycle every function slowly to both stops, twice, in circuit order (boom → stick → bucket → auxiliary). Spongy or juddering motion = air remaining; repeat the slow cycles and re-top the tank (the purged air’s volume is replaced by oil).
8. Verify idle and relief pressures against the step-2 photo ($\pm 10\%$); investigate any relief drift before release.

9. Wipe down; new leak tile; dispose of waste oil per the park environmental procedure (Volume 11); log volumes and condition to `maintenance_logs`.

8.6 Maintenance schedule — Kabolite-class machines

Interval	Task
Every session	Rod wipe-down; visual hose scan; quick-coupler cleaned and capped; park over leak tile
Daily (open)	Reservoir level at sight window; gauge sanity check (idle + relief); cycle all functions gently to stops once
Weekly	Log tank temperature after last session; torque scan on valve block and cylinder clevises; track tension (SOP-CD-001)
Monthly	Oil condition vs reference vial; strainer inspection; servo-linkage wear check on all spools
100 run-hours / 6 months	Full fluid change + bleed (SOP-CD-003); rod-seal inspection; relief setting verified
Annually	Pump flow test (timed full-stroke vs commissioning baseline; >20% slowdown = pump rebuild); full boom hose-set replacement

8.7 Why hydraulics stay out of the rental fleet — the math

The doctrine, now in numbers rather than principle. Compare one rental-duty year (assume 600 Shifts/year/machine at Year 2 volumes) for a hypothetical hydraulic rental against a 1580:

Line	Huina 1580 (electromech.)	Hydraulic machine on rental duty
Capital landed	~\$500	~\$8,000 (K970-class) — or ~\$2,000 even for a K336GC
Consumable failure, typical	\$1 micro-switch, \$6 gear set	\$12 seal kit + oil, \$150 pump events
Maintenance cost/Shift (observed class data)	\$0.50–0.90	\$3–6 conservatively, before pump events
Turnaround between customers	< 5 min (none needed beyond battery logic)	5–10 min rod-wipe/level/temp check every session
Downtime per major failure	20–45 min (shelf parts)	Days (seal kits, pump parts, oil flush)
Annual maintenance @600 Shifts	~\$400	~\$2,500–4,000 + one pump event likely
Capital + Year 1 maintenance at risk	~\$900	~\$11,000+

An 18× capital-at-risk gap and a 5–7× maintenance-per-Shift gap, for an experience difference most casual customers cannot articulate past “it sounds cooler.” The premium format captures that difference honestly instead: the customer who can articulate it pays \$22–38 for a supervised session, the Artisan’s presence protects the machine, and scarcity does the marketing. This table is the division’s constitutional law; any future proposal to rent hydraulic machines unsupervised must first defeat it in writing.

9. Fleet Planning & Capacity

9.1 Canonical fleet table

Duty ratings: **RD-1** rental unrestricted · **RD-2** rental, license-gated · **PD** premium/display, supervised · **SD** service/staff only.

Class	Model	Scale	Actuation	Count	Wholesale each	Fleet line	Duty
Excavator	Huina 1580 V4	1/14	Elec-tromechanical (lead-screw)	6	\$350-420	~\$2,300	RD-1
Dump truck	Huina 1582	1/14	Elec-tromechanical	12	\$150-200	~\$2,100	RD-1
Dump truck	Huina 1573	1/14	Elec-tromechanical	6	\$120-160	~\$850	RD-1
Wheel loader	Huina 1583	1/14	Elec-tromechanical	3	\$180-220	~\$600	RD-2 / SD
Dozer	Huina 1554/1569 class	1/16	Elec-tromechanical	2	\$45-90	~\$150	RD-1
Premium showcase	Kabolite K970 class	1/14	Hydraulic (brushless pump)	1	\$6,500-9,000 direct (\$10.5-12.5k Western retail)	~\$8,000	PD
Mining subtotal				30		~\$14,000	
Tractor	Double E E351 class	1/16	Elec-tromechanical	12	\$80-120	~\$1,200	RD-1
Implement library	Plows, discs, trailers, rakes	1/16	Towed / data-socket powered	~20 pcs	\$15-60	~\$700	—
Division total				42 powered		~\$15,900	

Spares provisioning on top — one “ghost machine” of high-turnover parts per six fleet units — is specified in Chapter 12.4 and priced in Volume 8.

9.2 Session capacity per zone per hour

Definitions: Shift = 20 min; turnaround (transmitter hand-off, battery decision, staging) budgeted 5 min → **2.4 sellable Shifts/seat/hour**.

Zone	Concurrent customer seats (peak config)	Shifts/hour	Shifts/day (10 h)
Mining — standard	16 (4 pods: 4 excavators + 12 trucks) + 2 loader/dozer seats = 18	~43	~430
Mining — premium pad	1 (supervised)	max 2, practical 1 (cool-down)	4-8 offered slots
Agriculture	8 (6 field tasks + 2 haul circuits)	~19	~190
Division ceiling	27 seats	~62-64	~620

The reserve (2 excavators, 6 trucks, 4 tractors, 1 loader) is not idle capital: it is the 33% rotation pool that absorbs battery swaps, PM slots and defects without a visible hole in the circuit.

9.3 Utilization targets and the worked revenue check

Blended reality against theoretical ceiling, consistent with Volume 10's Year 1 canon ($\approx \$1.28$ M park revenue):

- **Year 1 target: ~1.75 paid Shifts/machine/day blended** across the 360-day year (weekday daytime is school time); **Year 2: 3.0** as leagues and memberships build weekday base load.
- Effective revenue per block at canon pricing (\$15 Casual / \$26 Operator, Operator $\approx 25\%$ of sales at \$13/block): $\sim \$14.50$.
- **Standard fleet (41 machines): $1.75 \times \$14.50 \times 360 \approx \$9,100/\text{machine}/\text{year} \rightarrow \sim \$375,000$.** Realistically discounted for the tractor fleet's child-heavy pricing mix: **\$330-380k**.
- **Premium K970: 3-5 sold sessions/day at \$22-38 $\rightarrow$ \$90-150/day $\rightarrow$ \$35-50k/year** — from one machine, matching roughly a tenth of the whole standard fleet on 2% of its Shift volume.
- **Events and campaigns** (Chapter 11): \$40-60k Year 1.
- **Division Year 1 total: $\approx \$410-470k$** — about a third of park revenue, per the Volume 10 model.

Investor Note. The gap between the ~620-Shift/day ceiling and the ~75-Shift/day Year 1 blended reality is not waste — it is option value. The division can absorb roughly 8 $\times$ demand growth (leagues, school programs, corporate weekdays) with zero additional fleet capex. Volume 10's upside case is built on filling exactly this gap.

9.4 Battery logistics — the 3:1 doctrine applied

Canon: **3 packs per vehicle minimum** — one in the machine, one charging, one rested and ready; 2S 7.4 V heavy-fleet standard, XT60 connectors (Deans legacy acceptable on acquired stock); LiPo window **3.4-4.2 V/cell** enforced by the Heavy-Node kill-switch; all charging in the bunkered charging room on SkyRC T1000 / ISDT K4-class multi-port balance chargers.

Division pool: 41 common-pool assets $\times 3 =$ **123 packs** (excavators/trucks/loaders/dozers on 2S 2,000-3,000 mAh; tractors on 2S 1,500-2,200 mAh), plus the K970's own pair of 3S 11.1

V 15,000 mAh packs outside the common pool. Every pack carries a QR and a cycle log in RC WORLD OS.

Swap cadence from measured duty: excavators draw 25–40% per Shift → swap at every Operator pit stop and every second Casual Shift; trucks draw evenly (continuous driving) → swap every second Shift without exception; tractors ≈ trucks. The 90-second pit-stop swap at the service-spine bench:

☐

Inbound pack voltage read and logged (any cell < 3.5 V → machine was over-run; investigate before reissue)

☐

Pack body inspected: no puffing, no dented corners, XT60 pins bright

☐

Outbound pack rested ≥ 30 min post-charge and ≥ 4.15 V/cell balanced

☐

Battery hatch latched; wiring clear of the pinch line

☐

Swap logged against pack QR + machine asset tag in RC WORLD OS

9.5 Growth triggers for adding machines

Fleet additions are triggered by the dashboard, never by enthusiasm. Each trigger must hold for a rolling month:

Trigger	Threshold	Action
Standard-fleet blended utilization	> 4.5 paid Shifts/machine/day	Add one pod (1 × 1580 + 3 × 1582) — always in ratio ; never add excavators or trucks alone
Peak-day queue abandonment	> 10% of mining queue walks	Same as above, or extend hours first (cheaper)
Premium occupancy	> 70% of offered K970 slots sold	Add second premium machine (K336GC or K988-100S loader, ~\$1,700–4,600); budget +0.25 FTE Artisan
Agriculture party bookings	> 80% weekend slot occupancy	Add 4 tractors + implement set (~\$500 total)
League demand	Waitlist > 4 teams	Add evening league night before adding machines
Any addition	—	Pack pool grows 3:1 with the machine, same purchase order, no exceptions

10. Site & Track Design for Construction Play

10.1 Zone dimensions and layout

Mining Zone: 60 m × 40 m (2,400 m²), long axis east-west, sited per the Volume 11 master plan with the site's 2–4% natural grade falling to the southwest corner (drainage exploits this). Coordinates in metres from the southwest corner (x east, y north):

- **Operator Rim (y = 36–40, full width):** elevated customer terrace, Section 10.3.
- **Main Pit (x = 4–44, y = 8–32):** two benches (–300 mm and –600 mm), six dig faces DF1–DF6 under their stations, in-pit ramp.
- **Haul loop (~110 m lap):** faces → south wall → main ramp (x = 44–54, –600 mm → +400 mm over 4.0 m run = 14°) → hopper apron → return ramp (north side, 14°) → faces. All per the Chapter 5.4 standard.
- **Central Processing Hopper (x = 50–56, y = 18–26):** tipping apron west, conveyor discharging east to the stockpile.
- **Premium Pad (x = 50–58, y = 30–38):** the K970's fenced 8 × 8 m dig cell with hardstand, leak tile and a rail-side viewing edge — the showcase digs where everyone can watch.
- **Service Spine (y = 0–6, full width):** staff-only strip behind a 1.1 m fence — swap bench, staging racks, SD loader bay, tool locker, and the single marshal gate at x = 30.
- **Dozer Push Boxes (x = 8–16, y = 8–12):** two 4 × 4 m sand cells inside the pit but fenced from the haul loop, so the youngest operators never enter traffic.

Agriculture Zone: 50 m × 30 m (1,500 m²) adjacent, per the Chapter 7.4 grid, sharing the service spine's swap bench.

10.2 Pit depths safe for retrieval

Maximum excavation depth is **600 mm below datum** — set not by the machines (a 1580 bottoms out at ~200 mm below its own tracks) but by **retrieval**: the 1/10 recovery crawler must be able to reach, rig and winch any casualty, and its practical climb limit against the pit's 30–40° faces bounds the geometry. Design rules:

- No point on the pit floor may be more than 600 mm below an adjacent surface reachable by the recovery crawler at ≤ 35°.
- Bench crests carry a 500 mm guard line: faces are never cut closer than this to the crest, preventing undercuts that could slump onto a machine below.
- The in-pit ramp always connects both bench levels to the haul loop — no machine can ever be geometrically stranded.
- Nothing in the zone requires a human to step below datum during operating hours; every retrieval is machine-performed (Section 10.5) or, at day's end, an Artisan task behind a closed gate.

10.3 Operator stations and sight lines

Twelve numbered stations at 4 m centres along the rim — a designed workplace, not a fence gap:

- Standing rail at 1.1–1.2 m (comfortable forearm rest for adults; chest rail for children on the fold-down 250 mm step — the under-8-with-guardian pairing the station is dimensioned for).
- Transmitter shelf angled 15°; station number and assigned-face placard; QR opening that station's live telemetry card in the customer app.
- Shade sails over every station (4 × 4 m HDPE, 3.2 m posts): transmitter LCDs and summer dwell both demand it.
- **Sight-line rule (non-negotiable):** every operator must see their machine and the entire haul-road segment their trucks use, unaided, from their station, at a 15–25° downward viewing angle — steep enough to see into truck beds (judging fill is the skill), shallow enough to read machine attitude at 20 m. No station may operate into the hopper's visual shadow; this constraint fixed the hopper at the east end.

10.4 Netting, barriers and separation

Customers operate from the rim and the field line, full stop; pit floor, haul loop and field grid are machine space. Three enforcement layers:

1. **Physical.** Rails and fences per above; the single interlocked marshal gate; the hopper's captive-key lockout — one key, two mutually exclusive locks, so the conveyor cannot run while the bin access panel is open, and cleaning and running are mechanically impossible to combine. Unlike the Aviation Division, no overhead netting is required — nothing in this division leaves the ground; barriers are waist-height and sight-friendly. Low mesh (300 mm) along the rim toe catches the one projectile the zone produces: spilled aggregate kicked by a track.
2. **Procedural.** Toolbox Talk induction, marshal authority, and the Retrieval Protocol replacing every “can I just grab it” impulse.
3. **Technical.** Geofence on every Heavy-Node: a machine crossing the customer line is PWM-killed before it reaches a shoe; under-voltage (<3.4 V/cell) throttles to 20% and orders the machine to the pit.

Spectator viewing: a 2 m-deep lean-rail terrace behind the operator line, deliberately generous — construction is the park's best spectator product, and a loaded 1/14 truck climbing the ramp at golden hour is its single best marketing image. The premium pad's viewing edge doubles as the crowd magnet during Master Operator sessions.

Safety Warning. The most dangerous object in this division is not the 31 kg K970 — it is a charging LiPo. All charging occurs in the bunkered charging room without exception; no pack charges unattended anywhere in the zone, workshop bench included. The 3.4–4.2 V/cell window and the battery-bench checklist are safety controls first and asset controls second.

10.5 Retrieval, ergonomics and the marshal gate

A dead machine triggers its Heavy-Node's **85 dB localized buzzer**. The customer surrenders their transmitter at the marshal gate and receives the controls of the division's permanently staged **1/10 winch-equipped recovery crawler**; under marshal guidance they

drive out, rig the winch (Artisan-supervised hook-up — customers never pass the gate), and tow the casualty to the service spine. Shift clock pauses during recovery; a fresh machine is issued; the Tow-Truck Hero badge posts on first completion. Pilot feedback is unambiguous: a meaningful share of children try to get recovered on purpose. The friction point is the feature.

Ergonomics addendum from pilot testing: transmitters hang on station lanyard hooks (dropped-transmitter rate fell to ~zero), the rim surface is compacted gravel rather than pavers (standing comfort over a full Operator Shift), and stations 1-2 nearest the gate are reserved for accessibility — wheelchair-height rail section and a widened terrace bay.

11. Earthmoving Competitions & Skill Programs

11.1 Why competition matters here

Racing has a century of borrowed formats; earthmoving has none — RC WORLD gets to invent its competitive canon, and owning formats means owning the community that plays them. Competition also solves the division's weekday problem: leagues and corporate bookings are scheduled demand aimed exactly at the utilization gap in Chapter 9.3. Every format below runs on standard rental machines (the parity doctrine borrowed from Motorsport: skill decides, not hardware) and scores automatically through the hopper load cells, RFID tokens and Heavy-Node telemetry — no manual judging beyond a marshal's penalty flag.

11.2 Precision digging challenges

- **Golf-Ball Pickup (the signature event).** Six golf balls on 60 mm tees in a 2 m arc around a 1580. Score = balls placed unbroken into a bucket-width target box; time is tiebreaker only. Championship variant swaps balls for eggs — boiled for juniors, raw for finals night. Theatrical, photogenic, and a genuine multi-channel finesse test.
- **Bucket Curling.** Push a 500 g puck along a 3 m painted lane with the bucket; nearest-to-rings scores 5/3/1; team relay format.
- **Trench & Backfill.** Cut a 400 mm trench to a depth template, place the “pipe” (300 mm dowel, grapple attachment), backfill and dress flat. Judged on template fit plus surface flatness under a straightedge. The trade-skill event — real operators enter this one.

11.3 Timed load-and-haul contests

One excavator + one truck per two-person team (or solo iron-man with transmitter hand-offs). Move **20 kg net through the hopper** from a marked face; the clock stops at the twentieth weighed kilogram. Spillage is not swept — it is simply tonnage you no longer have. Course-record pace from pilot testing: ~11 minutes (versus ~26–31 kg per four-person pod per casual Shift — the skill headroom is the sport). Variants: uphill-only routing, loader class (1583 direct to hopper), and Phase 3 night format under work lights.

11.4 ‘Site foreman’ team scenarios — corporate events

The premium B2B product (from **\$1,400 / 2 h / 20 pax**, canon): the group becomes a contractor. A brief is issued — deliver 60 kg to the hopper, cut and backfill one trench, zero safety flags — and roles are assigned: operators, logistics planner, and a **site foreman** hold-

ing the marshal tablet's read-only dashboard. The debrief hands every participant their personal telemetry card. Add-ons: K970 demo with the Artisan (\$150), catered boardroom Toolbox Talk, branded hi-vis. The format sells because it is a genuine operations-management exercise wearing a toy's clothes — the 1:3 queueing logic of Chapter 5 becomes the customer's problem for two hours, and mixed-seniority groups reliably discover their best operator is the intern.

11.5 Skill-badge tasks (feeding Volume 9)

The division contributes a defined badge set to the park-wide RC WORLD License ladder; each badge is a machine-verified task, not a marshal's opinion:

Badge	Task	Verification
First Furrow	Plow one field pass within 100 mm of the chalk line	Marshal photo overlay
Clean Loader	10 truckloads with < 10% spillage	Hopper weight vs bucket-pass count
Ramp Master	100 loaded ramp climbs without a retrieval	Heavy-Node geofence + retrieval log
Steady Hands	Golf-Ball Pickup: 6/6 in under 5 min	Event scoring
Tow-Truck Hero	Complete one supervised retrieval	Protocol log
Harvest Hand	50 kg delivered in one Harvest Campaign	Weighbridge ledger
Master Operator	K970 assessment session passed	Artisan sign-off (tier 3 license)

License gating (division view): tier 1 (post-Toolbox-Talk) unlocks RD-1 machines; tier 2 (tonnage + the Clean Loader practical) unlocks the 1583, tandem trailers and league entry; tier 3 plus a booked assessment unlocks premium K970 sessions. Full tier/badge matrix: Volume 9, Chapter 3.

League structure mirrors the park ladder: **Open evenings** (casual, monthly) → **Division League** (8-week season, tier 2; teams of four running 45-minute campaign sessions, aggregate-tonnage table, playoff weekend; \$120/team/season against ~\$60 of Shift value consumed — margin-accretive by design) → **RC WORLD Championship** (annual, invitational), to which the division contributes three disciplines: Precision, Load-and-Haul, and the Team Campaign final.

12. Maintenance & Reliability

12.1 Division doctrine

One sentence governs everything: **electromechanical machines are maintained by scheduled replacement of cheap discrete parts; the hydraulic machine is maintained by condition monitoring and cleanliness (Chapter 8.6)**. All work logs to `maintenance_logs` (vehicle, mechanic, parts_used JSONB, description, date_resolved); predictive flags fire from `fleet_inventory.total_hours_run`, which increments on live PWM activity only — a machine that sat staged all day accrues nothing. Park-wide workshop standards, tooling and training live in Volume 7; this chapter is the division-specific layer.

12.2 Daily / weekly / monthly PM matrix

Task	1580 excavator	1582/1573 truck	1583 loader	1554/69 dozer	E351 tractor
Visual + function test, connector check	Daily	Daily	Daily	Daily	Daily
Blow-down (in The Works), track/tire debris pick	Daily	Daily	Daily	Daily	Daily
Track tension (SOP-CD-001)	Weekly	—	—	Weekly	—
Steering/articulation linkage & servo check	—	Weekly	Weekly	—	Weekly
Pivot-bushing slop gauge (< 2 mm at teeth/blade)	Weekly	—	Weekly	Weekly	—
Lead-screw lubrication (SOP-CD-005)	Monthly / 40 h	—	Monthly (lift arms)	Monthly	—
Micro-switch test bank (all limits)	Monthly	Monthly (tip limit)	Monthly	Monthly	—
Gearbox open-inspect, gear wear	Quarterly / 120 h	Quarterly	Quarterly	Quarterly	Quarterly
Motor brush / current-draw baseline	Quarterly	Quarterly	Quarterly	Quarterly	Quarterly
Slip-ring clean (continuous slew)	Annual	—	—	—	—
Full refurbishment pass (\$12.6)	150 h or defect	200 h	150 h	100 h	200 h

K970 follows the Chapter 8.6 schedule; battery pool audits run weekly by pack QR rotation, fleet-wide.

12.3 Sand ingress management

Sand is the division's business model and its universal solvent of machinery. The ingress-control stack, cheapest layer first:

1. **Material spec** — washed <3% fines aggregate and 4–6% face moisture (Chapter 6.6) prevent most airborne fines at source.
2. **Grease discipline** — printed in red on every bench card: nothing gets grease that touches sand directly. Exposed pivots get oil-pen film only; grease on an open pivot is grinding paste by lunchtime. Grease belongs inside screw tubes, gearboxes and sealed ports.

3. **Dry blow-down only, in The Works** (≤ 2 bar, outdoors) — never wet-wash an electromechanical machine; water carries fines into screw tubes and switch bodies.
4. **Bearing quarantine** — any wheel or idler bearing with grit-crunch is binned, never cleaned and re-oiled; a \$0.60 bearing is not worth a \$60 axle.
5. **Connector hygiene** — XT60s and JST XH latches wiped at every swap; the refurbishment electrical pass pull-tests every connector.

12.4 Spares stocking and reorder points

The division's shelf in The Works, sized so no single failure becomes downtime (par levels for the 42-machine fleet; suppliers and pricing in Volume 8, with 4-8 week China-direct lead times already priced into the reorder points):

Part	Par stock	Reorder at	Turnover
Micro-switches (KW11-class limit switches)	200	80	Medium-high — the #1 heavy-fleet consumable
Excavator track pins + pads	300 / 40	120 / 15	Continuous (canon consumable)
Gear sets, nylon + metal, per actuator model	12 each type	5	High — nylon stocked for its sacrificial role
Lead screw + nut pairs (1580 boom/stick/bucket)	6 each	2	Medium
Brushed 380/540/550 motors	15	6	Medium
Steering servos 15/25 kg digital	10	4	High (trucks + tractors)
Brushed ESCs, fleet standard	6	2	Medium
XT60 pairs / JST XH housings + pins	100 / 200	40 / 80	Continuous
2S LiPo packs (pool attrition ~15%/yr)	12	4	Continuous
Lithium grease (NLGI 2), Loctite blue, IPA	4 / 6 / 4 units	1 / 2 / 1	Continuous
Hydraulic seal kits (K970 full set)	2 sets	1	Low — long lead, never zero
ISO VG 32 oil (dyed)	4 L	1 L	Low
Tractor hitch pins / trailer axles	30 / 6	10 / 2	Medium (party-day attrition)

RC WORLD OS decrements stock from `parts_used` logging and fires reorder alerts automatically at the levels above.

12.5 Key procedures

SOP-CD-001 — Track Tension Check & Adjustment (tracked fleet) SOP ID: SOP-CD-001 · **Rev:** 1.0 · **Owner:** Heavy Fleet Artisan Lead · **PPE:** shop standard · **Tools:** 3 mm

steel rule, hex drivers, lithium grease, torque driver (0.6 N·m) · **Frequency:** weekly per tracked machine, or after any thrown track

1. Bench the machine, power off, bucket/blade grounded.
2. Brush the track runs; pick embedded stones from between pads (the #1 cause of false tension readings and pad wear).
3. Measure mid-span sag under one-finger pressure (~10 N): spec **3-5 mm** on 1580/1583-class, 2-4 mm on 1/16 dozers.
4. Adjust the idler-yoke tension screw a quarter-turn at a time. Never tension drum-tight — over-tension raises drive current 15-20% and eats sprocket bushings; the quarterly current baseline will betray you.
5. Check pin staking both runs; re-stake proud pins; bin any pin that walks twice.
6. Torque idler lock screws 0.6 N·m; one grease shot to the idler axle port.
7. Run a 60 s figure-eight in the workshop dig box; confirm no walk-off in tight turns; log sag before/after.

SOP-CD-005 — Lead-Screw Lubrication SOP ID: SOP-CD-005 · **Rev:** 1.0 · **Owner:** Heavy Fleet Artisan Lead · **PPE:** nitrile gloves · **Tools:** lithium grease (NLGI 2), brush, lint-free wipes, IPA, bench PSU · **Frequency:** monthly or 40 run-hours per actuator

1. Open the actuator access cover (boom, stick, bucket in turn on the 1580; lift arms on the 1583; blade screw on dozers).
2. Run to full extension on bench power; wipe the exposed screw with IPA and read the wipe: gray paste = normal wear; glitter = nut shedding — replace the screw/nut pair now, not mid-season.
3. Brush a rice-grain bead of grease along the full thread. Do not pack the tube — excess grease is a dust magnet; thin film, full coverage.
4. Cycle full travel three times; verify both limit switches click and cut at the ends.
5. Wipe squeeze-out, close, log.

SOP-CD-002 — Boom Micro-Switch Replacement SOP ID: SOP-CD-002 · **Rev:** 1.0 · **Owner:** Heavy Fleet Artisan Lead · **PPE:** safety glasses · **Tools:** soldering iron, solder, heat-shrink, flush cutters, multimeter, replacement KW11-class switch · **Frequency:** on defect or monthly test-bank failure

1. **Diagnose first.** Actuate the suspect switch by hand under a continuity meter — confirm fail-open (no click-through) or fail-closed (welded contacts) before touching solder. If the switch tests good, the fault is a misadjusted crush-stop; fix that instead.
2. Photograph the wire routing; note lever type (plain/roller) and throw direction — Huina uses both orientations in one machine.
3. Desolder or cut back to bright copper; strip 3 mm; slide heat-shrink on now (the step everyone forgets).
4. Solder the new switch NO/NC/COM exactly per photo; shrink the joints.
5. Set the trip point: the switch must click **1-2 mm before** the hard mechanical stop — that margin is the entire purpose of the switch. Verify by hand-cranking the screw to the stop.
6. Bench-run five full cycles; clean cut-off both ends, no stall audible.
7. Log the switch position (“boom, upper limit”) in parts_used — position-level data keeps the FMEA below honest.

Gearbox servicing (quarterly open-inspect): photograph gear train before touching; check nylon gears for tooth-root cracks under magnification (they crack before they strip); re-grease metal meshes sparingly; verify output-shaft end-float < 0.5 mm; replace any gear showing polished wear-through of the molding line. A stripped-gear symptom in the field — motor audibly spinning, function dead — is a 20-minute gear-set swap from shelf stock.

12.6 The 45-minute refurbishment turnaround

Every rental machine cycles through The Works on PM schedule or defect. Staged so one Artisan finishes inside 45 minutes from ghost-machine bins: intake scan and photo → dry blow-down → bench function test at 7.4 V (a healthy lead screw hums, a dry one squeals, a worn nut knocks) → open only the failed/scheduled actuator (every opening is a dust opportunity) → replace the discrete part → re-grease → pivot/track/torque pass → electrical pass (XT60 pull test, chafe scan, JST XH latches, conformal-coating visual on the Heavy-Node) → 5-minute load test in the workshop dig box → log and release to active.

Trade Hack. Keep each actuator's screws in their own magnetic tray, photographed before disassembly. The 1580 uses four visually similar screw lengths, and a long screw in a shallow boss cracks the casting — the single most common self-inflicted workshop injury to these machines.

12.7 FMEA — excavators and dump trucks

Scored 1–5 (severity × occurrence × detection difficulty = RPN; act on RPN ≥ 27):

Machine	Failure mode	Effect	S	O	D	RPN	Mitigation
1580	Limit switch fail-closed (welded)	Gearbox stall, possible gear strip mid-Shift	3	4	3	36	Monthly test bank; ESC current fold-back; SOP-CD-002
1580	Slip-ring oxidation	Intermittent slew/electrics, hard to reproduce	3	2	4	24	Annual clean; symptom triage card in The Works
1580	Lead-screw nut wear	Boom droop under load, slop	2	3	2	12	SOP-CD-005 wipe-read; nut pairs on shelf
1580	Nylon gear strip (shock load)	Function dead, Shift refund	3	3	1	9	Site geometry caps drop energy; 20-min gear swap
1580	Thrown track in pivot turn	Machine immobile → retrieval event	2	3	1	6	SOP-CD-001 tension spec; retrieval is gameplay

Machine	Failure mode	Effect	S	O	D	RPN	Mitigation
1580	Pack over-discharge (hidden by customer)	LiPo damage; storage fire risk	4	2	2	16	Node kill at 3.4 V/cell (canon); in-bound bench check
1582/73	Steering servo failure	Truck uncontrollable → circuit blockage	3	4	2	24	Weekly linkage check; servo par stock; marshal pulls to shoulder
1582/73	XT60/wiring chafe at bed pivot	Intermittent power; phantom “dead battery”	2	3	4	24	Refurb electrical pass targets this loom explicitly
1582/73	Wheel bearing seizure (fines)	Dragging wheel, motor heat	2	3	3	18	Daily blow-down; bearing quarantine rule
1582/73	Tip-linkage bent (overload + slam)	Bed won't seat; spillage all lap	2	3	2	12	Bucket-pass-count load discipline; linkage on shelf
1582/73	Drive gearbox strip (berm jump)	Truck dead on ramp	3	3	1	9	Berm geometry caps launch energy; gear sets on shelf

The two highest-RPN items — welded limit switches and steering servos — are exactly the two parts stocked deepest in Section 12.4. That is not a coincidence; it is the FMEA doing its job. Escalation, tooling standards, technician training and the cross-park failure database live in **Volume 7** (see especially its workshop layout and rebuild-training chapters), which treats this chapter's SOPs as division-local instances of the park-wide system.

Volume summary & cross-references

The Construction Division is RC WORLD's signature: a working 1/14 open-pit mine and a 1/16 commercial farm, run as a genuine production system. Its engineering constitution is the two-tier fleet — 29 rugged, lead-screw Huina earthmovers and 12 Double E tractors earning volume fees at \$15–26 per 20-minute Shift, plus one Kabolite K970-class hydraulic flagship earning supervised premium fees at \$22–38 — bound together by the 1:3 excavator-to-truck ratio (derived, Chapter 5.2), 15°-max haul roads built at 14°, a closed material-recir-

culatation loop metered through a load-cell hopper, and a maintenance program built on discrete-part replacement, lithium grease, and micro-switches stocked by the hundred. Division fleet capex is ~\$16k wholesale; the civil works and hopper are the real build cost; Year 1 division revenue target is ~\$410-470k inside the park's \$1.28 M canon, with roughly 8× demand headroom on the existing fleet.

Scope note (living document): Volume 3 references a crawler-program “adventure annex” under this volume (Track C, the 16-machine crawler park). That annex is deferred to Revision 1.1 of this volume; until then the crawler fleet is governed by the canon fleet table (Volume 1) and the Track C specification in Volume 3.

Cross-references: park-wide fleet, phasing and capex canon — **Volume 1**. Competitive landscape and Chinese diorama/miniature-equipment suppliers — **Volume 2**. The Motorsport parity doctrine this division's competition rules borrow — **Volume 3**. The Works: fabrication drawings (hopper, grading blade, pin hitch), RCW Heavy-Node hardware, rebuild training and the hydraulic technician curriculum — **Volume 7** (which should honour this volume's SOP numbering CD-001...CD-005). Suppliers, wholesale channels, spares pricing and the 4-8 week reorder mechanics — **Volume 8** (LESU and JDModel carried as second-source hydraulic suppliers; Siku Control as retail stock, not fleet). Licenses, badges, Gears, Toolbox Talk and the tier matrix this division's gates plug into — **Volume 9, Chapter 3**. Division financials and the utilization upside case — **Volume 10**. Site master plan, drainage, charging bunker and fencing standards — **Volume 11**. Franchising the Mining Zone as the flagship differentiator — **Volume 12**.

Volume 5 — Aviation Division

Document	RC WORLD — Master Development Plan, Volume 5 of 13
Title	Aviation Division
Revision	1.0 — July 2026
Status	Planning baseline — Phase 2 build (Months 13–30)

Purpose of this volume. This volume is the complete concept, fleet, operations, training and safety plan for the RC WORLD Aviation Division: the fully netted airfield that opens in Phase 2 (Months 13–30) with a 30 m paved runway, elevated helipads, and a fleet of approximately 20 aircraft spanning fixed-wing trainers, electric ducted-fan (EDF) jets, helicopters, and FPV multirotors. It explains why the full net enclosure is the single decision that makes the division insurable, licensable and operable by walk-in customers; specifies the flight programs discipline by discipline; defines the flight school, the simulator lab and the RC WORLD Wings certification ladder; sets the airspace-management doctrine (one airspace, one controller); and closes with the fleet table, crash-repair economics and capacity mathematics that drive the division’s P&L.

Intended readers. Investors and lenders assessing Phase 2 (Chapters 1, 10); the architect and structural engineer detailing the enclosure with Volume 11 (Chapters 1, 8); the General Manager, Aviation Lead and duty managers (all chapters); flight instructors and Academy staff (Chapters 7, 9); Artisans assigned to flight-line and workshop duty (Chapters 2–6, 8, 10); the safety officer, insurers and local regulators (Chapters 1, 8).

Chapters

- 1. Division Overview
- 2. Fixed Wing Program
- 3. Jets
- 4. FPV Program
- 5. Helicopters
- 6. Drone Racing
- 7. Flight School & Simulators
- 8. Safety & Airspace Management
- 9. Pilot Certification
- 10. Fleet, Maintenance & Economics
- 11. Volume Summary & Cross-References

1. Division Overview

1.1 The netted airfield concept

Every RC WORLD division before this one operates vehicles bound to the ground. A runaway buggy is stopped by a barrier; a mis-driven excavator simply stalls against a berm. Failure is two-dimensional and contained by default. Aviation removes that default. A model aircraft that departs controlled flight can leave the property, descend onto a spectator, or — in the worst historical pattern of the hobby — “fly away” beyond radio range and land where nobody intended. The Aviation Division’s founding decision, recorded in the original Omni-Zone blueprint and carried unchanged into this plan, is therefore **total containment**: the entire airfield — 30 m paved runway, taxiways, elevated helipads, pilot boxes and racing course — is enclosed in high-tensile netting on all sides and overhead, forming a single continuous flight cage. Nothing that flies at RC WORLD can physically exit the enclosure.

The reference enclosure (final structural design in Volume 11, Chapter 5) is a **60 m × 40 m footprint with a 15 m clear flight ceiling** under a crowned roof net carried on 16–17 m galvanised steel masts with tensioned perimeter cables — the same engineering family as golf driving-range and baseball-field barrier netting, a mature industry with established suppliers, load tables and inspection norms. The 30 m runway runs along the long axis with 15 m of paved overrun and grass margins inside the net; two elevated helipads (1.0 m platforms) sit on the eastern edge; the drone-racing gate course occupies the western half and is reconfigurable; a smaller 13 mm fine-mesh **Whoop Arcade** sub-enclosure sits beside the main cage for micro-drone flying that does not consume main-airspace capacity (Chapter 6). Spectators view through the net from a standoff rail and from the Phase 2 grandstand sightline (Volume 11).

The cage is not merely a safety device — it is the product. Full-size aviation is defined by the discipline the sky imposes; the RC WORLD cage lets us impose that discipline deliberately, in a space small enough that a first-time visitor can be flying a stabilised trainer within one visit, and structured enough that a hobbyist can earn jet and FPV type ratings over a season.

1.2 Why full enclosure changes the regulatory and safety posture

Uncontained outdoor model flying sits squarely inside national airspace regulation. In the United States the default regime for sub-25 kg (55 lb) uncrewed aircraft is **FAA Part 107**, with a statutory recreational exception (49 USC 44809) that requires flyers to pass the free **TRUST** online test, follow the safety code of a recognised community-based organisation such as the **Academy of Model Aeronautics (AMA)**, and register aircraft over 250 g. In Europe, the **EASA “open” category** (subcategories A1–A3) governs sub-25 kg operations up to 120 m with operator registration and online competency tests. Comparable regimes exist in most jurisdictions (UK CAA, Australia CASA/MAAA, etc.).

A fully enclosed structure changes the analysis fundamentally. The FAA has stated in published guidance that its rules apply to operations in the **National Airspace System** — operations conducted indoors are outside Part 107’s scope, and a fully netted enclosure is widely treated in the same way, because the aircraft cannot enter navigable airspace. EASA’s UAS framework similarly excludes indoor operations. **RC WORLD’s planning position is therefore that flights inside the completed cage are enclosed-structure operations,**

not open-airspace operations — which is what makes it feasible to let unlicensed walk-in customers fly at all.

Safety Warning. The regimes above are cited as reference frameworks only. Whether a netted outdoor enclosure is treated as “indoors,” whether staff instruction constitutes commercial operation, and whether the masts themselves need aviation obstruction review (they will exceed 15 m, and proximity to any aerodrome matters) are jurisdiction-specific questions. **Written confirmation from the local civil aviation authority and from the insurer must be obtained before enclosure design is frozen — this is a Month 13 gating task, not an opening-week formality.** Nothing in this volume is legal advice; verify locally.

The enclosure also transforms the insurance conversation. An uncontained model-flying operation open to the public is close to uninsurable at reasonable premium; a certified structure with documented netting inspection (Chapter 8), instructor supervision, a written airspace doctrine and telemetry-logged sessions is a conventional leisure-facility risk. Finally, the cage collapses the two failure modes that dominate RC aviation incident history — fly-away and third-party overflight injury — to zero and near-zero respectively, leaving on-site energy management (Chapter 8) as the residual risk to engineer against.

1.3 Phased opening inside Phase 2

The Aviation Division is deliberately sequenced into **Phase 2 (Months 13-30, “Full Park”)**. Phase 1 establishes the park’s operating culture — Toolbox Talk induction, Shift billing, Artisan maintenance discipline, RC WORLD OS telemetry — on ground vehicles where mistakes are cheap. Aviation inherits that culture rather than having to invent it. Within Phase 2 the division itself opens in staged tranches:

Stage	Months	Milestone
A — Regulatory & design	13-15	Authority/insurer confirmations; enclosure structural design frozen (Volume 11); netting supplier contracted (Volume 8)
B — Simulator lab opens	16	Flight School sim lab (Chapter 7) opens in the entry pavilion before the cage exists — building a certified-pilot pipeline and early revenue
C — Civil works & enclosure	15-19	Runway, aprons, helipads poured; masts and netting erected; commissioning inspection
D — Fixed-wing opening	19-20	Trainer program and buddy-box instruction open; Wings certification live in RC WORLD OS
E — Rotorcraft & FPV	21-22	Helipad operations, cinewhoop experiences, Whoop Arcade
F — Jets & racing league	23-24	EDF demo program; first drone-racing league season

Investor Note. Aviation is allocated $\approx$ **\$364 k of the \$1.15 M Phase 2 capex** (Chapter 10.6): roughly \$190 k enclosure, \$85 k civil works, \$45 k simulator lab, \$32 k fleet/radios/FPV/spares, \$12 k timing and AV. The expensive assets (cage, pavement, sim lab) last a decade or more; the aircraft customers actually crash are foam, cheap, and stocked with spare airframes. The division is the park's premium-price, lower-throughput, highest-halo product: it will not match Motorsport's Shift volume, but it commands premium-tier pricing (\$22/\$38), sells instruction at instructor margins, and gives the park its most photogenic marketing asset after the Mining Zone.

1.4 Division at a glance

Parameter	Value
Opens	Phase 2, staged Months 16–24 (sim lab first)
Enclosure	60 × 40 m footprint, 15 m clear flight ceiling (16–17 m masts), high-tensile knotless netting, fully roofed
Runway	30 m paved, with 15 m overrun; two elevated helipads
Fleet	~20 powered aircraft: 8 fixed-wing, 2 EDF jets, 4 helicopters, 7 FPV multirotors (Chapter 10)
Disciplines	Fixed wing, jets, helicopters, FPV freestyle/cinewhoop, drone racing
Pricing	Premium tier per canon: Casual Shift \$22; Operator Shift \$38; instruction and checkrides priced in Chapter 10.4
Airspace doctrine	One airspace, one controller (Chapter 8.2)
Certification	RC WORLD Wings ladder, integrated with the RC WORLD License (Chapter 9; Volume 9)

2. Fixed Wing Program

2.1 Trainer doctrine: stabilised, high-wing, foam

The fixed-wing program is the division's backbone: it is where customers learn, where instructors earn, and where the Wings ladder (Chapter 9) begins. The fleet doctrine mirrors the park-wide philosophy — standardised, repairable platforms differentiated by configuration rather than exotic hardware — applied to the three properties that matter most in a rental trainer:

1. **High-wing layout.** Pendulum stability: the aircraft wants to return to level flight. Every primary trainer in the fleet is high-wing.
2. **EPO/EPP foam construction.** Crashes end in glue, not write-offs. A foam trainer's fuselage survives impacts that would total a balsa airframe, and a broken one is repaired at the bench in minutes (Chapter 10.3).
3. **SAFE-class electronic stabilisation.** Horizon Hobby's SAFE (Sensor Assisted Flight Envelope) system — and the equivalent gyro/envelope systems on Chinese airframes such as FMS's Reflex and the XPilot-class stabilisers used across the Volantex range — provides three switchable modes: a beginner mode that limits bank/pitch angles and self-levels when sticks are released ("panic recovery"), an intermediate mode with wider envelopes, and an experienced mode with no assistance. This single feature is what makes a 20-

minute walk-in trainer Shift viable: the instructor can hand a total novice the sticks in beginner mode knowing the aircraft cannot be rolled inverted into the ground.

2.2 Trainer fleet

The trainer fleet pairs the Western reference platforms (HobbyZone/E-flite, whose SAFE implementation and spares network are the industry benchmark) with Chinese-sourced equivalents that cut airframe cost roughly in half — the same dual-sourcing logic used across Volumes 3 and 4 and executed in Volume 8.

Class	Model (reference)	Wingspan	Power	Street price (July 2026, approx.)	Role
Primary trainer	HobbyZone AeroScout S 2 1.1 m (SAFE, pusher prop)	1,100 mm	3S, ~1300-2200 mAh	\$200-270 RTF	First-flight workhorse; pusher layout protects the propeller in nose-over landings
Club trainer	E-flite Apprentice STS 1.5 m (SAFE)	1,500 mm	3S 3200 mAh	~\$360 RTF	Larger, smoother, more wind-tolerant; the buddy-box instruction ship
Value trainer	FMS Ranger 1220 mm V2 (Reflex V3 gyro)	1,220 mm	3S 2200 mAh	~\$180 PNP	Chinese-sourced equivalent; spares stocked identically
Sport/FPV platform	Volantex Ranger 1600 (pusher glider)	1,600 mm	3S 2200 mAh	~\$95-165 PNP	Slow, floaty camera/sport platform; blow-moulded fuselage shrugs off abuse

All PNP (plug-and-play) airframes are completed in The Works with standard receivers bound to fleet transmitters, an **RCW Micro-Node** on the balance lead for voltage telemetry and session logging (GPS geofencing is unnecessary inside the cage; the node's kill function is retained as a throttle-limit intercept), and XT60 battery leads per park battery doctrine.

2.3 Buddy-box instruction

All customer trainer flights below solo certification are flown on a **buddy-box (trainer link)**: two transmitters linked so the instructor's master radio can reclaim control instantly. The fleet standardises on wireless trainer linking (Spektrum's wireless trainer function on the master radio binding directly to the student's transmitter; on the OpenTX/EdgeTX side, RadioMaster radios support equivalent wired and wireless trainer modes). Operating rules:

- The **instructor's radio is master**; the student flies only while the instructor holds the trainer switch. Releasing the switch returns control in under half a second — faster than any verbal "I have control" exchange.
- Instructor:student ratio is **1:1 in the air**. One instructor may run a rotation of up to 3 students per hour (one aloft, one briefing, one debriefing).

- Beginner mode stays engaged until the Wings syllabus (Chapter 9.3) prescribes otherwise.
- Every buddy-box Shift is logged in RC WORLD OS against the student's Wings record — sim hours, buddy-box minutes and solo flights accumulate automatically toward certification.

2.4 Runway operations

Thirty metres of pavement is short by full-scale instinct and generous by foam-trainer reality: an AeroScout lifts off in under 10 m into any headwind and lands in 15 m with a normal approach. Standard pattern inside the cage:

- **Left-hand circuit**, 8–20 m altitude band for trainers, oriented into the prevailing wind; the circuit is flown inside the net with a 5 m lateral buffer from the netting maintained at all times (the “soft wall” rule — the net will stop an aircraft, but a net strike is logged as an incident, Chapter 8.6).
- Take-offs and landings are called to the airspace controller (“RC Tower”) by the instructor: “Trainer 3, departing runway 27.” One aircraft on the runway at a time; helis and FPV yield per the airspace matrix (Chapter 8.2).
- Nose-wheel steering taxi to and from the apron; no hand launches or hand catches by customers, ever.

Pre-flight checklist — fixed wing (laminated at each pilot box, mirrored in the app):

☐

Airframe: wings latched, control surfaces attached, hinges sound, no unrepaired cracks

☐

Linkages: clevises secure, no slop at horns

☐

Propeller: no chips or cracks; spinner tight (pusher prop guard intact on AeroScout)

☐

Battery: charged pack (storage-charge packs are marked), connector fully seated, strap tight, hatch latched

☐

Radio: correct model selected; range check at reduced power on first flight of the day

☐

Controls: surfaces move correct direction (elevator up = stick back); stabiliser mode confirmed

☐

Failsafe: transmitter off → motor stops within 1 s (first flight of day; Chapter 8.7)

☐

Trainer link: master override verified before student takes sticks

☐

Tower: session slot confirmed with airspace controller

2.5 Aircraft classes: trainer to sport

The fixed-wing ladder gives certified pilots something to climb toward, mirroring Motorsport's class ladder (Volume 3):

Class	Aircraft	Access requirement (Chapter 9)
FW-T (Trainer)	AeroScout S 2, FMS Ranger, Apprentice STS in beginner/intermediate mode	Any customer, buddy-box; solo with Bronze Wings
FW-S (Sport)	Apprentice STS experienced mode; Volantex Ranger 1600; four-channel sport handling, basic aerobatics (loops, rolls, stall turns)	Silver Wings
FW-J (Jet)	EDF jets (Chapter 3)	Gold Wings + Jet type rating, instructor-supervised

Field Note. Resist the temptation to add a low-wing warbird class in Year 1. Warbirds are the most-requested and most-crashed airframes in club flying — high wing loading, tip-stall behaviour on landing, and cosmetic damage that customers notice. The sport tier on the existing high-wing fleet delivers 90% of the progression satisfaction at a fraction of the repair load. Revisit at the Phase 3 fleet review.

3. Jets

3.1 EDF jets as the spectacle tier

Electric ducted-fan (EDF) jets are the division's spectacle product: a 70 mm-class EDF at full throttle produces a genuine turbine-like howl and 130+ km/h flypasts that stop foot traffic across the park. They are deliberately positioned as an **advanced and demonstration tier, not a rental tier**. The reference fleet is two airframes:

Model	Fan	Power	Street price (approx.)	Notes
E-flite Habu SS 70 mm	70 mm 10-blade	4S-6S 3200-4000 mAh	~\$280-300 BNF/PNP	The benchmark "easy" sport EDF: docile low-speed handling, SAFE Select envelope, tri-cycle gear
FMS or Freewing 70 mm scale jet (e.g. F-16 class)	70 mm 12-blade	6S 3300-4000 mAh	~\$250-350 PNP	Scale demo star; less forgiving at low speed — instructor-flown in demos, checkride ship for the type rating

Both are EPO foam and repairable at the bench, but a jet crash is an order of magnitude costlier than a trainer crash: fan units, 6S packs and retracts concentrate value. Spare-airframe policy in Chapter 10.3.

3.2 Energy management — the jet discipline

Jets are hard for one reason: **energy**. A trainer flies at 8 m/s and stops flying gently; a 70 mm EDF approaches at 20+ m/s, carries three times the kinetic energy per kilogram, and — critically — has no propwash over the tail, so control authority collapses with airspeed. In-

side a 60 m cage this compresses further: the pilot has perhaps three seconds of straight line before the far net. Jet operations therefore run under their own rules:

- **Racetrack pattern only** — continuous oval at the 12–18 m band, no hovering discipline sharing the airspace, no student traffic aloft (jets get exclusive airspace slots, Chapter 8.2).
- **Throttle is a planning tool, not a reflex.** The syllabus teaches energy budgeting: power set on downwind, speed bled progressively, approach flown on a stabilised glide-path with power carried to the flare. Chopping throttle on final in an EDF is the canonical beginner crash.
- **Flight duration 3–4 minutes** against a timer; 6S packs land at ≥ 3.5 V/cell per the park LiPo window (canon 3.4–4.2 V/cell, Chapter 8.4).
- **Wind limits tighter than trainers** (Chapter 8.5): crosswind on a 30 m runway with a fast, short-coupled airframe is an instructor-only condition.

3.3 Who may fly jets

Jet access requires **Gold Wings plus the Jet type rating** (Chapter 9.3), and every jet flight — including those by rated customers — is **instructor-supervised**: the instructor stands at the pilot box with a linked master radio for the customer's first three rated flights, and remains physically present as spotter/controller-liaison thereafter. There is no casual jet rental. This is the same premium-supervised logic the park applies to the Kabolite hydraulic excavator (Volume 4): the machine is the marketing; the supervision is the margin protection.

3.4 Demo-day programming

Because most visitors will never fly a jet, the jets earn their keep on the schedule:

Slot	Content	Duration
Daily 12:00 & 16:00	"Jet Blast" flypast demo — instructor-flown scale jet, commentary over PA, 2 flights	10 min
Weekend 14:00	Type-rating showcase: rated customer flights, Habu SS, announced by name (aspirational marketing for the Wings ladder)	20 min
Monthly	Twilight demo on events lawn sight-line, paired with Motorsport night-race events (Phase 3 lighting permitting)	30 min

Safety Warning. EDF intakes are the division's only exposed-rotor hazard that cannot be caged: a 70 mm fan at partial throttle will ingest a finger. Jets are armed **only on the runway centreline**, never on the apron; the arming procedure in SOP AV-03 applies to every jet movement, and customers never carry a live jet.

SOP AV-03 — EDF jet arming & movement (ID: AV-03 · Rev 1.0 · Owner: CFI · PPE: none beyond standard · Tools: throttle-lock tag, battery caddy · Frequency: every jet flight):

1. Jet is carried to the centreline **without a battery installed**; battery travels separately in the caddy.

2. Pilot and instructor confirm transmitter on, correct model, throttle stick low, **throttle-cut/lock engaged**, rates and failsafe verified per the fixed-wing checklist.
3. Battery installed and hatch latched at the centreline; aircraft pointed down-runway, intake and exhaust clear of all persons by ≥ 3 m.
4. Instructor calls “JET LIVE” to the tower; tower confirms exclusive slot active and airspace otherwise cold.
5. Throttle lock released only when both pilot and instructor are behind the wing line. Any interruption (spectator question, radio anomaly, weather call) → throttle lock re-engaged before anything else is discussed.
6. After landing: throttle lock on, “JET COLD” call, battery out at the centreline before the aircraft is carried anywhere. Battery voltage logged; pack to the flight-line cache or quarantine (Chapter 8.4) as condition dictates.

4. FPV Program

4.1 The FPV product family

First-person-view flight — piloting through goggles fed by an onboard camera — is the fastest-growing segment of RC aviation and the division’s strongest draw for the 16–35 demographic. The program runs three distinct products:

1. **Cinewhoop experiences (rental core).** Ducted 2–2.5” quadcopters in the BetaFPV Pavo20-class (~\$300–460 depending on video system) flown smoothly through a decorated course. Prop ducts make them near-harmless on contact; flight character is slow and cinematic. This is the walk-in FPV product: after a sim introduction (Chapter 7), a customer flies a real cinewhoop on their first visit with an instructor on a second set of goggles.
2. **Freestyle progression (certified tier).** 3.5”–5” freestyle quads for customers holding the FPV type rating (Chapter 9.3), flown in dedicated airspace slots. Acro-mode flight — no self-levelling — is the discipline’s summit and is gated accordingly.
3. **Tiny Whoop / micro (Whoop Arcade).** 65–75 mm brushless 1S whoops (Meteor65-class) flown in the fine-mesh side enclosure. Because the Arcade is its own sealed airspace, it runs in parallel with main-cage operations and is the division’s highest-throughput product (Chapter 6.4).

Video standard: the park standardises on **digital HD FPV (DJI O4-class air units)** for cine-whoops and analog 5.8 GHz for racing (analog remains the racing norm for latency, cost and multi-pilot channel management). Spectator screens at the FPV pit repeat the pilot’s feed — the single best conversion tool the division has, because bystanders see exactly what the pilot sees.

4.2 Goggles hygiene

Goggles are shared headwear, and the park treats them with barbershop discipline:

- Disposable single-use **face-foam liners** (fleet-standard faceplate pads) issued per customer; used liners binned in sight of the customer.
- Lens and shell wiped with alcohol-free optical disinfectant between users; UV-C sanitising cabinet at the FPV pit runs a 5-minute cycle per returned set.

- Focus/diopter adjustment briefed by staff; customers with glasses get the box-style goggle option (DJI Goggles N3-class, ~\$270, accepts spectacles).
- Goggle sets are fleet assets with asset tags in RC WORLD OS; foam-liner stock is a standing consumable in Volume 8's procurement schedule.

4.3 Video frequency management — the 5.8 GHz channel plan

Control links (2.4 GHz frequency-hopping) coexist essentially without management, but **5.8 GHz analog video does not**: two pilots on the same or adjacent video channels will wipe out each other's picture, and a pilot who powers up on the wrong channel mid-race blinds everyone. The park therefore runs a fixed channel plan, administered by the airspace controller, using **RaceBand** — the eight-channel band with 37 MHz spacing designed for multi-pilot racing:

Slot	Band/Channel	Frequency	Standing assignment
1	R1	5,658 MHz	Race heat slot 1
2	R2	5,695 MHz	Race heat slot 2
3	R3	5,732 MHz	Race heat slot 3 / freestyle A
4	R4	5,769 MHz	Race heat slot 4 / freestyle B
5	R6	5,843 MHz	Race heat slot 5
6	R7	5,880 MHz	Race heat slot 6 / instruct- or cinewhoop
7	R8	5,917 MHz	Race heat slot 7 (verify legal in-country)
8	Digital (O4-class, auto- managed)	5.8 GHz DFS	Cinewhoop rentals — di- gital system self-coordin- ates

Rules of the plan: maximum **7 simultaneous analog pilots** (R5 at 5,806 MHz is skipped as an interference guard for the digital system, and top-of-band channels must be verified against local spectrum law — several RaceBand channels are not legal in all jurisdictions; verify locally and amend the plan); **25 mW transmit power** standard (the MultiGP norm — sufficient inside a 60 m cage and the key to clean multi-pilot video); no video transmitter is powered up anywhere on site except at the FPV pit with a channel token physically in hand. Channel tokens are numbered tags hanging at the controller's desk — an old club trick that makes spectrum occupancy visible at a glance.

Trade Hack. Smoke-stop every analog channel dispute with a \$60 handheld 5.8 GHz spectrum scanner at the controller desk. When a pilot swears they are on R3 and the screen shows energy on R4, the argument is over in five seconds. Log recurring offenders' equipment — a mis-programmed VTX that "sticks" a channel is a maintenance ticket, not a discipline issue.

4.4 Spotter requirement

A pilot under goggles is blind to the world. **Every FPV flight requires a designated spotter** standing at the pilot's shoulder: eyes on the aircraft, ears on the tower, hand on the pilot's shoulder for urgent physical alerts. For rental cinewhoop sessions the instructor is the

spotter; for certified freestyle slots customers may spot for each other only if both hold the FPV rating; for racing the heat marshal spots the group collectively with lane spotters at large events. The spotter — not the goggled pilot — owns collision avoidance with other airspace users and executes the “LAND LAND LAND” call (Chapter 8.6).

4.5 Simulator-first pathway

FPV in acro mode is the only park discipline in which a novice’s first sixty seconds are near-guaranteed destruction: there is no self-levelling, no envelope, and control response is instantaneous. The park’s rule is absolute: **no customer touches a real acro-mode quad until they have logged simulator hours** (Chapter 7.3 sets the numbers — 3 sim hours for the cinewhoop-to-freestyle step, with gate-navigation proficiency demonstrated in Velocidrone). The sim-to-real transfer in FPV is the best of any RC discipline — muscle memory built in Liftoff or Velocidrone transfers almost 1:1 because the control loop is identical — which makes the simulator lab both a genuine training tool and an honest gatekeeper. Stabilised cinewhoop rentals are exempt from the sim gate (angle mode plus ducts plus instructor-spotter makes first-flight risk acceptable), which preserves the walk-in product.

5. Helicopters

5.1 The hardest discipline — and why the park teaches it anyway

A collective-pitch helicopter is the most demanding machine in all of RC. It is unstable in every axis simultaneously; hovering demands continuous correction on four controls at once; and orientation reverses when the nose faces the pilot. Full-scale pilots routinely report that a 400-class RC heli is harder to hover than the real aircraft, because the real one gives vestibular feedback and the model gives none. The park teaches helicopters precisely because of this: the discipline is the Wings ladder’s Everest, and “I can hover nose-in” is the kind of earned skill that turns a customer into a member. Modern flybarless stabilisation systems have also softened the entry: today’s micro collective-pitch machines carry gyro-stabilised rescue modes that recover to level flight at the flip of a switch — a panic button the discipline never used to have.

5.2 Trainer doctrine: coax first, collective-pitch second

Tier	Machine class	Reference model	Street price (approx.)	Character
H-1 Coax	Coaxial fixed-pitch micro	Blade 70 S-class coax/fixed-pitch micro (segment is thin in 2026 — source current equivalents at procurement, Volume 8)	\$60–120 RTF	Twin counter-rotating rotors self-cancel torque; machine hovers hands-off; teaches orientation only
H-2 CP trainer	Micro collective-pitch, stabilised	OMPHobby M2 EVO (MK2)	~\$330 PNP / ~\$470 RTF	Full collective-pitch response with GPS-free attitude rescue mode; brushless direct-drive; crash-cheap by heli standards

Tier	Machine class	Reference model	Street price (approx.)	Character
H-3 CP sport	380-450-class collective-pitch	OMPHobby M4/M4 Max-class or equivalent	\$500-800	Instructor and demo machine; checkride ship for the Heli type rating

The teaching sequence is fixed: **orientation on the coax → hover discipline on the sim (Chapter 7) → collective-pitch on the M2-class with rescue mode armed → circuits and autorotation awareness on the 380-class under buddy supervision**. Skipping the sim step on helis is the most expensive mistake a flight school can make; a single botched CP hover costs a full set of blades, main shaft, feathering spindle and often a tail boom.

5.3 Helipad operations

The two elevated helipads (1.0 m high, 1.5 m square, rubber-surfaced, railed on the spectator side) exist for three reasons: they put the hover at eye level for spectators; they keep rotor wash off runway dust; and they physically separate rotorcraft from the fixed-wing flow. Operating rules:

- Helis fly in the **eastern hover box** (helipad side), a 15 × 15 m column to 10 m altitude, deconflicted from the fixed-wing circuit by the airspace matrix (Chapter 8.2).
- Spool-up only on the pad, throttle-hold engaged until the pilot box confirms tower clearance; approach and departure paths never cross the runway centreline below 10 m.
- Customers stand at the marked pilot box 4 m from the pad — never beside a spooled machine. Rotor-disc contact is the heli program's defining hazard: even a 200-class machine's blades cut, and a 450-class disc is treated with the same respect as the EDF intake.
- Maximum **two helis airborne** in the hover box simultaneously, and only if both pilots hold the Heli rating; instruction sessions get the box exclusively.

Pre-flight checklist — rotorcraft:

☐

Blades: no cracks, chips or delamination; blade grips tight to spec

☐

Head: feathering spindle straight, linkages secure, swashplate level (leveller tool on first flight of day)

☐

Tail: belt/shaft drive intact, tail blades sound, tail servo linkage secure

☐

Frame: no cracked frames or loose motor mount; canopy latched

☐

Battery: charged, strapped, connector seated; hover-check voltage telemetry live

☐

Radio: correct model; throttle hold ON before power-up; rescue mode armed and switch verified

☐

Failsafe: hold-to-throttle-off verified (first flight of day)



Tower: hover-box slot confirmed

5.4 Maintenance intensity

Helicopters are the fleet’s maintenance outlier. A crashed foam trainer is a glue job; a crashed CP heli is a teardown — the industry’s “crash kit” (main blades, main shaft, feathering spindle, tail boom, servo gears) exists because those parts fail together. Budget doctrine (fed into Chapter 10 and Volume 7’s workshop planning):

- **Maintenance hours per flight hour: ~0.6 for CP helis vs ~0.1 for foam fixed-wing.** One Artisan bench-shift per week is standing heli allocation.
- Crash kits stocked at **2 per airframe** at all times; blades treated as consumables like Motorsport tires.
- After any ground contact, the machine is grounded until an Artisan has checked shaft runout and spindle straightness — a bent spindle flown on vibrates the airframe to death in ten flights.
- Skid-and-blade inspection every 10 flights; head teardown inspection every 50; motor and ESC thermal check every 25 (logged in `maintenance_logs` per park schema).

Field Note. The M2-class direct-drive brushless micros have quietly transformed heli-school economics: no gears to strip, no belt to snap, and a full crash repair often under \$30 in parts. Ten years ago a training fleet of nitro 450s would have needed a dedicated technician. Do not let nostalgia for big helis creep into the rental tier — the 380/450-class machines are instructor and demo ships only.

6. Drone Racing

6.1 Course design inside the net

The western half of the cage is the racing volume: a reconfigurable course of **air gates and flags** in the style established by MultiGP club racing and the Drone Racing League (DRL) — the two reference formats for course grammar, heat structure and spec philosophy. Physical inventory:

Element	Spec	Qty	Note
Square air gates	1.5 m × 1.5 m, push-fit poles + fabric flags, LED-taped	8	MultiGP-standard 5 ft gate class
Large start/finish gate	2.4 m arch, LED chase lighting	1	Houses the timing loop (6.2)
Flags/pylons	2.5 m flexible pylons	10	Turn markers; harmless on strike
Hurdles/ladders	0.6 m low gates	4	“Under” elements; forces altitude changes
Dive gate	Gate mounted at 6 m on mast, dive to ground gate	1	Signature spectacle element

Design rules learned from club practice: gates never point a race line at the spectator net; consecutive gates are ≥ 8 m apart at 5” speeds; the course crosses its own line at most once

(marshalling clarity); and every course revision is flown in Velocidrone first — the sim’s track editor doubles as the park’s course-design CAD (Chapter 7.2). Track resets take one staff-hour; the league publishes a new layout monthly.

6.2 Timing systems

Racing without timing is practice. The park runs RF-based lap timing, which detects each quad’s own 5.8 GHz video transmitter as it passes the start/finish gate — no transponders to install:

- **Primary: RotorHazard** — the open-source race-timing platform that has become the de facto club standard: multi-node 5.8 GHz receiver hardware (~\$150–300 self-built or assembled), per-pilot channel lock, voice callouts, and an API that The Works’ software team bridges into RC WORLD OS so laps, holeshots and personal bests post to the park-wide leaderboard alongside Motorsport lap times (Volume 13 carries the integration spec).
- **Backup/portable: ImmersionRC LapRF** 8-way class (~\$600) for pop-up events on the events lawn (inside temporary netting only) and as hot spare.
- Results feed the same `live_telemetry`-adjacent event tables as Motorsport timing; a customer’s racing history lives on their RC WORLD License profile (Volume 9).

6.3 Class tiers

Class	Airframe	Battery	VTX power	Venue	Who
Whoop (Tiny Whoop-class)	65–75 mm ducted, brushless 1S	1S 300–450 mAh	25 mW	Whoop Arcade	Any customer post-sim-intro; the racing on-ramp
3-inch	3–3.5" props, park spec ~250 g	4S 650–850 mAh	25 mW	Main cage, racing slots	FPV type rating
5-inch (league spec)	5" spec quad per league rules	6S 1,100–1,300 mAh	25 mW	Main cage, exclusive slots	FPV rating + league membership

The 5-inch tier runs as a **spec class** — identical park-supplied or homologated airframes, following the MultiGP spec-class logic (and its 25 mW VTX rule, adopted verbatim): parity keeps racing about the pilot, exactly mirroring the Motorsport doctrine of lap-time parity within 2–3%. Customer-owned quads are welcome in open practice if they pass a pit inspection (weight, VTX power, failsafe, battery condition) and fly on assigned channels.

6.4 The Whoop Arcade

The 12 × 8 m fine-mesh sub-enclosure beside the main cage is the racing program’s nursery and the division’s volume product: six LED-lit micro gates, park Meteor65-class whoops (~\$100–130 each for analog trainer spec; the HD-camera variants used for premium sessions run higher), goggle or FPV-monitor option for nervous first-timers, and — because it is a separate sealed airspace — **it runs continuously regardless of what the main cage is doing**, with up to 4 simultaneous 1S pilots on the channel plan. A whoop session is priced as a standard-tier Casual Shift (\$15): the deliberate cheap first taste of FPV.

6.5 League structure

The racing league follows the season template proven by MultiGP chapters worldwide, adapted to a commercial venue:

Layer	Format	Cadence
Open practice	Channel-managed sessions, RotorHazard running, personal bests logged	Weekly standing slots
Club night	2-minute qualifying heats ("laps in time" format), then triple-run mains by seeded groups of up to 7	Fortnightly, entry ~\$15-20
Season	8 rounds, points table, Whoop/3"/5" tiers; season champion earns a season pass + trophy	2 seasons/year
Marquee event	Invitational with visiting pilots, spectator programming, sponsor activation (Volume 9 events calendar)	1-2/year

League nights are also the division's community engine: the pilots who race Tuesday nights are the same enthusiasts who become weekend instructors, marshals and the park's most credible advocates. Treat the league as marketing that pays for itself.

7. Flight School & Simulators

7.1 The simulator lab as gateway

The Flight School's front door is not on the airfield — it is the **simulator lab**: six stations in the entry pavilion (relocating to the RC Academy classrooms in Phase 3), open from Month 16, three months before the cage itself. Each station is a mid-range gaming PC (~\$1,200), a 27" monitor, and real transmitter hardware — RadioMaster-class radios (~\$130-230) and a Spektrum-class radio matching the trainer fleet, connected as USB controllers — because the entire point is that the controller in the customer's hands at the sim is the controller they will hold at the flight line.

Station	Software	Approx. licence	Role
1-2	RealFlight Evolution (~\$100)	per-seat	Fixed-wing and heli primary; SAFE-style trainer models mirror the real fleet
3-4	Liftoff (~\$20)	per-seat	FPV freestyle and cine-whoop introduction; approachable physics, strong content
5-6	Velocidrone (~£17 base + packs)	per-seat	Race training and course design — the competitive-racing standard sim

A 15-minute sim block is bundled free into every aviation booking; standalone sim Shifts sell at the standard \$15 Casual rate — meaning the lab produces revenue from Month 16, a full quarter before the first real aircraft flies, and continues to produce it in weather that grounds the cage.

7.2 Why the sim is doctrine, not garnish

Three economics drive the sim-first rule. First, **crash cost transfer**: every orientation mistake made in RealFlight is a \$0 repair. Second, **instructor leverage**: one staff member supervises six sim stations vs one buddy-box student. Third, **honest gating**: sim hours are logged automatically per station login against the customer's Wings record in RC WORLD OS, so certification prerequisites (Chapter 9.3) are verifiable data, not self-declaration. Velocidrone earns its third role as the course-design tool: every real gate layout is built and flown virtually before a pole is moved (Chapter 6.1), and league pilots can practise the coming month's track from home — a retention hook no ground division can match.

7.3 The curriculum ladder: sim → buddy-box → solo

Rung	Content	Exit standard
1. Ground school (45 min)	Aviation Toolbox Talk: cage rules, airspace doctrine, LiPo handling, parts of the aircraft, radio basics	Quiz ≥ 80% in app
2. Sim foundation (2 h logged)	RealFlight: level circuits, climbing/descending turns, landing pattern on the trainer model	5 consecutive sim landings on the runway, no crash
3. Buddy-box (3 Shifts min)	Real AeroScout/Apprentice, instructor or master radio: straight-and-level, turns, then full circuits, then landings	Instructor sign-off per manoeuvre in OS
4. Pre-solo check	Instructor-observed circuit set incl. simulated motor-out glide from 15 m	Checkride pass (Chapter 9.4)
5. Solo (Bronze Wings)	First solo Shift, instructor on the line, beginner mode available but not required	Bronze Wings issued
6. Type extensions	Jet / FPV / Heli tracks per Chapter 9.3, each with its own sim prerequisite	Type rating endorsed

Typical calendar time for a motivated customer from first visit to Bronze Wings: 4–6 visits. The ladder is deliberately purchasable as a package (Chapter 10.4) — “Learn to Fly” is the division's flagship SKU.

7.4 Instructor staffing

Role	Headcount (steady state)	Profile
Chief Flight Instructor (CFI)	1 full-time	Owns syllabus, checkride standards, airspace doctrine; deputises for Aviation Lead
Flight instructors	2 part-time equivalent, scaling to 3–4 in season	Club-experienced pilots trained in the park syllabus; many recruited from league nights
Sim lab supervisor	Front-of-house cross-trained	Runs the lab, logs hours, teaches ground school
Flight-line Artisan	1 per operating day (shared with The Works)	Field repairs, battery logistics, pre-flight fleet checks

Instructor pay is structured as base plus per-instruction-Shift commission, making instruction margin-positive from the first booking (Chapter 10.4). All instructors hold current park certifications in every class they teach, plus the marshal-grade first-aid and LiPo-incident training from Volume 7's staff manual. The one-airspace rule means the airspace controller (Chapter 8.2) is a role, not necessarily an extra head: on quiet weekdays the duty instructor wears the controller hat; on peak days and league nights it is a dedicated position.

8. Safety & Airspace Management

8.1 Netting specification and inspection

The net is the division's licence to operate; it is engineered, inspected and documented like the structural asset it is.

Specification (procurement detail in Volume 8; structural design in Volume 11, Chapter 9):

Parameter	Main cage	Whoop Arcade
Mesh	25 mm (1") knotless HDPE square mesh	13 mm fine mesh
Twine strength	≥ 2.0 mm high-tenacity, UV-stabilised; breaking strength verified by supplier certificate against smallest/fastest containable aircraft (design case: 5" quad at 130 km/h and 70 mm EDF at 140 km/h)	1S micro only
Colour	Black (best spectator see-through; best UV life)	Black
Structure	Galvanised steel masts, tensioned perimeter and catenary cables, netting hung with breakaway margin — the net absorbs strikes by deflecting, never drum-tight	Framed panels
Ground seal	Netting skirt buried/ballasted at grade; zero gap ≥ 25 mm anywhere	Full seal
Doors	Two personnel airlocks (double-door vestibules — outer door closed before inner opens), one vehicle gate (openable only when airspace is cold)	Single vestibule

Inspection regime (SOP AV-01):

Check	Frequency	Owner
Walkdown: visual scan of all panels, doors, ground seal; pull-test any suspect panel	Daily, before first flight	Duty instructor
Detailed: every seam, shackle, tension point; mast bolts torque-checked; UV degradation sample test	Monthly	Artisan + Aviation Lead

Check	Frequency	Owner
Engineering: full structural inspection, tension measurement, netting-condition report	Annual + after any storm exceeding site wind design case	Contracted structural engineer
Strike log review	Weekly	CFI (every net strike is logged with location; repeat-strike panels get early replacement)

☐

Daily walkdown complete and logged in RC WORLD OS before airspace opens

☐

All doors closed and latched; vestibule discipline briefed to any new staff

☐

No unrepaired holes, detached panels, or slack exceeding placard limits

☐

Wind/weather check recorded (8.5)

Safety Warning. A hole in the net is a full airspace stop, not a judgement call. The airspace goes cold the moment a breach is found and stays cold until an Artisan has patched and the duty instructor has signed the log. There is no “we’ll keep the pattern on the other side” exception — the entire regulatory and insurance posture of the division (1.2) rests on containment being absolute.

8.2 One airspace, one controller

The cage is one volume of air. However many pilot boxes are occupied, **exactly one person owns the airspace at any moment**: the airspace controller (“RC Tower”). The rule exists because mid-air deconfliction by mutual glancing — the informal club method — fails exactly when traffic gets interesting. The controller:

- Runs the session board in RC WORLD OS: who is aloft, in which zone, on which frequency slot, until when.
- Issues and terminates every airspace slot; nobody powers up without one.
- Owns the emergency vocabulary (8.6) and the weather hold (8.5).
- Enforces the **compatibility matrix** — which disciplines may share the air:

Aloft together?	Trainer FW	Sport FW	Jet	Heli (hover box)	Cinewhoop	5"/3" FPV
Trainer FW	✓ (max 3)	✓	✗	✓	✓	✗
Sport FW	✓	✓ (max 2)	✗	✓	✗	✗
Jet	✗	✗	exclusive	✗	✗	✗
Heli (hover box)	✓	✓	✗	✓ (max 2)	✓	✗
Cinewhoop	✓	✗	✗	✓	✓ (max 2)	✗
5"/3" FPV	✗	✗	✗	✗	✗	exclusive (max 7)

Absolute ceiling: **4 aircraft aloft in the main cage** in mixed operations; the Whoop Arcade is separate and always-on. Jets and racing get exclusive slots — they are scheduled products, not walk-ups, which is also what makes them programmable spectacle (3.4, 6.5).

8.3 Flight-line discipline

The flight line is the painted boundary between the pilot boxes and the live airfield. The rules are few, absolute, and identical to the park's ground-division doctrine (customers never walk onto live operating areas):

1. Pilots fly from numbered pilot boxes only; transmitters stay in the box.
2. Nobody crosses the flight line while any aircraft is live. Retrieval of a downed aircraft happens only after the controller calls "AIRSPACE COLD" — all aircraft landed and throttle-locked.
3. Downed-aircraft retrieval is performed by staff. (The Tow-Truck Retrieval Protocol does not extend to aviation: the comedy of a recovery crawler is not worth customers fixating on entering the cage. The aviation equivalent that is offered: the customer keeps their Shift clock frozen during retrieval.)
4. Arming discipline: fixed-wing armed on the apron facing the runway; jets on the centreline only (3.4); quads armed on the launch pads; helis spool on the pads with hold engaged (5.3).
5. Props and rotors are treated as live whenever a battery is connected, without exception.

8.4 LiPo handling for aircraft

Aviation inherits the park battery doctrine (3.4–4.2 V/cell window, 3:1 battery:asset ratio, XT60, bunkered charging — Volume 7 carries the master procedures) with three aviation-specific extensions:

- **Higher energy, stricter margins.** Jet 6S packs land at ≥ 3.5 V/cell (3.2); all aircraft packs use timer + telemetry rather than "fly till it sags," because an aircraft that browns out doesn't coast to a stop — it arrives.
- **Crash quarantine.** Any pack that has been in a significant impact goes into the sand-filled quarantine bin at the flight-line charging cache for 24 h observation before inspection — impact-damaged LiPos can enter thermal runaway minutes to hours later. Puffed, dented or punctured packs are salt-water discharged and disposed per Volume 7.
- **Field cache limits.** The flight line holds max one operating day of charged packs in steel ammo-box caddies; all charging happens in the Phase 1 charging bunker, which Phase 2 extends with aviation shelving. Whoop 1S packs charge on multi-port 1S boards at the Arcade desk under staff eyes — the only charging permitted outside the bunker, capped at 6 packs simultaneously on a fireproof mat.

8.5 Wind and weather limits

The cage kills flyaway risk, not weather risk. Limits are placarded at the tower desk and enforced by the controller from the on-site anemometer (mast-mounted, feeding RC WORLD OS):

Condition	Trainer FW	Sport FW	Jet	Heli	Whoop/ Cinewhoop	5" FPV
Sustained wind	≤ 20 km/h	≤ 25 km/h	≤ 18 km/h	≤ 20 km/h	Arcade: n/a (sheltered); cage ≤ 15 km/h	≤ 25 km/h
Gust spread	≤ 10 km/h over sustained	≤ 10	≤ 8	≤ 8	≤ 8	≤ 10
Rain	Stop (electronics + visibility)	Stop	Stop	Stop	Arcade continues	Stop
Lightning within 10 km	Full stop, cage evacuated — the masts are the tallest steel on site					

Weather holds pause Shift clocks automatically; the sim lab absorbs displaced bookings — a deliberate redundancy that keeps weather days revenue-positive.

8.6 Incident procedures and emergency vocabulary

Three standard calls, briefed in every ground school, exercised monthly:

- **“LAND LAND LAND”** — controller’s general recall. Every aircraft lands immediately at its designated area; FPV spotters physically relay to goggled pilots.
- **“STOP STOP STOP”** — imminent danger (person inside the cage, structural issue). All motors stop now; fixed-wing ditches straight ahead. An aircraft is always sacrificed before a person is risked.
- **“FIRE FIRE FIRE”** — LiPo or other fire: airspace cold, evacuate per park emergency plan, LiPo-rated extinguishing media (sand/Class D at the cache; CO₂ for electrical) — staff fight only incipient fires.

SOP AV-02 — Incident response & reporting (ID: AV-02 · Rev 1.0 · Owner: CFI · PPE: gloves for wreck handling · Tools: incident kit, quarantine bin · Frequency: per event):

1. Controller declares the appropriate call; airspace cold.
2. Injuries first: first-aid per park emergency plan (Volume 7); serious injury = emergency services + GM notification within 15 min.
3. Battery from any crashed aircraft to quarantine (8.4).
4. Scene photos, telemetry snapshot (session log auto-preserved in RC WORLD OS), witness names before the crowd disperses.
5. Incident record filed same day: category (net strike / mid-air / ground strike / injury / battery / structural), aircraft, pilot certification level, conditions, narrative.
6. CFI weekly review; any injury or structural incident triggers a stand-down review before the affected program resumes.

8.7 Lost-link failsafe configuration

Even inside a net, a lost radio link must produce a predictable, minimum-energy outcome. Fleet standard, verified at build and re-verified at the first flight of each day (transmitter-off test on the checklists):

Type	Failsafe on link loss
Fixed-wing (SAFE/stabilised)	Throttle to idle/off; stabiliser to self-level; controls neutral — aircraft mushes down within the cage
EDF jet	Throttle off, wings-level command; accepted outcome is a firm arrival — energy off is the only priority
Helicopter	Throttle hold (autorotation state); rescue-mode level if system supports it
FPV quad (Betaflight-class firmware)	Failsafe stage 2 = drop (disarm) after 1.5 s signal loss — the configured and tested park standard; GPS-rescue modes are disabled (pointless and unpredictable inside a cage)
All types	Receiver failsafe positions physically verified after any receiver rebind (bind-time capture is the classic silent misconfiguration)

8.8 Spectator protection

Spectators are the net's second constituency. Standoff rail 2 m outside the netting on all public sides (a slowed aircraft can still deform the mesh inward ~0.5 m); viewing windows at the grandstand sightline use the same 25 mm mesh — no openable panels on public faces; the FPV pit's repeater screens (4.1) sit outside the rail so crowds gather at the screens, not against the net; PA callouts precede jet demos so the flypast startles nobody. The insurer's site file (Volume 10) carries the enclosure certificate, inspection log and this section's rail drawings.

9. Pilot Certification

9.1 The RC WORLD Wings program

Wings is the Aviation Division's implementation of the park-wide **RC WORLD License** progression (Volume 9 owns the overall scheme, tiers, badges and rewards; this section defines the aviation ladder that plugs into it). The design principle: every rung is earned by logged, verifiable evidence — sim hours from the lab logins, buddy-box minutes from instructor sign-offs, checkrides from the CFI — all recorded against the customer profile in RC WORLD OS, and every rung unlocks something a customer visibly wants (solo access, new aircraft classes, league eligibility, Gears bonuses).

9.2 Certification card and system integration

The Wings credential lives on the customer's RC WORLD License digital card in the app (physical collector cards printed for each tier — deliberately covetable, per Volume 9's badge doctrine). At booking, RC WORLD OS enforces prerequisites automatically: a customer without the FPV rating cannot book a freestyle slot; the tower session board shows each pilot's tier so the controller staffs supervision accordingly. Wings tiers also map to License

points and Gears earning multipliers defined in Volume 9 — aviation achievements feed the same park-wide status economy as Motorsport lap records and Mining tonnage.

9.3 The ladder

Tier	Prerequisites (logged)	Assessment	Unlocks
Ground Wings	Aviation Toolbox Talk + ground-school quiz $\geq 80\%$	In-app quiz	Sim lab; buddy-box booking; Whoop Arcade (post sim intro)
Bronze Wings	2 h sim + 3 buddy-box Shifts + manoeuvre sign-offs	Solo checkride (9.4)	Solo trainer (FW-T) rental
Silver Wings	Bronze + 10 solo Shifts + aerobatics sign-off	Sport checkride	FW-S sport class; spotter privileges
Gold Wings	Silver + 25 logged Shifts + currency (9.5)	CFI review flight	Type-rating eligibility; league officer roles
Type: Jet	Gold + 1 h EDF sim + energy-management ground module	Jet checkride with CFI (instructor-supervised flying thereafter, 3.3)	Habu SS under supervision; demo-day slots
Type: FPV	Ground Wings + 3 h logged FPV sim (gate proficiency in Velocidrone) + cinewhoop session	Acro checkride (line flight + course flight)	3" freestyle slots; league racing; self-spotting pairs
Type: Heli	Silver + coax session + 2 h heli sim	Hover checkride (four-point hover + circuit, rescue armed)	M2-class CP rental in hover box

9.4 Checkrides

Checkrides are flown to a published standard — customers see the exact card the examiner scores, removing all mystery:

- **Solo (Bronze):** preflight from memory; taxi; two circuits each direction; go-around on demand; simulated motor-out glide from 15 m to a survivable touchdown; landing within the marked 15 m zone. Two attempts included in the checkride fee; further attempts after a remedial buddy-box Shift.
- **Jet:** energy-managed circuit set; stabilised approach with power carried to flare; missed-approach; verbal walkthrough of failsafe and net-strike procedure.
- **FPV Acro:** line-of-sight hover and orientation recovery; goggled course flight with ≤ 2 gate misses; controlled disarm-and-drop on command; spotter-protocol test.
- **Heli:** pads-off four-point hover (nose-out, both sides, nose-in) within a 3 m box; controlled circuit; rescue-mode recovery demonstration; throttle-hold response on command.

9.5 Currency and renewal

Skills decay; the program prices that honestly rather than pretending a 2024 checkride guarantees a 2026 pilot:

Rule	Standard
Rolling currency	≥ 1 logged flight (real or examiner-witnessed sim session) per 90 days per tier

Rule	Standard
Lapse $\leq$ 12 months	One refresher buddy-box Shift restores currency
Lapse $>$ 12 months	Re-checkride at current standard (ground school not repeated)
Type ratings	Jet: currency flights are always supervised anyway (3.3). FPV/Heli: 90-day rule applies per type
Annual admin renewal	Free with any active membership; \$10 standalone — keeps records honest and nudges membership conversion (Volume 9)

Field Note. Publish the currency rule at sign-up, not at lapse. Customers accept “flying skills expire like gym fitness” cheerfully when told upfront, and resentfully when discovered at the booking screen. The 90-day nudge notification in the app (“Your Silver Wings go stale in 14 days — book a Shift”) is one of the division’s best re-booking triggers; Volume 9 wires it into the loyalty messaging calendar.

10. Fleet, Maintenance & Economics

10.1 Fleet table (~20 aircraft, Phase 2 baseline)

Prices are July 2026 street-price ranges for planning; Volume 8 executes actual procurement (expect 25–40% below Western street price on the Chinese-sourced lines via wholesale channels, per park sourcing doctrine).

#	Role	Model (reference)	Qty	Unit street price	Line total (mid)	Batteries (3:1)
1	Primary trainer	HobbyZone AeroScout S 2 1.1 m	3	\$200–270	\$705	3S 2200 $\times$ 9
2	Club trainer	E-flite Apprentice STS 1.5 m	2	~\$360	\$720	3S 3200 $\times$ 6
3	Value trainer	FMS Ranger 1220 mm (Reflex)	2	~\$180	\$360	3S 2200 $\times$ 6
4	Sport/utility	Volantex Ranger 1600	1	~\$95–165	\$130	3S 2200 $\times$ 3
5	Sport jet	E-flite Habu SS 70 mm	1	~\$280–300	\$290	6S 3300 $\times$ 3
6	Scale demo jet	FMS/Freewing 70 mm scale	1	~\$250–350	\$300	6S 3300 $\times$ 3
7	Coax heli trainer	70–120-size coax (current equivalent)	2	\$60–120	\$180	1S–2S $\times$ 6
8	CP heli trainer	OMPHobby M2 EVO MK2	2	~\$330 PNP	\$660	3S 550 $\times$ 6
9	Whoop fleet	Meteor65-class 1S brushless	4	\$100–130	\$460	1S 450 $\times$ 12

#	Role	Model (reference)	Qty	Unit street price	Line total (mid)	Batteries (3:1)
10	Cinewhoop	BetaFPV Pa-vo20-class (digital HD)	2	\$300-460	\$760	3S 550 × 6
11	3" freestyle	3.5" BNF park spec	1	~\$250	\$250	4S 650 × 3
Total air-frames			21		≈ \$4,815	

Supporting equipment: fleet transmitters (6 × Spektrum-class for the SAFE fleet, 6 × Radio-Master-class for FPV/sim, ~\$1,900); goggles (6 sets mixed digital/analog + monitor, ~\$1,900); RotorHazard timer + LapRF backup (~\$900); gates/flags/LED (~\$1,800); battery fleet per 3:1 doctrine (~\$2,600); tools/crash kits/initial spares (~\$2,400). **Fleet & equipment subtotal ≈ \$16.3 k** at street prices — wholesale execution should land near \$12-13 k, comfortably inside the \$32 k Chapter 1.3 allocation with the balance held as Year-1 spares and attrition reserve.

10.2 Crash-repair economics

Aviation's maintenance profile is the inverse of Motorsport's: fewer, cheaper assets, but a higher event rate per Shift, dominated by trivially repairable foam damage. Planning rates (to be replaced by logged data after Season 1, per the park's failure-data doctrine):

Event	Expected rate	Typical cost (parts + Artisan minutes)
Trainer hard landing / nose-over	1 per 6 rental Shifts	\$0-4, 10 min (glue, prop)
Trainer structural foam repair	1 per 25 Shifts	\$8-20, 30 min
Trainer write-off (airframe swap)	1 per 200 Shifts	\$80-140 replacement airframe (motor/servos transfer)
Whoop prop/frame	1 per 4 sessions	\$1-5, 5 min
Cinewhoop duct/camera event	1 per 20 sessions	\$10-40
5" race crash	~1 per 2 race heats	\$5-30 (props, arm) — the class is designed to crash
CP heli crash	1 per 15 CP sessions (post-sim-gate)	\$25-60 crash kit, 45-90 min bench
Jet event	Rare (supervised) but \$100-250 when it happens	Fan/ESC/retract concentration

The division's whole training doctrine — sim gates, buddy boxes, stabilisers, supervised jets — is visible in this table as suppressed right-hand columns. Every sim hour sold is also avoided repair cost.

10.3 Foam repair techniques and spare-airframe policy

The Works' aviation bench (full procedures in Volume 7, Chapter on airframe repair) runs on four techniques that between them fix 90% of fixed-wing damage: **foam-safe CA with kicker** for clean breaks (medium CA melts EPS; stock foam-safe plus activator); **5-minute epoxy + carbon spar splints** for wing folds and firewall tears; **fibre-reinforced tape hinging** for control-surface repairs; and **hot-glue skinning** for cosmetic crush zones (fast, peelable, sur-

prisingly aerodynamic at trainer speeds). Colour-matched packing tape over repairs keeps the rental fleet looking professional — customers judge airworthiness by cosmetics.

Spare-airframe policy: for every rental fixed-wing type, hold **one bare spare airframe** (no electronics) per two operating airframes. Electronics — motor, ESC, servos, receiver, RCW Node — are the durable value and transfer to the spare in under an hour, so a written-off AeroScout costs an \$80–140 airframe kit, not a \$250 aircraft. Helis hold 2 crash kits per airframe (5.4); quads hold frames, arms and props in depth (props are consumables at Motorsport-tire intensity); jets hold one complete spare fan/motor unit per type.

10.4 Session pricing

Per canon, aviation is a **premium class**: Casual Shift \$22, Operator Shift \$38. The aviation-specific price card:

SKU	Price	Notes
Sim Shift (20 min)	\$15	Standard tier; bundled free with aviation bookings
Whoop Arcade Shift	\$15	Standard tier; the FPV on-ramp
Trainer/cinewhoop Casual Shift	\$22	Premium tier; includes instructor-as-spotter where required
Operator Shift (2 blocks + battery-swap pit stop)	\$38	Premium tier
Instruction Shift (buddy-box, 1:1)	\$45	Instructor commission inside
“Learn to Fly” package (ground school + 4 sim h + 3 instruction Shifts + solo checkride)	\$249	Flagship SKU; ~\$310 of parts sold as a journey
Checkride / type-rating assessment	\$35	Includes one re-fly
Jet supervised experience (rated pilots)	\$55 per flight slot	Exclusive airspace priced in
League night entry	\$15–20	Chapter 6.5

10.5 Capacity under the one-controller constraint

The airspace matrix (8.2) caps mixed operations at 4 aircraft aloft. Realistic peak-hour throughput, with 20-minute Shifts and staged starts:

Stream	Simultaneous	Shifts/hour
Main cage mixed (trainers, heli box, cinewhoop)	4	12
Whoop Arcade (parallel airspace)	4	12
Sim lab	6	18
Peak total	14 flying + 6 sim	≈ 42

Blended peak revenue ≈ \$800–900/hour across the three streams — roughly half of Motorsport’s peak, which is exactly why aviation is priced premium and programmed with high-margin instruction, league nights and demo-anchored spectating rather than chasing raw Shift volume. The Arcade and sim lab exist precisely to add throughput that costs no main-cage airspace. Exclusive slots (jets, 5” racing) are scheduled off-peak or as ticketed programming so they displace the minimum of rental capacity.

10.6 Division economics summary

Item	Value (planning)
Capex allocation (within Phase 2's \$1.15 M)	≈ \$364 k: enclosure \$190 k; civil (runway, aprons, pads, drainage) \$85 k; sim lab \$45 k; fleet/radios/FPV/spares \$32 k; timing & AV \$12 k
Steady-state revenue (Year 3, division)	\$280–340 k/yr: rental & sim Shifts ~55%, instruction & certification ~25%, league/events/demos ~12%, aviation retail attach ~8% (retail P&L sits in Volume 9/10)
Direct operating cost	Instructor & controller labour dominant; parts & attrition ~ \$14–18 k/yr at planned Shift volumes (10.2)
Contribution posture	Margin-accretive from opening at Phase 2 volumes; division carries the park's premium-brand halo and the Academy pipeline into Phase 3

Full financial integration — division P&L, sensitivity to instruction attach rate, and the Phase 3 indoor-arena upside case (an all-weather netted volume multiplies aviation capacity) — is carried in Volume 10.

10.7 Division KPIs

The Aviation Lead reports the following on the RC WORLD OS division dashboard (park-wide KPI architecture in Volume 10; dashboard implementation in Volume 13):

KPI	Target (Year 1 of division)	Why it matters
Cage utilization (airspace-slot hours sold / hours open)	≥ 55% weekends, ≥ 30% weekdays	The single scarce resource is airspace, not aircraft
Sim-to-flight conversion	≥ 40% of sim-only customers book a real Shift within 60 days	Validates the gateway model; drives Learn-to-Fly attach
Learn-to-Fly completion rate	≥ 70% of package buyers reach Bronze Wings	Incomplete ladders create refund pressure and dead pipeline
Wings holders (cumulative)	150 Bronze+, 25 type ratings by Month 12 of operation	The certified base is the league, membership and Phase 3 Academy pipeline
Repair cost per Shift	≤ \$1.20 blended (10.2 rates)	Early-warning on training-gate erosion or fleet fatigue
Net strikes per 100 Shifts	≤ 3, trending down	Leading indicator for both pilot standards and course design
Incident rate (reportable per 1,000 Shifts)	≤ 1, zero injuries	The licence-to-operate metric; insurer-visible
Weather-displaced revenue recovered via sim lab	≥ 60%	Tests the redundancy design of 8.5
League retention (season-over-season re-entry)	≥ 65%	Community health; instructor recruitment pool

Investor Note. Watch sim-to-flight conversion and Learn-to-Fly completion before anything else in the division's first two quarters. Cage utilization will look poor early — it is capped by instructor supply and certification pipeline, not demand — and both metrics recover mechanically as the certified base grows. A division that converts sim customers at 40%+ is compounding; one that doesn't has a pricing or programming problem, not an aviation problem.

11. Volume Summary & Cross-References

The Aviation Division converts the most dangerous branch of the RC hobby into a walk-in leisure product through one structural decision and three doctrines. The decision is **total containment**: a 60 × 40 m high-tensile netted cage with a 15 m clear flight ceiling around the 30 m runway and helipads, which moves flying out of national-airspace regimes (FAA Part 107/TRUST and EASA open category treated strictly as reference frameworks — local written confirmation is a Month 13 gating task) and into the insurable world of certified structures. The doctrines: **one airspace, one controller**, with a published compatibility matrix and exclusive slots for jets and racing; **simulator-first training**, with a six-station lab (RealFlight, Liftoff, Velocidrone) opening three months before the cage and gating every high-risk discipline; and **earned progression**, the RC WORLD Wings ladder from ground school through buddy-box to solo and type ratings, enforced automatically by RC WORLD OS. The ~21-airframe fleet pairs SAFE-equipped Western trainers with Chinese-sourced equivalents, positions EDF jets as supervised spectacle, helicopters as the prestige discipline, and FPV — cinewhoop, freestyle, Tiny Whoop and a MultiGP-pattern racing league — as the division's growth engine. At premium-tier pricing with instruction margins layered on ~42 peak Shifts/hour across cage, Arcade and sim lab, the division earns its ≈ \$364 k share of Phase 2 capex while giving the park its strongest aspirational brand asset.

Cross-references:

- **Volume 7 (Engineering & Workshop)**: master LiPo procedures, foam and heli repair bench practice, staff training and first-aid standards extended by Chapters 5.4, 8.4 and 10.3.
- **Volume 8 (Procurement)**: wholesale execution of the Chapter 10.1 fleet table, netting supplier contract, goggle-liner and prop consumable schedules.
- **Volume 9 (Customer Experience & Loyalty)**: the RC WORLD License scheme that Wings (Chapter 9) plugs into; badges, Gears multipliers, membership hooks, events calendar and the currency-nudge messaging.
- **Volume 10 (Finance)**: division P&L, Phase 2 capex staging, insurance file, sensitivity cases.
- **Volume 11 (Architecture & Park Design)**: structural design of the enclosure, masts and helipads per Chapter 1.1 and 8.1; grandstand sightlines; Whoop Arcade placement.
- **Volume 12 (Franchise Manual)**: jurisdiction-specific regulatory verification protocol generalised from Chapter 1.2.
- **Volume 13 (RC WORLD OS)**: Wings record schema, sim-hour logging, session board and airspace-slot management, RotorHazard timing integration, weather-hold automation.

Volume 6 — Marine Division

Document	RC WORLD — Master Development Plan, Volume 6 of 13
Title	Marine Division
Revision	1.0 — July 2026
Status	Planning baseline — Phase 2 build (Months 13–30)

Purpose of this volume. This volume is the complete concept, engineering, fleet, operations and economics plan for the RC WORLD Marine Division: a two-pond waterspace complex added in Phase 2, operating approximately 24 vessels across four programs — a working miniature harbour with scale tugs, barges and fire boats; a fast-electric speed pond; a one-design sail fleet; and a small premium submarine program. It gives investors the business case and capacity mathematics, gives architects and pond contractors the waterspace engineering brief (pond geometry, depth profiles, liner selection, filtration hydraulics and safety barriers), and gives Artisans and duty managers the fleet specifications, water-specific maintenance doctrine, SOPs and checklists needed to run the division day to day.

Intended readers. Investors and lenders assessing Phase 2 (Chapters 1, 9); architects and pond contractors detailing the complex with Volume 11 (Chapter 6); the Marine Division lead, harbour masters and duty managers (Chapters 2–5, 8); Artisans assigned to marine duty (Chapters 5, 7); the safety officer and insurers (Chapters 5, 6 and the SOPs throughout).

Chapters

1. Division Overview
2. Scale Boats & the Working Harbour
3. Fast Electric Boats
4. Sail Program
5. Submarines
6. Pond & Harbour Design
7. Boat Maintenance
8. Marine Operations & Programs
9. Fleet & Economics
10. Volume summary & cross-references

1. Division Overview

1.1 The harbour concept

Every RC WORLD division is built on the same founding thesis: a **miniaturized industrial complex, not a hobby sandbox**. The Construction Division proved the formula — customers do not merely drive an excavator, they execute a production chain (dig, load, haul, dump) governed by real logistics constraints, which is why the canonical 1:3 excavator-to-dump-truck ratio exists. The Marine Division applies the identical doctrine to water. Its centerpiece is not a pond with boats on it; it is a **working miniature port** — quay walls, fendered berths, a loading wharf, hopper barges, a container terminal with a quayside crane, tugs that assist “ship movements”, and fire boats that answer incident calls from a harbour master’s console. Around that centerpiece sits a second, entirely different body of water: a **speed pond** running fast-electric monohulls and catamarans on a buoyed oval, delivering the rooster-tail spectacle that reads from anywhere in the park.

The two waterspaces are deliberately opposite in character. The harbour is slow, precise and narrative — the marine equivalent of the Mining Zone. The speed pond is loud (by water standards), fast and competitive — the marine equivalent of Track B. Between them, a sail fleet and a small submarine bay complete a division that covers the full emotional register of watercraft: work, speed, silence and mystery.

Investor Note — The Marine Division is a **Phase 2 addition** (Months 13–30, “Full Park”), funded inside the Phase 2 capex envelope of $\approx$ \$1.15 M alongside the Aviation Division, restaurant, expanded retail and grandstand (canon: Volume 1 and Volume 10). Its $\sim$ 24 vessels are the cheapest powered fleet in the park — roughly \$9,500 in hulls, under \$13,500 with batteries and first-year spares (Chapter 9) — but the ponds are the real asset: a $\approx$ \$212,000 civil work with a 25–40-year liner life that anchors the Phase 2 F&B precinct, extends average visit length, and captures family, senior and photographer demographics the land divisions under-serve.

1.2 Why marine broadens demographics

The land divisions skew toward drivers aged 8–35 who progress on skill and speed. Water pulls in everyone else:

Attribute	Motorsport / Construction	Marine
Dominant customer	8–35, progression-driven	Families with young children, seniors, couples, photographers
Session tempo	High arousal, competitive	Calm, methodical, narrative
Physical demand	Fast reactions	Low — seated operation, gentle inputs
Noise signature	Motor and impact noise	Near-silent (sail), low hum (scale)
Spectator draw	Grandstand bursts	Continuous terrace ambience
Shared-generation play	Limited (parallel driving)	Strong (tug + crane + harbour master roles)

Three design consequences follow. First, the harbour is **the park’s most accessible experience**: a 1/50 tug at scale speed is operable by a seven-year-old with a grandparent beside them, and both are genuinely engaged — it is the shared-generation product the rest of the park lacks. Second, the division **extends dwell time rather than competing for it**: marine

Shifts book naturally as the cooldown block after racing or mining, which is why Volume 11 sites the pond complex beside the restaurant terrace. Third, **weekday mornings finally have a product**: senior sail regattas and harbour-club sessions fill hours the land divisions leave empty (Chapter 8).

1.3 Why marine calms the park's pace

A theme park composed entirely of adrenaline products exhausts its guests by mid-afternoon and empties early. The Marine Division is RC WORLD's deliberate **pace regulator**. Water is intrinsically theatrical at low speed: a 600 mm tug pushing a loaded barge throws a real bow wave; a sail fleet heeling together on a beat is silent, which in a park full of brushless motors is itself an attraction; a submarine sliding under and reappearing thirty metres away is, for most guests, the first time they have ever watched one move. The pond complex works as spectacle even for guests who never book a marine Shift — and the restaurant terrace overlooking the harbour converts that spectacle directly into F&B revenue.

Weather works differently on water, and the plan is honest about it in both directions. Light rain is playable: scale boats and sailboats run happily in drizzle, and marine electronics are waterproofed to division doctrine anyway (Chapter 7), so marine is the park's designated soft-weather division when Tracks A/B close. Wind is the binding constraint: sail needs roughly 1.5–8 m/s; above ~8 m/s scale docking precision collapses; above ~10 m/s the water-space closes (wind sheltering in §6.6 widens that window). Lightning closes all water instantly — the ponds are the most exposed conductors on site.

1.4 Division scope, phasing and headline numbers

- **Opens**: Phase 2, targeted for Month 20–24 (pond civil works begin Month 14; water commissioning and fleet shakedown run 8–10 weeks before public opening).
- **Waterspace**: two ponds — harbour pond $\approx 30 \times 18$ m with quay walls and a 1.5 m-deep submarine bay; speed pond $\approx 60 \times 25$ m (Chapter 6).
- **Fleet**: ~24 powered vessels (9 harbour including the staff cargo ship, 8 fast electric, 6 sail, 1 submarine) plus a second show/spare submarine, 6 unpowered barges, a container terminal crane, and a dedicated retrieval tug as support assets (Chapter 9).
- **Staffing**: one harbour master per operating session (customer-facing controller role), one marine Artisan on duty, marshals shared with adjacent divisions.
- **Billing**: standard 20-minute Shift; Casual \$15 standard classes / \$22 premium (submarine, fire-monitor operations, 8S show boats); Operator Shift \$26/\$38 with pit-stop battery swap, per park canon.
- **Telemetry**: every powered surface vessel carries an **RCW Node** (Micro-Node footprint) in its radio box, reporting GPS position and pack voltage over the park Wi-Fi mesh, with the geofence redrawn as **virtual harbour limits** (Chapter 8). Submarines are the single documented exception (radio physics, Chapter 5).

Field Note — Public model-boating ponds have existed for over a century (the Kensington Round Pond in London and Central Park's Conservatory Water in New York still host weekly model yacht racing), so the pond is proven public infrastructure. What no municipal pond or hobby club offers is the port: missions, scoring, a crane, a harbour master, and a fleet a customer can rent for twenty minutes without owning anything. That is the moat.

2. Scale Boats & the Working Harbour

2.1 The working-harbour gameplay loop

The harbour pond is run as a live port under the direction of a **harbour master** — a staff role (Chapter 8) with a console at the quayside marina hut. Customers book a vessel class and receive missions; completed movements score points that feed the RC WORLD License badge system (Volume 9). The core loop mirrors the Mining Zone’s dig-haul-dump chain:

1. **Load** — a hopper barge is loaded at the bulk wharf (gravel chute fed with the same graded aggregate used in the Mining Zone) or a container barge is loaded by the quayside crane.
2. **Move** — a tug or push boat takes the barge across the basin, holding the marked fairway, giving way per the harbour master’s calls.
3. **Berth** — the barge is pushed onto a fendered berth and held while it is “worked”.
4. **Discharge** — bulk cargo is tipped into the receiving hopper; containers are lifted off by the crane and stacked on the quay.
5. **Score** — the harbour master logs the movement in RC WORLD OS; clean berthings (no fender strikes above walking pace) score bonus points.

2.2 Signature harbour challenges

Challenge	Vessels	Skill tested	Format
Precision docking	Tug, solo	Throttle discipline, prop-walk anticipation	Berth inside a marked box between two fendered piers; scored on time + contact count
Barge push	Tug + hopper barge	Push-coupling control, momentum management	Move a loaded barge across the basin and land it on the bulk berth
Container run	Tug + container barge + quay crane	Two-customer coordination	Boat customer berths under the crane; crane customer transfers 4 scale containers barge-to-quay
Tanker assist	2 tugs + the “ship” (staff-driven cargo vessel)	Team tug work, communication	Two customers nudge the 900 mm cargo ship onto the long berth, one at the bow, one at the stern
Fire response	Fire boat with working monitor	Navigation under time pressure, monitor aiming	Harbour master calls an “incident”; customer navigates to the incident float and knocks down a target flag with the water monitor

The **container run is the park’s first cross-division bridge**: the quayside crane uses the same control interface and lead-screw electromechanical doctrine as the Construction Division’s cranes (Volume 4), so a customer holding a Construction crane badge is immediately qualified on the harbour crane. The combined **Pit-to-Port mission** — ore is dug in the Mining Zone, weighed at the hopper, credited to the customer’s account, then “shipped” at the harbour by barge and crane — is the flagship corporate-event format (Chapter 8).

2.3 The harbour fleet — real vessels

The harbour fleet follows the park’s two-tier doctrine: a **rental workhorse tier** (cheap, tough, fast to repair) and a **premium/showcase tier** (higher detail, supervised or staff-operated).

Role	Model	Scale / LOA	Street price (2026)	Tier	Notes
Workhorse tug ×4	Heng Long 3810 work tug	1/50, 600 mm	\$100–150	Rental	ABS hull, 380 brushed motor, splash-proof 10 A ESC, 7.4 V 1800 mAh pack, working water cannon ; ducted prop guarded against fingers and lines
Fire boat ×1	Heng Long 3810 (fire-livery variant)	1/50, 600 mm	\$110–160	Rental	Same platform re-dressed; monitor pump upgraded in The Works for ~2 m throw
Heavy harbour tug ×2	Pro Boat Horizon Harbor 30” tug	~760 mm	≈ \$400	Rental-premium	700-size motor, functional water cannon; announced 2026 at \$399.99 — the natural “big tug” upgrade class
Scale flagship tug ×1	Bancroft Fairplay VI	1/50, 650 mm	≈ \$700–760	Premium	RTR harbour tug with brass photo-etch detail, dual motors; used for tanker-assist and photography sessions
Cargo ship (staff vessel)	Arkmodel-class ABS/FRP civil-ship kit, built in The Works	~1/72–1/100, 900–1,000 mm	\$350–700 built	Staff	The “ship” that customers’ tugs assist; slow, heavy, ballasted
Hopper/container barges ×6	Shop-built (plywood/epoxy or ABS sheet)	1/50, 500–700 mm	≈ \$120 each to build	Unpowered	Push-knees and container twist-lock studs; The Works builds these as apprentice projects (Volume 7)

Field Note — The Heng Long 3810 is the harbour’s Huina 1580: a sub-\$150 platform with an enormous parts ecosystem, tolerable to lose and trivial to fix. Its factory water cannon is weak; The Works replaces the pump with a standard 370-motor centrifugal unit for a proper 2 m stream — a 20-minute, \$8 upgrade. Buy two hulls per four in service as rolling spares.

Display and heritage tier. The marina hut doubles as a small maritime gallery. Two or three **static display builds** — e.g. a Billing Boats **Smit Rotterdam** 1:75 ocean tug (kit ≈ \$350–450) and a Türkmodel wooden fishing vessel — are built by Artisans in winter and displayed behind glass, exactly as the Kabolite K970 anchors the Construction showroom. Legacy **Graupner and Robbe** harbour models (both brands went through insolvency and restructuring in the mid-2010s; Robbe continues under new ownership) appear regularly on the second-hand market at \$150–400 and make excellent gallery pieces, but they are **not bought for the rental fleet**: spares availability does not meet the park’s repair-time doctrine.

2.4 Scale-speed discipline

The single rule that makes the harbour look real is **scale speed**. Model speed should follow Froude scaling: the model’s realistic speed is the full-size speed divided by the square root of the scale factor.

Worked example — a 1/50 harbour tug whose prototype makes 12 knots (6.2 m/s):

- Scale factor $\lambda = 50$, $\sqrt{\lambda} \approx 7.07$
- Scale speed = $6.2 \div 7.07 \approx \mathbf{0.87 \text{ m/s}}$ (about **1.7 knots — brisk walking pace across the pond**)

Anything faster reads as a toy and, worse, makes docking impossible and wakes destructive. Enforcement is technical, not verbal: rental harbour boats are delivered with **ESC throttle end-points limited to scale speed** (a one-time setting in The Works), and the RCW Node flags sustained over-speed to the harbour master’s console. Customers who want speed are sold the speed pond; the harbour sells precision.

Trade Hack — Set the throttle limit by measuring, not guessing: run the boat one pond length (30 m) at full stick and time it. 30 m in ~35 s = 0.86 m/s = correct for a 1/50 tug. Log the end-point percentage on the boat’s fleet card so a replacement ESC is configured identically in two minutes.

2.5 The quayside crane and harbour furniture

The container terminal’s **quay crane** is a Works-built machine, not a catalogue item, and it deliberately follows Construction Division doctrine (Volume 4): **lead-screw electromechanical actuation, no hydraulics**, because the rental fleet’s repair-speed rule applies on the waterfront too. Specification:

Parameter	Value
Type	Gantry crane on a 2.4 m quay rail, manual traverse lock per berth
Hoist / luff / slew	Three lead-screw axes, 25 kg-class digital servos + 550 brushed gearmotors from standard park spares stock
Spreader	Magnetic twist-lock spreader picking 1/50 scale containers (steel top-plate insert in each container)
Payload	1.2 kg rated — three loaded scale containers
Control	Same two-stick console mapping as the Construction Division cranes; customers keep their crane badge qualification
Build cost	≈ \$850 in materials; two Artisan-weeks

Harbour furniture completes the set: fendered berth faces (dock-bumper rubber strip on the quay capping), a bulk-loading chute gravity-fed from a quay-top hopper, mooring bollards (turned aluminium, glued to the capping — decorative but photograph like the real thing), berth number boards that match the harbour master's console display, and a marked **fairway** of small green/red lateral buoys so traffic rules can be taught with real-world conventions. None of this furniture costs more than a few hundred dollars, and together it is what makes the harbour read as a port instead of a pool.

2.6 Harbour session SOP

SOP M-01 — Harbour session turn-around · Rev 1.0 · Owner: Marine Lead · PPE: none (staff: waders for water entry only) · Tools: retrieval pole, towel, fleet tablet · Frequency: every Shift

1. Receive returning vessel at the landing stage; hoist by lifting the hull, never the superstructure.
2. Visual check: rudder linkage free, prop clear of weed/line, hatch seals seated, no water sloshing in the hull (tilt test).
3. Swap battery (3:1 rotation, park canon); log outgoing pack voltage in RC WORLD OS.
4. Wipe deck and superstructure; check fender strips and push-knees for damage.
5. Confirm RCW Node heartbeat on the fleet tablet before reissue.
6. Any water found inside the hull → tag **status: maintenance**, route to The Works (SOP M-20, Chapter 7). Never reissue a wet boat.

3. Fast Electric Boats

3.1 Hull types and what the park runs

Fast-electric ("FE") boating divides into three hull families, and the park runs all three in carefully separated roles:

- **Deep-V monohulls** — self-righting versions are the backbone of the rental fleet. A V-hull is forgiving in chop, turns predictably in both directions, and modern RTR models right themselves after a capsize, which eliminates 80% of retrieval events. This is the only hull type customers rent unsupervised.
- **Catamarans** — faster and flatter-cornering, but they trap air under the tunnel and **blow over** backwards when a gust gets under the bow at speed. Cats are premium-tier, marshal-supervised.
- **Hydroplanes / outriggers** — the fastest and most fragile class; sponsons touch the water at three points and any wake upsets them. Hydros are staff-demonstration and club-race hardware only, never rented.

3.2 Powertrain doctrine: brushless and water-cooled

All FE boats run brushless motors — the rental tier mostly **outrunners in the 2000-2300 Kv class on 3S** (e.g. the Spektrum 2300 Kv outrunner in the Pro Boat Recoil 2), the premi-

um tier 4-pole inrunners on higher voltage (the Sonicwake 36 V3 runs a 1350 Kv 4-pole on 8S with a 160 A ESC). Two facts drive marine powertrain maintenance doctrine:

1. **Water cooling is structural, not optional.** FE motors and ESCs are cooled by pond water picked up at the rudder or a hull scoop and pumped through jackets. A kinked or blocked line cooks a \$60 ESC in one Shift. Cooling-loop flow check is a daily-issue item (Chapter 7).
2. **The flex-shaft drivetrain is a wear system.** Power reaches the prop through a flexible steel cable running inside a stuffing tube. The cable must be pulled, cleaned and re-greased on a fixed schedule or it wears, whips and eventually snaps — the single most common FE failure in rental service (Chapter 7).

3.3 The speed fleet — real boats

Role	Model	LOA	Power	Speed	Street price (2026)
Rental trainer ×2	Pro Boat Sonicwake 24 (self-righting deep-V)	610 mm	Brushless, 3S	~40 km/h	≈ \$320 (battery + charger incl.)
Rental standard ×3	Pro Boat Recoil 2 V2 26" (self-righting deep-V)	660 mm	2300 Kv outrunner, 100 A, 3S	~55 km/h	≈ \$420
Rental value ×1	Volantex Vector SR80 Pro (self-righting deep-V)	800 mm	1800 Kv, 60 A water-cooled, 4S	~65–70 km/h	\$180–340 depending on channel
Premium mono ×1	Pro Boat Sonicwake 36 V3 (self-righting deep-V)	915 mm	1350 Kv 4-pole, 160 A, 8S	95+ km/h	≈ \$500
Premium cat ×1	Pro Boat Blackjack 24 V2 catamaran	610 mm	Brushless, 3S–4S	~50 km/h	≈ \$400
Show/club hardware	Pro Boat Blackjack 42 8S cat; TFL-class fiberglass race hulls	1,070 mm	Twin/big brushless, 8S	100+ km/h	\$730+ (Blackjack 42 at \$729.99)

Total speed-pond powered fleet: **8 vessels** (2+3+1+1+1 rental/premium) with show hardware owned by the park's race club program, not the rental pool.

Safety Warning — An 8S catamaran at 100 km/h carries more kinetic energy than anything else a customer can touch in this park, and unlike a car it cannot be fenced. **8S hardware never goes to unsupervised customers.** It runs only in marshal-controlled demonstration and club-race windows, with the pond cleared of all other traffic and spectators behind the setback line (\$6.8). This is division law.

3.4 Speed-pond rules and racing formats

The park adapts the established national model-powerboat frameworks (NAMBA and IMPBA publish the reference rulebooks) to a small commercial pond:

- **Course.** A regulation NAMBA electric oval — 30 ft (9.1 m) turn radii and a 170 ft (51.8 m) straight, 528.5 ft (161 m) per lap — is the design reference; the park's pond runs a **scaled oval of 25 m straights and 8 m turn radii ($\approx$ 100 m per lap)**, which suits 60–90 km/h rental hardware. Turns are driven in one fixed direction per the posted course card.
- **Buoys.** Per national practice: bright fluorescent red/orange turn buoys, contrasting green entrance buoys, 150–300 mm diameter, soft (non-injurious foam/plastic), anchored taut so they cannot walk in wind.
- **Heat format.** Three to six boats per heat; one-minute pit time, 30-second mill time to the flying start, then 4–6 laps; a heat never exceeds five minutes of running — which nests two heats plus turnaround comfortably inside one 20-minute Shift.
- **Right of way.** Overtaking boat stays clear; a boat that goes dead in the water is retrieved under yellow (all boats to idle lane) by the retrieval tug — the marine adaptation of the Tow-Truck Retrieval Protocol (Chapter 8).
- **Buoy courses for solo Shifts.** Outside race windows, the oval is supplemented by a slalom string and a timed “hot lap” gate pair wired to the RCW Node lap timer, so a solo customer always has a scored task.

3.5 Wake and shoreline erosion management

Wake is the speed pond's pollution. A 90 km/h monohull throws a wash that will chew an unprotected earth bank into a turbid soup within a season, and turbidity kills the harbour pond's clear-water submarine viewing next door. Controls, in order of importance:

1. **Hard edge on the racing perimeter.** The speed pond's long banks are finished with concrete block quay edging or rip-rap over liner (Chapter 6), not soft planted shores. Wave energy breaks on rock, not soil.
2. **Setback of the course.** Turn buoys sit $\geq$ 4 m off the banks so the highest-energy wash arrives attenuated.
3. **Wave-absorbing beach at the downwind end.** A 1:4 sloped cobble beach dissipates rather than reflects wake — reflected standing chop is what makes small ponds undriveable during long race sessions.
4. **Throttle governance between events.** Rental FE boats are software-limited (ESC profile) to $\sim$ 55 km/h; full power is a race-window setting applied by Artisans.
5. **Floating turbidity curtain** available for deployment between the ponds' shared service channel during dredging or bank repairs.

4. Sail Program

4.1 One-design or nothing

Sail racing rewards boat development — which is precisely what a rental fleet must suppress. The park therefore runs a strict **one-design fleet**: identical hulls, rigs and sails,

maintained to identical trim, so results are decided by the customer's hands, mirroring the Motorsport doctrine of lap-time parity within 2-3%.

The class decision:

Candidate class	LOA	Street price (2026)	Assessment
DragonForce 65 (DF65) V8 — RG65-class one-design	650 mm	≈ \$375 PNP + ~\$60 radio	Selected. The largest one-design RC sail class in the world, with an active international class association; ABS hull shrugs off dock knocks; three rig sizes cover 1.5-10 m/s; parts are catalogue items
DragonFlite 95 (DF95) V3	950 mm	≈ \$520-580 PNP	Club/flagship tier: two boats for staff demonstrations, match-race finals and the sailing badge examination — more presence on the water, same control layout
Generic RG65 / Volantex RG65-class RTR	650 mm	≈ \$120	Rejected for rental (rig quality and parts consistency below fleet standard); one kept as a try-out loaner

Fleet: **6 × DF65 V8** in rental service (plus 2 × DF95 club boats owned under the events budget, outside the 24-vessel count).

4.2 Wind on small ponds

Small-pond sailing is a wind-reading game more than a boat-speed game. Wind arriving over buildings and trees fragments into shifting lanes and holes; the sail area of the pond is deliberately positioned with the longest available **fetch aligned to the prevailing wind** (Volume 11 site plan) and with low planting — not walls — on the windward shore, because a solid windbreak creates a dead zone exactly where the beat begins. Operating envelope:

- **Below ~1.5 m/s:** drifting conditions; sessions convert to coaching/tuning clinics.
- **1.5-5 m/s:** A-rig conditions — the standard rental envelope.
- **5-8 m/s:** B-rig (smaller) fitted by Artisans; rental continues for badge-holders.
- **Above ~8 m/s:** sail rental closes with the rest of the scale waterspace.

Rig changes on the DF65 take under five minutes and are an Artisan task, never a customer task — rig tune is the fleet-parity variable.

4.3 Match-racing format

Sail sells best as **short-course match racing**: two boats, one umpired duel, first across the line. The format fits the Shift perfectly — a windward-leeward course of 40-60 m legs gives 3-4 minute races, so a 20-minute Shift delivers a best-of-three series with coaching between races. Fleet racing (all six boats) runs in club windows and weekend regattas under simplified RRS-style rules: port gives way to starboard, windward gives way to leeward, touch a mark = one penalty turn. The harbour master doubles as race officer with a horn and a course board.

Field Note — Junior sailors take to RC match racing startlingly fast because the feedback loop is visual and immediate. The park's schools program (Volume 9) uses the sail fleet as its STEM anchor: one hour teaches wind awareness, vectors and apparent wind with zero physical risk and near-zero running cost.

4.4 Why sail is the cheapest fleet to run

The sail program's unit economics are the best in the park, and the plan exploits them deliberately:

- **No propulsion energy.** The DF65 runs two servos from a 4×AA/NiMH pack (or a small LiFe) — a “battery cost per Shift” measured in fractions of a cent, versus \$0.15–0.40 of LiPo cycle depreciation on a powered boat. The 3:1 battery doctrine is trivially cheap to honour.
- **Almost no drivetrain.** No motor, ESC, cooling loop, flex shaft or stuffing box — the entire Chapter 7 powertrain maintenance regime reduces to servo checks, rigging inspection and hull care.
- **Slow failure modes.** Sails stretch and lines chafe over months, not minutes; a season's consumables per boat (a sail set, rigging line, a servo) come to under \$80.
- **Weather complementarity.** Sail is at its best in exactly the fresh-breeze conditions that degrade scale-harbour precision, keeping the waterspace earning across a wider weather band.

Result: after capital recovery, a sail Shift at \$15 carries the highest contribution margin of any product on the water (Chapter 9).

5. Submarines

5.1 Static divers versus dynamic divers

Two fundamentally different machines share the name “RC submarine”:

- **Dynamic divers** are positively buoyant boats that dive only by driving: forward speed plus dive planes push them under, and the moment power stops they float back up. Cheap (\$40–150 toy-grade), intrinsically self-recovering, and unserious-looking underwater. The park uses two as **training/taster hardware** in the viewing bay.
- **Static divers** carry a **ballast system** that takes on water to destroy reserve buoyancy, so the boat submerges and hovers like the real thing. This is the premium product — and the recovery problem (\$5.5).

Ballast technology, in ascending order of control fidelity:

System	How it works	Character
Pump/flood tank	A roll pump floods and empties a tank	Simple, proven (Thunder Tiger Neptune SB-1); coarser depth control
Piston tank	A motor-driven piston draws in a metered water volume	Precise, repeatable trim; the enthusiast standard (Arkmodel Dragon Shark II ships with one)

System	How it works	Character
Compressed-gas blow	Stored gas blows the tank dry instantly	An emergency-surface device more than a control system; fitted as backup on premium builds

5.2 The park's submarine hardware

Role	Model	Type	Price (2026)	Notes
Premium experience boat	Arkmodel Dragon Shark II 1/72 (piston tank), built in The Works	Static diver	Kit ≈ \$430; ≈ \$700–800 complete with electronics	866 mm; kit includes the piston-ballast system and sealed running gear; the definitive rental static diver
Premium/show boat	Thunder Tiger Neptune SB-1-class static diver	Static + dynamic	\$535–950 (new-old-stock/second-hand)	774 mm, 7.7 kg; pump-ballast; factory auto-surface protection on low battery, weak signal or leak detection — exactly the fail-safe doctrine the park requires
Trainer/taster x2	Toy-grade dynamic divers (27/40 MHz)	Dynamic diver	\$40–80	Run inside the viewing bay only; also cannibalized for low-frequency receiver boards

Both static divers use a **watertight cylinder (WTC)** architecture — all electronics and ballast machinery live in one sealed acrylic tube inside a free-flooding outer hull — which is what makes them maintainable: one o-ring-sealed cylinder to service instead of a whole “waterproof” hull to chase leaks in.

5.3 Radio physics in water — the honest constraint

Radio attenuation in water rises steeply with **frequency** and with **conductivity**. The practical consequences, well established across the RC-submarine community and consistent with basic EM physics:

- **2.4 GHz — the modern standard band — barely penetrates water at all.** A 2.4 GHz-equipped boat is controllable only with its antenna at or above the surface (periscope depth). This is why the RCW Node cannot ride a submerged hull, and why submarines are the telemetry exception in the park's fleet doctrine.
- **The legacy low bands (27 MHz, 40 MHz, 72/75 MHz)** penetrate fresh water usefully: with an insulated receiver antenna, positive control to roughly **2–6 m depth** at short range is routine, and experienced operators report more in very clean water. These frequencies are why serious submariners hoard legacy FM radio gear — production largely ceased when the industry moved to 2.4 GHz.
- **Salt water is effectively opaque** at all hobby frequencies — irrelevant to the park's fresh ponds but worth stating in staff training so nobody “tests” a sub on a seaside event hire.

- **Antenna discipline:** the submerged receiving antenna must be fully insulated from the water, with the cut end sealed; an unsealed nick in the antenna wire reads as “random signal loss at depth” and wastes hours of fault-finding.

Park doctrine: submarines run **75 MHz or 40 MHz FM surface-frequency gear** (region-dependent; legal hobby allocations vary by country and the local band plan is confirmed in Volume 8 procurement) with a frequency board at the sub bay because legacy FM has no automatic frequency hopping — two subs on one channel is a guaranteed double sinking. Both boats also carry the fail-safe described next.

5.4 Fail-safe and depth doctrine

- **Loss-of-signal = surface.** Every static diver is configured so that signal loss, low battery or leak detection drives the ballast system to blow/empty and the planes to rise — the Neptune SB-1 ships with this “auto-surface” logic, and the Arkmodel build replicates it with a ballast-control failsafe unit.
- **Depth is limited by design, not discretion:** the submarine bay is 1.5 m deep (\$6.3); at that depth even 40 MHz control margin is generous and a bottomed boat is visible and reachable with a pole.
- **One sub in the water at a time** during rental sessions; the second boat is the hot spare on charge.

5.5 Clear-water viewing windows

A submarine you cannot see is a radio-shaped hole in the water. The sub bay is therefore built for **spectating**:

- An **acrylic viewing window** (~1,200 × 600 × 40 mm marine-grade cast acrylic panel, ≈ \$900–1,500 supplied) is set into the concrete wall of the sub bay at child-eye height, framed like an aquarium panel and shaded by the quay canopy to kill surface glare.
- Water clarity in the bay is held by the division’s filtration and UV clarifier loop (\$6.5) plus a strict **no-disturbance rule**: the bay is hydraulically separated from the harbour basin by a baffle wall, so barge traffic cannot stir sediment into it.
- A fixed **underwater camera** feeds the marina hut screen and the customer app — the customer pilots partly by direct view through the window, partly by the feed, which is itself the premium experience’s signature.

5.6 Recovery procedures for sunk assets

Sinkings will happen; the plan treats them as routine, not crisis.

SOP M-05 — Submerged-asset recovery · Rev 1.0 · Owner: Marine Lead · PPE: waders, grip gloves · Tools: reach pole set, magnet rig, hook rig, retrieval tug, tag line · Frequency: as required

1. Mark the last-seen position from the harbour master’s console (RCW Node last fix for surface boats; viewing window/camera for subs).
2. Attempt self-recovery first: for a static diver, command emergency blow and wait 60 s — most “sinkings” are trim errors that resolve themselves.
3. Shallow water (< 600 mm): recover with the 4 m reach pole and hook head from the bank or landing stage. Never wade during public sessions.

4. Deep bay (1.5 m): deploy the **magnet rig** (a 75 mm neodymium recovery magnet, ~90 kg pull class, ≈ \$25, on a throw line) for steel-ballasted hulls, or the weighted **hook rig** drawn across the bottom on two lines for non-ferrous hulls.
5. If the asset cannot be raised in 15 minutes, buoy the position, log it, and recover after close with waders — the bay is 1.5 m deep and this is a two-person, feet-on-bottom job with a spotter, done under the park's shallow-water-entry rules (§6.8), not a diving operation.
6. Any submerged LiPo/Li-ion pack goes to the quarantine bunker per park battery doctrine before the hull enters The Works for strip-down (Chapter 7).

Trade Hack — Epoxy a 30 mm mild-steel washer inside the bow and stern of every non-ferrous hull in the marine fleet during commissioning. It costs nothing, is invisible, and turns every future magnet-rig recovery from a 20-minute dredge into a 2-minute lift.

6. Pond & Harbour Design

6.1 The two-pond strategy

One pond cannot serve both programs. Scale harbour work needs calm, clear, shallow water with hard vertical edges; fast-electric needs length, fetch-managed open water and energy-absorbing banks. Mixing them costs both: wake ruins docking and turbidity ruins the viewing window, while a harbour full of quay furniture leaves no racing line. The division therefore builds **two hydraulically independent ponds** sharing one plant room:

Parameter	Harbour pond	Speed pond
Plan dimensions	30 × 18 m (540 m ² water)	60 × 25 m (1,500 m ² water)
Shape	Rectangular basin with quay walls, one soft corner	Rounded-rectangle “stadium” plan matching the oval course
General depth	400–500 mm	500–600 mm
Special depth	Submarine bay 1.5 m deep, 6 × 4 m , baffled off the basin	None (uniform)
Edge treatment	Vertical concrete-block quay walls, fendered berths, timber-capped	Concrete edge beam + rip-rap on straights; cobble wave beach at downwind end
Water volume	≈ 250 m ³ (incl. bay)	≈ 820 m ³
Programs	Scale harbour, sail (shared, scheduled), submarines	Fast electric, retrieval training

Depth doctrine: **400–600 mm** is the working band across both ponds. It is deep enough that keels, rudders and water pickups never ground, deep enough to stay cool and resist algal soup, but shallow enough that a staff member in waders can walk every square metre — which converts asset recovery, buoy maintenance and liner inspection from a diving problem into a housekeeping task. The sole exception is the submarine bay, whose 1.5 m depth exists to give static divers a genuine water column under the viewing window.

6.2 Ground works and structure

- Cut/fill balanced excavation into the site's 2–4% natural grade; the harbour pond floor is laser-screeded to ± 20 mm, falling gently to a **silt sump** at the deepest corner where the bottom drain sits.
- Quay walls are **hollow concrete block on a strip footing, rendered**, built inside the liner line with the liner dressed up behind them — the liner does the waterproofing, the wall does the shape and the abuse-taking. Timber capping gives the quays their port character and a warm edge for leaning customers.
- The submarine bay is a **cast concrete box** (the acrylic window demands a rigid, crack-free frame), waterproofed with a cementitious tanking slurry and mechanically joined to the EPDM of the main basin with a clamped, sealed flange detail — the one genuinely specialist joint in the build, executed by the liner supplier's approved installer.

6.3 Liner selection: EPDM versus concrete

Criterion	45–60 mil EPDM rubber	Reinforced concrete throughout
Material cost	≈ \$8–22 /m ² (45 mil ≈ \$12–17 /m ² typical 2026) + underlayment \$3–4 /m ²	≈ \$500–1,400 /m ² installed for structural water-retaining work
Life	20–30 years typical; 40+ with underlayment and covered edges	50+ years, if it doesn't crack
Freeze-thaw / ground movement	Flexes, unaffected	Principal failure mode; needs joints, sealants, maintenance
Repair	Patch kit, same-day, ≈ \$20	Saw-cut, resin injection, specialist
Fish/plant safe	Yes (standard for koi ponds)	Needs curing washdown/sealer
Verdict	Selected for both pond bodies	Used only where structure is needed: quay footings, sub bay, plant chambers

The hybrid — **EPDM membrane basins with concrete structures standing inside them** — is standard best practice in commercial water-feature construction and gives the park concrete's shape-making where it matters and rubber's forgiveness everywhere else. Underlayment is non-negotiable: 300 g/m² (8–10 oz) non-woven geotextile under every square metre of liner, carried up behind walls, per liner-industry practice.

Trade Hack — Order the EPDM as few, huge factory panels rather than many small ones: field seams are the only part of a rubber pond that ever fails young. A 30 × 18 m basin can be lined with two factory panels and one taped seam if the supplier is told the panel plan at quotation time.

6.4 Filtration, skimmers and turnover

The clarity standard is set by the submarine window, and clarity is bought with turnover, mechanical filtration and UV:

- **Turnover:** harbour pond full volume every **4 hours** (≈ 62 m³/h duty); speed pond every **8 hours** (≈ 100 m³/h) — the speed pond needs less biology-fighting because wake aeration suppresses stratification, but more leaf/silt handling.

- **Intake:** floating-weir **skimmers** (4 on the harbour pond, 6 on the speed pond) take the top millimetre — where pollen, dust, leaves and sunscreen film travel — plus one bottom drain per silt sump.
- **Plant:** a shared plant room between the ponds houses drum or bead **mechanical filters**, a pumped **UV clarifier** loop sized for the harbour pond (algae bloom prevention; the window bay water passes the UV bank every cycle), and low-head, high-flow pumps (≈ 1.2 – 1.8 kW total connected load — small ponds reward oversized pipe and low-speed pumps).
- **Make-up water:** municipal top-up through an air-gap break tank; evaporation in a warm summer month can run 100–150 mm across both ponds (≈ 25 – 30 m³), which is a budget line, not a surprise.
- **No fish, minimal planting** in the operating basins: fish waste is a clarity tax, and planted margins tangle props. Planting lives in a separate ornamental margin strip outside the operating water, sharing the look but not the water body.

Worked example — harbour-pond pump sizing. Volume ≈ 250 m³; target turnover 4 h $\rightarrow$ duty flow $Q = 250 \div 4 = \mathbf{62.5\ m^3/h}$ ($\approx \mathbf{17.4\ L/s}$). Head: static lift skimmer-to-filter ≈ 0.8 m; friction in 40 m of 110 mm PVC at 17.4 L/s ≈ 1.8 m/s velocity $\rightarrow \approx 2.6$ m friction loss (Hazen-Williams, $C=150$, incl. fittings at +30%); filter and UV pressure drop ≈ 2.5 m $\rightarrow$ **total dynamic head ≈ 5.9 m**. Hydraulic power = $\rho g Q H = 1000 \times 9.81 \times 0.0174 \times 5.9 \approx \mathbf{1.0\ kW}$; at 65% wire-to-water efficiency, a **1.5 kW low-head circulation pump** (or two 0.75 kW units for redundancy — preferred) carries the duty. Two smaller pumps also allow a half-flow night/winter economy mode, cutting the plant's annual energy cost by roughly a third. The speed pond's 100 m³/h loop is sized identically and shares the plant room but not the water.

6.5 Wind fetch and shelter

Fetch — the open-water distance wind blows across — sets wave height, and on the speed pond 60 m of fetch in a 10 m/s blow builds genuine chop. Design responses:

- The speed pond's **long axis is set across, not along, the prevailing wind** where the site plan allows (final orientation is resolved with Volume 11's master plan), trading a little racing headwind for much shorter effective fetch.
- The harbour pond hides behind the marina hut, quay canopy and a 2.5–3 m evergreen shelter belt on the windward side — solid-enough shelter is appropriate here because the harbour doesn't need wind, unlike the sail area (§4.2), which gets a deliberately open, low-planted windward shore.
- The wave beach (§3.5) sits at the downwind end of the speed pond where wind-driven and wake energy both arrive.

6.6 Retrieval fleet and recovery tools

- **Retrieval tug:** one dedicated, staff-only workboat (a third Heng Long 3810 hull, stripped of scale detail, fitted with a bow push-pad, a towing horn and a Heavy-Node) lives on permanent standby at the marina hut. Dead rental boats are towed in by staff — or, in the marine version of the **Tow-Truck Retrieval Protocol**, by the customer under harbour-master supervision, converting the breakdown into gameplay exactly as the land divisions do.
- **Reach poles:** 4 m telescopic poles with interchangeable hook, net and magnet heads, racked at four stations around each pond (they double as the safety reach poles, §6.8).

- **Magnet and hook rigs** per SOP M-05 (\$5.6).
- **Waders and a throw line** at the marina hut complete the recovery kit; total tooling budget $\approx$ \$600.

6.7 Winter and algae management

- **Algae:** the UV clarifier loop plus 4-hour turnover prevents green-water blooms; string algae on warm quay walls is brushed weekly in season (checklist, Chapter 7). No algaecides in the operating water — they foam under prop wash and cloud the window bay.
- **Leaf season:** skimmer baskets emptied daily; a seasonal leaf net over the harbour pond's soft corner cuts the load by half.
- **Winter (temperate site assumption):** the ponds do not close, they slow — scale harbour and sail run on fair winter days (marine is the park's soft-weather division), while FE rental pauses below $\sim 5^\circ\text{C}$ when LiPo sag and customer appetite both fade. Pumps run continuously to keep water moving; in hard-freeze climates an air-bubble de-icer keeps the skimmer throats open, and the franchise localization annex (Volume 12) handles full winterization for cold-market sites.

6.8 Safety barriers, water depth and public protection

Water is the park's only drowning hazard and is treated with layered controls despite the shallow design depth:

1. **Barrier line:** a 1.1 m post-and-rail balustrade with mesh infill runs wherever the public can approach water deeper than 300 mm, broken only at the supervised landing stages and the viewing window bay. The quay edge zones inside the barrier are staff/supervised-customer areas.
2. **Depth honesty:** 400–600 mm of water is a genuine hazard only to unsupervised toddlers — which is exactly who theme parks host. The barrier rule is therefore absolute, and the landing stages have self-closing gates.
3. **Reach poles and throw line** stationed per §6.6; all marine staff hold basic water-rescue awareness training; the marina hut holds the division first-aid kit.
4. **Staff water entry** only in waders, only two-on-site, never during public sessions except emergency (SOP M-05).
5. **Electrical:** every mains circuit serving the ponds (pumps, UV, lighting, crane) is on 30 mA RCD protection with equipotential bonding of quay metalwork, per wet-location electrical code; the plant room is lockable and the ponds carry no mains equipment below water except the bonded, double-insulated pump intakes.
6. **Lightning:** 30/30 rule — thunder within 30 s of flash closes all water for 30 minutes, same doctrine as the Aviation Division (Volume 5).

Safety Warning — The single most dangerous object in this division is not a boat: it is a **spinning FE propeller in a boat being carried**. A 2300 Kv outrunner that bumps to full throttle in someone's hands will cut to the bone. Division law: transmitter throttle-locked and hull cradled prop-outward for every carry; boats are armed only on the stand at the landing stage, facing open water.

6.9 Pond complex capex summary

Item	Estimate
Excavation, grading, compaction (both ponds)	\$28,000
EPDM liner + underlayment, supplied & installed (~2,600 m ² incl. walls/overlap)	\$52,000
Quay walls, footings, timber capping, fendered berths	\$34,000
Submarine bay (cast box, tanking, acrylic window, frame)	\$18,000
Filtration plant, UV, pipework, skimmers, drains, plant room fit-out	\$38,000
Edges: rip-rap, wave beach, paths, balustrade	\$22,000
Marina hut, quay canopy, landing stages, course buoys & ground tackle	\$14,000
Contingency (~10%)	\$6,000
Pond complex total	≈ \$212,000

This sits inside the Phase 2 envelope of ≈ \$1.15 M alongside the Aviation Division's netted airfield (Volume 5) and the F&B/retail expansion; Volume 10 carries the consolidated schedule.

7. Boat Maintenance

7.1 Why water rewrites the maintenance manual

The land fleet's enemies are impact and dust. The marine fleet's enemies are **ingress, corrosion and galvanic decay** — slow, invisible, and fatal to electronics. Volume 7 carries the park-wide workshop doctrine; this chapter defines the marine-specific regime layered on top of it. Three principles govern:

1. **Every boat is assumed wet inside until proven dry.** Water finds a way; the regime is built around detecting it early, not preventing it perfectly.
2. **Water-side wear parts are consumables, not repairs.** Flex shafts, prop-shaft seals, stuffing-box grease, sacrificial hardware — replaced on schedule, never run to failure.
3. **Electronics are conformal-coated at commissioning, per park doctrine.** The mandatory acrylic/epoxy conformal coating specified for every RCW Node PCBA is extended in this division to every receiver and ESC board edge the factory left exposed, plus dielectric grease in every connector.

7.2 The post-session flush and dry-out

SOP M-20 — Marine end-of-day service · Rev 1.0 · Owner: Marine Artisan · PPE: nitrile gloves · Tools: fresh-water wash bottle, air line (low pressure), grease gun, fleet tablet · Frequency: daily, every boat that touched water

1. Rinse hull exterior and running gear with fresh tap water (pond water is clean, but the habit matters — event hires off-site will not be).

2. **Flush cooling loops** (FE boats): squeeze fresh water through inlet-to-outlet until it runs clear; blow residual water out with low-pressure air. A blocked outlet found now is a saved ESC tomorrow.
3. Open every hatch and radio box; boats dry overnight **open**, on their stands, bow-up so bilge water finds the drain point.
4. Tilt-test and inspect: any moisture in a radio box → dry, identify the ingress path (hatch seal, antenna exit, pushrod boot), fix before reissue.
5. Wipe prop shafts and rudder pivots with a lightly oiled cloth (corrosion film prevention).
6. Log completion per hull in RC WORLD OS; the system blocks next-day issue of any boat without a closed M-20 entry.

7.3 Drivetrain: stuffing boxes and flex shafts

The FE drivetrain is the division's highest-wear system. The flexible shaft runs inside a brass **stuffing tube** whose grease does three jobs: lubrication, vibration damping and — critically — **water sealing**, because the stuffing box is a deliberate slow-leak point that grease plugs.

- **Weekly (or every ~10 run-hours)**: pull each FE flex shaft; wipe old grease; inspect for kinked strands, bright wear bands or “birdcaging”; re-grease with marine-grade water-resistant grease (a \$12 tube services the fleet for a season); reinstall with correct end-float (the shaft must not bottom in the coupler — 1–2 mm float, or it walks forward under load and eats the prop-shaft/strut interface).
- **Replace on condition or 100 run-hours**, whichever first. A flex shaft is a \$10–18 part; a snapped one at speed usually holes nothing but always ends the Shift and sometimes whips the stuffing tube oval.
- **Scale boats** (fixed prop shafts): monthly shaft pull, grease, and check of the prop-shaft bushings for slop; a clacking shaft on a tug is bushing wear announcing itself.

7.4 Radio-box waterproofing and corrosion control

- **Radio boxes** on FE boats are sealed at commissioning: lid gasket checked, pushrod exits fitted with rubber bellows boots, antenna exit sealed with a dab of neutral-cure silicone, and a **fresh silica-gel sachet** placed inside at every weekly service (the sachet is also the moisture tell-tale — a colour-changed sachet means the box breathed water even if none pooled).
- **Connectors**: XT60s and servo plugs get dielectric grease; any connector that has been submerged is flushed with isopropyl alcohol, dried and re-greased — the failure is never the dunking, it is the green crust three weeks later.
- **Galvanic pairs**: stainless prop shafts in brass tubes with aluminium struts form classic galvanic cells. Doctrine: rinse (SOP M-20), oil-film wipe, and inspect the strut for white aluminium-oxide bloom monthly. On the two 8S show boats, a small zinc/magnesium **sacrificial pencil anode** on the strut is cheap insurance.
- **Ferrous fasteners are banned** from the marine fleet; replacements are stainless A2 or better from The Works' standard stock (Volume 7).

7.5 Hull repair: ABS and fiberglass

Hull material	Fleet examples	Field repair	Workshop repair
ABS (vacuum-formed/ blow-moulded)	Heng Long tugs, Volantex SR80, DF65, Recoil/Son- icwake canopies	Crack-stop drill + ABS ce- ment (solvent weld) in- side; tape outside; return to service same day	Solvent-weld ABS backing plate inside; fill, sand, re- spray. ABS glues to itself chemically — repairs are as strong as parent mater- ial
Fiberglass/FRP	TFL-class race hulls, Ark- model cargo ship, premi- um builds	Tape and retire from water that day (glass cracks wick water into the lamin- ate)	Grind back 20:1 scarf, wet-lay glass cloth + epoxy, fair, paint; 24 h cure minimum

Hull integrity is checked by the tilt/slosh test daily and by an annual **bathtub soak test** (loaded hull sits in the test tank for 30 minutes; interior inspected) during winter overhaul.

7.6 Prop balancing and running-gear tune

Out-of-balance props are the hidden tax on the whole drivetrain: vibration wears stuffing boxes, blurs radio boxes' seals, and on FE boats costs real speed. Every metal prop is **balanced on a magnetic prop balancer** ($\approx$ \$25 tool) at commissioning and after any strike: sharpen/dress the blades with a fine file, remove burrs, balance by sanding the heavy blade's back face, then polish. Plastic scale props are replaced, not dressed. Rudders and turn fins are checked square and sharp-edged monthly; a nicked turn fin on a cat is a handling fault customers will report as "the boat pulls".

7.7 Preventive-maintenance matrix

Task	Daily (per session day)	Weekly	Monthly	Season/annual
SOP M-20 flush & dry-out	●			
Cooling-loop flow check (FE, at issue)	●			
Hatch/radio-box seal visual	●			
Battery rotation & voltage log (3:1, park canon)	●			
Flex-shaft pull, in- spect, re-grease		●		
Silica-gel sachet swap		●		
Rigging & sail in- spection (DF65/95)		●		
Quay-wall algae brush, skimmer bas- kets	● (baskets)	● (walls)		
Prop condition & bal- ance check			●	

Task	Daily (per session day)	Weekly	Monthly	Season/annual
Scale-boat shaft pull & bushing check			●	
Rudder/strut fastener torque & anode check			●	
Ballast-system service (subs: piston seals, pump, failsafe test)			●	
Bathtub soak test, full strip, respray as needed				●
ESC/receiver conformal-coat inspection & renewal				●
Liner/quay underwater inspection (waders, closed day)				●

7.8 Failure-mode table (marine FMEA extract)

Failure mode	Typical cause	Detection	Effect	Prevention / response
ESC burnout (FE)	Blocked/kinked cooling line	No tell-tale outlet stream at issue	Boat dead on water, \$60-150 part	Daily flow check; flush per M-20; retrieve under yellow
Flex-shaft snap	Grease neglect, strike-kinked strand	Vibration rising over sessions	Instant power loss; possible stuffing-tube damage	Weekly pull/grease; 100 h replacement
Radio-box flood	Perished hatch seal, unbooted pushrod	Silica sachet colour; slosh test	Receiver/servo corrosion, delayed failure	Weekly seal check; conformal coating limits damage
Stuffing-box leak	Grease washed out	Bilge water after session	Progressive flooding aft	Re-grease on schedule; grease is the seal
Prop strike / thrown blade	Debris, buoy line contact	Audible; vibration	Imbalance cascade if run on	Line-free course discipline; balance after every strike
Sail-winch servo failure (DF65)	Water in servo, load spikes in gusts	Sluggish sheeting	Boat unsailable, drifts downwind	Weekly rig check; B-rig above 5 m/s; retrieval tug
Submarine fails to surface	Ballast jam, low battery under load	Auto-surface did not trigger	Sunk asset (recoverable)	Monthly ballast service; failsafe test; SOP M-05
Galvanic strut decay	Dissimilar metals, no rinse	White oxide bloom monthly check	Strut weakens, shaft misaligns	Rinse, oil film, anodes on 8S hulls
Battery over-discharge on water	Customer ignores sag	RCW Node under-voltage alert	LiPo damage; boat dead mid-pond	Node kill-to-idle at 3.4 V/cell + harbour-master recall, park canon

8. Marine Operations & Programs

8.1 Session formats on the 20-minute Shift

All marine products are sold in the park's canonical 20-minute Shift, with the ~30% battery buffer returned as per billing doctrine:

Product	Shift structure	Price (Year 1 canon)	Notes
Harbour Casual	1 Shift, one vessel, missions from the harbour master	\$15	Entry product; tug or fire boat
Harbour Operator	2+ Shifts with pit-stop battery swap	\$26	Battery swap staged as a "crew change" at the landing stage
Speed-pond Casual	1 Shift, rental mono, buoy course + hot-lap gate	\$15	Throttle-managed fleet
Speed-pond Premium	1 Shift, Sonicwake 36/ Blackjack 24 class, marshal present	\$22	Requires speed badge (Volume 9)
Sail Casual / Match	1 Shift, DF65; best-of-three match when paired	\$15	Highest-margin product on water
Submarine Piloting	1 Shift, static diver, instructor beside customer	\$22 (premium)	Two customers per hour maximum; window + camera view
Fire-boat Show slot	Scheduled demo, fire boat + monitor targets	Included / events	Also sold into birthday packages

The **harbour master** is the division's signature staff role: part air-traffic controller, part game master. One harbour master runs the harbour pond's traffic, missions and scoring from the marina hut console; the speed pond runs under a marshal during premium/race windows and under passive Node monitoring otherwise. Minimum marine staffing is two (harbour master + Artisan), matching the division's modest throughput.

8.2 RCW Node adaptation for boats

The land fleet's telemetry doctrine carries to water with three adaptations:

- **Fit:** Micro-Node in the sealed radio box, conformal-coated per park canon, powered downstream of the main switch. GPS antenna under the ABS deck or canopy — plastic hulls are RF-transparent; the one placement rule is never under a carbon or metallized deck.
- **GPS telemetry on water** is actually better than on land: no fences, walls or vehicle clutter, so position fixes are clean. Node reports position, speed and pack voltage over the park Wi-Fi mesh; the customer app shows a live harbour chart, and speed-pond hot laps are timed by geofence gates exactly as Track B laps are.
- **Geofence as virtual harbour limits:** each pond's operating area is a Node geofence. Breach (a boat wedged behind a landing stage, or headed for the skimmer throat) triggers the standard PWM-intercept response — throttle to idle, alert to the harbour master — rather than a hard kill, because unlike a car, a killed boat is a stranded boat; idle-home keeps steerage. Under-voltage below 3.4 V/cell triggers the same throttle-to-idle plus a recall order, per park canon.

- **The exception:** submarines run legacy low-band radio and carry no Node (2.4 GHz does not penetrate water, Chapter 5). Mitigation: one-at-a-time operation, instructor supervision, camera coverage and the auto-surface failsafe. This exception is documented here and must be honoured in Volume 13's fleet-coverage tables.

8.3 Regattas and events

- **Weekly club nights** (sail Wednesday, FE Friday): entry-fee racing under NAMBA/IMPBA-style heat rules (§3.4), building the resident community every division needs.
- **Monthly Harbour Cup:** the five harbour challenges (§2.2) run as a scored gauntlet; leaderboard in the app.
- **Quarterly regatta weekend:** invited hobby clubs bring their own boats (scrutineered: no IC/nitro, insurance-listed classes only, park frequencies respected) — the marine version of the Motorsport open meet, and a proven low-cost acquisition channel.
- **Pit-to-Port corporate format** (from \$1,400 / 2 h / 20 pax, park canon): teams rotate through excavator, dump truck, barge tug and quay crane stations; the scoring chain runs end-to-end through RC WORLD OS. This is the park's best demonstration that the divisions are one industrial system, not four attractions.
- **Fire-boat shows** and **submarine windows** are scheduled spectacles (11:00 / 14:00 / 16:00) that cost almost nothing and anchor the terrace dwell.

8.4 Badge tasks feeding the RC WORLD License

The division contributes a marine ladder to the tiered license program (structure and rewards are Volume 9's canon; the tasks are defined here):

☐

Deckhand — Toolbox Talk (marine module) passed; one clean harbour Shift, no barrier contacts

☐

Coxswain — precision docking under 4 minutes with ≤ 1 fender touch; barge push completed

☐

Tug Master — two-tug tanker assist as lead tug; container run with crane partner

☐

Race Skipper — speed-pond license: 10 clean hot laps + heat finish without a buoy cut; unlocks premium FE hire

☐

Sail Grade 1-3 — laps of the windward-leeward course; match-race win; B-rig session above 5 m/s

☐

Submariner (premium badge) — supervised static-diver Shift with a hover held ± 0.2 m for 60 s at the window

8.5 Weather-hold and emergency procedure

SOP M-30 — Waterspace weather hold & emergency clear · Rev 1.0 · Owner: Duty Manager ·
PPE: n/a · Tools: PA, harbour master console, anemometer feed · Frequency: as triggered

1. **Wind:** at sustained 8 m/s, close scale harbour and sail rental (badge-holders on B-rig may continue at harbour master's discretion); at 10 m/s, close all water. RC WORLD OS pushes rebooking offers to affected Shifts automatically.
2. **Lightning:** on the 30/30 trigger, sound the water-clear tone. All boats return at best speed; boats that cannot make the landing stage in 90 s are throttled to idle via Node and abandoned at their position — **staff do not remain at the water's edge to retrieve during a lightning hold.** Recovery after the all-clear.
3. **Person in the water:** harbour master calls all-stop (fleet-wide throttle-to-idle via console), one staff member effects reach-pole/throw-line rescue, second staff member clears the barrier line and calls per the park emergency plan. The ponds' 400–600 mm depth makes this a stand-up assist in almost every case; the drill is practised monthly regardless.
4. **Fuel/battery incident afloat** (smoking pack): do not bring the boat to the landing stage; steer it to the designated open-water quarantine corner, let it burn out or sink, recover with the magnet rig after 30 minutes, quarantine the pack per park battery doctrine.
5. Log every hold and clearance in RC WORLD OS with timestamps; the weather-hold dataset feeds the park's seasonal scheduling model (Volume 10).

8.6 Daily operations checklist

- ☐ Pond visual: water level, clarity, debris; skimmer baskets emptied
- ☐ Plant room: pump pressures, UV lamp hours, no leak alarms
- ☐ Course check: buoys on station, lines taut and submerged, landing-stage gates self-close
- ☐ Retrieval tug: charged, Node heartbeat, push-pad secure
- ☐ Reach poles, throw line, first-aid kit at stations
- ☐ Fleet issue check: cooling tell-tales, hatch seals, M-20 closed for every hull
- ☐ Frequency board set (submarine days); wind forecast reviewed against sail/scale limits
- ☐ Harbour master console: mission list loaded, berth board reset, leaderboard live

9. Fleet & Economics

9.1 The canonical 24-vessel fleet

Phase 2 canon assigns the Marine Division **~24 powered vessels**. The build-out:

#	Class	Model	Unit street price	Qty	Line total
1-4	Harbour tug (rental)	Heng Long 3810 work tug, monit-or-upgraded	\$130	4	\$520
5	Fire boat (rental)	Heng Long 3810 fire variant, pump-upgraded	\$140	1	\$140
6-7	Heavy tug (rental-premium)	Pro Boat Horizon Harbor 30"	\$400	2	\$800
8	Flagship tug (premium)	Bancroft Fairplay VI 1/50	\$730	1	\$730
9	Cargo ship (staff)	Arkmodel-class civil hull, workshop-completed	\$600	1	\$600
10-11	FE trainer	Pro Boat Son-icwake 24	\$320	2	\$640
12-14	FE standard	Pro Boat Recoil 2 V2 26"	\$420	3	\$1,260
15	FE value	Volantex Vector SR80 Pro (ARTR + radio)	\$260	1	\$260
16	FE premium mono	Pro Boat Son-icwake 36 V3 (8S)	\$500	1	\$500
17	FE premium cat	Pro Boat Black-jack 24 V2	\$400	1	\$400
18-23	Sail one-design	Joysway Dragon-Force 65 V8 + radio	\$435	6	\$2,610
24	Submarine (premium)	Arkmodel Dragon Shark II, completed	\$780	1	\$780
24a	Submarine (show/spare)	Thunder Tiger Neptune SB-1-class (NOS/used)	\$650	1	\$650
—	Retrieval tug (staff)	Heng Long 3810, stripped workboat	\$110	1	\$110
—	Barges (un-powered)	Shop-built hopper/container	\$120	6	\$720

#	Class	Model	Unit street price	Qty	Line total
—	Quay crane	Lead-screw electromechnical, The Works build (Volume 4 doctrine)	\$850	1	\$850
	Powered-vessel subtotal (the ~24)			24	\$9,890
	Division fleet total incl. staff/support assets				≈ \$11,570

(The second submarine and the retrieval tug are counted as one rental asset plus support assets; the roster is managed in RC WORLD OS `fleet_inventory` like every other division's.)

Investor Note — The entire marine powered fleet costs less than one Kabolite K970 showcase excavator. The division's capital is 95% civil works, 5% boats — the inverse of the land divisions — which is why the fleet can be refreshed aggressively (Chapter 9.4) while the pond amortizes over decades.

9.2 Battery logistics at 3:1

Park canon: three packs per powered vessel (one running, one charging, one rested). Sail's tiny receiver packs make the ratio trivially cheap; the FE premium boats dominate the line:

Fleet segment	Pack type	Packs (3:1)	Unit	Cost
Harbour tugs & fire boat (7 hulls)	2S 1800–2800 mAh Li-ion/LiPo, XT60	21	\$14	\$294
Cargo ship + retrieval tug	2S 5000 mAh LiPo	6	\$28	\$168
FE trainer/standard/value (6 hulls)	3S 2200–3200 mAh LiPo	18	\$22	\$396
FE premium (2 hulls, 8S = 2×4S)	4S 3200–5000 mAh 100C pairs	12	\$45	\$540
Sail (6 hulls)	5-cell NiMH / 2S LiFe RX packs	18	\$9	\$162
Submarines (2 hulls)	Sealed 12 V/2S service packs	6	\$30	\$180
Battery fleet total		81		≈ \$1,740

All charging happens in the park's bunkered charging room per canon (3.4–4.2 V/cell window, smart balance chargers); the marina hut holds only the day's rested packs in an ammo caddy. Any submerged pack quarantines before reuse (\$5.6).

9.3 Capacity and revenue per hour

Waterspace capacity per operating hour (3 Shifts/hour per station):

Water	Concurrent stations	Shifts/hour	Blended price	Revenue/hour (full)
Harbour pond (tugs, fire, crane pair)	6	18	\$15.50	\$279
Speed pond	4 (2 in premium windows)	12	\$16.75	\$201
Sail area (shared water, scheduled)	4	12	\$15.00	\$180
Submarine bay	1	2/hour (30-min premium format)	\$22.00	\$44
Division total	15	~44		≈ \$704/h full

At a planning-realistic 35% annual utilization across ~2,900 operating hours, the division books ≈ **\$210-260 k of Shift revenue in its first full year**, before events (Pit-to-Port corporates, regattas, birthday fire-boat packages) and its measurable uplift to F&B on the terrace — consistent with the Year 3 park revenue ramp to ≈ \$2.6 M carried in Volume 10, which owns the consolidated model.

9.4 Operating cost, spares and refresh

- **Spares budget:** 18% of fleet value/year (≈ \$2,100) — flex shafts, props, seals, servos, sail sets, ESCs — held in The Works' marine bin per the high-turnover spares doctrine.
- **Water/energy:** filtration plant ≈ 1.5 kW continuous ≈ \$1,600/yr; make-up water ≈ \$400/yr.
- **Fleet refresh:** rental workhorses (Heng Long, Recoil class) are replaced on a 2-season cycle from revenue; retired hulls become training airframes-equivalents, parts donors or gallery pieces.
- **Premium experiences carry the margin:** submarine piloting at \$22 with near-zero marginal cost, fire-boat show add-ons in birthday packages (\$349 base, park canon), and the Pit-to-Port corporate format are the division's profit concentrators; the \$15 harbour and sail Shifts are the volume base and the demographic gateway.

10. Volume summary & cross-references

The Marine Division converts a ≈ \$212,000 two-pond civil work and a ≈ \$11,600 fleet into the park's calmest, broadest-demographic product: a working miniature harbour run under a harbour master, a NAMBA-style speed oval, a one-design DF65 sail fleet with the best unit economics in the park, and a two-boat static-diver submarine program with an aquarium-grade viewing window. It opens in Phase 2 per park canon, honours every fleet doctrine (20-minute Shift, 3:1 batteries, 3.4-4.2 V/cell, RCW Node telemetry with geofence-as-harbour-limits, Tow-Truck Retrieval adapted afloat), and documents the park's one telemetry exception — submarines on legacy low-band radio, because 2.4 GHz does not penetrate water.

Cross-references:

- **Volume 4** — quay-crane control interface and lead-screw doctrine shared with the Construction Division; Pit-to-Port mission chain.

- **Volume 5** — Phase 2 sequencing partner; shared lightning (30/30) and weather-hold doctrine.
 - **Volume 7** — park-wide workshop doctrine under this volume's marine regime (SOP M-20, flex-shaft schedule, conformal coating); barge-building and Arkmodel/Billing kit builds as Artisan winter projects.
 - **Volume 8** — marine procurement: Horizon/Pro Boat dealer terms, Joysway class-legal DF65 supply, Arkmodel direct, legacy 75/40 MHz radio sourcing and regional frequency legality.
 - **Volume 9** — marine badge ladder (\$8.4) feeding the RC WORLD License; senior and schools programs.
 - **Volume 10** — consolidated Phase 2 capex (pond complex $\approx$ \$212 k civil inside Volume 10's \$231 k marine division line) and the division's revenue contribution in the park model.
 - **Volume 11** — pond siting beside the restaurant terrace; prevailing-wind orientation, shelter belts and fetch; barrier and path integration.
 - **Volume 12** — winterization and cold-climate pond annex for franchise sites.
 - **Volume 13** — Node fleet-coverage tables must record the submarine exception; harbour-master console, mission scoring, berth board and geofence-as-harbour-limits as RC WORLD OS modules.
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Volume 7 — Engineering & Workshop Manual

RC WORLD — Master Development Plan · Volume 7 of 13 **Revision:** 1.0 · **Date:** July 2026
· **Status:** Living document — bump revision on material change

Purpose of this volume. This is the technician's handbook for **The Works**, RC WORLD's central engineering and workshop facility, and the single most operationally load-bearing document in the master plan. Roughly 150 powered assets take the field in Phase 1, each crashed, over-driven and rained on by strangers up to a dozen times a day; the difference between a park that feels like a precision motorsport operation and a park that feels like a broken toy bin is entirely decided at the workbench. This volume specifies the workshop itself (floor plan, benches, ESD, air, ventilation), the tools at three investment tiers, the craft standards every Artisan is trained and audited against, a compendium of RC trade hacks, the full library of Standard Operating Procedures, the preventive-maintenance matrices driven by RC WORLD OS telemetry, repair scheduling mathematics, inventory management, FMEA-based failure analysis, the battery and charging rooms operated under the park's LiPo doctrine (3.4–4.2 V/cell operating window, 3:1 battery ratio, bunkered charging), the QC lane, a complete tutorial on building custom RC cars — required reading by founder's brief — and the bench-side of the RCW Node telemetry program. A competent RC hobbyist hired as a trainee Artisan should be able to reach productive competence from this volume plus supervised bench time.

Intended readers. Every Artisan and the Workshop Lead (this is their primary manual); the General Manager; procurement staff executing the tool and spares purchases (with Volume 8); division leads whose fleets this workshop keeps alive (Volumes 3–6); the RC WORLD OS team implementing the maintenance, inventory and telemetry modules (Volume 13); investors assessing whether the fleet-availability assumptions in Volume 10 are engineered or hoped for.

Chapters

1. The Works — Workshop Design
2. Tool Lists: Three Investment Tiers
3. Workshop Behaviours & Craft Standards
4. RC Trade Hacks
5. Standard Operating Procedures
6. Preventive Maintenance System
7. Repair Scheduling & Workflow
8. Inventory & Parts Management

9. Failure Analysis & FMEA
10. The Battery Room
11. The Charging Room
12. Quality Control
13. Building Custom RC Cars
14. Telemetry Hardware Bench Work
15. Volume Summary & Cross-References

1. The Works — Workshop Design

1.1 Design philosophy: a service department, not a hobby bench

The Works is sized and organized like a small vehicle dealership’s service department, because that is functionally what it is: a fleet of revenue-earning vehicles arrives broken and must leave billable. Three principles govern the layout:

1. **Flow, not rooms.** A vehicle enters at intake, moves through triage, repair, QC and back to the fleet staging shelf in one direction. No vehicle ever moves “backwards” past the QC lane without a new job ticket. One-way flow is what makes the RC WORLD OS fleet status trustworthy — the physical position of a vehicle always matches its database state (a seven-state machine: available → bound → on_track → pit → maintenance → charging → retired; see Volume 13 §6.3).
2. **Segregation of energy.** Batteries are the park’s dominant fire risk and are physically segregated: a **Battery Room** for storage and logging (Chapter 10) and a separate bunkered **Charging Room** (Chapter 11), both isolated from the main workshop by masonry. No LiPo is ever charged at a repair bench.
3. **Visibility as theatre.** The Works has a public-facing **guest viewing window (6.0 × 1.5 m)** along the pit lane. Customers watching an Artisan rebuild a differential is free marketing, reinforces the “miniaturized industrial complex” brand, and — usefully — keeps bench discipline honest. The glazing is specified in Volume 11, Chapter 6; this volume only requires that the wrenching bays face it.

1.2 Floor plan and zone schedule

Volume 11 (Chapter 6), the geometry authority, fixes The Works at **450 m² GFA (≈4,840 ft²)**. Inside that envelope, this volume’s working layout is a **~180 m² wrenching/electronics core** — the zones an Artisan moves between hourly — wrapped by the larger support rooms whose dimensions Volume 11 owns. Core allocation:

Core zone (~180 m ² total)	Area	Function	Key requirement
Intake & triage	20 m ²	Receiving shelf, triage bench, wash-down sink, air-blow station	Door to the intake yard off the pit lane; RC WORLD OS kiosk terminal
Wrenching bays (×4)	48 m ²	Mechanical repair benches, one Artisan each	1.8 m benches, tool shadow boards, parts trays; face the viewing window

Core zone (~180 m ² total)	Area	Function	Key requirement
Electronics bench (x2)	24 m ²	Soldering, ESC/servo work, RCW Node assembly & flashing	ESD-protected area, fume extraction
Machining corner	18 m ²	Lathe, drill press, grinder, rotary tool, vice work	Separated by partition; chip containment; eye-wash
3D printing & jig shelf	8 m ²	2 printers, filament dry-box, jig library	Ventilated enclosure exhaust
Paint & gluing booth corner	12 m ²	Spray booth, body prep, batch tire gluing	Ducted extraction (§1.7)
Fleet staging shelves	20 m ²	QC-passed vehicles awaiting return to service	Adjacent to QC lane exit
Circulation, lockers, docs	30 m ²	Artisan lockers, manual library, whiteboard, coffee	—

Support rooms inside the 450 m² GFA, per Volume 11 Chapter 6:

Support room	Area	Key requirement
Parts store	45 m ²	Bin shelving, kanban racks, consumables cabinet; adjacent to wrenching bays; single controlled entry
Battery Room	24 m ²	Storage-charge LiPo racks, IR/logging bench; masonry separation (Chapter 10)
Charging Room (bunker)	30 m ² internal (6.0 × 5.0 m)	Inside the building: filled-cell CMU, blast venting, FD90 door (Chapter 11)
QC lane	35 m ²	Rolling-road bench, scales, radio range check; direct exit to a 25 m fenced test strip
Intake yard access, plant, ancillary + wall/GFA allowance	≈136 m ²	Balance of the 450 m ² GFA — Volume 11 carries the dimensioned plan

The four wrenching bays are the capacity constraint of the whole park (see the loading math in Chapter 7); Volume 11's 450 m² shell reserves core floor area for a Phase 2 expansion to six bays without structural change.

1.3 Bench specification

All repair benches are built to one standard so any Artisan can work at any bay:

- **Dimensions:** 1,800 × 750 mm work surface, 900 mm working height (standing/high-stool), load rating ≥150 kg.
- **Surface:** 25 mm laminated hardwood or phenolic top, overlaid with a replaceable 600 × 900 mm cutting/soldering mat at each station. Electronics benches add a grounded ESD rubber mat (see 1.4).
- **Under-bench:** one 5-drawer roller cabinet per bay (personal kit lives here), one shelf for the in-progress vehicle's tote.

- **Over-bench:** shadow board for shared tools (outline-painted so a missing tool is visible at a glance), LED task light, power rail with 6 outlets + 2 USB-C PD, magnetic strip for hex drivers.
- **Each bay carries:** a bench vice (100 mm jaws, machining corner gets the 125 mm), a car stand (rotating, e.g. the ubiquitous 1/10 aluminum stands, \$10–15), two magnetic parts trays, and a labelled “job tote” system — one plastic tote per vehicle under repair, so a job can be shelved mid-repair without losing parts.

1.4 ESD protection

Everything downstream of the receiver — ESCs, gyros, RCW Nodes, telemetry PCBAs — is static-sensitive. The two electronics benches are run as an ESD-protected area to the intent (not full certification) of ANSI/ESD S20.20:

- Grounded dissipative rubber bench mats (Bertech or SCS class, \$40–80 per bench) bonded to building earth through a 1 MΩ resistor.
- Wrist straps at both benches; wearing one is mandatory whenever a bare PCBA (RCW Node, ESC with case opened) is handled. Handling a cased, plugged ESC during mechanical work does not require a strap.
- Field-service parts (spare ESCs, receivers, Nodes) stored and transported in metallized shielding bags, never in pink poly alone.
- No polystyrene cups, cling film, or tape dispensers on the electronics benches — the classic hidden charge generators.
- Humidity in the electronics zone kept $\geq 40\%$ RH where climate allows; dry-season operation adds an ionizing fan (\$150–300) if RH cannot be held.

Safety Warning. ESD damage is almost never immediate failure — it is latent damage that kills an ESC three weeks later on a customer Shift. You will never trace that failure back to the day someone handled the board in a fleece jacket. Follow the strap rule even though you have “never had a problem”.

1.5 Lighting

Lighting is specified by task, following general illumination-engineering practice for fine bench work:

Area	Maintained illuminance	Fixture guidance
General circulation & stores	300 lux	LED battens, 4000 K
Wrenching bays	750–1,000 lux at bench surface	Overhead + adjustable task lamp per bay
Electronics benches	1,000–1,500 lux at work point	Task lamp with diffuser; add a lighted magnifier (3-diopter ring light, \$40–80)
Machining corner	750 lux + machine-mounted lamp	No stroboscopic flicker near rotating machinery — specify high-frequency LED drivers
QC lane	750 lux, high CRI (≥ 90)	Paint/body inspection needs honest color rendering

Choose 4000 K neutral white throughout; 6500 K “cool daylight” causes eye fatigue on long shifts and 3000 K hides solder-joint defects.

1.6 Compressed air

A quiet workshop compressor feeds a ring main with drops at intake (blow-down station), each wrenching bay, the machining corner and the QC lane:

- **Compressor:** 1.0–1.5 kW ultra-quiet unit, ≥ 30 L receiver, e.g. California Air Tools 8010 class (\$180–280). Two smaller units beat one large one for redundancy.
- **Treatment:** water trap + coalescing filter at the manifold, regulator per drop. Vehicle blow-down at **max 3 bar (≈ 45 psi)** with a rubber-tipped nozzle — full pressure drives grit into bearings instead of off them.
- **Uses:** dust blow-down at intake (always outdoors or at the extracted intake station, never over an open bench), drying after wash, airbrush supply in the paint corner, clearing drilled holes in the machining corner.

Trade Hack. Fit a cheap inline “dust gun” with a fan nozzle at intake and keep a stiff 25 mm paintbrush hanging next to it. Brush-agitate while blowing and a post-Shift buggy is clean in 90 seconds; blow alone takes five minutes and redistributes half the dust into the electronics tray.

1.7 Ventilation: soldering and paint

Two extraction problems, two systems:

1. **Solder fume.** Rosin flux fume is a sensitizer; chronic exposure causes occupational asthma. Each electronics bench gets a fume extractor — either a filtered bench unit (Hakko FA-400 class, \$60–110, filters replaced monthly) or, preferred, an extracted arm ducted outdoors (Weller/Quick fume arm systems, \$250–600). The RCW Node assembly bench, which solders daily, must have the ducted option.
2. **Paint & solvents.** Polycarbonate body spraying, conformal coating and CA-glue tire work all release solvents. The paint corner is a bench with a spray booth — a hobby extraction booth with ducting to outside (Vevor/Master Airbrush class, \$90–180) is adequate for rattle-can and airbrush volumes. Bulk tire gluing happens at this bench, not at the wrenching bays. Conformal coating spray (Chapter 14) always happens in the booth.

The machining corner adds a shop-vac with cyclone separator for chips and grinding dust; never blow grinding dust with compressed air near the electronics zone.

1.8 Services summary for the architect

Hand Volume 11’s designer this checklist:

☐

450 m² GFA per Volume 11 Chapter 6, one-way flow intake → QC around a ~ 180 m² wrenching/electronics core; guest viewing window (6.0 × 1.5 m) to the pit lane at the wrenching bays; intake yard access

☐

Masonry-separated Battery Room (24 m²) and internal bunkered Charging Room (30 m² internal, 6.0 × 5.0 m: filled-cell CMU, blast venting, FD90 door)

☐

- ☐ 3-phase power to machining corner and Charging Room; 20+ double outlets distributed
- ☐ Dedicated circuits: charging banks (Chapter 11 load calc), compressor, extraction
- ☐ Ducted extraction: solder arm (electronics), spray booth (paint corner), printer enclosure
- ☐ Compressed-air ring main, 5 drops; compressor in acoustic cupboard
- ☐ Wash-down sink with silt trap at intake; eye-wash station at machining corner
- ☐ Floor: sealed concrete, light grey (dropped M2 screws must be findable)
- ☐ Wi-Fi mesh AP inside The Works (RC WORLD OS kiosks, Node flashing, telemetry bench)
- ☐ Door widths ≥ 1.2 m throughout (fleet trolleys), no steps on the vehicle flow path

2. Tool Lists: Three Investment Tiers

2.1 How the tiers work

The tool budget is split into three tiers with different ownership and replacement rules:

- **Tier 1 — Per-Artisan personal kit.** Issued to each Artisan on hiring, engraved/color-banded, kept in their bay drawer. The Artisan is accountable for it; a complete kit is checked at quarterly audit. Budget **\$420-600 per Artisan**.
- **Tier 2 — Shared bench equipment.** Lives on shadow boards and the electronics benches; anyone may use it, everyone returns it to its outline. Budget **\$4,500-6,500** for Phase 1.
- **Tier 3 — Capital equipment.** Machines with asset tags in RC WORLD OS `fleet_inventory` (yes — machines are assets too, with maintenance logs). Budget **\$6,000-9,000** Phase 1.

Prices below are July 2026 street-price ranges for the named class of product; procurement executes actual purchasing per Volume 8 and may substitute equivalents that meet spec.

2.2 The metric sizes that dominate RC

Before the lists: RC is a **metric hex world**. Better than 90% of fasteners across the RC WORLD fleet fall on seven tool sizes. Buy depth in these; everything else is occasional.

Tool size	Typical use across the fleet
1.5 mm hex	M3 set screws (pinions, drive cups), micro linkages, 1/14 fleet body hardware
2.0 mm hex	M2.5/M3 cap screws — the single most-used driver in the building
2.5 mm hex	M3 cap/button screws on 1/10 platforms, shock caps, motor mounts

Tool size	Typical use across the fleet
3.0 mm hex	M4 hardware on crawlers, construction fleet frames, tractor hitches
4.0 mm nut driver	M2 lock nuts (small linkages)
5.5 mm nut driver	M3 lock nuts — suspension pivots, servo horns
7.0 mm nut driver	M4 wheel nuts on 1/10 fleet (the daily driver of the tire bench)

Add 5.0 mm and 8.0 mm nut drivers for the construction and crawler heavy hardware, and 4 mm/5 mm/5.5 mm turnbuckle wrenches for camber/toe links. Phillips: PH1 and JIS #1 (Japanese-standard cross-head — many Far-East fasteners cam out under a Western PH1; a JIS driver grips them correctly).

2.3 Tier 1 — Per-Artisan personal kit (\$420-600)

Item	Example product	Price range
Hex driver set 1.5/2.0/2.5/3.0 mm	MIP Gen 2 metric set (\$51-55) + 3.0 mm, or ProTek “TruTorque SL” 4-pc (\$50-60)	\$55-110
Ball-end hex 2.0/2.5/3.0 mm	MIP Gen 2 ball set	\$55-60
Nut drivers 4.0/5.5/7.0 mm (+5.0/8.0)	MIP, ProTek or Arrowmax	\$45-90
Turnbuckle wrench set	ProTek or Yeah Racing multi-size	\$10-20
JIS #1 + PH1 screwdrivers	Vessel or Hozan JIS drivers	\$12-25
Precision side cutters (flush)	Knipex 78 61 125 or Hakko CHP-170	\$12-45
Needle-nose + bent-nose pliers	Knipex or generic ESD	\$20-50
Hobby knife + blades	Excel/X-Acto #1 with #11 blades	\$8-15
Curved lexan scissors	ProTek/Dubro body scissors	\$10-18
Body reamer (0-14 mm)	ProTek or HPI	\$10-15
Digital calipers 150 mm	iGaging (\$30-40) or Mitutoyo 500-196-30 (\$120-170) — one Mitutoyo per shop minimum	\$30-60
Shock/E-clip tools	Dental pick set + shock shaft pliers	\$15-30
Magnetic parts trays ×2	Generic 100 mm round	\$8-14
Small steel rule 150 mm, thread files	Generic	\$10-15
Headband magnifier or clip loupe	Generic 3.5×	\$10-20
Tweezers set (straight, curved, reverse)	ESD stainless set	\$8-15
Kit bag/roll + engraving	—	\$15-30

Field Note. Do not cheap out on the four hex drivers. A \$6 hex set rounds the first over-tightened M3 it meets, and every rounded screw costs 15 minutes and a drill-out. The MIP/ProTek tips are hardened tool steel and last years; this is the highest-leverage \$100 in the entire tool budget.

2.4 Tier 2 — Shared bench equipment (\$4,500-6,500)

Electronics benches:

Item	Example product	Price range	Qty
Soldering station 70-100 W	Hakko FX-888DX (\$110-150) or Weller WE1010 (\$150-190); budget alternative YIHUA 939D+ (\$70-90)	\$110-190	2
High-power iron for XT60/ battery leads	Hakko FX-601 or 100 W cartridge station; chisel tips $\geq$ D24	\$60-120	1
Hot-air rework station	Quick 861DW (\$230-300) or YIHUA 959D (\$80-120)	\$80-300	1
Bench power supply 30 V/ 10 A	RIDEN RD6006 (\$100-130) or Korad KA3005D (\$80-110)	\$80-130	2
Multimeter — reference	Fluke 117 (\$290-310)	\$290-310	1
Multimeter — bench spares	Fluke 101 or ANENG/UNI-T class	\$25-60	3
Battery checker / cell meters	ISDT BattGo BG-8S or SkyRC cell checker	\$25-40	4
Servo tester + PWM signal source	Generic CCPM testers (\$8-12) + one G.T.Power digital servo analyzer (\$30-50)	\$50-80	3+1
Watt meter / inline power analyzer	G.T.Power 130 A or generic 150 A watt meter	\$20-35	2
ESC programming cards	Hobbywing LED program card + brand cards for fleet ESCs	\$10-20 ea	4
Motor analyzer	SkyRC BMA-01 (KV, RPM, timing, current)	\$85-110	1
USB-UART flashing rigs (RCW Node)	CP2102/CH340 adapters + pogo-pin jig (Chapter 14)	\$10-20	2
Fume extraction	Ducted arm (Node bench) + Hakko FA-400 class (second bench)	\$60-600	2
Solder & consumables	63/37 leaded for bench work (0.8 mm), SAC305 where lead-free required; flux pens, wick, tip tinner	\$150/yr	—
PCB vice / third hand	Panavise 201 (\$40-55) + silicone-arm third hands	\$60-90	2

Wrenching bays (shared, on shadow boards):

Item	Example product	Price range
Torque screwdriver 0.1-1.2 N·m	Wiha TorqueVario-S or Wera Series 7400 with hex bits	\$110-180

Item	Example product	Price range
Pinion puller / gear puller	Robinson Racing or generic puller set	\$15–30
Bearing press/removal kit	Stepped-drift set + arbor blocks	\$25–50
Shock oil stand, syringes, cups	Generic + graduated cups	\$20–35
Digital scale 0.1 g (setup/ballast)	Generic 3 kg jewelry class	\$15–25
Corner scales (parity program, Vol. 3)	SkyRC corner weight system	\$80–120
Camber gauge + setup wheels 1/10	ProTek or Arrowmax setup system	\$60–150
Ride-height gauge, droop blocks	ProTek stepped gauge	\$15–40
Tire truer (optional Phase 2)	Hudy or generic truer	\$150–400
Heat gun (heat-shrink, body work)	Wagner/Steinel class	\$25–50
Threadlock: blue (243) — bulk	Loctite 243 10 ml ×6	\$60–90
Retaining compound (609/638)	Loctite 638 10 ml	\$15–25
CA glue thin/medium + activator	Bob Smith Industries range	\$30/mo
Greases & oils	See lubrication chart \$6.5	\$150 initial
Ultrasonic cleaner 3 L	Generic 40 kHz heated	\$60–120
Drybox cabinet + silica	Sealed cabinet, indicating silica gel bulk	\$60–100
Label printer	Brother P-touch PT-D610 or Zebra class	\$70–140

2.5 Tier 3 — Capital equipment (\$6,000–9,000)

Item	Example product	Price range	Purpose
Benchtop lathe 180–400 mm between centers	Sherline 4400 package (\$730–1,300) or SIEG SC2/7×14 class (\$700–1,000) with metric change gears	\$700–1,300	Custom driveshafts, spacers, shock pistons, hex adapters, boring wheel hexes, facing motor mounts
Commutator lathe (brushed fleet)	Hudy professional comm lathe or equivalent dedicated comm skimmer	\$250–400	Skimming 540/550 brushed motor commutators — the construction and tractor fleets run brushed motors; a \$6 skim revives a \$25 motor
Drill press 350–500 W	WEN 4208 class or bench Bosch	\$130–250	Accurate holes in chassis plates, jigs, servo mounts
Bench grinder + wire wheel	150 mm, 250–400 W	\$60–120	Dressing tools, cleaning shafts, shaping steel links
Rotary tool + stand	Dremel 4300 kit	\$100–150	Cutting, porting, cleanup — the RC workshop's universal tool
3D printers ×2	Bambu Lab P1S (\$370–700 depending on promo) ×2	\$740–1,400	Jigs, fixtures, body posts, custom mounts, prototype parts; two printers so a failed unit never blocks the jig library

Item	Example product	Price range	Purpose
Filament drybox + stock	PLA+ / PETG / ABS/ASA, ~20 kg rolling stock	\$250–400	PETG for functional park parts (UV + toughness), ASA for outdoor fixtures
Air compressor (×2 small)	California Air Tools 8010 class	\$360–560	§1.6
Spray booth + airbrush kit	Extraction booth + Iwata Eclipse or NEO	\$200–400	Body painting (Chapter 13), conformal coating
Charging infrastructure	See Chapter 11 (charger bank is costed there)	—	—
Rolling-road / dyno bench (QC)	Custom-built rollers + brake, or SkyRC chassis dyno if available locally	\$150–500	Post-repair load test without track time (Chapter 12)
Fleet trolleys ×3	3-shelf workshop trolleys	\$180–300	Moving 6–8 vehicles between pit lane and The Works

Investor Note. Total Phase 1 tooling — five Artisan kits (4 Artisans + Workshop Lead) plus one un-committed float kit as training/loaner stock, shared benches, capital equipment, charging infrastructure — lands at **\$28,000–38,000** including the charger bank costed in Chapter 11. This is under 2% of Phase 1 capex and is the least discretionary line in the budget: every dollar of fleet availability flows through these benches.

2.6 What we deliberately do not buy in Phase 1

CNC mill (jigs come off the printers; true machining is jobbed out), laser cutter (fire risk in a LiPo facility; outsource), injection molding (fantasy at this scale), a second lathe, and any tool that exists to make one specific job slightly faster before that job has proven its frequency in the RC WORLD OS repair data. The failure-analysis loop (Chapter 9) tells us what to buy in Phase 2.

3. Workshop Behaviours & Craft Standards

This chapter is the skills curriculum. Every Artisan is trained against it during onboarding, and the monthly bench audit (Chapter 12) samples work against these standards. The standards are not preferences; a repair that violates them fails QC even if the vehicle runs.

3.1 Bench discipline: how a professional RC tech works

One vehicle, one tote, one tray-map. Every job gets a labelled tote. Screws go into **magnetic parts trays**, and they go in mapped: tray positions mirror the vehicle — front-left screws in the tray's front-left quadrant, shock hardware grouped, diff hardware grouped. For deep teardowns, use a printed **screw map** — a laminated exploded-view card per fleet platform with adhesive-magnet zones — or the low-tech version: a strip of masking tape, sticky-side up, screws in removal order with pencil notes. An Artisan who dumps thirty screws into one tray is signalling that reassembly will be archaeology.

Photograph before you disturb. Two phone photos into the RC WORLD OS job ticket before teardown: linkage geometry, wire routing, shock positions. Thirty seconds now saves twenty minutes of “which hole was the camber link in”.

Clean before you open. Never open a gearbox on a dusty chassis. Blow down at intake, brush, then crack cases. One grain of Track B grit inside a fresh diff undoes the whole rebuild.

Finish the state, not just the job. A job ends when: the vehicle passes QC, the tote is empty, the tools are back on their outlines, the parts used are logged in `maintenance_logs` (parts_used JSONB — this drives reordering, Chapter 8), and the vehicle status flips out of `maintenance`. Half-logged repairs poison the failure data that Chapter 9 and fleet purchasing depend on.

3.2 Torque discipline on small fasteners

RC fasteners are small enough that wrist-feel spans a 4:1 error band, which is the difference between loose and stripped. The Works’ rules:

1. **Feel is calibrated, not assumed.** Every trainee spends a session with the Wiha/Wera torque screwdriver on scrap chassis until their “snug” reliably lands in band. Re-check yearly.
2. **The torque table.** Nominal values for steel screws; **into plastic or aluminum threads, use the lower bound and stop at first firm seat:**

Fastener	Into metal (nut/steel insert)	Into aluminum	Into plastic (self-tap or molded boss)
M2	0.15–0.20 N·m	0.12–0.15 N·m	seat + 1/8 turn max
M2.5	0.35–0.45 N·m	0.25–0.35 N·m	seat, no extra
M3	0.55–0.80 N·m	0.45–0.60 N·m	seat, no extra
M4	1.2–1.6 N·m	0.9–1.2 N·m	seat, no extra
M4 wheel nut (nylock)	0.9–1.1 N·m	—	—

1. **Plastic threads are consumable.** A self-tapper into ABS survives perhaps 10 cycles before the boss strips. On high-service fasteners (body mounts, gearbox cases on the WLtoys fleet), the hardening pass converts to machine screws with threaded inserts or nuts wherever the platform allows.
2. **Torque-critical points get the torque driver, not feel:** motor mount screws (thermal cycling loosens them), pinion grub screws, flywheel/slipper nuts, and anything into a \$50+ aluminum part.

Trade Hack. When a plastic boss does strip, don’t bin the part: a drop of thin CA in the hole, let cure fully, re-drive the screw — the CA shell re-forms a thread good for several more cycles. For a permanent fix, drill and fit the next screw size or a brass insert set with a soldering iron.

3.3 Threadlock rules: blue, not red

- **Blue (Loctite 243 or equivalent medium-strength) on every metal-into-metal joint:** grub screws on drive cups and pinions, motor screws, steel screws into aluminum

bulkheads, shock caps with metal threads, wheel hex grub screws. One small drop on the leading threads; more is not better — excess migrates into bearings.

- **Red (Loctite 263/271) is banned from the fleet.** Red is a permanent grade that needs >230 °C to release; on an M3 in an aluminum part you will destroy the part before the bond lets go. There is no fastener on an RC vehicle that legitimately needs it. The one red-adjacent exception: **green retaining compound (Loctite 638)** for bearing outer races spinning in worn aluminum bores — applied by the Workshop Lead only, logged on the ticket.
- **Never threadlock into plastic.** Most anaerobic threadlockers stress-crack polycarbonate, ABS and nylon. Plastic joints rely on correct torque and nylock nuts.
- **Purple (222) for M2 and smaller** where blue's breakaway torque would round the tiny socket before the bond releases.

3.4 Bearing craft

- **Seat with a press, never a hammer.** Use the stepped-drift kit; press on the race being fitted (outer race into a housing, inner race onto a shaft) — pressing through the balls brinells the races and the bearing is dead before it turns.
- **Square is everything.** Start the bearing by hand, feel it sit square, then press. A bearing that goes in crooked and is forced will gall an aluminum bore permanently.
- **Cleaning protocol:** post-water/mud Shifts, bearings out, dunk-and-swish in isopropyl or the ultrasonic cleaner, spin-dry with air at low pressure (hold the races — free-spinning a dry bearing with an air jet over-revs and pits it), then re-oil (light bearing oil for racing fleet, marine grease pack for boats and anything in the water-splash zone).
- **Rubber-sealed (2RS) for the fleet, metal-shielded (ZZ) only indoors.** The 0.5% drag penalty of rubber seals is irrelevant in a rental fleet; the grit exclusion is not.

3.5 Gear mesh: the paper-strip method

Pinion-to-spur mesh is the most frequently botched adjustment in RC. Too tight destroys bearings and eats the spur; too loose strips teeth on the first jump landing.

1. Cut a strip of ordinary printer paper (~80 gsm, ≈0.1 mm).
2. Lay it between pinion and spur; rotate the gears to feed it into the mesh.
3. Set the motor position so the gears **lightly grip** the paper, tighten the motor screws (blue threadlock).
4. Remove the paper. Correct mesh has just perceptible backlash — rock the spur with the pinion held; you should feel a tick of free play, not a clunk, not silence.
5. **Check at four points around the spur.** Spurs are seldom perfectly concentric; set mesh at the tightest point or the tight spot will bind.

Metal-gear transmissions on the construction fleet run slightly looser mesh (two paper thicknesses) with grease; sealed lead-screw actuator gears are set at the factory and are not adjusted, only inspected and greased.

3.6 Servo craft: center before you install

The cardinal rule: **a servo is centered electronically before the horn goes on, every time.**

1. Connect the servo to the bench servo tester, set to neutral (1,500 μ s).
2. Fit the horn as close to the required angle as the spline allows; correct residual error with the tester's sub-trim display, and record the offset on the job ticket if it exceeds $\pm 3^\circ$.
3. Install, connect linkage, then set end points (EPA) on the transmitter/receiver so the servo **never stalls against a mechanical stop** — a stalled 25 kg servo draws amps until something (gear teeth, BEC, wiring) gives.
4. Steering servos on the rental fleet get a **servo saver** verified at every service: compress by hand; if it doesn't give before the servo does, re-shim or replace the spring.

Construction fleet note: boom/bucket functions on the lead-screw machines use limit micro-switches, not servo EPA — test both limits under no load before buttoning up (SOP-WS-011).

3.7 Wiring craft

Strain relief is the standard, not the exception. Every wire that leaves a PCB or connector gets mechanical support within 20 mm: a zip-tie anchor, a dab of silicone (Shoe Goo/E6000 class — never CA, which wicks and embrittles), or heat-shrink bridging onto the harness. The RCW Micro-Node spec mandates silicone strain relief at the solder pads (Chapter 14) because vibration kills solder joints by fatigue, not by force.

Polarity conventions — park-wide law:

- Battery main: **XT60**, red/positive to the beveled corners per Amass standard; Deans T-plug legacy packs accepted at intake but converted to XT60 at first service.
- Silicone wire colors: red = positive, black = negative, **never repurposed**. Signal wiring follows JR/Futaba servo convention: brown/black = ground, red = center positive, orange/white/yellow = signal.
- Balance leads: JST-XH, keyed; never extended with unkeyed pin headers.
- RCW Heavy-Node field connectors: JST-XH 2.5 mm positive-latch per the telemetry spec.

Heat-shrink standards: 2:1 ratio adhesive-lined shrink on every splice and battery-lead joint; wall thickness matched so the shrunk sleeve grips the insulation, not just the joint; shrink extends ≥ 5 mm past exposed conductor each side; heat gun, not lighter (soot hides inspection). No electrical tape anywhere on the fleet — it unwraps in heat and marks an amateur repair.

Routing: power (battery/ESC/motor) and signal (receiver/Node) looms separated where the chassis allows; wires never touch the motor can (it hits 70–90 °C); no wire crosses a pivoting suspension member without a service loop.

3.8 Solder-joint quality criteria

A joint passes if **all** are true:

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Bright (lead) or uniformly satin (lead-free) surface — a grainy, lumpy joint is disturbed or cold

☐

Concave fillet wetted to both surfaces — solder flowed, not blobbed



Conductor outline faintly visible through the solder on wire-to-pad joints



No wicking of solder up stranded wire beyond the joint (stiffened wire fatigues and snaps at the wick line — this is the #1 field failure of amateur battery-lead repairs)



No flux charring, no melted insulation, no stray strands



Passes a firm tug (wire joints) — the tug test is mandatory, not optional

Technique standards: 63/37 leaded at 320–350 °C for bench work on wiring and Nodes (easier inspection, lower temp); chisel tip sized to the joint — **XT60 and 12 AWG battery leads take the 100 W iron with a ≥ 5 mm chisel**, never the fine-tip station (a small iron held on for 20 seconds heats the whole connector, melts the housing and cooks the wire; a big iron finishes the joint in 3 seconds). Tin both parts first; join; hold still through solidification.

Safety Warning. Battery-lead soldering is the one soldering job with a short-circuit hazard. One lead at a time, the other lead's connector taped or capped; never cut both battery leads in one scissor stroke — the cutter blades become a dead short across a charged pack.

3.9 The Artisan skill ladder

The craft standards above map to the three-grade internal ladder (HR detail in Volume 1; customer-facing titles unchanged — everyone is an “Artisan”):

Grade	Gate (assessed at bench audit)
Artisan I (trainee)	Tier-1 kit fluency; daily inspections; tire, body, servo, wheel/hub swaps; solder joints pass §3.8 on wire splices
Artisan II	Full drivetrain rebuilds (diff, shocks, transmission); ESC programming; battery-lead work; RCW Node installation; owns SOPs solo
Artisan III (lead)	Lathe work; FMEA facilitation; QC sign-off authority; custom builds (Chapter 13); Node assembly QC (Chapter 14); trains I & II

4. RC Trade Hacks

Everything in this chapter is a sanctioned, field-proven technique. Each entry states when to use it and why it works. Hacks marked ⌚ are pit-lane legal (fast enough for a 20-minute Shift turnaround); the rest are bench hacks. None of them replaces the proper repair when the SOP calls for one — a hack buys time or saves a part; it does not close a job ticket unless the ticket says so.

1. **Toothbrush + low-pressure air for dust.** ⌚ Agitate with a stiff toothbrush while blowing at ≤ 3 bar. Why: the brush breaks static cling and packed dust in corners the air jet skates over; low pressure avoids driving grit past bearing seals.

2. **Dental picks for e-clips.** Hook-pattern dental pick to roll e-clips off shock shafts and hinge pins — roll them around the groove, don't pry outward. Why: prying launches the clip into orbit; rolling keeps it captive on the tool.
3. **Fit e-clips inside a clear bag.** Cheap insurance for the smallest clips: work inside a zip-lock; when the clip pings, it hits plastic, not the floor. Why: a lost 2 mm e-clip is a 10-minute search or a stalled job.
4. **Fuel tubing as a screw gripper.** Slip 20 mm of silicone fuel tubing over a driver tip; it grips the screw head enough to start screws in deep recesses one-handed. Why: friction sleeve beats magnetizing drivers near sensed motors and receivers.
5. **Fuel tubing as shock-shaft protector.** Slide tubing over the shock shaft threads before pushing through the seal head. Why: threads cut seals; tubing gives the seal a smooth ramp.
6. **CA + baking soda structural fill.** ⌚ Pack baking soda into a crack (chassis brace, body post base), drip thin CA — it flashes off instantly into a hard, sandable composite. Why: the soda accelerates cure and bulks the CA into a gap-filler; a field repair strong enough to finish the day. Ventilate — the flash reaction fumes sting.
7. **Shoe Goo tire saves.** A torn tire sidewall or unglued section re-attached with Shoe Goo/E6000, zip-tie or rubber-band clamped overnight. Why: stays flexible, unlike CA which makes a hard hinge point that re-tears; saves \$15 tires by the dozen in a rental fleet.
8. **Zip-tie temporary body mounts.** ⌚ Snapped body post → zip-tie loop through the shell hole to any chassis anchor; foam pad under the shell. Why: returns the vehicle to the Shift roster instantly; proper post fitted at end-of-day.
9. **Drybox with indicating silica for electronics.** Any receiver, ESC, servo or Node that took water goes into the silica cabinet overnight after an isopropyl rinse of accessible boards. Why: IPA displaces water from board surfaces; silica finishes trapped moisture. Rice is a myth — it barely absorbs.
10. **Pencil graphite on sliding plastics.** Rub a soft pencil on battery-strap slides, body-clip channels, telescoping antenna tubes. Why: dry graphite lubricates without attracting the dust that grease-lubed slides collect in a dirt park.
11. **Freezer trick for stuck bearings/pins.** Frozen assembly (−18 °C, 30 min) shrinks the steel inner components faster than the aluminum housing warms — press the part out immediately on removal. Reverse trick: warm the aluminum hub with the heat gun to 80–100 °C to drop bearings out. Why: differential thermal expansion, ~2× coefficient difference between aluminum and steel.
12. **O-rings as shock preload spacers.** ⌚ No clip-on spacers at hand? Stretch appropriately sized O-rings over the shock body. Why: instant, reversible ride-height tweak in ~1 mm steps; also quiets a rattling preload collar.
13. **Wheel nut in a nut driver, wheel-first.** ⌚ Drop the nylock into the 7 mm driver, offer the driver to the axle. Why: one-handed wheel changes; the driver holds the nut captive so pit-stop tire swaps take seconds and nuts stop vanishing in the pit-lane gravel.
14. **Masking-tape screw timeline.** For unfamiliar teardowns: screws stuck to sticky-side-up tape in removal order with pencil notes. Why: reassembly is the tape in reverse; beats memory on a machine you see twice a year.

15. **Body shell as a parts bowl — never.** Anti-hack that earns a mention because everyone does it: shells flex, screws slide to the floor. Totes and magnetic trays exist. Why listed: audit item.
16. **Turnbuckle length by calipers, not counting flats.** Set one side's link, measure center-to-center with calipers, clone to the other side. Why: faster and more accurate than counting turns; keeps toe/camber symmetric to $<0.5^\circ$.
17. **Hot-glue balance weights.** Stick-on chassis weights secured with a hot-glue bead over the top. Why: stick-on adhesive fails at the first wet Shift; hot glue peels off cleanly when re-balancing.
18. **Heat-shrink servo-lead labels.** ⌚ White heat-shrink + fine marker on every servo/Node/receiver lead ("ST", "TH", "NODE"). Why: five seconds at build time saves every future disconnect/reconnect from the plug-swap lottery.
19. **Grease the balance-plug rubber.** A whisper of dielectric grease on JST-XH balance plugs. Why: they seize into charger boards and the wires pull out of the housing when yanked; grease means the housing comes out, not the pins.
20. **The "second pair of hands" is a vice + tape.** Solder XT60s with the connector mated to a scrap opposite-gender plug held in the Panavise. Why: the mated plug heat-sinks the pins in correct alignment — no melted housings, no misaligned pins.
21. **Diff fluid by scale, not eye.** Fill diffs by grams on the 0.1 g scale (record per platform, e.g. "1/10 touring gear diff: 1.3 g of 3,000 cSt"). Why: repeatable across the fleet; "fill to the gears" varies $\pm 30\%$ by Artisan.
22. **Old spur gears become paint stands.** Screw a used spur to a stick — instant rotating mount for painting small parts. Why: free fixtures from the scrap bin; the machining corner keeps a jar.
23. **CA-hardened thread rescue.** Stripped plastic boss → drop of thin CA, cure, re-tap with the screw (see §3.2). Why: CA shell re-forms a serviceable thread; costs nothing.
24. **Toothpaste as plastic polish.** Non-gel toothpaste polishes scuffed polycarbonate windscreens and lexan. Why: it's a fine abrasive paste; mild cutting compound in a pinch, and it's already in the staff kitchen.
25. **Velcro battery "gate".** ⌚ Adhesive hook-and-loop strip on the battery tray floor plus the strap. Why: stops pack sliding under crash loads that a strap alone allows; ejected packs rip their own leads.
26. **Track-pin keeper wire.** On excavator tracks, a 0.8 mm stainless safety wire twisted through the master-pin hole. Why: Huina-class master pins walk out under side-load; the wire converts a track-throw field failure into a QC-lane observation.
27. **Sacrificial XT60 "keys".** A box of XT60 plugs with the leads cut short and soldered into a loop, painted red, hung at the charging room door. Why: inserted into every stored pack as a "this pack is at storage charge" flag that's visible from across the room — process state made physical.
28. **Hot water for tight tires.** Soak new tires/inserts in hot ($\sim 70^\circ\text{C}$) water for 2 minutes before mounting on rims. Why: softened carcass seats into the bead channel without stretching or tearing; also helps remove glued tires non-destructively along with a heat-gun pass.
29. **Motor-spray nothing — use IPA and air.** "Motor spray" solvents strip commutator lubricant films and attack some insulation. Brushed motor cleaning = IPA dip for accessible

parts, air, then re-oil bushings; real restoration = comm skim on the lathe. Why: the aggressive aerosol shortcut costs motor life; the lathe pays for itself (\$2.5).

30. **Binder clips as wing/wire clamps.** Small binder clips clamp glued tire sections, hold wiring looms while silicone cures, and act as third hands on body work. Why: \$3 buys twenty clamps sized perfectly for RC-scale work.

Field Note. The hacks list is a living document. Any Artisan may propose a new hack; the Workshop Lead trials it, and accepted hacks get a number, an entry here, and — the real prize — the proposer's name in the revision log. Rejected hacks get a line in the "do not do" appendix with the reason, which is how institutional scar tissue is supposed to work.

5. Standard Operating Procedures

5.1 SOP system rules

- SOP IDs: SOP-WS-### (workshop), owned by the Workshop Lead; revisions logged in RC WORLD OS; the printed laminate at the bench must show the current revision number or it is invalid.
- Every SOP execution that touches a vehicle creates/updates a `maintenance_logs` row. No log, no work.
- PPE baseline everywhere in The Works: safety glasses at any bench with rotating machinery, cutting, springs/clips under load, soldering or chemicals. SOP blocks list only additional PPE.
- Steps marked **QC** require sign-off by a second Artisan (grade II+) before the vehicle leaves `maintenance` status.

5.2 SOP-WS-001 — Vehicle Intake & Triage

SOP-WS-001 · Rev 1.0 · Owner: Workshop Lead · PPE: baseline · Tools: air station, brush, cell checker, RC WORLD OS kiosk · Frequency: every vehicle entering The Works

1. Scan asset tag; kiosk opens a job ticket and flips status to `maintenance`.
2. Remove battery; check per-cell voltage. <3.4 V/cell → tag pack and route to Battery Room quarantine shelf with the ticket number (over-discharge protocol, \$10.4).
3. Blow down and brush at the intake station (≤ 3 bar).
4. 2-minute triage: wheels/steering free, visible damage, radio bind check, drivetrain rotation by hand, telemetry last-fault review from the Node feed (under-voltage events, kill-switch triggers, impact G if reported).
5. Assign triage class A/B/C (Chapter 7 definitions) and estimated bench minutes; place vehicle + tote on the matching queue shelf.
6. Photograph damage into the ticket.

5.3 SOP-WS-002 — Daily Fleet Inspection (pre-open)

SOP-WS-002 · Rev 1.0 · Owner: Duty Artisan · PPE: baseline · Tools: Tier-1 kit, cell checker, transmitter · Frequency: daily, before park open; ~90 s per vehicle, split across Artisans by zone

Per vehicle:

- ☐ Asset tag legible; body clips present; shell crack-free at mounts
- ☐ Wheel nuts snug (7 mm driver pass); tires glued, no tears; wheel hexes not wallowed
- ☐ Steering: full lock both ways under power, returns to center, no servo buzz at neutral
- ☐ Drivetrain: hand-roll — smooth, no clicking (clicking = CVD or diff, pull it)
- ☐ Suspension: drop test — settles without binding; shock shafts not bent, no oil film
- ☐ Battery bay: strap + velcro secure; XT60 housing intact, no browned pins
- ☐ Radio: bind, throttle/steering direction, failsafe holds neutral on TX power-off
- ☐ Node heartbeat visible on the RC WORLD OS grid map; voltage telemetry matches cell checker ± 0.05 V/cell
- ☐ Construction fleet add: boom/bucket full travel, limit switches stop the lead-screw at both ends; tracks tensioned (excavators: 5–8 mm sag mid-span)
- ☐ Log pass/fail; failures open a ticket automatically

5.4 SOP-WS-003 — Post-Crash Inspection

SOP-WS-003 · Rev 1.0 · Owner: any Artisan II+ · PPE: baseline · Tools: Tier-1 kit, setup gauges · Frequency: after any marshal-reported major impact or Node impact flag

1. Confirm battery seated and leads undamaged **before** any power-on test.
2. Squeeze-test each suspension arm and knuckle (hairline cracks whiten under flex on nylon).
3. Check hinge pins straight (arms swing free), shock shafts straight (extend/compress by eye), turnbuckles straight (roll on the bench).
4. Wheel run-out check on a spin; bent hexes/axles replaced, not straightened.
5. Chassis flex test: hold diagonal corners, twist gently, watch for tub cracks at bulkhead screws.
6. Gear mesh re-check (impacts shift motor mounts): paper-strip method §3.5.
7. Steering trim on the bench $\pm$: if sub-trim needed >5 μ s from the setup sheet, geometry is bent somewhere — find it.

8. **QC** if any structural part replaced; otherwise return to service and log “post-crash, NFF” (no fault found) — NFF rates per platform feed Chapter 9.

5.5 SOP-WS-004 — Brushed Motor Replacement (540/550 fleet)

SOP-WS-004 · Rev 1.0 · Owner: Artisan I+ · PPE: baseline · Tools: 2.0/2.5 mm hex, torque driver, watt meter, blue threadlock · Frequency: on failure or PM flag (§6)

1. Photograph wire routing; desolder or unplug motor tabs (note polarity — swapped tabs on brushed = reversed rotation).
2. Remove pinion (note tooth count on ticket), motor screws, motor.
3. New/refurbished motor: bench-spin at 3 V on the PSU — current draw within table (§12.3) confirms brushes seated; if refurbished, comm was skimmed and bushings oiled per §2.5.
4. Fit pinion at recorded position (grub screw on the shaft flat, blue threadlock, 1.5 mm hex).
5. Mount motor, set mesh (paper strip), torque motor screws 0.55–0.8 N·m with threadlock.
6. Reconnect; full-throttle free-rev ≤ 5 s with watt meter inline — log free current.
7. **QC**: load test on rollers, temperature by IR gun after 2 min ≤ 45 °C above ambient.

5.6 SOP-WS-005 — ESC Replacement & Programming

SOP-WS-005 · Rev 1.0 · Owner: Artisan II+ · PPE: baseline; ESD strap when case open · Tools: 100 W iron, program card, laptop, watt meter · Frequency: on failure

1. Pull the fleet ESC profile sheet for the platform from the RC WORLD OS platform register (racing profiles are specified in Volume 3, Chapter 5; construction profiles in Volume 4, Chapter 4).
2. Replace unit; battery-lead and motor-lead joints per §3.8 (100 W iron, adhesive shrink; one lead at a time).
3. Program: cell-count/LVC **hard cutoff 3.4 V/cell** (park doctrine — the ESC is the last line behind the Node’s soft intervention), punch/drag-brake/reverse per profile, thermal cutoff ON.
4. Calibrate throttle range to the transmitter (full/neutral/brake endpoints).
5. Verify Node PWM intercept still in circuit — receiver → Node → ESC, not receiver → ESC (Chapter 14 wiring).
6. **QC**: rollers under load; confirm LVC triggers at 3.4 V/cell using the bench PSU ramped down through the threshold; confirm kill command from RC WORLD OS drops throttle to neutral < 1 s.

5.7 SOP-WS-006 — Servo Replacement

SOP-WS-006 · Rev 1.0 · Owner: Artisan I+ · PPE: baseline · Tools: servo tester, Tier-1 kit · Frequency: on failure or PM flag

1. Center new servo on the tester (§3.6); confirm current draw at neutral < 10 mA (buzzing/hunting servo = reject).

2. Match spline horn; transfer servo-saver; install with rubber isolation grommets and brass eyelets correctly stacked (eyelet flange under the grommet, screw torque to seat only).
3. Connect linkage; set EPA so neither lock stalls the servo; verify at full lock the current draw <1.5 A on the bench PSU.
4. Heat-shrink label the lead; route clear of the motor.
5. **QC** only if steering geometry parts were also replaced; otherwise self-check drive test on the strip.

5.8 SOP-WS-007 — Differential Rebuild (gear diff, 1/10 fleet)

SOP-WS-007 · Rev 1.0 · Owner: Artisan II+ · PPE: nitrile gloves (silicone fluid) · Tools: Tier-1 kit, 0.1 g scale, diff fluid set, paper towels · Frequency: 100-cycle PM or on failure

1. Pull diff per platform manual; totes and tray-mapping mandatory — diff shims are position-critical.
2. Open case; inspect: gear teeth (chipping = replace set), cross-pins (grooving), O-rings and gasket (always replace — \$0.40 parts protecting a 45-minute job).
3. Clean all parts in the ultrasonic; dry.
4. Refill by weight from the platform sheet (e.g. touring gear diff 1.3 g of 3,000 cSt; buggy front 2.0 g of 5,000 cSt — see register). Pump the gears to burp air, top to weight.
5. Reassemble with new gasket; torque case screws in a cross pattern to lower-bound M2.5 values (plastic case).
6. Reinstall; reset ring/pinion shims to factory backlash where applicable; grease outdrives' CVD cups (black CV grease).
7. **QC**: wheels-off spin test — both wheels drive, diff action smooth; road test on the strip with two tight figure-8s; no clicking.

5.9 SOP-WS-008 — Shock Rebuild

SOP-WS-008 · Rev 1.0 · Owner: Artisan I+ (pair build with II first time) · PPE: nitrile gloves · Tools: shock pliers, fuel-tubing shaft sleeve, oil stand, dental pick · Frequency: 100-cycle PM (racing), season (crawler/construction), or leak

1. Shocks off in pairs (front pair / rear pair — never mix rebuilt with unrebuilt on one axle).
2. Strip: cap, spring, seal cartridge; e-clip technique §4-hack-2; sleeve the shaft threads (hack 5) before pulling through seals.
3. Inspect shafts (pitting/bend = replace), pistons (wear ovality), seals & O-rings (replace every rebuild — kit cost \$1-2).
4. Fill: platform oil weight from the register (e.g. touring 400 cSt, buggy 550 front/450 rear, crawler 300); bleed method per shock type (emulsion vs bladder); set rebound per setup sheet (touring: 2-3 mm rod push-out).
5. Match the pair: both shocks compressed slowly side-by-side must feel identical; if not, re-bleed.
6. Refit; drop-test the corner; ride height check against setup sheet.

5.10 SOP-WS-009 — Tire Mounting & Gluing

SOP-WS-009 · Rev 1.0 · Owner: Artisan I+ · PPE: nitrile gloves, ventilation (paint bench), glasses · Tools: CA thin, activator, rubber bands, hot-water bath · Frequency: continuous (highest-volume SOP in the building)

1. Wash rims and tire beads with IPA; dry fully (CA + damp = white bloom and weak bond).
2. Insert foams seam-opposite the tire seam; hot-water soak for tight carcasses (hack 28).
3. Seat tire; check bead fully home all round both sides.
4. Glue in 4 pulls per side: pull bead back, one drop of thin CA run 90° of circumference, release, press 10 s. Kicker spray only on the last pass (kicker embrittles if soaked into the joint).
5. Band the wheels or use mounting rings for 20 min minimum before service.
6. Batch rule: tires are glued in fleet batches at the paint bench, ≥8 wheels per session — per-wheel gluing at wrenching bays is banned (fume + quality variance).
7. Vent hole check: each tire has its vent open (blocked vents balloon at speed).

5.11 SOP-WS-010 — Track-Pin Replacement (excavators & dozers)

SOP-WS-010 · Rev 1.0 · Owner: Artisan I+ · PPE: baseline · Tools: pin punch set, small hammer + anvil block, safety wire, lithium grease · Frequency: on thrown/kinked track or 100-cycle PM finding

1. Slacken track tensioner fully; note shim/adjuster position on ticket.
2. Drive out the master pin with the matched punch on the anvil block — support the link, strike the pin.
3. Remove damaged links; inspect neighbours for elongated pin bores (light through the bore shows ovality) — replace any link with visible ovality; they cascade.
4. Rebuild with new pins from the spares bin (track pins are a CONTINUOUS-class consumable, Chapter 8); grease pins sparingly — a wet film attracts pit sand; “damp, not dripping”.
5. Master pin gets the safety-wire keeper (hack 26).
6. Re-tension: 5–8 mm sag mid-span top run; sprocket alignment sight-check.
7. **QC**: 5-minute figure-8 in the test yard including one 15° ramp climb; watch tracking straightness.

5.12 SOP-WS-011 — Lead-Screw Actuator Service (construction fleet)

SOP-WS-011 · Rev 1.0 · Owner: Artisan II+ · PPE: baseline · Tools: Tier-1 kit, PSU, lithium grease, micro-switch spares · Frequency: 100-cycle PM per \$6; on symptom (stall, chatter, drift)

1. Cycle the function full-travel on the bench PSU with the watt meter inline; log stall/running current against the platform baseline (rising running current = dry or fouled screw).
2. Strip the actuator cover; clean old grease and pit dust from the screw with brush + IPA (never air-blast — drives grit into the nut).
3. Inspect: screw straightness, nut backlash (axial shake at the carriage >0.5 mm = replace nut), thrust bearing feel.
4. Re-grease with white lithium, thin film full length; cycle 5× to distribute.

5. Test both **limit micro-switches**: manual trigger stops the motor; measure switch continuity; replace at the first sign of intermittence (MEDIUM-turnover spare — boom-limit switches are a known Huina weak point).
6. Reassemble; re-run current log; must be within 10% of baseline. **QC** on the premium showcase machines only.

5.13 SOP-WS-012 — Waterproofing Treatment

SOP-WS-012 · Rev 1.0 · Owner: Artisan II+ · PPE: gloves, ventilation (booth) · Tools: conformal coating, dielectric grease, balloon/finger cots, silicone sealant · Frequency: at fleet hardening (new vehicles) and after any electronics replacement on wet-zone fleets (crawler, marine support, agriculture)

1. Receivers: open case, acrylic conformal coat the board both sides masking the antenna pad and connector pins (booth only); reassemble with a silicone bead at the case seam and case-exit grommets.
2. Servos: case seams and wire exit sealed with silicone; do not coat the pot/board of budget servos — buy waterproof-rated units for wet fleets instead (spec in Volume 8).
3. ESCs: fleet standard is waterproof-rated ESCs on wet fleets; add dielectric grease in connector housings.
4. Node: RCW Nodes arrive conformal-coated from bench assembly (Chapter 14); verify coating date on the Node label.
5. Motor note: brushed cans tolerate water but flush + re-oil after immersion; bearings replaced per §3.4 cleaning protocol.
6. Log treatment date; wet-fleet vehicles carry a blue dot on the asset tag.

5.14 SOP-WS-013 — RCW Node Installation & Test

SOP-WS-013 · Rev 1.0 · Owner: Artisan II+ · PPE: ESD strap · Tools: soldering station, JST-XH crimp set, zip anchors, laptop, RC WORLD OS test screen · Frequency: fleet induction; Node swap

1. Confirm Node type: Micro-Node (~25×25 mm, solder pads) for racing platforms; Heavy-Node (~40×30 mm, JST-XH latching) for construction/agriculture.
2. Wire the PWM intercept: receiver throttle channel → Node input; Node output → ESC signal lead (full pinout in §14.3). Battery sense leads to the main pack terminals **downstream of the vehicle's main switch** (no parasitic overnight drain — doctrine).
3. Micro-Node: solder pads per §3.8, then silicone strain-relief blob bridging harness to board edge. Heavy-Node: latch all JST connectors, tug-test each.
4. Mount: double-sided foam + zip anchor, antenna area clear of carbon/metal, GPS patch (ATGM336H) sky-facing where the shell allows.
5. Power on; confirm on the bench test screen: heartbeat, MAC bound to asset tag in `fleet_inventory`, voltage reads pack ± 0.05 V/cell, GPS lock within 120 s outdoors.
6. **QC**: kill-test — issue KILL from the admin screen; throttle must drop to neutral/brake <1 s; release restores control. Under-voltage test with bench PSU ramp: soft-limit intervention at 3.4 V/cell (throttle to 20%, pit order push) per doctrine.

5.15 SOP-WS-014 — Transmitter Binding & Failsafe Setting

SOP-WS-014 · Rev 1.0 · Owner: Artisan I+ · PPE: none · Tools: platform TX, bind plug/procedure card
· Frequency: at induction, after RX/TX replacement, at any unexplained control glitch

1. One vehicle powered at a time on the bind bench (a live twin on the same model memory is how “possessed vehicle” reports happen).
2. Bind per platform card; confirm model memory name = asset tag.
3. **Failsafe: throttle → neutral (brake where supported), steering → hold**; set on the RX per manufacturer procedure.
4. Test failsafe by TX power-off at 30% throttle on the rollers: wheels must stop.
5. Range check: TX low-power mode, 30 m walk-out, full control authority.
6. Log firmware/model-memory version. Note: failsafe is independent of the Node kill-switch — both must pass, they cover different failure modes (RF loss vs park-command).

5.16 SOP-WS-015 — End-of-Day Shutdown

SOP-WS-015 · Rev 1.0 · Owner: Duty Artisan (closing) · PPE: baseline · Tools: cell checker, kiosk ·
Frequency: daily, after last Shift

☐

All vehicles scanned back; RC WORLD OS grid shows zero assets on-field

☐

Every battery out of every vehicle — no pack sleeps in a vehicle, ever

☐

Packs sorted: ≥ 3.9 V/cell resting → discharge queue shelf; 3.80–3.85 → storage racks;
 < 3.7 → charge-to-storage queue (Chapter 10 logic)

☐

Chargers: all cycles complete or attended-charging ended; bunker power master OFF
except storage-mode smart chargers on the interlocked circuit

☐

Soldering stations, heat guns, PSUs off at the power rails; compressor drained (moisture)

☐

Benches clear, tools on outlines, totes shelved with tickets

☐

Repair queue board reconciled with RC WORLD OS for tomorrow’s loading (Chapter 7)

☐

Battery Room and Charging Room locked; thermal camera/alarm armed

☐

Walk the floor for stray packs — the closing Artisan signs the shutdown log; this signature
is the one the insurer will read

Safety Warning. The two non-negotiable lines in SOP-WS-015 are “no pack sleeps in a vehicle” and the signed walk-through. Nearly every RC facility fire in hobby-community memory traces to an unattended pack left connected or charging overnight.

6. Preventive Maintenance System

6.1 Cycle-based, not calendar-based

Calendar PM wastes labor on garage queens and misses the weekend workhorses. RC WORLD OS gives every asset two odometers — **battery cycles** (packs issued against the asset) and **runtime hours** (Node telemetry, `total_hours_run`) — and PM triggers on whichever threshold arrives first. A battery cycle approximates one Shift; the conversion for planning is ~0.33 h runtime per cycle.

PM tiers:

Tier	Trigger	Bench time	Who
Daily	park open	90 s/vehicle	Duty Artisans (SOP-WS-002)
Minor ("25-cycle")	25 cycles or 8 h	10–15 min	Artisan I+
Major ("100-cycle")	100 cycles or 33 h	45–90 min	Artisan II+
Season	6 months or 600 cycles	half day	Artisan II/III

RC WORLD OS raises a PM flag at 90% of threshold and hard-blocks Shift assignment at 115% — a vehicle cannot be booked past its overdue PM (fleet-management module, Volume 13).

6.2 Minor service (25-cycle) — all families

☐

Full blow-down and brush; shell off; chassis visual

☐

Fastener pass with torque driver on the platform's "loosens in service" list (motor mount, servo mount, shock towers, wheel nuts, pillow balls)

☐

Bearings: spin-feel at each wheel; gritty = pull and clean or replace

☐

Gear mesh check; slipper clutch setting where fitted (push-test: slips on wheel-held throttle blip, doesn't slip on rolling launch)

☐

Connectors: XT60 pin browning, servo plugs seated, Node leads secure

☐

Tires: wear, glue joints, vent holes; tread depth vs class minimum

☐

Node: heartbeat, voltage-agreement check vs cell meter

☐

Log in `maintenance_logs`; 6-line form, target 12 min

6.3 Major service (100-cycle) — family-specific matrix

Task	Touring/ GT	Drift	Buggy/SC	Crawler	Excavator	Dump/ Loader/ Dozer	Tractor
Shock rebuild (SOP-WS-008)	●	●	●	●	—	—	—
Diff service (SOP-WS-007)	●	●	●	● (+lock-ers check)	—	● (axles)	●
Drivetrain teardown: CVDs, cups, spur/pinion wear	●	●	●	●	drive sprockets	●	●
Bearing full pass (clean/re-place)	●	●	●	●	●	●	●
Motor: brushless — sensor lead + bearing check; brushed — brush length, comm condition, skim if grooved	●	●	●	●	●	●	●
Steering rack/servo-saver wear	●	●	●	●	—	●	●
Lead-screw service (SOP-WS-011)	—	—	—	—	● boom+bucket+swing	● bed lift / bucket	hitch lift
Track service: pins, links, tension, idler bearings	—	—	—	—	●	dozers	—
Body/roll-cage fastener + mount inspection	●	●	●	●	cab	bed pivots	●
ESC profile audit vs register	●	●	●	●	●	●	●

Task	Touring/ GT	Drift	Buggy/SC	Crawler	Excavator	Dump/ Loader/ Dozer	Tractor
Waterproof- ing integ- rity (wet- zone fleets)	—	—	—	●	—	—	●
Setup sheet re- baseline (parity pro- gram, Vol. 3)	●	●	●	—	—	—	—

Phase 2 families: **aircraft** follow a flight-hours regime (25 flights minor / 100 flights major: prop + spinner torque, control-linkage slop, elevator/aileron servo current draw, netting-strike airframe inspection) — detail lives in Volume 5’s airworthiness annex; **boats** follow runtime (flex-shaft grease every 4 h — the dominant marine PM item, hull integrity, water-cooling loop flush, sacrificial-anode check on pond hardware) — detail in Volume 6. Both fleets’ PM flags run through the same RC WORLD OS engine.

6.4 Season service (6-month)

Frame-off teardown to the bare chassis; every bearing out; every fastener re-threadlocked; wiring loom re-shrink where chafed; Node re-flash to current firmware + coating inspection; ESC thermal paste/heatsink renewal; full setup re-baseline and parity lap verification (racing); repaint/retire decision on shells; formal retire/rebuild economic call per asset (>60% of replacement cost in cumulative parts = retire to the donor shelf, Chapter 8).

6.5 Lubrication chart

Point	Product (example)	Interval	Note
Gear diffs	Silicone diff fluid 1k-10k cSt (Team Associated/TLR/Traxxas lines)	100-cycle	By weight, per platform register
Shocks	Silicone shock oil 250-600 cSt	100-cycle	Pairs matched; log batch
CVD/dogbone cups	Black molybdenum CV grease (MIP/ProTek)	100-cycle	Sparingly — attracts grit
Lead-screws	White lithium grease	100-cycle	Thin film; never air-blast the nut
Track pins	Lithium grease, damp film	on service	“Damp, not dripping”
Wheel/transmission bearings	Light bearing oil (Team Associated FT or sewing-machine grade)	25-cycle spot / 100-cycle full	After solvent clean only
Servo gears (metal)	Clear servo grease (Traxxas 1647 class)	on rebuild	Micro amount on load faces
Sliding plastics (battery slides, clips)	Pencil graphite	as needed	Hack 10 — no wet lube
Boat flex-shafts	Marine/flex-shaft grease	every 4 h runtime	Volume 6 owns detail

Point	Product (example)	Interval	Note
O-rings & seals	Green slime / silicone O-ring grease (Associated 6588 class)	at rebuild	Assembly lube, prevents seal nicks
Motor bushings (brushed)	Light machine oil, 1 drop	100-cycle	Never on the commutator

Trade Hack. Buy diff fluids and shock oils in one brand family and never mix brands within a class: “3,000 cSt” is not perfectly standardized across manufacturers, and fleet parity (Volume 3’s 2–3% rule) depends on repeatability more than on the absolute number.

6.6 PM capacity load

At steady state the PM engine generates, per 100 fleet-wide daily cycles: ~4 minor services (48 min) and ~1 major (75 min) — call it **2 bench-hours per 100 cycles**. A 230-cycle average day (Chapter 7’s utilization case) therefore books ~4.5 bench-hours of PM on top of corrective work. This number sizes the Artisan roster in Chapter 7 and is the reason PM is scheduled into the morning trough, before the park’s afternoon peak loads the corrective queue.

7. Repair Scheduling & Workflow

7.1 Triage classes and SLAs

Every intake ticket (SOP-WS-001) gets a class:

Class	Definition	Examples	Turnaround SLA	Queue rule
A — Pit-fixable	≤10 bench-minutes, no QC sign-off parts	Tire swap, body clip, wheel nut, re-bind, zip-tie body mount (hack 8)	Same Shift where possible; ≤2 h	Worked immediately at the triage bench
B — Bench repair	10–90 min, standard spares on shelf	Servo, motor, diff, shocks, arms/ knuckles, ESC swap	Back in fleet ≤24 h	FIFO within class, priority bump if class availability <buffer
C — Major / waiting	>90 min, parts on order, structural, or economic-retire assessment	Chassis tub, transmission case, water-damaged electronics set, premium hydraulic showcase faults	≤7 days or formal retire decision	Workshop Lead schedules; donor-shelf harvest allowed

The SLA is measured by RC WORLD OS from status-change timestamps and reported on the workshop KPI dashboard (\$7.5). The class-availability buffer rule: each vehicle class must keep **≥85% of its fleet count available** during operating hours; when a class dips below buffer, its class-B tickets jump the queue.

7.2 Bench loading: the worked math

Assumptions (base case, Year 1 steady state; sources: Volume 10 revenue model, Volume 3/4 utilization):

- Fleet $\approx$ 150 powered assets; average utilization $\approx$ **230 Shifts/day** park-wide (85,000 Shifts/year $\div$ 365, smoothed), peak weekend days $\approx$ 420.
- Corrective ticket rate (industry-informed planning figure, to be replaced by our own `maintenance_logs` data within one quarter): **1 ticket per 25 Shifts** overall = 4% per Shift, distributed A/B/C $\approx$ 55/38/7%.

Average day:

- Tickets: $230 \div 25 \approx$ **9.2/day** $\rightarrow$ 5.1 class A, 3.5 class B, 0.6 class C.
- Bench minutes: A $\approx 5.1 \times 8 \text{ min} = 41$; B $\approx 3.5 \times 45 \text{ min} = 158$; C $\approx 0.6 \times 150 \text{ min} = 90$.
Corrective total $\approx$ 4.8 h.
- PM load (\$6.6): 230 cycles $\rightarrow$ **$\approx$ 4.5 h.**
- QC lane + intake/triage overhead $\approx$ 20% of the above $\approx$ 1.9 h.
- **Total $\approx$ 11.2 bench-hours/day** average; peak day scales to $\approx$ 20 bench-hours.

Roster consequence: an Artisan yields $\sim$ 5.5 productive bench-hours per 8-hour shift (the rest is pit-lane support, customer-facing work, 5S). Average day needs **2.0 bench-Artisans**, peak day **3.6**. Phase 1 roster: **4 Artisans + Workshop Lead** covering a 7-day rota gives 2–3 on duty weekdays and 4 on weekends — consistent with the payroll model in Volume 10. The four wrenching bays therefore run at $\sim$ 60% occupancy on peaks, which is the correct margin: a bay-saturated workshop converts every bad crash-day directly into lost Shifts.

Investor Note. The 1-per-25-Shifts failure rate is the plan's most consequential engineering assumption after battery life. It is deliberately conservative (hobby-forum rental operators report anywhere from 1/15 unhardened to 1/40 well-hardened); the hardening pass (Volume 3 §12.2) and this volume's PM system exist to push the real number toward 1/40. Every point of improvement is $\sim$ 1 bench-hour/day, i.e. most of an Artisan salary at scale.

7.3 Queue management in RC WORLD OS

The repair queue is a Kanban board in the artisan interface (Volume 13): columns **Intake $\rightarrow$ Triage $\rightarrow$ Queued (A/B/C swimlanes) $\rightarrow$ On Bench $\rightarrow$ Waiting Parts $\rightarrow$ QC $\rightarrow$ Fleet Staging**. Rules enforced by software, not memory:

- A ticket cannot enter On Bench without an Artisan and a bay assigned; a bay holds one active ticket.
- Waiting Parts auto-links to the inventory module; when the bin scan or PO receipt lands, the ticket bounces back to Queued at the head of its class.
- Vehicles in QC block their asset from booking until sign-off (the fleet state stays `maintenance`; Volume 13 §6.3).
- The board's aging alarm turns any B ticket amber at 18 h and red at 24 h (SLA), and pushes a notification to the Workshop Lead.

The physical mirror: queue shelving at intake is labelled by class and the shelf tag holds the tote + ticket. Physical position and board column are reconciled at end-of-day (SOP-WS-015).

7.4 Spares kanban

Repair speed is inventory speed. High-turnover spares run a **two-bin kanban**: two labelled bins per SKU on the wrenching-bay rack; when the front bin empties, the Artisan drops the bin card into the reorder slot and pulls the back bin forward. The storekeeper (a rotating Artisan duty, 30 min/day) scans dropped cards into RC WORLD OS, which raises the purchase requisition against the Volume 8 supplier register. Min/max levels in Chapter 8.

7.5 Workshop KPI dashboard

KPI	Target	Source
Fleet availability (operating hours)	≥92% overall, ≥85% per class	fleet_inventory status history
Class-B SLA hit rate	≥95% ≤24 h	ticket timestamps
Repeat-failure rate (same subsystem, 10 cycles)	≤5%	maintenance_logs join
PM compliance (flags cleared before hard-block)	≥98%	PM engine
QC first-pass yield	≥90%	QC module
Cost of parts per 100 Shifts	trend ↓, budget \$85/100 Shifts Year 1	parts_used × price

8. Inventory & Parts Management

8.1 The canonical high-turnover spares — min/max register

Source doctrine: the high-turnover spares list from the original blueprint, extended by platform. Min = reorder trigger; Max = shelf cap. Quantities set for the Phase 1 fleet mix and ~2-week supplier lead time (fast lanes via Volume 8's distributor tier; slow boat for bulk).

SKU class	Turnover	Min / Max	Note
Pinions (brass, per fleet pitch/counts)	HIGH	10 / 30 per count	48P (1/10), Mod 0.6/0.7 (1/14)
Spur gears (nylon + metal)	HIGH	8 / 24 per platform	Impacts kill spurs; nylon is the fuse — that's intentional
CVD/dogbone driveshafts	HIGH	8 / 20 per platform	Buggy fronts deplete 3× rears
Suspension A-arms	HIGH	12 / 36 per platform-corner	Front-left ≠ front-right consumption; track direction bias is real — let the data set the split
Steering knuckles & hubs	HIGH	8 / 24 per platform	
Shock shafts + full seal kits	HIGH	10 / 30	Seals: 20 / 60 kits
Servos 15 kg & 25 kg digital (metal gear)	HIGH	6 / 15 each rating	Waterproof-rated for wet fleets
Wheel bearings (per size: 5×11, 10×15, 8×16 etc.)	HIGH	40 / 120 per size	Rubber-sealed only

SKU class	Turnover	Min / Max	Note
Wheel hexes + pins, wheel nuts	HIGH	30 / 100	Nylocks are one-way parts: 3 uses max
Tires: knobby (Track B), drift (A), slick/touring (A), crawler	CONTINUOUS	24 / 80 sets per class	Pre-glued spare wheel-sets: 8 per class minimum
Excavator track pins & links	CONTINUOUS	100 / 400 pins; 30 / 100 links	Bag-of-pins economics; never stock out
Brushed motors 540/550	MEDIUM	6 / 15	Plus refurbished pool from comm lathe
Brushless motors (fleet KV per class)	MEDIUM	3 / 8 per class	
ESCs (fleet standard per class)	MEDIUM	3 / 8 per class	Pre-programmed to profile before shelving
Receivers (fleet protocol)	MEDIUM	6 / 15	
Boom-limit micro-switches	MEDIUM	10 / 30	Known weak point
RCW Nodes (assembled, coated, flashed)	MEDIUM	10 / 25	Chapter 14 bench keeps the pool topped
Body posts, clips, shells	MEDIUM	posts 20/60; clips 200/600; shells 2/6 per class	Clips are the park's confetti
XT60 pairs, 12-16 AWG silicone wire, heat-shrink	CONTINUOUS	40 / 120 pairs; 10 m / 30 m per gauge	
Loctite 243, CA (thin/med), activator, Shoe Goo	CONTINUOUS	4 / 10 units each	
Silicone fluids (diff + shock, per weight)	CONTINUOUS	2 / 6 bottles per weight	One brand family (\$6.5)
Lithium & CV greases, bearing oil	CONTINUOUS	2 / 6 each	
Fasteners: M2/M2.5/M3/M4 assortments, e-clips, nylocks	CONTINUOUS	1 / 3 refill boxes per size	Sorted cabinet, never loose bags

8.2 Bin system and labeling

- **Location code:** RACK-SHELF-BIN (e.g. B2-3-07), printed with the P-touch on both bin faces plus a QR that opens the SKU in RC WORLD OS.
- Two-bin kanban for HIGH/CONTINUOUS SKUs (\$7.4); single bin + monthly count for MEDIUM.
- **Donor shelf:** retired vehicles are stripped to a labelled donor tote per platform; the donor shelf is checked before the new-parts bin for class-C repairs. Harvested parts log at zero cost, which keeps the parts-per-100-Shifts KPI honest about scavenging.
- Electronics SKUs live in shielding bags inside the bins; batteries never live in the parts store (Battery Room only).
- Quarterly full count; HIGH SKUs cycle-counted weekly (10 SKUs per day rotation — 5 minutes).

8.3 Consumables burn-rate table (Year 1 planning)

Consumable	Burn rate @230 Shifts/day	Annual cost est.
Tires (all classes, incl. pre-glued sets)	~1 set / 60 Shifts	\$8,000-11,000
CA glue + activator	1 bottle / 4 days	\$700-900
Fasteners & clips	steady trickle	\$600-900
Silicone fluids & greases	monthly bottles	\$500-700
Threadlock	1 × 10 ml / 3 weeks	\$250-350
Solder, flux, shrink, wire	Node bench dominates	\$600-900
Track pins/links	continuous	\$400-600
Shop consumables (IPA, gloves, towels, blades, silica)	—	\$900-1,200
Total consumables		\$12,000-16,500

8.4 Reorder triggers and Volume 8 linkage

Kanban card scans raise requisitions automatically; the procurement rules — supplier register, wholesale-vs-distributor split, landed-cost math, order batching to hit freight breaks, and the strategic reserve of one-model-year buyouts when a fleet platform faces discontinuation — are Volume 8's territory. The workshop's obligations to that system are exactly three: scan cards same-day, log `parts_used` on every ticket, and flag rising-consumption anomalies (a SKU whose burn doubles is a fleet defect signal for Chapter 9, not just a bigger order).

9. Failure Analysis & FMEA

9.1 Structured failure reporting

Every class-B/C ticket closes with a structured failure record: **subsystem** → **component** → **failure mode** → **suspected cause** → **cycles-at-failure**, picked from controlled vocabularies (free text allowed only in the notes field — you cannot Pareto free text). The monthly failure review (Workshop Lead + division leads, 45 min) reads three standard cuts from RC WORLD OS: top 10 components by ticket count, by bench-hours, and by parts cost; repeat-failure assets (3+ tickets/30 days — candidates for teardown or retirement); and cycles-at-failure distributions vs the PM thresholds (if diffs fail at 80 cycles, the 100-cycle PM is mis-set — move it).

9.2 Root-cause discipline

The workshop standard is **5-Why with physical evidence** — a claimed root cause must point at an artifact (the part, a photo, a telemetry trace). Two worked examples from commissioning-period expectations:

- Symptom: buggy spur gears stripping at 3× fleet rate on Track B. Why? Mesh too tight on inspection → Why? Motor mounts shifting → Why? Mount screws loosening → Why? No threadlock found on sampled screws → Why? Batch of vehicles hardened before the

threadlock step was added to the induction checklist. **Action:** recall pass on the batch; checklist gate added; spur burn returns to baseline in 3 weeks — verified in the KPI cut.

- Symptom: excavator boom stalls, cluster of tickets. Trace: running-current logs (SOP-WS-011) show 40% rise over 20 cycles before each stall. Root cause: pit sand ingress at a cover seam on one hull revision. **Action:** foam seal added at hardening; current-trend alert added to the Node telemetry rules so RC WORLD OS flags the next one at +20%, before the customer feels it.

9.3 FMEA method

Scoring 1-10 per standard practice: **Severity** (1 = cosmetic ... 10 = safety/fire), **Occurrence** (1 = rare ... 10 = near-certain per 100 cycles), **Detection** (1 = caught by existing checks ... 10 = undetectable before failure). **RPN = S × O × D**; anything ≥100 requires a documented action; anything with S ≥ 8 requires action regardless of RPN. Reviewed quarterly; scores re-baselined against real ticket data.

9.4 FMEA — 1/10 touring car (rental-hardened)

Item / function	Failure mode	Effect	S	Cause	O	Current controls	D	RPN	Action
Front A-arm	Crack/snap at hinge boss	Vehicle down mid-Shift	4	Wall impact	7	Daily flex check; spares depth	3	84	Accept; monitor per-corner consumption
Steering servo	Gear strip	No steering — run-away risk until kill	7	Impact through servo-saver, stall at EPA	4	Servo-saver verify at PM; EPA rule §3.6; Node kill	3	84	Keep; audit saver spring quarterly
Spur gear	Stripped teeth	No drive	3	Tight mesh / debris	5	Paper-strip method; mesh at PM	3	45	Accept — spur is the designed fuse
ESC	Thermal shut-down mid-Shift	Customer stops; refund risk	4	Overgearing, blocked airflow, hot day	4	Rollout register; thermal cutoff ON; IR gun at QC	4	64	Summer gearing table; log shut-downs via Node
Battery connector (XT60)	Browned/loose pins → resistance heating	Power loss; melt hazard	7	Cycle wear, poor solder	3	Daily connector check; §3.8 solder criteria	4	84	Connect or replaced at 300 cycles preventively

Item / function	Failure mode	Effect	S	Cause	O	Current controls	D	RPN	Action
Wheel bearing	Seizure	Drag, pull, DNF	3	Grit + water	6	25-cycle spin-feel; sealed 2RS	4	72	Accept; ultrasonic clean batches
Chassis tub	Crack at bulkhead	Class-C repair	5	Repeated big hits	3	Post-crash SOP-WS-003	5	75	Skid/brace kit on high-hit cars
RCW Node solder pads	Fatigue fracture	Telemetry loss → no kill authority	8	Vibration, missing strain relief	3	\$3.8 + silicone blob QC at install; heart-beat watch	2	48	Heart-beat-loss auto-flags vehicle to intake

9.5 FMEA — Huina 1580-class excavator (lead-screw)

Item / function	Failure mode	Effect	S	Cause	O	Current controls	D	RPN	Action
Boom lead-screw	Stall / chatter	Function loss; motor overheat	5	Grit in nut, dry screw	6	SOP-WS-011 current logging; 100-cycle grease	3	90	Current-trend alert at +20% (done, §9.2)
Boom-limit micro-switch	Fails to open at limit	Motor stalls at end-stop → burnout	7	Switch wear, misadjustment	5	PM switch test; spares MEDIUM	4	140	Action open: add software current-limit kill in Node firmware as redundant stop (Volume 13 back-log)
Track master pin	Walks out	Thrown track, field recovery	4	Side-load vibration	6	Safety-wire keeper (hack 26); daily tension check	3	72	Keeper made standard at hardening
Slew gear	Tooth wear → slop	Sloppy rotation, play in digging	3	Sand ingress	5	Season tear-down	6	90	Felt seal retrofit trial on 2 units

Item / function	Failure mode	Effect	S	Cause	O	Current controls	D	RPN	Action
Bucket pivot pins	Ovalized bores	Slop, in-accurate digging	2	Wear	7	Season inspection	5	70	Accept; bushing upsize at re-build
Drive motor (brushed 550)	Brush wear-out	Track dead	4	Runtime	5	100-cycle brush check; comm lathe re-furb pool	3	60	Accept
Cab glazing/body	Cracking	Cosmetic → brand damage	2	UV + impacts	6	Monthly fleet audit	2	24	Accept; repaint rotation

9.6 FMEA — 2S/3S LiPo pack (fleet battery)

Item / function	Failure mode	Effect	S	Cause	O	Current controls	D	RPN	Action
Cell	Over-dis-charge <3.4 V/ cell	Capacity loss; internal damage; later swelling	6	ESC LVC mis-set; Node bypass; storage drain	3	Dual cutoff (Node soft 3.4 V + ESC hard); intake voltage check; no-pack-sleeps-in-vehicle rule	2	36	Keep dual-control doctrine
Cell	Internal short → thermal runaway	Fire	10	Puncture/ crush in crash; charge after damage	2	Post-crash pack quarantine; charge only in bunker, in bags, attended; thermal watch	3	60	S=10 → action regardless: bunker doctrine (Ch. 11), quarterly fire drill
Pack	Swelling ("puffing")	Retirement; jammed battery bays	4	Age, heat, abuse cycles	6	IR tracking + physical inspection at every charge (Ch. 10)	2	48	Retirement thresholds §10.3

Item / function	Failure mode	Effect	S	Cause	O	Current controls	D	RPN	Action
Balance lead	Wire pull-out at housing	Charger error; mis-wired-charge risk	6	Yank removal	4	Grease hack 19; handling training; keyed JST-XH	3	72	Balance-lead repair is Artisan II bench job only
Main lead / XT60	Solder-joint failure, wick-line fracture	Power loss; short risk during handling	7	Amateur repair, vibration	3	\$3.8 criteria; one-lead-at-a-time rule	3	63	All battery-lead work at electronics bench only
Wrong charge profile	LiPo charged as NiMH / wrong cell count	Over-charge → fire	10	Operator error	2	Smart chargers with auto cell-count sanity + operator double-confirm; settings table posted (Ch. 11)	3	60	S=10: two-person rule for any non-standard charge

9.7 Feeding fleet purchasing

FMEA outputs and ticket Paretos flow into Volume 8's purchasing decisions in two ways: **spares mix** (min/max levels re-set quarterly from real burn) and **platform selection** (a candidate fleet model inherits the FMEA of its class; if a new buggy platform can't beat the incumbent's top-3 RPNs on paper — metal drive cups, serviceable diffs, available A-arms — it doesn't get bought, whatever the unit price). This is the loop that makes the fleet cheaper to run every year it operates.

10. The Battery Room

10.1 Role and stock

The Battery Room — **24 m², geometry per Volume 11 Chapter 6** — is the storekeeping half of the LiPo estate: storage racks, logging bench, quarantine shelf. Charging happens next door in the bunker (Chapter 11); the two rooms share a pass-through hatch and nothing else. Stock under the 3:1 doctrine: $\sim 150 \text{ vehicles} \times 3 = \approx \mathbf{450 \text{ packs}}$ (2S and 3S, XT60), each with a laser-etched or heat-stamped pack ID linked to a battery record in RC WORLD OS (chemistry, capacity, C-rating, purchase date, cycle count, IR history).

10.2 State-of-charge storekeeping

The park's LiPo doctrine sets the operating window at 3.4–4.2 V/cell; the storage discipline sits inside it:

- **Storage state of charge: 3.80–3.85 V/cell** ($\approx 50\text{--}60\%$ SoC). Every pack not scheduled to run within ~ 24 h is brought to storage voltage — up via storage-charge, down via storage-discharge on the chargers' storage mode. Full packs held full for days lose capacity fast and swell early; empty packs risk drifting under 3.4 V.
- **Rack layout mirrors state:** READY racks (charged, 4.15–4.20 V/cell, ordered by class, oldest-charge-first issue), STORAGE racks (3.80–3.85), DISCHARGE-QUEUE shelf (came back high, waiting storage-discharge), QUARANTINE (locked metal cabinet, Chapter 10.4). Sacrificial red XT60 flags mark storage state (hack 27).
- READY packs not consumed within 48 h go back through storage mode. The overnight sort is SOP-WS-015's battery checklist.

10.3 Cycle logging, internal resistance, and retirement

Every charge event logs pack ID, cycles, per-cell voltages, and **internal resistance (IR)** as reported by the smart chargers (SkyRC/ISDT class report per-cell IR). IR is the pack's honest odometer:

Metric (per cell, 25 °C reference)	Healthy 2S/3S 5,000 mAh class	Watch	Retire
IR per cell	3–8 m Ω	12–15 m Ω or +50% vs new-baseline	>20 m Ω or any cell 2 $\times$ its siblings
Cell divergence at full charge	<0.02 V	0.03–0.05 V	>0.05 V persistent after balance
Capacity vs label (annual discharge test)	>85%	75–85%	<75%
Cycle count	—	150	200–250 typical economic life

Retirement is triggered by **any** retire condition, not consensus. Retired packs are logged, physically marked (corner cut off the shrink label), storage-discharged, then routed to disposal (§10.5). Replacement tempo follows Volume 10's throughput-derived model, which is canonical: **57,720 pack-cycles/year** across the ~ 450 -pack pool $\approx$ **128 cycles per pack per year (~ 0.35 cycles/day)** — not every Shift consumes a full charge cycle ($\sim 30\%$ buffer returned; multi-block Operator Shifts). Against the conservative 200-cycle retire trigger (economic band 200–250 cycles, or earlier on the IR thresholds above), a pack reaches retirement in roughly 19 months of cycling, bounded by a **~ 2 -year calendar horizon**; plan **$\approx 40\text{--}50\%$ pool replacement per year** in the Volume 10 opex line.

10.4 Physical inspection and quarantine

At every charge hookup, a 5-second look and squeeze:

☐

No puffing — pack flat, shrink not drum-tight (light squeeze: a healthy pack has no “give-then-firm” balloon feel)

☐

No dents, punctures, split shrink, exposed foil

☐

Leads: insulation intact, no wick-stiffened zones, XT60 pins bright

☐

Balance lead: all wires seated in housing

☐

No sweet/solvent smell (electrolyte)

Any failure → **QUARANTINE**: the locked steel cabinet (or LiPo bunker box) outdoors-vented, logged, assessed by an Artisan II+ within 24 h. Post-crash packs from vehicles with battery-bay deformation are quarantined automatically for 24 h observation before any recharge — crush damage can take hours to become a short. Over-discharged packs (<3.0 V/cell) are quarantined and only recovered by the Workshop Lead at 0.5 C-limited low-current charge with continuous observation, once, then IR-tested; below 2.5 V/cell they are not recovered, they are retired.

Safety Warning. A puffed pack is not “still fine because it works”. Puffing is electrolyte decomposition gas — the cell chemistry is already failing and its abuse tolerance is gone. Puffed packs never return to the fleet and never get charged again except a supervised storage-discharge for disposal.

10.5 Disposal: the salt-water myth vs correct practice

The classic hobbyist advice — “soak the pack in salt water for a few days, then bin it” — is **obsolete and wrong**: salt water corrodes the terminals long before it discharges the cells, the current path stops, and you’re left with a corroded pack that still holds charge, plus a bucket of contaminated electrolyte-tainted brine. RC WORLD’s protocol:

1. Storage-discharge, then full discharge to ~3.0 V/cell on the charger; finish to near-0 V with a resistive discharger or 12 V bulb rig at the bench (attended), over hours.
2. Tape the connector (XT60 capped, balance lead taped), corner-cut label, log disposal in the battery record.
3. Deliver to a **certified battery recycler / municipal battery collection point** (Li-ion stream). Never general waste, never landfill skips.
4. Damaged packs that cannot be safely discharged go to the recycler in a sand-filled metal container, declared.

10.6 Fireproof storage

- Storage racks are open steel shelving (heat cannot build unnoticed) with packs in **LiPo-safe bags** or **steel ammunition boxes with the rubber door-seal removed** (pressure must vent, not build) — bags for singles, ammo boxes for class-batches of 4-6.
- No pack stored within 1 m of the door or blocking egress; no flammables (CA, IPA, paint) in the room — those live in the flammables cabinet in the main shop.
- Room fit-out: masonry walls, self-closing fire-rated door, smoke + heat detection wired to the park alarm, thermal camera or spot IR sensor watching the racks after hours (feed into RC WORLD OS alerting), dry-sand buckets and a Class ABC extinguisher outside the door (for surrounding materials — see §11.6 for the fire doctrine itself).

11. The Charging Room

11.1 Bunker design per doctrine

The charging room is the park's engineered fire cell, built per the original blueprint's doctrine and dimensioned by Volume 11, Chapter 6: **30 m² internal (6.0 × 5.0 m), inside the building**, in **filled-cell CMU (cinderblock) construction, sandbag-supplemented**, with blast venting to the exterior wall, an **FD90 fire-rated self-closing door**, and the pass-through hatch to the Battery Room:

- Filled-cell CMU walls on all sides, blockwork to ceiling; sandbags stacked to 1.2 m against the interior face of the working wall as spall/heat mass behind the charge racks.
- Ventilation: engineered blast/pressure vents through the exterior wall, high-level relief + low-level intake (thermal-runaway gas is hot and voluminous; the room must vent outwards), no recirculation into the workshop HVAC.
- Charge racks: steel shelving; every charging pack sits **inside a LiPo-safe bag or open sand tray / ammo box**; 100 mm sand-filled steel trays under each shelf level catch and smother a dropping burning pack.
- Floor: bare concrete; a 20 L dry-sand bucket per rack bay plus a long-handled scoop; nothing combustibile in the room — no cardboard, no spare shrink, no curtains on the hatch.
- Electrical: dedicated circuits (below), emergency power-off (EPO) mushroom button at the door cutting all charger outlets, smoke/heat detection + thermal camera on the racks with RC WORLD OS alerting, door signage: occupancy rules and the fire card (\$11.6).

11.2 Charger bank and electrical load

Daily demand at 230 Shifts/day: ≈230 pack-charges. Fleet packs are 2S-3S, 1,500-5,000 mAh; average ≈35 Wh per charge including losses. Bank specification:

Position	Unit	Channels × power	Qty	Street price
Main bank	ISDT K4 (dual, AC 400 W / DC 600 W×2, 1-8S)	2 × 200 W (AC)	6	\$220-295 ea
High-throughput	SkyRC T1000 (dual, AC 450 W, 1-6S)	2 × 225 W	2	\$200-300 ea
Storage/utility	SkyRC or ToolkitRC compact duals (e.g. ToolkitRC M6D class)	2 × 200-350 W (DC via PSU)	4	\$70-120 ea
Bench/field spares	SkyRC iMAX B6 Evo class	1 × 60 W	2	\$40-55 ea

= **24 charge channels**. At a fleet-average ~45 min per 1 C-class balance charge, the bank turns ~30 packs/hour flat out; the realistic duty pattern (morning top-up wave, midday rolling, evening storage wave) clears 230 charges in ~8 attended hours with ~60% channel occupancy — the correct margin for peak days. Electrical: worst-case simultaneous AC draw ≈ 6×400 + 2×450 + 4×250 + 2×60 ≈ **4.4 kW** plus ventilation; specify two dedicated 16 A circuits (or local equivalent) with the EPO upstream, and 30% spare breaker capacity for Phase 2 (Volume 11 electrical schedule).

11.3 Standard charger settings tables

Posted as laminates above each rack; operators select by pack label color:

Pack class (label)	Chemistry/cells	Capacity	Charge mode	Current	Cutoff / balance	Storage mode
RED — 1/14 racing (144010-class, LDRC)	LiPo 2S	1,500–2,200 mAh	Balance charge	1.5–2.2 A (1 C)	4.20 V/cell, balance to <0.01 V	3.85 V/cell, ±1.0 A
BLUE — 1/10 touring/drift/buggy	LiPo 2S	4,000–5,200 mAh	Balance	4.0–5.0 A (1 C)	4.20 V/cell	3.85 V/cell
ORANGE — short-course / premium crawler	LiPo 3S	5,000 mAh	Balance	5.0 A (1 C)	4.20 V/cell	3.85 V/cell
GREEN — construction/agriculture	LiPo 2S–3S	2,200–5,000 mAh	Balance	1 C by label	4.20 V/cell	3.85 V/cell
GREY — small crawler (MN99S/C24 class)	LiPo/Li-ion per pack label	350–1,500 mAh	Balance / Li-ion per label	1 C (≤1.0 A small packs)	LiPo 4.20 / Li-ion 4.10 V/cell	3.85 / 3.70

Rules: **1 C is the fleet ceiling** — faster charge rates are available on modern packs but trade cycle life we're not willing to sell; balance charge is the default mode (plain “fast charge” without balance ports connected is banned); cell-count on the charger display is read aloud and matched to the pack label before start (two-point check; the FMEA S=10 line).

11.4 Parallel charging: restricted practice

Parallel boards multiply throughput and multiply consequences. Permitted only for the RED 1/14 class on the T1000 positions, by Artisan II+, under these rules:

1. **Same chemistry, same cell count, same capacity class** on one board — no mixing.
2. **Voltage matching before connection: all packs within 0.05 V/cell** (checked with the cell meter; packs further apart equalize through the board at uncontrolled current).
3. Connect main leads first pack to last, then balance leads; charger current = 1 C × sum of capacities; per-pack share verified sane.
4. Maximum 4 packs per board; board fused per position (use fused parallel boards only, \$25–45).
5. Attended for the full cycle — parallel charging is never “start and walk”.

11.5 Charging SOP and occupancy rules

SOP-WS-016 — Charging Operations · Rev 1.0 · Owner: Workshop Lead · PPE: baseline; no loose lanyards over racks · Tools: cell meter, IR thermometer · Frequency: continuous during operating hours

1. Inspect pack (§10.4 checklist) — failures to quarantine, not to charge.
2. Select settings from the laminate by label color; read cell count aloud against the label.

3. Pack into bag/tray on the rack; balance lead connected; start; verify first-minute current and detected cell count.
4. Log start in RC WORLD OS (scan pack QR; charger channel auto-associates where supported, else manual).
5. The room is **attended-adjacent**: a designated Artisan is within sight/alarm range whenever any channel runs; the thermal camera + smoke alarm cover gaps; overnight charging is prohibited except storage-mode cycles on the interlocked circuit, which self-terminates.
6. On completion: pack to READY rack (oldest-first issue), cycle + IR auto-logged.
7. Spot-check pack surface temperature by IR gun once per session: >45 °C at 1 C = stop, quarantine, investigate.

11.6 Fire response doctrine

Posted on the door as the **Fire Card**:

- **Never water on a burning LiPo pack.** Water conducts and spreads burning electrolyte; a lithium-polymer (Li-ion chemistry) fire is self-oxidizing during runaway — you cannot smother the reaction, only contain it and let it burn out.
- **Class D extinguishers are for lithium-metal fires** (machining swarf, primary lithium cells) — they are not the tool for LiPo packs and we do not stock one for the charging room; this is a common and expensive misunderstanding.
- **The correct response to a pack in runaway:** hit the EPO; if safely reachable with the scoop, lower the tray/bag to the concrete floor clear of the racks (never bare hands); **bury in dry sand** from the rack buckets; close the FD90 door behind you; let it complete; ventilate via the blast vents. The bunker's whole design assumes the pack finishes burning where it sits.
- The Class ABC extinguisher outside the door exists for **secondary fires** (packaging, wiring, adjacent materials) — never as the primary response to the pack itself.
- Any thermal event, including a pack that merely vented, triggers the incident report, quarantine of every pack that shared the rack shelf, and a review of the charge log within 24 h.
- Quarterly drill: full walkthrough with a dummy pack, timed; the marshal team's park-wide emergency procedures are in Volume 1's safety annex.

12. Quality Control

12.1 Pre-rental QC lane (fleet staging gate)

Every vehicle passes the QC lane between **maintenance** and **available** — after any repair, PM, or overnight if flagged. The lane checklist (2–4 min, printed + kiosk):

☐

Visual: shell, mounts, tires, no missing fasteners on the visible plane

☐

Rollers: throttle response clean at 25/50/100%, no drivetrain noise signature, brake/reverse per class profile



Steering sweep at speed on rollers: full lock both ways, returns true, no chatter



Failsafe: TX off under power → neutral <1 s



Node: heartbeat, voltage agreement ± 0.05 V/cell, kill command drop <1 s



Current draw at free-run vs platform table ($\pm 20\%$ band) — the cheapest hidden-fault detector in the building



Construction: full function sweep, limit stops, current trace within band



Battery bay: retention, connector condition



Sign-off scan → state `available`, vehicle to staging shelf

12.2 Post-repair road test protocol

Class-B/C repairs additionally get the **strip test** on the 25 m fenced strip: two full-throttle passes, two figure-8s each direction, one panic stop; construction machines get the yard test (SOP-WS-010/011 patterns). The tester is not the repairing Artisan for QC-flagged steps — fresh eyes are the point. Failures bounce the ticket to the head of its class queue with the QC note attached; **QC first-pass yield $\geq 90\%$** is a bench-quality KPI, and repeat bounces route to the Workshop Lead as a coaching signal, not a blame signal.

12.3 Reference tables and instrument calibration

QC depends on reference numbers (free-run current per platform, temperature-rise bands, rollout per class) held in the **platform register** in RC WORLD OS and reviewed at season service. The instruments that generate them are calibrated on a schedule:

Instrument	Check	Frequency	Method
Multimeters (Fluke 117 = shop reference)	vs reference	Spares monthly vs the 117; 117 externally every 2 yrs	3-point V/I check
Cell checkers	vs Fluke 117	Monthly	Known-pack ladder
Charger IR/voltage reporting	vs Fluke + reference pack	Quarterly	Same reference pack, logged drift
Torque screwdriver	vs test values	Quarterly	Calibration fixture or external cert annually
Scales (0.1 g, corner scales)	Check weights	Monthly	3-point
Calipers	Gauge block / known pin	Quarterly	Zero + 25 mm point
IR thermometer	Ice bath / boiling ref	Quarterly	2-point
Rollers/dyno	Reference vehicle	Monthly	Golden-car run, logged

12.4 Monthly fleet audit

One half-day per month, the Workshop Lead + one Artisan audit a **20% rotating sample of the fleet** (whole fleet covered every 5 months) against: PM currency, setup-sheet conformity, hardening-spec conformity (threadlock witness marks, Node strain relief, connector condition), cosmetic grade (customer-facing paint/shell standard — brand matters), and telemetry-data sanity (odometer vs logs). Output: audit score per division on the KPI dashboard, a punch list of tickets, and — quarterly — the FMEA re-baselining data for Chapter 9. The audit is also the bench-quality audit: sampled vehicles' most recent repairs are inspected against Chapter 3's craft standards.

13. Building Custom RC Cars

This chapter is required by the founder's brief: a complete tutorial on building custom RC vehicles, from kit assembly through scratch design, closing with two worked park builds. It serves three business purposes: it is the Artisan III training text; it is the engineering basis for the park's own special vehicles (recovery crawlers, marshal utilities, showcase builds); and it seeds the future RC Academy curriculum (Phase 3, Volume 1).

13.1 Three routes to a custom car

1. **Kit build** — a hobby-grade kit (bag-by-bag assembly, ~6-12 bench hours) with your choice of electronics. Best learning-to-effort ratio; every Artisan III must have completed one.
2. **Platform conversion** — start from a cheap RTR (ready-to-run) and re-engineer it to a specification. This is the park's bread and butter (worked example §13.9).
3. **Scratch build** — design the chassis yourself around purchased drivetrain modules (axles, transmissions, motors). Reserved for vehicles no market product serves (worked example §13.10).

13.2 Chassis selection and design

The chassis answers four questions before any parts are chosen:

- **Mission:** speed surface, jump loads, crawl articulation, payload? Everything trades against everything: a stiff low tub that corners flat will drag its belly on Track C; a flexy crawler frame wanders at speed.
- **Layout:** where do the heavy masses (battery, motor) sit? Aim for the mission's weight split — touring $\approx$ 50/50 F/R with mass low and central; crawler $\approx$ 55-60% front (weight over the steering axle climbs); buggy slightly rear (traction on loose surfaces, jump attitude).
- **Structure:** tub (bathtub molded plastic — impact-tough, dirt-shedding, rental-friendly), plate (flat FRP/carbon deck — tunable flex, racing), or ladder frame (steel/aluminum rails + cross-members — crawlers, trucks, easiest to scratch-build and to 3D-print brackets for).
- **Serviceability:** the park test — can the diff come out without removing the shock tower? A custom build that fails the 15-minute-diff test fails the design review, whatever else it does well.

Scratch-design rules of thumb: wheelbase/track ratio ≈ 1.5 – 1.7 for stable trucks (shorter = agile, tippy); keep battery mass inside the wheelbase midpoint; design in mm around purchased hard points (bearing ODs, servo case standard sizes, 12 mm hex hubs) so failures are replaceable from the fleet bins.

13.3 Drivetrain layout: FWD / RWD / 4WD

Layout	Behaviour	Park use
RWD	Throttle steers the rear; simplest, cheapest; demands throttle finesse	Drift fleet (deliberately loose), formula class, learning tool at the Academy
FWD	Pulls itself straight; understeers at the limit; forgiving under panic-lift	Rare in RC (some touring classes); good teaching contrast; not fleet-relevant
4WD shaft	Neutral, fast, launches hard; one propshaft ties both diffs; more parts to service	Standard for touring, buggy, crawler fleets
4WD belt	Smooth, quiet, tunable slip; belts hate grit	Indoor/asphalt only; avoid for dirt fleets

Within 4WD, the differential stack sets character: open diffs (fluid-filled gear diffs, tune by cSt) for racing; **locked axles or one-way spools have no place in rental racing** (they punish smooth driving) but **lockers are mandatory in crawlers** (both axles turn everything, always). Ball diffs are banned fleet-wide as maintenance-intensive.

13.4 Motor and ESC matching: the mathematics

The chain: **motor KV $\times$ pack voltage $\rightarrow$ RPM; gearing $\rightarrow$ wheel RPM; wheel diameter $\rightarrow$ speed**. Define **rollout** = distance travelled per motor revolution = wheel circumference $\div$ total drive ratio.

Worked example (1/10 touring, park spec):

- Motor: 13.5T sensored brushless $\approx$ **2,850 KV**. Pack: 2S LiPo, nominal 7.4 V.
- Free RPM = $2,850 \times 7.4 \approx$ **21,090 RPM**; under load assume $\sim 85\% \approx 17,900$ RPM.
- Gearing: pinion 27T, spur 78T, internal ratio 2.0 $\rightarrow$ total ratio = $(78/27) \times 2.0 =$ **5.78**.
- Wheel $\varnothing$ 64 mm $\rightarrow$ circumference 0.201 m. Rollout = $201 \div 5.78 =$ **34.8 mm/rev**.
- Speed = $17,900 \div 5.78 \times 0.201 \div 60 =$ **10.4 m/s $\approx$ 37 km/h** — right for Track A's 180 m line.

Matching rules: hotter motor (higher KV / lower turns) demands shorter gearing (smaller rollout) or it cooks — the thermal check is the IR gun after 5 minutes' running: **motor $\leq 75^\circ\text{C}$, ESC $\leq 70^\circ\text{C}$** , above that drop 2 pinion teeth and re-test. ESC current rating $\geq 1.5\times$ the motor's expected burst draw; sensored systems for anything a customer throttles from zero (smooth low-speed control); the fleet's detune levers live in the ESC (punch, throttle cap, drag brake), not in motor swaps — one motor SKU per class, always.

13.5 Servo torque calculation

Steering torque scales with vehicle weight, tire grip and scrub geometry. Bench approximation: required stall torque (kg·cm) $\approx$ vehicle mass (kg) $\times$ grip factor (asphalt 2.5, dirt 2.0, crawl 4.0) $\times$ scrub radius factor (≈ 1.0 for typical 1/10 geometry). Examples:

- 1.7 kg touring car on asphalt: $1.7 \times 2.5 \approx$ **4.3 kg·cm minimum** $\rightarrow$ fit 9–12 kg·cm for margin and crash loads.
- 2.9 kg crawler, high-grip rock, big scrub: $2.9 \times 4.0 \approx$ **11.6 kg·cm** $\rightarrow$ fit 20–25 kg·cm waterproof (this is why the spares register standardizes 15 and 25 kg servos).

Speed matters too: racing wants ≤ 0.10 s/60°; crawlers tolerate 0.15+. Always feed high-torque servos from an adequate BEC — a 25 kg servo can pull 3–4 A stalled, above many built-in ESC BECs; the scratch-build spec (§13.10) uses an external 5 A switching BEC.

13.6 Battery selection: capacity and C-rating math

A pack's deliverable current = capacity (Ah) $\times$ C-rating. Work the requirement backwards from the powertrain:

- Touring example: 13.5T system bursts ~ 55 A, sustains ~ 25 A. A 5.0 Ah pack needs $55 \div 5.0 =$ **11 C minimum burst**. Any honest 30 C pack covers it 3 $\times$; buy 30–50 C from reputable lines (Gens Ace / CNHL per Volume 8) and ignore three-figure marketing C-ratings.
- Runtime sanity: 25 A sustained from 5.0 Ah $\approx$ 12 min flat-out; a rental Shift's real duty cycle ($\sim 40\%$ throttle time-average) $\rightarrow$ 25–30 min — matching the 20-minute Shift with the doctrinal $\sim 30\%$ buffer returned.
- Weight is a design input: a 2S 5,000 mAh hard case $\approx$ 265 g — about 15% of a touring car's total mass; capacity beyond the Shift + buffer is dead ballast that slows the car and lengthens charge cycles. Fit the smallest pack that funds the Shift plus 30%.

13.7 Body prep and painting polycarbonate

RC bodies are clear polycarbonate (Lexan) painted **inside-out** — the paint lives protected behind the plastic and the outside stays glossy:

1. Trim the shell on the molded lines (curved lexan scissors), ream body-post holes 1 mm oversize (paint later takes the slack), drill the roof vent holes.
2. Wash the inside with warm water + dish soap — mold-release residue is the #1 cause of paint peel. Dry lint-free; from now on touch the inside with gloves only.
3. Mask the windows (window masks or masking film) and the outside overspray zones; leave the protective outer film ON until the build is done.
4. Paint **polycarbonate-specific paints only** (Tamiya PS line, Spaz Stix; airbrush or rattle-can) — normal enamels/acrylics crack off the flexing shell. Spray in the booth (§1.7).
5. Order: details and shadows first, then main color in 2–3 light coats, then **back the whole job with white or silver** — backing coat gives colors their depth and blocks light. Dark-to-light layering, always; each coat flash-dries 15–20 min.
6. Peel the window masks, apply decals to the outside, peel the outer film last, edge-seal decals with a heat gun waft. Fleet shells add a strip of lexan tape (drift/touring) or shoe-goo bead (off-road) inside the front lip — cheap crack insurance where every shell dies first.

13.8 Setup from first principles

A custom build is set up in this order, one variable at a time, on the platform setup sheet:

1. **Neutral mechanicals:** ride height per class (touring 5–6 mm, buggy 22–25 mm), droop symmetric, camber -1 to -2° , toe: front 0 to -1° (out), rear $+2$ to $+3^\circ$ (in) for stability.
2. **Diffs and shocks:** start at the class register's fluid weights; verify shock pairs matched (§5.9).
3. **Rollout:** gear to the track's longest straight (§13.4 math), thermal-check, lock the pinion count in the register.
4. **Drive:** three-lap feel test on the strip/track, changing exactly one thing per stop. Push (understeer) → softer front oil / more front droop / less front toe-out; loose (oversteer) → softer rear springs / more rear toe-in / smaller rear sway bar.
5. **Freeze and document.** The finished sheet enters the platform register; from here the parity program (Volume 3, Chapter 4) owns any racing-class deviations.

13.9 Worked build #1 — WLtoys 144010 → park-spec rental buggy

The 144010 is the canonical Phase 1 rally/buggy donor: ~\$60–90 wholesale, metal-ish chassis, brushless power, absurd parts economy. Stock, it is a 60 km/h hand grenade in rental service. The conversion (target ≤ 2.5 bench-hours, parts $\approx$ \$38–55):

Teardown findings the conversion addresses: plastic outdrive cups that split under abuse; self-tapping screws into soft bosses; marginal servo-saver; no telemetry; full-power ESC profile; body clips customers lose in minutes.

1. **Driveline hardening:** replace plastic drive cups and diff outdrives with the **metal drive cup/outdrive kit** (aftermarket steel, \$12–18/set); metal center driveshaft if not already fitted; blue threadlock every grub screw; re-shim diffs and set mesh (§3.5).
2. **Fastener pass:** convert body-mount and gearbox self-tappers to M2.5 machine screws + nuts/inserts where bosses allow; nylock the wheel nuts (stock ones back out).
3. **Suspension:** stock shocks re-filled with 450 cSt (they arrive with inconsistent fluid), spring collars set; steering servo-saver spring shimmed one washer (stock is too soft to steer, too stiff to save).
4. **Electronics detune:** fleet ESC profile — **throttle cap 55%** (≈ 33 km/h from the §13.4 math run backwards on measured KV: bench KV on sampled units reads $\approx 3,900$ with the SkyRC BMA-01; $3,900 \times 7.4 \times 0.55$ duty $\approx 15,900$ RPM loaded ≈ 9.2 m/s ≈ 33 km/h at the stock 5.7:1 and 62 mm wheels), punch level 3/9, drag brake 8%, reverse locked out on track classes, **LVC hard cutoff 3.4 V/cell**. Where the stock ESC lacks programmability, it is replaced by the fleet-standard programmable unit (\$18–25) — the conversion budget assumes 50% need it.
5. **RCW Micro-Node install** per SOP-WS-013: PWM intercept, pads soldered, silicone strain relief, GPS clear of the metal top deck, bind to asset tag.
6. **Rental furniture:** tethered body clips (fuel-tubing lanyards — hack family), asset-tag decals, front-lip shoe-goo bead, velcro battery gate (hack 25), tires glued per SOP-WS-009 (factory glue is decorative).
7. **QC:** full lane + strip test; parity check against the class golden car (lap delta $\leq 3\%$); registration in `fleet_inventory`; setup sheet frozen.

Field Note. The 144010 conversion is the best Artisan II training job in the building: it touches every craft standard in Chapter 3 in one afternoon, on a platform cheap enough that mistakes are tuition, not losses.

13.10 Worked build #2 — scratch-built heavy recovery crawler with winch

The Tow-Truck Retrieval Protocol (customers pilot a recovery vehicle to retrieve breakdowns) needs a machine no catalog sells: crawler-slow, stable under side-load from a winch line, near-indestructible, and telemetry-supervised. Phase 1 fields four; they are scratch-built on purchased modules. Specification and build:

- **Frame:** ladder frame from 3 mm aluminum flat bar rails + 3D-printed PETG cross-member/skid modules (printed at 60% gyroid infill, 4 walls; jig files in the library). Wheelbase 313 mm on 1.9" wheels — long for line-pull stability; track widened +10 mm per side with hex extensions.
- **Axles/transmission:** portal axles and a 2-speed transmission from the TRX-4-class aftermarket ecosystem (fleet commonality with the premium crawlers — same spares bins). Both diffs locked. Overall crawl ratio in low $\approx 100:1$.
- **Power:** 540 brushed 20T motor (torque-dense, cheap, comm-lathe serviceable) on a crawler ESC with drag brake 100%; 3S 5,000 mAh pack low and central; external 5 A BEC feeding a 25 kg waterproof steering servo (\$13.5 math: $2.9 \text{ kg} \times 4.0 \approx 12 \text{ kg}\cdot\text{cm}$ minimum, doubled for winch-era side loads).
- **Winch:** hobby-class electric winch ($\varnothing 1.5 \text{ mm}$ synthetic line, $\sim 4 \text{ kg}$ pull — sized to drag any 1/10 fleet vehicle up a 15° haul-road grade: worst case $3.2 \text{ kg} \times (\sin 15^\circ + 0.35 \text{ rolling/drag}) \approx 1.9 \text{ kg}$ line pull, $2\times$ margin), driven through a servo-signal winch controller on channel 3, **line-out limit switch** and a printed fairlead. Free-spool clutch so marshals can pull line by hand.
- **Telemetry:** RCW **Heavy-Node** (JST-XH field connectors — this vehicle gets opened often), geofence set to the active recovery corridor only: the customer piloting it can physically drive only where the protocol allows; kill authority with the marshal.
- **Customer detune:** throttle cap 60% in high range, high range software-locked during retrievals, steering rate limited — piloting stays easy and theatrical, per the experience design in Volume 9.
- **Build sequence:** frame + axles dry-fit $\rightarrow$ drivetrain shakedown before electronics (roll it down the strip on a temp servo) $\rightarrow$ electronics + Node $\rightarrow$ winch + limit switch $\rightarrow$ waterproofing pass (SOP-WS-012) $\rightarrow$ ballast to 55/45 front bias at 4.1 kg all-up $\rightarrow$ QC yard test including a full retrieval rehearsal towing a dead 1580 excavator's dump-truck partner.

Documentation duty: both worked builds' full BOMs, print files and setup sheets live in the platform register; the recovery crawler's BOM is the template every future special build copies.

14. Telemetry Hardware Bench Work

The RCW Node's electrical design, firmware and cloud side belong to Volume 13; this chapter is the workshop's share: assembling, coating, testing, flashing and installing the boards. Canon recap: ESP32-C3-MINI-1 (or SuperMini module) + ATGM336H GPS, 10 k Ω /2.2 k Ω voltage-divider battery sense, TPS54202 (or LM2596-class) buck from 2S/3S to 3.3 V, PWM intercept between receiver and ESC; **Micro-Node** ~25×25 mm direct-solder for racing platforms, **Heavy-Node** ~40×30 mm JST-XH latching for the heavy fleet; unit cost target <\$15; boards fabricated/assembled in JLCPCB-class batches per Volume 8.

14.1 Incoming PCBA QC (per batch)



Visual at the magnifier lamp: solder bridges, tombstoned passives, module seating, cleanliness (no flux lakes)



Sample 10% of batch: power-on current at 8.0 V bench supply — sleep and active currents within firmware datasheet band; buck output 3.30 ± 0.05 V



Voltage-divider sanity: injected 8.40 V reads 8.40 ± 0.05 on the debug console (divider ratio 10 k/2.2 k $\rightarrow$ ADC ~ 1.52 V)



GPS: cold-fix under open sky <60 s on sampled units



Reject rate >3% $\rightarrow$ batch hold, supplier ticket via Volume 8

14.2 Conformal coating and verification

Boards are coated at the paint booth after flashing and before connectors/harness (mask connector pads, antenna areas, and the ATGM336H patch):

1. Acrylic conformal spray (MG Chemicals 419D/422B class), two light cross-coats, 30 min between; epoxy coating reserved for marine-fleet Nodes.
2. **Verification:** most acrylic coatings fluoresce — a \$10 UV torch pass shows coverage as a uniform glow; voids re-coated. Cured coating passes a fingernail-drag without flaking. Coating date + operator initials on the Node label.
3. Coated spares stored in shielding bags in the parts store (10/25 min-max, Chapter 8).

14.3 PWM intercept wiring (textual diagram)

Described left-to-right along the signal path:

- **Receiver throttle channel (CH2 typical)** $\rightarrow$ 3-wire servo lead $\rightarrow$ **Node input header** (signal / +5 V / GND). The Node's MCU reads the incoming PWM (1,000–2,000 μ s).
- **Node output header** $\rightarrow$ 3-wire lead $\rightarrow$ **ESC signal input**. In PASS mode the Node reproduces the input pulse; in LIMIT mode it clamps the pulse toward 1,500 μ s (e.g. under-voltage soft limit = max 20% throttle authority); in KILL it holds neutral/brake.
- **Battery sense pair** (red/black, fused 100 mA inline) $\rightarrow$ main pack terminals **downstream of the vehicle master switch** — the doctrinal no-parasitic-drain point. Divider on-board.

- **Power:** Node logic feeds from its own buck off the battery sense pair, not from the receiver's BEC rail — a browning-out BEC must not take the telemetry down with it.
- Heavy-Node adds channel-3 intercept (boom/winch functions) and the JST-XH latching versions of all of the above; Micro-Node terminates all four looms on solder pads with the silicone strain-relief blob bridging loom to board edge.

Failsafe philosophy: the Node fails **transparent** — watchdog or brown-out puts the intercept into hardware PASS, so a dead Node never strands a customer; the RX failsafe (SOP-WS-014) still covers RF loss.

14.4 Bench-test jig

The flashing/test jig (3D-printed cradle + pogo pins onto the programming pads, USB-UART on the bench PC, PWM generator + oscilloscope-lite USB analyzer):

1. Node clicks into cradle; pogo pins land on TX/RX/EN/IO9/3V3/GND.
2. Jig PSU emulates a 2S pack (7.4–8.4 V sweep) on the sense pair; servo tester feeds a known PWM into the input header; the analyzer watches the output header.
3. Automated script (bench PC, part of the Volume 13 toolchain) walks: flash → boot → Wi-Fi join to the bench AP → heartbeat seen in RC WORLD OS test tenant → voltage read-back at 3 sweep points → PASS/LIMIT/KILL commands verified on the analyzer (KILL latency logged; must be <300 ms bench-side) → GPS fix (window bench antenna) → label print.
4. Jig throughput ~6 min/Node; the Chapter 8 spares pool (10–25 units) is topped up in monthly bench sessions.

14.5 Firmware flashing procedure

SOP-WS-017 — Node Firmware Flash · Rev 1.0 · Owner: Artisan III / Node bench · PPE: ESD strap · Tools: test jig, bench PC · Frequency: at assembly; fleet-wide OTA campaigns are software-side (Volume 13); bench re-flash on OTA failure only

1. Pull the **released** firmware build from the RC WORLD OS artifact registry — never a developer's laptop build onto a fleet Node.
2. Jig-seat the Node; hold BOOT (IO9) via jig button; flash over UART at the toolchain's standard baud; verify checksum readback.
3. First-boot provisioning: Node generates/receives its identity, is bound to MAC in `fleet_inventory`, joins the bench AP.
4. Run the §14.4 automated test to PASS; label with firmware version + date.
5. Log flash event; a Node that fails flash twice is quarantined for the electronics bench, not retried indefinitely.

Safety Warning. The kill-switch chain (park command → Supabase Realtime → Node → PWM clamp) is a safety function. Any firmware or wiring change that touches the intercept path requires the full §14.4 KILL/PASS verification and the SOP-WS-013 in-vehicle kill test before the vehicle re-enters service — no exceptions for “one-line changes”.

15. Volume Summary & Cross-References

The Works is where RC WORLD's economics are physically defended. This volume specified the facility (450 m² GFA per Volume 11 Chapter 6, organized around a ~180 m² wrenching/electronics core with one-way flow, a 24 m² battery room and a 30 m² internal charging bunker, ESD-protected electronics benches, task-graded lighting, treated compressed air, extracted soldering and paint stations); the tooling in three tiers (~\$420-600 per Artisan, \$4.5-6.5 k shared bench, \$6-9 k capital including lathe, comm lathe and two 3D printers — \$28-38 k all-in with charging infrastructure); the craft standards every repair is audited against (screw mapping, torque tables, blue-not-red threadlock, paper-strip gear mesh, servo centering, strain-relieved wiring, solder-joint criteria); a 30-entry trade-hacks compendium; seventeen SOPs from intake to end-of-day shutdown; a cycle-based PM system driven by RC WORLD OS odometers with family matrices and a lubrication chart; repair workflow math showing ~11 bench-hours/day at 230 Shifts/day and a 4-Artisan + Lead roster; min/max inventory with two-bin kanban and a \$12-16.5 k consumables budget; FMEA tables for the touring car, excavator and LiPo pack with one open action (redundant current-limit stop for boom lead-screws); battery storekeeping at 3.80-3.85 V/cell with IR-based retirement and correct (non-salt-water) disposal; the charging bunker with a 24-channel charger bank, settings tables, restricted parallel-charging rules and the dry-sand-never-water fire doctrine; the QC lane and calibration schedule; the full custom-build tutorial with the 144010 park-spec conversion and the scratch-built recovery crawler; and the RCW Node bench program from incoming QC to firmware flash.

Cross-references.

- **Volume 3** — racing platform specs, ESC parity profiles, setup sheets and the 2-3% parity audit this workshop executes.
- **Volume 4** — construction fleet duty cycles, lead-screw doctrine and the 1:3 excavator:dump-truck logistics the PM system protects.
- **Volumes 5 & 6** — aviation and marine PM annexes (flight-hours and flex-shaft regimes) that run on this volume's PM engine.
- **Volume 8** — supplier register, landed-cost purchasing, kanban requisition handling and the platform-selection gate fed by Chapter 9's FMEA data.
- **Volume 9** — Tow-Truck Retrieval Protocol experience design served by the §13.10 recovery crawler; Academy curriculum seeded by Chapter 13.
- **Volume 10** — the opex lines this volume quantifies: Artisan roster, parts per 100 Shifts, ≈40-50%/year battery pool replacement (from the canonical 57,720 pack-cycles/year model), tooling capex.
- **Volume 11 (Chapter 6)** — geometry authority for The Works (450 m² GFA, battery room 24 m², charging bunker 30 m² internal); architectural execution of Chapter 1's services checklist, bunker construction, and electrical schedules.
- **Volume 13** — RC WORLD OS modules this volume consumes: the seven-state fleet status machine (§6.3), `maintenance_logs`, PM engine, repair Kanban, battery records, Node firmware pipeline and kill-switch chain.

Volume 8 — Procurement Handbook

RC WORLD — Master Development Plan · Volume 8 of 13 **Revision 1.0** • July 2026 · Status: Living document — bump revision on material change

Purpose of this volume. This volume is the buying manual for the entire park. RC WORLD's business model rests on a single procurement thesis: the park buys commercial-grade, modifiable RC equipment **direct from Chinese factories at wholesale**, cutting out two to three distribution margins, and converts that cost advantage into fleet depth, spares abundance, and pricing headroom that no competitor buying at hobby-shop retail can match. This volume turns that thesis into an operating system. It sets the procurement doctrine (Chapter 1), maps the Chinese manufacturing geography the park will buy from (Chapter 2), and delivers the core asset of the volume — a directory of more than seventy real Chinese (plus Taiwanese and Hong Kong) manufacturers and brands, profiled by location, specialty, price tier, OEM capability, MOQ class, lead time, and RC WORLD relevance (Chapter 3). It then builds the machinery around the directory: a factory rating scorecard (Chapter 4), OEM and custom-program strategy including park-spec variants and RCW Node pre-installation (Chapter 5), MOQ and lead-time planning with worked buffer-stock math (Chapter 6), a China-specific negotiation playbook (Chapter 7), import and logistics strategy including the LiPo dangerous-goods constraints that shape the whole battery supply chain (Chapter 8), inspection and quality assurance with AQL tables and container checklists (Chapter 9), supplier scorecards and relationship management (Chapter 10), and the SOPs and templates that make the whole system executable by a two-person procurement function (Chapter 11). Where a specific commercial figure (exact MOQ, exact factory-gate price, exact tooling cost) could not be verified from public sources at the time of writing, this volume gives a realistic range and marks it **[verify at factory audit]** — the plan never fabricates precision it does not have.

Intended readers. The founder and General Manager (who own supplier strategy in Year 0–1); the Procurement & Logistics Lead (the volume's day-to-day user); The Works' Head Artisan (who co-signs technical evaluations and spare-parts plans, Volume 7); the Finance Lead (landed-cost and payment-terms discipline, Volume 10); investors performing supply-chain diligence on the funding ask.

Chapters

1. Procurement Strategy Overview
2. The Chinese RC Manufacturing Landscape
3. Manufacturer Directory
4. Factory Rating System

5. OEM & Custom Programs
6. MOQ & Lead-Time Planning
7. Negotiation Strategy
8. Import & Logistics Strategy
9. Inspection & Quality Assurance
10. Supplier Scorecards & Relationship Management
11. Procurement SOPs & Templates
12. Volume Summary & Cross-References

1. Procurement Strategy Overview

1.1 The direct-wholesale doctrine

Every RC vehicle that reaches a Western hobby shop has typically passed through two or three margin layers: factory → exporter/trading company → national distributor → retailer. Each layer adds 25–60%. A WLtoys 144010-class buggy that retails at \$120–140 in the West leaves the factory in Chenghai at roughly \$60–90; a Huina 1580 excavator retailing at \$600–800 wholesales at \$350–420 (source digest, fleet matrix). RC WORLD’s doctrine is to buy at or near the factory gate:

1. **Buy factory-direct or one step removed.** For fleet vehicles, spares, electronics and batteries, purchase from the manufacturer’s own export department, its official Alibaba/Made-in-China storefront, or (for small brands) its designated export agent. Accept a trading company only where it demonstrably adds value: consolidation of many small brands, English-language engineering support, or credit terms.
2. **Buy at wholesale quantity from day one.** The Phase 1 fleet (~150 powered assets, 3:1 battery ratio, and a spares depth specified in Volume 7) is a genuine wholesale order book — roughly \$55,000–75,000 of vehicles and \$20,000–30,000 of batteries, chargers, and spares in the first buying wave alone [verify at contract]. That volume commands wholesale pricing, factory attention, and OEM conversation rights that a hobbyist or small shop never gets.
3. **Convert margin into depth, not savings.** The money saved against retail is not banked; it is spent on fleet redundancy (spare complete vehicles), a 3:1 battery ratio, and a spares inventory sized so that no common failure ever idles a revenue asset for more than one working day (Volume 7, Chapter 6).

Investor Note. Direct sourcing is not merely a cost tactic; it is a moat. A competitor buying the same fleet at Western retail needs roughly 1.8–2.4× RC WORLD’s fleet capex and pays 2–3× for every replacement part, forever. The procurement function described in this volume is therefore a permanent structural advantage, and the supplier relationships it builds compound over time (Chapter 10).

1.2 Total cost of ownership, not unit price

The cheapest quote is frequently the most expensive purchase. Every sourcing decision in this volume is made on **total cost of ownership (TCO) per operating Shift**, not on unit price. The TCO of a fleet vehicle is:

TCO = landed unit cost + (spares consumption × parts landed cost) + (artisan labour hours × loaded rate) + (downtime × contribution margin per Shift) – residual/salvage value, all over the vehicle's service life in Shifts.

Worked contrast (numbers are planning-grade; Volume 10 carries the canonical financial model):

Metric	Buggy A ("cheapest")	Buggy B (WLtoys 144010-class)
Landed unit cost	\$52	\$78
Spare-parts ecosystem	Sparse, slow (single seller)	Deep, cheap, multi-seller
Parts cost / 100 Shifts	\$28	\$14
Artisan hours / 100 Shifts	3.5 h	1.6 h
Expected service life	~700 Shifts	~1,400 Shifts
TCO per Shift (labour at \$12/h)	≈ \$0.20	≈ \$0.09

The doctrine that follows: **a vehicle is only purchasable if its spare-parts ecosystem is purchasable**. Before any platform is fleet-approved, procurement must demonstrate three independent sources for its top ten failure parts (Volume 7's failure-parts list: control arms, knuckles, driveshafts, gears, shock shafts, servos, tires, bearings, ESCs, motors).

1.3 Dual-sourcing rules

Single-source dependencies are treated as risks with owners and review dates (risk register, Section 9.4 of Volume 1). The rules:

- **Rule DS-1 — every fleet category has two qualified platforms.** Example: entry crawler = MN Model MN99S and WPL C24; if MN Model's line stops, the category does not.
- **Rule DS-2 — every consumable and high-turnover spare has two qualified sellers**, even if both sell the same OEM part (e.g., factory store + a large Chenghai parts consolidator).
- **Rule DS-3 — electronics standards are multi-vendor by design.** XT60 connectors, 2S/3S packs, 540/550 motors, and standard-pinout ESCs are commodity interfaces; any compliant vendor can supply them. Proprietary connectors and sealed packs are banned by battery doctrine (style-guide canon).
- **Rule DS-4 — premium showcase assets are exempt.** The Kabolite-class hydraulic showcase machine is a single-source purchase by nature; it is a display/premium asset, not a rental workhorse, so its downtime does not break the park.
- **Rule DS-5 — failover is pre-negotiated.** The second source is not a name on a list; it has been sampled, audited at least remotely, and holds a dormant price agreement refreshed annually (Chapter 10 failover triggers).

1.4 Fleet standardization as procurement leverage

Volume 3's racing doctrine (one or two standardized chassis platforms, personalities via tuning) and Volume 4's construction doctrine (electromechanical Huina-class fleet) are also procurement weapons:

- **Order concentration.** Buying 24 identical touring cars, 18 identical dump trucks, or 60 identical LiPo packs moves the park up every factory's customer-priority list and across MOQ thresholds for OEM asks (Chapter 5).
- **Spares compression.** One chassis platform = one spares bin architecture = fewer SKUs at higher reorder quantities = better pricing and simpler AQL inspection (Chapter 9).
- **Negotiating theatre.** A buyer who says "we will standardize our park on your platform for three years" is offering a factory something rare: predictable, repeating B2B volume in a business dominated by one-shot consumer orders. This is the park's single strongest card and it must be played deliberately (Chapter 7).

1.5 The annual procurement calendar

Chinese manufacturing has a heartbeat, and orders must be timed to it. The two dominant events are **Chinese New Year (CNY)** — factories close entirely for 2–4 weeks in the late-January-to-February window, with degraded capacity for 2–3 weeks on either side as migrant workforces travel and re-form — and the **Canton Fair** (China Import and Export Fair, Guangzhou, April and October, with toys/hobby product concentrated in the second phase of each session).

Month	Procurement action	Rationale
Jan	Final pre-CNY deliveries land; freeze new POs; annual supplier reviews (Ch. 10). Hong Kong Toys & Games Fair (early Jan) for scouting	Factories quote long or not at all pre-CNY
Feb	CNY shutdown. Consume buffer stock; audit inventory; refresh RFQs	No production; plan, don't buy
Mar	Place post-CNY "restock wave" POs the week factories re-open; expect first-batch QC risk (new line workers)	Beat the post-CNY backlog; inspect harder
Apr	Canton Fair Phase 2 (toys, ~mid-late Apr) — annual supplier tour #1: meet incumbents, scout alternates, collect samples	Every major Chenghai/Shenzhen supplier exhibits or hosts visits
May-Jun	Negotiate annual agreements off fair meetings; place main fleet-expansion POs	Factories quietest, most flexible
Jul	Mid-year scorecard round; sample-test next-year platforms	Low-season attention
Aug	Place all Q4 and pre-CNY buffer POs now	Western Christmas production peaks Aug-Oct; space and airfreight tighten
Sep	Pre-shipment inspections of Q4 orders; book sea freight early (pre-Golden-Week squeeze)	Rates and space degrade into October

Month	Procurement action	Rationale
Oct	China National Day “Golden Week” (Oct 1–7, factories slow); Canton Fair Phase 2 (late Oct) — supplier tour #2; China Toy Expo (Shanghai) scouting	Second annual face-to-face round
Nov	Final POs that must ship before CNY; confirm production slots in writing	Slots sell out 8–10 weeks before CNY
Dec	Receive/inspect pre-CNY wave; build 8–10 week buffer stock on all HIGH-turnover spares	Cover the Feb–Mar supply hole

Trade Hack. The pre-CNY buffer is not optional. A park that runs out of touring-car control arms in February will not see replacements until late March at the earliest — production restart + backlog + transit. Rule of thumb: on 1 December, on-hand quantity of every HIGH-turnover spare must be ≥ 10 weeks of forecast consumption plus safety stock (worked math in Chapter 6).

1.6 Procurement organization and authority

Phase 1 runs procurement with **one Procurement & Logistics Lead** (full-time from Month –6) supported by the Head Artisan (technical evaluation) and the Finance Lead (payments, landed-cost accounting). Authority matrix:

Decision	Authority	Countersign
New supplier onboarding	Procurement Lead	GM
PO $\leq$ \$2,000 (catalogue spares)	Procurement Lead	—
PO \$2,000–15,000	Procurement Lead	GM
PO $>$ \$15,000 or any OEM/tooling commitment	GM	Founder/Board
Supplier suspension / failover trigger	Procurement Lead	GM
Payment release against inspection	Finance Lead	Procurement Lead

All POs, quotes, supplier records, landed-cost calculations and scorecards live in RC WORLD OS procurement module (Volume 13, Chapter 6) — no spreadsheet shadow systems after Month 3.

1.7 What the doctrine is not

Four boundaries keep the direct-wholesale doctrine honest:

- **It is not “cheapest wins”.** Section 1.2’s TCO rule outranks any unit price; a supplier that fails the spares-ecosystem test is unpurchasable at any discount.
- **It is not grey-market importing.** The park declares honestly, pays duties, ships batteries as dangerous goods, and keeps compliance files an insurer can audit (Chapter 8). The margin comes from removing distribution layers, not from removing legality.
- **It is not brand-hostile.** Where a Western-branded platform is genuinely superior for a premium class (TRX-4-class crawlers in the fleet matrix), the park buys it through the cheapest legitimate channel and accepts the margin stack — a handful of premium assets do not justify a parallel-import adventure.

- **It is not a one-person memory.** Every relationship, price, defect, and promise lives in the OS supplier record from day one, because the doctrine must survive staff turnover and franchise replication (Volume 12 licenses this volume to franchisees as an operating asset).

2. The Chinese RC Manufacturing Landscape

2.1 Why China — and why these four clusters

China manufactures the overwhelming majority of the world's RC products — including most vehicles sold under Western brand names — and, decisively for RC WORLD, it is where the **value hobby-grade** band (serviceable, tunable, spare-parted vehicles at near-toy prices) was invented between roughly 2015 and 2024 (Volume 2, Chapter 2 carries the full market analysis). Four geographic clusters matter to this plan, and knowing which cluster a supplier sits in predicts a great deal about its capabilities, pricing, and behaviour.

2.2 Shantou / Chenghai — the toy capital

Chenghai District, Shantou, Guangdong is the densest toy-manufacturing cluster on earth — thousands of factories and workshops within roughly 20 km, producing a large share of the world's toys and the majority of the RC vehicles in RC WORLD's fleet plan. Huina, MN Model, WPL, WLtoys, MJX, Remo Hobby, HBX, Double E, Heng Long and many others are Chenghai companies (directory, Chapter 3). Cluster characteristics:

- **Strengths:** unbeatable cost engineering; complete local supply chain (mold shops, motor winders, plastic compounders, PCB assemblers within minutes); fast product cycles; deep RC-specific tooling experience; used to export compliance (EN71, ASTM F963, CE).
- **Weaknesses:** quality ceilings — most Chenghai factories engineer to a price, not to a duty cycle; English-language engineering communication is thin outside the biggest firms; brand/factory boundaries blur (one factory may produce for three “brands”, and one “brand” may buy from three factories).
- **Behaviour:** Chenghai firms are volume merchants. They respond to concrete quantities and quick decisions, not to long technical correspondence. Site visits are cheap to arrange — dozens of relevant factories can be audited in a three-day trip from the Shantou high-speed-rail station.

2.3 Shenzhen (and Huizhou) — the electronics engine

Shenzhen supplies the fleet's nervous system: ESCs and motors (Hobbywing, Surpass), radio systems (Flysky, Radiolink), chargers (SkyRC, ToolkitRC, ISDT), batteries (Grepow/Gens Ace), FPV systems (BetaFPV, GEPRC, HGLRC, SpeedyBee), and the PCBA ecosystem (JLCP-CB class) that fabricates the RCW Node (source digest; Volume 13). Neighbouring **Huizhou** hosts overflow manufacturing (iFlight; Hobbywing's motor plant). Characteristics:

- **Strengths:** genuine engineering departments; firmware competence; OEM/ODM as a native business model; international-grade communication; certifications (FCC/CE/UN38.3) handled routinely.

- **Weaknesses:** higher labour and land costs than Chenghai (reflected in pricing); consumer-drone regulatory turbulence periodically distracts the FPV firms; fast product churn means spare-part longevity must be contracted, not assumed.
- **Behaviour:** Shenzhen firms think in projects and platforms. They will engage seriously with the RCW Node pre-install proposition (Chapter 5) and with custom firmware asks (detuned ESC profiles) — provided the volumes are honest and the specification is written like an engineering document.

2.4 Dongguan, Foshan, and the Pearl River Delta machining belt

Between Shenzhen and Guangzhou, **Dongguan** (ZD Racing, Joysway, Flysky's and Radiolink's production plants, RGT's southern factory) and **Foshan** (LESU) represent the Delta's precision-manufacturing tier: CNC machining, aluminium and zinc die-casting, hydraulics, higher-grade injection molding. This is where the park's metal-intensive premium assets (LESU/Kabolite-class hydraulics, CNC upgrade parts) and its most exacting plastic tooling come from. **Guangzhou** anchors the region with the Canton Fair, HSP's trading networks, and the deepest freight-forwarding market in south China. Characteristics: mid-to-premium pricing, strong OEM culture inherited from decades of contract manufacturing for global brands, and the best factories for "toy platform, industrial expectations" upgrades — exactly RC WORLD's niche.

2.5 Yiwu and the trading layer

Yiwu (Zhejiang) hosts the world's largest small-commodity wholesale market. Almost nothing is manufactured there; nearly everything can be bought there. For RC WORLD, Yiwu and its online equivalents (Alibaba trading companies, AliExpress consolidators, Taobao agents) are the channel for **long-tail purchases**: diorama and scenery materials (with Volume 2 Chapter 4's supplier study), track furniture, pit-lane consumables, retail-counter merchandise, staff uniforms and giveaway stock. Doctrine: the trading layer is permitted for items where (a) no single item exceeds \$500/order, (b) failure is an inconvenience rather than a fleet stoppage, and (c) consolidation into one shipment saves more than the trading margin costs.

2.6 Factory or trading company? How to tell

Chinese B2B marketplaces blur the line deliberately; many storefronts marked "Manufacturer" are traders with a related factory, and many honest traders provide real value. The park's verification protocol:



Business licence check. Request the licence (营业执照) and read the registered business scope (经营范围): a factory's scope includes 生产/制造 ("production/manufacturing"); a pure trader's scope reads 贸易/批发 ("trade/wholesale"). Cross-check the registered capital and address on the National Enterprise Credit Information system or via a \$30-60 third-party company report.



Product coherence test. A factory's catalogue is narrow and deep (forty excavator SKUs, one category); a trader's is wide and shallow (excavators, hoverboards, kitchenware). Incoherent catalogues = trader, regardless of the badge.



Ask for the tooling. Request photos/video of your product’s injection molds and assembly line with a dated note in frame. Factories comply in days; traders stall or send stock footage.



VAT invoice test. Ask whether they can issue a 13% VAT special invoice for the goods from the manufacturing entity itself, and whether the exporting entity matches the manufacturing entity. Mismatches are normal (export agents are common) but must be explained coherently.



Video factory audit. A 45-minute live video walk-through (line, warehouse, QC benches, mold shop) is now a standard, free pre-qualification step — script in Section 4.4.



Third-party verification. For any supplier expected to take >\$10,000/year, commission a supplier-verification report (QIMA/V-Trust class, \$100–350) or a full on-site audit (Chapter 9) before the first large PO.

Field Note. Do not treat “trading company” as a slur. Several of the directory’s most useful counter-parties — Chenghai parts consolidators, LESU’s export resellers, Hong Kong upgrade-parts houses — are traders that hold stock, speak fluent English, ship DDP, and answer at 11 p.m. The sin is not trading; it is hidden trading, where the buyer prices a factory relationship and gets a middleman’s lead times.

2.7 OEM/ODM capability levels

The directory (Chapter 3) grades each supplier’s customization capability. The scale used throughout this volume:

Level	Label	What the supplier can actually do	Typical evidence
L0	Stock only	Sells catalogue product as-is; no changes	Refuses colour/packaging asks
L1	Cosmetic OEM	Custom colours, decals, logo printing, custom packaging at modest MOQs	Existing white-label customers
L2	Configuration OEM	Component swaps within existing tooling: different ESC/servo/motor, connect-or standard (XT60), firmware settings, pre-installed third-party modules (RCW Node)	Engineering contact who discusses BOMs
L3	Structural ODM	New or modified tooling: reinforced arms, metal replacement parts, custom bodies; co-developed variants	In-house mold shop or captive mold partner
L4	Full ODM/JDM	Designs to a customer specification from a blank sheet	Design department, patent portfolio, brand-name OEM references

RC WORLD's Phase 1 asks live at **L1-L2** (park colours, XT60 standardization, detuned ESC profiles, RCW Node pre-install); Phase 2-3 asks reach **L3** (reinforced control arms, park-branded bodies). L4 is out of scope this planning horizon — the park is a fleet buyer, not a product company.

2.8 The Taiwan and Hong Kong lanes

Two directory sub-populations sit outside mainland logistics. **Taiwan** (MST, Team Magic) means separate freight lanes and customs origin, typically courier or air for the modest volumes involved, and a business culture closer to Japanese suppliers: slower to discount, faster to document, extremely reliable on spec. Do not bundle Taiwanese POs into mainland consolidations; the paperwork friction eats the freight saving. **Hong Kong** (3Racing, Yeah Racing, GPM, JDModel's trading entity) is a commercial layer over PRC manufacturing: English-fluent, small-MOQ-friendly, credit-card/PayPal-tolerant, courier-native — the ideal counterparties for The Works' fast-turn upgrade orders, at a structural 10-20% premium over factory-gate. Both lanes are also the park's hedge against any future mainland export friction on specific categories [monitor annually with the forwarder].

3. Manufacturer Directory

3.1 How to read this directory

This chapter profiles **76 manufacturers and brands** across ten categories. It is the park's sourcing map: every fleet platform in Volumes 3-6 traces to at least one entry here, and every entry carries enough structure to start an RFQ (Chapter 11) the same day. Reading conventions:

- **HQ / factory:** principal verified location; where brand and factory differ, both are noted.
- **Founded:** from company statements or public registries where available; otherwise marked unverified.
- **Price tier:** Entry (toy-adjacent) / Value (the park's core band) / Mid / Premium / Flagship.
- **OEM:** capability level L0-L4 per Section 2.7. Levels asserted from public OEM/ODM offers are marked "(claimed)" until audited.
- **MOQ class:** A = off-shelf wholesale, <50 units; B = 50-200; C = 200-1,000; D = 1,000+ or tooling-gated. Classes are planning estimates from published wholesale behaviour — **all MOQs [verify at factory audit]**.
- **Lead time:** indicative production lead time for a wholesale order, ex-works, outside CNY season; add freight per Chapter 8.
- **RCW rating:** 5 = strategic core supplier (fleet doctrine names it); 4 = planned supplier; 3 = qualified alternate / dual-source; 2 = watchlist, sample before relying; 1 = reference/context only.

Field Note. A directory is a snapshot of a fast-moving industry. Chinese RC brands appear, merge, and vanish within two product cycles; the brand on the box is not always the factory behind it. Treat every entry as true as of July 2026, to be re-verified at order time, and re-audit the whole directory annually at the Canton Fair rounds (Section 1.5).

3.2 Cars, buggies & trucks

The Motorsport Division's volume fleet (Volume 3) comes from this table. The category is dominated by Chenghai, with Dongguan and Shenzhen supplying the performance tier and Hong Kong the upgrade-parts layer.

#	Supplier	HQ / factory	Founded	Specialty	Tier	OEM	MOQ	Lead time	RCW
1	WLtoys (Weili)	Cheng-hai, Shantou	co. reg. 2011	High-speed value buggies/tourers (144010, 124017)	Value	L1-L2 (claimed)	B	3-6 wk	5
2	MJX (Meijiixin)	Cheng-hai, Shantou	1983 (re-named 2001)	Hyper Go brushless 1/16-1/14 line	Value-Mid	L2 (claimed)	B	4-6 wk	5
3	ZD Racing	Feng-gang, Dong-guan	~2011 [verify]	1/10-1/8 buggies, truggies, DBX-10	Mid	L2 (claimed)	B-C	4-8 wk	4
4	HBX / Haiboxing	Cheng-hai, Shantou	1990s [verify]	1/18-1/12 entry buggies (16889 class)	Entry-Value	L1-L2	B-C	3-6 wk	4
5	Remo Hobby	Cheng-hai, Shantou	~2011 [verify]	1/16-1/8 trucks, Smax/Mmax lines	Value	L1-L2	B	4-6 wk	3
6	JLB Racing	Longhua, Shen-zhen	~2015 [verify]	Cheetah 1/10 brushless tourer/truggy	Value-Mid	L1	B	4-8 wk	3
7	SG / Pinecone Forest	Cheng-hai, Shantou [verify]	2010s [verify]	1/16 SG-1603/1604 budget drift & tourers	Entry-Value	L1	B-C	3-6 wk	3
8	Hosim / Xinle-hong (XLH)	Cheng-hai, Shantou	2010s [verify]	Amazon-optimized 1/16-1/10 trucks (9125 class)	Entry-Value	L1	B-C	3-5 wk	2
9	LC Racing	Guang-dong (Dong-guan area) [verify]	~2011 [verify]	1/14 competition-grade buggies/truggies (EMB)	Mid	L2 [verify]	A-B	4-8 wk	4
10	HSP Racing	Guang-zhou	~1990s [verify]	Legacy 1/10 nitro/electric platform ecosystem	Entry-Value	L1-L2	C	4-8 wk	2
11	Team Magic	Taiwan HQ; PRC production [verify]	2003 [verify]	Competition touring (E4 series)	Mid-Premium	L1	A-B	4-8 wk	2

#	Supplier	HQ / factory	Founded	Specialty	Tier	OEM	MOQ	Lead time	RCW
12	3Racing	Hong Kong; PRC factories	2001 [verify]	Budget competition kits (Sakura line), option parts	Value-Mid	L2 (claimed)	A-B	3-6 wk	3
13	Yeah Racing	Hong Kong	2004 [verify]	Aluminium/steel upgrade & option parts	Value-Mid	L2	A	2-4 wk	4
14	GPM Racing	Hong Kong	1980s [verify]	CNC aluminium upgrade parts, brand-specific	Mid	L2	A	2-4 wk	3

WLtoys (Shantou Chenghai Weili Toys Industrial Co., Ltd.) is canon: the 144010-class buggy is the named Track B platform (fleet matrix). Weili is a genuine Chenghai manufacturer with 500-1,000 staff and a 10,000-30,000 m² plant per marketplace registry data, selling through a dense reseller network at \$60-90 wholesale for the 144-series. Strengths: extraordinary price-to-speed; the deepest cheap-spares ecosystem in RC (arms ~\$2-4, complete driveshafts ~\$5-8 from multiple sellers); every Artisan already knows the platform. Weaknesses: QC variance between batches (motor/ESC lottery), soft stock servos, marketing-grade speed claims; factory-direct communication is transactional, so most buyers work through large Chenghai consolidators. Buy strategy: order vehicles + a 25% spares basket per PO; inspect to the Chapter 9 car checklist; standardize on the brushless 144010 spec for rental duty. **RCW 5.**

MJX / Meijiixin Toys — Chenghai veteran (roots 1983 as Jiaxin Toys, renamed Meijiixin 2001, per company history) whose **Hyper Go** brushless line (14209/14210 1/14, 16207-class 1/16) redefined entry performance pricing and is a named touring/GT platform in the fleet doctrine. Strengths: modern brushless + gyro spec at \$80-130 wholesale, improving parts support, licensed-brand experience (Ferrari, Lamborghini historically), real factory scale. Weaknesses: hobby-grade parts network younger than WLtoys'; some Hyper Go plastics engineered close to their limits — fleet duty needs the metal-upgrade kit basket. **RCW 5.**

ZD Racing (Dongguan ZD Hobby Co., Ltd.) — Dongguan factory (Fenggang Town) with 10+ years in RC, brand storefronts in Chenghai. The DBX-10 1/10 desert buggy and 9106 truggy sit one durability class above the WLtoys band at \$110-180 wholesale [verify]. Use case: Track B upper class and staff/marshal vehicles. Strengths: metal-gearred diffs, real 45-70 km/h performance, factory OEM posture on Made-in-China. Weaknesses: parts pricing ~2× WLtoys; brand support thinner in the West. **RCW 4.**

HBX / Haiboxing — long-standing Chenghai maker of entry buggies; the 16889/16890 1/16 class (\$45-70 wholesale) is a candidate for the junior/training fleet where crash energy is low. Strengths: cost, simplicity, tolerant electronics. Weaknesses: plastic drivetrain at the margin of rental duty; verify current corporate form at audit. **RCW 4** (junior fleet).

Remo Hobby — Chenghai factory (Yongxin Industrial Area, Lianshang) building 1/16-1/8 trucks and buggies with unusually good metal content for the price. Qualified alternate to ZD Racing for Track B upper class. **RCW 3.**

JLB Racing (Shenzhen Golden Cheetah) — Shenzhen-based maker of the Cheetah 1/10 brushless platform (\$130-190 wholesale [verify]); known for speed, less for refinement. Qualified alternate; sample before fleet commitment. **RCW 3.**

SG / Pinecone Forest — budget 1/16 drift and touring line (SG-1603/1604) popular as gyro-equipped entry drifters at \$40-60 wholesale; a candidate for the Casual drift taster class where MST would be wasted. Brand/factory structure opaque — **[verify at factory audit]**. **RCW 3.**

Hosim / Xinlehong Toys — Chenghai manufacturer behind the Amazon-dominant Hosim brand (9125/9135 class). Useful as a price benchmark and emergency alternate; the park's inspection burden for XLH batches is higher. **RCW 2.**

LC Racing — Guangdong maker of genuinely competition-grade 1/14 buggies/truggies (EMB/PTG series, \$150-220 wholesale [verify]); the platform used by several national-level small-scale race series. Candidate for the park's racing-league upper class and staff race fleet. Verify factory location and OEM depth at audit. **RCW 4.**

HSP Racing (Guangzhou) — the classic first-generation Chinese 1/10 export platform (94122 tourers etc.). Aging designs but a gigantic, cheap, standardized parts ecosystem, and corporate kinship with RGT (crawlers, Section 3.4). Relevance now is mostly parts-layer and context. **RCW 2.**

Team Magic / 3Racing / Yeah Racing / GPM Racing — the Taiwan/Hong Kong performance-and-parts layer. Team Magic (competition touring) matters if the racing league professionalizes. **3Racing** (Hong Kong, PRC tooling) supplies the Sakura kit line — the cheapest credible competition chassis — plus vast option-part catalogues; **Yeah Racing** and **GPM** are the park's named channels for aluminium upgrade parts (knuckles, C-hubs, links) for WLtoys/MJX/Traxxas-pattern fleets at MOQ class A with 2-4 week availability. All communicate in fluent English and ship small orders fast — ideal for The Works' running upgrades. Ratings **2-4** as tabled.

Trade Hack. For Chenghai volume brands (WLtoys, MJX, HBX), the fastest wholesale route is often not the factory itself but two or three of the big Chenghai/Shenzhen RC consolidators that hold official distribution. Play them against each other for the same SKU basket: same goods, real competition, and they will bundle mixed-brand spares into one shipment — which the factories will never do.

3.3 Drift & touring specialists

#	Supplier	HQ / factory	Founded	Specialty	Tier	OEM	MOQ	Lead time	RCW
15	MST (Max Speed Technology)	Taiwan	2000s [verify]	RWD drift chassis (RMX 2.5/4.0, FXX)	Premium	L1	A-B	4-8 wk	5
16	LDRC	Guangdong [verify]	2020s [verify]	1/18 gyro drift RTRs, 1/14 scale tourers	Value	L1-L2 [verify]	B	3-6 wk	5

#	Supplier	HQ / factory	Founded	Specialty	Tier	OEM	MOQ	Lead time	RCW
17	Sino-hobby	Shenzhen [verify]	2010s [verify]	1/28 mini touring/drift (Q-series)	Value-Mid	L1	B	3-6 wk	2
18	Firelap	Shenzhen [verify]	2000s [verify]	1/28 Mini-Z-class racers, timing ecosystems	Value	L2 [verify]	B	4-6 wk	2

MST is the named premium drift platform (source digest): Taiwanese engineering, purpose-built RWD geometry (RMX 2.5 RTR ~\$180-260 wholesale [verify]; kits less), superb parts machining. The 12-car drift fleet standardizes on one MST platform with Killerbody/Pandora shells (Section 3.10). Strengths: the best steering geometry and tuning depth in the class; strong global reputation that markets the park's drift program by itself. Weaknesses: Taiwanese pricing; RTR supply allocation can lag demand; parts must be stocked deep because substitutes don't fit. Note for logistics: Taiwan origin changes the freight/customs lane vs PRC goods (Chapter 8). **RCW 5.**

LDRC is the named entry-scale platform: 1/18 gyro-equipped drift RTRs (1803 NSX, 1804 MX-5 class, \$45-70 wholesale [verify]) with alloy upgrades and LED shells that photograph beautifully — the Casual-tier drift/touring taster. The brand's corporate structure (and its distinctness from the similarly named FPV maker LDARC of Dongguan) must be pinned down at audit — **[verify at factory audit]**. Strengths: instant fun, low crash cost, strong social-media presence. Weaknesses: young parts chain; verify continuity of supply before fleet-scale commitment; dual-source with SG (Section 3.2). **RCW 5.**

Sinohobby and **Firelap** cover the 1/28 micro class — relevant only if Phase 3 adds an indoor arena micro league (Volume 3 options list). Watchlist. **RCW 2.**

3.4 Crawlers & scale trail

Track C and the Crawler Park (Volume 3) plus the recovery fleet draw from here. This is the fastest-innovating value category in RC.

#	Supplier	HQ / factory	Founded	Specialty	Tier	OEM	MOQ	Lead time	RCW
19	MN Model (Chile Toys)	Cheng-hai, Shantou	2010s	1/12 scale crawlers (MN99S, MN86)	Entry-Value	L1-L2	B-C	3-6 wk	5
20	WPL	Cheng-hai, Shantou	~2016 [verify]	1/16-1/10 mini trucks/crawlers (C24, D12)	Entry-Value	L1-L2	B-C	3-6 wk	5

#	Supplier	HQ / factory	Founded	Specialty	Tier	OEM	MOQ	Lead time	RCW
21	RGT / Ruitai Model	Kun-shan, Jiangsu + Dong-guan plant	2013	1/24-1/10 crawlers (EX86 line); OEM/ODM	Value-Mid	L2-L3 (stated)	B-C	4-8 wk	5
22	FMS Model	Shen-zhen	2007	Licensed scale crawlers (FCX24/FCX10), planes	Mid	L2-L3	B	4-8 wk	4
23	RocHobby	Shen-zhen (FMS brand)	2010s	Scale crawlers/vehicles under FMS group	Mid	L2	B	4-8 wk	3
24	Eazy RC	Shen-zhen [verify]	~2021 [verify]	1/18 RTR scale crawlers (Triton, Arizona)	Value	L1	B	4-6 wk	3
25	Cross RC	Shen-zhen [verify]	~2013 [verify]	Premium scale truck/crawler kits (8x8, trial)	Premium	L2 [verify]	A-B	4-8 wk	3
26	Traction Hobby	China (Guang-dong) [verify]	~2017 [verify]	1/8 large-scale crawlers (Founder/KM)	Mid-Premium	L2 [verify]	A-B	4-8 wk	2
27	Yikong Model	China [verify]	~2020 [verify]	1/10-1/8 RTR scale crawlers (4102/4103)	Value-Mid	L1-L2 [verify]	B	4-6 wk	3
28	Capo Racing	China [verify]	2000s [verify]	Ultra-de-tailed crawler kits (JK Max)	Flagship	L1	A	6-12 wk	1
29	Orlando Hunter	Guang-dong [verify]	~2015 [verify]	1/32-1/24 micro scale kits	Value-Mid	L1	A-B	4-6 wk	2

MN Model (brand of Chile Toys, Xinxiang Road Industrial Zone, Chenghai) is canon: the MN99S 1/12 crawler anchors the entry crawler fleet at \$45-60 wholesale (fleet matrix). Strengths: absurd value, huge aftermarket of metal upgrades, simple electronics that Artisans can service in minutes, iconic Defender-style looks customers recognize. Weaknesses: toy-grade stock electronics (upgrade servo + ESC on fleet units), plastic transmission gears (fit the brass upgrade at induction — Volume 7 build sheet), batch QC variance. **RCW 5.**

WPL (Fengxia Street, Chenghai) is the C24/C34/D12 mini-truck factory — the second leg of the entry-crawler dual source and the park's scale-utility comedy fleet (D12 drift vans are a social-media weapon). Wholesale \$30-60. Same strengths/weaknesses profile as MN with an even bigger modding culture. **RCW 5.**

RGT (Ruitai Model Technology) — established 2013, HQ Kunshan (Jiangsu), with a modern Dongguan factory (2020) running CNC, injection molding, and ABB body-cutting; explicitly offers OEM/ODM (company statements). The EX86 1/10 crawler line (\$120–200 wholesale [verify]) is the named premium-tier crawler candidate alongside TRX-4-class imports, and RGT's OEM posture makes it the leading candidate for a park-spec crawler (park colours, XT60, RCW Node pre-fit — Chapter 5). Strengths: real factory, stated OEM/ODM openness, trademark registrations in EU/US, scale range 1/24–1/10. Weaknesses: brand less known to Western customers than Traxxas — mitigated by park context where customers ride the experience, not the badge. **RCW 5.**

FMS Model (Shenzhen, est. 2007) — 300+ staff, licensed-scale specialist (Land Rover, Toyota licences) whose FCX24/FCX18/FCX10 crawlers bring showroom-grade detail at Value-Mid pricing; also a first-rank foam-plane maker (Section 3.6). The FCX24 (\$60–90 wholesale [verify]) is the premium mini crawler option; licensing means bodies the public instantly recognizes. **RCW 4**, with **RocHobby** (its scale sub-brand) at **3**.

Eazy RC / Cross RC — related Shenzhen scale-crawler operations [corporate linkage **verify at factory audit**]: Eazy RC sells accessible 1/18 RTRs; Cross RC sells premium multi-axle truck and trial kits (\$250–900) beloved by advanced hobbyists. Eazy RC is a Casual-fleet candidate; Cross RC is a Works showcase/build-class platform (Volume 7 custom-build curriculum). **RCW 3 / 3.**

Traction Hobby, Yikong, Capo, Orlando Hunter — respectively: 1/8 large-scale crawlers with presence (premium supervised class candidate, **2**); fast-rising value RTR crawlers to dual-source the premium tier (**3**); the Rolls-Royce of crawler kits, one display/build unit at most (**1**); micro-kits for retail/Academy build classes rather than fleet (**2**).

3.5 Construction, mining & agriculture

The Construction Division (Volume 4) is the park's differentiator, and this supplier set is its backbone. Doctrine reminder: the rental fleet is electromechanical (lead-screw) — Huina class; hydraulics (Kabolite/LESU/JDModel class) are premium supervised experiences and display only.

#	Supplier	HQ / factory	Founded	Specialty	Tier	OEM	MOQ	Lead time	RCW
30	Huina (Guangdong Huina Model)	Cheng-hai, Shantou	2012	1/14 electromechanical construction fleet (1580/1582/1583)	Value	L1–L2	B	4–8 wk	5
31	Kabolite	Cheng-hai, Shantou (Huina premium line)	~2018 [verify]	Full-hydraulic 1/14–1/16 machines (K970, K961/963, K336)	Premium-Flagship	L1	A	6–12 wk	5
32	LESU Model	Danzao, Foshan	2000s [verify]	1/14 hydraulic excavators, trucks, metal option parts	Premium	L2–L3	A–B	6–12 wk	4

#	Supplier	HQ / factory	Founded	Specialty	Tier	OEM	MOQ	Lead time	RCW
33	JModel (Eagle Machinery)	HK entity; Guangdong production [verify]	2010s [verify]	CNC hydraulic excavators/loaders/dozers	Flagship	L2 [verify]	A	8-16 wk	3
34	Tongde	China [verify]	[unverified]	1/14 hydraulic loaders/dozers, aggressive pricing	Mid-Premium	[unknown]	A-B	6-12 wk	2
35	XDRC	China [verify]	[unverified]	1/14 construction models & metal parts	Mid	[unknown]	B	6-10 wk	2
36	Diecast Masters (RC line)	US brand; PRC manufacture	2010s	Officially licensed Cat® RC/diecast machines	Mid-Premium	L0-L1	A	via distribution	2
37	Double E (Doubleeagle Industry)	Chenghai, Shantou	1988	Licensed RC toys & E351 tractors; CaDA bricks; 300+ molding machines	Entry-Value	L2-L3 (stated)	C	4-8 wk	5
38	Heng Long	Chenghai, Shantou	1986 (workshop roots)	1/16 RC tanks, metal chassis sets; some marine	Value-Mid	L1-L2 (claimed)	B	4-8 wk	2
39	Huina parts ecosystem (consolidators)	Chenghai/ Shenzhen	—	Tracks, pins, motors, lead-screws for 15xx fleet	Value	—	A	1-3 wk	4

Huina (Guangdong Huina Model Co., Ltd., Chenghai, founded 2012) is the single most important supplier in this plan. The 1580 V4 excavator (\$350-420 wholesale), 1582/1573 dump trucks (\$150-200), and 1583 wheel loader (\$180-220) are the Mining Zone fleet (fleet matrix; 1:3 excavator:truck ratio). Huina pivoted from electronics gifts to RC construction around 2015 and now leads the global hobby segment, with in-house mold making and export compliance experience. Strengths: metal structures with lead-screw (fluid-free) actuation exactly matching the park's electromechanical doctrine; spares availability; V-generation improvements each cycle; a factory used to container-quantity buyers. Weaknesses: control feel is steppier than hydraulics (accepted trade-off); electronics are the weak point under rental duty (stock 15% spare mainboards/ESCs); documentation is thin, so The Works writes its own service manuals (Volume 7). OEM asks in play: park colours, XT60, Heavy-Node pre-wiring [negotiate at audit]. **RCW 5.**

Kabolite — Huina's premium hydraulic line from the same Chenghai group: K961/K963 excavators (\$1,750-1,900 retail class), K336GC, K970 flagship (~\$10,500-12,500 retail, 31 kg, brushless, 18-channel radio) and K5701 dump (~\$670) (research anchors, July 2026). Role: the **premium supervised experience and showcase machine** — one to two units in Phase 1, expanding with the Premium Shift program (\$22/38 pricing tier). Strengths: near-industrial hydraulic realism that anchors marketing and justifies premium pricing; shared parentage

with Huina simplifies commercial relations. Weaknesses: hydraulic maintenance burden (seals, fluid — exactly why doctrine bans it from open rental); five-figure flagship capex; allocation queues on new releases. **RCW 5** (premium tier).

LESU Model Technology (Danzao Town, Nanhai District, Foshan) — the metal-machining champion of 1/14: hydraulic excavators (Aoue-SK500, PC360 class), mining trucks, and the deepest catalogue of CNC option parts and truck components in the industry; one of the earliest 1/14 truck-parts makers in China (company statements). Role: premium display/experience alternates to Kabolite; source of metal upgrade assemblies for the truck fleet; long-term partner for Phase 3 showcase builds. Kits demand skilled assembly — a feature for The Works' build program, a cost everywhere else. **RCW 4**.

JDModel / Eagle Machinery — flagship-grade CNC hydraulic machines (excavators, wheel loaders, the D575 dozer) retailing \$1,900–7,400 via Western resellers; HK-registered trading/manufacturing entity with PRC production [structure **verify at factory audit**]. Watch for Phase 3 destination showcase pieces; not a Phase 1 purchase. **RCW 3**. **Tongde** and **XDRC** are lower-priced hydraulic challengers seen through reseller channels; data is thin and quality reports mixed — sample and audit before any commitment. **RCW 2 / 2**.

Diecast Masters (RC) — the officially Cat-licensed line (US brand, Chinese manufacture). Relevance: licensed realism for retail shelf and possibly a licensed premium experience; but park fleet purchases route to Huina/Kabolite economics. **RCW 2**.

Double E (Doubleeagle Industry (China) Ltd., Chenghai, founded 1988) is canon for Agriculture: the E351 tractor (1/16, \$80–120 wholesale) with hitch-standardized implement library (Volume 4, Agriculture Zone). One of Shantou's largest toy groups: 300+ injection-molding machines, in-house mold production, BSCI/CE/ASTM certification, and licensing relationships (Mercedes-Benz, Volvo, JCB, Land Rover among others per company statements) — meaning L2–L3 OEM capability is credible if the park later wants park-branded tractors. Weaknesses: toy-grade electronics on entry SKUs (fleet units get servo/ESC upgrades per Volume 7); MOQ class C for custom work. **RCW 5**.

Heng Long (Chenghai, roots 1986) — the world's dominant RC tank maker (1/16 battle tanks, metal chassis options, IR combat systems), with marine SKUs. Not in current fleet doctrine, but tanks are a recurring customer request and a plausible Phase 3 “armour arena” event class; keep warm. **RCW 2**.

Field Note. Entry 39 is not one company but a function: two or three Chenghai/Shenzhen consolidators that keep Huina 15xx tracks, idlers, pins, lead-screw assemblies, and mainboards on the shelf. The Works' continuous consumables (track pins, grease, motors — source digest spares list) flow through them weekly; the factory itself only sees quarterly volume POs. Qualify two, scorecard them like factories (Chapter 10).

3.6 Aviation & FPV

Phase 2 fleet (~20 aircraft, netted airfield — Volume 5). Procurement begins Month 10–12; directory maintained from Day 1 because electronics cross-pollinate (radio links, chargers, batteries).

#	Supplier	HQ / factory	Founded	Specialty	Tier	OEM	MOQ	Lead time	RCW
40	Volantex / Ex-hobby	Shantou	2000 (brand 2005)	Foam trainers, warbirds, gliders; RTF eco-systems	Value	L2-L3 (stated ODM)	B-C	4-8 wk	5
41	XK / WLtoys air	Cheng-hai, Shantou	2010s	Entry fixed-wing (A250/A430), helis	Entry-Value	L1	B-C	3-6 wk	4
42	Eachine	Guangzhou/Shenzhen (Bang-good house brand)	~2014 [verify]	Entry FPV quads, RTF bundles	Entry-Value	L0-L1	A-B	stock	3
43	BetaFPV	Long-gang, Shenzhen	2017	Tiny Whoop/cine whoops, RTF FPV kits	Value-Mid	L2	B	4-6 wk	5
44	iFlight	Huizhou	2014	FPV quads (Nazgul/Cidora), motors, frames	Mid	L2-L3	B	4-8 wk	4
45	GEPRC	Shenzhen	brand 2012	FPV quads (Cinelog), OEM/ODM offered	Mid	L2-L3 (stated)	B	4-8 wk	4
46	Flywoo	Shenzhen [verify]	~2019 [verify]	Micro long-range quads (Explorer)	Mid	L2 [verify]	B	4-6 wk	3
47	HGLRC	Shenzhen	2016	FPV components, budget quads	Value-Mid	L2	B	4-6 wk	3
48	Speedy-Bee	Shenzhen	2010s [verify]	Flight controllers, app-configurable stacks	Value-Mid	L2	B	4-6 wk	3

#	Supplier	HQ / factory	Founded	Specialty	Tier	OEM	MOQ	Lead time	RCW
49	Happy-model	Shenzhen/Dongguan [verify]	~2016 [verify]	Ultralight whoops (Mobula series)	Value	L1-L2	B	4-6 wk	4
50	EMAX	Shenzhen [verify]	~2004 [verify]	Motors, Tinyhawk RTF line	Value-Mid	L2	B	4-6 wk	3
51	T-Motor	Nanchang, Jiangxi	~2009 [verify]	Premium UAV/FPV motors	Premium	L2-L3	B	4-8 wk	2
52	Radio-Master	Shenzhen [verify]	~2019 [verify]	EdgeTX radios (TX16S/Pocket), ELRS	Value-Mid	L1-L2	B	4-6 wk	4
53	LDARC	Dongguan	2010s	FPV drones/parts; drone-soccer ecosystems	Value	L2 (claimed)	B	4-6 wk	2

Volantex (Shantou VolantexRC / Exhobby, est. 2000; VolantexRC brand 2005) is the anchor fixed-wing supplier: crash-tolerant foam trainers (Trainstar, Sport Cub, ASW28 gliders) at \$40-120 wholesale, plus the Racent marine line (Section 3.7). Company states ODM services and ~\$3.35 M annual export on its Alibaba profile. Strengths: RTF completeness, gyro-stabilized trainers ideal for tethered/instructed Shifts, one commercial relationship covering air and water. Weaknesses: foam airframes are consumables under rental duty (order 30-40% airframe-only spares); electronics adequate but not premium. **RCW 5.**

BetaFPV (Shenzhen Longgang, est. February 2017) anchors the indoor/netted FPV program: Tiny Whoop-class quads (Meteor65/75) and Cetus RTF kits (\$25-60 wholesale per quad [verify]) that bounce off nets and people harmlessly — the only FPV class permitted for uninstructed customers under Volume 5 safety doctrine. Strengths: the most complete beginner-to-intermediate ecosystem in micro FPV (quad + goggles + radio + simulator in one carton), deep spares (frames, canopies, motors sold in fleet-friendly multipacks), and a brand customers already know from YouTube. Weaknesses: product churn — a given Meteor revision lives 12-24 months, so fleet buys must include end-of-life spares. **RCW 5.**

Happymodel (Shenzhen/Dongguan [verify]) builds the Mobula6/Mobula7 ultralight whoops that dominate the sub-25 g class — the qualified dual source to BetaFPV under Rule DS-1, with near-identical economics and the lightest airframes (least sting on contact, a safety selling point). Parts interchange partially with BetaFPV stock (motors, cameras), which compresses the spares bin. **RCW 4.**

iFlight (Huizhou, 2014) is the largest of the freestyle/racing makers (Nazgul Evoque and Chimera lines, plus motors and frames sold as components) with a genuine factory campus in the Greater Bay Area and consistent global distribution. Role: the 3-5-inch instructor, display, and cinematic tier — flown by staff pilots and licensed advanced customers only

(Volume 5 licensing gates). **GEPRC (Shenzhen, brand 2012)** — legal entity Shenzhen Beizao Innovation Technology, ~100–200 staff, 3,000–5,000 m² plant, explicit OEM/ODM offer per marketplace data — is the credible counterparty if the park later wants a park-liveried cinewhoop fleet (Cinelog class) for the spectator-feed drone that films races (Volume 9 media plan). **Flywoo** (Explorer long-range micros) and **HGLRC** (Shenzhen, 2016; budget components) are qualified alternates and the component layer for The Works' FPV repair bench. **SpeedyBee** matters for a different reason: its app-configurable flight controllers let Artisans tune and diagnose without a laptop — a workflow win worth standardizing on for staff quads. **RCW 4 / 4 / 3 / 3 / 3** respectively.

EMAX (Shenzhen [verify], one of the oldest motor houses in FPV) supplies the Tinyhawk RTF line — the strongest single-box alternative to BetaFPV's Cetus for the training fleet — and motors with long revision lives, which fleet maintenance loves. **T-Motor (Nanchang, Jiangxi)** is the premium motor tier: relevant only if Phase 3 fields a sponsored demo/race team where failure is reputational. **RadioMaster (Shenzhen [verify])** standardizes the airfield's radio fleet: EdgeTX open-source firmware, multi-protocol RF, and per-student model profiles — one TX16S/Pocket pool serving every trainer protocol on the field, with lockable profiles that enforce beginner rates (the airborne cousin of the ESC detune doctrine). **XK/WLtoys air** (A250/A430 class, \$30–60) equips the “first flight” bulk tier where write-offs are budgeted, riding the existing WLtoys commercial relationship. **Eachine**, Banggood's house brand, is a stock-availability convenience for odd items rather than a factory relationship — useful, never load-bearing. **RCW 3 / 2 / 4 / 4 / 3**.

3.7 Marine

Phase 2 pond complex (~24 vessels — Volume 6). Small category, but the venue-starved boat niche gives it outsized community value.

#	Supplier	HQ / factory	Founded	Specialty	Tier	OEM	MOQ	Lead time	RCW
54	Joysway Hobby	Dong-guan	~2009	RC sail-boats (Dragon-Flite 95), speed-boats	Value-Mid	L2-L3 (stated)	B-C	4-8 wk	5
55	Volantex marine (Racent)	Shantou	2012 (line)	Value speed-boats (Vector series)	Value	L2	B-C	4-8 wk	4
56	TFL Hobby	Shen-zhen [verify]	2000s [verify]	CNC-aluminium/fiber-glass race hulls	Premium	L2 [verify]	A-B	6-10 wk	3
57	Feilun	Shantou/Shen-zhen [verify]	2000s [verify]	Entry boats (FT011/FT012)	Entry-Value	L1	B-C	3-6 wk	3

#	Supplier	HQ / factory	Founded	Specialty	Tier	OEM	MOQ	Lead time	RCW
58	Hen-glong marine	Cheng-hai, Shantou	(see #38)	Entry/mid boats from the tank house	Entry-Value	L1	B	4–8 wk	2

Joysway (Dongguan, est. ~2009) is the strategic marine partner, and the reason is one product family: the **DragonForce 65 / DragonFlite 95** one-design racing sailboats with organized global class associations. One-design means every hull is identical by rule — which is the park’s racing doctrine (skill decides, hardware doesn’t) already institutionalized by someone else. Volume 6 selects the DF65 V8 as the rental sail fleet with two DF95s as club/flagship boats; either way the park plugs straight into an existing worldwide ranking culture, giving the sailing league instant legitimacy. Around it, Joysway’s electric hulls (Bullet, Magic series, \$40–110 wholesale [verify]) cover the powerboat Shifts, and the company’s stated OEM/ODM posture makes park-liveried hulls a realistic L1–L2 ask. Weaknesses: sailboats demand wind and coaching (staffing implication, Volume 6); electric hull electronics need the same waterproofing audit as every marine brand. **RCW 5.**

Volantex Racent (Vector SR48/SR65/SR80, \$30–90 wholesale) is the volume speedboat tier — self-righting hulls, tolerant electronics, and it rides the existing Volantex air relationship, so one supplier meeting covers two divisions. **TFL Hobby (Shenzhen [verify])** machines CNC-aluminium and fiberglass race hulls one class above anything else in the table — the premium demo/race tier and a source of hardware (struts, rudders, couplers) for The Works’ marine bench. **Feilun** (FT011/FT012 class, \$25–60) is the budget dual source and a retail-shelf line; treat its included electronics as replaceable. **Heng Long marine** exists mainly as an alternate quote. Marine-specific procurement notes: order 2× propeller sets and flex-shafts per hull per season (the category’s tires); UV and water ingress make ESC potting/conformal coating a receiving-inspection checkpoint (Chapter 9); pond-weed tolerance testing is a sample-evaluation criterion (Chapter 11 form); and every marine PO includes hull-only spares at 15–20% of fleet count, because hulls fatigue at dock impacts long before electronics die. **RCW 4 / 3 / 3 / 2.**

3.8 Electronics: power systems, radios, chargers

The standardization layer — these choices bind the whole fleet (XT60, 2S/3S, common radio protocols, bunker charging doctrine).

#	Supplier	HQ / factory	Founded	Specialty	Tier	OEM	MOQ	Lead time	RCW
59	Hobby-wing	Long-gang, Shenzhen (+ Huizhou motor plant)	2005	ESCs & brushless systems (QuicRun/EzRun/XeRun)	Mid-Premium	L2–L3	B	4–8 wk	5

#	Supplier	HQ / factory	Founded	Specialty	Tier	OEM	MOQ	Lead time	RCW
60	Flysky	Shenzhen (Dongguan production)	2006	Radio systems (NB4/ NB4+, GT5, i6); custom RF solutions	Value-Mid	L2-L3 (stated)	B-C	4-8 wk	5
61	Radiolink	Futian, Shenzhen (Dongguan plant)	2003	Radios (RC4GS/ RC6GS/ AT10), receivers, GPS modules	Value-Mid	L2-L3	B	4-6 wk	4
62	Surpass Hobby / Rocket	Longgang, Shenzhen	2013	Budget-mid brushless motors/ ESC combos	Value	L2	B-C	3-6 wk	5
63	SkyRC	Shenzhen	2008	Chargers (T1000 quad-channel class), analyzers	Mid	L2	B	4-6 wk	5
64	ToolkitRC	Bao'an, Shenzhen	2018	Compact smart chargers (M6D/ Q6AC), tools; custom charging OEM (stated)	Value-Mid	L2-L3 (stated)	B	4-6 wk	4
65	ISDT	Shenzhen	~2015 [verify]	Smart chargers (K4 dual-channel class)	Mid	L2 [verify]	B	4-6 wk	4
66	HOTA	Shenzhen [verify]	[unverified]	Value multi-channel chargers (D6 Pro class)	Value-Mid	[unknown]	B	4-6 wk	3
67	Power HD	Shenzhen [verify]	2000s [verify]	Digital servos 15-45 kg class	Value-Mid	L2	B-C	3-6 wk	4

Hobbywing (Shenzhen, 2005) is the named ESC standard (source digest): QuicRun 10BL120/16BL30-class ESCs for the racing fleet, with programmable profiles that implement the park's tuning-personality and detune doctrine (Volume 3). A national "Little Giant" enter-

prise with real firmware depth — the counterparty for the Chapter 5 ask of **park-spec ESC profiles** (thermal + current limits tuned for rental abuse, locked customer profiles). **Flysky (Shenzhen, 2006)** is the named radio standard: NB4/GT5-class surface radios, i6-class air; the company advertises customized wireless solutions, which is exactly the Phase 2–3 conversation about park-locked transmitter firmware and QR-bound handsets (Volume 13 binding flow). **Surpass Hobby (Shenzhen Surpass Tech, 2013)** supplies fleet motors (540/550 brushed for construction; 2838/3650-class brushless for cars) at \$6–25 wholesale — the named motor source. **SkyRC (2008)** and **ISDT** are canon for the charging bunker (T1000 / K4-class multi-channel balance chargers per battery doctrine), with **ToolkitRC** (2018; states custom charging OEM work) the value alternate and field-tool source, **HOTA** the budget dual source [audit before reliance]. **Radiolink** doubles as an RCW Node component source (its GPS modules are drop-in candidates for the ATGM336H position in the Node BOM — Volume 13 engineering note). **Power HD** standardizes the 15 kg/25 kg servo spec from the source digest’s spares list. All entries here communicate in professional English and are accustomed to B2B/OEM terms; MOQs for custom firmware asks are the binding constraint (Chapter 5).

3.9 Batteries

The most safety- and logistics-critical category (UN3480/3481 rules, Chapter 8; bunker doctrine, Volume 7). Fleet standard: 2S/3S LiPo, XT60, 3:1 ratio.

#	Supplier	HQ / factory	Founded	Specialty	Tier	OEM	MOQ	Lead time	RCW
68	Grepow (Gens Ace / Tattu)	Longhua, Shenzhen (4 plants)	1998	Full-stack LiPo maker; RC + UAV packs; custom cells	Mid-Premium	L3–L4	C	6–10 wk	5
69	CNHL (China Hobby Line)	Shenzhen [verify]	~2005 [verify]	Direct-sale value LiPo (Black/MiniStar)	Value	L1–L2 [verify]	B	3–6 wk (stock)	5
70	Ovonic (Ampow)	Guangdong [verify]	~2017 [verify]	Value LiPo, strong 2S/3S car range	Value	L1 [verify]	B	3–6 wk	4
71	Zeee Power	Guangdong [verify]	~2010s [verify]	Value LiPo, market-place-optimized	Value	L1 [verify]	B	3–6 wk	3
72	Auline	Shenzhen [verify]	~2018 [verify]	FPV-focused Li-ion/LiPo packs	Value-Mid	L1 [verify]	B	4–6 wk	2

Grepow (Shenzhen Grepow Battery Co., Longhua, founded 1998) — parent of **Gens Ace** (RC, brand from 2010) and **Tattu** (UAV): ~3,500 staff, four factories totalling ~250,000 m², 300+ R&D engineers, and genuine custom pack and BMS programs (company statements) — one of very few directory entries that is a cell maker, not a pack assembler buying cells elsewhere. This is the park's quality anchor: Gens Ace packs for the racing and premium fleets, where sag consistency directly affects the lap-parity doctrine, and the long-term counterparty for a custom park pack — park-branded, XT60, capacity tuned so a full Casual Shift lands at the doctrinal ~30% buffer (realistic custom MOQ 500–2,000 packs [verify]; the park crosses that line in Year 1 at 3:1 ratios). Weaknesses: premium pricing (30–60% over value brands) and Class C MOQs for anything custom. **RCW 5.**

CNHL (China Hobby Line) is the named value source (source digest): a factory-direct e-commerce operation whose Black Series and MiniStar lines price roughly 30–50% under Western-branded equivalents with well-regarded consistency and — critically — a decade of routine experience shipping DG worldwide, including sea-freight bulk. The volume workhorse for construction, crawler, and marine fleets where a few percent of sag consistency is invisible. Verify the manufacturing entity behind the storefront at audit (pack assembly vs cell origin). **RCW 5.**

Ovonic (Ampow) and **Zeee Power** are the qualified value alternates under Rule DS-2 — both Guangdong pack assemblers optimized for marketplace retail, both credible on 2S/3S car packs, both to be batch-tested harder (Chapter 9 capacity sub-sampling) because marketplace-tier QC is statistically noisier. **Auline** matters when Phase 2 FPV wants Li-ion endurance packs for staff camera quads. Battery procurement rules, non-negotiable: every PO requires current **UN38.3 test summaries and MSDS**; every batch gets the Chapter 9 LiPo inspection including per-cell voltage and IR sampling; suppliers must ship via DG-certified forwarders (Chapter 8); and no pack enters service without an RC WORLD OS asset record binding it to cycle-count tracking and the bunker charging log (Volume 13). **RCW 4 / 3 / 2.**

3.10 Bodies, upgrade parts & accessories

#	Supplier	HQ / factory	Founded	Specialty	Tier	OEM	MOQ	Lead time	RCW
73	Killer-body	Shenzhen [verify]	~2011 [verify]	Licensed hard/poly bodies, scale accessories	Mid-Premium	L2–L3	B	4–8 wk	5
74	Pandora RC	Japan (design); PRC-fac-ing supply chain [verify]	2000s [verify]	1/10 drift body kits (JDM styles)	Premium	L1	A–B	4–8 wk	4

#	Supplier	HQ / factory	Founded	Specialty	Tier	OEM	MOQ	Lead time	RCW
75	Team C	Brand (EU-linked); PRC manufacture [verify]	2000s [verify]	Polycarbonate bodies, wide catalogue	Value-Mid	L2 [verify]	B	4–6 wk	3
76	Injora	Shenzhen [verify]	~2016 [verify]	Crawler upgrade parts, wheels/tires, scale accessories	Value	L1-L2	A-B	2–4 wk	4

Killerbody (named in source digest) supplies the drift/touring fleet’s licensed body shells and scale detail parts — the visual identity of Track A. Buy pre-painted park-livery batches (L2 ask: custom paint/decal MOQ realistically 50–200 shells [verify]). **Pandora RC** is the premium Japanese-designed drift-shell line for league and showcase cars (its “China ops” are effectively distribution/production interfaces — treat commercially as an import via HK/JP channels [verify]). **Team C** provides value polycarbonate volume. **Injora** is the crawler-fleet accessory engine — cheap, fast, vast catalogue — ideal for retail-shelf stock (Volume 9 retail plan) as well as fleet upgrades.

3.11 Diorama & scale-scenery suppliers

Volume 2, Chapter 4 carries the commissioned deep study of Chinese diorama and miniature-environment makers; this section holds only the procurement interface. Three buying channels: (1) **architectural-model-supply manufacturers** (scale trees, figures, textured sheet — the Guangzhou/Shenzhen model-shop supply belt; Evemodel-class online brands [verify entities at order time]); (2) **Chenghai/Yiwu généraliste suppliers** for signage, fencing, cones, barrels, container props at toy pricing; (3) **brick-construction makers** (Double E’s CaDA line, MouldKing class) for modular buildings the Academy can rebuild seasonally. Procurement treats scenery as MOQ-class A-B consumables ordered twice yearly with the fleet waves; landed-cost discipline still applies (it is bulky, low-value freight — sea LCL only, never air).

Trade Hack. The park’s most photogenic scenery — graded aggregate, boulders, timber — is local (quarry and landscaping supply), not imported. Import only what must be to-scale and manufactured (figures, signage, structures). This halves scenery freight and makes the Mining Zone’s material replenishable in a pickup truck (Volume 4 site operations).

4. Factory Rating System

4.1 Purpose and scope

Every supplier expected to receive more than \$5,000/year, or to supply any fleet-critical item at any spend, must carry a **Factory Rating** before the first volume PO. The rating is pro-

duced once at onboarding (desk research + video audit, upgraded to on-site audit at the next China trip) and refreshed annually. It feeds the quarterly scorecard (Chapter 10), which tracks performance; the Factory Rating tracks capability.

4.2 The six dimensions and weights

Dimension	Weight	What is measured	Evidence sources
Quality system	25%	Incoming/in-process/final QC existence and records; defect rates on samples; AQL literacy; calibration of test gear	Audit walk-through, sample inspection results, ISO 9001 (bonus, not required)
Communication	15%	Response time, English/technical clarity, one named account contact, honesty about problems	90-day correspondence log
Engineering depth	20%	Real R&D staff, mold shop access, BOM transparency, ability to discuss changes at L2-L3	Engineering call, tooling photos, change-request test
Capacity & flexibility	15%	Lines, shifts, realistic monthly output vs. promises; small-batch tolerance; peak-season behaviour	Audit, order history, peak-season test order
Compliance & ethics	15%	Business licence coherence; export record; safety/cert files (EN71/ASTM/CE/FCC/UN38.3 as applicable); labour conditions observed	Registry checks, certificate verification, audit observation
Financial stability	10%	Registered capital, years trading, litigation flags, credit report; does it survive a bad season?	Third-party company report (\$30-60), payment-term behaviour

Scoring: each dimension 1-5 (anchored rubrics in the audit workbook, Chapter 11); weighted sum × 20 gives a 0-100 score.

Score	Class	Procurement consequence
≥ 80	A — Strategic	Eligible for OEM programs, annual agreements, 30/70 terms, sole-PO status within dual-source rules
65-79	B — Qualified	Volume POs allowed; 100% pre-shipment inspection until two clean quarters
50-64	C — Conditional	Sample/small POs only; corrective-action plan required to grow
< 50	D — Not approved	No POs; re-audit possible after 12 months

4.3 Audit question bank (excerpt)

The full 90-question bank lives in the procurement workbook; the twenty highest-yield questions:

1. Show me the production line building this SKU today. (If it isn't running: when did it last run?)
2. Who owns the injection molds for this product — you, or a customer? May I photograph them?
3. What is your incoming QC procedure for motors/ESCs you buy in? Show the last rejection record.
4. What percentage of last month's output failed final QC, and what were the top three defects?
5. Show me your repair/rework station. (No rework station = defects ship.)
6. Which export markets do you ship to, and under which certifications? Show current test reports for this SKU.
7. What is your realistic monthly capacity for this SKU in October? In March?
8. How many workers return after CNY, on average? (Honest answers: 70–90%.)
9. If we standardize on this platform for three years, what changes for us? (Tests imagination and B2B literacy.)
10. Can you fit customer-supplied electronics (a 25×25 mm PCB) inline between receiver and ESC on your line? What would you charge per unit?
11. What is the smallest custom-colour batch you have actually produced? For whom (category, not name)?
12. Walk me through what happens when a customer rejects a batch at pre-shipment inspection.
13. Which components in this product have single-source suppliers upstream of you?
14. Show me how finished goods are packed for a sea container. Drop-test standard?
15. What are your payment terms for a first order, and when did you last give 30/70 to a new customer?
16. Who is my engineering contact by name, and what are their working hours?
17. What spare parts do you commit to producing after this SKU is discontinued, and for how long?
18. Has your company changed its registered name or legal entity in the last five years? Why?
19. What happens to our tooling and our spec if we stop ordering for a year?
20. What do you not make well? (The only honest factories answer this one.)

Field Note. Question 20 is the audit inside the audit. A factory that names a real weakness (“our painting line is inconsistent, we outsource metallic colours”) is managing quality; a factory that answers “no weakness” is managing you.

4.4 The 45-minute video audit script

The free pre-qualification step referenced in Section 2.6, run over WeChat/WhatsApp video with the sales contact walking and the buyer directing. Insist the call is live (ask them to pan

to a window, a clock, today's newspaper equivalent — a dated production board works). Sequence:

1. **Gate and signage (2 min).** Company name on the building vs. the licence name. Photograph.
2. **The line running your SKU or its family (15 min).** Count stations and workers; watch one unit assembled end-to-end; ask the line worker (via the guide) what the most common assembly fault is. Look for torque drivers with set clutches vs. raw electric screwdrivers — a small tell with high signal.
3. **Incoming-material and QC benches (10 min).** Ask to see today's incoming inspection record and the reject shelf. An empty reject shelf is a bad sign, not a good one.
4. **Warehouse and packing (8 min).** Carton stacks, humidity, drop-test rig if claimed, how packed goods are staged by customer.
5. **Mold shop or mold storage (5 min).** For factories claiming L2-L3: see the tools, ask which are customer-owned.
6. **Close (5 min).** Ask the three hardest bank questions (Q4, Q8, Q20 from Section 4.3) face-to-face; note whether answers match the earlier email answers.

Score the six rating dimensions provisionally the same day, while impressions are fresh; the on-site audit later confirms or corrects. Cost: zero. Suppliers who refuse a video walk with two weeks' notice are auto-classed D.

5. OEM & Custom Programs

5.1 What “park-spec” means

RC WORLD's fleet is standard product, modified. The park's OEM program converts the highest-value modifications from Works labour (paid at artisan hours) into factory options (paid pennies at assembly). The Phase 1-2 ask list, in priority order:

Ask	Level	Target suppliers	Realistic MOQ [verify]	Cost effect	Value
XT60 connector standard, no proprietary plugs	L2	All vehicle makers	often 50-100/SKU	±\$0	Kills connector chaos; battery doctrine
Park colours + decals, custom carton	L1	Huina, WLtoys, MJX, MN, WPL, Double E	100-500/SKU	+\$0.5-2/unit	Brand coherence; theft deterrence
Detuned ESC profile (current/thermal caps, soft throttle curve)	L2	Hobbywing, Surpass; vehicle makers using own ESCs	100-500 units or firmware fee \$500-2,000	+\$0-1/unit	Fleet lifetime + lap-parity doctrine
RCW Node pre-install (harness tap, Node mounting, JST XH pigtail for Heavy-Node)	L2	Huina, RGT, MJX, WLtoys [negotiate]	200-500/SKU	+\$1-3/unit labour	Deletes ~20 min/vehicle of Works labour

Ask	Level	Target suppliers	Realistic MOQ [verify]	Cost effect	Value
Reinforced high-failure parts (metal knuckles/arms as line-fit)	L2-L3	MJX, RGT, ZD, Huina	500-1,000 or tooling	+\$2-6/unit	Halves top failure mode
Upgraded servo line-fit (15/25 kg Power HD-class)	L2	Construction & crawler makers	100-300/SKU	+\$3-8/unit	Deletes the #1 Works retrofit
Park-branded battery pack (label + capacity spec)	L1-L2	Grepow, CNHL	500-2,000 packs	±\$0 at volume	Asset control, doctrine-tuned capacity
Custom body shells (park livery, sponsor panels)	L2-L3	Killerbody, Team C	50-200 shells	+\$2-5/shell	Sponsorship inventory (Volume 9)

5.2 Tooling economics

Structural asks (L3) mean tooling money. Planning figures for Chinese injection tooling [all verify at quotation]: simple single-cavity tool for a small arm/knuckle **\$1,500-5,000**; multi-cavity production tool **\$5,000-20,000**; a full vehicle body tool **\$8,000-30,000**. Amortization rule: an L3 ask is approved only when $(\text{tooling cost} \div \text{expected 3-year unit volume}) + \text{unit-price delta} < 60\%$ of the Works retrofit labour it replaces. In practice this means Phase 1 buys **zero custom tools**; the first plausible L3 case is a reinforced touring-car arm at Phase 2 fleet scale, and even that competes against simply stocking more \$2 arms (Chapter 6 math).

5.3 Who owns what

Every OEM engagement states in writing: (1) tooling paid by RC WORLD is RC WORLD property, tagged and photographed, removable on 30 days' notice; (2) park livery, logos, and the RCW Node design are RC WORLD IP, usable only on park POs; (3) factory improvements to their own base product remain theirs. Do not pay for tooling on another customer's mold, and do not accept "free tooling" amortized invisibly into unit prices without a stated buyout schedule.

5.4 IP protection basics: NNN before NDA

A Western NDA is close to unenforceable against a Chinese factory. The standard instrument is the **NNN agreement — Non-Disclosure, Non-Use, Non-Circumvention** — drafted (a) in Chinese as the governing text, (b) under PRC law, (c) with jurisdiction in the supplier's local court or CIETAC arbitration, and (d) with liquidated damages in RMB per breach event. Cost from a China-focused firm: \$800-2,500 per template class [verify]. Doctrine:

- NNN **before** any RCW Node schematic, park-spec ESC parameter set, or customer-data-adjacent integration is shared. Fleet colour schemes and decals do not need NNN theatre.
- The Node's real IP protection is architectural: factories receive a mounting/harness spec and a black-box unit, never gerbers or firmware (Volume 13 security posture).

- Register the RC WORLD trademark in China (classes 28 toys, 41 entertainment, 9 software) **before** the first branded OEM order — China is first-to-file; cost \$300–600/class via agent [verify].
- Assume anything customer-visible will be imitated if the park succeeds. The moat is the operation (telemetry, ERP, venue, brand), not the plastic.

Trade Hack. The cheapest OEM concession in China is paint, and the most expensive is firmware. Sequence asks accordingly: win colours/connectors/cartons in the first negotiation (they cost the factory nothing and build the habit of saying yes), and bring the ESC-profile and Node-pre-install asks only once you are a proven repeat buyer with a face — typically at the second Canton Fair meeting, not the first email.

5.5 Case study: the detuned ESC, end to end

The single highest-value L2 ask, walked through as a template for all others. **Problem:** stock 144010-class buggies pull ~55–60 km/h — thrilling, and lethal to consistency: crash rates and drivetrain wear scale super-linearly with top speed, and lap-parity doctrine (2–3% across the fleet) is impossible when ESC batches vary. **Specification:** a locked customer profile at ~70% throttle ceiling, softened initial punch, thermal cutback at 85 °C, low-voltage cutoff aligned to the 3.4 V/cell doctrine — and a PIN-protected staff profile at full power for marshal vehicles and events. **Routes:** (a) vehicle factory flashes its stock ESC supplier's firmware variant — cheapest, but opaque and batch-risky; (b) the park standardizes on Hobbywing QuicRun-class ESCs fleet-wide and orders them pre-programmed from Hobbywing at 100–500-unit MOQ, then has the vehicle factory line-fit them (the recommended route: one ESC vendor, auditable parameters, and the swap doubles as the QC uplift the value platforms need anyway); (c) Works flashes everything by hand — the Phase 0 fallback, ~6 min/vehicle. **Economics (route b):** ESC delta +\$9–14/vehicle over stock [verify], offset by measured reductions in crash damage and ESC failures; break-even at roughly one avoided control-arm-plus-labour incident per vehicle per season — comfortably cleared by Volume 3's damage assumptions. **Verification:** every batch sampled at PSI with a programming-card readback against the parameter sheet (Chapter 9 checklist line). The general lesson: every OEM ask needs this same chain — problem, spec, route options, economics, verification — before it goes in front of a factory.

6. MOQ & Lead-Time Planning

6.1 MOQ bands by category

Planning bands compiled from published wholesale behaviour of the Chapter 3 directory — **confirm per supplier at RFQ:**

Category	Stock wholesale (class A/B)	Custom colour (L1)	Config OEM (L2)	Structural (L3)
Value cars/buggies (WLtoys/MJX class)	20–100 units	100–500	200–500	1,000+
Crawlers (MN/WPL/RGT)	20–100	100–300	200–500	1,000+

Category	Stock wholesale (class A/B)	Custom colour (L1)	Config OEM (L2)	Structural (L3)
Construction (Huina 15xx)	10-50	100-200	200-300	500+
Premium hydraulics (Kabolite/LESU)	1-5	10-50	20-50	negotiated
Foam aircraft (Volantex/XK)	20-100	100-500	300-500	1,000+
FPV quads (BetaFPV class)	10-50	50-200	100-300	500+
Boats	20-100	100-300	200-500	1,000+
ESCs/motors/radios	50-200	100-500	200-1,000 (fw fee alt.)	1,000+
LiPo packs	50-200	500-1,000	500-2,000	2,000+
Spares/upgrade parts	10-100 lots	n/a	n/a	tooling-gated

The Phase 1 fleet order (64 cars, 30 construction, 12 tractors, 16 crawlers incl. 4 premium plus 4 recovery crawlers, batteries at 3:1) clears stock-wholesale MOQ everywhere and reaches L1 thresholds on the four biggest SKUs (touring cars, buggies, dump trucks, entry crawlers) when initial + first replenishment orders are combined on one annual agreement — which is exactly how procurement should present it (Chapter 7).

6.2 The lead-time stack

Total replenishment lead time = **production + QC + freight + customs + inland**, and each stage has a season:

Stage	Courier route	Sea LCL route	Notes
Production (stock SKU)	1-3 wk	3-6 wk	0 if consolidator holds stock
Production (L1/L2 batch)	—	4-8 wk	Post-CNY add 2-4 wk
Pre-shipment inspection	2-4 d	3-5 d	Book at 80% completion
Freight	3-7 d	30-45 d port-to-door	Air freight middle option 7-12 d
Customs & inland	1-3 d	3-10 d	DG (batteries) adds 3-7 d
Planning total	2-5 wk	9-17 wk	Use 12 wk sea planning figure

6.3 Buffer-stock math — worked example: touring-car control arms

Doctrine: no common failure idles a revenue asset >1 working day. Control arms are the #1 impact casualty (source digest failure list). Inputs (Volume 3/7 planning figures):

- Fleet: 24 touring/GT cars; average 30 Shifts/car/week in season.
- Failure rate: 1 arm per 90 Shifts (planning figure from rental-duty assumptions; recalibrate from RC WORLD OS failure logs after Month 3).
- **Demand d** = $24 \times 30 \div 90 = 8 \text{ arms/week}$; observed weekly σ assumed 4.

- Replenishment: consolidator stock via courier, **L = 3 weeks**; deep restock via sea PO, 12 weeks.
- Service target 98% ($z = 2.05$).

Safety stock = $z \times \sigma \times \sqrt{L} = 2.05 \times 4 \times \sqrt{3} \approx \mathbf{14 \text{ arms}}$. Reorder point = $d \times L + SS = 8 \times 3 + 14 = \mathbf{38 \text{ arms}}$. Order quantity: arms cost ~\$2.5 landed in 10-packs; carrying cost is trivial against a \$15 lost Shift, so order in 100-unit lots (≈ 12 weeks of demand) and let the pre-CNY rule (Section 1.5) override: on 1 Dec, on-hand $\geq 10 \text{ wk} \times 8 + 14 = \mathbf{94 \text{ arms minimum}}$.

The same arithmetic runs for every HIGH-turnover spare in Volume 7's inventory list; RC WORLD OS computes d and σ live from maintenance logs and flags reorder points automatically (Volume 13, inventory module). Two structural lessons: (1) cheap parts get deep stock — the math almost never justifies scarcity on sub-\$5 items; (2) the battery buffer is set by the 3:1 doctrine and DG lead times (sea, 12-17 wk), so battery POs are placed two seasons ahead, not from reorder points.

6.4 The expedite ladder

When buffer math fails anyway — a demand spike, a lost carton, a failed lot — procurement climbs a pre-priced ladder rather than improvising:

Rung	Action	Typical cost premium	Buys back
1	Pull from dual-source consolidator stock (courier)	+15-30% on parts	2-4 wk
2	Air-freight the in-production balance (non-DG only)	+\$3.5-6/kg	3-5 wk vs sea
3	Cannibalize the spare-vehicle pool (Volume 7 authorizes)	internal	immediate
4	Cross-class substitution (run adjacent fleet class at higher rotation)	revenue mix effect	immediate
5	Local retail purchase at destination pricing	+100-180%	days

Each rung is legitimate; the failure mode is skipping to rung 5 by panic. The OS shortage alert names the rung the playbook recommends, and every rung-2+ event triggers a Chapter 10 review of why the buffer failed.

7. Negotiation Strategy

7.1 The relationship premise

Chinese factory negotiation is repeat-game, not one-shot. Price is remembered for a quarter; behaviour is remembered for a decade. The park's posture: a **small but permanent customer** — modest orders by export standards (\$75k-120k/year across the directory) but repeating, predictable, technically literate, and publicity-generating. That posture, played consistently, buys more than aggressive price-grinding: priority slots before CNY, honest defect disclosure, engineering favours, first allocation on new releases.

7.2 Price anchoring around published wholesale

Never open a negotiation blind. For every SKU the RFQ pack (Chapter 11) records: (a) the factory's own Alibaba/Made-in-China listed band; (b) two consolidator quotes; (c) the Western street price $\div 2.2$ as a sanity anchor. Open at the bottom of the evidenced band, not below it — quoting under the credible band signals ignorance and invites the classic counter: a “yes” delivered with silent component downgrades. Target outcomes: 8–15% under listed wholesale on volume SKUs, plus non-price wins (spares bundles, carton customization, inspection access) that are worth more than the last 3%.

Trade Hack. Ask every vehicle factory for a “**spares ratio**” line in the PO: 5–10% of order value in a factory-picked assortment of that SKU's fastest-moving parts, at parts-wholesale prices, packed in the same cartons. Factories price this generously (it's their own inventory), it ships freight-free inside the vehicle shipment, and it seeds The Works' bins with exactly what the factory knows breaks.

7.3 Payment terms

- **Norm: 30/70 T/T** — 30% deposit at PO, 70% against copy of bill of lading or, better, **against passed pre-shipment inspection** (fight for this sequencing; A-class suppliers grant it).
- First orders with unproven suppliers: **Alibaba Trade Assurance** even at 1–2% platform cost — the dispute mechanism is real leverage while trust is zero.
- Never 100% up front, at any discount. Never Western Union/private accounts; company-to-company bank transfer only, matching the licence name (mismatch = the Section 2.6 protocol failed).
- Letters of credit are overkill below ~\$50k/PO; revisit for Phase 2 container-scale orders.
- Currency: quote and pay USD Year 1; consider RMB pricing later if the bank spread + factory USD-hedging premium exceeds ~2% [review with Finance, Volume 10].

7.4 The Canton Fair playbook

Twice-yearly rhythm (Section 1.5). Tactics that pay:

1. **Book incumbents before the fair** — booth meetings are for renewal signatures and new-model previews; discovery happens in the aisles.
2. **Walk Phase 2 (toys/hobby) with the fleet matrix printed.** A buyer holding a real order book is visibly different from a browser; booth staff escalate to owners fast.
3. **Collect quotes on a standard one-page RFQ card** (SKU, quantity band, target spec, inspection expectation) so post-fair comparison is mechanical.
4. **Visit factories the same week** — Chenghai is 4 hours from Guangzhou by high-speed rail; a fair trip that skips factory floors wastes half its airfare.
5. **Bring the telemetry demo.** A two-minute live RCW Node + dashboard demo does more for OEM credibility than any deck; factories decide whether you're technically real within minutes.
6. Off-years alternates: Hong Kong Toys & Games Fair (Jan) for scouting; China Toy Expo Shanghai (Oct) for Chenghai brands' domestic lines.

7.5 Consolidated-freight leverage and sample protocol

Because the park ships consolidated (Chapter 8), it can dangle **carton-level add-ons** to smaller suppliers: “join our monthly LCL from Shenzhen and your effective MOQ to us drops.” This wins better pricing from accessory makers (Injora/Yeah Racing class) who otherwise gate on courier costs. Sample protocol, fixed: pay full sample price + courier without negotiation (it marks you as serious); order samples from two suppliers minimum per category; run samples through the Volume 7 teardown rubric and the Chapter 11 sample-evaluation form; photograph everything into the supplier record; then negotiate volume with sample defects as the agenda. Refuse “free samples against future order promises” — they create soft obligations that cost more than the sample.

7.6 When it goes wrong

Escalation ladder for disputes (defect batch, silent spec change, delay): (1) evidence pack within 48 h — photos, video, inspection report, PO clause; (2) proposed remedy sized to the harm (rework, discount, freight credit, replacement in next shipment) — factories settle fast when the ask is proportionate; (3) withhold the 70% only if inspection-gated terms allow; (4) Trade Assurance / CIETAC per contract; (5) failover to the dual source (Chapter 10 triggers) — and say so plainly. Do not threaten what you will not execute; in a repeat-game market, empty threats price you as noise.

8. Import & Logistics Strategy

8.1 Incoterms for this business

Incoterms (2020) define where the supplier’s risk and cost end. The four that matter here:

Term	Supplier delivers...	RC WORLD pays/handles...	When to use
EXW (Ex Works)	at their factory door	everything: China pickup, export clearance, freight, import, delivery	Only with a strong forwarder; maximum control, maximum admin
FOB (Free On Board, named port)	loaded on vessel at (e.g.) Shenzhen/Yantian	ocean freight, insurance, import, inland	Park default for sea shipments — clean split, forwarder-friendly, prices comparable across suppliers
CIF (Cost, Insurance, Freight to named port)	to destination port, minimally insured	import clearance, duties, inland	Acceptable for one-off buys; hides freight margin, weakens control — avoid as default
DDP (Delivered Duty Paid)	to the park gate, duties paid	nothing but receiving	Small courier orders and sample flows; verify duties were genuinely paid, not smuggled under-declared

Doctrine: **FOB Shenzhen/Yantian (or Shantou) for consolidated sea; DDP courier for samples and urgent spares; EXW only when consolidating multiple small Chenghai suppliers via our own forwarder’s warehouse.**

8.2 Freight modes and when

Mode	Cost (planning)	Door-to-door	Use for
Courier (DHL/FedEx/SF class)	\$5–9/kg	3–7 d	Samples, urgent spares <30 kg, documents
Air freight (forwarder, airport-to-airport + handling)	\$3.5–6/kg	7–12 d	Season-saving replenishment 30–300 kg; never batteries without DG lane
Sea LCL (less than container)	\$30–80/m ³ + \$150–350 fixed fees per shipment	5–8 wk	Park default: monthly/bi-monthly consolidation
Sea FCL 20 ft (~28 m ³ usable)	\$1,500–4,000 lane-dependent (volatile post-2024; verify at booking)	5–8 wk	Fleet waves ≥ ~15 m ³ — initial fleet, annual expansion
Rail/truck (China–Europe lanes, if applicable)	between air and sea	3–5 wk	Verify by destination country

Rule of thumb: below ~2 m³ LCL fixed fees dominate — batch orders until a consolidation clears 3 m³. The Phase 1 initial fleet (~150 vehicles + spares + scenery) is realistically **one 20 ft FCL plus one DG battery shipment plus 2-3 LCL follow-ups** [verify volumes at packing-list stage].

8.3 Freight forwarders

Appoint **one primary freight forwarder with proven dangerous-goods capability and a South-China consolidation warehouse** (Shenzhen/Guangzhou), plus one backup. Selection checklist:

- ☐ Licensed for DG sea (IMDG) and DG air (IATA CAT-certified staff) — ask for battery shipment references, not assurances
- ☐ Consolidation warehouse accepting multi-supplier cartons with photo-on-receipt service
- ☐ Destination-country customs brokerage in-house or partnered
- ☐ All-in quotes to door showing every fee line (watch: fuel, CIC, peak, chassis, DO fees)
- ☐ Cargo insurance at ~0.2–0.4% of CIF value arranged per shipment
- ☐ Weekly status reporting the Procurement Lead can pipe into RC WORLD OS

8.4 Customs classification and duties

- **RC models and vehicles:** heading **9503** (toys, including reduced-scale models, whether or not working) covers most RC cars, construction models, and parts under the WCO nomenclature; some destination regimes push hobby-grade or camera-equipped items elsewhere (e.g., drones toward aircraft/UAS headings post-2022 WCO updates). **Classification and duty rates are destination-specific — verify locally with the customs**

broker before the first shipment; many jurisdictions apply 0% to modest duty on 9503 plus local VAT/GST on CIF + duty.

- **Batteries:** lithium cells/packs classify under **8507.60** (lithium-ion) with their own duty profile, and their transport is governed by DG rules regardless of duty (next section).
- Declare honestly. Under-declaration to save a few percent of VAT puts the entire container — and the operating licence narrative in front of investors — at risk. The landed-cost model (8.6) treats duty as a planning line of 0–5% pending local verification.

8.5 LiPo dangerous-goods reality — a major planning factor

Lithium batteries are Class 9 dangerous goods, and this constraint shapes the park's battery logistics more than price does:

- **UN3480** — lithium-ion batteries shipped alone. Air: IATA PI965, **cargo aircraft only**, state of charge $\leq 30\%$, Class 9 fully regulated (no passenger-flight uplift). Sea: IMDG Class 9, DG declaration, UN-spec packaging.
- **UN3481** — batteries **packed with or contained in** equipment (PI966/PI967). Slightly gentler handling; this is how RTR vehicles with included packs typically move.
- Every shipment needs the supplier's **UN38.3 test summary** and MSDS on file; carriers and forwarders will ask, and so will the park's insurer (Volume 10 risk schedule).
- Practical doctrine: **batteries travel by sea, in consolidated DG shipments, two seasons ahead of need** (lead time 12–17 wk). Air-freighting LiPo is possible via DG lanes at \$6–12/kg equivalent but is the emergency channel, not the plan. Vehicles ship with batteries removed to a separate DG consignment where feasible — it simplifies both inspection and any non-DG air contingency for the vehicles themselves.
- Small-parcel “battery included” e-commerce shipments that dodge DG paperwork are common in the hobby and **prohibited for the park**: uninsurable, seizable, and indefensible in a safety-first business that operates a public charging bunker.

Safety Warning. DG compliance is not only a shipping formality. Receiving inspection (Chapter 9) must verify packs arrive at storage charge ($\sim 3.80\text{--}3.85\text{ V/cell}$), undamaged and unswollen, before they enter the bunker inventory. A pack that shipped abused enters service pre-damaged — and fails in a customer's hands.

8.6 Worked landed-cost example

Consolidated LCL replenishment from Shenzhen, FOB, destination a mid-size international city (planning rates; rebuild with real quotes per shipment — template in Chapter 11):

Line	Basis	Amount
40 × MJX Hyper Go 1/14 @ \$95 FOB	invoice	\$3,800
20 × MN99S @ \$52 FOB	invoice	\$1,040
Spares basket (arms, gears, servos, tires)	invoice	\$1,150
FOB subtotal		\$5,990
Sea LCL 4.2 m ³ @ \$55/m ³ + \$260 fixed	freight	\$491

Line	Basis	Amount
Cargo insurance 0.3% × (FOB+freight)	insurance	\$19
CIF value		\$6,500
Import duty (planning 3% on CIF; verify locally)	duty	\$195
VAT/GST (planning 15% on CIF+duty; verify locally ; often recoverable)	tax	\$1,004
Broker + DO + inland delivery	fees	\$310
Pre-shipment inspection (1 man-day)	QA	\$290
Landed total (ex-recoverable VAT)		\$7,285

Landed-cost factor = $7,285 \div 5,990 \approx 1.22$ on **FOB** (ex-VAT). Planning factors adopted: **1.20-1.30 sea LCL; 1.12-1.18 FCL; 1.35-1.60 courier/DDP small orders; batteries add 3-6 points for DG handling**. Every unit cost quoted anywhere in this master plan as “landed” uses these factors until real lanes are priced.

9. Inspection & Quality Assurance

9.1 Third-party inspection layer

For every PO $\geq$ \$3,000, and every battery PO of any size, commission a **pre-shipment inspection (PSI)** at the factory when goods are $\geq$ 80% complete and packed. Providers in the QIMA / V-Trust / HQTS / SGS / Bureau Veritas / TÜV class operate across Guangdong on 48-hour booking at **\$230-350 per man-day** [verify current rate cards]; one man-day covers one location and roughly one AQL sample of the sizes below. Annual budget: 14-18 man-days $\approx$ **\$3,500-5,500** — noise against the fleet capex it protects. Factory audits (Chapter 4) from the same firms run \$300-700/day. Never let the supplier book or pay the inspector.

The service menu, and when the park uses each:

Service	When performed	Park usage
Pre-shipment inspection (PSI)	$\geq$ 80% produced & packed	Default on every qualifying PO
During-production inspection (DUPRO)	20-50% produced	First OEM batch of any L2+ ask; first post-CNY batch from B/C-class suppliers
Initial production check (IPC)	Materials/first articles	New tooling (L3) only
Container loading supervision (CLS)	At loading	FCL fleet waves and every DG battery consignment (verifies packaging, marks, quantities into the sealed box)
Supplier verification / audit	Onboarding, annual	Per Chapter 4

One tactical note: rotate inspection firms occasionally, and never announce which units the inspector will sample. Factories learn individual inspectors' habits; variety keeps the sampling honest.

9.2 AQL sampling, explained and tabled

PSIs use ISO 2859-1 / ANSI-ASQ Z1.4 sampling. The park's standard: **General Inspection Level II, AQL 0 critical / 2.5 major / 4.0 minor**. Critical = safety (battery damage, exposed mains, sharp edges); Major = function or durability loss (won't run, cracked chassis, wrong ESC); Minor = cosmetic. Working table:

Lot size	Sample size	Critical (0): Ac/Re	Major (2.5): Ac/Re	Minor (4.0): Ac/Re
51-90	13	0 / 1	1 / 2	1 / 2
91-150	20	0 / 1	1 / 2	2 / 3
151-280	32	0 / 1	2 / 3	3 / 4
281-500	50	0 / 1	3 / 4	5 / 6
501-1,200	80	0 / 1	5 / 6	7 / 8
1,201-3,200	125	0 / 1	7 / 8	10 / 11
3,201-10,000	200	0 / 1	10 / 11	14 / 15

Ac = accept at or below; Re = reject at or above. A rejected lot triggers the Chapter 10 corrective-action process: 100% factory rework and re-inspection at supplier cost is the standard remedy written into every PO (Chapter 11 checklist).

9.3 Park-specific PSI checklists

Container/lot of RC cars (touring/buggy class):

☐

Quantity, SKU, colour/livery vs PO; carton markings and drop-test spec

☐

Per sample: visual (chassis cracks, flash, paint), fastener torque spot-check, wheel/hex tightness

☐

Function: bind to radio, full throttle/brake/steer sweep on stand, 2-minute floor run, steering trim within spec, gyro (drift class) engages

☐

Electrical: connector is XT60 per park spec; ESC model matches BOM (photograph!); servo spec matches; no hot ESC/motor after run

☐

Spares basket present and matching packing list; manuals/decals in carton

☐

Serial capture: photograph model plate + MAC/ID where present for RC WORLD OS asset pre-registration

Lot of LiPo packs:

☐

UN38.3 summary + MSDS accompany shipment; cartons carry correct UN3480/3481 marks

☐

Per sample: voltage per cell 3.80–3.85 V (storage), cell IR within datasheet band, no swelling/dents/leaks, XT60 + JST-XH balance lead orientation correct

☐

Capacity test on sub-sample (2–5 packs): discharge at 1C to 3.5 V/cell $\geq$ 95% of rated

☐

Label: capacity, C-rating, batch code present; batch recorded for OS cycle tracking

☐

Reject lot on ANY critical (swollen/damaged cell) regardless of AQL arithmetic

Lot of excavators (Huina 15xx class):

☐

Metal structure: boom/stick/bucket pins seated, no casting cracks, track tension in spec, no missing track pins

☐

Function: full boom/stick/bucket travel under load (sample digs prepared sand tray), slew smooth, lead-screw current draw within spec (no stall clicking), limit behaviour at end stops

☐

Electronics: mainboard version noted, connectors seated, radio range spot-check 30 m

☐

15% spare mainboard/ESC line present per PO; grease points pre-lubricated

☐

Livery/decals per park spec; Heavy-Node pre-wiring present if the L2 ask applies (JST XH pigtail, correct polarity — meter every sample)

9.4 Incoming-goods inspection at the park

SOP PRC-01 — Incoming Goods Inspection. Rev 1.0 · Owner: Procurement & Logistics Lead · PPE: gloves, safety knife · Tools: multimeter, cell checker, scale, camera, OS terminal · Frequency: every shipment, within 24 h of arrival.

1. Photograph consignment sealed; verify carton count vs packing list; note damage on carrier POD before signing.
2. Quarantine batteries to the bunker ante-room immediately; no LiPo waits on the receiving dock.
3. Sample-check per Section 9.2 sizes against the relevant checklist (cars / packs / machines); PSI report open alongside — you are auditing the audit.
4. Register accepted assets in RC WORLD OS (fleet_inventory for vehicles, battery batch records for packs); print asset tags; route vehicles to The Works induction queue (Volume 7 build sheets: metal-gear fits, Node install, detune verification).
5. Log defects with photos into the supplier record; if the lot fails, freeze payment steps per PO terms and open corrective action (Chapter 10) within 48 h.
6. File customs docs, UN38.3, MSDS, inspection certificates into the compliance folder (insurer + authority audits, Volume 10).

10. Supplier Scorecards & Relationship Management

10.1 Quarterly scorecard

Every active supplier $\geq$ \$2,500/year is scored quarterly in RC WORLD OS on delivered performance (distinct from the Chapter 4 capability rating):

Metric	Weight	Data source	5 looks like	1 looks like
On-time delivery	25%	PO vs receipt dates	$\geq$ 95% lines on time	< 70%
Quality (lot acceptance + field failures)	30%	PSI results + Works failure logs	$\geq$ 98% lots accepted, field failures at plan	Rejected lots, systemic field failures
Price stability vs agreement	15%	Invoice audit	Honoured bands, honest surcharges flagged early	Silent increases, spec downgrades
Responsiveness	15%	Correspondence SLA log	< 24 h substantive answers	Ghosting, holiday blackouts unannounced
Documentation & compliance	15%	Shipment file completeness	Every doc right first time	Chasing UN38.3s at the port

Sample filled-in scorecard — Q3 (illustrative), supplier: Huina (vehicles + spares):

Metric	Score	Note
On-time delivery	4	1583 loader batch slipped 9 days post-Golden-Week; flagged in advance
Quality	4	2 lots accepted clean; field main-board failures 1.8%/quarter vs 2.5% plan
Price stability	5	Annual band honoured; freight surcharge documented
Responsiveness	4	Engineering answers in 36 h avg; sales in 12 h
Documentation	5	Complete, first time
Weighted total	4.3 / 5	Class A trajectory — renew annual agreement; raise Node pre-install ask at October meeting

10.2 Corrective action and failover triggers

Corrective-action process (CAP): any rejected lot, field-failure spike ($>150\%$ of plan for two consecutive months), or documentation breach opens a CAP: written problem statement with evidence $\rightarrow$ supplier root-cause + containment within 10 working days $\rightarrow$ agreed remedy and verification method $\rightarrow$ verified closure recorded on the scorecard. Two open CAPs = new POs require GM sign-off.

Failover triggers (Rule DS-5 executes): score < 2.5 for two consecutive quarters; a critical safety defect reaching customers; insolvency signals (asked for $>30\%$ deposit changes, licence changes, factory visits refused); discontinuation of a fleet platform without the con-

tracted spares tail. On trigger: dormant dual-source agreement activates, open POs complete under escrowed terms, and the supplier moves to Class C pending re-audit.

10.3 The annual review

Each January (pre-CNY freeze, Section 1.5): consolidate four quarters of scorecards, re-run the Chapter 4 capability rating for A/B suppliers, re-verify licences and certificates, refresh dormant dual-source pricing, and issue next year's annual agreements ahead of the April Canton Fair signing round. Output: a one-page **Supplier Portfolio Map** (category × class × spend) for the GM and board pack — the procurement function's annual report card.

11. Procurement SOPs & Templates

11.1 RFQ template (one page per SKU)

1. **Header:** RFQ ID, date, buyer contact, validity requested (60 d).
2. **Item:** SKU/model, photos, target spec table (scale, motor/ESC, servo, connector = XT60, battery = 2S/3S none-included option, colour/livery code).
3. **Quantities:** three bands (e.g. 50 / 150 / 400) — always ask three; the curve reveals cost structure.
4. **Terms requested:** FOB port, 30/70 T/T with 70% against passed PSI, spares-ratio line 7% of order value, lead time ex-works, carton spec + drop test.
5. **Compliance:** certifications required for destination; battery documentation if applicable.
6. **Evaluation notice:** "Order subject to sample approval and pre-shipment inspection at AQL 0/2.5/4.0."

11.2 Purchase-order checklist

- ☐ Supplier legal name matches licence and bank account
- ☐ SKU, spec table, and photographed sample reference attached to PO ("as approved sample #S-042")
- ☐ Price, currency, Incoterm, port named
- ☐ Payment terms with inspection gate sequenced explicitly
- ☐ Lead time with penalty/remedy clause (e.g. 1%/week credit after grace week; capped)
- ☐ Spares-ratio line included (vehicles)
- ☐ Rejected-lot remedy: 100% rework + re-inspection at supplier cost
- ☐ Spares-tail clause: parts availability commitment (target 24 months post-discontinuation)
- ☐

Battery lines: UN38.3 + MSDS + DG shipping responsibility assigned

☐

Tooling/IP clauses if OEM content (Chapter 5.3)

☐

PO entered in RC WORLD OS with milestone dates before deposit released

11.3 Sample-evaluation form (summary)

Header (SKU, supplier, date, evaluator) then scored sections, 1-5 each: build quality on tear-down (fastener quality, plastics, bearing feel); electronics inspection (ESC/servo/motor identity vs BOM, solder quality, connector spec); performance test (speed/function per class rubric, temperature after 10-minute duty cycle); durability proxy (drop/crash test per Volume 7 rubric, water/dust splash for outdoor classes); spares check (top-10 failure parts priced and sourced from 3 sellers); serviceability (time for an Artisan to reach motor, servo, arms). Verdict: Approve / Approve with conditions / Reject — two evaluator signatures (Procurement Lead + Head Artisan). Filed to the supplier record; referenced by every subsequent PO.

11.4 Landed-cost calculator layout

Columns per shipment line: FOB unit price × qty → FOB subtotal → +allocated freight (by volume share) → +insurance → CIF → +duty (rate by HS line) → +local tax (flag recoverable) → +fees allocated → +inspection allocated → **landed unit cost** → landed factor vs FOB. Maintained as an OS worksheet; every fleet asset's landed cost writes to `fleet_inventory` for Volume 10's cost-per-Shift model. Planning factors from Section 8.6 pre-fill until real quotes replace them.

11.5 Supplier onboarding checklist

☐

Directory entry created/updated (Chapter 3 format)

☐

Business licence + registry cross-check (Section 2.6) filed

☐

Video factory audit completed and scored (Chapter 4)

☐

Third-party verification or on-site audit for spend > \$10k/yr

☐

Samples evaluated and approved (11.3)

☐

NNN signed if any OEM/IP exposure (5.4)

☐

Bank details verified by test payment + callback

☐

Dual-source identified for the category (DS rules, 1.3)

☐

Scorecard shell opened in RC WORLD OS; account contact named both sides

☐

First PO ≤ 30% of intended steady-state volume

12. Volume Summary & Cross-References

This volume converted the park's founding procurement thesis — direct-from-China whole-sale as a structural moat — into an operating system: a doctrine (TCO per Shift, dual-sourcing rules DS-1-5, fleet standardization as leverage, a CNY-and-Canton-Fair-shaped calendar); a map of the four manufacturing clusters and a factory-vs-trader verification protocol; a directory of **76 profiled manufacturers and brands** across ten categories, each with location, tier, OEM level, MOQ class, lead time, and an RCW relevance rating, with unverifiable specifics honestly flagged for factory audit; and the machinery around it — a six-dimension weighted factory rating with an audit question bank, an OEM program that sequences cheap asks (colours, XT60) before expensive ones (ESC firmware, RCW Node pre-install) under NNN protection, MOQ/lead-time planning with worked buffer-stock math, a China-specific negotiation playbook, an import strategy dominated by the LiPo dangerous-goods constraint and a 1.20-1.30 sea landed-cost factor, AQL-based inspection with park-specific checklists, quarterly scorecards with failover triggers, and the five templates that make a two-person procurement function executable.

Cross-references:

- **Volume 2, Chapters 2 & 4** — the market context for the Chinese manufacturing ecosystem and the commissioned diorama-supplier study this volume's Section 3.11 interfaces with.
 - **Volume 3 / 4 / 5 / 6** — the division fleet doctrines (platforms, classes, tuning parity, electromechanical-vs-hydraulic rule) that this volume's directory and OEM program serve.
 - **Volume 7** — The Works: spares lists, failure data, induction build sheets, and the tear-down rubrics used in sample evaluation; failure logs feed Chapter 6 buffer math and Chapter 10 scorecards.
 - **Volume 9** — retail and sponsorship inventory (bodies, accessories, branded merchandise) sourced through Sections 3.10-3.11.
 - **Volume 10** — landed costs, payment terms, duty verification, and insurance schedules; the financial source of truth this volume's planning factors roll into.
 - **Volume 12** — franchise procurement: franchisees buy through the park's supplier agreements; this volume's directory and scorecards become licensed franchise assets.
 - **Volume 13** — RC WORLD OS procurement, inventory, and fleet modules that operationalize the POs, scorecards, buffer-stock triggers, and asset registration described here; RCW Node BOM sourcing.
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Volume 9 — Customer Experience & Loyalty

Document	RC WORLD — Master Development Plan, Volume 9 of 13
Title	Customer Experience & Loyalty
Revision	1.0 — July 2026
Status	Living document — bump revision on material change

Purpose of this volume. This volume is the design authority for everything a customer sees, feels, earns and keeps at RC WORLD. It defines the experience philosophy (immersion-as-product, friction-to-gameplay conversion), the end-to-end customer journey with owners and KPIs per touchpoint, and — as its core chapter — the complete **RC WORLD License** tiered progression system the founder’s brief mandates: XP earning rules, six license tiers with numeric thresholds and equipment unlock gates, division endorsements, a catalogue of 60 named achievement badges with telemetry triggers, the Gears loyalty economy, leaderboards and season structure with a two-track Season Pass, the three canonical membership tiers, and the corporate, birthday, school and university product lines built on top of them. It closes with the customer-facing mobile app walkthrough and the service standards and recovery doctrine. Volumes 3-6 gate equipment and score events against the tiers defined here; Volume 13 builds the software that runs them. If a number in this volume and a division volume ever disagree, this volume is the source of truth for licenses, badges, Gears and memberships.

Intended readers. The General Manager and front-of-house team (all chapters); the marketing lead (Chapters 2, 5-9); the RC WORLD OS product team (Chapters 3-5, 10 — build detail in Volume 13); division leads whose equipment is license-gated (Chapter 3); the sales lead for corporate and education products (Chapters 7-9); investors assessing retention economics (Chapters 1, 3, 5, 6).

Chapters

1. Experience Philosophy
2. The Customer Journey
3. The RC WORLD License & Progression System
4. Achievement Badges
5. Gamification Mechanics
6. Memberships & Season Passes
7. Corporate Packages
8. Birthday Parties & Family Products

- 9. Schools, Universities & STEM
- 10. The Mobile App Experience
- 11. Service Standards & Recovery
- 12. Volume Summary & Cross-References

1. Experience Philosophy

1.1 Immersion is the product

RC WORLD does not sell radio-controlled vehicles by the minute. It sells the experience of being someone — a race driver on a timed grid, an excavator operator with a production quota, a harbour pilot with a barge to berth, a pilot on a buddy-box working toward solo wings. The vehicles are the instruments of that identity, and every design decision in this volume follows from one rule inherited from the original Omni-Zone blueprint: **the park is a miniaturized industrial complex, not a hobby sandbox**. A sandbox hands a customer a machine and leaves the meaning to them. An industrial complex hands them a job — with a shift clock, a work order, a score, a supervisor, and a career.

The industrial frame is not set dressing; it is the load-bearing structure of the commercial model:

- **The Shift** (the 20-minute billing block) reads as a work shift, not a rental countdown. Customers “clock in”, they don’t “start the timer”. Operator Shifts include a pit-stop battery swap performed by an Artisan as refuelling theatre.
- **Staff titles carry the frame.** Technicians are **Artisans**; the induction is a **Toolbox Talk**; the progression program is a **License**, not a “loyalty scheme”; the loyalty currency is **Gears**, machined tokens of work done.
- **Scores are production numbers.** Lap times, career tonnage, mission completions and smoothness indices are the customer’s professional record, displayed the way a real site displays safety days and output boards.
- **Progression is vocational.** You do not “level up” at RC WORLD; you get licensed on equipment — and the license genuinely unlocks machines the unlicensed cannot touch (Chapter 3).

Investor Note. The industrial frame is retention economics wearing a costume. Industry analysts (2025–2026) consistently find that competitive-socializing venues (karting, TopGolf-class formats) out-earn passive attractions on repeat visitation because scoring converts visitors into competitors. RC WORLD’s frame goes one step further: it converts competitors into careerists. A customer three badges short of their Operator license has an unfinished identity, and unfinished identities re-book. The 90-day second-visit KPI (operating target $\geq 22\%$ Year 1, building to $\geq 28\%$ by Year 3 — Volume 10 \$10.4) is owned by this volume’s mechanics.

1.2 The three audiences of every moment

Every experience moment is designed for three audiences at once, and is reviewed against all three before it ships:

1. **The operator** — the customer holding the transmitter. Their needs: clear controls, honest scoring, visible progress, no dead air.
2. **The spectator** — the family member, friend or passer-by watching. Their needs: legible drama (the hopper dump, the pit stop, the tow-truck rescue), comfortable sightlines, a reason to become an operator. Volume 2's competitive analysis is blunt: the non-driving half of every group is where F&B and conversion margin live.
3. **The camera** — the customer's phone. RC content is unusually algorithm-friendly (short, kinetic, legible without sound — Volume 2, Chapter 7), so every set piece is composed to be filmed: name-labelled leaderboard screens, overhead camera angles, automatic Shift highlight clips pushed to the customer's phone.

1.3 Friction-to-gameplay conversion

The doctrine that most distinguishes RC WORLD from every rental operation it was benchmarked against: **when an operational problem touches a customer, convert it into gameplay before you convert it into an apology.** Three canonical conversions, all inherited from the source blueprint and already engineered into Volumes 3–6:

Friction point	Naive handling	RC WORLD conversion
Vehicle dies mid-Shift (flat pack, breakdown)	Staff walks onto track, customer waits, refund argument	Tow-Truck Retrieval Protocol: 85 dB localized buzzer marks the casualty; customer hands in their transmitter and receives a 1/10 winch-equipped recovery crawler; they pilot the rescue themselves, earn Gears for a clean tow, then continue on a fresh vehicle. Customers never walk onto live tracks — the safety rule is the game
Battery life shorter than session ambition	Mid-session battery anxiety, “it died early” complaints	Industrial Shift pricing: billing decoupled from battery life; the Casual Shift returns the vehicle with ~30% buffer; the Operator Shift's mandatory 20-minute pit stop is staged as F1 refuelling theatre with the Artisan on camera
Aggressive driving drains packs and breaks fleet	Speed governors, staff telling customers off	F1-style power management: telemetry reads voltage sag live; throttle abuse visibly burns the customer's energy allocation; smooth drivers post better scores. The scoreboard does the discipline (Chapter 5.3)
Queue at popular classes	Standing in line, walk-aways	Live queue top-ups: at T-5 minutes the app checks the class queue; if nobody waits, a one-tap “Top-Up Shift” wallet extension appears. Scarcity is honest, and empty capacity sells itself (Chapter 10.6)

Friction point	Naive handling	RC WORLD conversion
Waiting for a booked slot	Dead time	Spectator missions in the app (predict the sprint-race podium for Gears), retail/F&B placement on the walking line, The Works viewing window

The test for any new friction point discovered in operations: can a customer be given agency inside the problem? If yes, design the game before designing the compensation. Only when agency is impossible (weather closure, our fault entirely) does the recovery doctrine in Chapter 11 take over.

Field Note. The Tow-Truck Retrieval Protocol polls as a highlight, not a failure. In pilot-format testing of comparable recovery gameplay, a majority of first-time customers who experienced a breakdown-plus-retrieval rated the visit higher than customers with no breakdown at all — the rescue is a story they tell. Do not “optimize away” breakdown theatre with over-cautious battery margins; the ~30% buffer doctrine already protects the fleet, and the retrieval fleet (4 recovery crawlers, park canon) exists to be used.

1.4 Progression must be earned, honest and legible

Three promises the progression system makes to customers, which every chapter of this volume must keep:

- **Earned:** license tiers cannot be bought. Memberships (Chapter 6) buy access and value, never tier. A Foreman-tier driver earned it on telemetry, and everyone on the leaderboard knows it.
- **Honest:** scores come from measured reality — the RCW Node’s voltage and position stream, the MYLAPS timing loop, the hopper load cells (± 20 g). Class-normalized leaderboards (Chapter 5.1) keep an entry-class child’s pace index comparable with a touring racer’s. The parity doctrine (lap-time parity within 2–3% per class, Volume 3) is what makes “the driver won, not the car” a true sentence.
- **Legible:** a customer can always answer “what do I do next?” in one glance at their license card: next tier, XP remaining, the specific badges or practical checks still open.

2. The Customer Journey

2.1 The journey spine

The journey is designed as a loop, not a line: every stage ends by loading the next stage, and the final stage (results, badges, re-booking) is deliberately the strongest. The spine below is the day-visitor version; Section 2.4 gives the member and party variants.

Discovery → App download & Toolbox Talk → Arrival & wristband/QR → Vehicle binding → The Shift → Pit stop / battery-swap theatre → Results & badges → Retail & F&B → Re-booking.

2.2 Journey moments table

Each touchpoint has a single accountable owner and one primary KPI reviewed monthly on the GM dashboard (KPI definitions and instrumentation: Volume 13).

#	Moment	What happens	Owner	Primary KPI (target)
1	Discovery	Social clips, search, word of mouth; landing page sells the job fantasy, not the vehicles	Marketing lead	Cost per booked first visit ($\leq \$9$)
2	App download & account	Store listing → account → child profiles under guardian account	OS product team	Download→account conversion ($\geq 70\%$)
3	Toolbox Talk induction	8-12 min interactive module: battery limits, collision liability, track rules, tow-truck doctrine; doubles as signed digital waiver; per-division modules added later	Safety officer (content), OS team (build)	Completion rate pre-arrival ($\geq 60\%$)
4	Booking & wallet	Pick division, class, Shift type, slot; wallet loaded; Learner license issued on first booking	Front-of-house lead	Pre-booked vs walk-up ratio ($\geq 55\%$ pre-booked)
5	Arrival & entry pavilion	QR at gate → NFC wristband issued (doubles as license card token); park map + today's events on lobby screens	Front-of-house lead	Gate-to-first-Shift time (≤ 12 min)
6	Vehicle binding	At the counter: QR scan on transmitter pairs customer → transmitter → asset tag in RC WORLD OS; transmitter profile auto-set to license tier; OS refuses pairings above the customer's tier	Duty marshal	Binding errors per 100 Shifts (≤ 1)
7	The Shift	20-minute block; live dashboard on phone/track screens: battery %, time, position, score; marshals per zone SOPs	Division lead on duty	Shift utilization ($\geq 65\%$ of open slots)
8	Breakdown (if any)	Tow-Truck Retrieval Protocol (Chapter 1.3); fresh vehicle issued; Gears for clean tow	Duty marshal	Retrieval-to-resume time (≤ 6 min)

#	Moment	What happens	Owner	Primary KPI (target)
9	Pit stop theatre (Operator Shifts)	At minute 20, car called to pit; Artisan swap staged front-of-house: fresh pack, tire glance, “you’re clear” send-off	Pit-lane Artisan	Swap time (≤ 90 s)
10	Results & badges	Session summary in-app within 60 s: laps/tonnage/mis-sion score, smoothness index, XP, Gears, any badge pops (with sound, on the big screen if gold-class); highlight clip attached	OS product team	Summary open rate ($\geq 80\%$)
11	Retail & F&B	Exit route passes re-tail counter and kiosk; badge pins, park merch, starter RC kits; Gears burnable here	Retail/F&B lead	Attach rate — % of visits with F&B or re-tail spend ($\geq 45\%$)
12	Re-booking hook	Before the customer leaves Wi-Fi range: “next unlock” screen (XP to next tier, one suggested badge), one-tap re-book with off-peak discount	Marketing lead	90-day second-visit rate ($\geq 22\%$ Y1 / $\geq 28\%$ Y3, Volume 10 §10.4)
13	Post-visit	NPS ping at +24 h; highlight clip share prompts; win-back ladder if dormant (Chapter 6.5)	Marketing lead	NPS (≥ 55 Y1, ≥ 60 by Y3)

2.3 Designing the two “wow” windows

Journey research across FEC formats (Volume 2) is consistent: visits are remembered by their peak moment and their ending. RC WORLD engineers both:

- **The peak** is division-specific and staged: the first hopper dump registering on the tonnage board (Construction), the first timed lap posting to a named leaderboard (Motorsport), the pit-stop swap (any Operator Shift), the tow-truck rescue (any breakdown). Marshals are trained to announce peaks (“new personal best on Track A — bay 4”), because a witnessed score is worth two private ones.
- **The ending** is the results-and-badges screen plus the physical exit line past retail. The rule: **no customer leaves without knowing their next unlock**. The app’s closing screen is a career prompt, not a receipt.

2.4 Journey variants

Stage	Day visitor	Member (any tier)	Birthday party	Corporate group
Induction	Toolbox Talk on first booking	Already inducted; delta modules only (new divisions)	Guardian completes for minors in invite flow	Bulk induction link pre-event; 5-min live briefing on site
Entry	QR → day wristband	Member wristband/card, dedicated lane at peak	Host meets group at pavilion	Event coordinator meets organizer
Booking	App or walk-up	Priority windows (7–14 days by tier), member hours	Fixed run-sheet (Chapter 8.3)	Fixed run-sheet (Chapter 7.2)
The Shift	Standard queue	Streak shields, block allowances (Chapter 6)	Party classes only (entry class, crawlers, tractors)	Format-specific fleets reserved
Results	XP/Gears/badges	+ streak progress, season track levels	Paper license certificates + pin for each child	Team scoreboard, framed podium photo to organizer
Exit hook	Next-unlock + off-peak offer	Renewal health nudges (Chapter 6.5)	Party-guest voucher: first solo visit discount	B2B follow-up: league/venue offer within 48 h

Field Note. The single most operationally fragile moment in the spine is #6, vehicle binding — it happens at the counter, at peak, with a queue watching. Rehearse it like a pit stop: transmitter pre-staged by class, QR label on the transmitter and the car stand, and a fallback manual pairing code printed under the counter. If binding takes more than 40 seconds, the queue feels it; the ≤ 1 -per-100 error KPI exists because every misbind is a wrong-profile safety event, not just a delay.

3. The RC WORLD License & Progression System

3.1 System architecture: one license, two axes, four endorsements

The RC WORLD License is the park's single progression spine. Everything a customer earns anywhere in the park — laps, tonnage, missions, flight-school stages, badges — feeds one career. The system has deliberately simple bones:

- **One park-wide License tier** (six tiers, Section 3.3) driven by **XP**, a skill-and-activity score. Tier is permanent: once earned, never lost. Tier gates equipment, transmitter power profiles, and league eligibility.
- **Two axes of progression.** Performance (XP → tier) and frequency (visit streaks → Gears multipliers and season-track boosts, Section 5.4). The axes never mix: streaks cannot unlock equipment, and skill cannot substitute for the loyalty economics of showing up. A brilliant driver who visits twice a year is a Foreman with no streak; a devoted weekly visitor of modest skill is an Apprentice with a 20-week flame. Both are valuable customers, and each axis pays the behaviour it measures.
- **Four division endorsements** (Section 3.5) — Motorsport, Construction, Aviation, Marine — which sit on the license like categories on a real driver's license and gate division-specific equipment. Endorsements let an aviation-only customer progress without ever touching a race car, while the park-wide tier still rewards breadth.

Terminology note: the membership products (Chapter 6) reuse the Apprentice/Operator/Foreman vocabulary deliberately — the brand speaks one industrial language — but membership is purchased access and license tier is earned rank. The app renders them in different visual systems (membership = card colour; license = embossed tier seal) and staff scripts never say “level” for either: you hold a membership and you carry a license.

3.2 The XP model

XP is earned from four sources, all machine-measured through RC WORLD OS (schema and event pipeline: Volume 13):

Source	Base XP	Notes
Casual Shift completed (1 block)	50	Any division, any class
Operator Shift completed (2+ blocks)	120	Includes pit-stop; +30 per additional block beyond two
Premium-class Shift (\$22/\$38 classes)	+30 on top of base	Hydraulics, FPV, big crawlers, flagship vessels
Smoothness bonus	up to +50% of Shift base	Telemetry smoothness index $\geq 70 \rightarrow +25\%$; $\geq 85 \rightarrow +50\%$ (Section 3.4)
Sprint race / mission / regatta finish	40	Every scored event finish, any placing
Podium: 3rd / 2nd / 1st	+40 / +60 / +100	Class-normalized events only (Chapter 5.1)
League round completed	60	Plus podium bonuses; league season champion +500
Skill task / practical check passed	75–150	Marshal-verified tasks from the endorsement syllabi (Section 3.5)
Badge earned	50–500	Per badge value in the catalogue (Chapter 4)
Clean tow (Tow-Truck Retrieval Protocol)	40	Winch hook, recovery, no geofence breach
Toolbox Talk module (each)	25	Core + one per division + premium-equipment modules

Design rules that keep XP honest:

- **XP is never sold, gifted or multiplied by spend.** Gears (Chapter 5.4) carry all monetary generosity; XP carries only demonstrated activity and skill. This is the wall that keeps the license credible.
- **Daily earn softcap of 600 XP** (excess earns at 25%) — protects the ladder from binge-grinding and keeps tier ages meaningful.
- **Anti-gaming:** Shifts abandoned before 10 minutes earn no XP; smoothness bonuses require $\geq 60\%$ of the Shift under throttle; repeated same-badge attempts are unlimited but the XP pays once.

3.3 The six License tiers

#	Tier	XP threshold	Additional gates (all required)	Expected profile
L1	Learner	0	Core Toolbox Talk complete (includes waiver)	Every customer on first booking
L2	Apprentice	500	One clean full Shift (no kill-switch event, no geofence breach); 1 badge	Typically visit 2-3
L3	Operator	2,000	Smoothness average ≥ 70 over last 10 Shifts; 5 badges; 1 division endorsement at Basic	The committed regular; visit 6-10
L4	Foreman	6,000	12 badges incl. 1 Gold-class; 1 endorsement at Advanced; 1 scored event finish	Top ~15% of active customers
L5	Site Manager	15,000	25 badges; 3 endorsements at Basic+ (breadth gate); marshal-shadow session completed	Top ~3%; the community's senior figures
L6	Legend	40,000	Site Manager for ≥ 1 full season; 1 championship podium or 40 badges incl. 3 Gold; ratified by GM each season	Deliberately rare; the park's hall of fame

Tier ceremonies are staged: Apprentice through Foreman pop in-app with a leaderboard-screen announcement if the customer is on site; Site Manager and Legend are awarded at the monthly Community Night with a physical metal license card (Legend cards are individually numbered). Legends get their name on the entry-pavilion Legends board — permanent, like a topping-out plaque.

3.4 The smoothness index — F1-style power management as a skill score

The smoothness index is the park's signature telemetry statistic, computed per Shift by the RCW Node stream: a 0-100 score derived from voltage-sag variance (aggressive throttle spikes sag the pack; smooth loading doesn't), throttle-input smoothness, geofence discipline, and — on construction machines — boom/load shock events. It is the same doctrine that runs the F1-style energy-management races (Chapter 5.3): **the pack is your fuel; brutality is visible; smooth is fast**. The index appears live on the Shift dashboard as an analogue "operator gauge", so a customer can watch their own inputs move it. It gates the Operator tier (average ≥ 70) because smooth driving is precisely the skill that protects the fleet — the license system pays customers to behave like professionals, which is cheaper than repairing the alternative (fleet-damage economics: Volume 7).

3.5 Division endorsements

Each endorsement has three grades — **Basic** → **Advanced** → **Master** — earned through the division’s own syllabus of skill tasks and badges. Existing division ladders map onto the grades as follows (this volume ratifies and, where needed, supersedes earlier drafting):

Division	Basic	Advanced	Master	Syllabus anchor
Motorsport	Track-craft check: 10 clean laps, pit-lane entry, flag signals	Sub-110% of class reference lap; race-start procedure; 1 sprint podium	Sub-105% reference lap; league season completed; drift or formula check	Volume 3, Ch. 13
Construction	Load-cycle check: 10 weighed loads, no spillage penalty; haul-road rules	1 t career tonnage; loader check; Precision event finish	5 t tonnage; hydraulic orientation; Team Campaign season	Volume 4, Ch. 8–9
Aviation	= Ground Wings + Bronze Wings (ground school, buddy-box, solo checkride)	= Silver Wings (solo currency, aerobatics sign-off)	= Gold Wings + one type rating (Type: Jet / Type: FPV / Type: Heli)	Volume 5, §9.3
Marine	= Deckhand + Coxswain (clean harbour Shift; precision docking, barge push)	= Tug Master or Race Skipper (two-tug assist / speed-pond license)	= Sail Grade 3 or Submariner (B-rig weather sailing / precision hover)	Volume 6, Ch. 8 §8.4

3.6 Equipment unlock gates

The unlock table below is the canonical park-wide gate list. Division volumes referenced “the second license tier” and “tier 3” during drafting; **this table supersedes those phrasings** — gates are now expressed as named tiers plus endorsement grades.

Equipment class	Price band	Minimum license tier	Endorsement grade	Supervision
Entry classes: 1/14 racing, dump trucks, tractors, entry crawlers, DF65 discover sail, buddy-box trainers	\$15	Learner	—	Standard marshal cover
Standard classes: 1/10 touring, buggy/rally, excavators, harbour tugs	\$15	Apprentice	Division Basic	Standard
Drift, short-course, wheel loader, dozer, fast-electric boats	\$15	Apprentice	Division Basic	Standard
Premium crawlers (TRX-4 class), premium tractors + implement library	\$22	Operator	Division Basic	Standard
Premium hydraulics (Kabolite K961/ K963 class supervised Shifts)	\$22/\$38	Operator	Construction Advanced	One-on-one Artisan

Equipment class	Price band	Minimum license tier	Endorsement grade	Supervision
FPV (drone racing, FPV crawler cams)	\$22	Operator	Aviation Type: FPV rating (Ground Wings prerequisite — Volume 5 §9.3)	Spotter required
Flagship hydraulic showcase (K970-class seat time)	event-priced	Foreman	Construction Master	One-on-one Artisan, listed slots
Formula RC / drag events; jet-class and scale-heli aircraft (Phase 2+)	event-priced	Foreman	Division Master	Event marshals / instructor
Recovery crawlers (outside a retrieval event, i.e. recreational towing Shifts)	\$15	Apprentice	—	Standard

Transmitter power profiles bind to tier automatically at vehicle pairing (the OS refuses over-tier pairings — Volume 3, Section 12.1): **Learner/Apprentice → Rookie profile** (steering 75%, throttle 60%, softened expo), **Operator → Standard profile** (90/80), **Foreman and above → Licensed profile** (100/100 within class caps). The tier is therefore not symbolic: a Foreman’s identical car is measurably faster in their hands because the electronics trust them more.

3.7 Tier privileges and discounts

Privilege	Learner	Apprentice	Operator	Foreman	Site Manager	Legend
Gears earn multiplier (license axis)	×1.0	×1.0	×1.1	×1.2	×1.3	×1.5
Off-peak Shift discount	—	—	5%	10%	10%	15%
Priority booking window	3 days	3 days	5 days	7 days	10 days	10 days
League eligibility	—	Open evenings	Division League	Division League	Championship invitational	Championship invitational
Transmitter profile	Rookie	Rookie	Standard	Licensed	Licensed	Licensed
The Works bench access (supervised hour)	—	—	—	1/quarter	1/month	1/month
Guest privilege	—	—	—	bring-a-Learner: guest’s first Shift 50% off	+ guest skips Toolbox Talk queue (staff-assisted)	+ 2 comp guest Shifts/season
Physical card	app only	app only	printed card	printed card	metal card	numbered metal card

License-axis multipliers and discounts stack with membership benefits (Chapter 6) multiplicatively for Gears and additively-capped for discounts (maximum combined Shift discount 25% — protects Shift yield, per Volume 10 pricing guardrails).

Trade Hack. Print the practical-check criteria on the back of every paper license certificate. Customers photograph the card, and the photo is the marketing: it tells their group exactly what to come back and attempt. The cheapest acquisition channel the park owns is a twelve-year-old showing their Apprentice card to a sibling.

3.8 Progression pacing model

The ladder is tuned so that meaningful state change happens on almost every visit. A typical enthusiastic customer (one Operator Shift + one event per visit, decent smoothness) earns 250–350 XP per visit: Apprentice on visit 2, Operator around visits 6–8, Foreman inside the first year at 20–25 visits. That pacing is deliberate against the membership price ladder — the Apprentice→Operator XP window is where the \$29 membership pays for itself in included Shifts, and the Operator→Foreman grind is where league nights and the \$59 membership become rational. Pacing is reviewed each season against the funnel KPIs (Section 5.6); thresholds may be re-tuned $\pm 20\%$ by the GM with a revision note here, never mid-season.

4. Achievement Badges

4.1 Badge system design

Badges are the granular texture of progression: the license answers “what rank am I”, badges answer “what have I done”. System rules:

- **Three classes:** Bronze (achievable in a visit or two), Silver (deliberate practice over multiple visits), Gold (genuine accomplishment; $\leq 10\%$ of active customers should hold any given Gold). Class sets the XP value: Bronze 50, Silver 150, Gold 400 (event and secret badges vary, per catalogue).
- **Machine-verified wherever possible.** Every badge lists its trigger. **T:** marks a fully automatic telemetry/OS trigger; **M:** marks a marshal or Artisan verification tap on the staff app (two-tap: staff ID + customer wristband). Automatic triggers are the default; marshal triggers are reserved for judgment calls (form, technique, conduct).
- **Badge pops are theatre.** In-app animation with sound; Silver and Gold pops also flash the customer’s name on the nearest leaderboard screen; Gold pops are announced by the duty marshal.
- **Five divisions of the catalogue:** Skill Mastery, Equipment Mastery, Milestones, Event badges, Secret badges.

4.2 The badge catalogue

Skill Mastery — cross-cutting craft, mostly telemetry-triggered:

#	Badge	Class	Criteria & trigger
S01	Drift Initiate	Bronze	Hold a scored drift line through the annex’s 3-gate sector once. T: gyro/Node trace + timing gates

#	Badge	Class	Criteria & trigger
S02	Drift Sensei	Gold	Judge-scored drift run $\geq 80/100$ at a drift night. M: judge panel entry
S03	Smooth Operator	Silver	Smoothness index ≥ 85 on 5 consecutive Shifts. T: index history
S04	Feather Foot	Bronze	Complete a full Shift with zero voltage-sag spikes above threshold. T: Node sag analysis
S05	The Surgeon's Hands	Silver	Construction precision course (Volume 4, Ch. 8.2) clear with zero penalty touches. T: course sensors + marshal confirm
S06	Hydraulic Surgeon	Gold	Premium hydraulic precision task (egg-lift protocol on K961-class) completed intact. M: supervising Artisan
S07	Line Perfect	Silver	10 consecutive laps within 102% of your own best. T: timing loop
S08	Night Flier	Silver	3 night-session flights (or night FPV heats) completed clean. T: session timestamps + clean-flight flag
S09	Crosswind Landing	Silver	Landing check passed in measured crosswind ≥ 3 m/s. M: instructor
S10	Winchmaster	Silver	10 clean tows under the Tow-Truck Retrieval Protocol. T: retrieval log
S11	First Responder	Bronze	First clean tow. T: retrieval log
S12	Harbour Pilot	Silver	Berth a loaded barge inside the marked box, no wall contact. T: mission console
S13	Deadstick	Gold	Sailing: win a regatta race; or Aviation: power-off landing check. M: race officer / instructor
S14	Load Whisperer	Bronze	10 weighed hopper loads in one Shift, zero spillage penalty. T: load cells
S15	The Long Haul	Silver	100 kg weighed tonnage in a single day. T: tonnage ledger

Equipment Mastery — one per machine class; earned by a clean qualifying Shift plus the class practical check:

#	Badge	Class	Machine class & trigger
E01	Touring Card	Bronze	1/10 touring: 10-lap clean check. T+M
E02	Buggy Broken-In	Bronze	1/14 buggy: Track B lap check incl. both jumps landed. T+M
E03	Short-Course Certified	Bronze	SCT class check. T+M
E04	Formula Steward	Gold	Formula RC event participation + staged-launch drag pass. M: event marshal
E05	Dig Card	Bronze	Excavator: load-cycle check (Section 3.5). T+M
E06	Truck Card	Bronze	Dump truck: 10 hauls, haul-road rules clean. T
E07	Loader License	Silver	Wheel loader check (the strongest rental machine — gated per Volume 4). M
E08	Dozer Operator	Silver	Dozer grading task to marshal standard. M
E09	Iron Rancher	Bronze	Tractor + one implement hitched, field task complete. T+M
E10	Implement Master	Silver	All four implement types used across any number of Shifts. T: implement RFID
E11	Crawler Scout	Bronze	Entry crawler: Trail C bridge section clean. T: geofence
E12	Summit Club	Silver	Premium crawler: full Trail C including water hazard, no tow. T
E13	Tug Ticket	Bronze	Harbour tug basics check. M
E14	Sail Trim	Bronze	= Marine Sail Grade 1 task (Volume 6 §8.4). M
E15	Submarine Qualified	Silver	Sub Hunt mission scored ≥50%. T: mission console
E16	Wings	Silver	= Bronze Wings solo checkride (Volume 5 §9.3–9.4). M: instructor
E17	FPV Cleared	Silver	FPV module + first clean FPV heat. T+M
E18	Heavy Ticket	Gold	Premium hydraulic orientation + supervised Shift complete (Kabolite class). M: Artisan

Milestones — pure accumulation, all automatic:

#	Badge	Class	Criteria (T: OS ledgers)
M01	Clock-In	Bronze	First completed Shift
M02	Ten Shifts Deep	Bronze	10 career Shifts
M03	Centurion	Silver	100 career Shifts
M04	Five-Hundred Club	Gold	500 career Shifts
M05	First Tonne	Bronze	1 t career tonnage (name-plate unlock, per Volume 4)
M06	Five Tonnes	Silver	5 t career tonnage
M07	Quarry King	Gold	25 t career tonnage
M08	Lap 1,000	Silver	1,000 career timed laps
M09	Four Corners	Silver	At least one Shift in all four divisions (Phase 2+; Phase 1 variant: all three Phase 1 zones)
M10	Anniversary Wrench	Bronze	Active in the park 12 months after first Shift
M11	Streak 5 / 10 / 25	Bronze/Silver/Gold	Visit streak milestones (Section 5.5)
M12	Gears Millionaire	Silver	100,000 lifetime Gears earned (earned, not held)

Event badges — issued only during the named window; permanent proof of having been there:

#	Badge	Class	Criteria
V01	Harvest Hand (seasonal)	Bronze	Any scored delivery during a Harvest Campaign fortnight (Volume 4)
V02	Harvest Champion (seasonal)	Gold	Age-class campaign winner
V03	Season One Veteran (etc. per season)	Bronze	10+ Shifts inside the named season
V04	Championship Week-end	Silver	Competed at the annual RC WORLD Championship
V05	Podium	Gold	Any championship-event podium
V06	Opening Day	Bronze	On site during park opening weekend — never reissued
V07	Night League Regular	Silver	5 night-league rounds in one season
V08	Treasure Week	Bronze	Found a Sub Hunt treasure puck during Treasure Week (Volume 6)

Secret badges — undocumented in the app until earned (the catalogue below is staff-confidential; customers discover them):

#	Badge	Class	Criteria & trigger
X01	Good Samaritan	Silver	Tow another customer's casualty when the protocol offers it to the queue. T: retrieval log
X02	The Closer	Bronze	Book the last Shift of the day and complete it. T
X03	Rain Operator	Silver	Complete a Shift in officially logged rain (open divisions only). T: weather flag
X04	Photo Finish	Silver	Win a sprint race by <0.1 s. T: timing loop
X05	Zero to Hero	Gold	Win a class-normalized event within 30 days of holding the Learner tier. T
X06	The Understudy	Bronze	Watch a full Works window rebuild (dwell ≥15 min at the viewing window during a live teardown). M: Artisan wave-in
X07	Full House	Gold	Hold every Bronze equipment badge in the catalogue. T

Catalogue total at launch: **60 badges** (15 skill, 18 equipment, 12 milestone incl. the three-stage streak badge, 8 event, 7 secret — event badges rotate). The catalogue is a living annex: divisions propose additions each season; the GM ratifies; retired badges remain on customer records forever.

4.3 Physical collectibles

Every Bronze badge has a **35 mm enamel pin** (\$4.50 retail, or 900 Gears); Silver and Gold pins are earn-only — they cannot be bought, only claimed at the retail counter after the digital badge is verified on the wristband (first Silver/Gold pin free, replacements \$6). Kids' party packages include a **starter badge board** (A5 printed card with 12 pin slots matching the entry-class badges). The pin wall behind the retail counter displays the full catalogue with "earned by N drivers" counters under each — a physical progression advertisement positioned exactly where every customer exits.

Field Note. Enforce the earn-only rule for Silver/Gold pins ruthlessly, including for staff and VIPs. The moment one unearned Gold pin is seen on a lanyard, the entire physical program deflates to merchandise. The Bronze tier being purchasable is deliberate — it seeds collecting behaviour in children — but the wall between bought and earned is the program.

5. Gamification Mechanics

5.1 Leaderboards and class normalization

Every scored surface in the park posts to the leaderboard framework (one framework, many boards — Volume 13):

- **Boards:** per-track lap boards (A/B/C), tonnage boards (daily/seasonal/all-time, per Volume 4), mission boards (harbour, Sub Hunt), event boards, and the park-wide **Season Board** (Section 5.5).
- **Windows:** today / this week / this season / all-time. Weekly boards reset Monday 06:00 local; the reset is the retention heartbeat — a fresh board is a winnable board.
- **Class normalization.** Raw lap times only compare within a class. Cross-class boards use the **Pace Index**: the customer's best lap divided by the class reference lap (the calibrated Artisan-driven benchmark maintained by the weekly parity audit, Volume 3, Chapter 4), expressed as a percentage. A 108% pace index in the 1/14 entry class and a 108% in 1/10 touring rank identically — an eight-year-old and a club racer can share a board honestly. Construction normalizes by machine class against reference cycle rates; marine missions are scored 0-100 natively.
- **Identity:** boards display the customer's handle, license-tier seal and division endorsement chips. Real names only with explicit opt-in; child accounts display handle + first name initial (Chapter 11.6).

5.2 Fair-play surface

All boards apply the same three filters: (1) telemetry-verified sessions only (no manual score entry except marshal-judged formats); (2) kill-switch or geofence-breach events void the affected lap/cycle; (3) statistical outlier review — any score $>3\sigma$ better than the customer's rolling baseline is flagged for marshal review before posting (protects against timing-loop glitches and cut lines). The appeal path is Chapter 11's service desk, answered within 24 hours.

5.3 The Energy Cup — F1-style race scoring

The park's signature race format, run weekly per track, converts the power-management doctrine into sport:

1. Each car starts the heat with a fixed **energy allocation** (a Node-computed budget derived from pack state — not raw voltage, so pack age doesn't bias grids).
2. Telemetry deducts from the allocation in real time as a function of throttle demand and voltage sag: aggressive driving visibly burns budget on the customer's live gauge and the track screen.
3. Exhausting the allocation before the flag triggers a soft limit (throttle capped to 40% — the "limp home" penalty), not a kill; strategic drivers manage to the flag like an F1 fuel-saving stint.
4. Scoring: race position points (10/8/6/5/4/3/2/1) **plus** an efficiency bonus (energy remaining $\times$ position multiplier) — the winning play is fast and smooth, never one alone.

The Energy Cup is the format most cited in staff scripts when explaining “why smooth matters”, and it is the qualifying surface for the seasonal championship. It runs identically in a construction variant (the Load-and-Haul enduro: tonnage per energy budget).

5.4 Gears — the loyalty currency

Gears are the monetary axis of the reward system: earned generously, burned on real value, and kept strictly separate from XP (Gears never advance the license). Ledger, wallet and expiry mechanics: Volume 13. Economics guardrail (with Volume 10): **1 Gear ≈ \$0.005 redemption value; target overall redemption cost ≤4% of gross customer revenue.**

Earn table:

Action	Gears
Per \$1 spent (Shifts, F&B, retail, parties, events)	10
Completed Shift bonus	25
Clean tow (retrieval protocol)	50
Sprint race / mission / regatta entry	25
Event podium	100
Badge earned (Bronze/Silver/Gold)	50 / 150 / 400
License tier-up	500
Referral: referred friend completes first paid Shift	500
NPS survey completed	25
Birthday month bonus (once)	250
Visit-streak multiplier	×1.1 at 3-visit streak, ×1.25 at 6, ×1.5 at 10+ (applies to earn, Section 5.5)
License-tier multiplier	×1.0 → ×1.5 per Section 3.7
Membership multiplier	×1.1 / ×1.2 / ×1.3 (Apprentice/Operator/Foreman, Chapter 6)

Burn table:

Redemption	Gears
Casual Shift, standard class (\$15)	3,000
Casual Shift, premium class (\$22)	4,400
Shift top-up block (when the queue-top-up offer appears)	2,500
F&B kiosk item (up to \$5)	1,000
Bronze badge pin	900
Park cap / small merch (up to \$12)	2,400
Party upsell: extra child slot	5,000
Coaching session discount (\$10 off)	2,000
Season Pass paid track (Section 5.5) — partial payment allowed	up to 50% of price

Redemption	Gears
Charity option: donate to the schools-program bursary fund	any amount

Gears expire 12 months after earning (rolling, FIFO), paused for any month with a visit. Expiry warnings at 60/14 days are retention touches, not threats — each includes a one-tap off-peak booking suggestion sized to the expiring balance.

5.5 Streaks and the season structure

- **Visit streaks** count consecutive calendar months with ≥ 1 completed Shift (a monthly cadence is honest for a destination venue; weekly streaks would punish normal families). Streak rewards are the Gears multipliers above, the M11 streak badges, and a **streak shield**: one missed month per 6-month streak is auto-forgiven. Members' streaks are shielded for the life of the membership.
- **Seasons** run 13 weeks, four per year, each with a name and a light theme (e.g. "Season 3: Night Works" carries the night-league push). At season end: seasonal boards archive to the customer's career page, the Season Board crowns per-division and overall champions at Community Night, and the Season Pass track resets. License tier and badges never reset — seasons reset competitions, not careers.
- **The Season Pass** is a battle-pass-style reward track fed by **Track Points** (1 TP per 10 XP earned that season — so the pass rides on the same honest activity measure, and nothing on it requires new grinding surfaces). Thirty levels; rewards alternate between the free and paid tracks:

Track	Price	Sample reward ladder (30 levels)
Free track	\$0, everyone enrolled	L2: 250 Gears · L5: seasonal badge V03 progress · L8: F&B voucher · L12: 500 Gears · L18: Bronze pin voucher · L24: off-peak Shift 50% voucher · L30: seasonal trophy (digital) + 1,000 Gears
Paid track: Season Pass	\$24/season (or 12,000 Gears; included free in Operator & Foreman memberships)	Everything on free track, plus — L1: exclusive seasonal body-shell livery on rental cars bound to your Shifts · L6: 1 free standard Shift · L10: seasonal cap · L15: priority-queue token $\times 2$ · L20: 1 premium Shift · L25: exclusive seasonal pin (never re-issued) · L30: guaranteed championship-weekend grid slot + name on the season plaque

A committed customer completing ~2 visits/month with events finishes the free track comfortably and the paid track with headroom; the paid track is priced under two Casual Shifts and returns more than that in vouchers by design — it is a retention product, not a margin product (yield modelling: Volume 10).

5.6 Mechanics dashboard

The GM reviews one funnel monthly: actives $\rightarrow$ % with a streak $\geq 3 \rightarrow$ % Apprentice+ $\rightarrow$ % Operator+ $\rightarrow$ % on paid Season Pass $\rightarrow$ 90-day repeat rate. Targets by end of Year 1: 40%

streak $\geq$ 3, 55% Apprentice+, 18% Operator+, 12% paid pass, 22% repeat (rising to 28% by Year 3 per Volume 10 §10.4). Any mechanic that doesn't move this funnel within two seasons is retired — gamification is subject to the same utilization discipline as the fleet.

Field Note. Resist the temptation to add currencies. Every gamified venue that decays into a “points, stars, tickets, crowns and crystals” bazaar trains customers to value nothing. RC WORLD runs exactly three numbers — XP (career), Gears (wallet), Track Points (season) — each derived from real behaviour, each spent on a different axis, and the third derived from the first. That is the entire economy. Hold the line.

6. Memberships & Season Passes

6.1 The three canonical tiers

Memberships convert enthusiasm into predictable monthly revenue and are the park's single most important financial smoothing instrument (deferred-revenue and churn modelling: Volume 10). Canonical pricing per the style guide; monthly, cancel any time, with a 12-month prepay option at two months free.

Included Shifts / month	3 Casual (standard classes)	6 Casual or 3 Operator equivalents (block-based: 6 blocks)	12 blocks, any mix; premium classes count 1.5 blocks
Additional Shifts	10% off	15% off	20% off
Gears earn multiplier	×1.1	×1.2	×1.3
Booking window	7 days	10 days	14 days
Member hours (Tue/Thu 17:00–19:00, Sun 08:00–10:00)	✓	✓	✓ + guaranteed slot with 48 h notice
Season Pass paid track	—	included	included
League entry	standard fees	1 division league/season included	all division leagues + championship qualifier entry
Guest passes	—	1 guest Casual Shift / month	3 guest Casual Shifts / month
The Works	—	member workshop evening 1/quarter	monthly bench hour with an Artisan
Streak shield	✓ (full)	✓ (full)	✓ (full)
Birthday-party discount	5%	10%	15% + priority dates
Retail & F&B	5% off	10% off	10% off + members' merch drops

Blocks unused in a month roll over one month, then lapse (prevents hoarded liability while forgiving a busy month). Included Shifts earn XP and Gears normally — a member's activity is real activity.

6.2 Fair-use rules

- Included blocks are personal and non-transferable (guest passes are the sharing mechanism); wristband identity is checked at binding.
- Peak-day cap: maximum 2 included blocks redeemed on any Saturday/public holiday (paid Shifts uncapped) — protects walk-up capacity on the days that recruit new customers.
- Premium hydraulic and Foreman-gated equipment remain license-gated regardless of membership (the earned/bought wall, Section 3.1).
- Membership pauses: up to 2 months/year self-service in the app (streak shield holds); medical/relocation pauses by service desk without limit.

6.3 The Annual Season Pass product

Distinct from the quarterly Season Pass reward track (Section 5.5), the **RC WORLD Annual Pass — \$499/year** (family add-on: +\$299 per additional household driver) is the day-visitor's unlimited-ish product: 1 Casual Shift per visit day, any standard class, unlimited visit days, 10% F&B/retail discount, and all four quarterly paid Season Pass tracks included. It deliberately does not include Operator Shifts, premium classes or leagues — heavy users are worth more as Operator/Foreman members, and the pass's job is frequency for the family segment, not value transfer to enthusiasts. Break-even against door prices sits near 34 visit days; modelled utilization is far lower, and the pass's real payload is the F&B/retail attach of frequent family visits (Volume 10).

6.4 Sales choreography

Membership is sold at three journey moments and nowhere else pushily: (1) the results screen when the app can show “this month would have cost \$X less as an Apprentice member” from real history; (2) the front desk at a customer's third visit in a rolling 60 days (the OS flags it discreetly on the staff app); (3) the league sign-up flow, where Operator membership bundles the entry. Staff earn a small spiff on activations that survive 90 days — never on day-one signups alone, which buys churn.

6.5 Churn management and win-back

Churn is managed as a pipeline with named states, automated in RC WORLD OS (CRM detail: Volume 13):

State	Definition	Automated action	Human action
Healthy	≥2 block redemptions/month	none	none
Cooling	1 redemption, or booking-window use stops	“your blocks roll over” nudge + one-tap booking	none
At-risk	0 redemptions in a month	pause offer (before they cancel themselves), off-peak suggestion sized to their history	Foreman tier: personal note from a named Artisan or league marshal
Cancelled	churned	exit micro-survey (one question); Gears and license status confirmation (“your career is safe”)	logged reason code

State	Definition	Automated action	Human action
Win-back 30/90/180	30/90/180 days post-cancel	30 d: first month back 50% · 90 d: free Casual Shift + season preview · 180 d: “your streak shield is waiting” + event invite	90 d cohort reviewed monthly by GM

Two doctrines: **pause beats cancel** (a paused member keeps identity, streak and habit), and **the career never lapses** — license tier, badges and XP are explicitly permanent, so every win-back message can truthfully say the customer is returning to a career in progress, not starting over. Target Year-1 monthly churn $\leq 4.5\%$ blended; win-back reactivation $\geq 12\%$ of the 90-day cohort.

Investor Note. At the Year-1 modelled mix (Volume 10) memberships contribute roughly 20–25% of revenue at materially higher margin than walk-up Shifts (no acquisition cost, off-peak-skewed usage, F&B attach). Every mechanic in Chapters 3–5 ultimately serves this line: the license creates the identity, the identity justifies the subscription.

7. Corporate Packages

7.1 Product logic

Corporate events monetize weekday daytime capacity that walk-up demand never fills, at pricing anchored from **\$1,400 (2 hours, 20 pax)**. The park’s pitch against karting, escape rooms and axe venues: RC WORLD formats are genuinely collaborative (a supply chain needs five people doing five jobs), skill floors are low (everyone can drive a dump truck in two minutes), and the telemetry produces objective team scoreboards HR can’t get elsewhere. Sales collateral leads with the format, not the facility.

7.2 The three flagship formats

Site Foreman Challenge (Construction, 2–3 h, 12–40 pax) — the signature product. Teams of 4–5 run a scored construction campaign: excavator, three dump trucks (the 1:3 doctrine becomes a teamwork lesson — the excavator is idle without its trucks), and a loader keeping the haul road clean. Weighed tonnage per energy budget scores the board; a mid-session “site incident” (staged breakdown → team must execute a Tow-Truck Retrieval while production continues) tests role reallocation. Debrief includes each team’s smoothness and coordination telemetry — the closest thing to an objective teamwork metric on the corporate market.

Relay Grand Prix (Motorsport, 2 h, 10–30 pax) — teams share one car per team in a driver-change relay on Track A; Energy Cup scoring makes pace and energy strategy a team decision; pit-stop battery swaps are performed with Artisan supervision as the relay’s exchange ritual. Class-normalized handicapping lets mixed-ability groups compete honestly.

Engineering Challenge (The Works + track, 3 h, 8–20 pax, premium) — teams receive identical baseline cars and a constrained tuning menu (gearing, springs, ballast, ESC profile per Volume 3’s parity toolkit), guided by an Artisan; a workshop tour covers the RCW Node and fleet doctrine; the afternoon ends in a time-trial shootout where the tuning choices are

the differentiator. Sold to engineering firms, universities and technical teams; the strongest recruiting-event format in the portfolio.

7.3 Corporate pricing

Package	Duration	Pax	Price	Includes
Toolbox (base)	2 h	up to 20	\$1,400	1 format, event host, team score-board, photo pack
Site Works	3 h	up to 40	\$2,600	2 formats or extended campaign, catering slot (F&B billed separately), branded leader-board
Full Site Takeover	4 h + exclusive zone	up to 80	from \$4,800	Zone exclusivity, 2-3 formats, event coordinator, framed podium photos, custom event badge
Add-ons	—	—	Engineering Challenge +\$40/pax · evening/night session +15% · catering per Volume 10 F&B menu · branded body shells \$18/car	

Deposits 50% at booking; weekday-daytime slots discounted 10% (they are the inventory this product exists to fill). Every attendee leaves with a Learner license already provisioned and a personal “come back” voucher — corporate events are also a customer-acquisition channel with negative CAC.

7.4 Sales pipeline and B2B calendar

Pipeline stages in the OS CRM: Lead → Qualified (date + budget) → Proposal → Confirmed (deposit) → Delivered → Rebook-ask (within 48 h of delivery, with the event’s scoreboard attached). One part-time B2B coordinator owns the pipeline from Month 4; targets ramp to 6 events/month by Month 12 and 12/month in Year 3 (Volume 10). The B2B calendar follows corporate budget rhythm: January-February (kick-offs, sales-team energizers), June-July (mid-year offsites), September (team rebuilds), November-early December (year-end parties — premium pricing, book by September). Standing partnerships: hotel concierge desks, DMC/event agencies (10% commission), and co-working/tech-park community managers (hosted taster evenings each quarter).

8. Birthday Parties & Family Products

8.1 Package ladder

Parties are the park’s highest-margin repeatable product and its most efficient child-acquisition channel: every party imports 8-15 prospective license-holders with a parent watching. Canonical anchor **from \$349 (10 children)**.

Package	Price	Children	Includes
Crew Party	\$349	up to 10 (each extra \$28)	90 min: 2 entry-class Shifts each (entry racing + trucks/tractors rotation), party room 45 min, dedicated host, paper license certificates + Bronze pin each, badge board for the birthday child
Foreman's Party	\$499	up to 10 (extra \$38)	2 h: 3 Shifts incl. crawler trail, tow-truck rescue staged for the birthday child, party room 60 min, themed cake slot, host + assistant, group photo on the podium
Site Manager's Party	\$749	up to 14 (extra \$45)	2.5 h: mini Energy Cup with podium ceremony, all Foreman's inclusions, custom event badge in-app, birthday child's name on the leaderboard screens all session

Ages 6–7 run tethered/passenger formats with a guardian; 8+ per the entry-class canon. Food from the standard party menu (F&B pricing: Volume 10); outside cakes welcome (candle policy per fire SOP).

8.2 Party room and logistics

The Phase 1 entry pavilion includes one party room (Phase 3 adds two more with the event centre — Volume 11): seats 16 children + 8 adults, wipeable surfaces, direct sightline to Track A so parents watch from the table. Turnover standard: 30 minutes between parties (reset checklist below). Peak capacity: 3 parties/Saturday, 2/Sunday off the single room; the constraint is host staffing, not space — never run a party without its dedicated host.

☐

Room reset: tables wiped, floor swept, bins cleared, decorations neutralized

☐

Party fleet staged: correct class count + 2 hot spares, packs at storage-to-full per charging SOP

☐

Certificates printed with children's names; pins counted; badge board for birthday child

☐

Host briefed: birthday child's name, allergies flagged from booking form, photo consent list

☐

Tow-truck rescue pre-staged (Foreman's+): casualty vehicle and recovery crawler positioned

☐

Guardian waivers verified complete in OS (Toolbox Talk child-flow, Section 2.4)

8.3 The host role

The party host is a trained front-of-house role (not a marshal pulled off a track): owns the run-sheet minute by minute, holds the group's energy, executes the ceremony beats (first Shift countdown, the rescue, the podium), and hands the parent a one-tap rebooking offer at farewell. Hosts are scripted but not robotic — the scripts (Chapter 11.2) define beats, not lines. One host per 10 children; assistant above 10.

8.4 Upsells and family products

In-flow upsells (offered at booking, never mid-party): extra Shift round +\$8/child, premium cake slot, group photo prints, goodie-bag upgrade (park cap + extra pin), and the strongest performer — the **Birthday License Pack** (+\$59): the birthday child gets a printed license card with photo, 3 return Casual Shifts, and 1,000 Gears. Family bundle (park canon: 4 drivers × 2 shifts, \$89) is the standing family product outside parties; school-holiday **Family Mornings** (Sunday member hours extended to bundle-holders) keep the family segment on the calendar between birthdays.

Field Note. The single highest-leverage moment of any party is the staged tow-truck rescue. Brief the birthday child privately (“a machine is going to break down out there — you’re the recovery driver”), then let it “happen” in front of their friends. The rescue converts the party’s social hierarchy for one afternoon — the birthday child is the hero with the winch — and it is the moment parents film. Never skip it to save schedule; cut a Shift instead.

9. Schools, Universities & STEM

9.1 Why education is a product line, not CSR

School programs fill weekday mornings (dead inventory), build the license pipeline years ahead of spending power, and give the park civic standing that pays off in permits, press and partnerships (Volume 2’s stakeholder analysis). They are priced to margin, not charity — bursary places are funded by the Gears charity-burn option (Section 5.4) and a fixed 5% of program revenue.

9.2 School programs (ages 8-18)

All programs are curriculum-mapped (local mapping per deployment; the structure below maps cleanly to most physics/technology syllabi) and run 09:30-13:00 weekdays, 20-60 students, \$22/student (min \$440), teacher resources included:

- **Gears & Gearing (ages 8-12):** the physics of gear ratios taught on the machines that embody them — students measure how pinion changes alter a buggy’s speed and a tractor’s pulling force, then test predictions on track. Worksheet: ratio arithmetic; takeaway: every student earns the Clock-In badge on a real profile.
- **Electrons at Work (ages 11-15):** circuits, batteries and motors via the park’s own doctrine — why LiPo cells live between 3.4-4.2 V, what voltage sag is, why the charging room is a bunker. Live demo at the charging bunker window; students read real Node voltage traces from their own Shift.
- **Data Drives (ages 14-18):** telemetry data science. Each team exports their session’s Node data (CSV from the customer app), computes their own smoothness index in a

spreadsheet, and competes on analysis — the best data story wins, not the fastest lap. This program is the park's flagship STEM asset because no competitor owns a comparable dataset.

- **Site Logistics (ages 12-16, Construction):** the 1:3 excavator-truck ratio as an operations-research problem — teams predict, then measure, throughput with 1, 2 and 3 trucks per excavator. Queueing theory taught with dirt.

Field-trip format: arrival briefing (Toolbox Talk school edition, teacher-supervised group waivers collected digitally in advance), two teaching blocks with Shift time embedded, results assembly with badge ceremony. Ratios: 1 park educator per 15 students plus teachers per school policy; accessibility provisions per Chapter 11.6.

9.3 University partnerships

- **Engineering-club leagues:** universities field standing teams in a dedicated **Varsity League** (season-long, Engineering Challenge rules — identical hardware, constrained tuning menu, telemetry-audited). Entry \$360/team/season; the league is deliberately cheap because its yield is weeknight utilization, brand-embedded future professionals, and recruiting-sponsor revenue (engineering employers sponsor the league table).
- **Capstone projects on RCW Node / RC WORLD OS:** the park offers real engineering problems as final-year projects — Node antenna performance in the mesh, predictive-maintenance models on the fleet's runtime logs, computer-vision lap validation, battery state-of-health estimation from sag signatures. The park provides data access under NDA, a named Artisan mentor, and test time; the university provides free R&D and a hiring pipeline for Artisan and OS roles (framework agreement template: Volume 13 appendix).
- **Formula-style RC competition sponsorship:** the park sponsors and hosts an annual **Formula RCW** intervarsity event — student-built cars to a published rules formula (motor/battery/budget caps, mandatory RCW Node for scrutineering telemetry), design-report judging plus on-track events, mirroring Formula Student's structure at 1/10 scale. Hosting cost is modest (one weekend, existing track); the return is regional press, a pipeline of the exact talent the park hires, and the strongest possible answer to "is this a toy park?"

9.4 Education KPIs

School-days sold/term (target: 20 by Year 2), weekday-morning utilization on school days $\geq 70\%$, student-family return conversion within 90 days $\geq 8\%$ (tracked via the take-home family voucher), 2 signed university partnerships by end of Year 2, Formula RCW inaugural event in Year 3 (aligns with Phase 3 classrooms — Volume 1 phasing).

10. The Mobile App Experience

10.1 Scope and platform

This chapter specifies the customer-facing app as an experience, screen by screen; the technical build — Kotlin Multiplatform / Compose Multiplatform on **both Android and iOS**, Supabase backend, role-based access inside RC WORLD OS — is Volume 13's territory and is only cross-referenced here. The design brief in one sentence: **the app is the customer's license, wallet and pit crew, in that order.** It must be excellent offline-tolerant on park Wi-

Fi, one-handed, and legible in sunlight; every screen below has a stated job and one primary action.

10.2 Onboarding & the Toolbox Talk

1. **Welcome / account** — email or social sign-in; guardian flow creates child sub-profiles under one wallet. Job: get to a booking in under two minutes for the impatient; everything else can wait.
2. **Toolbox Talk** — the interactive induction: battery limits (the 3.4–4.2 V/cell window, told as “why your machine protects itself”), collision liability, track rules, the tow-truck doctrine (“you never walk onto a live site — you drive the rescue”), closing with the digital waiver signature. Interactive checks, not a video to skip; 8–12 minutes; per-division delta modules unlock later (marine, aviation, premium hydraulics). Completion mints the **Learner license** with a small ceremony — the card animates onto the screen, first 25 XP posts.
3. **First-visit primer** — optional 5-screen visual of the journey spine (wristband, binding, pit stop) so nothing on site is a surprise.

10.3 Wallet & booking

- **Wallet** — payment cards, Gears balance with expiry-soonest shown honestly, membership blocks remaining (with rollover state), vouchers and guest passes as scannable tokens. One screen; no sub-menus for money.
- **Booking** — division → class (locked classes shown greyed with their unlock stated plainly: “Requires Operator + Construction Advanced — you’re 2 badges away”, turning every refusal into a goal) → Shift type (Casual/Operator) → slot grid with live availability and member-window shading → pay with card, blocks or Gears. Family bookings seat multiple profiles in adjacent slots in one flow.

10.4 The live Shift dashboard

The screen customers actually use on site, auto-launched at vehicle binding (QR scan on the transmitter):

- **Header:** bound vehicle’s name and asset tag, class, your handle and tier seal.
- **Battery/energy gauge** — the Node’s live allocation readout, styled as an industrial fuel gauge; sag spikes flash amber (the F1-style doctrine made visible). In Energy Cup sessions this becomes the race budget.
- **Session clock** with pit-stop countdown on Operator Shifts (“pit window opens 18:00”).
- **Live score strip:** laps and best lap (timing loop), or tonnage this Shift (load cells), or mission checklist (harbour/Sub Hunt).
- **Smoothness gauge** — the operator dial, live.
- **One action button, context-switching:** it is Pit In near the pit window, Retrieve when your vehicle dies (launches the tow-truck flow with a map pin on the casualty), and Top-Up when the queue logic offers an extension (next section).

10.5 Results, badge case & license card

- **Shift summary** (pushed within 60 s of session end): score, personal-best deltas, XP breakdown with smoothness bonus, Gears earned, badge pops, highlight clip when a

track camera caught one — with a share sheet, because the clip is marketing the customer distributes voluntarily.

- **License card** — the career home screen: tier seal, XP bar to next tier with the specific remaining gates listed (“Smoothness avg 66/70 · 4/5 badges”), endorsement chips with grades, streak flame, season Track Points. Tapping the card flips it to a QR for staff scanning. This is the screen the whole chapter exists to serve; it must answer “what’s next?” without scrolling.
- **Badge case** — the catalogue as a collection wall: earned badges in colour with date and the triggering session; unearned in silhouette with criteria; secret badges as unlabelled slots (counted, not described). Pin-claim status shown for physical collectibles.
- **Leaderboards** — every board from Section 5.1, filterable to “people I know” (friends list) because a rank of 3,407 motivates nobody but 2nd-among-friends motivates everyone.

10.6 The queue top-up flow

Implemented exactly per source doctrine: at T-5 minutes before a Shift ends, the OS queries the class `queue_roster`; **if and only if nobody is waiting for that class**, the dashboard’s action button becomes Top-Up — one tap, wallet-charged block extension (\$8 standard / \$12 premium per extra block, or 2,500 Gears), vehicle never comes off the track. If a queue exists, the button never appears and the session ends on time — the offer’s credibility rests on customers never seeing it while others visibly wait. Top-up revenue is pure utilization yield on already-staffed capacity; it is tracked as its own P&L line (Volume 10).

10.7 Notifications discipline

Push categories, each individually controllable: session/safety (cannot be disabled while on site), booking reminders, badge/tier events, streak & expiry warnings, offers (default off — earned trust, not claimed). Hard rule: no more than one marketing push per week, and every marketing push must contain a one-tap action worth taking. The app’s notification reputation is a park asset; it is spent, not free.

11. Service Standards & Recovery

11.1 The service posture

Staff play colleagues on the same site, not vendors: the customer is a fellow operator, addressed with workshop courtesy (“your machine’s ready, bay 3”). This is Disney-grade role discipline applied to an industrial frame, and it is trained, scripted and audited like any SOP. Front-of-house standards: greet within 10 seconds of reaching any counter; name the customer once their wristband is scanned; never say “no” without the path (“that class needs Operator — you’re 300 XP out, and today’s Shift gets you a third of the way”).

11.2 Scripts as beats

Scripts define mandatory beats, with staff free in wording (verbatim-scripted staff sound like machinery — the wrong kind):

Moment	Mandatory beats
Vehicle binding	Confirm name → confirm class & tier aloud → transmitter handover with one machine-specific tip → “clock in when ready”
Pit stop	Call the car in by name → narrate the swap (“fresh pack, tires look good”) → explicit release (“you’re clear — 20 more minutes”)
Breakdown	Reassure fast (“your machine’s down, not your Shift”) → offer the rescue (“recovery crawler’s yours — go get it”) → fresh vehicle promise stated before the tow starts
Badge/tier pop (Gold/tier-up witnessed)	Announce on PA or to the immediate group → physical acknowledgment (handshake, pin pointer to retail)
Complaint opening	Listen without interruption → restate the problem in one sentence → state what happens next and when
Farewell (desk)	Next-unlock mention → invite back to a specific thing (“league night Thursday — your class”)

11.3 Complaint handling & the empowerment ladder

Every front-line staffer holds **Level 1 authority** without approval: re-run a Shift, comp an F&B kiosk item, grant up to 2,000 goodwill Gears — logged in the staff app with a reason code, reviewed weekly for patterns (not for blame; unlogged generosity is the firing offence, not logged generosity). **Level 2** (duty manager): refunds to original payment, party/event make-goods, membership-month credits. **Level 3** (GM): legal-adjacent matters, injuries (which immediately follow the incident SOP, Volume 7 safety chapter), media-visible situations. Response-time standards: on-site issues resolved or escalated within 10 minutes; app/service-desk tickets first-response within 24 hours; leaderboard appeals (Section 5.2) within 24 hours.

11.4 Refund policy — tied to the breakdown protocol

The policy is mechanical, so staff never negotiate from scratch:

Situation	Remedy
Breakdown, retrieval + fresh vehicle inside 6 min	No refund due — the protocol is the remedy; clean-tow Gears paid; session clock paused during retrieval
Breakdown, no replacement vehicle available in class	Full Shift re-credit to wallet + 500 Gears; slot priority next visit
Second breakdown, same Shift	Full refund or re-credit (customer’s choice) + 1,000 Gears — two strikes means fleet readiness failed (fed back to the daily-readiness KPI, Volume 7)
Weather closure mid-Shift (open divisions)	Pro-rata re-credit for unshifted blocks; booked-ahead sessions moved free
Park-fault cancellation (staffing, systems)	Full refund + a comped equivalent booking
Customer no-show / late >10 min	Slot released; one courtesy rebook per rolling 6 months, then forfeit per T&Cs

Session-clock pause during retrieval is the load-bearing detail: it makes the “no refund due” line honest, because the customer genuinely lost no paid time.

11.5 Measurement: NPS and CSAT

- **NPS:** single-question push at +24 h post-visit (25 Gears for answering); target ≥ 55 by operating Month 12, building to ≥ 60 by Year 3; detractors (< 7) trigger a service-desk callback task within 48 h.
- **CSAT micro-pulses:** one-tap ratings embedded at three moments only — post-Shift summary, post-party (to the booking parent), post-corporate (to the organizer). Never more; survey fatigue is a real cost.
- Both post to the GM dashboard alongside the journey KPIs (Section 2.2), reviewed weekly; any touchpoint two weeks below target gets a named owner and a dated fix.

11.6 Accessibility and inclusion

RC operation is unusually accessible — the machine does the running — and the park commits to making that true in practice:



Step-free routes to all driver rostrums, pit lane, party room and viewing rails (build spec: Volume 11); seated-operation positions at every rostrum



Adaptive control library: wheel-to-stick conversions, single-handed transmitter mounts, switch-access trigger adapters — bookable free with any Shift, maintained by The Works



Sensory provisions: published quiet hours (first Sunday morning monthly — PA off, buzzers to visual-flash mode), ear defenders at the desk, sensory map in the app



Toolbox Talk available with captions, audio narration and simplified-language mode; staff briefing covers non-visible disabilities



Child protection: photo-consent flags on all party/school bookings honoured in every clip pipeline; child leaderboard identity rules (handle + initial, Section 5.1); staff working education/party programs hold local-equivalent background clearance



Financial inclusion: schools bursary fund (Section 9.1); off-peak community rates reviewed annually



Companion/carer enters free with a paying operator; the operator holds the license and the spotlight

Safety Warning. No service-recovery gesture ever overrides a safety rule. A customer denied a premium hydraulic Shift for lacking the tier is offered a path, never an exception; a goodwill re-run never skips vehicle binding or tier profiles. The empowerment ladder buys goodwill with money and Gears — never with the safety envelope.

12. Volume Summary & Cross-References

RC WORLD's customer experience is one machine with many faces. The industrial frame (Chapter 1) turns rentals into careers and friction into gameplay — the Tow-Truck Retrieval

Protocol being the emblem. The journey spine (Chapter 2) is instrumented end to end, with an owner and a KPI at every touchpoint and a hard rule that no customer leaves without knowing their next unlock. The **RC WORLD License** (Chapter 3) is the core asset: six earned tiers (Learner → Apprentice → Operator → Foreman → Site Manager → Legend, at 0 / 500 / 2,000 / 6,000 / 15,000 / 40,000 XP) on a skill axis, visit streaks on a frequency axis, four division endorsements (Basic/Advanced/Master), and equipment gates that make rank real — premium hydraulics, FPV and jet-class machines are unreachable without earning them. Sixty named badges (Chapter 4) give the career its texture; the Gears economy, class-normalized leaderboards, Energy Cup scoring and the two-track quarterly Season Pass (Chapter 5) give it a pulse. The three memberships (Apprentice \$29 / Operator \$59 / Foreman \$99) and the \$499 Annual Pass (Chapter 6) convert it to recurring revenue; corporate (from \$1,400), parties (from \$349) and education programs (Chapters 7–9) fill the off-peak calendar and feed the license pipeline. The app (Chapter 10) carries it all in the customer’s pocket, and the service doctrine (Chapter 11) keeps the promises — mechanically, with an empowerment ladder and a refund table tied to the breakdown protocol.

Cross-references

- **Volume 1** — Executive Master Plan: phasing of party rooms, classrooms and event centre; retention economics in the investment case.
- **Volume 2** — Market Research: the competitive lessons (score everything, feed the spectators, sell the next visit) this volume operationalizes.
- **Volume 3** — Motorsport Division: parity doctrine and reference laps behind the Pace Index; transmitter tier profiles; race formats, leagues and timing systems feeding XP and boards.
- **Volume 4** — Construction Division: career tonnage, hopper load-cell scoring, Harvest Campaigns, hydraulic premium-experience supervision — all gated per Section 3.6 (which supersedes earlier “tier 2/3” phrasing).
- **Volume 5** — Aviation Division: the Wings ladder (§9.3: Ground → Bronze → Silver → Gold Wings, plus Type: Jet / Type: FPV / Type: Heli ratings) mapped to the Aviation endorsement grades; FPV and jet-class gates; checkride and currency rules.
- **Volume 6** — Marine Division: the marine badge-task ladder (Ch. 8 §8.4: Deckhand, Coxswain, Tug Master, Race Skipper, Sail Grade 1–3, Submariner) mapped to the Marine endorsement grades; mission scoring and retrieval Gears.
- **Volume 7** — Engineering & Workshop Manual: fleet-readiness and safety SOPs that the refund table and breakdown protocol depend on; The Works bench-hour supervision.
- **Volume 10** — Finance: Gears redemption-cost guardrail ($\leq 4\%$ of gross revenue), membership and deferred-revenue modelling, top-up and party P&L lines, discount stacking cap; home of the 90-day second-visit KPI (§10.4: $\geq 22\%$ Year 1 / $\geq 28\%$ Year 3, Phase 2 gate floor $\geq 20\%$).
- **Volume 11** — Architecture & Park Design: party room, accessibility build specs, leaderboard screen placement, retail exit line.
- **Volume 12** — Franchise Manual: license/badge/Gears canon as a mandatory franchise standard — customer careers must be portable across RC WORLD sites.
- **Volume 13** — RC WORLD OS: the technical build of everything here — XP/Gears/Track Point ledgers, badge trigger pipeline, queue top-up logic, CRM churn states, leaderboard framework, app screens on Kotlin Multiplatform for Android and iOS.

Volume 10 — Finance

RC WORLD — Master Development Plan · Volume 10 of 13 **Revision 1.0 — July 2026** ·
Status: Living document — bump revision on material change

Purpose of this volume. This volume is the financial source of truth for the entire Master Development Plan. Every capex figure, revenue projection, cost assumption, and return metric quoted anywhere in Volumes 1–13 reconciles to the models built here. It constructs, line by line, the Phase 1 startup budget of $\approx \$1.85$ M; the $\$2.6$ M funding ask and its tranche structure; a bottom-up revenue engine that lands on the canonical Year 1 $\approx \$1.28$ M, Year 3 $\approx \$2.6$ M, and Year 5 $\approx \$3.4$ M; the operating cost stack that yields a 24–28% EBITDA margin from Year 3; the monthly cash model that shows operating break-even at Month 16–19; and the profitability, sensitivity, and scenario apparatus an investor’s analyst will want to stress. Every major claim is carried by a table, with the assumptions stated above it and the arithmetic shown so a CFO can audit it with a calculator. Where this volume and any other volume disagree on a number, this volume governs, and the other volume must be corrected.

Intended readers. Investors and lenders performing diligence on the $\$2.6$ M ask; the founder and General Manager, who own this model and update it monthly against actuals; the finance/administration hire, for whom Chapters 4–7 are the operating manual of the company’s numbers; franchise developers, who will rebuild Chapter 4’s engine with local inputs (see Volume 12, Chapter 3); the RC WORLD OS team, who implement Chapter 10’s KPI dashboard (see Volume 13).

Chapters

1. Financial Summary & Reading Guide
2. Startup Budget — Phase 1 Capex
3. Funding Plan
4. Revenue Model
5. Operating Cost Model
6. Cash Flow & Break-Even
7. Profitability & ROI
8. Pricing Strategy & Elasticity
9. Sensitivity Analysis
10. Growth Scenarios
11. Volume summary & cross-references

1. Financial Summary & Reading Guide

1.1 The headline dashboard

Everything an investor needs to hold in their head about RC WORLD's finances fits in one table. Every figure below is derived — not asserted — in the chapter cited, and each derivation reconciles to the canonical numbers fixed in the park's style guide.

Metric	Base-case value	Derived in
Phase 1 capex ("Core Park", Months 0–12)	≈ \$1.85 M (\$1,850 K)	Chapter 2
Phase 2 capex ("Full Park", Months 13–30)	≈ \$1.15 M (\$1,150 K)	\$2.12
Phase 3 capex ("Destination & Beyond", Months 31–60)	≈ \$0.9 M (\$900 K)	\$2.12
Five-year program capex	≈ \$3.9 M	\$2.12
Funding ask (equity/quasi-equity, staged tranches)	\$2.6 M	Chapter 3
Year 1 revenue	≈ \$1.28 M	Chapter 4
Year 3 revenue	≈ \$2.6 M	\$4.9
Year 5 revenue	≈ \$3.4 M	\$4.9
EBITDA margin, Year 3 onward	24–28% (24.2% Y3 → 27.4% Y5)	Chapter 7
Monthly operating break-even	operating Month 16–19 (base model: Month 16)	Chapter 6
Full payback of invested capital	Year 4–5 (base: operating Month ~53 on Phase 1 capital)	\$7.4
Minimum modelled cash position	\$476 K (Month 14) vs \$250 K policy floor	\$6.4

Investor Note. The single most important structural fact in this volume: RC WORLD is an **infra-structure-light, fleet-cheap, labor-and-experience business**. The entire Phase 1 powered fleet — roughly 150 vehicles including batteries, radios, and opening spares — costs \$80 K, i.e. **4.3% of Phase 1 capex**. The expensive things (earthworks, buildings, tracks) last 10–20 years; the things customers wear out (cars, gears, tires, batteries) are cheap, sourced wholesale from China, and replaced from operating cash flow. This inversion — durable capex, disposable fleet — is why the model survives the stress tests in Chapter 9.

1.2 Modelling conventions

Stated once here; they apply to every table in this volume.

Convention	Value	Rationale
Currency	USD throughout, nominal (no inflation indexing)	Style-guide canon; nominal keeps tables auditable. Local-currency conversion is a franchise-localization task (Volume 12)
Geography baseline	Mid-size international city, metro 1.0–2.5 M (worked at 1.5 M), peri-urban 4.8 ha leased site	Canon; demand analysis in Volume 2, Chapter 9 (\$9.4–9.5)

Convention	Value	Rationale
Month frames	Build phasing uses the project frame (Phase 1 = project Months 0–12, park opens at project Month 12; Phase 2 = project Months 13–30; Phase 3 = project Months 31–60). All financial statements and division opening schedules in this volume use the operating frame : M1 = the opening month. Frames are labelled wherever ambiguity is possible	Style-guide month-numbering convention
“Year 1”	First full operating year, opening assumed Month 1 = March (northern-temperate seasonality; shift the curve, not the totals, for other climates)	Style guide dating: plan written July 2026
Operating calendar	360 operating days/year (5 maintenance closure days), 8 revenue hours/day average (10 weekend, 6–7 midweek)	Division operating doctrine, Volumes 3–6
Tax	25% flat placeholder — verify locally. Loss carryforward assumed available	Corporate rates in candidate jurisdictions span ~15–35%; the placeholder is deliberately mid-range and flagged in every after-tax table
Units of account	Tables in \$ thousands (\$K) unless stated	Readability
Rounding	Line items rounded to \$0.1–1 K; columns are constructed to sum exactly as printed	Auditability
Depreciation	Fleet & RCW Nodes 2–3 yr; IT 4 yr; software 5 yr; FF&E 6 yr; tracks 10 yr; site works 15 yr; buildings 18 yr	\$7.3 policy table
Payment mix	~85% card/app, ~2.2% blended processing fee	Carried in software/payments opex line, \$5.5

1.3 How this volume reconciles — a reading guide

The volume is a single connected model, and the joins are explicit. Chapter 2 builds capex bottom-up to \$1,850 K and shows the Phase 2/3 envelopes summing the five-year program to \$3,900 K. Chapter 3 shows the \$2.6 M ask as Phase 1 capex plus \$750 K of working capital, operating funding, and reserve — the same \$750 K that appears as opening cash in Chapter 6’s monthly cash model. Chapter 4’s revenue engine multiplies visitor volumes by the canonical price card and reconciles monthly to \$1,280 K in Year 1 and annually to the five-year canon; its Year-1 visit base also reconciles with Volume 2’s independent ~37,000–43,000-visits Year-1 demand estimate (Chapter 9). Chapter 5’s cost stack subtracts from Chapter 4’s revenue to give the EBITDA row that Chapter 6 spreads monthly (break-even Month 16) and Chapter 7 carries into the P&L (margins 24.2–27.4% from Year 3, payback Year 4–5). Chapters 8–10 stress the same model rather than building new ones. A reader with one hour should read §1.1, §2.1, §3.1, §4.9, §6.3, and §7.1.

2. Startup Budget — Phase 1 Capex

2.1 Summary build-up to ≈\$1.85 M

Assumptions: 4.8 ha flat peri-urban leased site with municipal power and water at the boundary (site brief, Volume 11, Chapter 2); construction in a mid-cost international market (blended civil rates ~\$55–90/m² for external works, light industrial building rates ~\$450–560/m²); fleet and electronics sourced wholesale from China per Volume 8; all pre-opening costs capitalized. Contingency is held at 10.0% of the subtotal, consistent with a design-complete, competitively-tendered civil package.

#	Category	\$K	% of total	Detail
A	Site works & earth-works (per zone)	455	24.6%	\$2.2
B	Track & operating surfaces	172	9.3%	\$2.3
C	Buildings (entry pavilion, The Works, charging bunker)	418	22.6%	\$2.4
D	Fleet acquisition (~150 powered assets + batteries + spares)	80	4.3%	\$2.5
E	RCW Node telemetry program	25	1.4%	\$2.6
F	Wi-Fi mesh & IT infrastructure	87	4.7%	\$2.7
G	RC WORLD OS v1 development	165	8.9%	\$2.8
H	FF&E, workshop equipment & theming	118	6.4%	\$2.9
I	Pre-opening (staff, training, marketing, licences)	162	8.8%	\$2.10
	Subtotal	1,682	90.9%	A+...+I
J	Contingency @ 10.0% of subtotal	168	9.1%	\$2.11
	Phase 1 capex total	1,850	100.0%	≈ \$1.85 M ✓ can-on

Arithmetic check: $455 + 172 + 418 + 80 + 25 + 87 + 165 + 118 + 162 = 1,682$; $1,682 \times 10.0\% = 168.2 \rightarrow 168$; $1,682 + 168 = 1,850$.

2.2 Site works & earthworks — \$455 K

Assumptions: flat site with 2–4% natural grade (canon) minimizes cut/fill; the Mining Zone pit is the only deep excavation and its spoil is reused for Track B berms and spectator mounds (a deliberate double-use that Volume 11 details); haul roads graded to the canonical $\leq 15^\circ$ maximum incline; drainage sized for the pond-free Phase 1 (the Marine pond is Phase 2).

Line	Scope	\$K
Bulk earthworks & grading	Whole-site strip, cut/fill balance, Mining Zone pit excavation & benching, compaction	118
Drainage, stormwater & utilities	Perimeter swales, culverts, detention basin; power/water/data trenching from boundary to all zones	96
Internal roads, paths & parking	Gravel service road loop, pedestrian paths, 88-bay gravel car park (Phase 1; expands to 132 bays + 4 bus bays in Phase 2, \$2.12, per Volume 11), entrance apron	62
Mining Zone civil fit-out	Graded haul roads ($\leq 15^\circ$), aggregate beds (loose sand $\rightarrow$ 10 mm crushed stone), central hopper foundation & load-cell pad	54
Agriculture Zone civil fit-out	Field grading, imported soil & row formation, irrigation trenches, barn/silo slab	48
Fencing, gates & landscape	Full perimeter fence, zone barriers, spectator berms dressing, planting	47
Survey, geotech, permits & design fees	Topographic survey, geotechnical investigation, civil/structural design, planning fees	30
Site works total		455

Check: $118 + 96 + 62 + 54 + 48 + 47 + 30 = 455$.

2.3 Track & operating surfaces — \$172 K

Assumptions: Track A is the premium surface (fine asphalt, machine-laid, the park's most exacting civil spec — drift and touring parity in Volume 3 depends on it); Tracks B and C are shaped-earth and placed-rock works riding on \$2.2 earthworks, so their line items are surfacing and features only.

Line	Scope	\$K
Track A (asphalt/drift), $\sim 2,200 \text{ m}^2$	Fine asphalt wearing course on prepared base, kerbing, run-off zones, paint	88
Track B (baja/rally)	Packed-dirt surfacing, engineered jumps & berms, geotextile under high-wear corners	34
Track C (crawler trail)	Placed natural stone, timber bridges, water-hazard liner, anchor points	28
Pit lane & marshaling surfaces	Concrete pit apron, transmitter-handover counters, marshal walkways	22
Track surfaces total		172

Check: $88 + 34 + 28 + 22 = 172$.

2.4 Buildings — \$418 K

Assumptions: areas per the built-form canon (Volume 11 is the geometry authority): entry pavilion **490 m² GFA** (including the 120 m² retail area) at a blended $\approx$ \$425/m² pavilion-grade shell-and-fit-out rate $\rightarrow$ \$208 K; The Works **450 m² GFA** (workshop core $\sim$ 180 m², parts store, QC lane, viewing window, plus the 24 m² battery room and **30 m² charging bunker, both internal** to the building) at a blended $\approx$ \$350/m² industrial-shell rate $\rightarrow$ \$158 K. The engineering judgement: Volume 11's areas are larger but simpler than earlier massing studies — bigger clear-span shells at lower unit rates, with the services-dense zones priced separately — so the tendered category total is unchanged. The third line is not a building: it is the specialized fire-rated **fit-out** of the internal bunker and battery room to Volume 7's doctrine (masonry lining, sand-topped charge bays, forced ventilation, thermal monitoring, rated doors).

Line	Scope	m ²	\$K
Entry pavilion	Reception/POS, Toolbox Talk briefing room, F&B kiosk, 120 m ² retail, toilets, first aid	490 GFA	208
The Works (shell & standard fit-out)	Workshop core $\sim$ 180 m ² , parts store, QC lane, test bench row, staff room, GM/admin office, viewing window	450 GFA	158
Charging bunker & battery-room fit-out	Fire-rated fit-out of the 30 m ² bunker + 24 m ² battery room within The Works, per Volume 7 fire doctrine	(internal)	52
Buildings total		940 GFA	418

Check: $490 \times \$425 = \$208.25 \text{ K} \rightarrow 208$; $450 \times \$350 = \$157.5 \text{ K} \rightarrow 158$; $208 + 158 + 52 = \mathbf{418}$.

2.5 Fleet acquisition — \$80 K

Assumptions: quantities are the canonical Phase 1 fleet ($\sim$ 150 powered assets); unit costs are **landed** (wholesale ex-China per the Volume 8 procurement handbook + $\sim$ 22-28% freight/duty/QC blended in); batteries at the canonical 3:1 ratio (450 packs for 150 assets, blended \$23/pack across 2S/3S sizes); opening spares inventory sized to Volume 7's high-turnover spares doctrine at $\sim$ 90 days' cover. Rental construction machines are electromechanical (lead-screw) per canon; the single hydraulic showcase machine is the Kabolite-class flagship.

Class	Qty	Unit landed \$	\$K	Notes
Touring/GT — entry 1/14 (LDRC/MJX class)	12	110	1.32	Motorsport, Vol 3
Touring/GT — standard 1/10	12	230	2.76	Motorsport
Drift (MST-based)	12	340	4.08	Premium chassis class

Class	Qty	Unit landed \$	\$K	Notes
Buggy/rally (WLtoys 144010-class, up-graded)	12	125	1.50	Metal-gear/bearing upfit included
Short-course	8	270	2.16	
Formula/drag (special events)	8	300	2.40	Event & media fleet, Vol 3 Ch 14
Excavators (Huina 1580 class)	6	520	3.12	Landed incl upfit; 1:3 ratio anchor
Dump trucks (Huina 1582/1573 class)	18	230	4.14	3 per excavator ✓ canon
Wheel loaders (Huina 1583)	3	270	0.81	
Dozers	2	330	0.66	
Premium hydraulic showcase (Kabolite class)	1	11,800	11.80	K970-class flagship; supervised premium sessions only
Tractors (Double E E351 / 1/16)	12	135	1.62	Agriculture
Implement library (hitches, trailers, discs, plows)	1 lot	2,600	2.60	Custom fab share in Vol 7
Crawlers — standard (MN99S/WPL C24 class)	12	80	0.96	
Crawlers — premium 1/10 (TRX-4 class)	4	430	1.72	
Recovery crawlers (1/10, winch-equipped)	4	390	1.56	Tow-Truck Retrieval Protocol fleet
Utility/marshal vehicles	4	360	1.44	Track service
Spare rolling hulls (rotation pool)	18	—	2.82	10 × \$150 + 4 × \$250 + 4 × \$80
Batteries (3:1 canon: 450 packs, 2S/3S, XT60)	450	23	10.35	Consumable thereafter (\$5.2)
Smart balance chargers (SkyRC T1000/ISDT K4 class) + bunker racks	1 lot	4,540	4.54	
Standardized transmitters (rental-hardened)	30	45	1.35	Fleet radios beyond bundled units
Opening spares inventory (~90 days)	1 lot	16,000	16.00	Gears, arms, servos, motors, ESCs, tires — Vol 7/8 lists
Commissioning consumables	1 lot	290	0.29	Loctite, grease, coating, labels

Class	Qty	Unit landed \$	\$K	Notes
Fleet acquisition total	~150 powered + spares		80.00	

Check (in \$K): $1.32 + 2.76 + 4.08 + 1.50 + 2.16 + 2.40 + 3.12 + 4.14 + 0.81 + 0.66 + 11.80 + 1.62 + 2.60 + 0.96 + 1.72 + 1.56 + 1.44 + 2.82 + 10.35 + 4.54 + 1.35 + 16.00 + 0.29 = \mathbf{80.00}$. Powered-asset count: $12+12+12+12+8+8+6+18+3+2+1+12+12+4+4+4 = 130$ fleet vehicles + 4 utility + 18 spare hulls $\approx \mathbf{\sim 150 \text{ powered assets } \checkmark \text{ canon}}$.

Investor Note. Note the shape of this table: the **single Kabolite-class showcase machine costs more than the other 29 construction machines combined** (\$11.8 K vs \$8.7 K), and opening spares plus batteries (\$26.4 K) cost more than the entire motorsport fleet (\$14.2 K). Both are deliberate. The showcase machine is a marketing asset earning premium supervised fees (Volume 4); the spares depth is what keeps 150 assets at rental duty-cycle actually running (Volume 7). A competitor who “saves” on spares buys downtime, not margin.

2.6 RCW Node telemetry program — \$25 K

Assumptions: every revenue vehicle carries an RCW Node (ESP32-C3 + GPS, PWM-intercept kill-switch, per canon); production unit cost lands under the \$15 canonical target; the budget is dominated by firmware/backend engineering and installation labor, not the PCBAs themselves.

Line	Scope	\$K
Firmware & telemetry-backend engineering	Node firmware, OTA update path, Supabase Realtime ingest, kill-switch logic integration	9.00
PCBA design, NRE & prototype runs	Unified schematic, two footprints (Micro-Node $\sim 25 \times 25$ mm; Heavy-Node $\sim 40 \times 30$ mm), 3 prototype spins (JLCPCB class)	2.70
Production run — 220 units @ \$14.50 landed	130 installs + growth/attrition stock; $< \$15/\text{unit} \checkmark \text{ canon}$	3.19
Harnesses, connectors, conformal coating, mounts	JST-XH sets (Heavy), silicone strain relief (Micro), acrylic coating per canon	2.31
Installation & commissioning — 190 installs @ \$28 labor	Fleet + spare hulls + utility vehicles; bench test + geofence verification each	5.32
Test jigs, calibration rigs & program spares	Go/no-go jig, voltage-calibration rig, RMA float	2.48
RCW Node program total		25.00

Check: $9.00 + 2.70 + 3.19 + 2.31 + 5.32 + 2.48 = \mathbf{25.00}$.

2.7 Wi-Fi mesh & IT infrastructure — \$87 K

Assumptions: outdoor mesh (TP-Link Omada class) blanketing all Phase 1 zones so ESP32 nodes hand off between APs without dropping telemetry (canon); CCTV coverage of all customer areas and the charging bunker; POS/kiosk hardware for the pavilion.

Line	Scope	\$K
Outdoor Wi-Fi mesh	14 outdoor APs, poles, fiber back-haul ring, controller	33
Core network, servers, UPS & CCTV	Rack, switches, on-prem edge server, UPS, 24-camera CCTV incl bunker thermal camera	32
POS, kiosks, screens & timing	4 POS stations, 2 self-serve kiosks, leaderboard screens, lap-timing loops	22
Wi-Fi & IT total		87

Check: $33 + 32 + 22 = 87$.

2.8 RC WORLD OS v1 development — \$165 K

Assumptions: v1 scope is the opening-day minimum defined in Volume 13 — bookings & wallet, Toolbox Talk induction/waiver, queue roster & top-ups, fleet inventory & maintenance logs, live telemetry map with kill-switch, POS integration, and the web admin console — on the canonical stack (Supabase/PostgreSQL + Kotlin services; Kotlin Multiplatform/Compose Multiplatform apps for Android **and** iOS). Budget: 14 developer-months blended at \$9.5 K (mixed senior contract + founder-team rates) = \$133 K, plus \$32 K for design, security review, device lab, app-store and tooling costs. Post-launch development is an opex line (\$5.5), not capex.

Line	\$K
Development — 14 dev-months @ \$9.5 K blended	133
UX design, security review, device lab, tooling & launch costs	32
RC WORLD OS v1 total	165

Investor Note. \$165 K is modest for an ERP because v1 is scope-disciplined: it ships only what opening day needs and rides on Supabase primitives rather than custom infrastructure (Volume 13, Chapter 3). The strategic payoff is out-of-proportion: RC WORLD OS is the moat competitors cannot buy off the shelf, and it is the franchisable asset priced in Volume 12. Treat this line as protected — it is the last line to cut in any value-engineering exercise.

2.9 FF&E, workshop equipment & theming — \$118 K

Line	Scope	\$K
Entry pavilion fit-out & F&B kiosk equipment	Counters, coolers, coffee machine, warmers, retail shelving, briefing-room AV	46
The Works equipment	Benches, ESD stations, soldering/re-work, compressor, parts bins, battery test rigs, hand tools x4 sets	38
Shade, furniture & viewing	Shade sails, spectator benches, picnic sets, bag storage	22
Signage & theming	Wayfinding, zone theming (scale props per Volume 11 world-building spec), safety signage	12

Line	Scope	\$K
FF&E total		118

Check: $46 + 38 + 22 + 12 = 118$.

2.10 Pre-opening — \$162 K

Assumptions: core team hired 8 weeks before opening (GM 16 weeks) for build supervision, SOP training (Volume 7), fleet commissioning, and soft-opening rehearsals; launch marketing per Volume 9's launch plan; licences and insurance from first occupation.

Line	Scope	\$K
Pre-opening staff & training	Salaries before Day 1 (GM 16 wks; ops/workshop leads 10 wks; 12 staff 8 wks), training materials, soft-opening events	62
Launch marketing	Brand assets, launch campaign, creator/media program, opening event	55
Licences, insurance & professional	Public-entertainment licence, LiPo storage compliance, legal, accounting setup, insurance from occupation	45
Pre-opening total		162

Check: $62 + 55 + 45 = 162$.

2.11 Contingency — \$168 K (10.0%)

Held as a single unallocated line under GM + founder joint control, releasable only against documented scope events (latent ground conditions, tender overruns, FX movement on the fleet order). Ten percent is appropriate for a design-complete package on a flat serviced site; Chapter 9 stresses the case where it is not enough (capex +20%).

2.12 Phase 2 and Phase 3 envelopes; five-year program — ≈\$3.9 M

Phase 2/3 budgets are planning envelopes ($\pm 15\%$), refined to line-item level in the Phase 2 investment memo gated at operating Month 12 (\$3.5). They carry the division volumes' own build-ups: the aviation line is Volume 5's bottom-up budget (enclosure \$190 K + runway/apron civil \$85 K + sim lab \$45 K + fleet & radios \$32 K + timing/AV \$12 K = **\$364 K**, including the ~20-aircraft fleet), and the marine line carries the pond civil work at ≈\$212 K inside its \$231 K (Volumes 6 and 11). The aviation uplift versus earlier drafts (+\$128 K, plus its fleet moving inside the division line) is funded by value-engineering the restaurant/grandstand/retail/lawn package (−\$64 K, per Volume 11's staged fit-out sequence) and by trimming Phase 2 contingency from ~10% to ~6.2% — defensible because the largest Phase 2 line is now a bottom-up engineering budget rather than an envelope.

Phase 2 line (project Months 13-30)	\$K	Phase 3 line (project Months 31-60)	\$K
Aviation Division (Volume 5 build-up: enclosure 190, runway/apron civil 85, sim lab 45, fleet & radios 32, timing/AV 12)	364	Indoor all-weather arena (~1,200 m ² hall, indoor track, lighting, HVAC)	452
Marine Division (pond civil ≈212, decks/stations, water treatment, theming)	231	Corporate event centre	132
Full restaurant (build 208, kitchen 58, furniture 24)	290	RC Academy classrooms	108
Grandstand viewing	96	Night-racing lighting	78
Expanded retail	50	Franchise pilot support (systems, documentation, pilot fit-out share)	48
Events lawn & parking expansion (+44 bays + 4 bus bays, Volume 11)	34		
Marine fleet (~24 vessels ≈ 8) + Phase 2 RCW Nodes, batteries & spares (10)	18		
Subtotal	1,083	Subtotal	818
Contingency @ ~6.2%	67	Contingency @ ~10%	82
Phase 2 total	1,150	Phase 3 total	900

Checks: $364+231+290+96+50+34+18 = 1,083$; $+67 = \mathbf{1,150 \text{ } \checkmark \text{ canon}}$. Aviation internal: $190+85+45+32+12 = 364 \text{ } \checkmark$. Restaurant internal: $208+58+24 = 290 \text{ } \checkmark$. Phase 3: $452+132+108+78+48 = 818$; $+82 = \mathbf{900 \text{ } \checkmark \text{ canon}}$. Five-year program: $1,850 + 1,150 + 900 = \mathbf{3,900 \approx \$3.9 \text{ M } \checkmark \text{ canon}}$.

3. Funding Plan

3.1 The \$2.6 M ask — uses of funds

The ask is Phase 1 capex plus everything needed to reach sustained monthly break-even (Month 16–19) without a further raise, plus an unallocated reserve. The non-capex components below reappear, to the dollar, as the \$750 K opening cash position in Chapter 6's monthly model.

Use	\$K	Basis
Phase 1 capex	1,850	Chapter 2 (includes its own 10% construction contingency)
Working capital at opening	130	F&B/retail opening stock, consumables float, deposits, till/wallet float
Operating funding through break-even	350	Covers the cumulative Year 1–2 EBITDA trough of –\$118 K (Month 15, \$6.2) plus sustaining capex (\$3 K/month Y1, \$6 K/month Y2) with margin for a slow ramp
Corporate, legal & transaction costs	90	Entity setup, raise legal, audit/tax setup, founder-office overhead pre-opening
Unallocated reserve	180	Board-controlled; released only against Chapter 9 downside triggers
Total funding ask	2,600	✓ canon

Check: $1,850 + 130 + 350 + 90 + 180 = \mathbf{2,600}$. Non-capex funds at opening: $130 + 350 + 90 + 180 = \mathbf{750}$ (= opening cash, \$6.2).

3.2 Tranche structure with milestone gates

Capital is drawn in three tranches so investors never fund the next stage of risk before the previous one has been retired. Gates are objective and verifiable in a site visit plus an RC WORLD OS data pull.

Tranche	\$K	Timing	Milestone gate (all must be true)	Risk retired
T1	1,200	At close ($\approx$ operating Month –12, i.e. project Month 0)	Lease executed; planning permits granted; civil tender within budget; GM contracted	Site & permitting risk
T2	850	$\approx$ operating Month –5	Buildings weather-tight; Tracks A/B/C substantially complete; $\geq 80\%$ of fleet landed and commissioned; RC WORLD OS beta operating end-to-end (booking $\rightarrow$ telemetry $\rightarrow$ kill-switch demo); opening date fixed	Construction & technology risk
T3	550	At opening (operating Month 1)	Soft opening complete; safety sign-off (Volume 7 audit); first 1,000 paid Shifts sold; insurance in force	Launch risk
Total	2,600			

Check: $1,200 + 850 + 550 = \mathbf{2,600}$. If a gate fails, the tranche pauses and the pre-agreed remediation ladder applies (descope per §2.11 hierarchy, re-tender, or — worst case — orderly wind-down with land improvements as residual value).

3.3 Instrument options

The model is instrument-agnostic; three structures bracket the negotiation space. All returns math in §7.5 uses Structure A for clarity.

Structure	Composition	Investor economics	Founder economics	When it fits
A — Straight equity	\$2.6 M common/preferred	65% fully diluted (see §3.4); pro-rata dividends from Year 4	30% + control provisions	Single lead investor or small syndicate wanting clean alignment
B — Convertible + equity	\$1.6 M equity + \$1.0 M convertible note (8% PIK, converts at Phase 2 gate at 15% discount)	Downside seniority on ~40% of capital; equity upside on conversion	Less immediate dilution; conversion priced on Year-1 actuals	Investors wanting Year-1 proof before pricing the whole position
C — Equity + asset debt	\$2.0 M equity + \$0.6 M equipment/fit-out facility (9%, 4-yr amortizing, secured on buildings/FF&E)	Smaller equity cheque for same park	More dilution-efficient; adds fixed service cost from Year 1	Markets with accessible venue-finance; note Chapter 6 already models a separate \$0.6 M Phase 2 facility — Structure C would replace, not stack with, it

Investor Note. The plan deliberately does **not** assume venture-style follow-on rounds. The \$2.6 M is sized to reach self-funding: Phase 2 (\$1.15 M) is financed from operating cash flow plus an optional \$0.6 M facility (§3.5), and Phase 3 (\$0.9 M) from cash flow alone. An investor's exposure is therefore capped at the ask in every scenario except the deep-downside cases of Chapter 9, where the pre-agreed responses are descoping and the cost-flex ladder (§6.5) — not a rescue round priced against the existing holders.

3.4 Illustrative cap table (Structure A)

Pre-money value of \$1.4 M reflects contributed IP: the complete Master Development Plan, RC WORLD OS architecture and codebase progress, RCW Node designs, and the supplier network of Volume 8 — assets a competitor would need ~\$300-500 K and 18 months to replicate, valued here with a market-entry premium.

Holder	Basis	Post-money %
Investor(s)	\$2.6 M new money into \$4.0 M post (\$1.4 M pre)	65.0%
Founder	Contributed IP, plan, and execution role	30.0%
Employee option pool	Reserved at close (GM and early key hires)	5.0%
Total		100.0%

3.5 Funding Phases 2 and 3

Phase 2 (\$1,150 K — project Months 13-30 in the phasing canon; in this model's cash flow the spend lands in **operating** Months 13-30, §6.1) is funded by: retained operating cash flow

(cumulative EBITDA turns positive during Year 2 — \$154 K generated in Year 2, \$7.1) plus an optional **\$600 K equipment-and-fit-out facility** (modelled at 9% p.a., drawn in four \$150 K tranches Months 15–21, amortizing over Years 4–5 — the exact draws, interest, and repayments appear in Chapter 6 and \$7.1). The Phase 2 commitment itself is **gated on Year-1 actuals**: revenue $\geq$ 80% of plan and 90-day second-visit rate $\geq$ 20% — the gate **floor**, deliberately set below the KPI's operating targets of $\geq 22\%$ Year 1 / $\geq 28\%$ Year 3 (the KPI's home is \$10.4). Below gate, Phase 2 compresses to a “Marine-first” descope (~\$620 K) or delays 6–12 months — the conservative scenario of Chapter 10. Phase 3 (\$900 K, spent operating Months 37–60) is funded entirely from operating cash flow and is optionality, not obligation: each Phase 3 element must clear a standalone hurdle ($\geq 25\%$ ROIC on incremental EBITDA) at the operating Month-30 review (\$10.2).

4. Revenue Model

4.1 Architecture of the engine

Revenue is built bottom-up in four stages — (1) visitor volume by day-type and season, (2) capacity and utilization from canonical fleet counts, (3) yield from the canonical price card, (4) attach and program revenue — and then reconciled three ways: monthly to Year 1's \$1,280 K, annually to the five-year canon, and externally against the demand side: Volume 2's Chapter 9 works from catchment and per-visit spend to a Year-1 estimate of **~37,000–43,000 visits at a blended \$30–35 per visitor-visit**, and this engine's 38,920 paid visits (~42,000 gate visits) at \$30.0 blended sit inside that band — the agreement you want from two models built from opposite ends.

4.2 Visitor volume by day-type and season

Assumptions: 360 operating days; Year 1 opens March (operating M1); day-types calibrated to ~110 visitors/operating-day average — the midpoint of Volume 2's ~37–43 K-visit Year-1 estimate (Chapter 9, \$9.4) — with peak days 220–300 and midweek 40–60.

Day-type	Days/yr	Avg paid visitors/day	Paid visits
Peak (weekends, public/ school holidays)	118	195	23,010
Midweek standard	190	55	10,450
Shoulder & event days (league nights, twilight events, corporate blocks)	52	105	5,460
Total individual paid visits	360	108 avg	38,920

Check: $118 \times 195 = 23,010$; $190 \times 55 = 10,450$; $52 \times 105 = 5,460$; sum = **38,920** ($\div 360 = 108$ /day; inside Volume 2's ~37–43 K band ✓). Member visits, party guests, and corporate attendees add $\approx 3,100$ gate visits $\rightarrow$ **$\approx 42,000$ total visits**. Blended on-site spend = $(1,280 - 6 \text{ sponsorship} - 14 \text{ education}) \div 42.0 =$ **\$30.0/visitor-visit**, at the conservative bottom edge of Volume 2's \$30–35 band (Chapter 9).

4.3 Capacity and utilization from fleet counts

Assumptions: capacity is **concurrent operating stations**, not fleet size (cars queue for customers, not customers for cars — Volume 3, Chapter 2 doctrine); 2 Shift-slots per station-hour (20-min Shift + turnaround inside a 25–30 min cycle); 8 average revenue hours across 360 days.

Division	Concurrent stations	Basis (fleet canon)
Motorsport (Tracks A/B)	22	64 cars; track slots bind: 8 touring + 6 drift (Track A), 8 off-road (Track B) — Volume 3
Construction — Mining	12	30 machines; 3–4 excavator stations (of 6) + circulating dump trucks + loaders — 1:3 ratio
Agriculture	8	12 tractors, implement-limited
Crawler park (Track C)	13	16 crawlers, trail-section-limited
Premium supervised (Kabolite show-case + premium crawler line)	2	Artisan-supervised
Phase 1 total	57	

Annual slot capacity = 57 stations × 8 h × 2 slots/h × 360 d = **328,320 Shift-slots**. Year 1 consumes 57,720 blocks (\$4.4) → **17.6% slot utilization** — deliberately low, leaving ~5× headroom so growth to Year 3–5 requires marketing and retention, not capex (the Volume 4 “option value” argument, park-wide).

4.4 Shift & session revenue — Year 1 build

Assumptions: canonical price card (Casual \$15/\$22; Operator \$26/\$38; Day Pass \$59; Family bundle \$89); premium share of casual volume ~14% (hydraulics, FPV-prep, big crawlers); operator-conversion ~25% of session buyers; blocks per product as shown. Volumes follow from §4.2 visits × observed products-per-visit (~0.9 sessions/visit for individual visitors, the rest arriving inside passes/bundles/groups).

Product	Unit price \$	Units sold	Revenue \$K	Blocks consumed
Casual Shift — standard	15	21,400	321.0	21,400
Casual Shift — premium	22	3,500	77.0	3,500
Operator Shift — standard (2 blocks + pit stop)	26	7,200	187.2	14,400
Operator Shift — premium	38	1,250	47.5	2,500
Day Pass (avg 4 blocks used)	59	1,440	85.0	5,760
Family bundle (4 drivers × 2 shifts)	89	770	68.5	6,160
Member/party/corporate included blocks	in other lines	—	—	4,000

Product	Unit price \$	Units sold	Revenue \$K	Blocks consumed
Shift & session revenue / blocks			786.2	57,720

Checks: $321.0 + 77.0 + 187.2 + 47.5 + 85.0 + 68.5 = 786.2$ (\$84,960 and \$68,530 shown rounded to 85.0/68.5). Blocks: $21,400 + 3,500 + 14,400 + 2,500 + 5,760 + 6,160 + 4,000 = 57,720 \checkmark \4.3 . Average yield = $786.2 \div 53.72 \text{ K revenue-bearing blocks} = \$14.6/\text{block}$ (\$13.6/block across all blocks including bundled).

4.5 Membership MRR build

Assumptions: canonical tiers (Apprentice \$29, Operator \$59, Foreman \$99); sales start Month 2; net adds accelerate with the license ladder and league seasons (Volume 9); churn 3.5%/month netted within the adds shown.

Milestone	Apprentice	Operator	Foreman	Members	MRR \$K
End Month 3	45	10	2	57	2.1
End Month 6	105	30	7	142	5.5
End Month 9	160	52	13	225	9.0
End Month 12	240	82	22	344	14.0

Check (Month 12): $240 \times 29 + 82 \times 59 + 22 \times 99 = 6,960 + 4,838 + 2,178 = \$13,976 \approx \$14.0 \text{ K MRR}$. Year 1 recognized membership revenue (monthly curve in §4.8) = **\$70 K**; exit run-rate \$167 K/yr, which is why Year 2 membership revenue steps to \$160 K. Year 3 assumes ~620 average members (mix 420/150/50 → \$25.98 K MRR) → **≈\$300 K** $\checkmark \$4.9$.

4.6 Corporate, parties, education

Assumptions: canonical anchors (corporate from \$1,400 for 2 h/20 pax; parties from \$349 for 10 children); group-demand pools sized from the catchment segmentation in Volume 2, Chapter 9 (employers of 20+ staff, children turning 7–13, addressable schools in a 1.5 M metro); averages exceed anchors because packages upsell (catering, extra Shifts, premium classes).

Program	Year 1 volume	Avg ticket \$	Revenue \$K	Catchment-pool sanity band
Corporate events	45 events	1,600	72	\$70–140 K $\checkmark$
Birthday parties & groups	190 parties	420	80	\$80–150 K $\checkmark$
Education/schools	35 visits	400	14	\$8–25 K $\checkmark$
Program revenue			166	

4.7 F&B, retail, repairs, sponsorship

Assumptions: F&B kiosk-only in Phase 1 at **\$3.76/gate visit** (42.0 K visits — 52% purchase incidence $\times$ \$7.20 average ticket); retail counter at \$2.00/visit; paid repair/upgrade services for customer-owned vehicles priced per Volume 7's service menu; sponsorship is deliberately token in Year 1 (two local partners) and grows with audience (Volume 9 sponsorship strategy).

Line	Driver arithmetic	Revenue \$K
F&B	42.0 K visits × \$3.76	158
Retail	42.0 K visits × \$2.00	84
Repairs & services	≈ 490 jobs × \$20 avg	9.8
Sponsorship	2 local partners	6.0
Attach & other revenue		257.8

4.8 Year 1 monthly ramp — reconciles to \$1,280 K

Assumptions: March opening; ramp Months 1–4; July–August peak; September–November shoulder; December spike from holiday demand plus corporate party season; January–February soft but cushioned by memberships, leagues, and the maturing corporate pipeline. All figures \$K.

Month	Shifts & sessions	Memberships	Corporate/parties/education	F&B + retail	Repairs + sponsorship	
M1 (Mar)	59.6	0.0	8.0	15.9	0.5	84
M2 (Apr)	66.1	1.0	10.0	18.1	0.8	96
M3 (May)	70.3	2.0	11.0	19.7	1.0	104
M4 (Jun)	74.6	3.0	12.0	21.2	1.2	112
M5 (Jul)	83.8	4.0	13.0	23.8	1.4	126
M6 (Aug)	79.5	5.0	13.0	23.1	1.4	122
M7 (Sep)	59.6	6.0	14.0	18.9	1.5	100
M8 (Oct)	54.9	6.5	15.0	18.1	1.5	96
M9 (Nov)	49.6	7.5	16.0	17.4	1.5	92
M10 (Dec)	73.8	9.5	22.0	25.0	1.7	132
M11 (Jan)	56.2	11.5	15.0	19.7	1.6	104
M12 (Feb)	58.2	14.0	17.0	21.1	1.7	112
Year 1	786.2	70.0	166.0	242.0	15.8	1,280

Checks: column sums — 786.2 ✓ \$4.4; 70.0 ✓ \$4.5; 166.0 ✓ \$4.6; 242.0 = 158 F&B + 84 retail ✓ \$4.7; 15.8 = 9.8 + 6.0 ✓. Every row sums across (e.g. M10: 73.8 + 9.5 + 22.0 + 25.0 + 1.7 = 132.0). Grand total = **\$1,280 K ≈ \$1.28 M ✓ canon.**

4.9 Five-year revenue — reconciles to canon

Assumptions by year (all months **operating** frame): **Y2** — frequency growth (leagues, licenses), membership base ×2.3, aviation opens in stages across operating Months 16–24 (sim lab M16, fixed-wing M19–20, full program M23–24) and marine across operating Months 20–24, per the canonical opening schedule; **Y3** — first full year of the complete Phase 2 park (Aviation + Marine + restaurant: F&B steps up, per-cap rises); **Y4–Y5** — Phase 3 elements (indoor arena from operating Month ~44 removes weather ceiling; Academy scales education; night racing lifts peak yield). All figures \$K.

Revenue line	Y1	Y2	Y3	Y4	Y5
Shifts & sessions	786	1,120	1,390	1,585	1,730
Memberships	70	160	300	370	430
Corporate events	72	115	170	205	240
Parties & groups	80	112	150	170	190
Education & Academy	14	32	60	80	100
F&B	158	220	310	360	410
Retail	84	112	150	170	190
Repairs & services	10	18	30	38	45
Sponsorship & media	6	16	40	52	65
Total revenue	1,280	1,905	2,600	3,030	3,400

Column checks: Y1 $786+70+72+80+14+158+84+10+6 = 1,280$ ✓; Y2 $1,120+160+115+112+32+220+112+18+16 = 1,905$; Y3 $1,390+300+170+150+60+310+150+30+40 = 2,600$ ✓ **canon**; Y4 $1,585+370+205+170+80+360+170+38+52 = 3,030$; Y5 $1,730+430+240+190+100+410+190+45+65 = 3,400$ ✓ **canon**. Growth sanity: Y2 exit run-rate (Months 23-24 at \$186-204 K/month, \$6.2) annualizes to ≈\$2.3-2.4 M, making Y3's \$2.6 M a ~10% step on Phase 2 completion — frequency and new divisions, not a larger catchment (consistent with the frequency assumptions in Volume 2, §9.5).

Division attribution (Year 1, Shift + program revenue of \$952 K): Motorsport ≈ \$470 K, Construction & Agriculture ≈ \$430 K, Crawler park & recovery ≈ \$52 K — honouring both Volume 3's "highest single revenue engine" and Volume 4's "≈\$410-470 K, roughly a third of park revenue" ($430 \div 1,280 = 34\%$). Day-pass and bundle value is attributed by observed block consumption in RC WORLD OS.

Investor Note. Three properties make this revenue model conservative rather than optimistic. First, **utilization is 17.6% of slot capacity in Year 1** and only ~24% by Year 3 — the park never needs to be busy in absolute terms to hit canon. Second, **recurring revenue compounds quietly**: memberships grow from 5% to 13% of revenue across five years, and every point of that mix shift stabilizes the winter months that drive break-even timing. Third, the model **excludes franchise income entirely** (royalties, fees, and the Volume 12 program are upside outside canon, pointed to in §10.3).

5. Operating Cost Model

5.1 Staffing plan & payroll

Assumptions: salaries are full-year costs in \$K for a mid-size international city (adjust locally with the Volume 12 localization index); **on-costs 15%** (employer taxes, workers' compensation, basic benefits) applied to base; casual/seasonal flex staff cover peak days and are

shown as a separate line; headcounts step with phases. Artisans are the customer-facing technician corps (canon).

Year 1 establishment (18.5 FTE):

Role	FTE	Base \$K each	Base \$K total
General Manager	1	70	70
Operations Manager	1	52	52
Workshop Lead (Head Artisan)	1	48	48
Artisans (technicians)	4	36	144
Track Marshals	5	26	130
Front-of-house / bookings	3.5	24	84
F&B kiosk staff	2	24	48
Marketing & community	1	38	38
Admin / bookkeeper	1	30	30
Base payroll	18.5		644
On-costs @ 15%			97
Casual peak flex + recruitment & training			26
Year 1 people cost			767

Check: $70+52+48+144+130+84+48+38+30 = 644$; $644 \times 1.15 = 740.6 \approx 741$; $741 + 26 = 767$.

Year 3 establishment (≈ 24.5 FTE + restaurant team): GM 74; Ops Manager 56; Workshop Lead 52; Artisans $6 \times 37 = 222$; Marshals $6 \times 26 = 156$; Front-of-house $4 \times 24 = 96$; restaurant/F&B team 125; Marketing 48; Admin/finance 38 $\rightarrow$ base 867; $\times 1.15 = 997$; + casual flex 43 = **1,040**. Headcount growth over five years tracks divisions, not overhead: the Year 5 additions are marshals for Aviation/Marine, restaurant staff, and two Artisans — management stays at three salaried leaders plus the GM.

People cost (\$K)	Y1	Y2	Y3	Y4	Y5
Total people	767	940	1,040	1,217	1,351
As % of revenue	59.9%	49.3%	40.0%	40.2%	39.7%

The people ratio falling from 60% to 40% is the core of the margin story (\$7.2): a safety-credible venue must be fully staffed on Day 1, so revenue growth outruns headcount growth.

5.2 Fleet maintenance, parts & batteries

Assumptions: Year 1 fleet duty = $57,720 \text{ blocks} \times 20 \text{ min} = 19,240 \text{ vehicle-hours} + \sim 760 \text{ staff/test/practice hours} \approx \mathbf{20,000 \text{ vehicle-hours}}$. Parts cost per vehicle-hour is derived from the failure-rate table below (component MTBF from the Volume 7 failure library and Source-2 failure-point lists), then cross-checked top-down. Batteries are consumables costed on the merged cycle-life model shared with Volume 7: one pack-cycle per block; retirement at **200-250 logged cycles or on breaching the internal-resistance threshold**, whichever comes first; **$\approx 40\text{-}50\%$ of the pool replaced per year**.

Component class (examples)	MTBF (veh-h)	Landed cost/event \$	Cost per veh-h \$
Tires & wheels (knobby, drift slicks)	60	14	0.23
Suspension & steering (arms, knuckles, shock shafts, links)	45	11	0.24
Drivetrain (pinions, spurs, CVDs, diffs, bearings)	55	16	0.29
Servos (15/25 kg digital)	220	24	0.11
Motors & ESCs (540/550, brushless ESC events)	400	52	0.13
Bodies, mounts & cosmetic	90	12	0.13
Construction-specific (track pins, lead-screws, micro-switches)	150	20	0.13
Fasteners, lubricants, coating, misc consumables	—	—	0.84
Blended parts cost per vehicle-hour			2.10

Check: $0.23+0.24+0.29+0.11+0.13+0.13+0.13+0.84 = 2.10$ (motorsport runs $\approx \$2.60/h$, construction $\approx \$1.30/h$, crawlers $\approx \$1.10/h$; the blend reflects the block mix). Battery math (merged model, per Volume 7): Year 1 throughput of 57,720 pack-cycles spreads across the ~ 450 -pack pool as **≈ 128 cycles per pack per year** ($57,720 \div 450$); with retirement at 200–250 cycles or the internal-resistance threshold, ≈ 40 – 50% of the pool retires each year — 180 – 225 packs $\times \$23 =$ **$\$4.1$ – 5.2 K** (budgeted $\$6$ K including crash-damaged packs). The ~ 2 -year average pack life this implies matches the 2–3-year fleet depreciation policy ($\$7.3$).

Fleet cost line (\$K)	Y1	Y2	Y3	Y4	Y5
Parts (veh-h $\times$ \$/veh-h)	42	64	78	92	104
Batteries (cycle-life consumption)	6	9	11	13	15
Tires/consumables uplift & event wear	10	15	18	22	25
Tools, warranty freight, external services	8	10	11	13	14
Fleet maintenance total	66	98	118	140	158

Check Y1: $42+6+10+8 = 66$. Maintenance cost per vehicle-hour (total basis) = $66,000 \div 20,000 =$ **$\$3.30/veh-h$** — a headline KPI ($\10.4). Fleet renewal (replacing worn-out vehicles) is sustaining **capex**, not opex: $\$36$ K (Y1), $\$72$ K, $\$150$ K, $\$170$ K, $\$190$ K — rising with Phase 2/3 fleet size and utilization.

5.3 Occupancy & utilities

Assumptions: land lease at $\$2.20/\text{m}^2/\text{yr} \times 48,000 \text{ m}^2 = \$105.6 \text{ K}/\text{yr}$ (peri-urban serviced-land band $\$1.50\text{--}\$3.50/\text{m}^2/\text{yr}$; verify locally — Chapter 9 stresses this). Utilities note the counterintuitive fact: fleet charging energy is trivial — $57,720 \text{ charge cycles} \times \sim 0.055 \text{ kWh} \approx 3.2 \text{ MWh} \approx \$640/\text{year}$; utility spend is actually F&B refrigeration, workshop power, lighting, offices, water/irrigation.

Occupancy line (\$K)	Y1	Y2	Y3	Y4	Y5
Land lease	106	110	114	118	122
Utilities (power, water, comms, gas)	52	68	80	88	94
Waste, security, site services	32	40	38	40	42
Occupancy total	190	218	232	246	258

Check Y1: $106+52+32 = 190$ (lease shown with modest contractual escalation).

5.4 Cost of goods sold (F&B, retail, events direct)

Assumptions: F&B COGS at the $\sim 30\%$ food-cost norm (32% Y1 kiosk, 34% from Y3 with the restaurant's broader menu); retail COGS 52% (hobby-retail keystone-minus); repairs parts at 50% of service revenue; direct event costs (catering, consumables, party hosts' variable hours) at $\sim 15\%$ of program revenue in Year 1, peaking $\sim 20\%$ in Year 2 (Phase 2 launch events and a heavier catered corporate mix), then declining with scale and in-house restaurant catering ($12\% \rightarrow 10\% \rightarrow 7\%$).

COGS line (\$K)	Y1 arithmetic	Y1	Y2	Y3	Y4	Y5
F&B COGS	$158 \times 32\%$	50.6	70	105.4	122.4	139.4
Retail COGS	$84 \times 52\%$	43.7	58	78.0	88.4	98.8
Repairs parts	$9.8 \times 50\%$	4.9	9	15.0	19.0	22.5
Events direct	$166 \times 15\%$	24.8	51	45.6	44.2	39.3
COGS total		124	188	244	274	300

Check Y1: $50.6+43.7+4.9+24.8 = 124.0 \checkmark$. Check Y3: $310 \times 34\% + 150 \times 52\% + 30 \times 50\% + (170+150+60) \times 12\% = 105.4 + 78.0 + 15.0 + 45.6 = 244 \checkmark$.

5.5 Marketing, insurance, software, admin

Assumptions: marketing at 8% of revenue Year 1 (launch-heavy) declining to $\sim 5\%$ at maturity — never below 4.5%, because Chapter 9 shows visitor volume is the model's biggest lever; insurance covers public liability (the dominant premium for a participation venue), property, and specialty lines (drone/aviation endorsement from Phase 2, open-water endorsement per Volume 6); software/hosting includes payment processing at 2.2% of the $\sim 85\%$ card share, RC WORLD OS hosting, and continuing development (post-v1 development is expensed here, not capitalized).

Line (\$K)	Y1 arithmetic	Y1	Y2	Y3	Y4	Y5
Marketing	8.0% of revenue	102	133	140	155	168
Insurance	liability 26 + property 12 + specialty 10	48	56	60	64	68
Software, hosting & payments	processing 24 + hosting/ SaaS 19 + continuing dev 14	57	78	96	108	119
Admin & misc (accounting, legal, bank, office)		26	40	40	44	48

Check Y1 payments line: $1,280 \times 85\% \times 2.2\% = 23.9 \approx 24$; $24+19+14 = 57 \checkmark$.

5.6 Consolidated operating cost stack

Opex category (\$K)	Y1	Y2	Y3	Y4	Y5
People (\$5.1)	767	940	1,040	1,217	1,351
Occupancy & utilities (\$5.3)	190	218	232	246	258
Fleet maintenance & batteries (\$5.2)	66	98	118	140	158
COGS (\$5.4)	124	188	244	274	300
Marketing (\$5.5)	102	133	140	155	168
Insurance (\$5.5)	48	56	60	64	68
Software, hosting & payments (\$5.5)	57	78	96	108	119
Admin & misc (\$5.5)	26	40	40	44	48
Total operating costs	1,380	1,751	1,970	2,248	2,470
EBITDA (revenue \$4.9 – opex)	–100	154	630	782	930
EBITDA margin	–7.8%	8.1%	24.2%	25.8%	27.4%

Column checks: Y1 $767+190+66+124+102+48+57+26 = 1,380$; Y3 $1,040+232+118+244+140+60+96+40 = 1,970$; Y5 $1,351+258+158+300+168+68+119+48 = 2,470$. EBITDA: $1,280-1,380 = -100$; $2,600-1,970 = 630$ (24.2%); $3,400-2,470 = 930$ (27.4%) — **EBITDA margin 24-28% from Year 3 ✓ canon.**

Investor Note. The Year 1 EBITDA of –\$100 K is a **feature of honest modelling, not a flaw of the business**: the park opens fully staffed and fully insured against a revenue base that takes 12–16 months to build its membership and corporate layers. The funding plan carries this loss explicitly (\$3.1's \$350 K operating-funding line). Beware any comparable plan showing positive EBITDA in the opening year of a staffed venue — it is usually understaffing safety roles or amortizing pre-opening costs into later years.

6. Cash Flow & Break-Even

6.1 Conventions of the monthly model

All months in this chapter are **operating** months (M1 = opening; §1.2 convention). Opening cash = \$750 K (the \$2.6 M ask less \$1,850 K Phase 1 capex, per §3.1 — all Phase 1 capex assumed spent by opening day). Sustaining fleet-renewal capex \$3 K/month in Year 1 and \$6 K/month in Year 2. Phase 2 capex spends \$55 K/month in operating Months 13–18 and \$75 K/month in operating Months 19–24 (\$780 K in Year 2; the remaining \$370 K falls in operating Months 25–30). The optional Phase 2 facility (§3.5) draws four \$150 K tranches in Months 15, 17, 19, 21; interest at 0.75%/month (9% p.a.) on the drawn balance, paid monthly in arrears. Working-capital movements are treated as neutral (membership prepayments and gift-wallet float offset inventory growth; stated as a simplification). Tax: no cash tax in Years 1–2 (losses; §7.1).

6.2 Monthly cash flow, Years 1–2 (\$K)

Mo	Revenue	Cash opex	EBITDA	Sust. capex	Ph 2 capex	Facility draw	Interest	Net cash	Closing cash
1	84	110	–26	–3	—	—	—	–29	721
2	96	111	–15	–3	—	—	—	–18	703
3	104	112	–8	–3	—	—	—	–11	692
4	112	114	–2	–3	—	—	—	–5	687
5	126	117	9	–3	—	—	—	6	693
6	122	117	5	–3	—	—	—	2	695
7	100	114	–14	–3	—	—	—	–17	678
8	96	114	–18	–3	—	—	—	–21	657
9	92	115	–23	–3	—	—	—	–26	631
10	132	119	13	–3	—	—	—	10	641
11	104	118	–14	–3	—	—	—	–17	624
12	112	119	–7	–3	—	—	—	–10	614
Y1	1,280	1,380	–100	–36	—	—	—	–136	614
13	118	128	–10	–6	–55	—	—	–71	543
14	124	130	–6	–6	–55	—	—	–67	476 (min)
15	134	136	–2	–6	–55	150	—	87	563

Mo	Revenue	Cash opex	EBITDA	Sust. capex	Ph 2 capex	Facility draw	Interest	Net cash	Closing cash
16	156	144	12	−6	−55	—	−1.1	−50.1	512.9
17	166	147	19	−6	−55	150	−1.1	106.9	619.8
18	176	150	26	−6	−55	—	−2.2	−37.2	582.6
19	160	151	9	−6	−75	150	−2.2	75.8	658.4
20	152	150	2	−6	−75	—	−3.4	−82.4	576.0
21	154	151	3	−6	−75	150	−3.4	68.6	644.6
22	175	153	22	−6	−75	—	−4.5	−63.5	581.1
23	186	154	32	−6	−75	—	−4.5	−53.5	527.6
24	204	157	47	−6	−75	—	−4.5	−38.5	489.1
Y2	1,905	1,751	154	−72	−780	600	−26.9	−124.9	489

Checks: Year 1 revenue column sums to 1,280 ✓ \$4.8; opex to 1,380 ✓ \$5.6; EBITDA to −100 ✓. Year 2: revenue 1,905 ✓, opex 1,751 ✓, EBITDA 154 ✓; facility draws $4 \times 150 = 600$ ✓; interest $1.1+1.1+2.2+2.2+3.4+3.4+4.5+4.5+4.5 = 26.9$ ✓ (0.75% on the stepped balance). Every closing-cash cell = prior cell + net cash. The Year-2 revenue ramp carries the staged Phase 2 openings (all operating months): aviation sim lab from Month 16, fixed-wing Months 19–20, marine from Months 20–24, and the aviation full program in Months 23–24 adding ≈\$25–35 K/month — the bridge to Year 3's full-park revenue.

6.3 Break-even — shown and explained

Monthly operating break-even = the month from which monthly EBITDA is positive **and stays positive**. Two arithmetic facts drive its timing. First, the steady Year-2 cost stack is ≈\$118–125 K/month of effectively fixed cost (people, lease, insurance, software, base marketing) plus ≈17–19% of revenue in variable cost (COGS, fleet wear, payment fees, casual labor) — so break-even revenue $\approx 122 \div 0.82 \approx$ **\$149 K/month**. Second, the revenue ramp (membership MRR compounding, corporate pipeline maturing, league base load) crosses that line at **Month 16** (\$156 K revenue vs \$144 K opex = +\$12 K) and never goes negative again in the base model (the shallowest subsequent month is Month 20 at +\$2 K, the post-summer shoulder). The canonical band **Month 16–19** exists because the crossing month is sensitive to opening-date seasonality and ramp speed: an autumn opening, or a ramp 10% slower, slides the crossing to Months 18–19 without changing annual totals materially. (First isolated positive months occur earlier — Months 5–6 in the summer peak — but the plan does not call that break-even; sustained coverage of the winter trough is the test.)

6.4 Cumulative cash curve and minimum-buffer policy

The cash curve has three regimes visible in §6.2: a **shallow Year-1 glide** from \$750 K to \$614 K (the −\$136 K Year-1 burn: −\$100 K EBITDA − \$36 K sustaining capex); a **Phase-2 construction trough** in early Year 2, bottoming at **\$476 K in Month 14**, with the facility draws deliberately timed after the trough so equity cash never falls further; and a **self-funding climb** from Month 22 as monthly EBITDA outruns Phase 2 spend. Policy: **minimum cash ≥ \$250 K at all times** — approximately three months of the survival-floor cost stack (§6.5) — with a board-notification trigger at \$350 K. The base model's worst point (\$476 K)

holds a \$226 K cushion above the floor; Chapter 9's downside scenarios are tested against exactly this policy.

6.5 The cost-flex ladder (downside machinery)

If revenue underperforms, costs flex in pre-agreed stages rather than ad-hoc cuts. The survival floor ($\approx \$0.9$ M/year) is the minimum annualized cost stack that keeps the park open safely, $\sim \$74$ K/month; it is the reference line for the downside cases in Volume 2's demand analysis (Chapter 9) and the master risk register (Volume 1, Chapter 8).

Stage (trigger)	Actions	Annualized stack \$K
Plan (base)	Full Year-1 establishment	1,380
Flex 1 (revenue <85% of plan for 8 rolling weeks)	Freeze hiring; casual hours –30%; marketing remixed not cut; defer non-safety maintenance projects	$\approx 1,250$
Flex 2 (<75% for 8 weeks)	Midweek hours compressed; F&B menu simplified; sustaining capex paused; one FOH and one F&B role held vacant	$\approx 1,100$
Survival floor (<60%, board decision)	Skeleton crew 8 FTE (~ 420), lease 106 + minimum utilities/site 69, insurance 45, software 40, fleet safety-minimum 45, marketing 55, COGS at reduced volume ~ 90 , admin 20	≈ 890 (the “\$0.9 M line”)

Check (floor): $420+106+69+45+40+45+55+90+20 = \mathbf{890}$. Even the survival floor keeps every safety-critical function (marshals within the 8 FTE, insurance, battery-room compliance) fully funded.

7. Profitability & ROI

7.1 Five-year P&L

Assumptions: revenue \$4.9; opex \$5.6; D&A per the \$7.3 policy applied to Chapter 2 asset categories plus phased and sustaining capex; interest on the Phase 2 facility (drawn \$600 K by Month 21, interest-only through Year 3, amortizing \$300 K/year across Years 4–5); tax at the **25% placeholder (verify locally)** with loss carryforward.

P&L (\$K)	Y1	Y2	Y3	Y4	Y5
Revenue	1,280	1,905	2,600	3,030	3,400
Operating costs	1,380	1,751	1,970	2,248	2,470
EBITDA	–100	154	630	782	930
EBITDA margin	–7.8%	8.1%	24.2%	25.8%	27.4%
Depreciation & amortization	268	289	354	402	461
EBIT	–368	–135	276	380	469
Interest expense	0	27	54	40	14

P&L (\$K)	Y1	Y2	Y3	Y4	Y5
Earnings before tax	−368	−162	222	340	455
Tax (25% place-holder, after carryforward)	0	0	0	8.0	113.8
Net income	−368	−162	222	332.0	341.2

Checks: EBIT = EBITDA − D&A each year (e.g. Y3: 630 − 354 = 276); EBT = EBIT − interest (276 − 54 = 222). Tax: cumulative losses 368 + 162 = 530 carried forward; Y3's 222 fully sheltered (remaining carryforward 308); Y4 taxable = 340 − 308 = 32 → tax 8.0; Y5 taxable 455 → 113.8. Interest ties to \$6.2 (Y2 = 26.9 ≈ 27) and the amortization schedule (Y3 full-drawn \$600 K × 9% = 54; average balances of \$450 K and \$150 K give 40 and 14). D&A ties to \$7.3's schedule.

7.2 EBITDA bridge, Year 1 → Year 3

The move from −\$100 K to +\$630 K decomposes exactly; the striking feature is that **revenue grows \$1,320 K while costs grow only \$590 K** — operating leverage from a venue staffed and insured for capacity on Day 1.

Bridge element	\$K effect	Running EBITDA \$K
Year 1 EBITDA		−100
Revenue growth (frequency, memberships, Phase 2 divisions, restaurant)	+1,320	1,220
People (restaurant team, +2 Artisans, +1 marshal, +0.5 FOH, wage drift)	−273	947
COGS (scales with F&B/retail/events volume)	−120	827
Fleet maintenance (utilization + Phase 2 fleet)	−52	775
Occupancy & utilities (escalation + restaurant load)	−42	733
Software, hosting & payments (volume-linked fees)	−39	694
Marketing (absolute growth; falls as % of revenue, 8.0% → 5.4%)	−38	656
Insurance (Phase 2 endorsements)	−12	644
Admin & misc	−14	630
Year 3 EBITDA		630 (24.2%) ✓ canon

Check: $-100 + 1,320 - (273 + 120 + 52 + 42 + 39 + 38 + 12 + 14) = 590 = 630$.

7.3 Depreciation policy

Canonical split: **fleet-class assets 2-3 years; infrastructure 10-20 years**. The policy per category, applied to Chapter 2 costs (contingency capitalized pro-rata as a 10-year line for simplicity):

Asset class	Cost \$K	Life (yrs)	Annual D&A \$K
Site & civil works	455	15	30.3
Track surfaces	172	10	17.2
Buildings	418	18	23.2
Fleet (vehicles, batteries, radios, spare hulls)	80	2.5	32.0 (Y1-2), 16.0 (Y3)
RCW Node program	25	2	12.5 (Y1-2)
Wi-Fi & IT	87	4	21.8 (Y1-4)
RC WORLD OS v1 (capitalized software)	165	5	33.0
FF&E & theming	118	6	19.7
Pre-opening (amortized)	162	3	54.0 (Y1-3)
Contingency (as spent, blended)	168	10	16.8
Year 1 D&A from Phase 1 assets	1,850		260.5

Check: $30.3 + 17.2 + 23.2 + 32.0 + 12.5 + 21.8 + 33.0 + 19.7 + 54.0 + 16.8 = 260.5$; plus sustaining-fleet capex depreciating from mid-Year 1 (+7.2) = **268 ✓ \$7.1**. Later years add the growing sustaining-fleet layer (annual charges 7.2 / 28.8 / 73.2 / 122.8 / 166.0 as the renewal program scales at 2.5-year life, half-year convention), Phase 2 assets (in service ~Month 27; full-year charge ≈\$117.4 K: civil/buildings/FF&E ≈\$1,098 K at a blended ~11.4-yr life ≈ 96.6, + fleet & nodes ≈\$52 K incl. contingency share ÷ 2.5 = 20.8), and Phase 3 assets (first elements in service early Year 5; part-year charge ≈\$37 K), while the original fleet (fully depreciated Month 30), Nodes (Month 24), Wi-Fi/IT (Month 48), and pre-opening (Month 36) roll off — producing the **268 / 289 / 354 / 402 / 461** sequence. Checks: Y3 = 232.0 remaining Phase 1 + 73.2 sustaining + 48.8 part-year Phase 2 ($117.4 \times 5/12$) = 354; Y5 = 140.2 remaining Phase 1 + 166.0 sustaining + 117.4 Phase 2 + 37.4 Phase 3 = 461. The short fleet life is honest economics, not conservatism: a rental car at Year-3 utilization genuinely is consumed in 2-3 years, and the sustaining-capex line (\$5.2) is its cash twin.

7.4 Payback

Cumulative EBITDA runs $-100 \rightarrow 54 \rightarrow 684 \rightarrow 1,466 \rightarrow 2,396$ (\$K) across Years 1-5 (each year adds \$7.1's EBITDA row; e.g. $684 = -100 + 154 + 630$). Against **Phase 1 invested capital (\$1,850 K)**: the crossing occurs when $1,850 - 1,466 = 384$ of Year 5's 930 has accrued $\rightarrow 384 \div 930 \times 12 \approx 5$ months into Year 5 = **operating Month ~53, early Year 5**. Against **capital actually consumed** (the \$2.6 M ask less the ~\$221 K still held as cash at end-Year 5, \$7.6 — i.e. \$2,379 K): $2,379 - 1,466 = 913 \div 930 \times 12 \approx 12$ months $\rightarrow$ **Month ~60, end of Year 5**. The ambitious scenario (\$10.1) pulls Phase-1 payback to **Month ~44 (Year 4)**; the conservative case pushes it beyond Year 5. Hence the canonical statement: **full payback Year 4-5, base case** — Year 5 on the base numbers, Year 4 achievable on the upside case, with Phase 2/3 self-funded throughout so payback is never diluted by follow-on equity.

7.5 Investor returns — IRR / MOIC ranges

Assumptions: Structure A cap table (investor 65%, \$3.4); exit at end of Year 5 (≈ 6 years from close, which precedes opening by ~ 12 months); exit value = EV/EBITDA multiple $\times$ Year-5 EBITDA + closing cash – debt (facility fully repaid by end-Year 5); LBE/FEC venue transactions historically clear 6–9 $\times$ EBITDA for proven single sites (industry transaction ranges, 2024–2026) — 6.5 $\times$ /7.5 $\times$ /8.5 $\times$ used. Franchise value (Volume 12) excluded.

Scenario (Y5 EBITDA / closing cash)	6.5 $\times$ exit	7.5 $\times$ exit	8.5 $\times$ exit
Conservative (520 / 100, no cash tax paid): equity value \$K	3,480	4,000	4,520
— investor proceeds (65%) / MOIC / IRR	2,262 / 0.87 $\times$ / –2.3%	2,600 / 1.00 $\times$ / 0.0%	2,938 / 1.13 $\times$ / +2.1%
Base (930 / 221): equity value \$K	6,266	7,196	8,126
— investor proceeds (65%) / MOIC / IRR	4,073 / 1.57 $\times$ / 7.8%	4,677 / 1.80$\times$ / 10.3%	5,282 / 2.03 $\times$ / 12.5%
Ambitious (1,280 / 560): equity value \$K	8,880	10,160	11,440
— investor proceeds (65%) / MOIC / IRR	5,772 / 2.22 $\times$ / 14.2%	6,604 / 2.54 $\times$ / 16.8%	7,436 / 2.86 $\times$ / 19.1%

Check (base/7.5 $\times$): $7.5 \times 930 = 6,975 + 221 = 7,196$; $\times 65\% = 4,677$; $\div 2,600 = 1.80\times$; $1.80^{(1/6)} - 1 = 10.3\%$. **Stated ranges for cross-volume citation (Volume 1 quotes these exactly): base case ≈ 1.6 – $2.0\times$ MOIC / ≈ 8 – 12.5% IRR; ambitious case ≈ 2.2 – $2.9\times$ MOIC / ≈ 14 – 19% IRR; conservative case ≈ 0.9 – $1.1\times$ MOIC.** Two upsides sit outside this table deliberately: **dividend capacity** of $\sim \$300$ – 400 K/year from Year 4 (post-facility-amortization free cash) adds ≈ 2 – 4 points of IRR if distributed rather than reinvested, and the **franchise program** (\$10.3) is pure option value on the same equity.

Investor Note. Read the base case honestly: **1.6–2.0 $\times$ MOIC and high-single to low-double-digit IRR** is an infrastructure-flavored return, not a venture return — earned with real assets, a self-funding expansion path, capped follow-on exposure, and Year-4 dividend capacity. What makes the risk-reward attractive is the shape: the conservative case still returns ~ 0.9 – $1.1\times$ (assets, cash, and a functioning business retain value), while the ambitious case and the franchise option carry venture-like upside. Investors seeking $>20\%$ IRR should price the franchise program into their thesis (Volume 12) or negotiate Structure B's downside seniority.

7.6 Five-year sources & uses (integrity check)

Sources \$K	Uses \$K
Equity ask	2,600
Phase 2 facility drawn	600
Cumulative EBITDA Y1–Y5	2,396
	Program capex (Phases 1–3)
	Sustaining fleet capex (36+72+150+170+190)
	Facility interest (0+27+54+40+14)
	Facility repayment

Sources \$K		Uses \$K	
		Cash taxes (8.0 + 113.8, \$7.1)	122
		Closing cash, end Year 5	221
Total	5,596	Total	5,596

Check: $2,600 + 600 + 2,396 = 5,596 = 3,900 + 618 + 135 + 600 + 122 + 221$. The model is closed: every dollar in is a dollar out or a dollar held. (The \$221 K closing position sits below the \$250 K policy floor only because the model repays the full facility inside Year 5; in practice the final \$150 K amortization payment slides one quarter, an immaterial timing choice.)

8. Pricing Strategy & Elasticity

8.1 How the canonical prices were set

The price card was benchmarked against the competitor and precedent set of Volume 2, Chapter 8, positioned on a simple rule: **price the Casual Shift below a karting heat and above a trampoline hour, because the experience sits between them in intensity and above both in novelty** — then let premium classes, time-extension, and bundles do the yield work.

Benchmark (Volume 2, Ch 8)	Typical price	RC WORLD position
Karting, single heat (10–15 min)	\$25–35	Casual Shift \$15 undercuts at 20 min — deliberate trial-friendly entry
Trampoline/adventure park, per hour	\$15–25	Operator Shift \$26 $\approx$ 40+ min of operation + pit-stop theatre
VR venue, per session/hour	\$30–60	Premium Operator \$38 with physical, photographable machines
RC rental venues (small format), 15–20 min	\$10–20	\$15 at radically higher production value; \$22 premium classes above the band
Dig This-class real-machinery experience	\$170–330	Premium construction Shifts deliver the fantasy at 1/14 scale for $\sim$ 1/10 the price
Cinema family night (4 people)	\$60–80	Family bundle \$89 for a two-hour active outing — the head-to-head the marketing plan targets

8.2 Price ladders and bundle logic

Every step of the ladder is priced off the \$15 base with a visible, defensible ratio — customers should feel the logic, and staff should be able to explain it in one sentence.

Product	Price	Ratio logic (arithmetic)
Casual Shift — standard	\$15	Base unit: 1 block

Product	Price	Ratio logic (arithmetic)
Casual Shift — premium	\$22	1.47× base: premium classes carry supervision and higher-value machines (hydraulics, FPV, big crawlers)
Operator Shift — standard	\$26	2 blocks à la carte = \$30; bundle saves \$4 (13%) and includes the battery-swap pit stop — rewards commitment, smooths queues
Operator Shift — premium	\$38	2 × \$22 = \$44 — \$6 (14%) — same discount discipline
Day Pass	\$59	≈ 5–6 casual blocks of value for 3.9× base; average redemption 4 blocks (\$4.4) keeps per-block yield at \$14.75 while capping the customer's spend anxiety
Family bundle	\$89	4 drivers × 2 shifts = 8 blocks; à la carte \$120; 26% family discount targets the highest-LTV segment
Memberships \$29/\$59/\$99	monthly	Priced at ≈2/4/6.6 casual blocks; pays back for the customer at 2–3 visits/month, exactly the frequency Volume 9's loyalty ladder engineers
Corporate from \$1,400	2 h / 20 pax	= \$70/head against a \$90–150/head corporate-event market norm; catering and premium classes upsell to the \$1,600 realized average (\$4.6)
Parties from \$349	10 children	= \$34.90/child vs \$25–45 party-market band; includes host, blocks, and party room

8.3 Dynamic and peak pricing options

Phase 1 opens with **fixed prices** (trust-building and operational simplicity), with three instrumented options held ready in RC WORLD OS, activated only on evidence: (1) **peak/off-peak spread** — ±\$3 on the Casual Shift (Sat-afternoon \$18, midweek-morning \$12), projected +4–6% Shift revenue at neutral volume based on observed FEC peak-pricing outcomes; (2) **Top-Up dynamic offers** — the existing T-5-minute top-up mechanic (canon) priced 10–20% below counter rate when the queue roster is empty, converting perishable slot inventory; (3) **event-yield pricing** — league entries and night events priced by demand tier. Elasticity beliefs used in Chapter 9 (category benchmarks, to be replaced by A/B evidence from Month 6): casual/family segments ≈ −0.8 around the \$15 anchor; enthusiast/premium ≈ −0.4; corporate/parties ≈ −0.3.

8.4 Discount governance

Discounting is the fastest way to destroy a yield model, so it is governed: **headline prices never discount** (no “50% off Shifts” — it reprices the anchor permanently); all discounts flow through **defined channels with owners and caps** — schools/education 20% (owner: GM), corporate volume tiers to 15% (sales), member benefits as included blocks rather than % off (product design, Volume 9), promotional trials as time-boxed bundles (marketing, ≤500 units/quarter). Every comp and discount posts to a **comp register in RC WORLD OS** re-

viewed monthly against a ceiling of **2.5% of gross revenue**; Gears loyalty redemptions are budgeted separately at $\leq 1.5\%$ of revenue as a marketing cost (Volume 9 mechanics).

Investor Note. The pricing architecture is built so that **yield grows without headline price rises**: premium-class mix, operator conversion, day-pass redemption depth, membership penetration, and (later) peak spreads are five independent yield levers, each worth 2–6% of Shift revenue. The plan needs none of them heroically; Chapter 9 shows what $\pm 15\%$ on price itself would do, and §8.3's elasticity beliefs are why the model prefers mix to price.

9. Sensitivity Analysis

9.1 Method

Each stress is applied to the base model in isolation (*ceteris paribus*), then read out on the two management-critical outputs: **break-even month** (base: Month 16, band 16–19) and **Year 3 EBITDA** (base: \$630 K, 24.2%). Flow-through logic: revenue-volume changes carry $\approx 82\%$ contribution (variable costs $\approx 18\%$ scale away); price changes carry $\approx 95\%$ contribution net of payment fees, before elasticity feedback; volume-linked share of revenue $\approx 92\%$ (memberships and sponsorship lag volume in-year).

9.2 Tornado table

Scenario	Mechanics (Karithmetic on Year3) Y3 EBITDA (K / margin)	Break-even month	
Visitors +20%	Rev +20% $\times 92\% \times 2,600 = +478$; $\times 82\%$ contribution $\rightarrow$ +392	1,022 / 33%	~Month 12–13
Visitors –20%	Mirror: -478 revenue $\rightarrow$ –392	238 / 11%	~Month 24–28
Price +15% (elasticity -0.4 enthusiast-weighted)	Price-bearing rev $2,360 \times 15\% = +354$; volume gives back $2,360 \times 6\% \times 82\% \approx -116 \rightarrow$ +238	868 / $\sim 30\%$	~Month 13–14
Price –15%	Mirror with weaker volume recovery (families don't fully backfill): –260	370 / $\sim 16\%$	~Month 20–23
Fleet maintenance +25%	$\$5.2$ Y3 line $118 \times 25\% =$ –30	600 / 23.1%	+1 month (~ 17 –20)
Capex overrun +20% (Phase 1 +\$370)	No P&L effect; cash effect -370 : opening cash $750 \rightarrow 380$; Month-14 trough $476 \rightarrow 106$, breaching the \$250 K floor	630 / 24.2% (unchanged)	Unchanged — but funding action required (\$9.4)
FX/shipping shock +30% on China sourcing	Opex: parts+batteries+consumables ≈ 107 of Y3 fleet line $\times 30\% \approx -32$ /yr; capex: sustaining fleet $150 \rightarrow \sim 195$ (cash, not EBITDA)	598 / 23.0%	+1 month (~ 17 –20)

Ordering confirms the tornado shape: **visitor volume ($\pm\$392$ K) > price ($\pm\$238$ - 260 K) > FX ($-\$32$ K) $\approx$ maintenance ($-\$30$ K), with capex overrun a cash risk rather than a P&L risk. This is why marketing spend is protected in the cost-flex ladder (\$6.5) while maintenance over-run is absorbed: the model is demand-fragile and cost-robust — the right shape for a venue, and the reason the KPI dashboard (\$10.4) leads with demand indicators.**

9.3 Compound downside

The honest nightmare is correlated: -20% visitors and $+25\%$ maintenance and a capex overrun (a rushed build causing quality problems suppressing demand). Stacked: Year 3 EBITDA $\approx 630 - 392 - 30 \approx \208 K (8%); break-even drifts to Months 26-30; the Month-14 cash trough breaches the floor. Survival mechanics: the cost-flex ladder holds the stack near Flex-2 ($\sim\$1.1$ M) buying $\approx\$180$ K/year back; Phase 2 defers per its gate (releasing the \$780 K Year-2 spend and its facility); the \$180 K unallocated reserve deploys. Modelled through, the park survives at roughly the conservative scenario of Chapter 10 with Phase 2 delayed 12 months — impaired returns ($\approx 1\times$ MOIC), not insolvency. That resilience is bought by three design choices: phase gating, the fixed-cost-light fleet, and the reserve inside the \$2.6 M.

9.4 Downside triggers and mitigations

Trigger (monitored in RC WORLD OS, monthly board pack)	Threshold	Pre-agreed mitigation
Rolling 8-week revenue vs plan	$<85\%$ / $<75\%$ / $<60\%$	Cost-flex Stages 1/2/floor (\$6.5); marketing remix before any price action
Month-end cash vs policy floor	$<\$350$ K notify / $<\$250$ K act	Freeze sustaining capex; draw reserve; if pre-opening: descope ladder (\$2.11) — theming $\rightarrow$ grandstand-class scope $\rightarrow$ never OS, safety, or spares
Capex committed vs budget (pre-opening, monthly)	$>95\%$ of category budget with work remaining	Contingency release protocol; re-tender remaining packages; T2/T3 gate review with investors
Maintenance cost per vehicle-hour	$>\$4.00$ for 2 consecutive months	Volume 7 failure-data review; supplier/QC audit (Volume 8); fleet-class retirement review
Price test results (any §8.3 experiment)	Contribution-negative at 95% confidence	Revert within one week; elasticity belief updated in this chapter's next revision
FX (USD-RMB) or freight index	$\pm 15\%$ vs plan	Forward-order 2 quarters of spares ($\approx\$25$ K working capital); re-quote across Volume 8's second-supplier matrix

10. Growth Scenarios

10.1 Conservative / base / ambitious — five-year view

Assumptions differ only on the demand side and phasing pace; the cost machinery (Chapter 5) and cost-flex ladder (\$6.5) apply in all three. The base case **is** the canon of Chapters 4–7.

Assumption	Conservative	Base	Ambitious
Year 1 visits	≈31 K (–20%)	≈39 K	≈46 K (+18%)
90-day second-visit rate	17%	22%	28%
End-Y1 members	220	344	470
Phase 2 timing	Gated: delayed to Months 19–36, Marine-first de-scope	Months 13–30	On schedule, restaurant pulled forward 3 months
Phase 3	Deferred indefinitely (arena case fails hurdle)	Months 37–60 selective	Full program + franchise pilot at Month 36
Blended per-cap spend trend	Flat \$28	\$30 → \$33	\$31 → \$36

Output (\$K)	Y1	Y2	Y3	Y4	Y5
Conservative — revenue	1,020	1,450	1,950	2,250	2,500
— EBITDA (margin)	–230	20 (1%)	330 (17%)	430 (19%)	520 (21%)
Base (canon) — revenue	1,280	1,905	2,600	3,030	3,400
— EBITDA (margin)	–100	154 (8%)	630 (24%)	782 (26%)	930 (27%)
Ambitious — revenue	1,450	2,300	3,200	3,900	4,500
— EBITDA (margin)	–40	320 (14%)	870 (27%)	1,090 (28%)	1,280 (28%)

Derived markers: break-even Month 22–26 / **16–19** / 12–14; Phase-1 payback beyond Year 5 / **Month ~53** / Month ~44 (ambitious cumulative EBITDA: $-40 + 320 + 870 = 1,150$ by end-Y3; $1,850 - 1,150 = 700 \div 1,090 \times 12 \approx 8$ months into Year 4). Note the conservative case still reaches a **21% margin business worth roughly its invested capital** — the downside is mediocrity, not wipeout — while the ambitious case is deliberately not heroic: +18% Year-1 visits (≈46 K) barely exceeds the top of Volume 2's ~37–43 K demand band (Chapter 9), and its margins are capped at 28% (canon) rather than allowed to inflate.

10.2 Phase 3 optionality

Phase 3 is a menu, not a bundle: each element clears a standalone hurdle (incremental EBITDA ÷ element capex ≥ 25%) at the operating Month-30 review, funded from cash flow only. Planning estimates: **indoor arena** (\$452 K) — removes the weather ceiling on ~30% of currently-lost days and unlocks winter leagues; projected +\$260–340 K revenue at ~45% incremental margin → ROIC ≈ 26–34%, the strongest and most strategic case (it is also the structural fix for the weather-compression risk carried in the master risk register, Volume 1,

Chapter 8). **Night-racing lighting** (\$78 K) — +\$60-90 K high-margin event revenue → ROIC ≈ 35-55%, cheapest yes. **Academy classrooms** (\$108 K) — education line 60 → 100+ with school-day utilization of empty midweek capacity → ROIC ≈ 20-30%, borderline standalone but strategically loaded (feeds licensing and franchise training). **Corporate event centre** (\$132 K) — corporate 205 → 240+ at high margin → ROIC ≈ 20-26%. Elements failing their hurdle at operating Month 30 are re-tested annually; the base case assumes all four proceed across operating Months 37-60 within the \$900 K envelope.

10.3 Franchise upside (pointer to Volume 12)

Nothing in this volume's canon includes franchising, deliberately. Volume 12 develops the model — franchise fee, royalty on gross revenue, RC WORLD OS SaaS fee per site, and procurement margin via the Volume 8 supply chain — with its pilot gated on three Volume-10 facts being demonstrated at the flagship: sustained 24%+ EBITDA margin (Year 3), a documented operating system (Volumes 3-9 SOPs live in RC WORLD OS), and a replicable capex bill of materials (Chapter 2 of this volume, which doubles as the franchise build budget template). For scale: each franchised site paying a mid-single-digit royalty on revenues comparable to this park's Year-3 canon would contribute high-five to low-six figures of near-pure-margin annual royalty — economics equivalent to adding a division for zero capex. Treat every franchise dollar as upside to \$7.5's returns, not as plan.

10.4 KPI dashboard definition

The park is managed on ten numbers, computed continuously by RC WORLD OS (implementation: Volume 13, Chapter 9; review cadence: weekly ops, monthly board). Definitions are fixed here so every volume and every site computes them identically.

KPI	Definition (exact)	Y1 target	Y3 target
RevPASH (revenue per available Shift-hour)	Shift & session revenue ÷ (concurrent stations × operating hours). Y1: 786.2 K ÷ (57 × 8 × 360) = \$4.79	≥ \$4.75	≥ \$6.50 (74 stations post-Phase 2: 1,390 K ÷ 213,120 = \$6.52)
Slot utilization	Blocks consumed ÷ slot capacity (\$4.3). Y1: 57,720 ÷ 328,320 = 17.6%	≥ 17%	≥ 24%
Membership count / churn	Active members at month-end / cancellations ÷ opening members	344 / ≤3.5%/mo	~620 avg / ≤2.5%/mo
90-day second-visit rate	First-time visitors returning within 90 days (cohort-tracked). This table is the KPI's canonical home; the Phase 2 gate uses a floor of ≥ 20% (\$3.5)	≥ 22%	≥ 28%
Maintenance cost per vehicle-hour	Total \$5.2 fleet cost ÷ telemetry-logged vehicle-hours. Y1: 66,000 ÷ 20,000 = \$3.30	≤ \$3.50	≤ \$3.40

KPI	Definition (exact)	Y1 target	Y3 target
Fleet availability	Fleet-hours in active status ÷ scheduled fleet-hours (from fleet_inventory); gate floor ≥ 90%	≥ 92%	≥ 94%
F&B attach	F&B revenue ÷ gate visits. Y1: 158 K ÷ 42 K = \$3.76	≥ \$3.75	≥ \$4.40 (restaurant)
Per-cap spend	(Total revenue – sponsorship – education) ÷ gate visits	≥ \$30	≥ \$33
Labor ratio	People cost ÷ revenue (\$5.1)	≤ 60%	≤ 41%
NPS	Standard survey, pushed post-visit via app	≥ 55	≥ 60

Investor Note. Insist on this dashboard in the monthly pack from opening week, and on the definitions above rather than management-friendly variants. The first two KPIs (RevPASH and utilization) are the earliest honest indicators of whether the Chapter 4 engine is performing — they move weeks before revenue misses become visible — and the second-visit rate is the single best predictor of whether Year 2–3 canon is reachable, because everything in the growth model (§4.9) is frequency, not reach.

11. Volume summary & cross-references

This volume built RC WORLD’s complete financial system and reconciled every output to the canonical headlines. **Capex:** Phase 1 builds bottom-up to \$1,850 K across nine audited categories plus 10% contingency, with the striking structural fact that the entire ~150-asset fleet is only \$80 K (4.3%) of it; Phase 2 (\$1,150 K) and Phase 3 (\$900 K) envelopes complete the ≈\$3.9 M five-year program. **Funding:** the \$2.6 M ask = \$1,850 K capex + \$750 K of working capital, operating funding, and reserve, drawn in three milestone-gated tranches (\$1,200/\$850/\$550 K), with Phases 2–3 self-funded from cash flow plus an optional \$600 K facility. **Revenue:** a bottom-up engine (39 K paid visits, 57 stations, 17.6% utilization, the canonical price card) lands Year 1 at \$1,280 K monthly-reconciled, and the five-year table at \$1,905 K / **\$2,600 K** / \$3,030 K / **\$3,400 K** — sitting inside Volume 2’s independent ~37,000–43,000-visit, \$30–35-per-visit demand estimate (Chapter 9). **Costs and margin:** an \$1,380 K Year-1 stack (60% people-weighted) grows only \$590 K while revenue grows \$1,320 K to Year 3, delivering EBITDA of –\$100 K / \$154 K / \$630 K / \$782 K / \$930 K and the canonical **24–28% margin from Year 3**. **Cash:** the monthly model shows sustained operating break-even at **Month 16** (canonical band 16–19), a minimum cash of \$476 K against a \$250 K policy floor, and a closed five-year sources-and-uses at \$5,596 K. **Returns:** payback **Year 4–5** (base Month ~53 on Phase 1 capital), base-case investor MOIC 1.6–2.0× / IRR 8–12.5% with conservative-case capital protection and ambitious-plus-franchise upside to ≈2.9× / ≈19%.

Line items other volumes must treat as fixed (change them here first, then sweep): the Chapter 2 capex categories and totals; the \$750 K opening cash and tranche gates (Chapter 3); the \$4.4 price-volume table and §4.9 annual revenue lines; the \$5.1 staffing establishment

and \$5.6 opex stack; the operating-Month-16 break-even mechanics and \$6.5 survival floor ($\approx$ \$0.9 M); the \$7.3 depreciation policy; and the \$10.4 KPI definitions (including the second-visit operating targets $\geq 22\%$ / $\geq 28\%$ with the $\geq 20\%$ Phase 2 gate floor).

Cross-references: market and demand inputs — **Volume 2** (Chapter 8's competitor and precedent benchmarks feed §8.1; Chapter 9's demand analysis, §9.4-9.5, feeds §4.2 and reconciles with Chapter 4's visit totals; the master risk register lives in Volume 1, Chapter 8). Division revenue capacity and operating doctrine — **Volumes 3-6** (station counts in §4.3; division attribution in §4.9). Maintenance failure data and spares doctrine behind §5.2 — **Volume 7**. Procurement costs, landed-cost build-ups, and FX mitigation behind §2.5 and §9.4 — **Volume 8**. Membership, loyalty, parties, and the retention machinery that drives §4.5 and the second-visit KPI — **Volume 9**. Site works, buildings, and the world-building scope priced in §§2.2-2.4 — **Volume 11**. Franchise economics and localization of this model — **Volume 12** (Chapter 3 rebuilds Chapter 4 with local inputs). RC WORLD OS scope capitalized in §2.8 and the KPI dashboard implementation — **Volume 13**.

RC WORLD Master Development Plan — Volume 11: Architecture & Park Design

Volume 11 of 13 · Revision: 1.0 — July 2026 · **Status:** Living document — bump revision on material change

Purpose of this volume. This volume is the architects' and contractors' brief for the RC WORLD park: the master site plan for the canonical 4.8-hectare reference site, dimensioned zone-by-zone design concepts, building floor plans with room schedules, utilities and infrastructure concepts, and the safety, security and landscape systems that bind them together. It translates the operating doctrine of Volumes 3–7 — sight-line separation, the 15° haul-road limit, the bunkered charging doctrine, the fully netted airfield, the 20-minute Shift throughput model — into buildable geometry: areas in m², heights in mm, gradients in degrees and percent, electrical loads in kVA. It is written so that a design team can take it, a topographic survey, and the local code set, and proceed directly to schematic design. Where this volume states a dimension, every other volume must quote it; where another volume is named as source of truth (fleet counts, pond depth, cage envelope), this volume quotes it and says so.

Intended readers. The appointed architect, civil/structural engineer, MEP engineer and landscape architect; the main contractor and specialist net/pond/track subcontractors; the founder and General Manager acting as informed client; investors and lenders assessing the physical asset behind the Phase 1 capex (≈ \$1.85 M, Volume 10); franchisees adapting the reference design under Volume 12; the permitting consultant preparing planning and building submissions.

A note on codes. RC WORLD is planned for a mid-size international city not yet fixed; this volume therefore specifies performance and geometry, not statutory compliance. Every item flagged **[verify locally]** — occupancy classifications, fire separations, sanitary fixture counts, accessible-parking ratios, electrical standards, structural wind/seismic loads, pond safety barriers — must be confirmed against the governing local code by the appointed design team before construction documents. Nothing in this volume overrides local law.

Chapters

1. Design Philosophy & Site Selection
2. Master Site Plan
3. Zone Design — Motorsport
4. Zone Design — Construction & Agriculture
5. Zone Design — Aviation & Marine

6. The Works — Workshop Building
7. Pit Lane & Charging Station
8. Guest Buildings
9. Viewing Areas & Circulation
10. Parking & Access
11. Landscaping & Environment
12. Utilities & Infrastructure
13. Security & Safety Systems
14. Future Expansion

1. Design Philosophy & Site Selection

1.1 The park is the product: industrial-authentic, not theme-park

RC WORLD sells the fantasy of operating a working industrial complex at 1/14 scale. The architecture must therefore be industrial, not depict it. The design language is industrial-authentic: real materials doing real jobs, detailed honestly, weathering gracefully, photographing like a documentary rather than a stage set. Three consequences follow.

First, the materials palette is the working world's palette. Scaffold-tube guardrails, shipping containers as enclosed program, corten and galvanized steel, board-marked and power-floated concrete, timber sleepers, HDPE shade sails in plant colors (safety yellow, machine orange, gunmetal). Nothing is painted to look like something else. A container that houses the marine plant room is labeled MARINE PLANT ROOM in stenciled DIN lettering, because that is what a port operator would do — and the stencil is the theming.

Element	Canonical material & finish	Where used
Guardrails & operator rails	42 mm galvanized scaffold tube, key-clamp fittings, rail height 1.1 m (operator) / 1.1–1.2 m (spectator)	All zones, walkways, rostrums
Enclosed program (plant, stores, kiosks)	20 ft / 40 ft high-cube shipping containers, corten patina or RAL 7016/1023 repaint, container-code graphics retained	Marine plant, parts overflow, F&B kiosk, marshal posts
Feature cladding & signage plates	1.5–3 mm corten steel sheet, laser-cut lettering, unfinished	Zone gateways, wayfinding totems, hopper cladding
Walking surfaces	Compacted crushed stone (guest paths), power-floated concrete (buildings, pit lane), timber board-walk (crawler zone, pond quay)	Site-wide
Retaining & pit edges	New/reclaimed timber sleepers 200 × 100 mm, galvanized pin fixings	Mining rim, berms, planters
Shade	HDPE knitted shade sails, 90%+ UV block, on 76–114 mm galvanized posts	Operator stations, rostrums, queues

Element	Canonical material & finish	Where used
Roofs (buildings)	Trapezoidal profile steel sheet, 6° min pitch, exposed gutters and downpipes to visible rainwater channels	The Works, entry pavilion, restaurant

Field Note — The cheapest theming decision in the whole park is restraint. Every fake rock, cartoon graphic or plastic-log fence added “for the kids” dilutes the premium that Volume 2’s market research identified: adults pay because the place feels like a real site they are finally allowed to run. Kids do not need help pretending; adults do. Build for the adults’ disbelief and the kids come free.

Second, scale discipline. The park operates two scales simultaneously: 1:1 for humans (rails, counters, doors) and roughly 1/14 for the machine world (dig faces, crop rows, quay bollards). The two must never blur. Human furniture is deliberately plain and industrial so the eye reads it as “backstage”; the machine world is detailed to scale-model fidelity (Volume 2, Chapter 4 sources the dressing; Volume 4 specifies the mining scenography). Rule of thumb for designers: anything a camera at machine height can see is scenography and must be to scale; anything only a standing human sees is plant and must be honest.

Third, the sight-line doctrine: see everything, touch nothing live. This is the load-bearing safety idea of the entire park, inherited from the source doctrine (customers never walk onto live tracks; Tow-Truck Retrieval Protocol instead) and it is architectural before it is procedural:

- Every operator station gives an unaided, unobstructed view of the operator’s machine and its full working envelope — no station may require operating into a visual shadow (this rule fixed the mining hopper’s position, Volume 4, Chapter 6).
- Every live surface (track, pit, pond, cage, field) is separated from guests by a physical edge — rail, fence, level change, water — of at least 1.1 m effective height, with kick-plates where objects could roll under. Gates into live areas are staff-controlled and interlocked with zone status in RC WORLD OS.
- Spectating is a designed product, not a leftover: terraces, mounds, the grandstand, and the elevated mining walkway (Chapter 9) are placed so that a non-paying companion always has somewhere good to stand, in shade, with the action framed for a phone camera. Volume 2’s finding that “media is the marketing department” is implemented here as geometry.
- Nothing that is live is reachable, and nothing that is reachable is live. Dead vehicles are retrieved by the recovery crawlers (gameplay) or by staff; the architecture removes the temptation by making the reach physically impossible, not merely forbidden.

Safety Warning — The 1.1 m rail is the minimum effective barrier height and assumes level standing ground on the guest side. Wherever guests stand on raised terraces, mounds or walkways, the barrier must be re-derived from the local fall/reach condition and checked against the governing building code [**verify locally**]. The sight-line doctrine never justifies lowering a barrier; if a rail blocks a child’s view, the ground is raised (step, mound) — the rail is not lowered.

1.2 Design tenets for every consultant

1. **Throughput is geometry.** The 20-minute Shift model (canon) means every attraction is sized around simultaneous stations × turnover, not around track length or pond area for

their own sake. Queue, briefing, hand-over and retrieval space is program, not circulation.

2. **Maintenance access is a first-class user.** Every zone has a staff-side service edge reachable by the utility vehicles and, for mining/marine/airfield, by a light truck. If an Artisan cannot reach a feature with a toolbag in under two minutes, redesign it.
3. **Phase lines are clean lines.** Phase 2 and 3 works (Chapter 2.4) must be buildable behind construction fencing without closing Phase 1 zones or crossing guest circulation. Utilities are stubbed to phase boundaries at Phase 1 (Chapter 12).
4. **Design for the hose and the broom.** Dust, mud and water are the park's daily weather. Falls to drainage everywhere, no un-drained low points, washdown points at every zone, surfaces that a leaf-blower and a hose can reset in 30 minutes each morning.
5. **Wind is a program input.** The airfield's orientation, the pond's sheltering berms and the shade-sail engineering all begin from the site wind rose (Section 1.4 checklist; Volume 5, Chapter 2 and Volume 6, Chapter 2 state the operating wind limits this plan must protect).

1.3 The canonical reference site

All dimensions in this volume are laid out on the **canonical reference site** (style-guide canon): **4.8 hectares (~12 acres)** of leased peri-urban land in a mid-size international city; **flat, with a 2-4% natural grade** falling to the southwest corner; **road frontage** on the southern boundary; **municipal three-phase power and water** at the frontage. The reference plat is taken as a rectangle **240 m (E-W) × 200 m (N-S)** = 48,000 m²; a real site of different proportion re-flows the master plan (Chapter 2) but must preserve the adjacency logic, the noise gradient, and every zone's internal dimensions.

The 2-4% grade is an asset, not a defect: it drains the site to a single southwest detention point (Chapter 11), gives the mining rim its elevation for free at the north end, and is imperceptible to guests. Sites steeper than ~5% average impose retaining costs that break the Phase 1 budget; sites dead flat impose drainage pumping. The band is deliberate.

1.4 Site-selection criteria checklist

The following checklist is the go/no-go filter for the reference site and for every franchise site under Volume 12. Items marked ● are hard requirements; ○ are strong preferences that trade against each other.

☐

● **Area:** ≥ 4.5 ha contiguous and developable (4.8 ha reference); regular shape, no dimension < 150 m.

☐

● **Grade:** average natural grade 1-5%; no more than 15% of the site steeper than 8%.

☐

● **Frontage & access:** direct frontage to a public road able to take coaches and articulated delivery vehicles; sight distances at the access point adequate for a leisure use [verify locally].

☐

● **Power:** municipal three-phase supply at or near the frontage, upgradeable to ≥ 200 kVA demand (Chapter 12 load schedule); note the utility's connection lead time in the project program.

☐

● **Water & sewer:** municipal potable water at frontage; sewer connection or approved on-site treatment feasible **[verify locally]**.

☐

● **Planning/zoning:** leisure/recreation use permitted or realistically permissible; no residential facade within 100 m of the northern (operational) half of the plan, or acoustic report demonstrates compliance **[verify locally]**.

☐

● **Flood:** outside the locally defined high-risk flood zone; if in a moderate zone, finished floor levels and the detention strategy (Chapter 11) resolve it **[verify locally]**.

☐

● **Tenure:** lease ≥ 10 years with renewal option, or purchase; landlord consent for the netted airfield's 16–17 m poles obtained in principle before signing.

☐

○ **Catchment:** $\geq 500,000$ population within 45 minutes' drive (Volume 2, Chapter 7 demand model).

☐

○ **Wind:** 12-month wind rose available or commissioned; prevailing direction steady enough to orient the airfield within $\pm 20^\circ$ (Volume 5 canon); mean speeds compatible with marine operating windows (Volume 6).

☐

○ **Soil:** free-draining granular soils preferred (tracks, pond excavation, pole foundations); a Phase 0 geotechnical investigation with ≥ 6 trial pits and 2 boreholes at the airfield and arena positions is a pre-contract requirement.

☐

○ **Aviation constraints:** site outside airport obstacle-limitation surfaces and any UAS restriction zone that would prohibit even netted model flight; confirm the 17 m poles trigger no aviation-obstacle process **[verify locally]**.

☐

○ **Neighbours:** no dust-sensitive or noise-sensitive neighbour (hospital, kennel, precision industry) sharing the northern boundary.

☐

○ **Visibility:** frontage visible to passing traffic (the netted cage and corten gateway are the billboard).

☐

○ **Transit & coach:** public transport stop within 500 m; school-coach approach without reversing.

Investor Note — Volume 2 identified land as a structural barrier to entry: 4–5 contiguous peri-urban hectares with utilities are scarce exactly where demand is strongest. The checklist above is deliberately strict; a site that fails two ● items is a walk-away regardless of price, because every failure lands in Phase 1 capex or in the operating calendar (wind, flood) forever.

2. Master Site Plan

2.1 Allocation: 48,000 m² fully accounted

The master plan allocates every square meter of the 4.8 ha reference site. Areas are precinct areas — the zone plus its own internal service edge and immediate spectator edge; park-wide guest circulation and landscape are carried as their own lines. Dedicated guest circulation (lines 10 + 16) is **≈ 12% of the developed area** (site less land bank), inside the 12–15% band that keeps peak-day flows one-way capable once each precinct's own spectator edge is counted (Chapter 9).

#	Precinct	Phase	Area (m ²)	% of site	Key internal dimension
1	Entry pavilion, plaza & F&B kiosk	1	1,600	3.3%	Pavilion 490 m ² GFA (Ch. 8)
2	Parking, drop-off & access roads	1 (ext. 2)	6,200	12.9%	132 car + 4 bus bays at build-out (Ch. 10)
3	Track A precinct (asphalt/drift + drag straight)	1	3,600	7.5%	~180 m line, ≥ 55 m straight (Ch. 3)
4	Track B precinct (dirt rally)	1	3,200	6.7%	~160 m line (Ch. 3)
5	Track C crawler park	1	1,200	2.5%	40 × 30 m (Ch. 3)
6	Pit lane & customer charging station	1	450	0.9%	8 swap bays (Ch. 7)
7	Mining Zone	1	2,400	5.0%	60 × 40 m (Vol. 4 canon)
8	Agriculture Zone	1	1,500	3.1%	50 × 30 m (Vol. 4 canon)
9	The Works precinct (building + yard)	1	900	1.9%	Building 450 m ² GFA (Ch. 6)
10	Guest circulation & viewing terraces (Ph. 1)	1	4,300	9.0%	3 m main spine (Ch. 9)
11	Landscape berms, swales & detention (Ph. 1)	1	3,650	7.6%	SW detention basin (Ch. 11)
Phase 1 sub-total			29,000	60.4%	
12	Aviation precinct (cage + Whoop Arcade + promenade)	2	3,400	7.1%	Cage 60 × 40 × 15 m (Vol. 5 canon)

#	Precinct	Phase	Area (m ²)	% of site	Key internal dimension
13	Marine precinct (pond pair + marina + plant)	2	3,200	6.7%	Harbour 30 × 18 m + speed 60 × 25 m (Vol. 6 canon)
14	Restaurant & terrace	2	800	1.7%	120 covers + 60 terrace (Ch. 8)
15	Grandstand & events lawn	2	2,600	5.4%	200-seat grandstand (Ch. 9)
16	Phase 2 circulation & landscape extension	2	1,000	2.1%	—
Phase 2 sub-total			11,000	22.9%	
17	Indoor arena, RC Academy & corporate centre	3	2,600	5.4%	Arena 40 × 25 m clear-span (Ch. 14)
18	Phase 3 plant & lighting infrastructure	3	200	0.4%	—
19	Land bank (held in landscape)	3+	5,200	10.8%	Two parcels (Ch. 14)
Phase 3 + reserve subtotal			8,000	16.7%	
Total			48,000	100%	

Two ratios matter for review discipline. **Attraction : support** — precincts 3–9 and 12–13 (the paid experiences) total 19,850 m², 41.4% of site; if value-engineering ever pushes this below ~35% the park has become a car park with hobbies attached. **Hard : soft** — sealed and roofed surfaces total ≈ 31% of site at Phase 3 build-out, which the stormwater strategy (Chapter 11) is sized for; any increase re-opens the detention calculation.

2.2 Adjacency logic

The plan is an argument, and each adjacency is a clause of it:

- **Mining borders The Works** (precincts 7 and 9 share an edge) because the Construction Division has the park's heaviest assets, the highest per-Shift maintenance event rate (Volume 4), and the only bulk-material logistics: aggregate top-ups, hopper maintenance and the nightly face re-cut all run from The Works' yard through a single staff gate. The mining hopper sits at the zone's east end (fixed in Volume 4 by the sight-line rule) — which is exactly the Works-facing end, so the conveyor and weighbridge electronics are a 20 m walk from the electronics bench.
- **The Works sits on the eastern service spine**, front door facing the guest path (the viewing window, Chapter 6, makes Artisan work a spectacle), back door facing the service road, so parts deliveries and fleet movements never cross a guest route.
- **The Airfield is peripheral and downwind: the northeast corner.** Three independent reasons converge. (1) Risk geometry — although the cage makes flyaway physically contained (Volume 5), defense in depth wants maximum distance between flight operations

and both the entry crowds and the public road. (2) Noise — EDF and prop noise is the park's loudest continuous source; with the prevailing southwesterly of the reference wind rose, the northeast corner exports noise away from the entry, the restaurant and the frontage neighbours. (3) Structure — the 16-17 m poles (Volume 5 canon) want the longest possible setback from the boundary the planning authority cares about, which is the road frontage. The cage's long axis runs E-W, within $\pm 20^\circ$ of the prevailing wind — this volume's siting tolerance on Volume 5's requirement that the runway and circuit work into wind.

- **Marine and Agriculture are the calm band**, west-center, adjacent to each other: the two quietest, most family-skewed zones share the gentlest part of the noise gradient, the events lawn, and the afternoon shade of the western berm. The harbour pond sits in the wind shadow engineered by that berm plus its own shelter belt (Volume 6 makes wind shelter a revenue item), while the sail area's windward shore is deliberately kept low and open (Volume 6, §4.2 and §6.6 doctrine); the speed pond's long axis is set across the prevailing wind to cut fetch, and the restaurant sits between the harbour and the spine so its terrace overlooks the working port (Volume 6's siting requirement) with Track A's straight in the second sight line.
- **Track A fronts the social heart**. The asphalt track is the park's most legible spectacle, so it sits center-east where the pit lane and grandstand overlook it, and the (Phase 2) restaurant's second, east-facing terrace front catches its straight. The drag straight aligns with the grandstand's axis.
- **Track B and Track C occupy the northwest**, between Agriculture and Mining: dirt with dirt. Their dust sources sit downwind of the guest core on the prevailing wind and their service needs share the northern staff route.
- **Entry, parking and the Phase 3 arena hold the south frontage**. Everything a visitor does before paying — parking, drop-off, ticketing — stays south of the guest gateway; the arena (Phase 3) also fronts south so corporate-event traffic never enters the park proper.

2.3 The noise gradient

Sound is planned as a south-to-north crescendo. At the frontage: car doors and café murmur. Through the gateway: PA at conversation level, the pond's calm. Mid-park: Track A tire noise and commentary. North band: mining buzzers, Track B's jumps, the airfield's motors — all held against the northern landscape berm (Chapter 11), which is both acoustic screen and viewing mound. Design targets, to be confirmed by the acoustic consultant against local limits **[verify locally]**: ≤ 55 dB(A) LAeq at the southern boundary, ≤ 50 dB(A) at any residential facade. The gradient also works commercially: the entry experience stays unhurried (retail dwell), and anticipation builds as guests walk north into the machine noise.

2.4 Phased build overlay

Phase	Occupies	Construction access	Guest impact
Phase 1 (Months 0-12)	Precincts 1-11: entry + parking core (88 car bays), Tracks A/B/C, pit lane + charging, Mining, Agriculture, The Works, Phase 1 circulation and berms. Utilities trunk routes laid full-length with stubs at Phase 2/3 boundaries.	Full site is a construction site	—
Phase 2 (Months 13-30)	Precincts 12-16: airfield cage + Whoop Arcade, marine pond pair, restaurant, grandstand + events lawn, parking extension (+44 bays).	NE and W-center parcels, fenced; heavy deliveries (poles, liner, precast) via the eastern service road out of guest hours	Zero closure of Phase 1 zones; crane lifts for airfield poles scheduled on closed days
Phase 3 (Months 31-60)	Precincts 17-18: indoor arena + academy + corporate centre (SE), night-racing lighting retrofit on Tracks A/B, franchise-pilot support space.	SE parcel direct from frontage — never enters the park	Lighting works at night behind track closures

The phase lines were drawn so that each later phase is a parcel, not a scatter: one construction fence, one gate, one insurance boundary. Phase 1 tenders should price the Phase 2 duct and drainage stubs as provisional sums — they cost hundreds now and tens of thousands if retro-trenched through a live park.

Tender packaging. Phase 1 splits naturally into six packages, and splitting them is a cost decision, not just a program one — three of the six are specialist trades a general contractor would only mark up:

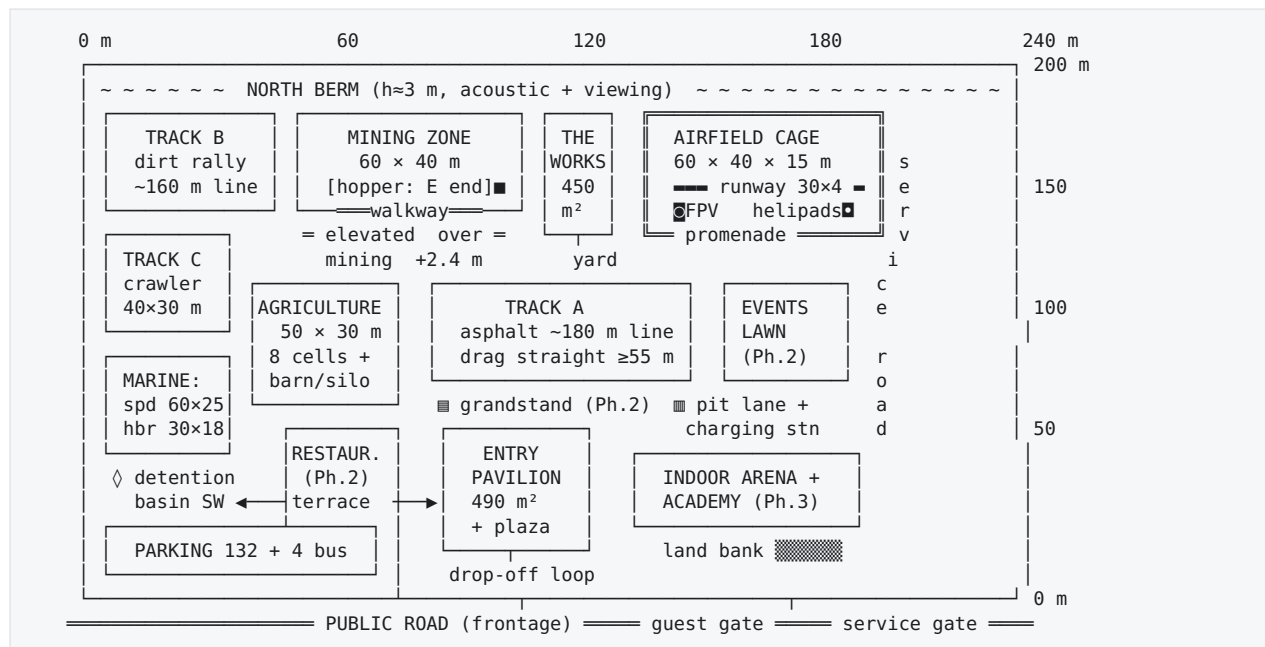
Package	Scope	Procurement route
P1-A Civils & earthworks	Bulk cut/fill, berms, pit excavation, roads, drainage, ducts, bases	Main contractor
P1-B Buildings	The Works, entry pavilion, pit building shell	Main contractor (same tender as P1-A)
P1-C Track surfaces	Track A asphalt/concrete + kerbs; Track B clay-gravel system	Specialist paving sub, direct — surface tolerance is the product
P1-D Zone fit-out & scenography	Sleeper works, hopper, barn/silo, dressing packages (Volume 2 supplier channel), rails, sails	Direct trades + Volume 8 procurement
P1-E MEP & data	Boards, sub-boards, bunker electrical, Wi-Fi/CCTV/PA, comms room	MEP contractor, designed by the MEP engineer
P1-F Loose & operational	Rostrum systems, barriers, signage, furniture	Direct purchase, Artisan-installed where possible

Phase 2 adds two more specialist directs: the **net system** (design-build by a netting engineer against Volume 5's performance spec — never let a general contractor "equal-or-approve" a life-safety net) and the **pond liner** (installed by the membrane manufacturer's approved crew, with the warranty tied to their installation). The client-side rule across all packages:

any substitution touching a hold-point dimension (handover checklist) routes through the founder and this volume's change register, not through site instructions.

2.5 Master plan diagram

Scale: 1 character $\approx$ 3 m horizontal. North is up. Road frontage at bottom (south).



Legend: netted airfield cage (helipads east, FPV gate course west, Whoop Arcade sub-enclosure beside cage) · runway · mining hopper · elevated walkway · grandstand · pit lane · stormwater detention · landscape berm · land bank · Ph.2/Ph.3 phase-2/3 parcels · MARINE = speed pond 60 × 25 m + harbour pond 30 × 18 m with 6 × 4 m submarine bay (Vol. 6) · east edge service road runs frontage → The Works → airfield, fenced from guests.

Field Note — Walk the plan as a family of four before approving it: park (2 min), tickets and Toolbox Talk check (5), first machine visible within 60 seconds of the gateway (the pond or Track A — by design), lunch overlooking Track A, mining in the afternoon shade, exit through retail. If any future revision breaks the “first machine in 60 seconds” rule, the revision is wrong.

3. Zone Design — Motorsport

The Speed & Off-Road Hub is three engineered surfaces plus the shared driver infrastructure. Volume 3 owns everything about the cars and the racing program; this chapter owns everything you can pour, compact, bolt or plant. Where Volume 3 stated interface requirements (its Sections 9.6, 10.4, 11.4, 13.2), they are restated here as built geometry.

3.1 Track A — asphalt/drift circuit

Layout geometry. Track A is a closed circuit of **~180 m racing line** within a 60 × 45 m precinct pad, designed bidirectional (Volume 3 canon: league nights may invert direction, so barrier profiles, kerb ramps and rostrum sight lines are symmetrical about the racing line). The canonical layout:

Parameter	Specification
Racing line length	180 m (-0/+10 m as built; report as-built to Volume 3 for lap-parity math)
Lane width	3.5 m minimum, 4.5 m on the main straight and outside of Turn 1 (widest where speeds are highest and passing is intended)
Main straight	58 m between corner tangent points — satisfies the ≥ 55 m drag requirement: 40.2 m strip + ≥ 15 m braking run-off (Volume 3 canon)
Corner set	8 corners: T1 R=6.0 m 90°; T2-T3 chicane R=3.0 m; T4 hairpin R=2.5 m (minimum radius on circuit); T5 R=8.0 m sweeper 120°; T6 R=4.0 m; T7-T8 esses R=3.5 m onto the straight
Minimum corner radius	2.5 m (hairpin) — below 2.0 m, 1/10 touring cars scrub instead of steer
Crossfall	1.5–2.0% uniform to the outside edge (drainage; imperceptible to handling at this scale)
Drift annex	15 × 12 m polished pad joined at T5, own entry gate, sharing the T5 sweep as a linked layout (Volume 3, Section 9.6)

Surface build-up (section, top down): 25–30 mm fine-graded asphalt wearing course (6 mm nominal aggregate, machine-laid — hand-laid asphalt leaves grip inconsistencies the parity program will find) or 120 mm power-floated concrete slab with a light mechanical polish on the drift annex; 150 mm compacted crushed-stone base (CBR ≥ 30); compacted subgrade; geotextile separator on soft spots. Joints in the concrete option are sawn, not tooled, and sealed flush — a 3 mm lip is a launch ramp at 1/10 scale. Surface tolerance: 3 mm under a 3 m straightedge anywhere on the racing line.

Kerbs. Corner apexes and exit edges carry **scale FIA-profile kerbs**: precast concrete or machined HDPE strips 300 mm wide, rising 15 mm at 1:10 chamfer, painted red/white in 250 mm alternations. Kerbs are glued and pinned, not just bedded — a loose kerb is a projectile. The kerb face never exceeds 15 mm: at 60 km/h a 1/10 car treats 15 mm as a rumble, 25 mm as a wall.

Run-off and barriers, in order of energy absorbed: (1) painted run-off apron 0.5–1.0 m wide in the same surface, (2) a 0.3 m crushed-rubber-granule strip at high-energy corners (T1, T4, end of straight), then (3) the **barrier system**: 200 mm diameter foam pool-noodle-over-rope barrier on 12 mm polypropylene rope threaded through galvanized eye stakes at 1.0 m centers — the industry-standard RC track barrier, cheap, continuous, and energy-absorbing with no hard edges. Behind it, at the spectator edge only, a rigid 1.1 m scaffold-tube rail with 150 mm timber kick-plate. The two-stage system means a 60 km/h drag car meets foam first, steel never. Marshal access gaps every 15 m, offset so no straight-line path from track to public exists.

Embedded services (cast in at construction, per Volume 3): MYLAPS RC4 timing-loop conduit (25 mm) across start/finish plus a spare loop conduit mid-straight for the drag trap; 50 mm power/data conduit ring for trackside cameras and (Phase 3) floodlighting; drainage channel at the low edge discharging to the swale network (Chapter 11).

Construction sequence and QA hold points. Track A is the one surface in the park where the contractor's habits from full-scale roadwork actively mislead: tolerances that pass on a highway fail here. Sequence: (1) subgrade proof-roll, witnessed; (2) base course laid and compacted to $\geq 98\%$ modified Proctor, surveyed on a 5 m grid, ± 10 mm; (3) timing and lighting conduits set and sleeved through, not under, the base; (4) wearing course machine-laid in maximum two passes with a hot longitudinal joint on the racing line's least-loaded axis — never mid-corner; (5) 72-hour cure/traffic hold; (6) acceptance survey: **3 mm under a 3 m straightedge anywhere on the racing line, no ponding after a hose test, joint lips ≤ 1 mm.** A failed panel is cut back to the nearest joint and re-laid — surface patching is prohibited on the racing line because the parity program (Volume 3, lap times within 2–3%) will detect a grip patch faster than any straightedge. The drift annex's mechanical polish is a separate visit after cure, with a skid-value target set jointly with the Motorsport Lead from test-car sessions: polish is sneaking up on a number, not executing a spec.

3.2 Track B — dirt rally circuit

Layout geometry. ~160 m racing line in a 55 × 42 m precinct, packed-dirt surface, 4.0 m nominal width (4.5 m on the jump straight), designed unidirectional (jumps have a direction; inverting Track B is forbidden — the landing ramps only work one way).

Surface engineering. 100 mm of screened clay-bound gravel ("road base with fines", 10–15% clay fraction) laid over 100 mm compacted crushed stone, watered and rolled to a hard, slightly plastic running crust. The clay fraction is the maintenance variable: too little and the surface ravel to marbles, too much and rain closes the track. The track crew's daily water-and-roll reset (Volume 3, Section 10.4 moisture doctrine: "dark but not shiny") is enabled by a dedicated hose ring with take-offs every 20 m and a walk-behind roller garaged at the zone.

Jump profiles and the safe-landing ratio. All jumps are **tabletops** (never doubles — a cased double destroys rental fleet at industrial rates; a cased tabletop is just a flat landing) built to Volume 3's canon of **≤ 0.5 m height, landing-matched:**

Element	Dimension	Rule
Take-off ramp	Rise 0.5 m over 1.5 m run (18°) with the last 300 mm relaxed to 12°	The relaxed lip stops rental buggies from over-rotating; a sharp lip is a backflip machine
Table (deck)	2.5 m level	Cars landing short land flat
Landing ramp	Fall 0.5 m over 2.5 m run (11°)	Landing length $\geq 5 \times$ jump height — the safe-landing ratio this volume fixes. At the 50 km/h rental cap, flight is 3–5 m; an 11° landing matches the descent angle so suspension absorbs mm, not the chassis
Transition radii	≥ 1.0 m at ramp toe and crest	No kinks: a kink is a load spike
Construction	Compacted fill core faced with 75 mm clay-gravel wearing layer over geogrid; timber-sleeper edge boards flush with surface	Jumps that shrink under traffic become doubles by erosion — the geogrid and edge boards hold profile

Two tabletops sit on the jump straight, one single “kicker” tabletop at 0.3 m height on the back section. A whoops section (five rollers, 150 mm high at 1.2 m pitch) provides the technical suspension test without airtime.

Berm construction. Bermed corners are the dirt track’s kerbs: compacted fill banks on the outside of corners T2, T5 and T7, **inner radius per corner (2.5–5 m), bank face rising to 400 mm at 30–35°**, faced with the same clay-gravel wearing course over a geogrid-wrapped core, sleeper-edged at the back. Berms are cut with a 100 mm flat shelf at the top so an overshooting car parks rather than launches. Rebuild profile is checked monthly against painted depth gauges set in the berm core.

Drainage. Dirt tracks die by water. The whole precinct is crowned at 2% to perimeter french drains (100 mm perforated pipe in 300 × 300 mm gravel trench, geotextile-wrapped) discharging to the northern swale; jump and berm cores are free-draining fill so they don’t slump; and the marshal posts stand on 100 mm raised concrete pads (Volume 3 requires them splash-safe). After the design storm the track must be rollable within 2 hours — that criterion, not a pipe size, is the spec the civil engineer must meet **[verify locally]** for local rainfall intensity.

3.3 Track C — crawler park

Track C serves the 16-crawler fleet (Volume 4 program ownership) on a **40 × 30 m (1,200 m²)** terrain garden: the park’s slowest zone and its highest dwell-per-square-meter. Design speed is walking pace; danger is near zero; the design problem is interest density and operator sight lines.

Layout. A figure-eight trail network (~90 m of marked lines, 600 mm lane width) through a landscape of feature stations, with operators following their crawlers along a raised timber boardwalk spine (1.2 m wide, 300 mm above grade, 1.1 m rail on the terrain side) — the one zone where guests move with their machines. Boardwalk and trail never cross at grade; the trail passes under boardwalk bridges.

Feature catalogue (each feature is numbered, difficulty-rated 1–5 in RC WORLD OS, and buildable/replaceable as a unit):

#	Feature	Dimensions	Difficulty	Construction
C1	Boulder garden	8 × 4 m field, stones 100–400 mm	2–4	Sorted natural stone, largest pieces pinned with dowel and epoxy
C2	Rock steps ascent	6 steps, 120–180 mm rise each, total climb 0.9 m	3	Flat-faced stone bedded in dry-pack mortar
C3	Timber bridge pair	Spans 1.8 m and 2.4 m, deck 350 mm wide, no rails (fall 300 mm to sand)	2/4	100 × 50 mm treated timber, wire-brushed for grip
C4	Log crossing	Half-round logs 150–250 mm dia., 3 m run	4	Fixed with hidden pins; replace when polished smooth
C5	Water hazard — ford	1.2 m wide crossing, max depth 80 mm	3	Lined concrete dish, textured invert

#	Feature	Dimensions	Difficulty	Construction
C6	Water hazard — pool skirt	2 × 3 m pool, trail on submerged shelf, max depth 120 mm at trail, 200 mm at center (no trail line through center)	4	EPDM-lined, pump-recirculated from the crawler zone sump
C7	Scree slope	4 m at 25°, loose 10 mm gravel over fixed base	5	Contained by sleeper edge; gravel topped up weekly
C8	Teeter bridge	2.0 m pivoting plank, 400 mm max tip	5	Steel pivot, rubber stops; the signature photo feature
C9	Sand wash	6 × 2 m fine-sand basin	2	The beginners' loop centerpiece
C10	Tunnel & culvert	600 mm dia. half-pipe, 2 m long	1	Concrete pipe section; kids' favourite, keep sight line through it

Water-hazard depth limits are a fleet-protection spec, not a safety-of-life spec (no water on Track C exceeds 200 mm): the rental crawler electronics (Volume 4 upfit: conformal-coated Nodes, splash-resistant ESCs) tolerate 80–120 mm fording indefinitely but not submersion, so **no trail line may exceed 120 mm water depth** and the 200 mm pool center is off-line by geometry. Both water features drain to the sump nightly in frost season **[verify locally]** for the climate's freeze regime.

3.4 Driver rostrums

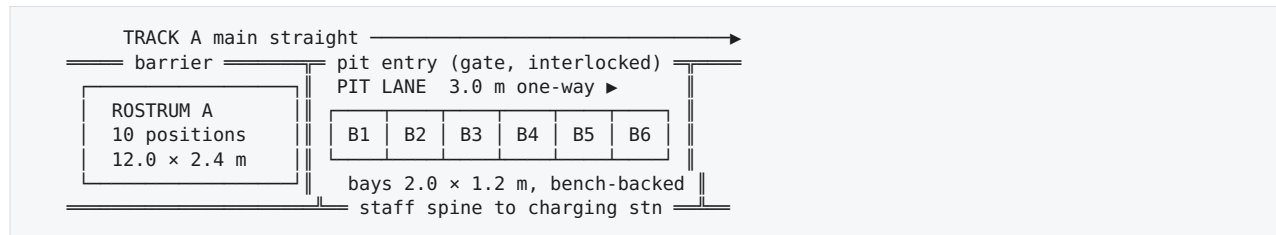
Each track has a **driver rostrum**: an elevated stand giving drivers the commanding view that separates racing from guessing.

Parameter	Track A	Track B	Track C
Capacity	10 drivers (Volume 3 can-on)	8 drivers	boardwalk-integrated (no fixed rostrum)
Deck height	1.2 m above track datum	1.5 m (jumps need more angle)	0.3 m boardwalk
Deck size	12.0 × 2.4 m	9.6 × 2.4 m	—
Position	Outside the main straight, mid-length, set back 2.0 m from the barrier	Outside the jump straight	—
Structure	Galvanized scaffold system or steel sub-frame, timber deck, 1.1 m front rail with padded transmitter ledge (300 mm wide, 15° tilt), rear rail and single 1.2 m-wide stair + accessible ramp 1:12	same	—
Shade	Full-cover HDPE sail, 90% block	same	sail clusters over feature stations

Driver positions are marked at **1.2 m centers** — elbow room for transmitter steering wheels. Rostrum sight-line check (mandatory at design review): from a 10th-percentile adult eye height at every marked position, the entire racing line is visible, including the far side of every berm and the landing of every jump. On Track A, bidirectional running means the check is done both ways.

3.5 Racing pit lane (trackside)

The racing pit lane serves Tracks A and B jointly, located between them at the rostrum end (the customer-facing pit theatre and charging station are Chapter 7; this is the working lane where Artisans stage, swap and re-issue fleet).



Element	Dimension	Note
Pit lane	3.0 m wide, one-way, power-floated concrete	Wide enough for a staff trolley + kneeling Artisan
Pit bays	6 bays, 2.0 m wide x 1.2 m deep each	One car + one Artisan + parts caddy; bay number painted, task lighting over each
Bench	Continuous 600 mm-deep bench at 900 mm height along the back of bays	ESD mat at bays 1-2 (electronics triage)
Power	Double GPO + USB-C at every bay on a dedicated 20 A circuit	No charging in the pit lane — packs move to the charging station (Ch. 7); doctrine, not convenience
Access	Interlocked gates to each track; roller door to the staff spine	Gate state visible in RC WORLD OS zone dashboard

Trade Hack — Cast a 100 mm-high, 300 mm-wide concrete plinth along the pit-bay front edge and set the bay floor mats on it. Cars roll in at bench-adjacent height, Artisans stop stooping, and dropped M3 screws land on mat, not gravel. Cost at pour time: nearly nothing. Retrofit cost: a weekend and an angle grinder.

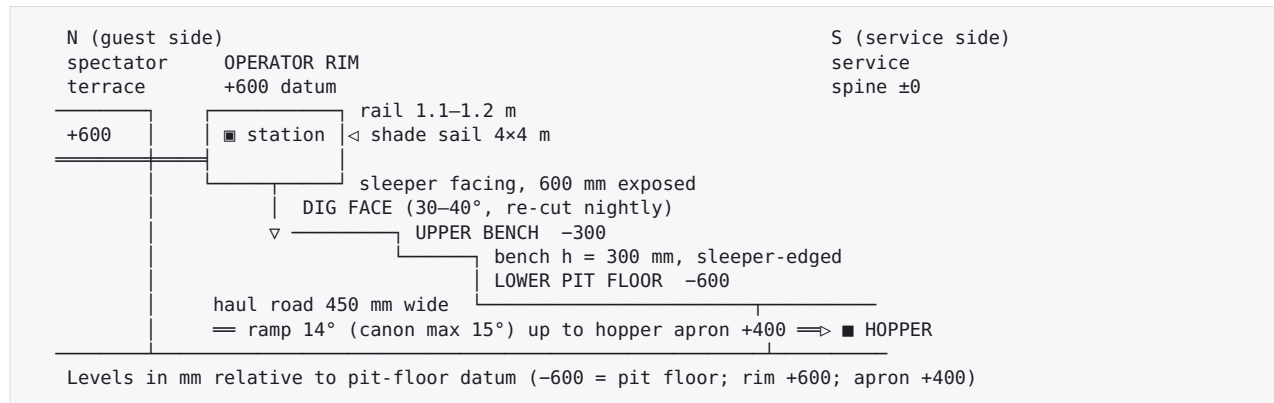
4. Zone Design — Construction & Agriculture

Volume 4 is the operating specification for both zones and has already fixed their internal geometry; this chapter restates that geometry as construction information, adds the civil/structural detail Volume 4 delegated here, and dimensions the guest edge. Where numbers below repeat Volume 4, Volume 4 governs.

4.1 Mining Zone — civil design

Envelope (Volume 4 canon): 60 × 40 m (2,400 m²), long axis E-W, operator rim on the north edge, service spine on the south, hopper at the east end, premium pad NE. The natural grade falls SW, exploited for drainage.

Pit cross-section. The pit is a two-bench excavation cut below a raised operator terrace:



Element	Specification
Bench heights	Two benches: upper at –300 mm, pit floor at –600 mm; bench step 300 mm , sleeper-edged (200 × 100 mm sleepers, pinned) so faces re-cut nightly against a fixed line
Dig faces	6 faces (DF1–DF6), 3 m wide, cut at 30–40°, one per operator pair of stations (Volume 4, Chapter 6)
Haul roads	450 mm running width single-lane, 700 mm at passing bays; wearing course 20 mm compacted crusher fines over the base aggregate; max grade 14° built (15° absolute canon limit) on the main and return ramps — the main ramp rises –600 → +400 mm over 4.0 m of run (Volume 4)
Retaining edges	All level changes ≥ 300 mm on the guest side retained with timber sleeper walls, galvanized pins at 600 mm centers into compacted base; walls > 600 mm exposed get a second pinned course and a drainage gravel backfill strip
Rim terrace	4 m deep compacted-gravel terrace at +600 mm, sleeper-faced; loads: crowd 5 kPa design [verify locally]
Pit floor drainage	Falls 1% to a SW sump with a lidded silt trap and a manual pump-out point; the pit is the site's lowest scenographic point but never its drainage low point

Hopper structure. The central processing hopper (Volume 4, Section 6.5 owns its mechanics: 600 × 600 mm grizzly, surge bin, 2.5 m conveyor, stockpile discharge at +1.4 m) needs from this volume: a **2.5 × 2.0 m reinforced concrete plinth** at apron level (+400 mm), cast-in M12 holding-down bolts for the steel support frame, a corten cladding skin (the hopper is the zone's visual landmark and must read as plant, not as a prop), a lockable ground-level electrics cabinet (IP65) fed from the zone sub-board, and maintenance clearance of 800 mm on three sides inside the service fence.

Operator rail stations — ergonomics. Twelve stations at **4.0 m pitch** (Volume 4 canon) along the rim:

Ergonomic parameter	Value	Rationale
Rail height	1.1 m (1.1–1.2 m band per Volume 4)	Forearm rest for adults; chest rail for children
Child step	Fold-down 250 mm step, full station width	Brings a 6-year-old's eye line over the rail; folds to clear crowds
Transmitter shelf	300 mm deep at 950 mm height, tilted 15°, drained slot at rear	Rests the transmitter without masking sticks; sheds rain
Station width	1.5 m marked zone within the 4.0 m pitch	One operator + one leaning parent
Viewing angle to assigned face	15–25° downward (Volume 4)	Deep enough to see into truck beds; shallow enough to read machine attitude at 20 m
Shade	4 × 4 m HDPE sail per station pair, posts 3.2 m	LCD readability + summer dwell
Station furniture	Number placard, assigned-face placard, QR to live telemetry card	Volume 4, Section 6.6

Material containment and recirculation. The zone's aggregates ($\approx 55 \text{ m}^3$ initial fill: sand, pea gravel, crushed stone — Volume 4) are a closed inventory. Containment is architectural: a continuous **150 mm-high concrete edge kerb around the entire pit perimeter** at the base of all sleeper walls (material cannot migrate out at floor level), the service-spine gate has a 100 mm bump sill, and the hopper stockpile sits on a concrete saucer that the wheel loader reclaims from (Volume 4's closed material cycle). A washdown apron (2 × 3 m, falls to the silt trap) at the service gate keeps aggregate out of the park's drains — the silt trap, not the swale, is where mining fines are allowed to stop.

4.2 Agriculture Zone — civil design

Envelope (Volume 4 canon): 50 × 30 m (1,500 m²), the grid of **eight field cells at 9.0 × 5.5 m** separated by **600 mm compacted tractor lanes**, barn/silo complex on the east edge, irrigation trenches as working scenery.

Element	Specification
Field cells	8 cells, 9.0 × 5.5 m each (396 m ² fields total); cells F1–F4 loose tillable media 150 mm deep over a compacted base with drainage falls; F5–F6 permanent row-crop cells (row spacing 250 mm, Volume 4 materials spec); F7–F8 rotating event cells
Cell edging	150 × 50 mm timber edge boards flush with lane level — a tractor must be able to cross into a field anywhere along the edge, so no upstand
Tractor lanes	600 mm wide, compacted crusher fines at lane level; lane grid drains at 1% to the western swale; passing squares 1.2 × 1.2 m at grid intersections
Irrigation trenches	200 mm wide × 100 mm deep lined channels along cell heads; wet by recirculating pump on show days, dry-scenic otherwise; never open water > 100 mm deep
Operator edge	South edge, 8 stations at 4.0 m pitch, same rail/shelf/shade ergonomics as mining (§4.1) — one detail differs: rail height 1.1 m but no child step fold-down; instead a continuous 200 mm raised timber curb-step (agriculture skews youngest, the step is always needed)

Element	Specification
Barn/silo complex	Volume 4 canon: 2.4 × 1.8 m scale barn with drive-through weighbridge bay + silo tower. This volume adds: concrete plinth 3.0 × 2.4 m, barn structure in timber at scale detail (a Volume 2 diorama-supplier package), silo in rolled steel 600 mm dia. × 2.0 m, both anchored for wind [verify locally] , weighbridge electronics conduit back to the zone sub-board
Perimeter	1.1 m post-and-rail (timber, agricultural character) with kick-plate on the guest sides; staff gate to the northern service route

Field Note — Resist the urge to “improve” the 600 mm lanes to something wider during setting-out. The lanes are gameplay: Volume 4 fixed between-row and lane clearances at ~1.4 tractor widths, and threading them clean is the zone’s core skill. A contractor’s generous extra 100 mm per lane quietly deletes the skill curve. The lane width is a hold-point dimension on the setting-out survey.

4.3 Shared service infrastructure

Both zones hang off the northern/eastern staff route: a 2.5 m-wide compacted service road linking The Works’ yard, the mining service spine, the agriculture staff gate and Track B’s maintenance corner. Along it: aggregate top-up bunkers (3 × 2 m sleeper bins for sand/pea/crushed — mining’s replenishment stock), the walk-behind roller and hand-tool garage (20 ft container), and hose points at 40 m spacing off the irrigation main (Chapter 12). The route is fenced from guests full-length (1.5 m mesh) and gated at both ends; it doubles as the emergency vehicle route to the park’s northern half (Chapter 10).

5. Zone Design — Aviation & Marine

Both divisions are Phase 2 builds on parcels reserved, graded and utility-stubbed in Phase 1 (Chapter 2.4). Volume 5 owns the aviation program and the containment-engineering rationale; Volume 6 owns the marine program and the water-quality doctrine. This chapter carries the construction geometry.

5.1 Airfield envelope and siting

The canonical enclosure (Volume 5, Chapter 1 — the numbers every volume quotes): **flight cage 60 × 40 m footprint, 15 m clear flight ceiling, ≈ 36,000 m³ enclosed volume**, long axis aligned within ±20° of the prevailing wind (this volume’s siting tolerance on Volume 5’s into-wind circuit requirement), sited in the northeast quadrant per the adjacency logic of Chapter 2.2. The aviation precinct (allocation table line 12) is 3,400 m²: the 2,400 m² cage, the **12 × 8 m Whoop Arcade** fine-mesh sub-enclosure beside it (Volume 5, Chapter 6 — its own sealed airspace, framed-panel construction), the southern spectator promenade (60 × 6 m) with its standoff rail 2 m outside the netting (Volume 5’s spectator rule: a slowed aircraft can still deform the mesh inward ~0.5 m), the pilot-access vestibule aprons, the mast maintenance strip (2 m clear around the full perimeter — scissor-lift access to every mast, non-negotiable), and the flight-line container store.

Siting inputs the architect must fix before Phase 2 documents: the 12-month wind rose (commissioned at Phase 1 if not already available — Volume 5 requires it and this

volume's site checklist demands it), a geotechnical confirmation at each pole position, and the aviation-obstacle check for 17 m structures **[verify locally]**.

5.2 Net engineering

Volume 5, Chapter 8 fixes the netting specification: main cage in **25 mm (1") knotless HDPE square mesh**, twine ≥ 2.0 mm **high-tenacity, UV-stabilised**, with breaking strength verified by supplier certificate against the design cases Volume 5 states (a 5" racing quad at 130 km/h and a 70 mm EDF at 140 km/h); the **Whoop Arcade in 13 mm fine mesh** as framed panels; netting hung with breakaway margin — it absorbs strikes by deflecting, never drum-tight; ground seal **zero gap ≥ 25 mm anywhere**. This volume adds the structure that carries it:

Element	Specification
Masts	Tapered galvanized steel, 16–17 m above ground, at 10–12 m centres (18–20 masts); base plate on reinforced concrete pad foundations, typically $\sim 1.5 \times 1.5 \times 1.5$ m in granular soils — final sizing by the structural engineer from site wind data and soil report [verify locally]
Roof structure	Tensioned catenary cable grid, 8–10 mm galvanized or Dyneema-core, ≤ 12 m spacing, crowned +1 m at centre for sag control and rain shedding; roof net in laced panels $\approx 12 \times 12$ m so a prop-strike hole means dropping and re-lacing one panel from a scissor lift, not a crane-and-crew replacement
Tension system	Each perimeter cable terminated with a galvanized turnbuckle (M16 class) and a marked witness gauge; walls laced to border ropes (≥ 10 mm braided) top and bottom; pretension set by the net engineer to preserve Volume 5's breakaway-margin doctrine while keeping working-wind deflection clear of the 5 m flight-path buffer
Ground seal	Bottom border rope buried 150 mm or threaded through ballast pipe full perimeter; any gap ≥ 25 mm is a defect (Volume 5: a whoop that fits under the net is not contained)
Access	Two personnel airlocks (double-door vestibules, sized $\geq 2 \times 2$ m by this volume) — outer door closed before inner opens — and one 3 m double-gated vehicle gate openable only when airspace is cold (Volume 5 door doctrine)
Whoop Arcade structure	12 $\times$ 8 m framed-panel enclosure on a galvanized steel frame at 3 m height, 13 mm mesh, single vestibule, fully sealed at grade
Inspection regime	Structural: annual mast/cable/termination inspection by the engineer; nets: Volume 5's daily walkdown and monthly detailed inspection schedule, with mast bolts torque-checked monthly

Safety Warning — The cage is a life-safety structure with a 15–25 year design life and it fails in storms, not in operation. The structural engineer's wind analysis must model the net **iced and fouled** (leaves, snow band per climate) as a load case, and the operations manual must include a drop-or-derig wind threshold for named storms **[verify locally]** for the local wind code. A net specified only for sunny-day containment is under-specified by roughly a factor of three.

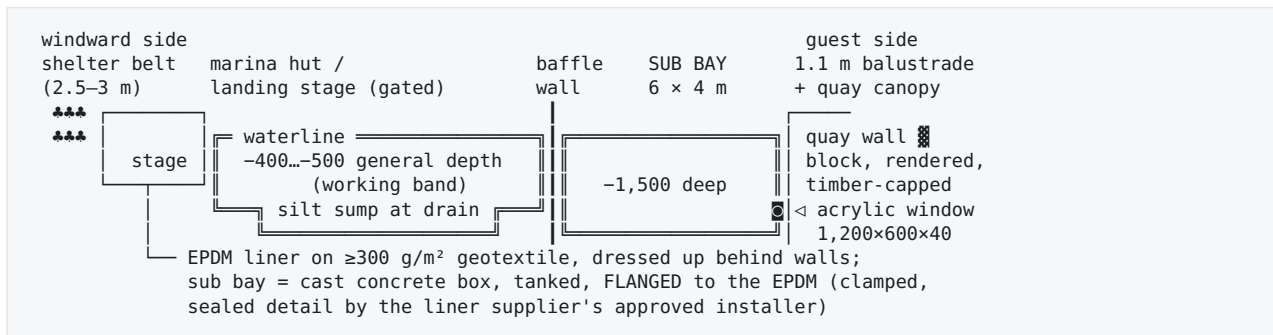
5.3 Runway, helipads and flight-line

Element	Specification
Runway	30 m paved (canon) at 4 m width (this volume's construction dimension), machine-floated concrete 120 mm on 150 mm base, or fine asphalt equivalent; runs along the cage's long axis; centreline and threshold markings in traffic paint; crossfall 1.5% to the north edge drain; surface tolerance 3 mm / 3 m
Overruns & margins	15 m paved overrun plus mown-grass margins inside the net (Volume 5); grass on 100 mm rootzone over free-draining base
Apron	12 × 8 m concrete pad at the runway's south-east, linking the vehicle gate, the pilot boxes and the helipad edge; fixed-wing arming happens here facing the runway (Volume 5 flight-line doctrine)
Helipads	Two, 1.5 × 1.5 m platforms at 1.0 m height (Volume 5 canon), steel frame, rubber-surfaced deck, railed on the spectator side, on the cage's eastern edge serving the 15 × 15 m heli hover box (Volume 5, §5.3)
Pilot boxes	Four, numbered, in line along the runway's south side, 3 m back from pavement (this volume's setback); 2.0 × 1.5 m each, 1.1 m rail, transmitter shelf per §4.1 ergonomics; a 2 m overhead net apron projects from the cage wall over the boxes so no aircraft can descend on a pilot; the painted flight line per Volume 5, Chapter 8 runs along their front edge
Drone-racing course	Western half of the cage (Volume 5): reconfigurable air-gate and flag course in MultiGP/DRL grammar, gates on movable ballasted bases — no fixed structure, so the volume reverts to general flight between race slots
Flight-line store	20 ft container inside the cage at the vehicle gate: aircraft racking, field-repair bench, steel ammo-box caddies for the day's packs; no charging except the Arcade desk's supervised 1S board per Volume 5 — all pack charging happens in the Phase 1 charging bunker, extended with aviation shelving in Phase 2

5.4 Marine ponds

Volume 6, Chapter 6 fixes the waterspace as a **pond pair — two hydraulically independent ponds sharing one plant room**: the **harbour pond, 30 × 18 m (540 m² water)**, a rectangular quay-walled basin at 400–500 mm general depth; and the **speed pond, 60 × 25 m (1,500 m² water)**, a rounded-rectangle “stadium” plan at a uniform 500–600 mm. The division's depth doctrine is a **400–600 mm working band across both ponds** — deep enough that keels, rudders and water pickups never ground, shallow enough that a staff member in waders can walk every square metre. The **sole engineered exception is the submarine bay: 6 × 4 m at 1.5 m deep**, a cast-concrete box baffled off the harbour pond, carrying the acrylic viewing window (~1,200 × 600 × 40 mm marine-grade cast panel per Volume 6, set at child-eye height and shaded by the quay canopy). This volume repeats all three depth figures as hold-points on the excavation drawings.

Section through the harbour pond (windward edge to quay, showing the submarine bay):



Element	Specification (Volume 6 governs; construction notes here)
Liner system	EPDM membrane basins with concrete structures standing inside them (Volume 6's hybrid doctrine); ≥ 300 g/m² non-woven geotextile under every square metre, carried up behind walls; ~2,600 m² liner total incl. walls/overlap; order as few, huge factory panels (Volume 6 Trade Hack: field seams are the only young failure); no chlorine ever
Quay walls (harbour)	Hollow concrete block on strip footings, rendered, built inside the liner line with the liner dressed up behind; timber capping; fendered berths (gameplay per Volume 6)
Submarine bay	Cast-concrete box, cementitious tanking, mechanically flanged to the EPDM with a clamped sealed detail — the one specialist joint in the build, executed by the liner supplier's approved installer ; acrylic window framed aquarium-style; baffle wall hydraulically separates the bay so barge traffic cannot stir sediment into it
Speed-pond edges	Concrete edge beam + rip-rap over liner on the straights (wake energy breaks on rock, not soil — Volume 6); cobble wave beach at the downwind end where wind and wake energy both arrive; turn buoys ≥ 4 m off the banks
Pond floors	Harbour floor laser-screeded ±20 mm, falling to a silt sump with bottom drain at the deepest corner; cut/fill balanced into the site's 2–4% grade
Landing stages	Supervised, staff-controlled, self-closing gates (Volume 6 barrier rule); boats armed only on the stand, facing open water
Public barrier	1.1 m post-and-rail balustrade with mesh infill wherever the public can approach water deeper than 300 mm (Volume 6's absolute rule), broken only at the supervised landing stages and the viewing-window bay [verify locally] against local pool/pond barrier law
Wind & siting	Speed-pond long axis set across the prevailing wind (fetch control); harbour pond sheltered by the marina hut, quay canopy and a 2.5–3 m evergreen shelter belt on the windward side; the sail area's windward shore stays low and open (sail needs wind); restaurant terrace overlooks the harbour (all Volume 6 siting requirements, resolved in this master plan)
Capex	Pond complex civil work ≈ \$212,000 (Volume 6, \$6.9 breakdown; carried in Volume 10's Phase 2 envelope)

5.5 Marine plant room

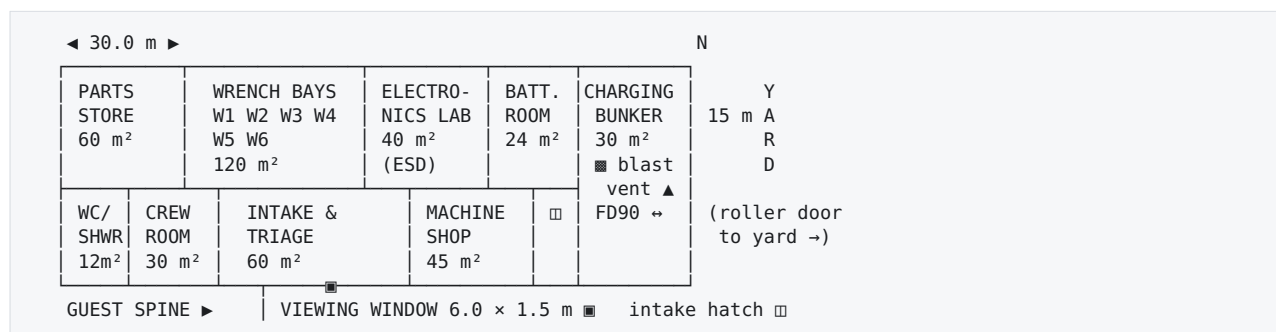
One shared plant room between the two ponds (Volume 6) — a 20 ft high-cube container beside the marina hut on a 100 mm bunded slab — serving two hydraulically separate loops:

Item	Specification
Harbour loop	Turnover every 4 h → duty flow 62.5 m³/h ; Volume 6's worked sizing: ≈ 5.9 m total dynamic head → 1.5 kW low-head pump, preferably 2 × 0.75 kW for redundancy and half-flow night/winter economy
Speed-pond loop	Turnover every 8 h → ≈ 100 m³/h , sized by the same method; shares the plant room, never the water
Filtration	Drum/bead mechanical filters both loops; UV clarifier loop on the harbour pond (the viewing-bay water passes the UV bank every cycle — clarity is the submarine product); total pump connected load ≈ 1.2–1.8 kW (Volume 6); backwash discharges to sewer, not the swales [verify locally]
Intakes	Floating-weir skimmers: 4 on the harbour pond, 6 on the speed pond , plus one bottom drain per silt sump; isolation valves inside the container
Top-up	Metered mains top-up through an air-gap break tank (backflow prevention [verify locally]); summer evaporation budget 100–150 mm/month across both ponds ≈ 25–30 m ³ (Volume 6)
Electrical	Fed from the marine sub-board; pumps on the essential-services bus (Chapter 12) so circulation survives a park power event
Access	Double doors to a 2 m hard apron; sumps and strainers serviceable by one Artisan without entering the water

6. The Works — Workshop Building

The Works is the park's engine room and, deliberately, one of its attractions: the building is placed on the guest spine with a viewing window so that guests watch Artisans rebuild machines — Volume 7's craft made visible. It is a single-storey steel portal-frame building, **450 m² GFA (30.0 × 15.0 m)**, eaves 4.2 m, on the eastern service spine between the Mining Zone and the airfield parcel, with a fenced 450 m² work yard behind (precinct total 900 m², allocation line 9).

6.1 Floor plan and room schedule



Room	Area (m²)	Function & key requirements
Intake & triage	60	Fleet reception from all zones via the intake hatch; QR scan-in to maintenance_logs ; 4 triage benches; blow-down booth (dust extraction) at the door
Wrench bays W1–W6	120	Six 2-post benches 2.0 × 0.9 m at 900 mm height; tool walls; the two bays nearest the window are the “show bays”
Electronics lab	40	ESD-floored and earthed; RCW Node assembly/repair, soldering stations with fume extraction, conformal-coating vent hood (Volume 7)
Battery room	24	LiPo storage: steel racking, packs at storage charge in ammo-can class boxes; ventilated; no charging in this room — storage and logistics only
Charging bunker	30	Per doctrine — see §6.2
Machine shop	45	Pillar drill, bench grinder, small lathe, parts washer; machine guarding per code [verify locally]
Parts store	60	High-turnover spares per Volume 8’s inventory: boltless racking, small-parts carousel, issue counter to the wrench bays
Crew room	30	12 lockers, kitchenette, briefing wall (daily fleet status from RC WORLD OS)
WC/shower	12	Staff sanitary + PPE wash [verify locally] for fixture counts
Circulation & walls	29	1.5 m main aisle, trolley-width throughout
Total GFA	450	

The viewing window: 6.0 × 1.5 m laminated glazing at 900 mm sill on the guest spine facade, looking into wrench bays W1–W2 and the intake floor. One-way sound (guests hear a mic’d bay feed via a push-button speaker, Artisans hear nothing), interpretive graphics naming the tools, and a live RC WORLD OS fleet-status screen beside it. The window converts maintenance from cost into content; Volume 9 builds a customer journey around it.

6.2 The charging bunker

The bunker implements the canonical battery doctrine (bunkered charging; 3:1 battery:vehicle ratio ⇒ ~450 packs cycling in Phase 1; LiPo window 3.4–4.2 V/cell) as architecture:

Parameter	Specification
Construction	Fully filled cinderblock (CMU) walls, 190 mm , to the underside of a reinforced-concrete lid slab; internal 30 m² (6.0 × 5.0 m); no combustible linings; concrete floor with 50 mm containment sill at the door

Parameter	Specification
Blast/fire venting	Deliberate weak panel venting upward and outward to the yard: two 600 × 600 mm blow-out panels high in the yard-side wall, lightweight sacrificial construction, discharging to a fenced 3 m exclusion strip — never toward an occupied room or guest area
Door	Single fire-rated door (90-minute class, FD90) with self-closer , opening outward to the corridor; signage per fire code [verify locally]
Charging plant	Industrial multi-port smart balance chargers (SkyRC T1000 / ISDT K4 class, per doctrine) on steel racking; each pack charges in an individual steel/ceramic tray; LiPo-sack policy for any suspect pack
Electrical	Dedicated sub-board (Chapter 7 sizing); every rack circuit RCD-protected; master EPO mushroom at the door; a timed master contactor de-energizes all charge circuits outside staffed hours
Detection & suppression	Smoke + heat detection zoned to the park panel; no water sprinkler over lithium — Class D/ABC extinguishers at door plus a sand/vermiculite quench bin and lifting tongs; ventilation ≥ 6 air changes/hour extracting at high level [verify locally]
Monitoring	Temperature and smoke telemetry into RC WORLD OS; charger fleet on networked monitoring; CCTV inside (Chapter 13)

Safety Warning — The bunker’s safety case rests on three assumptions that operations must never erode: (1) charging happens **only** in the bunker and the Chapter 7 customer charging station — never at pit bays, never in the battery room; the sole documented exception is Volume 5’s supervised 1S whoop board at the Arcade desk (max 6 packs, fireproof mat, staff eyes); (2) packs outside the 3.4–4.2 V/cell window or with any physical damage are quarantined in the sand bin, not racked; (3) the blow-out panels’ exclusion strip stays fenced and empty — it is not storage. All three appear on the daily close-down checklist (Volume 7).

6.3 MEP requirements

System	Requirement
Power	The Works sub-board: 3-phase, sized in Chapter 12 (≈ 45 kVA connected); bunker on its own metered sub-board; ESD lab on clean earth; welding outlet (32 A 3-ph) in machine shop
Lighting	500 lux maintained at benches, 750 lux task at electronics, 300 lux general; high-CRI (≥ 90) at the show bays — they are on camera all day
Ventilation	Solder-fume extraction (lab), dust extraction (blow-down booth, machine shop), bunker extract per §6.2; general ventilation per occupancy code [verify locally]
Compressed air	10-bar compressor in the yard lean-to; ring main with drops at all wrench bays and intake (blow-down)
Water	Sink at intake (parts washing, silt interceptor to drain), crew-room kitchenette, safety shower/eyewash at lab and bunker corridor [verify locally]

System	Requirement
Data	24-port switch; wired drops at every bench (telemetry flashing, diagnostics), lab and stores; Wi-Fi AP; the building also houses the park comms room (Chapter 12) in a rack cabinet at the electronics lab
Fire	Detection throughout; extinguisher schedule per Chapter 13; the bunker's specifics per §6.2

7. Pit Lane & Charging Station

Chapter 3.5 dimensioned the trackside working pit lane. This chapter is the customer-facing pit theatre and the park's second charging node — the operational heart of the Shift model, where every Operator Shift's battery swap (the “refueling” moment, canon) happens in public view, and where the park's highest continuous electrical load lives. Precinct: 450 m² (allocation line 6) between Track A and the guest spine.

7.1 Pit theatre layout

Element	Dimension	Function
Swap counters	8 counters, 1.5 m wide × 0.75 m deep at 950 mm height, in a 12 m colonnade facing the guest spine	The stage: customer hands machine/transmitter over, Artisan swaps pack, RC WORLD OS logs the swap; over-counter task light and a lap-timer style swap clock per counter (theatre)
Transmitter racks	Two lockable racks, 48 slots each, behind the counters	Charged, sanitized, class-tagged transmitters; QR on each slot binds to vehicle at issue (Volume 3/OS doctrine)
Queue	Serpentine, 1.2 m lane width, capacity 40 guests, under a full shade canopy, with the swap counters and Track A both visible from every point in the queue	Watching swaps is the queue entertainment; live queue state feeds the app's <code>queue_roster</code> (canon)
Staff spine	2.0 m corridor behind counters connecting to the trackside pit lane and the charging room	Fleet and packs move behind, never through, guests
Buffer racking	Vehicle staging racks for 30 machines	The ready line for the next Shifts

7.2 Charging station equipment room

A 30 m² masonry room at the pit building's north end, built to the same doctrine as the Works bunker (fully filled CMU, FD90 door, blow-out panel to a fenced service strip, no sprinkler over lithium, EPO at door): the park's second bunker in all but name. Equipment: 10 multi-port smart balance chargers (T1000/K4 class), each serving 4 channels, on two steel racks with per-pack trays; storage racking for 200 packs in rotation; pack-triage bench with cell checker and IR meter.

7.3 Electrical load — worked example

Circuit sizing for the charging room, as a worked example the MEP engineer scales for the Works bunker (this method, with local code factors, is the deliverable — the numbers are the reference case) **[verify locally]**:

1. **Pack energy.** Reference pack: 2S 7.4 V nominal $\times$ 5 Ah = **37 Wh** (3S 11.1 V $\times$ 5 Ah = 55.5 Wh). Fleet mix $\approx$ 70/30 2S/3S $\rightarrow$ weighted $\approx$ 42.6 Wh/pack.
2. **Charge power per channel.** Doctrine charge rate 1C (5 A) at the 4.2 V/cell termination voltage with charger losses ($\eta \approx 0.85$): 2S channel $\approx$ (8.4 V $\times$ 5 A)/0.85 $\approx$ **49 W**; 3S $\approx$ (12.6 $\times$ 5)/0.85 $\approx$ **74 W**. Weighted $\approx$ 57 W/channel.
3. **Connected load.** 10 chargers $\times$ 4 channels $\times$ 57 W $\approx$ **2.3 kW** charging + charger overhead/fans $\approx$ 0.3 kW + room services (lighting, extract fan, monitoring) $\approx$ 0.8 kW $\rightarrow$ **$\approx$ 3.4 kW connected.**
4. **Demand.** All channels genuinely run together at peak turnover (diversity 1.0 on charge circuits — never diversify a battery room), room services diversity 0.8 $\rightarrow$ **design demand $\approx$ 3.2 kW $\approx$ 3.5 kVA at PF 0.95.** Note the packs arrive $\sim$ 70% depleted at most (the canonical $\sim$ 30% buffer returns with every Casual Shift), so the 1C session runs $\approx$ 50 minutes including the CV tail — the number step 6 uses.
5. **Circuits.** Two 20 A radial circuits per rack (5 chargers/rack, $\leq$ 1.5 kW/circuit — 50% loaded, cool and expandable), one 16 A services circuit, all 30 mA RCD-protected; sub-board 3-phase 32 A/phase with 40% spare ways. The Works bunker, with $\sim$ 2.5 $\times$ the channel count for the 450-pack fleet rotation, scales to $\approx$ 9 kVA by the same method.
6. **Throughput check.** 40 channels $\times$ $\sim$ 50 min/pack $\Rightarrow$ $\approx$ 48 packs/hour through the customer station — comfortably ahead of peak swap demand (8 counters $\times$ $\sim$ 2 swaps/counter/Shift-block $\approx$ 48 packs/hour at full house), which is the point: the electrical design and the Shift model are the same calculation.

7.4 Fire suppression & controls

As §6.2: detection zoned to the park panel; extinguishers (ABC + Class D capability) both sides of the door; sand quench bin and tongs; thermal camera watchpoint over the racks feeding an RC WORLD OS alert; timed master contactor kills all charge power 30 minutes after close; the counters' public side needs nothing special — no pack is ever charged, opened or stored on the customer side of the wall. Signage: "LiPo CHARGING — AUTHORIZED STAFF ONLY", door held shut, never wedged.

8. Guest Buildings

8.1 Entry pavilion

The entry pavilion is the park's front door, wallet and classroom: **490 m² GFA**, single storey, steel frame with corten and glazed facades, fronting the arrival plaza at the guest gateway. Everything a guest must do before their first Shift — ticket, waiver, Toolbox Talk, transmitter familiarization — happens here, so the building is sized for the peak arrival pulse, not the daily average.

Room	Area (m ²)	Specification
Entrance hall & queue	90	4 POS positions + 4 self-service kiosks (RC WORLD OS bookings/wallet); queue capacity 60 inside + 60 under plaza canopy; gate line with QR scan
Retail	120	Park-branded and hobby retail (Volume 9 owns range); 2 POS, lockable after-hours shutter to the hall so retail trades on event nights independently
Briefing room (Toolbox Talk)	60	40 seats, projection wall, transmitter demo bench; the digital Toolbox Talk induction (canon) is completed in-app, but first-timers and groups get the live version here; doubles as party/school room off-peak
F&B kiosk	35	Phase 1 catering: counter service, coffee, packaged food; 3-phase supply and extract stub sized for a future fit-out upgrade
Guest toilets	70	Sized to peak-day load (\$8.3): indicative 6 F + 4 M + urinals + 2 accessible + 2 family rooms with change tables [verify locally — fixture counts are code-driven]
First-aid room	12	Per Chapter 13; street-accessible for ambulance handover
Staff & back-of-house	45	Cash office (Volume 10 controls), comms cabinet, cleaners' store, retail stock
Circulation & walls	58	—
Total	490	

The plaza (part of precinct 1) carries the corten RC WORLD gateway totem, the park map, shade trees, and the “first machine in 60 seconds” sight line: from the gate line, guests see straight up the spine to Track A’s main straight.

Sizing check against the arrival pulse. The pavilion is sized by the worst 30 minutes, not the day: on a Phase 1 peak day (~280 visitors), booking data from comparable timed-entry venues puts ~45% of arrivals inside one 30-minute pulse $\approx$ 126 guests. At 4 staffed POS $\times$ 90 s per group of 2.8 plus 4 kiosks $\times$ 60 s, gate throughput $\approx$ 300 persons/30 min — the pulse clears with half the positions down. The queue reservoir (60 inside + 60 canopy) holds the pulse entirely even at half throughput, which is the real design case (one trainee on POS, one kiosk faulted, it is raining). The briefing room’s 40 seats turn over on the 20-minute Shift clock by design: a Toolbox Talk cohort empties into the park exactly as the next Shift block opens, so first-timers arrive at the pit theatre in phase with the operating rhythm rather than against it.

8.2 Restaurant (Phase 2)

A 380 m² GFA pavilion sited between the harbour pond and the guest spine, with a dual-aspect terrace: the primary, west-facing front **overlooks the working harbour** — Volume

6's fixed siting requirement, because the terrace converts the port's continuous slow spectacle (tugs, barges, the fire-boat shows) directly into F&B dwell — while the east front catches Track A's T5 sweeper and main straight across the spine. The table with a view of a working machine is the product on both fronts.

Element	Specification
Covers	120 inside + 60 terrace = 180 at full service; sized against Phase 2 design-day dwell (Volume 10 F&B model)
Dining room	200 m ² ($\approx$ 1.65 m ² /cover incl. internal circulation)
Kitchen line	90 m ² : hot line (6 m cook suite), cold prep, pass, ware-wash, dry/cold stores (12 m ² walk-in), staff WC; grease interceptor and kitchen extract to roof [verify locally]
Servery/bar	30 m ² counter with direct terrace hatch
Guest WCs	40 m ² (supplements pavilion counts for the northern park)
Terrace	220 m ² external at +450 mm above spine level (sight line over standing spectators to the track), glass balustrade 1.1 m, shade sails, evening heaters conduit for Phase 3 night racing
Circulation/walls	20 m ²

8.3 Sanitary provision sized to peak load

Design population (Volume 2 demand model): Phase 1 design-day $\approx$ 140 simultaneous guests + 25 staff; Phase 2 $\approx$ 220 + 35; Phase 3 events $\approx$ 400. Provision strategy: the pavilion carries the Phase 1 load entirely; the restaurant block and a plumbed toilet pod at the grandstand carry Phase 2 growth; the arena carries its own event load. Family rooms (2 in the pavilion, 1 at the grandstand pod) reflect the park's family skew — stroller-parking bays beside every toilet block. All counts above are indicative for space-planning: **final fixture schedules come from the governing sanitary code applied to these populations [verify locally]**.

9. Viewing Areas & Circulation

9.1 Spectating as designed program

Roughly one companion watches for every driver on-station (Volume 2's visit-group data), so at Phase 2 design-day the park holds $\sim$ 110 spectators who are the marketing department's unpaid film crew. The hierarchy:

Viewing asset	Phase	Capacity	Specification
Grandstand at Track A	2	200 seated	5 tiers × 40; steel frame, timber seats, 400 mm row rise / 800 mm going (C-value sight-line check over a standing child at the rail — RC spectating looks down, so generous rise matters more than at a stadium); full shade canopy; front row +1.2 m above track datum behind the two-stage barrier; accessible platform for 6 wheelchairs + companions at tier 1 [verify locally]
Elevated mining walkway	1	60 standing	1.8 m wide × 30 m long at +2.4 m above pit datum, crossing the Mining Zone's west end; steel structure, mesh balustrade 1.3 m with kick-plate (elevated = taller rail), anti-drop mesh to 100 mm aperture full height — nothing drop-pable reaches the pit; ramped access 1:14 both ends; the park's signature photo spot, looking down the haul-road loop to the hopper
Viewing mounds (berm tops)	1	~150 standing total	The Chapter 11 berms double as terraced viewing: sleeper-edged terraces cut into the south faces at Track B and the airfield promenade
Spectator terraces	1	continuous	The 2 m terrace behind mining's operator line (Volume 4), Track C boardwalk widenings, the harbour's canopied quay and viewing-window bay (Volume 6) with the restaurant terrace above them
Airfield promenade	2	80 standing	60 × 6 m along the cage's south face, viewing through the 25 mm mesh (no openable panels on public faces — Volume 5), standoff lean rail 2 m outside the netting, interpretive graphics, FPV repeater screens set at the rail so crowds gather at the screens, not against the net

9.2 Circulation network

- **Main spine:** 3.0 m compacted stone with concrete bands at building thresholds, entry plaza → Track A/pit theatre → mining/agriculture junction → northern zones. 3.0 m passes two strollers plus a counter-flow group and admits the smaller utility vehicle for out-of-hours service.
- **Secondary loops:** 2.0 m paths to each zone's viewing edge; 1.8 m boardwalk in Track C; all gradients $\leq 1:20$ preferred, 1:14 max with landings **[verify locally]** for accessibility code.
- **One-way flow at peak:** the spine and the mining walkway are the two pinch candidates. At >150 guests on site (RC WORLD OS occupancy counter), duty management activates the posted one-way scheme: spine flows clockwise (north on the west side, south on the east), the mining walkway becomes west→east one-way, and the grandstand stair pair splits up/down. The scheme is signed with flip-panels at five decision points — no temporary barriers needed.
- **Capacity check:** at the leisure-comfortable planning rate of ~25 persons/min per metre of clear path width (well below crush thresholds — this is a strolling park, not a stadium concourse), the 3.0 m spine passes ≈ 75 persons/min two-way and roughly the same one-way with the counter-flow removed; the whole Phase 2 design-day population could traverse it in under four minutes. The binding constraint is never the path — it is stationary people: a crowd three deep at Track A's barrier narrows the adjacent spine to 1.2 m. Hence the standing rule for detailed design: **every barrier where crowds predictably form gets a 2 m spectator layby off the path edge** (built as paving band, drawn on the layout plans), so watchers stand out of the stream. The mining walkway (1.8 m) is capacity-managed instead: a marshal-clicked or camera-counted cap of 60, queue at the west ramp, one-way at peak.
- **Queue capacities (designed, not residual):** pit theatre 40 (\$7.1); entry 120 (\$8.1); mining operator wait 24 under the rim sails; airfield vestibule 12; restaurant wait 20 on terrace stair. Every queue has shade, sight of the action, and a posted "next Shift" screen fed by RC WORLD OS.

9.3 Wayfinding and themed naming

One family of corten totems (1.8 m blade, laser-cut lettering, zone pictogram, painted interior faces in zone colours) at every decision point; fingerposts at path junctions; zone gateways as scaffold-and-corten portals with the zone name in DIN stencil. Naming is industrial-diegetic and consistent across map, app and signage — the park reads as a worksite you've been badged into:

Asset	Public name
Main spine	The Haul Road
Entry plaza	The Gatehouse
Pit theatre	The Pits
Mining walkway	The Gantry
Grandstand	The Stand
Events lawn	The Yard
Restaurant	The Canteen (Phase 2)

Asset	Public name
Track C	The Trailhead
Marine precinct	The Docks (Phase 2)
Airfield	The Strip (Phase 2)

Field Note — Wayfinding failure shows up as staff interruptions: every “excuse me, where’s the crawler track?” is a sign in the wrong place. Budget a post-opening signage snag round at Month 2 — walk the park behind first-time visitors and put totems where they hesitated, not where the drawing looked balanced.

10. Parking & Access

10.1 Demand model and bay counts

From Volume 2’s peak-day model (≈ 220 – 300 visitors on Phase 1 peaks, ~ 400 at Phase 2 events): average vehicle occupancy 2.8 (family skew), $\sim 78\%$ arriving by car, dwell 3.5 h with two arrival pulses. Required peak accumulation ≈ 84 cars (Phase 1) / 122 (Phase 2 event day).

Bay type	Phase 1	Phase 2 build-out	Dimensions
Standard car	80	120	2.5 × 5.0 m, aisles 6.0 m, 90°
Accessible	5	8	3.6 × 5.0 m adjacent to plaza, level route [verify locally — ratio is code-driven]
Family priority	3	4	3.2 × 5.0 m, next band out from accessible
Coach/bus	2	4	3.5 × 15 m, drive-through — school coaches never reverse
Motorcycle/cycle	10 + 20 racks	16 + 30	at the plaza edge, covered cycle racks
Total car bays	88	132	surface, permeable gravel-grid on the outer bands (Chapter 11)

Worked accumulation model (Phase 1 peak day, 280 visitors). Arrivals split across two pulses (10:00–11:30 $\approx 55\%$, 13:00–14:30 $\approx 35\%$, trickle 10%); car share 78% at 2.8 occupancy = 78 cars/day. With 3.5 h mean dwell, peak accumulation occurs $\approx 13:30$ when the morning pulse has not yet fully departed: accumulation = $(0.55 \times 78 \times 0.85 \text{ retained}) + (0.35 \times 78 \times 0.60 \text{ arrived}) \approx 36 + 16 \approx 53$ cars typical peak, **84 at the 95th-percentile day** (longer dwell + event overlay). Phase 1’s 88 bays therefore run at $\approx 60\%$ typical / $\approx 95\%$ design-day — full is acceptable twice a month, queuing onto the public road is never acceptable, hence the drop-off loop is outside the parking circulation (a full car park must not block drop-off) and the L2 overflow protocol below exists from day one. The same model at Phase 2 event days (400 visitors, 4.5 h dwell) yields ≈ 122 cars against 132 bays. These coefficients

(car share, occupancy, dwell, pulse shape) are restated as a live tab in Volume 10's model — re-run them with real RC WORLD OS visit data at Month 6 and let the Phase 2 parking extension be sized by evidence, not by this page.

Overflow for Phase 3 events: the land-bank parcel L2 (Chapter 14) is graded and gated as marshalled grass overflow (~80 cars) before any permanent expansion is considered.

10.2 Drop-off and access geometry

- **Drop-off loop:** one-way counterclockwise loop off the frontage at the plaza, 6.0 m lane, 5 pull-in bays + 1 accessible, no parking, 2-minute rule signed; coach drop-off on the loop's outer edge with a 20 m level footway to the gate.
- **Guest access:** single signalized/priority junction per highway authority requirements **[verify locally]**; sight-distance splays kept clear in the landscape plan.
- **Service route:** a separate gated entry 90 m east of the guest junction feeds the eastern service road directly to The Works' yard, the airfield gate and the arena dock — **delivery vehicles never enter guest circulation**. Design vehicle: 12 m rigid truck (net poles and pond liner arrive on articulated vehicles twice ever — those movements are managed deliveries on closed days).
- **Emergency access:** two fire-appliance routes — the frontage/plaza and the eastern service road — plus the northern staff route (§4.3) giving a 4.5 m-wide hard route to within 45 m of every structure; gates on the emergency routes take the local brigade padlock/key system **[verify locally]**.

11. Landscaping & Environment

11.1 Berms: noise control that sells tickets

The park's earthworks balance goes into perimeter berms — spoil from the pond, pit and track excavations becomes the landscape (target: zero soil export):

Berm	Location	Profile	Functions
North berm	Full northern boundary, ~180 m	h = 3.0 m, crest 2 m wide, guest face 1:3, boundary face 1:2	Acoustic screen for the loud band (mining, Track B, airfield); wind shelter; terraced viewing on the south face at Track B
West berm	West boundary at the ponds, ~90 m	h = 3.0 m at the harbour end, tapering to ≤ 1 m with low planting along the sail area's windward shore; 2.5–3 m evergreen shelter belt windward of the harbour only	The harbour wind shelter Volume 6 prices as revenue — solid at the harbour, deliberately low and open where the sail fleet needs clean wind (Volume 6, §4.2/§6.6); afternoon shade for Agriculture
Frontage berm	South, broken at gates	h = 1.5 m, planted	Visual dress to the road; keeps the cage and gate-way visible above it by design

Acoustic performance to be verified by the consultant against the Chapter 2.3 targets **[verify locally]**; as a planning rule of thumb a 3 m berm breaking line-of-sight yields ~5–8 dB attenuation at the boundary — the gradient plan does the rest.

11.2 Planting palette

Three rules govern species selection (local ecologist adapts the list **[verify locally]**): drought-tolerant once established (irrigation is for turf and hedges only), no thorn species within reach of guest paths, and — critically — **low-FOD near the airfield**: within 30 m of the cage, no species that sheds cones, large seed pods, catkins or twiggy debris that ends up on the runway or in the nets. Evergreen, small-leaf, deep-rooting species (holm-oak/olive/pit-tosporum types in temperate-dry climates) around the cage; deciduous shade trees are welcome at the plaza, restaurant and southern spine where their leaf-fall lands on sweepable paving; ornamental grasses and gravel-garden planting dress the berms (industrial-authentic: the palette of a rehabilitated quarry, not a resort).

11.3 Dust suppression

Dust is the operating cost of a dirt park: Track B's watering ring (§3.2), mining's nightly wet-grooming (Volume 4), the northern service road dressed in crusher fines with a calcium-chloride/lignin binder refreshed each season, and hose points at 40 m spacing on the irrigation main. Wind-blown dust from the loose zones is intercepted by the berm-and-hedge line before it reaches the ponds (skimmer load and viewing-bay clarity) or the airfield (runway grip). The morning "hose-and-broom reset" (design tenet 4, §1.2) is a designed capability: every zone has a washdown point, falls, and a silt trap.

11.4 Stormwater and drainage strategy

The 2–4% SW grade organizes the whole strategy: a network of grass swales along the path edges collects zone runoff (each zone discharging through its own silt trap — mining's fines and Track B's clay never reach the network raw) and feeds the **SW detention basin** (~600 m³, side slopes 1:4, max design depth 600 mm, dry between storms, planted as a rain garden) which throttles discharge to the municipal system or watercourse at the permitted rate **[verify locally]**. Hard-surface ratio at build-out ≈ 31% (Chapter 2.1); car-park outer bands in permeable grid pavers reduce the peak. The pond is hydraulically independent (lined, top-up metered); its overflow weir discharges to the swale network. Design storm and discharge consent are jurisdictional **[verify locally]** — the layout reserves the basin footprint for the wettest plausible case.

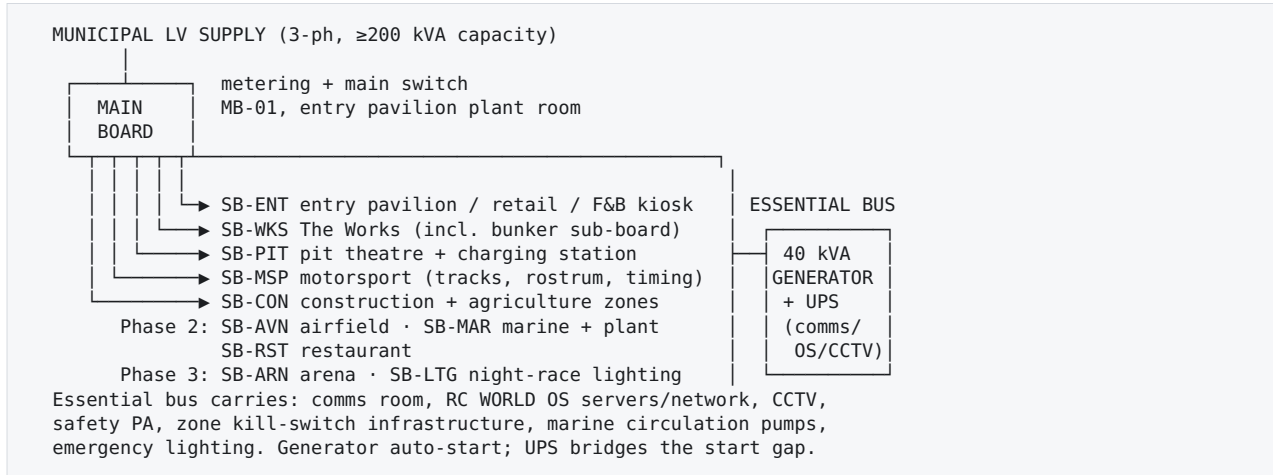
11.5 Shade strategy

Shade is a throughput input: dwell collapses on unshaded summer sites. Layered provision — engineered sails over every operator station, rostrum and queue (§§3–7); canopy trees at plaza, lawn and spine seating; built canopies at grandstand and spectator quay; and the buildings' own deep eaves on southern facades. Target: at 14:00 on the summer design day, **≥ 60% of all designed guest positions (stations, seats, queues) are in shade**. Sails come down for the storm season per the maintenance calendar (wind rating on every sail post is a structural item **[verify locally]**).

12. Utilities & Infrastructure

12.1 Electrical: single-line concept

The park takes a municipal three-phase LV supply at the frontage (upgradeable connection ≥ 200 kVA — a site-selection ● item). Concept single line:



Design intends the MEP engineer must preserve: each zone's sub-board is in its zone (fault isolation per zone; a track fault never darkens the pit); both charging rooms are separately metered sub-boards with timed master contactors (§6.2, §7.4); the essential bus keeps **RC WORLD OS, the network, CCTV, PA and the telemetry kill-switch chain alive through any supply event** — the safety system's power is never hostage to the park's; and every sub-board carries $\geq 30\%$ spare ways (this park adds circuits every season of its life).

12.2 Site-wide load schedule

Board	Serves	Connected (kVA)	Demand est. (kVA)	Phase
SB-ENT	Pavilion, retail, F&B kiosk, plaza	40	28	1
SB-WKS	The Works incl. bunker (≈ 9) + compressor	45	30	1
SB-PIT	Pit theatre + charging station (≈ 3.5)	12	8	1
SB-MSP	Tracks A/B/C: timing, PA, rostrum power, watering pump	15	8	1
SB-CON	Mining (hopper, conveyor) + agriculture + service route	12	7	1
Phase 1 total		124	≈ 81	
SB-AVN	Airfield: gate interlocks, PA, flight-line container, promenade	8	5	2

Board	Serves	Connected (kVA)	Demand est. (kVA)	Phase
SB-MAR	Marine plant (pumps, UV), marina, quay	18	12	2
SB-RST	Restaurant kitchen + dining	60	38	2
Phase 2 total		210	≈ 136	
SB-ARN	Arena, academy, corporate centre	80	45	3
SB-LTG	Night-racing flood-lighting (Tracks A+B)	30	26	3
Phase 3 total		320	≈ 207	

The Phase 1 connection is sized for Phase 2 (≈ 140 kVA demand); the Phase 3 step (arena + lighting) triggers the connection upgrade priced in Volume 10's Phase 3 capex. Diversity assumptions above are planning-grade; the MEP engineer recalculates to the local method **[verify locally]**.

12.3 Water

- **Mains:** municipal potable connection at the frontage; meter + backflow prevention **[verify locally]**; potable distribution to pavilion, Works, restaurant, first-aid, hose-union points.
- **Pond top-up:** metered branch with air-gap break tank at the marine plant room (\$5.5); budget summer evaporation of 100–150 mm/month across both ponds $\approx 25\text{--}30$ m³/month (Volume 6) plus splash losses — the meter reading is a Volume 6 checklist item.
- **Irrigation & dust suppression:** a dedicated non-potable ring main (fed from mains in Phase 1; a rainwater-harvest tank at The Works' roof is the Phase 2 upgrade) serving Track B's watering ring, the hose points at 40 m spacing, hedge drip lines and the turf overruns.
- **Sewer:** pavilion, Works and restaurant to the municipal sewer (or approved on-site treatment — site-selection item); kitchen via grease interceptor; workshop intake sink via silt/oil interceptor; pond backwash to sewer, never to the swales **[verify locally]**.

12.4 Data: the park is a radio site

RC WORLD OS and the telemetry chain make connectivity a life-safety utility (the kill-switch rides on it — canon). Per source doctrine: a **TP-Link Omada-class outdoor Wi-Fi mesh** blankets the park so ESP32 2.4 GHz Nodes hand off between APs without dropping telemetry.

Element	Specification
Comms room	Rack cabinet in The Works' electronics lab (\$6.3): router/firewall, PoE switches, OS edge server, UPS (essential bus); a second distribution cabinet at the entry pavilion

Element	Specification
AP plan	12 outdoor PoE APs Phase 1 on 4–6 m poles or building facades: 2 × Track A, 1 × Track B, 1 × Track C, 2 × Mining (rim + hopper), 1 × Agriculture, 2 × pit/spine, 1 × plaza, 2 × The Works/yard. Phase 2 adds 5 (cage corners ×2, promenade, marina, restaurant); Phase 3 adds arena indoor set. Design target: –65 dBm or better at machine height (0.3 m) over 100% of operating surfaces , validated by a post-build survey at machine height, not head height
Backbone	Single-mode fibre in a 100 mm duct ring following the spine and service road, drawn at Phase 1 full-length (including Phase 2/3 stubs); APs on Cat6A from zone poles
Segregation	Three SSIDs/VLANs: telemetry+ops (Nodes, kill-switch, timing), staff (OS clients, POS, CCTV), guest (captive portal); telemetry VLAN has priority QoS — a full guest Wi-Fi never delays a KILL command
Resilience	Comms room + all APs' PoE switches on the essential bus; 4G/5G failover router for payments and OS cloud sync

Duct schedule (Phase 1 civils — the cheapest insurance in the project):

Route	Ducts	Serves
Frontage → pavilion plant room	2 × 100 + 2 × 50 mm	Incoming supply, incoming telecom, spares
Pavilion → The Works (spine route)	2 × 100 + 4 × 50 mm	Fibre ring leg, LV interlink, CCTV/PA, spares
The Works → airfield parcel (service road)	2 × 100 + 2 × 50 mm	Phase 2 SB-AVN feed, fibre, cage systems
Spine → marine parcel	2 × 100 + 2 × 50 mm	Phase 2 SB-MAR feed, plant controls, fibre
Track A perimeter ring	2 × 50 mm + loop conduits	Timing, cameras, Phase 3 floodlighting
Spine → arena parcel (SE)	2 × 100 + 2 × 50 mm	Phase 3 SB-ARN feed, fibre
Zone pole positions (all)	1 × 50 mm each from nearest chamber	AP + CCTV + PA co-mounted per pole

Draw pits at ≤ 50 m spacing and every change of direction; every duct proved and roped at handover (checklist item). The rule stated once and enforced always: **ducts now, trenches never** — no open-cut through a live park is ever approved for a service that this schedule could have carried.

12.5 Lighting

- **Operational (Phase 1):** path and plaza lighting to 20 lux average (guest areas after dusk), building task lighting per §6.3, statutory emergency lighting **[verify locally]**. Zones are daylight operations in Phase 1 — zone lighting is close-down/security level only.
- **Night racing (Phase 3, SB-LTG):** Track A floodlit to **300 lux maintained average on the racing line, uniformity ≥ 0.5** (Volume 3's interface figure, uprated from event experience if league TV coverage demands), Track B to 200 lux; 8–10 m poles with full-cutoff LED floods aimed by a lighting designer to keep obtrusive spill below curfew limits

at the boundary **[verify locally]** and — a park-specific constraint — **no fitting within the drivers' rostrum sight cone may be aimed toward the rostrum** (glare on a driver is a crash). Conduit and pole bases are cast in at Phase 1 track construction (Chapter 2.4 doctrine: ducts now, trenches never).

13. Security & Safety Systems

13.1 Perimeter and fencing

Line	Specification
Site perimeter	1.8 m galvanized weldmesh (358-class anti-climb on the road frontage and service gates), concealed in the berm/hedge line where it crosses guest views
Zone edges (guest/live separation)	The 1.1 m rail-and-kick-plate family (§1.1); mesh infill ≤ 100 mm aperture wherever small children queue
Pond barrier line	1.1 m post-and-rail balustrade with mesh infill wherever the public can approach water deeper than 300 mm; self-closing gates at the supervised landing stages (Volume 6 canon) [verify locally]
Charging rooms, plant, bunker strips	2.0 m mesh enclosures, locked, signed
Construction phase lines	2.0 m temporary hoarding, graphics-wrapped (the hoarding sells Phase 2)

13.2 CCTV coverage logic

Coverage is designed by incident question, not camera count — every location where the operation would ask “what happened?” gets a view: all POS and cash handling; both charging rooms (interior, thermal watchpoint per §7.4); every zone's operating surface (the same masts as the Wi-Fi APs — one pole family carries AP + camera + PA horn); all gates and interlocks; both pond waterlines (full surface coverage); the car park and drop-off; The Works' intake (asset chain-of-custody). Phase 1 $\approx$ 32 cameras, PoE, recorded 30 days at the comms room NVR (essential bus), with RC WORLD OS integration limited to event bookmarks (a kill-switch trigger or gate alarm bookmarks the relevant streams). Privacy signage and retention per local law **[verify locally]**.

Coverage group	Cameras (Ph. 1)	View standard
POS / cash office / retail	6	Identify (face-recognizable at counter)
Charging rooms + battery room	4 + 2 thermal watchpoints	Detect + thermal alarm
Zone operating surfaces (Tracks A/ B/C, mining, agriculture)	10	Observe (incident reconstruction, machine-scale)
Gates, interlocks, service road	5	Recognize (badge-holder at gate)
Car park, drop-off, plaza	5	Observe + plate capture at both vehicle gates
The Works intake & yard	2	Recognize (asset chain-of-custody)

Phase 2 adds ≈ 11 (harbour pond + submarine bay window $\times 2$, speed pond $\times 2$, marina hut, cage interior $\times 2$, promenade, restaurant, grandstand, events lawn, extension parking); Phase 3 the arena interior set. One deliberate omission: no camera is pointed at the rostrum drivers' faces at eye level — telemetry already records what their vehicles did, and the park photographs its guests for joy (Volume 9's media program), not surveillance.

13.3 Access control tied to RC WORLD OS roles

Physical access mirrors the OS role model (canon: executive, finance, HR, operations, artisan/technician, marshal, front-of-house, franchisee, customer): staff badges (RFID + QR fallback) open what the role's OS permissions imply, and nothing else.

Access zone	Roles admitted
Guest areas	All (customers via ticket QR at the gate line)
Trackside pit lane, zone service edges	Marshal, Artisan, Operations, Executive
The Works (except bunker)	Artisan, Operations, Executive
Charging bunker & charging station room	Artisan (charging-trained flag in OS) + Operations only; door controller checks the training flag
Comms room, cash office	Named individuals; dual-badge for the cash office (Volume 10 controls)
Interlocked live-zone gates	Marshal badge + zone status "safe" in OS — the gate is the physical face of the software interlock

13.4 Emergency preparedness

- **First-aid points:** the pavilion first-aid room (§8.1, ambulance-accessible) plus wall stations (AED, trauma kit, burn kit) at the pit theatre, The Works and — Phase 2 — the marina. Burn kits acknowledge the LiPo risk profile.
- **Assembly points:** two — the events lawn (north assembly) and the car-park head (south assembly), signed from every zone; choice of two means no route past a plausible incident (fire at The Works blocks neither).
- **PA:** zone horns on the AP/CCTV poles, addressable by zone from the duty office and the OS duty console; the all-park evacuation tone is generator-backed.
- **Lightning protection:** the airfield's 17 m poles are the site's air terminals whether planned or not — so plan it: the cage poles are bonded and earthed as a lightning protection system by the electrical designer **[verify locally]**, the flight program's storm rules (Volume 5) close the cage well before risk, and the same weather trigger closes both ponds (Volume 6's 30/30 water-clear protocol) and clears the mining rim's sail line. A site lightning-risk assessment is a Phase 2 design deliverable.
- **Fire zones and extinguisher schedule:** the site is zoned to the detection panel as: Z1 pavilion/retail, Z2 The Works, **Z2B bunker**, Z3 pit/charging, Z4 restaurant (Ph. 2), Z5 arena (Ph. 3), plus external plant nodes. Extinguisher family **[verify locally — types/spacing are code items]**: ABC dry powder at all plant and vehicles; CO₂ at electrical boards and comms; wet chemical at kitchens; **Class D capability + sand quench bins at both charging rooms and the battery room**; travel distance to an extinguisher ≤ 20 m back-of-house.

Safety Warning — The interlock chain (OS zone status → gate release → marshal badge) is only as safe as its failure mode. Specify every live-zone gate **fail-secure for entry, fail-safe for exit**: a network or power failure must never trap a person inside a zone, and must never release public entry into one. This requirement is written on the door-hardware schedule, tested at commissioning, and re-tested quarterly (Volume 7 maintenance matrix).

14. Future Expansion

14.1 Phase 3 indoor arena

The arena makes RC WORLD weather-proof and books-corporate: a clear-span portal-frame hall on the SE parcel, fronting the service road and frontage so events traffic never enters the park.

Parameter	Specification
Clear span	40 × 25 m (1,000 m²) column-free , 7 m clear under haunches — reconfigurable indoor track (carpet/asphalt-tile modular), drone micro-cage, or 600-seat corporate floor
Total GFA	1,600 m ² : arena + 2 RC Academy classrooms (60 m ² each, 16 students — Volume 9's curriculum), corporate suite (150 m ² divisible, catering pantry served from the restaurant kitchen), stores, WCs, plant
Floor	Power-floated slab, 5 kPa, cast-in timing-loop conduit grid at 5 m centers (any layout finds a loop)
Services	SB-ARN (Ch. 12); its own entrance plaza and 20-bay evening parking overlay
Interface	A gated link path to the park spine — arena events can buy park access as an upsell, but default to self-contained

14.2 Land-banking within the 4.8 ha

Two parcels (allocation line 19, 5,200 m² total) are held in low-cost landscape: **L1 (north-east, 2,800 m²)** beside the airfield — the natural home of a future cage extension, a second crawler landscape, or franchise-training compound; **L2 (south-east, 2,400 m²)** behind the arena — events overflow parking (§10.1) and the arena's expansion room. Both are graded, drained and duct-served at Phase 1 (the standing rule: ducts now, trenches never), planted as meadow, and protected by a one-line covenant in every future drawing issue: no permanent structure on L1/L2 without a master-plan revision through this volume.

14.3 Second-site learnings

Everything this volume fixes — the adjacency logic, the noise gradient, zone envelopes, the room schedules, the load schedule — is the reference design that Volume 12 turns into a franchise kit-of-parts: which dimensions are doctrine (haul-road 15°, the 400–600 mm pond working band with the 1.5 m submarine-bay exception, cage envelope, bunker construction, sight-line rules), which are climate-adaptable (shade ratios, indoor-hybrid variants for Gulf and monsoon markets flagged in Volume 2), and which are free (track layout within the geo-

metric rules, palette species, building expression). As-built deviations from this volume at the reference site must be logged in the drawing register with reasons — the second site inherits the reasons, not just the drawings. See Volume 12, Chapter 3.

Handover checklist (design → construction → operations)

☐

Topographic + geotechnical surveys reconciled with the Chapter 2 allocation plan; setting-out grid issued.

☐

All **[verify locally]** items dispositioned in a code-compliance register signed by the local engineer of record.

☐

Hold-point dimensions verified on setting-out: haul-road grades $\leq 15^\circ$ (built 14°), lane widths (agriculture 600 mm), pond depths in the 400–600 mm working band with the 6 × 4 m submarine bay at 1.5 m as the sole exception (Vol. 6), cage envelope 60 × 40 × 15 m, jump landing ratio $\geq 5:1$, rail heights 1.1 m / 1.3 m elevated.

☐

Cast-in items confirmed before pours: timing-loop conduits (Tracks A/B, arena grid), lighting pole bases and ducts, pond pipework, hopper holding-down bolts, duct ring with Phase 2/3 stubs.

☐

Charging bunker + charging station: construction per §6.2/§7.2 witnessed (filled CMU, blow-out panels, FD90 doors, EPO, contactor timing tested).

☐

Interlock chain commissioned: every live-zone gate tested fail-secure-in/fail-safe-out with OS zone status; results in the O&M file.

☐

Wi-Fi survey at machine height (0.3 m) shows ≥ -65 dBm on 100% of operating surfaces; kill-switch latency test passed park-wide.

☐

Net installation certified by the net engineer: pretension witness gauges recorded with breakaway margin preserved, ground seal at **zero gap ≥ 25 mm** full perimeter (Vol. 5), panel lacing per spec, mesh supplier certificates on file (25 mm main cage / 13 mm Arcade).

☐

Rostrum and operator-station sight-line checks signed (every position sees its full envelope; Track A checked bidirectionally).

☐

Drainage proved: hose-test every zone to its silt trap; detention basin outlet at consented rate.

☐

As-built drawings, O&M manuals, and the deviations-with-reasons register lodged (feeds §14.3 and Volume 12).



Signage and wayfinding snag-round scheduled for Month 2 post-opening (\$9.3).

Volume summary & cross-references

Volume 11 converts RC WORLD's operating doctrine into buildable geometry on the canonical 4.8 ha reference site. Its fixed decisions: a fully allocated **48,000 m² master plan** (Phase 1 60.4%, Phase 2 22.9%, Phase 3 + land bank 16.7%) organized by a south-to-north noise gradient with Mining beside The Works and the airfield peripheral-downwind; an **industrial-authentic** materials and naming language governed by the sight-line doctrine (see everything, touch nothing live, 1.1 m rail family, interlocked gates); **Track A** at ~180 m/3.5–4.5 m with a 58 m straight, 15 mm kerbs and two-stage barriers; **Track B** at ~160 m with 5:1 landing-ratio tabletops and 400 mm berms; a ten-feature **Track C** with 120 mm water-depth limits; the Volume 4 mining/agriculture geometry carried into civil detail (300 mm benches, 14° ramps, 4.0 m station pitch at 1.1 m rails); the Volume 5 cage (60 × 40 × 15 m, 25 mm knotless HDPE mesh with the 13 mm Whoop Arcade beside it) and the Volume 6 pond pair (harbour 30 × 18 m + speed 60 × 25 m at 400–600 mm, with the 6 × 4 m submarine bay at 1.5 m as the sole depth exception, ≈ \$212 k civil work) given their structures, liners and plant; a **450 m² Works** with a doctrine-compliant 30 m² charging bunker and a 490 m² entry pavilion with 120 m² retail; a worked charging-load method (≈ 3.5 kVA customer station, ≈ 9 kVA bunker); a site load schedule reaching ≈ 207 kVA demand at Phase 3; a 12-AP telemetry-grade Wi-Fi plan; and role-keyed access control that makes RC WORLD OS's permissions physical.

Cross-references. Volume 1 carries the master-plan summary for investors. Volume 2 supplies the demand model behind capacity, parking and catchment (Chapters 7) and the diorama-supplier channel behind zone dressing (Chapter 4). Volume 3 governs everything on wheels on Tracks A/B and the racing program this geometry serves (its Sections 9.6/10.4 are closed out by Chapter 3 here). Volume 4 is source of truth for Mining and Agriculture internals restated in Chapter 4. Volume 5 owns the cage rationale and netting spec built in Chapter 5; Volume 6 owns the waterspace doctrine behind §5.4–5.5. Volume 7 operates The Works designed in Chapter 6 and owns the maintenance regimes this fabric assumes. Volume 8 procures the materials, nets, chargers and dressing packages. Volume 9 programs the guest journey through Chapters 8–9's spaces. Volume 10 carries the capex these drawings imply and the phasing economics of Chapter 2.4. Volume 12 inherits the reference design per §14.3. Volume 13 specifies RC WORLD OS, whose zone statuses, roles and telemetry this volume wires into gates, screens and the essential bus.

Volume 12 — Franchise Manual

RC WORLD — Master Development Plan · Volume 12 of 13 **Revision:** 1.0 · **Date:** July 2026 · **Status:** Living document — bump revision on material change

Purpose of this volume. This volume defines how the RC WORLD business is replicated: the franchise strategy and its rationale, the three licensed park formats, the international expansion framework, the legal and licensing architecture, the fee structure and unit economics for both franchisee and franchisor, and the complete operating machinery of a franchise system — recruitment, site approval, training through the RC WORLD Academy, brand standards, quality audits, and franchisor support operations. It is written before the first franchise is sold, deliberately: the founding thesis of this volume is that the thirteen-volume documented system you are holding — engineering specifications, SOPs, fleet doctrine, procurement channels, RCW Node hardware, RC WORLD OS software, and the customer-experience canon — **is the franchise product**. The flagship park proves it; this volume packages it. The franchise pilot is a Phase 3 activity (months 31-60); everything here is sequenced against that canon.

Intended readers. The founder and board; the Franchise Director hired in Phase 3; investors and lenders evaluating the long-term equity story beyond a single park; prospective franchisees and master franchisees conducting due diligence; legal counsel preparing disclosure documents; the RC WORLD OS team building multi-site capability (Volume 13).

Chapters

1. Franchise Strategy & Rationale
2. Franchise Models & Formats
3. International Expansion Strategy
4. Licensing & Legal Framework
5. Franchise Economics
6. Franchisee Recruitment & Selection
7. Site Approval & Build-Out
8. Training Program — The RC WORLD Academy
9. Brand Standards
10. Quality Systems & Compliance
11. Franchise Support Operations
12. Five-Year Franchise Roadmap
13. Volume Summary & Cross-References

Legal framing for the entire volume. Franchising is one of the most heavily regulated commercial formats in the world, and the rules differ materially by country and, in federal systems, by state or province. Every legal mechanism described in this volume — disclosure documents, fee structures, termination rights, territory protections, non-competes — is a **design intention to be verified and implemented with qualified local franchise counsel** in each target jurisdiction before any offer is made. Nothing in this volume is legal advice, and no franchise may be offered on the basis of this document alone.

1. Franchise Strategy & Rationale

1.1 Why franchise at all

RC WORLD's flagship park is designed to reach roughly \$3.4 M revenue by Year 5 at a 24–28% EBITDA margin (Volume 10). That is a good business, but it is one park. The strategic question for Phase 3 and beyond is how the concept scales, and there are only three honest answers: build more owned parks, franchise, or do nothing. This volume argues for a **flagship-plus-franchise** model, for four reasons.

Capital efficiency. A full park consumes $\approx$ \$3.9 M of capex over five years and $\approx$ \$2.6 M of external funding (canonical figures, Volume 10). Owned expansion at three parks means raising and risking roughly \$8 M of additional capital while management attention is split across construction projects in different cities. Franchising moves the site capex onto the franchisee's balance sheet and converts RC WORLD's growth into a royalty and technology-fee stream that carries software-like margins once the support organization exists.

The product is already written down. Most first-time franchisors fail at the same hurdle: the founder's business runs on tacit knowledge that was never documented, so the "system" being sold does not actually exist in transferable form. RC WORLD inverts this. The Master Development Plan was commissioned from day one as an investor prospectus, operations manual, and engineering handbook. Fleet balancing methodology (Volume 3), the excavator-to-dump-truck 1:3 ratio and haul-road doctrine (Volume 4), workshop rebuild procedures and trade hacks (Volume 7), the procurement network with named suppliers and landed-cost models (Volume 8), the RC WORLD License progression (Volume 9), the financial model (Volume 10), the park design standards (Volume 11), and the RC WORLD OS platform with role-based interfaces including a dedicated **franchisee role** (Volume 13) — these are the deliverables a franchisee is paying for. The license fee buys access to a documented, versioned, living system, not a handshake and a logo.

Hard-to-copy assets travel well. The defensible core of RC WORLD is not the land or the buildings — any competitor can lease a field. It is (a) the RCW Node telemetry hardware and kill-switch architecture, (b) RC WORLD OS and its operational data, (c) the wholesale procurement relationships built in Volume 8, (d) the engineering doctrine that keeps a rental fleet of $\sim$ 150 powered assets at high availability, and (e) the customer-experience mechanics — Toolbox Talk, the Shift, the Tow-Truck Retrieval Protocol, the License system. All five are portable: they cost the franchisor almost nothing to deliver to a second site, and they are exactly the things a would-be imitator cannot buy off a shelf.

Local knowledge is genuinely local. An RC park lives or dies on land acquisition, municipal relationships, local marketing, school and corporate networks, and weekend labor mar-

kets. A franchisee who owns those relationships in Kuala Lumpur or Dubai will outperform a head-office expatriate manager, and their own capital at risk aligns incentives in a way salaries never do.

1.2 What is genuinely replicable — and what is not

Franchise systems fail when they license things that do not transfer. The table below is the honest inventory, and it drives everything in Chapters 2 and 4.

Asset	Replicable?	Transfer mechanism
Park layout & civil design standards	Yes	Volume 11 design standards + franchisor design review gates (Chapter 7)
Fleet doctrine, models, ratios (1:3, 3:1 batteries, class ladder)	Yes	Volumes 3–6 + approved-equipment list, updated quarterly
Workshop practices, spares strategy, rebuild skills	Yes, with training	Volume 7 + Artisan certification at the flagship (Chapter 8)
Procurement channel (wholesale China network)	Yes	Central procurement desk (Chapter 11); Volume 8 as reference
RCW Node hardware & kill-switch logic	Yes, controlled	Nodes supplied only by franchisor; firmware signed; no design transfer
RC WORLD OS	Yes	SaaS license, multi-site tenancy (Volume 13, Chapter on franchise support)
Customer-experience canon (Toolbox Talk, Shift, Tow-Truck Retrieval Protocol, License, Gears)	Yes	Brand standards (Chapter 9) + OS enforcement
Brand & trademarks	Yes	License grant (Chapter 4)
Flagship's specific land economics & city demographics	No	Site criteria + local adaptation, never copied blindly
Founder's personal network & improvisation	No	Must be replaced by documentation and the Academy — this is the test of the whole plan

1.3 The flagship-first proof requirement

RC WORLD will not sell a franchise on projections. The self-imposed, non-negotiable gate is:

No franchise will be offered until the flagship park has a minimum of two full years of operating data, including at least one complete seasonal cycle at Full Park scope (Phase 2), audited financials demonstrating the unit economics claimed in disclosure materials, and a documented safety record.

Sequenced against the canonical phasing — Phase 1 (months 0–12) builds the Core Park, Phase 2 (months 13–30) completes the Full Park, Phase 3 (months 31–60) includes the franchise pilot — this means the earliest a franchise agreement is signed is around month 36, and the pilot park opens around months 52–58 after its own build (Chapter 12). This discipline costs perhaps a year against an aggressive timeline and buys three things: credible disclosure numbers (a legal necessity in most regimes — Chapter 4), a functioning RC WORLD Academy with real classrooms (a canonical Phase 3 build item), and a flagship that has already survived the failure modes a franchisee will meet.

1.4 Franchise versus owned expansion — the decision framework

Franchising is the default growth mode, not the only one. The board should apply this rule set per opportunity:

- **Own** when the market is the flagship's home region (support costs near zero, brand risk highest, learning value highest), when a strategic site becomes available at exceptional terms, or when a format experiment (e.g., the first indoor arena) needs full control to iterate.
- **Franchise (single-unit or area development)** when the market is proven analogous to the flagship's, a qualified operator exists, and the support organization can service the geography (Chapter 11).
- **Master franchise** only for distant, regulated, or culturally distinct territories where a capable local partner adds more than the ~50% royalty share they cost (Chapter 3.4), and never before the domestic pilot has operated for at least a year.

Investor Note. The franchise program converts RC WORLD from a single-asset leisure operator into a system business. At maturity, each full-format franchise contributes roughly \$230–250 k per year to the franchisor in royalties, technology fees, and procurement margin against a marginal support cost of \$60–90 k (Chapter 5.3). The equity story investors are buying in the \$2.6 M flagship raise includes this option value — but the option is only exercisable if the flagship performs, which is why the two-year proof gate is written into this volume rather than left to management discretion.

2. Franchise Models & Formats

2.1 Format philosophy

The flagship is a 4.8-hectare, four-division park in a mid-size international city. Very few franchise markets will replicate those conditions exactly: urban land is scarcer and dearer, northern climates cannot run outdoor divisions year-round, and some catchments cannot support a full park's revenue requirement. The franchise system therefore offers **three engineered formats**, each a defined subset or adaptation of the flagship specification — not improvisations. A franchisee licenses a format; deviations go through the local-adaptation process (Chapter 9.5).

All three formats share the invariant core: the Shift billing unit, Toolbox Talk induction, the Tow-Truck Retrieval Protocol, the RC WORLD License and Gears programs, RCW Node telemetry on every powered asset, the bunkered charging room, the 3:1 battery ratio, electromechanical (lead-screw) rental construction fleet with hydraulics reserved for premium supervised experiences, and RC WORLD OS as the sole operating platform.

2.2 Format A — Full Park (4-5 ha)

The reference format: a licensed replica of the flagship at Full Park scope. Motorsport Division (Tracks A/B/C), Construction Division with Mining and Agriculture Zones, Aviation Division (netted airfield), Marine Division (pond complex), The Works, entry pavilion, restaurant, retail, events lawn. The franchisee builds in two compressed phases (Core Park then Full Park, typically 24–30 months to full scope rather than the flagship's 30) because the design is already proven — no experimentation budget, no layout iteration.

Fleet at maturity $\approx$ **195 powered assets** (the flagship's $\sim$ 150 Phase 1 baseline plus $\sim$ 20 aircraft and $\sim$ 24 vessels). Mature revenue potential is benchmarked at 85–100% of the flagship's Year 5 canon (\$3.4 M), i.e., **\$2.4–2.9 M by the franchise's Year 3–4 and \$2.9–3.4 M at full maturity**, depending on catchment. Suits peri-urban land in cities of 1.5 M+ metro population.

2.3 Format B — Compact Urban ($\sim$ 1.5 ha)

A dense, two-division park for expensive urban and inner-suburban land: **Motorsport Division plus Construction Division only** — Tracks A and B (Track C crawler trail in a compressed 400 m² technical course), Mining Zone, a reduced Agriculture Zone (optional; default is Mining plus crawler course), The Works at 60% flagship floor area, entry pavilion with F&B kiosk and retail counter, no restaurant, no aviation, no marine. This is deliberately the flagship's proven Phase 1 "Core Park" scope, densified onto 1.2–1.8 ha.

The compact format trades breadth for footfall: an urban catchment of 1 M+ within 30 minutes supports higher utilization per asset than the flagship's peri-urban site. Fleet $\approx$ **115–125 powered assets** (motorsport 64, construction and mining $\sim$ 30, crawlers 16, recovery 4, plus utility). Single-phase build, 9–12 months. Mature revenue potential **\$1.6–1.9 M** — above the flagship's \$1.28 M Year 1 (which was a ramp year at the same scope) because the format assumes a denser catchment and opens with the benefit of every lesson the flagship learned.

2.4 Format C — Indoor Arena (cold climates)

For markets where an outdoor season shorter than $\sim$ 7 months breaks the outdoor formats' economics (Scandinavia, Canada, northern US, parts of Central Europe) — and equally viable in extreme-heat markets as a climate-controlled variant (Chapter 3.3). A leased industrial building of **6,000–9,000 m²** with 6 m+ clear height houses: an indoor asphalt/drift circuit (Track A specification, Volume 3 Chapter 10 indoor annex), an indoor off-road course on engineered soil trays, a Mining Zone under dust management, a crawler course exploiting vertical structures, The Works, charging bunker, F&B kiosk, party rooms. No aviation (ceiling and netting constraints limit it to micro/indoor classes as an optional premium module), no marine beyond an optional 100 m² shallow pool module.

Fleet $\approx$ **90–110 powered assets**. The format's structural advantage is **weather-independent, 12-month utilization and evening trade** (indoor lighting is cheap; the flagship needed Phase 3 night-racing lighting to unlock evenings). Its structural burdens are rent, HVAC, and dust control. Mature revenue potential **\$1.2–1.6 M**.

2.5 Format comparison table

All monetary figures are derived proportionally from the canonical flagship baseline (Volume 10) and stated as ranges pending each site's approved business plan. Headcount is mature-state FTE, with weekend casuals excluded.

Metric	A — Full Park	B — Compact Urban	C — Indoor Arena
Land / premises	4–5 ha leased peri-urban	1.2–1.8 ha urban/inner-suburban	6,000–9,000 m ² leased industrial building, $\geq$ 6 m clear

Metric	A — Full Park	B — Compact Urban	C — Indoor Arena
Divisions	Motorsport, Construction (Mining + Agriculture), Aviation, Marine	Motorsport + Construction (Mining) only	Indoor Motorsport + Construction; optional micro-aviation & pool modules
Build program	Two phases, 24–30 months to full scope	Single phase, 9–12 months	Single phase (fit-out), 6–9 months
Capex (ex-franchise fee)	\$3.4–4.0 M phased	\$1.3–1.6 M	\$1.6–2.1 M
Initial franchise fee	\$150,000	\$90,000	\$110,000
Total cash funding requirement ¹	\$2.7–3.1 M	\$1.6–1.95 M	\$2.0–2.45 M
Fleet (powered assets, mature)	~195	115–125	90–110
Batteries (3:1 doctrine)	~585 packs	~360 packs	~310 packs
Headcount (mature FTE)	26–34	14–18	12–16
Artisans among headcount	8–10	5–6	4–5
Mature revenue potential	\$2.9–3.4 M (Yr 5+)	\$1.6–1.9 M (Yr 3+)	\$1.2–1.6 M (Yr 3+)
Pre-fee EBITDA margin at maturity	24–28%	27–30%	25–28%
Post-fee EBITDA margin at maturity	15–18%	17–19%	15–18%
Base-case payback on cash funding	Year 6–7	Late Year 6	Year 6–7
Seasonality exposure	High (mitigated by the all-weather upgrade, Section 2.6)	High	Minimal
Primary catchment requirement	≥750 k within 45 min	≥1 M within 30 min	≥600 k within 30 min

¹ Cash funding requirement = phased capex funded externally plus initial fee, pre-opening costs, and working capital; later-phase capex in Format A is partly funded from operating cash flow, mirroring the flagship's \$2.6 M funding against \$3.9 M total capex (Volume 10).

2.6 Format upgrade paths

Formats are starting points, not cages. Two upgrade paths are pre-engineered so that a successful franchisee grows inside the system rather than outside it:

- **B → A (“Compact-to-Full”).** A compact urban franchisee whose unit sustains ≥85% of Format B's mature revenue band for four consecutive quarters may apply to develop a Full Park in an adjacent territory (their protected territory extends accordingly) at a reduced initial fee of \$100,000, crediting their proven operating record against the standard discovery process. The compact unit continues operating — the formats serve different visit occasions and cannibalization at ≥20 km separation is expected to be minimal, to be verified with pilot data.
- **A + indoor module (“all-weather upgrade”).** A Full Park franchisee in a seasonal market may add an indoor arena module (a scoped subset of Format C, typically 2,500–4,000 m²) on or adjacent to their site, mirroring the flagship's own Phase 3 indoor all-weather

arena. This is treated as a design change through gates G1–G4, not a new franchise; no additional initial fee, fees continue on the enlarged revenue base.

Downgrades are not offered: a park that cannot sustain its format’s economics is a support case (Chapter 10.4), not a re-scoping exercise.

Investor Note. The compact urban format is the expected workhorse of the system: lowest absolute ticket, fastest build, and the densest revenue per hectare — and it licenses only the flagship’s Phase 1 scope, which by franchise launch will have three-plus years of operating history. The full park is the flagship-equivalent statement piece for master franchisees; the indoor arena is the climate unlock. The pilot franchise (Chapter 12) should be a Format B unless the pilot candidate’s market argues decisively otherwise.

3. International Expansion Strategy

3.1 Market prioritization framework

Markets are scored before any franchisee conversation begins, on four weighted axes plus a regulatory modifier. Scoring is maintained in RC WORLD OS as a living dataset; the table below defines the rubric.

Axis	Weight	What is measured	Data sources (category)
RC culture depth	30%	Hobby-grade RC penetration: club counts, track counts, hobby-shop density, RC e-commerce volume, event calendars	Hobby trade associations, marketplace data, club registries (2025–2026)
Disposable income	25%	Household discretionary leisure spend; propensity to pay \$15–59 per experience visit	National statistics, leisure-industry analysts
Land cost & availability	20%	Achievable lease terms for 1.5–5 ha peri-urban or 6,000+ m ² industrial	Local brokerage surveys
Climate	15%	Outdoor operating days per year; heat/cold/monsoon profile; match to format A/B/C	Meteorological normals
Regulatory ease (modifier)	10%	Franchise-law burden, foreign-ownership rules, licensing of amusement operations	Local counsel briefings — verify locally

A market scoring below 60/100 is deferred regardless of franchisee enthusiasm. The global context supports patience: industry analysts (2025–2026) place the global RC car market between \$1.5 B and \$6.2 B depending on scope, growing 5.8–7.5% annually, with hobby-grade share rising — the addressable base is expanding in every priority region (Volume 2 carries the full market analysis).

3.2 Target region analysis

Gulf states (UAE, Saudi Arabia, Qatar, Kuwait). Highest disposable income in the candidate set, strong motorsport culture (full-scale and RC), deep corporate-events demand, and government appetite for entertainment investment. Constraints: summer heat makes June–September outdoor operation marginal, so the indicated formats are Format C (climate-controlled arena) or Format A with an indoor arena built in phase one rather than as a later option, plus night operations as the default summer mode. Land is available but often via government-linked developers, favoring a **master franchise or area-development deal with a local entertainment group**. Foreign-ownership and commercial-agency rules vary sharply by emirate/state — verify locally. Priority: **first international wave**.

Southeast Asia (Malaysia, Thailand, Indonesia, Vietnam, Philippines). The deepest grassroots RC culture per dollar of income in the world — crawler and drift scenes are mass hobbies — plus low land costs and year-round warmth. Constraints: monsoon seasons demand the Volume 11 drainage standards without compromise plus covered pit lanes; price points must localize toward the bottom of the pricing governance band (Chapter 9.3); currency and royalty-repatriation mechanics need structuring. Indicated formats: A and B. Entry mode: single-unit or area development with hands-on franchisor support; master franchise only for archipelago markets (Indonesia, Philippines) where geography defeats direct support. Priority: **first international wave (Malaysia/Thailand), second wave (others)**.

Europe (Germany, UK, Netherlands, Nordics, Poland). Mature hobby markets with strong club infrastructure and high willingness to pay, but high labor costs, short outdoor seasons in the north, and demanding planning/permitting regimes. Indicated formats: C in the north, A/B in Iberia and southern France. EU pre-contractual disclosure rules and national franchise laws (e.g., France) apply — verify locally. Priority: **second wave**, led by an indoor-arena pilot in Germany or the Netherlands.

North America (US, Canada). The largest RC market and the most franchise-literate consumer base, with abundant peri-urban land in secondary metros. Constraints: the US is the most heavily regulated franchise-disclosure regime on earth (FDD under the FTC Franchise Rule plus state registration states — Chapter 4.2), liability-insurance costs are materially higher, and competition for family leisure spend is intense. Indicated formats: A in the Sun Belt, C in the north and Canada. Entry: **area development with experienced multi-unit operators**; do not enter the US before the franchise system has at least three operating units elsewhere and a US-counsel-prepared FDD. Priority: **third wave**.

3.3 Localization requirements

Every franchise adapts along four sanctioned dimensions; everything else is standard.

Language. RC WORLD OS ships with full internationalization: customer and staff apps, Toolbox Talk modules, License theory tests, signage packs, and waivers are translated and legally re-reviewed per market (Volume 13). Toolbox Talk voiceover and safety signage must be in the local language and English; the waiver must be in the contract language local counsel specifies — verify locally.

Climate engineering — hot markets. Shade sails over all spectator and driver rostrum areas; misting lines at queue points; charging-bunker ventilation upgraded (LiPo charging in 45 °C ambient requires forced airflow and tightened charge-rate limits per Volume 7 battery doctrine); track-surface temperature monitoring (tire compound shifts above ~55 °C sur-

face); summer schedule pivots to 16:00–23:00 with night-racing lighting; battery storage air-conditioned to $\leq 28^{\circ}\text{C}$ (82°F).

Climate engineering — cold markets. Format C as baseline; humidity control 40–55% RH for static and dust management; floor-level heating economics reviewed at design gate G2; entrance airlocks; winter tire compounds for any outdoor shoulder-season operation; LiPo cold-storage doctrine (packs warmed to $\geq 15^{\circ}\text{C}$ / 59°F before charge or discharge).

Cultural calendar. The events and marketing calendar (Volume 9) re-anchors to local rhythms: Ramadan evening-hours mode in Gulf markets (peak trade after iftar); Lunar New Year and Songkran programming in Southeast Asia; European school-holiday matrices; local motorsport calendar tie-ins. Pricing promotions follow the local calendar within the governance bands of Chapter 9.3.

3.4 Entry modes

Mode	Structure	When used	Fee mechanics (summary; detail Ch. 5)
Single-unit franchise	One franchisee, one site, direct franchisor support	Domestic market and first units in supportable near markets	Standard fees per Chapter 5.1
Area development	One franchisee commits to 3–5 units on a schedule in a defined territory	Proven markets, capable multi-unit operators (esp. North America, Gulf)	Standard fees per unit; development fee = 50% of each committed unit's initial fee, paid upfront, credited at signing; units forfeit if schedule missed
Master franchise	Local partner buys territory rights, sub-franchises, delivers first-line support	Distant/regulated/culturally distinct territories; only after domestic pilot has ≥ 12 months of operation	Territory fee \$250 k–\$1.2 M by market size; unit initial fees and royalties split 50/50 franchisor/master; master must operate ≥ 1 unit itself first

Field Note. Several jurisdictions regulate who may franchise at all: China, for example, has historically required a franchisor to have operated two directly-owned units for at least one year before franchising (“2+1 rule”), and several Gulf states route foreign franchises through commercial-agency registration with strong agent protections. These rules can dictate entry mode regardless of commercial preference — another reason the master-franchise route exists in the toolkit. Verify each market with local counsel before any negotiation.

4. Licensing & Legal Framework

4.1 What the license grants

The franchise agreement grants, for the term and territory, a non-exclusive-brand / exclusive-territory license comprising:

1. **Brand.** Use of the RC WORLD name, logos, trade dress, division names (Motorsport Division, Construction Division, Aviation Division, Marine Division), and canonical program

names (The Works, Shift, Toolbox Talk, Tow-Truck Retrieval Protocol, RC WORLD License, Gears) strictly per the brand standards manual (Chapter 9).

2. **RC WORLD OS access.** A multi-tenant SaaS subscription covering every platform module — bookings, payments, POS, fleet management, live telemetry, maintenance logs, HR/payroll, finance, CRM, reporting — with the franchisee role-based interface and mandatory franchisor read-access to operational and financial KPIs (Volume 13). Software is licensed, never sold; no source code transfers.
3. **Procurement network access.** The right (not obligation, except for controlled items) to buy through the central procurement desk at disclosed terms (Chapter 5.5); the approved-equipment list; supplier introductions per Volume 8. **Controlled items — RCW Nodes, node firmware, and License/badge credential stock — may be sourced only from the franchisor.**
4. **Manuals and training.** Licensed access to the Master Development Plan operating volumes (versioned, watermarked, confidential), all SOPs and checklists, and the RC WORLD Academy programs (Chapter 8).
5. **Ongoing system improvements.** Engineering bulletins, OS updates, new formats and products, marketing kits (Chapter 11).

Explicitly **not** granted: equity in RC WORLD; RCW Node schematics, PCBA design files, or firmware source; RC WORLD OS source code; the right to sub-franchise (master franchisees excepted); the right to operate any competing RC venue during the term plus a post-term period counsel confirms is enforceable locally.

4.2 Disclosure obligations — verify locally

Most priority markets impose pre-sale disclosure duties on franchisors, and breach typically gives franchisees rescission rights. Treat the following as a planning map, not advice:

- **United States:** a Franchise Disclosure Document (FDD) under the FTC Franchise Rule, delivered at least 14 days before signing or payment, with 23 prescribed items including audited franchisor financials and litigation history; a dozen-plus states add registration or filing requirements. Financial performance representations (Item 19) may only present substantiated data — which is precisely what the flagship-first proof requirement (Chapter 1.3) exists to generate.
- **Canada, Australia:** province-level disclosure laws (e.g., Ontario) and the Australian Franchising Code of Conduct respectively, each with their own disclosure documents, cooling-off periods, and good-faith duties.
- **EU and UK:** no single franchise statute, but national laws (e.g., French pre-contractual disclosure), competition-law limits on pricing and territory clauses, and data-protection law (GDPR) governing the customer data RC WORLD OS processes.
- **Gulf and Asia:** a patchwork of franchise-specific statutes (e.g., Saudi Arabia's franchise law), commercial-agency regimes, foreign-investment approvals, and registration duties.

Budget line: the franchisor legal build (first FDD-equivalent, trademark filings, template agreements) is carried in the Phase 3 franchisor budget (Chapter 11.1) at \$120–180 k before the first international market, and each new country adds \$25–60 k — engage franchise counsel per market and never reuse a disclosure document across borders.

4.3 Term, renewal, termination

Design intentions, to be conformed to local law:

- **Term:** 10 years, matching the depreciation life of the major civil works, with one 10-year renewal at the franchisee's option conditional on: audit scores $\geq 850/1000$ average over the prior three years (Chapter 10.2), refurbishment to then-current standards, and re-signing the then-current agreement form. Renewal fee: 25% of the then-current initial fee.
- **Franchisor termination for cause:** unremedied safety-critical breach (immediate, after a single 7-day cure notice for anything not endangering people; zero cure period for knowing operation of a condemned asset class or disabling of RCW Node kill-switch functions); insolvency; abandonment (>14 consecutive unauthorized closed days); persistent under-reporting of gross revenue discovered by audit (Chapter 10); unauthorized transfer. Termination rights are among the most locally variable provisions in franchise law — several regimes require notice, cure, or even compensation regardless of contract wording; verify locally.
- **Franchisee exit:** sale of the business permitted with franchisor approval of the buyer (same selection standards as Chapter 6), a transfer fee of \$25 k covering re-training and due diligence, and a franchisor right of first refusal on any sale.
- **Post-termination:** de-branding within 30 days, OS access revoked (customer-data handling per data-protection counsel), controlled items repurchased at book value at franchisor option, non-compete as locally enforceable.

4.4 Territory protection

Each agreement defines a **protected territory** — the greater of a 30 km radius or the 45-minute drive-time catchment around the site, adjusted at signing for geography — within which the franchisor will neither operate nor license another RC WORLD park of any format. Carve-outs: temporary event activations (mall pop-ups, expo circuits) under 30 days; e-commerce and app sales, which are system-wide by nature (retail revenue attribution for inter-territory customers is credited per the OS attribution rules, Volume 13). Territory clauses interact with competition law in the EU and several other regimes — verify locally.

4.5 IP protection

- **Trademarks.** “RC WORLD,” the logo, “RCW Node,” “RC WORLD OS,” “Toolbox Talk,” “Tow-Truck Retrieval Protocol,” and “Gears” filed in home market before franchise launch, then internationally (Madrid Protocol where available) in every priority market before entering negotiations there — trademark squatting in first-to-file jurisdictions (notably China, where the supply chain already operates) is the single most common franchisor IP loss; file early, in local script where applicable.
- **RCW Node designs.** Layered protection: registered designs where cost-effective, but primarily **trade-secret handling plus supply-chain control** — franchisees receive assembled, conformally-coated, serialized nodes, never design files; fabrication remains with the franchisor's contract manufacturers (Volume 8); firmware is signed and encrypted, flashed only by the franchisor; nodes phone home to the franchisee's tenancy and cannot be re-provisioned outside RC WORLD OS.
- **Software.** RC WORLD OS is licensed SaaS; no on-premise deployment except where data-residency law forces it (then containerized, franchisor-managed, with counsel-ap-

proved terms). Master franchisees may negotiate source-code escrow releasable only on franchisor insolvency — grant reluctantly and narrowly.

- **Manuals.** Watermarked per licensee, confidentiality obligations surviving termination, and staged access: prospects see summaries; only signed franchisees see full volumes.

Investor Note. The licensing posture is deliberately asymmetric: the franchisee gets everything they need to operate and nothing they would need to defect. Brand, OS tenancy, node supply, and procurement access all terminate with the agreement, which converts the franchise network into a durable annuity rather than a training program for future competitors.

5. Franchise Economics

5.1 Fee structure

The fee architecture follows one principle: **the franchisor earns when the franchisee earns**. Initial fees are set to cover onboarding costs plus a modest margin — not as a profit center — so that franchisor incentives stay aligned with long-run unit performance rather than unit sales.

Fee	Amount	Basis & timing	What it pays for
Initial franchise fee	Format A \$150,000 · Format B \$90,000 · Format C \$110,000	One-time, at signing	License grant, site-approval process, design review, Academy training for franchisee + 2 leaders, pre-opening support team (Ch. 8.4)
Royalty	6% of gross revenue	Monthly, auto-calculated from RC WORLD OS revenue ledger	System license, ongoing support, engineering updates, manuals
Technology fee (RC WORLD OS)	\$2,000/month base + \$8/month per active powered asset	Monthly SaaS invoice	Full OS tenancy, hosting, telemetry backhaul, updates, support desk; scales with fleet: ≈\$3,560/mo (A), ≈\$2,960/mo (B), ≈\$2,800/mo (C)
Marketing fund	2% of gross revenue	Monthly, ring-fenced fund	System-wide brand campaigns, content library, app-store presence; audited annually, reported to the advisory council (Ch. 11.4)
Procurement handling margin	8-12% on landed cost, disclosed per order	Per purchase through the central desk	Consolidation, QC inspection, freight optimization (Ch. 5.5)
Renewal fee	25% of then-current initial fee	At 10-year renewal	Re-documentation, refresh training
Transfer fee	\$25,000	On approved sale of the franchise	Buyer due diligence, re-training

Total recurring load on the franchisee is therefore **8% of gross (royalty + marketing fund) plus a technology fee that lands at 1.4-2.4% of gross** depending on format and

maturity — roughly 10% all-in, at the moderate end of experience-economy franchising. Gross revenue is defined in the agreement as all revenue recognized in RC WORLD OS excluding sales taxes and the marketing fund is a pass-through, not franchisor income. Because every booking, POS transaction, and payment runs through RC WORLD OS (Volume 13), **royalty under-reporting is structurally difficult** — the ledger the royalty is computed from is the same ledger the park runs on.

5.2 Franchisee unit economics — worked example (Format B, Compact Urban, Year 3)

The example below scales the canonical flagship model (Volume 10) to the compact format: flagship Phase 1 “Core Park” scope, denser urban catchment, mature operation in its third year. All figures USD, rounded; local plans will vary and disclosure documents must use audited flagship data plus pilot-franchise actuals, not this illustration.

Revenue build (\$1,750,000):

Stream	Share	Amount	Notes
Shifts & sessions (Casual \$15/\$22, Operator \$26/\$38, Day Passes \$59)	62%	\$1,085,000	≈ 61,000 billed Shift-equivalents/yr across ~120 assets — ~1.6 revenue-hours/asset/operating day at canonical pricing
Memberships (Apprentice \$29 / Operator \$59 / Foreman \$99 per month)	12%	\$210,000	≈ 340 average active members at ≈ \$51 blended
Events: parties (from \$349), corporate (from \$1,400), leagues	14%	\$245,000	≈ 320 parties + 60 corporate events + league fees
F&B (kiosk format)	7%	\$122,000	No restaurant in Format B
Retail & repair services	5%	\$88,000	Counter retail + Artisan repair bench
Total revenue	100%	\$1,750,000	

Operating costs and franchisee P&L:

Line	% of revenue	Amount
Labor (14–18 FTE incl. 5–6 Artisans, plus weekend casuals)	34.0%	\$595,000
Rent (1.5 ha urban lease)	9.0%	\$157,500
Fleet parts, consumables, batteries (Volume 7 spares doctrine)	7.0%	\$122,500
F&B and retail COGS	5.0%	\$87,500
Utilities (power incl. charging bunker, water, connectivity)	3.5%	\$61,000
Insurance (public liability, property, fleet)	2.5%	\$44,000
Local marketing (franchisee’s own, beside the 2% fund)	3.0%	\$52,500

Line	% of revenue	Amount
Admin, professional fees, payments processing	4.5%	\$79,000
Facilities & grounds maintenance	2.7%	\$47,000
Pre-fee EBITDA	28.8%	\$504,000
Royalty (6%)	6.0%	\$105,000
Marketing fund (2%)	2.0%	\$35,000
Technology fee (\$2,960/mo)	2.0%	\$35,500
Franchisee EBITDA (post-fee)	18.7%	\$328,500

Investment and returns:

Item	Amount
Site capex (civil works, buildings, tracks, fleet, RCW Nodes, fit-out)	\$1,450,000
Initial franchise fee	\$90,000
Pre-opening costs (training travel, hiring, launch marketing)	\$110,000
Working capital	\$150,000
Total cash funding requirement	\$1,800,000

Ramp assumption: Year 1 $\approx$ \$1.2 M revenue at $\approx$ 12% post-fee EBITDA (\$145 k), Year 2 $\approx$ \$1.55 M at $\approx$ 16% (\$250 k), Year 3 onward per the table above. On those figures the franchisee's cumulative post-fee EBITDA recovers the \$1.8 M cash funding **during Year 6-7**, mature cash-on-cash yield is $\approx$ **18%**, and the unlevered project return over the 10-year term lands in the **low-to-mid teens before terminal value, mid-to-high teens including business resale or renewal at term end**. Franchisees using sensible site-level debt (the civil works and buildings are bankable assets) will see equity returns above those figures. These returns are consistent with, and slightly behind, the flagship's own canon (break-even months 16-19; payback Year 4-5) — as they should be: the franchisee pays fees the flagship does not, and buys a de-risked, pre-engineered system in exchange.

Sensitivity. The worked example moves most on three levers, in order: **utilization** (each $\pm 10\%$ on billed Shift-equivalents moves post-fee EBITDA by roughly $\pm \$95$ k, because Shift revenue carries minimal marginal cost), **labor ratio** (each ± 2 points of revenue moves EBITDA $\pm \$35$ k; the OS rostering module against demand curves is the control), and **rent** (fixed at signing — which is why site economics are interrogated at gate G0, not repaired later). A downside case at 85% of plan revenue and +2 points labor still clears $\approx$ \$190 k post-fee EBITDA at maturity — uncomfortable but solvent, provided the franchisee respected the 60% maximum-debt rule in Section 6.4. Fleet-parts cost is deliberately not a savings lever: the audit's fleet-condition domain (Chapter 10.2) exists precisely because deferring maintenance flatters one quarter and destroys the next year.

Investor Note. Read the worked example from the franchisee's chair before believing it from the franchisor's. An operator committing ~\$1.8 M wants three things this system genuinely offers: a proven design (no experimentation capex), structurally honest reporting (the OS ledger), and a franchisor whose income is 80%+ recurring and performance-linked rather than front-loaded in initial fees. Franchise systems with the opposite profile — fat initial fees, thin support — churn franchisees and die by lawsuit. The fee architecture above is a deliberate position on that spectrum.

5.3 Franchisor revenue model

Per-unit franchisor revenue at unit maturity (marketing fund excluded as pass-through):

Stream	Format A (\$3.1 M gross)	Format B (\$1.75 M gross)	Format C (\$1.4 M gross)
Royalty (6%)	\$186,000	\$105,000	\$84,000
Technology fee	\$42,700	\$35,500	\$33,600
Procurement margin (≈10% on \$110–160 k/yr of parts, consumables, fleet refresh, nodes)	\$16,000	\$11,000	\$10,000
Training & other (recertifications, transfers amortized)	\$6,000	\$4,500	\$4,500
Recurring franchisor revenue per unit	≈ \$250,000	≈ \$156,000	≈ \$132,000
Initial fee (one-time, at signing)	\$150,000	\$90,000	\$110,000

Marginal franchisor support cost per unit (field visits, audit, support-desk load, engineering time) is estimated at \$60–90 k for near markets and up to \$120 k for far markets before a master franchisee absorbs first-line support — implying **steady-state contribution of \$60–160 k per unit per year** plus procurement float. Under master-franchise deals the franchisor's shares halve but so does its support cost. The franchisor P&L consolidates into Volume 10's Phase 3 model; the five-unit Year 5 scenario appears in Chapter 12.2.

5.4 Pricing interaction

Fee percentages are levied on gross revenue as locally priced. Franchisees in lower-PPP markets pricing at the bottom of the governance band (Chapter 9.3) pay proportionally lower absolute royalties — the model self-adjusts and no per-market royalty renegotiation is entertained. Technology fees are dollar-denominated (they fund dollar-denominated infrastructure); master-franchise agreements may hedge this for volatile currencies with counsel and treasury advice — verify locally.

5.5 Procurement margin option via central purchasing

Volume 8's wholesale network (Huina/Kabolite, MST, MJX, LDRC, Double E, Hobbywing, Gens Ace/CNHL and peers, plus PCBA fabrication) is opened to franchisees through a **central procurement desk** (Chapter 11.2). The commercial logic is mutual: consolidated orders across the network reach volume tiers no single park can, the desk adds pre-shipment QC and freight consolidation, and the 8–12% handling margin still leaves the franchisee's landed cost **below what they could achieve alone** — typically by 10–20% against retail-adjacent

sourcing. Purchasing through the desk is mandatory only for controlled items (RCW Nodes, credential stock) and for any item on the approved-equipment list where the franchisee cannot evidence an equivalent-spec alternative; everything else is priced to win the order on merit. The desk's margin is disclosed in the franchise agreement — undisclosed supplier rebates are a classic franchisor trust-killer and are contractually forsworn.

6. Franchisee Recruitment & Selection

6.1 Ideal franchisee profile

The franchise is an **operator's business with an engineering heart**, not a passive investment. The ideal candidate is an owner-operator or a tightly-held partnership where one principal commits full-time, with: P&L management experience leading 10+ staff in a customer-facing operation (leisure, hospitality, fitness, automotive retail); comfort with technical operations (they need not solder, but they must respect the people who do); local roots in the target catchment; capital per Section 6.4 without over-leverage; and genuine enthusiasm for the product — a franchisee who has never held a transmitter will not survive the Academy, and should not.

Explicit disqualifiers: pure financial investors with hired “manager will run it” structures (permitted only in area-development deals with a certified operating partner); candidates whose capital plan depends on selling before break-even; anyone unwilling to complete the full Academy program personally.

6.2 Selection scorecard

Scored by two assessors independently at stage 4 of discovery; minimum passing score 70/100 with no axis below half marks.

Axis	Weight	Evidence examined
Operating track record	25	References, prior P&Ls, staff-retention history
Capital adequacy & plan	20	Verified liquidity, funding letters, stress-tested plan
Local market strength	15	Site pipeline, municipal and school/corporate network
Technical & systems aptitude	15	Academy pre-assessment, response to OS demo, workshop day performance
Culture & brand fit	15	Behavior with flagship staff and customers during park immersion day
Alignment on term & standards	10	Reaction to audit regime, pricing governance, controlled-item rules

6.3 Discovery process stages

1. **Inquiry & pre-qualification** — application, capital self-declaration, NDA (week 0–2).
2. **System overview** — format summaries, fee disclosure summary, market scoring review (week 2–4).

3. **Flagship immersion day** — full day at the park including a workshop session with Artisans and a shift as a marshal shadow (week 4–8).
4. **Formal assessment** — scorecard interviews, capital verification, background checks (week 8–12).
5. **Disclosure** — statutory disclosure document delivered; mandatory local cooling-off periods respected — verify locally (week 12+).
6. **Business-plan defense** — candidate presents site pipeline and 5-year plan to the franchise committee (week 14–18).
7. **Award & signing** — agreement executed, initial fee paid, Academy enrollment (week 18–24).

The franchisor may exit the process at any stage without cause; the candidate's costs are their own except the immersion day, which the franchisor hosts.

6.4 Capital requirements

Requirement	Format A	Format B	Format C
Total cash funding requirement (Ch. 2.5)	\$2.7–3.1 M	\$1.6–1.95 M	\$2.0–2.45 M
Minimum unencumbered liquid capital ($\geq 40\%$)	\$1.1 M	\$700 k	\$850 k
Minimum net worth ($\geq 1.5 \times$ funding)	\$4.2 M	\$2.7 M	\$3.1 M
Maximum debt share of funding	60%	60%	60%

Proof standards: bank confirmations, audited statements or accountant letters, and funding term sheets before stage 6. The franchisor does not lend to franchisees and does not guarantee franchisee debt.

6.5 Financing support short of lending

While the franchisor takes no credit risk, it materially improves franchisee bankability: lenders receive (with the candidate's consent) the disclosure document, the format design kit, the flagship's audited performance history, and — once the pilot operates — pilot actuals; the 10-year lease-plus-renewal requirement (Section 7.1) gives asset-backed lenders a security horizon; and the OS ledger gives lenders live covenant reporting no independent leisure startup can offer. In markets with government small-business loan schemes or tourism-investment incentives, the franchise committee maintains a per-market briefing note — prepared with local advisors, verify locally — flagging programs a candidate's plan should exploit.

Field Note. The single best predictor of franchisee failure across the leisure industry is not competence — it is under-capitalization meeting a slow ramp. A candidate who plans to open with the minimum liquidity and “grow into” working capital will make the one decision that kills an RC park: cutting Artisan hours and maintenance spend in month 8, just as the fleet's honeymoon ends. The 40% liquidity floor and the maintenance-protected audit exist to make that decision structurally difficult. Hold the line on capital adequacy even for otherwise outstanding candidates.

7. Site Approval & Build-Out

7.1 Site criteria as franchise standards

Volume 11's site-selection criteria for the flagship become **binding franchise standards**, parameterized per format:

- ☐ Land/premises per format: A 4-5 ha, B 1.2-1.8 ha, C 6,000-9,000 m² with ≥6 m clear height
- ☐ Topography: flat to 2-4% natural grade (A/B); level slab, ≥25 kN/m² floor loading zones for aggregate beds (C)
- ☐ Catchment: per Chapter 2.5 population thresholds, evidenced by drive-time analysis
- ☐ Road frontage and visibility, or (C) established destination-retail/industrial-leisure precinct
- ☐ Municipal power adequate for the charging bunker load profile (Volume 11 electrical schedule) plus (C) HVAC
- ☐ Municipal water + drainage compatible with track watering and (monsoon markets) Volume 11 storm standards
- ☐ Parking per Volume 11 ratios; coach/bus bay for school and corporate trade
- ☐ Lease: minimum 10-year term with renewal option matching the franchise term; landlord consent to signage and outdoor operations
- ☐ Zoning/permits for amusement/recreation use confirmed in writing — verify locally
- ☐ Noise assessment against local ordinance (electric fleet is quiet; PA systems and crowds are not)
- ☐ Environmental screen: no flood-plain siting without engineered mitigation; soil suitable for civil works (A/B)

7.2 Design review gates

No franchisee builds from scratch: the franchisor issues the **format design kit** (Volume 11 drawings parameterized to the site) and reviews at five gates. Franchisor sign-off at each gate is a condition of keeping the license; the franchisee's architect and engineer of record retain full statutory responsibility — the franchisor reviews for brand and system conformity, not as engineer of record.

Gate	Deliverable reviewed	Franchisor SLA
G0 — Site approval	Site dossier vs Section 7.1 checklist	15 business days
G1 — Concept layout	Adapted format kit: zone plan, track geometries, sightlines, pit lane, charging bunker location	15 days
G2 — Developed design	Civil/structural/electrical packages, drainage, Wi-Fi mesh RF plan, The Works layout	20 days
G3 — Pre-construction	Contractor selection, program, safety plan, procurement schedule (fleet + nodes lead times)	10 days
G4 — Pre-opening	Physical inspection: track commissioning, bunker certification, OS go-live, fleet scrutineering, staff certification	On-site, 5 days

7.3 Approved-contractor policy

Construction uses local contractors — the franchisor approves rather than supplies them. Requirements: demonstrated civil-works competence (track subgrade and drainage are the make-or-break trades, per Volume 11), licensed electrical contractor for the charging bunker and distribution, and acceptance of the franchisor's hold-point inspections (subgrade, drainage, slab pours for Track A, bunker fire-safety fit-out). Specialist items — track surface finishing, netting systems (Format A aviation), telemetry mesh commissioning — may be delivered by franchisor-nominated specialists at disclosed rates. The franchisor never takes construction risk or margin on the build itself.

7.4 Build timeline template (Format B reference)

Month	Workstream
1-2	G1/G2 design adaptation; permits lodged; long-lead procurement released (fleet, nodes, chargers, netting)
3-5	Civil works: earthworks, drainage, track subgrades; services trenching
4-6	Buildings: entry pavilion, The Works, charging bunker; Track A slab cure window
6-8	Track surfacing and features; Mining Zone aggregate beds; crawler course; fencing and barriers
7-9	Electrical fit-out; Wi-Fi mesh; OS tenancy provisioning; fleet arrival, hardening passes and node installation (Volume 7 procedures)
9-10	Staff hiring; Academy certifications; soft-systems commissioning; G4 inspection
10-12	Marshal rehearsals, invited-guest soft opening (2-3 weeks), grand opening

Format A runs the same template across two phases (Core Park months 1-12, remaining divisions months 13-26); Format C compresses to 6-9 months since it is fit-out, not civil construction.

8. Training Program — The RC WORLD Academy

8.1 The Academy

The RC WORLD Academy is the training institution operating from the flagship's Phase 3 classrooms (a canonical Phase 3 build item) and The Works. Its franchise mandate: **no franchise park opens, and no franchise park keeps operating, without currently-certified people in defined roles.** Certification is personal, non-transferable, recorded in RC WORLD OS against the individual's staff profile, and checked automatically by the OS rostering module — an uncertified Artisan cannot be rostered to fleet duties (Volume 13, role-based access control).

8.2 Curriculum

Program	Audience	Duration	Location	Core modules (source volume)
FLC — Franchisee Leadership Course	Franchisee principal + up to 2 senior leaders (mandatory)	120 h / 3 weeks	Flagship	Business model & unit economics, 16 h (Vols 1, 10, 12) · Division operations overview, 24 h (Vols 3–6) · RC WORLD OS executive & finance suite, 16 h (Vol 13) · Safety leadership & compliance, 12 h (Vols 7, 11) · Hiring, rostering, culture, 16 h (Vol 9 service standards) · Marketing & events, 12 h (Vol 9) · Procurement & supply chain, 8 h (Vol 8) · Supervised field rotation across all divisions, 16 h

Program	Audience	Duration	Location	Core modules (source volume)
ATC — Artisan Technical Certification	Every franchise Artisan (lead Artisan certifies at flagship; others may certify at a certified regional hub once one exists)	160 h / 4 weeks	Flagship (The Works)	Workshop skills: teardown, rebuild, tuning, trade hacks, 60 h (Vol 7) · Powertrain & electronics: ESC programming, motors, servos, 24 h (Vols 3, 7) · LiPo doctrine: 3:1 rotation, 3.4–4.2 V/cell window, bunker operations, 12 h (Vol 7) · RCW Node installation & service, 12 h (Vol 7 + controlled supplement) · Division fleet specialization: parity method, 1:3 mining logistics, aircraft/marine as applicable, 32 h (Vols 3–6) · OS artisan interface & maintenance logs, 8 h (Vol 13) · Safety & incident response, 12 h
FOC — Front-of-House & Marshal Certification	Franchise trainers (train-the-trainer)	40 h / 1 week	Flagship	Toolbox Talk delivery & waiver integrity · Shift operations & pit lane · Tow-Truck Retrieval Protocol facilitation · License/badge assessment integrity (Vol 9) · Emergency procedures
GMC — General Manager Certification	Franchise GM if not the principal	80 h / 2 weeks	Flagship	Condensed FLC + rostering, cash & reporting controls, audit-readiness
Annual recertification	Principal/GM 16 h; Artisans 24 h; FOC trainers 8 h	Annual	Regional hub or remote + practical assessment	Bulletin-driven updates, safety refresh, new-product modules

Academy fees for initial certifications of the opening team are included in the initial franchise fee; travel and lodging are the franchisee's; recertification is charged at cost ($\approx$ \$350–600 per person per cycle).

8.3 Assessment standards

Every program ends in assessment: written (OS-delivered, question banks versioned with the manuals), practical (e.g., ATC candidates perform a timed gearbox rebuild, a battery-bunker incident drill, and an RCW Node replacement with telemetry re-binding verified live), and behavioral (FOC candidates deliver a Toolbox Talk to real flagship customers under observa-

tion). Pass mark 80%; one resit; a failed second attempt requires a new candidate — the Academy protects the network, not the individual enrollment.

8.4 Pre-opening on-site support team

For each opening, the franchisor deploys a **4-person opening team** — an operations lead, a lead Artisan, an OS/telemetry engineer, and a front-of-house trainer — on site for **14 days before opening and 7 days after**, covering: fleet scrutineering and parity verification, bunker and charging commissioning, OS go-live and payment-rail testing, marshal rehearsals with full Tow-Truck Retrieval drills, soft-opening supervision, and first-week stabilization. A second, 2-person visit at week 6–8 tunes rostering and maintenance cadence against real demand data. Both visits are funded from the initial fee; extensions bill at disclosed day rates.

8.5 Ongoing training

Quarterly engineering and product bulletins (Chapter 11.2) each carry a micro-module in the OS learning system with completion tracked per certified role; annual recertification per the curriculum table; and an annual **Franchise Convention** at the flagship combining recertification intensives, the advisory council plenary (Chapter 11.4), and the system's championship finals — the event that turns a contract network into a community.

8.6 Regional certification hubs

Flying every Artisan to the flagship does not scale past a handful of units. Once a region hosts three good-standing parks, the franchisor designates one as a **regional certification hub**: its Works is audited to training standard, its lead Artisan completes a train-the-certifier program at the flagship (40 h), and ATC and FOC programs can then run regionally under flagship-issued question banks and practical-assessment rubrics, with the franchisor moderating a sample of assessments remotely via recorded practicals and OS telemetry evidence. FLC and GMC remain flagship-only for the plan horizon — the leadership courses are as much enculturation as instruction, and the flagship is the culture. Hub hosts are compensated per trainee at published rates; hub status is reviewed at each annual field audit and is itself an earned, revocable standard.

9. Brand Standards

9.1 Visual identity rules

The brand book (issued with the franchise agreement) governs: logo forms, clearspace and minimum sizes; the palette and typography; signage families (wayfinding, safety, sponsor inventory per Volume 9); staff uniform specifications by role (Artisan workshop wear is functional PPE first, brand second); vehicle livery rules — fleet liveries are format-standard, with a defined local-sponsor zone on each body shell; and digital assets (the customer app is franchisor-controlled; franchisee social accounts follow the template kit). Photography standard: machines working, hands on transmitters, dirt and detail — never sterile catalog shots.

9.2 Naming conventions

Canonical names are used exactly and are never translated, abbreviated, or localized without written approval: **RC WORLD** (all caps in headings), the park styled “RC WORLD [City]”; divisions as Motorsport Division, Construction Division (incl. Mining & Agriculture Zones), Aviation Division, Marine Division; **The Works**; **Artisans**; the **Shift** (Casual Shift, Operator Shift); **RCW Node**; **RC WORLD OS**; **RC WORLD License**; **Tow-Truck Retrieval Protocol**; **Toolbox Talk**; **Gears**. Local-language explanatory taglines may accompany, never replace, canonical names (approved via 9.5). The internal codename “Omni-Zone” is legacy and never appears customer-facing.

9.3 Customer-experience non-negotiables

Four mechanics define the brand promise and are contractually protected; disabling or diluting any of them is a material breach:

1. **Toolbox Talk.** Every new customer completes the digital induction and waiver before first operation — no verbal shortcuts, no paper bypass. Completion is enforced by the OS booking flow.
2. **Tow-Truck Retrieval Protocol.** Customers never walk onto live tracks. Breakdowns are recovered by the customer piloting a winch-equipped recovery crawler under marshal supervision, per the flagship SOP. This is simultaneously the park’s signature delight moment and its central safety control; it is audited hardest (Chapter 10).
3. **License and badge integrity.** RC WORLD License tiers, badges, and Gears (Volume 9) are earned, portable across every park in the network via the customer’s OS profile, and never sold, gifted, or locally re-specified. A Foreman-tier member from the flagship walks into any franchise and is recognized; that portability is a system asset no single park may debase.
4. **Pricing governance bands.** Local pricing is set by the franchisee within franchisor-approved bands: **70-130% of the canonical anchors** (Casual Shift \$15/\$22, Operator Shift \$26/\$38, Day Pass \$59, family bundle \$89, memberships \$29/\$59/\$99), adjusted per market via a published PPP-based index reviewed annually. Within the band, price relativities must be preserved (premium classes above standard; Operator Shift below 2× Casual — the pit-stop swap is part of the value). Outside-band pricing requires a written variance with market evidence.

9.4 Service standards

The flagship’s service canon (Volume 9) applies verbatim: greeting-to-transmitter time targets, pit-lane etiquette, marshal-to-track ratios, party and corporate-event run sheets, complaint-resolution empowerment limits, and cleanliness standards for The Works viewing window — the workshop is on show, and a chaotic bench is a brand breach, not a housekeeping issue.

9.5 Local-adaptation approval process

Any deviation from format kits, naming, CX mechanics, pricing bands, or approved equipment follows one process: the franchisee submits a **Variance Request** in RC WORLD OS (what, why, evidence, duration, rollback plan); the franchise support office responds within 10 business days with approve / approve-as-trial / decline; trials carry measurement plans and sunset dates; approved adaptations that outperform are considered for system-wide ad-

option with credit to the originating franchisee — the network is also an R&D engine, and the process is designed to harvest, not merely police, local intelligence. Safety-relevant variances additionally require engineering sign-off; CX non-negotiables (9.3) are not variance-eligible.

10. Quality Systems & Compliance

10.1 Audit program structure

Two instruments on fixed cadence: a **quarterly self-audit** conducted by the franchisee's GM in RC WORLD OS (checklist with mandatory photo/telemetry evidence attached per item; ~4 hours), and an **annual field audit** conducted unannounced-within-a-window by a franchisor auditor over two days, covering the full scored standard plus financial-reporting verification. Self-audit results that materially contradict field findings are themselves a scored integrity failure.

10.2 Scored standards checklist

The audit scores 1,000 points across four domains; ≥ 850 is good standing (and the renewal threshold, Chapter 4.3), 700–849 triggers an improvement plan, < 700 enters the escalation ladder. Safety items marked * are pass/fail regardless of aggregate score. Representative items (full instrument in the audit workbook):

Safety — 300 points

☐

* RCW Node kill-switch live-tested on sample vehicles from each class; under-voltage and geofence intercepts verified

☐

* Charging bunker to standard: fire-rated construction, smart balance chargers only, LiPo-rated extinguishing media, no unattended charging outside bunker

☐

* LiPo doctrine in force: 3.4–4.2 V/cell window enforced by OS, 3:1 rotation evidenced, damaged-pack quarantine and disposal log current

☐

* Tow-Truck Retrieval Protocol observed live: zero customer track incursions; marshal intervention drill passed

☐

Toolbox Talk completion rate 100% of first-time operators (OS report)

☐

Haul roads within 15° maximum incline (spot-measured); track barriers and netting (where applicable) to Volume 11 spec

☐

Incident log complete, injuries reported per SLA, corrective actions closed

Fleet condition — 250 points

☐

Fleet availability $\geq$ 90% of rated assets per operating day (OS trailing 90-day report)

☐

Motorsport parity within 2-3% lap-time spread per class (weekly parity audit records)

☐

Excavator:dump-truck ratio 1:3 maintained in live Mining Zone operation

☐

Preventive-maintenance compliance $\geq$ 95% of OS-scheduled tasks on time; maintenance logs complete with parts traceability

☐

Rental construction fleet electromechanical only; hydraulic (Kabolite-class) units confined to premium supervised experiences

☐

Critical-spares inventory at or above Volume 7 minimum lines; battery fleet within cycle-count retirement policy

Service & customer experience — 250 points

☐

Mystery-shopper composite $\geq$ 85% (Section 10.5)

☐

Net promoter / review scores at or above system floor; complaint resolution within empowerment SLAs

☐

License/badge assessments conducted per Volume 9 integrity rules (assessor certifications current)

☐

Pricing within governance bands; advertised offers approved

☐

Brand presentation: signage, uniforms, liveries, The Works viewing window to standard

Financial reporting & administration — 200 points

☐

All revenue channels transact through RC WORLD OS; no off-system POS or cash lanes

☐

Royalty and fee remittances current; reconciliation clean vs OS ledger

☐

Insurance certificates current and to required limits; statutory licenses/permits current — verify locally

☐

Staff certifications current per Academy matrix; rostering-lock respected

☐

Data-protection obligations met for customer data (per Volume 13 controls and local law)

10.3 KPI reporting through RC WORLD OS

Because every franchise runs on the same platform, KPI reporting is not a submission — it is a fact of the system. The franchisor's **multi-site dashboards** (Volume 13) surface, per park

and network-wide: revenue per available Shift-hour, fleet availability, Shift utilization by class and daypart, parity spread, maintenance compliance, battery-fleet health, Toolbox Talk completion, License progression rates, NPS, labor ratio, and royalty base. Franchisees see their own park benchmarked against anonymized network quartiles — the single most effective performance tool a franchise system owns, because no operator tolerates being fourth-quartile at fleet availability once they can see it.

10.4 Non-compliance escalation ladder

Level	Trigger	Response	Exit criteria
1 — Advisory	Single audit domain 70–84% or isolated finding	Written findings, franchisee action plan	Findings closed in 30 days
2 — Improvement plan	Score 700–849, or repeated Level 1	Joint improvement plan, monthly OS-tracked milestones, support visit at franchisee cost	≥850 at re-audit within 90 days
3 — Probation	Score <700, integrity finding, or failed Level 2	Formal default notice per agreement; expansion/renewal rights suspended; weekly reporting	Cure within contractual period
4 — Safety intervention	Any * item failure endangering people	Immediate: affected operation suspended (OS can disable bookings for an asset class remotely); franchisor engineer on site	Verified rectification before restart
5 — Termination track	Uncured Level 3/4, fraud, kill-switch tampering	Termination per Chapter 4.3, subject to local-law process — verify locally	—

10.5 Mystery-shopper program

Two visits per park per year by contracted independent shoppers using the network scorecard: booking friction, Toolbox Talk quality, transmitter handover, marshal attentiveness, a staged breakdown to experience the Tow-Truck Retrieval Protocol as a customer, F&B and retail interaction, and departure/loyalty prompts. Reports flow into the audit score and the network benchmark. Shoppers are rotated so no park is scored twice consecutively by the same individual.

11. Franchise Support Operations

11.1 Franchisor organization build-out (Phase 3)

The franchisor function is built inside the flagship company during Phase 3 (months 31–60), deliberately lean and sequenced against the roadmap in Chapter 12:

Hire / function	Timing	Notes
Franchise Director	Month 31–34	Owns this volume; leads legal build, pilot selection

Hire / function	Timing	Notes
Legal build (external counsel)	Months 32–40	Trademark portfolio, first disclosure document, agreement templates; \$120–180 k budget (Ch. 4.2)
Academy Manager (part-time from existing training lead)	Month 36	Formalizes flagship training into the certifiable curriculum
Procurement desk (extension of flagship purchasing, +1 coordinator)	Month 40	Volume 8 network opened to franchise orders
Franchise Field Manager (first)	Month 46–50	Opening-team lead for the pilot; becomes pilot's field auditor
OS multi-tenancy hardening	Months 31–44	Volume 13 roadmap item: franchisee role, multi-site dashboards, tenancy isolation, marketing-fund ledger

Total incremental Phase 3 franchisor opex before pilot revenue: $\approx$ \$280–360 k, carried within the Phase 3 envelope and offset by the pilot's initial fee and first royalties (Volume 10 Phase 3 model). Support scales with the network: one field manager per 6–8 units in-region is the staffing ratio planning assumption.

11.2 Support functions

Procurement desk. Consolidated ordering across the Volume 8 network; quarterly approved-equipment list updates (models enter and exit as suppliers change — the list, not the manual, is the live authority); pre-shipment QC; RCW Node supply with serialized provisioning; emergency spares express channel with 72-hour dispatch for grounded-fleet situations.

Engineering updates. Quarterly engineering bulletins from The Works: fleet hardening revisions, failure-data findings from network-wide maintenance logs (the OS aggregates every park's `maintenance_logs`, so the network learns at network speed), node firmware releases (pushed centrally), battery doctrine updates, and safety directives — the last being mandatory-compliance items with stated deadlines.

Marketing kits. Seasonal campaign kits (assets, media plans, offer templates pre-cleared against pricing bands), the content library, launch playbooks for new parks, and PR support for openings. The 2% marketing fund pays for system-level brand work and the customer app's market presence; local media buying remains the franchisee's 3% local obligation.

New-product rollouts. New experiences validated at the flagship first (e.g., night racing, drift leagues, seasonal build-workshops), then packaged as rollout kits: equipment list, pricing guidance, staff micro-module, OS configuration, and marketing assets. Franchisees adopt on their own schedule except where a rollout supersedes a safety-relevant standard.

11.3 Support-desk operations

A tiered support model through RC WORLD OS: Tier 1 (OS usage, bookings, payments — response same business day), Tier 2 (fleet engineering, telemetry, parity problems — 48 h, escalating to The Works), Tier 3 (safety incidents — immediate, 24/7 on-call rotation). All tickets and resolutions are searchable network-wide, so the knowledge base compounds.

11.4 Franchisee advisory council

From five operating units, a **Franchisee Advisory Council** convenes: elected franchisee representatives (one per region once regions exist), meeting quarterly (remote) and annually in person at the Franchise Convention. Mandate: review the marketing-fund accounts (audited annually and presented to the council), consult on standards changes with 30-day comment periods before non-safety standards take effect, prioritize the OS feature backlog's franchise lane, and channel variance-process innovations toward system adoption. The council is consultative, not governing — but a franchisor that routinely overrides its council is burning the trust the whole model runs on, and the board should treat council dissent as a leading indicator.

12. Five-Year Franchise Roadmap

12.1 Pilot franchise timeline

Sequenced against the canonical master phasing (flagship Phase 3 = months 31-60, measured from flagship project start):

Months	Milestone
31-36	Franchise Director hired; legal build begins; trademark filings extended; this volume revised against two full years of flagship operating data (proof gate satisfied ≈ month 36)
36-42	Pilot candidate recruitment (Chapter 6 process); target: Format B compact urban in a near market the flagship team can reach within a short flight
40-44	Disclosure delivered; pilot agreement signed ≈ month 42; initial fee received; Academy FLC for pilot principal
42-48	Pilot site approval (G0-G2); permits; long-lead procurement
48-54	Pilot build (Chapter 7.4 template); ATC/FOC certifications for pilot staff; OS tenancy live
52-58	Pilot park opens ≈ month 54 ; opening team deployed; week 6-8 stabilization visit
58-60+	Pilot's first two quarters of data feed the Gate F1 review (Section 12.2); disclosure documents updated with pilot actuals

12.2 Gates to scale: 1 → 3 → master territories

Scaling is gated, not scheduled — each gate is an evidence test, and the dates are planning markers only.

Gate F1 (pilot validation, ≈ pilot month 12 / program month 66): pilot audit score ≥850; pilot revenue and post-fee EBITDA within 15% of its approved plan; zero uncured safety findings; franchisee would re-invest (structured interview); franchisor support cost per unit within budget. **Pass → release up to 3 regional units** (mix of Formats B and C, at least one in a second country from the first-wave list) over the following 18-24 months. Fail → fix the system before selling anyone else a franchise; the pilot franchisee's remedies are

defined in the pilot agreement's side letter (build-out support extensions, fee holidays — negotiated once, at signing, with counsel).

Gate F2 (regional validation, ≈ 12 months after the third unit opens): three-plus units at good standing; multi-site dashboards and support desk proven at $n > 1$; procurement desk demonstrating landed-cost advantage; advisory council seated. **Pass → open master-franchise negotiations** for first-wave distant territories (Gulf, Southeast Asia archipelagos per Chapter 3), with Format A flagship-equivalents anchoring each master territory. This stage extends beyond the five-year plan horizon; it is sketched here so the earlier gates are built to serve it.

Illustrative Year 5+ franchisor economics (pilot mature + 3 regional units ramping, per Chapter 5.3 rates): recurring franchisor revenue ≈ \$450–600 k/yr against support costs ≈ \$280–350 k — roughly breakeven-to-modestly-positive, with the initial fees having funded the build. The franchise program's Year 5 contribution to the flagship company is therefore **strategic, not financial**: the financial payoff arrives in years 6–10 as units mature and count grows, which is exactly the honest message for investors and the reason this volume resists front-loading fees.

Investor Note. The gate design encodes the program's core discipline: RC WORLD sells its second franchise on the strength of its first, not on the strength of its deck. If Gate F1 fails, the loss is one pilot relationship and roughly \$300 k of franchisor build cost — survivable. If a 10-unit system fails, the brand fails everywhere at once. Slow is the cheap option.

12.3 Franchise program KPIs

The program is managed on its own dashboard in RC WORLD OS, reviewed monthly by the Franchise Director and quarterly by the board:

KPI	Target	Why it matters
Pilot audit score	≥850/1000	Gate F1 input; system-quality proof
Franchise revenue vs approved plan	Within 15% by unit month 12	Disclosure credibility; Gate F1/F2 input
Franchisor support cost per unit	≤\$90 k near-market (≤\$120 k far)	Protects the recurring-revenue contribution model
Support-desk resolution SLAs	Tier 1 same-day ≥90%; Tier 2 ≤48 h ≥90%	Leading indicator of franchisee satisfaction
Franchisee satisfaction (annual survey + “would re-invest”)	≥80% positive	The only marketing a franchise system really has
Safety *item failures network-wide	Zero uncured	Brand-existential
Royalty collection rate	100% within terms	The OS ledger makes less than 100% an integrity event
Candidate pipeline quality	≥3 qualified candidates per open slot	Selection discipline requires the luxury of choice

12.4 Franchise-specific risk register

Risks below are additional to the flagship's enterprise risk register (Volume 10) and the per-division registers. Scale: L/M/H likelihood × impact.

#	Risk	L	I	Mitigation	Owner
FR-01	Pilot underperforms plan, poisoning disclosure data and pipeline	M	H	Pilot format = proven Phase 1 scope (Format B); near market; heavyweight opening support; Gate F1 stops scaling on failure	Franchise Director
FR-02	Regulatory breach in a disclosure regime (rescission, fines)	M	H	Local franchise counsel per market before any offer; no cross-border re-use of documents; disclosure calendar in OS compliance module	Franchise Director + counsel
FR-03	Trademark squatting in first-to-file markets	M	M	File marks in all first-wave markets (incl. local script) before negotiations; watch service	Counsel
FR-04	Safety incident at a franchise damages the whole brand	L	H	Non-negotiable CX/safety canon (Ch. 9.3), * audit items, Level-4 intervention powers incl. remote booking disablement, network-wide insurance minimums	Franchise Field Manager
FR-05	Franchisee defection: operates a copycat post-term	M	M	Controlled items (nodes, OS, credentials) terminate with license; trade-secret handling of node/OS internals; locally-enforceable non-competes — verify locally	Franchise Director
FR-06	RC WORLD OS multi-tenancy failure (outage or data leak across tenants)	L	H	Volume 13 tenancy isolation, RLS architecture, staged rollout, disaster-recovery drills; contractual SLA with service credits	CTO

#	Risk	L	I	Mitigation	Owner
FR-07	Procurement desk disruption (supplier or freight shock)	M	M	Volume 8 dual-sourcing doctrine; approved-equipment list carries alternates; franchisee emergency self-sourcing protocol with retrospective approval	Procurement desk
FR-08	Currency/repatriation stress on dollar-denominated fees	M	M	Master-franchise treasury structuring; technology-fee hedging clauses; royalty in local currency converted at contract mechanics — verify locally	CFO
FR-09	Franchisor support capacity lags unit growth (service quality dilution)	M	H	Staffing ratio rule (1 field manager per 6–8 units) hard-coded into growth gates; support-cost KPI reviewed at every gate	Franchise Director
FR-10	Master franchisee underinvests in territory after paying territory fee	M	H	Development schedules with unit-count milestones and claw-back/reversion of territory rights; master must operate a unit itself	Franchise Director

13. Volume Summary & Cross-References

This volume converts the RC WORLD system into a replicable, licensable product on a disciplined timeline: **no franchise sold before the flagship has two full years of audited operating data** (earliest signing ≈ month 36, pilot opening ≈ month 54, inside the canonical Phase 3 window of months 31–60). Three engineered formats are offered — **Full Park (4–5 ha, ≈\$2.7–3.1 M funding, ~195 assets)**, **Compact Urban (~1.5 ha, motorsport + construction only, ≈\$1.6–1.95 M, ~120 assets)**, and **Indoor Arena (6,000–9,000 m², ≈\$2.0–2.45 M, ~100 assets)** — all carrying the invariant core: the Shift, Toolbox Talk, the Tow-Truck Retrieval Protocol, License/Gears portability, RCW Node telemetry, LiPo doctrine, and RC WORLD OS as sole platform. The fee model is deliberately recurring-weighted: **initial fees of \$90–150 k by format, 6% royalty, 2% marketing fund, and a technology fee of \$2,000/month plus \$8 per active asset**, with an 8–12% disclosed procurement

margin via central purchasing. A worked Format B example shows a franchisee earning $\approx$ 18.7% post-fee EBITDA at maturity on an $\approx$ \$1.8 M investment, while each mature unit contributes $\approx$ \$130-250 k of recurring revenue to the franchisor. Expansion proceeds through evidence gates — one pilot, then three regional units, then master-franchise territories — prioritizing the Gulf and Southeast Asia in the first international wave. All legal mechanics are design intentions to be implemented with local franchise counsel in every market.

Cross-references.

- **Volume 1** — Executive Master Plan: phasing canon this roadmap is sequenced against; the equity story the franchise option supports.
 - **Volume 2** — Market research: the market-sizing ranges and competitive landscape behind Chapter 3's prioritization.
 - **Volumes 3-6** — Division manuals: the operating content licensed to franchisees and taught in Academy division modules.
 - **Volume 7** — Engineering & Workshop Manual: the Artisan certification syllabus source; battery, spares, and hardening doctrine enforced by the audit.
 - **Volume 8** — Procurement Handbook: the supplier network the central procurement desk commercializes; RCW Node fabrication control.
 - **Volume 9** — Customer Experience & Loyalty: the License/Gears system whose network portability Chapter 9 protects; service standards audited in Chapter 10.
 - **Volume 10** — Finance: the canonical baseline every format's economics are derived from; the Phase 3 budget carrying the franchisor build-out.
 - **Volume 11** — Architecture & Park Design: the design standards parameterized into format kits and enforced at gates G0-G4.
 - **Volume 13** — RC WORLD OS: multi-tenancy, the franchisee role, multi-site dashboards, marketing-fund ledger, and the telemetry/compliance enforcement this volume's audit regime depends on.
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Volume 13 — IT, IoT & ERP Systems (RC WORLD OS)

RC WORLD — Master Development Plan · Volume 13 of 13 **Revision:** 1.0 · **Date:** July 2026 · **Status:** Living document — bump revision on material change

Purpose of this volume. This volume is the complete technical specification and development plan for **RC WORLD OS** — the single software platform that runs every sector of the business: finance and accounting, HR and payroll, online payments and the closed-loop wallet, bookings and queue management, fleet and maintenance management, live IoT telemetry from every powered asset, POS/retail/F&B, CRM and gamification, corporate events, access control, document management, and franchise multi-site operations. It specifies the architecture (Supabase/PostgreSQL + Kotlin services; Kotlin Multiplatform / Compose Multiplatform client apps on **both Android and iOS**; a web admin console), the founder-mandated **role-based access control** design under which every user class — executive, finance, HR, operations, division lead, Artisan, marshal, front-of-house, F&B, franchisee, customer — sees a different interface scoped to its permissions, the consolidated data model, the non-functional requirements, and the sprint-level implementation roadmap that delivers a working v1 before Phase 1 opening. A CTO or lead engineer hired into RC WORLD should be able to scope, staff, and ship the platform from this volume alone, honouring the telemetry and schema doctrine established in the original engineering blueprint.

Intended readers. The CTO / lead engineer and the RC WORLD OS development team; the founder and General Manager (as product owners); the finance manager (modules 5 and 8); investors performing technical due diligence; franchise operators evaluating the platform they will licence (with Volume 12); The Works engineering staff who maintain the RCW Node fleet.

Chapters

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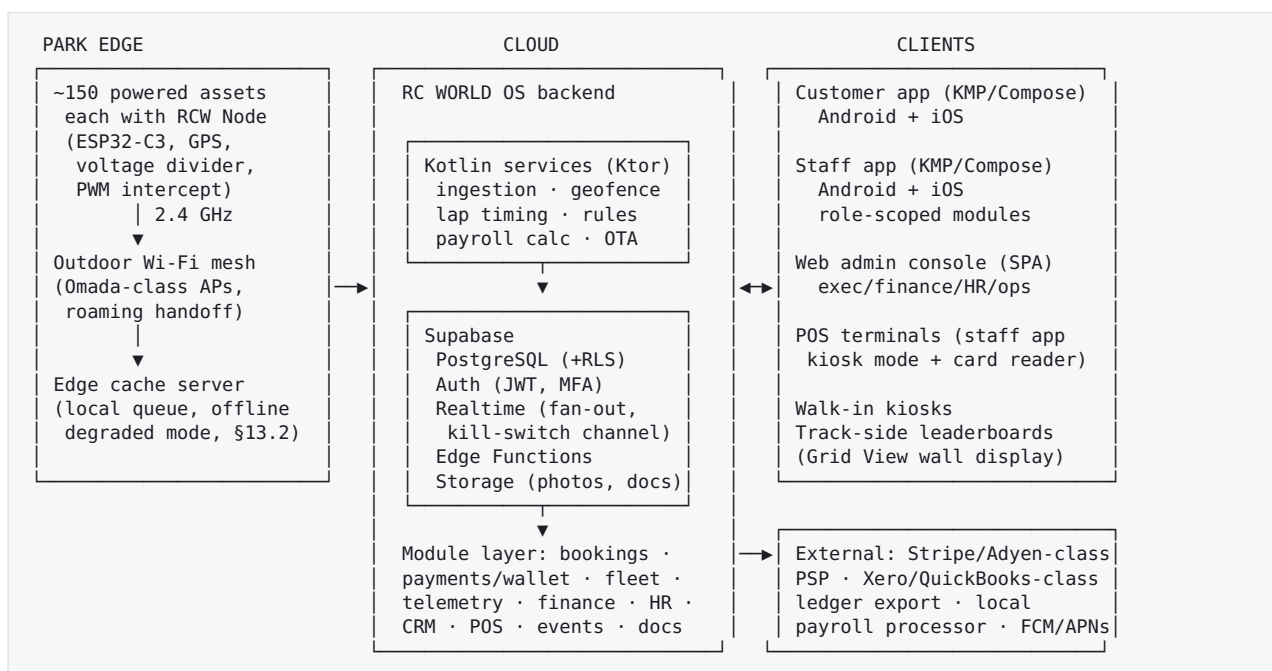
1. Platform Vision & Architecture Overview

1.1 One operating system for the whole business

RC WORLD is a miniaturized industrial complex, and it is run like one: by a single integrated operating system, not a patchwork of disconnected SaaS subscriptions. RC WORLD OS is that system. The same platform that streams a 1 Hz voltage feed from a \$60 buggy on Track B also computes the marshal roster for Saturday, recognizes the revenue on a \$99/month Foreman membership, decrements a brass pinion from The Works' spares shelf when an Artisan closes a work order, and shows the founder a live P&L. This is a deliberate strategic position, inherited from the original engineering blueprint: **high utilization of commercial-grade equipment is achieved through software** — telemetry that knows where every asset is and how healthy it is, bookings that keep every track slot sold, and gamification that brings the customer back.

The integration is the moat. A generic booking tool cannot offer a Top-Up button at T-5 minutes because it cannot see `queue_roster`. A generic maintenance CRM cannot open a work order because a motor's runtime crossed a greasing threshold, because it cannot see `live_telemetry`. A generic loyalty app cannot award a "Smooth Operator" badge for a low voltage-sag score, because it never sees a voltage curve. RC WORLD OS can do all three, on one customer record, one asset record, and one ledger.

1.2 Architecture at a glance



Data flows left to right for telemetry (RCW Nodes → mesh → Kotlin ingestion → Postgres → Realtime fan-out → apps) and right to left for commands (admin app → Realtime kill-switch channel → node → PWM intercept → ESC to neutral). Every module reads and writes the same PostgreSQL schema under Row Level Security, so “different interfaces for different users” (Chapter 3) is enforced in the database, not merely hidden in the UI.

1.3 Build-vs-buy policy

The platform’s scope is large; the engineering team is small. The policy that reconciles the two:

Capability	Decision	Rationale
Telemetry ingestion, kill-switch, lap timing, geofencing	Build	Core differentiator; no product on the market intercepts PWM on a 1/14 excavator
Bookings, Shift inventory, queue_roster, Top-Up flow	Build	The 20-minute Shift and T-5 Top-Up are proprietary revenue mechanics
Gamification: XP, licenses, badges, leaderboards	Build	Driven by telemetry events only we have; the loyalty moat
Closed-loop wallet + Gears ledger	Build (double-entry core)	Wallet liability accounting must integrate with our finance module
Card processing, Apple Pay / Google Pay	Buy — Stripe-class / Adyen-class PSP	Never touch PANs; tokenization keeps us out of PCI-DSS SAQ D (\$5.6)
Accounting ledger (statutory books)	Buy — Xero/QuickBooks-class, fed by our GL-lite export	Tax filing and audit belong in a mature accounting product
Payroll processing (tax tables, filings)	Buy — local payroll processor; we compute hours and export	Jurisdiction-specific tax logic is a liability, not a differentiator
Email/push delivery	Buy — FCM/APNs, SES/Postmark-class email	Commodity infrastructure
Auth, database, realtime transport	Platform — Supabase	Managed Postgres + Auth + Realtime is the fastest safe path
CCTV VMS, card readers, receipt printers	Buy + integrate	Hardware ecosystems; we integrate via APIs (\$11.4)

Field Note. The test we apply before writing any line of code: does this feature touch a Shift, an asset, or a customer’s progression? If yes, build — it will need our data and our doctrine. If no, it is commodity; buy it and spend the saved sprint on telemetry reliability, which is the one system that can close the park if it fails.

1.4 Platform principles

1. **One schema, one truth.** Every module writes to the same PostgreSQL database. No CSV shuttling between systems inside the park; exports exist only at the boundary (accounting, payroll).
2. **RLS is the security perimeter.** UI hiding is a courtesy; Row Level Security is the enforcement. Every table carries policies; no client ever holds a service-role key.
3. **The park must run when the internet does not.** Telemetry safety (kill on under-voltage/geofence) executes on the node itself; check-in, POS, and queue continue from the edge cache server (§13.2).

4. **Site-scoped from day one.** Every business table carries `site_id`. Site 1 is the flagship; the franchise network (Volume 12) is a configuration change, not a rewrite.
5. **Auditability over cleverness.** Money, roster, and safety actions are append-only, attributed, and timestamped. If a regulator, insurer, or franchisee asks “who did what, when” the answer is one query.

2. Technology Stack & Development Approach

2.1 Client apps: Kotlin Multiplatform + Compose Multiplatform, Android and iOS

The original blueprint specified a Kotlin + Jetpack Compose native Android app. This volume extends that decision — explicitly, per the living-document rule — to **Kotlin Multiplatform (KMP) with Compose Multiplatform (CMP)**, producing native apps for **both Android and iOS from one shared codebase**. The founder’s requirement is unambiguous: client apps must be supported on both platforms, because roughly half of the target family and hobbyist demographic in a mid-size international city carries iPhones, and a customer who cannot install the app cannot complete the Toolbox Talk, hold a wallet, or join the queue.

Why KMP/CMP rather than two native teams or a JavaScript-based cross-platform stack:

- **Continuity.** The backend services are Kotlin; the original app plan was Kotlin; the team hires from one language pool. Domain logic — Shift timing rules, wallet arithmetic, XP evaluation, telemetry parsing — is written once in `commonMain` and unit-tested once.
- **Real native UI performance** for the live parts of the product: the Grid View map, the lap-timer HUD, and the live voltage gauge redraw at interactive frame rates; Compose Multiplatform renders via Skia on iOS with native performance characteristics.
- **Two apps, one codebase, shared modules.** We ship a **Customer app** and a **Staff app** as separate store listings (different review cycles, different permission requests), both built from the same repository with shared `core`, `design-system`, `telemetry`, `wallet`, and `api` modules. The staff app adds role-gated feature modules (Chapter 3).

What stays platform-specific (in `androidMain` / `iosMain` via `expect` / `actual`):

Concern	Android	iOS
Push notifications	Firebase Cloud Messaging	APNs
Wallet-sheet payments	Google Pay API	Apple Pay (PassKit)
QR/NFC scanning	CameraX + ML Kit-class scanner; NFC via Android NFC API	AVFoundation; Core NFC
Geofenced clock-in location	FusedLocationProvider	Core Location
Background telemetry viewing	Foreground service (staff app)	Background modes (audited narrowly for App Store review)
Secure credential storage	EncryptedSharedPreferences / Key-store	Keychain
Store distribution	Play Console (internal → closed → production tracks)	App Store Connect + TestFlight

Field Note. Budget App Store review friction into the schedule. The staff app’s kill-switch button and the customer app’s closed-loop wallet both attract reviewer questions (remote device control; stored value). Prepare a reviewer-facing demo video and a written explanation of the closed-loop wallet (single-merchant stored value, not a money-transmitter product) before the first TestFlight external build — this paperwork has a two-sprint lead time if handled reactively.

2.2 Web admin console

Deep back-office work — permission matrices, finance close, payroll review, franchise dashboards — belongs on a large screen. The web admin console is a single-page application: **TypeScript + React-class SPA** (Next.js-class framework), using the Supabase JavaScript client for auth/RLS-scoped queries and the Kotlin service APIs for heavy operations. Kotlin/JS and Compose for Web were evaluated and declined for v1: the web hiring pool, component ecosystem (tables, charts, form-heavy CRUD), and admin-dashboard tooling are materially stronger in the TypeScript ecosystem, and the console shares data contracts (generated from the OpenAPI spec and the Postgres schema) rather than UI code with the apps. Revisit at v2 if Compose for Web matures.

2.3 Supabase: what we use and how

Supabase component	Use in RC WORLD OS
PostgreSQL	The one schema (Chapter 12). Partitioned <code>live_telemetry</code> ; JSONB for flexible payloads (<code>parts_used</code> , <code>badge criteria</code>); <code>pgcrypto</code> for column encryption of sensitive HR fields
Auth	Email/phone + OAuth sign-in for customers; staff accounts provisioned by HR; JWT carries <code>app_role</code> and <code>site_id</code> custom claims (§3.4); TOTP MFA enforced for privileged roles
Realtime	Fan-out of telemetry rows to Grid View and customer dashboards; the kill-switch command channel (doctrine: KILL over Supabase Realtime); queue position updates; leaderboard pushes
Row Level Security	The enforcement layer for the entire RBAC design (Chapter 3)
Edge Functions	PSP webhooks (payment confirmed, refund settled), booking-confirmation email triggers, scheduled jobs (downsampling, cert-expiry alerts) where a full Kotlin service is overkill
Storage	Damage photos, staff documents, SOP PDFs, firmware binaries for OTA (signed, versioned)

2.4 Kotlin backend services

Everything that is compute-heavy, latency-sensitive, or long-running lives in **Kotlin services (Ktor-class framework)** deployed as containers alongside Supabase:

- **Ingestion service** — terminates node connections, validates/batches telemetry inserts (Chapter 7).
- **Rules engine** — geofence evaluation, under-voltage detection, lap-timing state machines, predictive-maintenance threshold evaluation, XP/badge triggers. One event-driven service consuming the telemetry stream.

- **Ledger service** — the only writer to wallet/Gears ledger tables; enforces double-entry invariants in one transactional code path (§5.3).
- **Payroll calculator** — hours/overtime computation runs (§9.4).
- **OTA manager** — firmware manifest signing and staged rollout (§7.8).
- **Export service** — accounting journal export, payroll export, BI extracts.

Services are stateless where possible, horizontally scalable, and communicate through Postgres (LISTEN/NOTIFY and outbox tables) rather than a separate message broker in v1 — one fewer system to operate at a single site.

2.5 Environments, CI/CD, testing

Three environments, each a separate Supabase project and container stack: **dev** (developers, seeded synthetic data), **staging** (park-identical config, a bench of 6 physical RCW Nodes in The Works wired to test vehicles), **prod**. Database schema migrates via versioned SQL migrations (sqldelight/Flyway-class tooling); RLS policies are migrations too, and are code-reviewed like application code.

CI/CD on a GitHub Actions-class pipeline:

1. Every PR: Kotlin compile for all targets, unit tests (`commonTest`), lint, RLS policy tests (pgTAP-class suite that asserts each role can/cannot read canary rows), SQL migration dry-run.
2. Merge to main: deploy dev; run integration suite (API contract tests, Maestro-class UI smoke flows on Android emulator + iOS simulator).
3. Tagged release: deploy staging; hardware-in-the-loop test against the bench nodes (telemetry round-trip, kill-switch latency assertion $\leq$ budget in §7.4); manual QA sign-off; then prod, with app binaries promoted through Play internal track / TestFlight.

Testing strategy pyramid: heavy unit coverage of domain logic in `commonMain` (wallet arithmetic, Shift-state transitions, XP rules — these are pure functions by design); pgTAP-class tests for every RLS policy; contract tests between apps and services from the shared OpenAPI spec; a small, ruthless set of end-to-end flows (book → pay → check in → drive → telemetry → Shift close → wallet debit) that run nightly against staging with bench hardware.

2.6 API design and offline-first client data

Two API surfaces, deliberately kept distinct. **Direct Supabase access** (PostgREST-style queries through the Supabase client, always under RLS) serves simple reads and own-record writes — browsing slots, reading a wallet balance, acknowledging a document. **Kotlin service APIs** (REST, OpenAPI-specified, JWT-authenticated with the same claims) serve anything transactional or multi-table: taking a payment, closing a work order, starting a Shift clock, issuing a kill command's sibling actions. The rule for choosing: if the operation must be atomic across tables or enforce an invariant, it goes through a service; if RLS alone expresses the rule, the client may query directly. The OpenAPI spec generates the Kotlin client in `commonMain` and the TypeScript client for the console, so contract drift is a compile error, not a production incident.

Clients are **offline-tolerant by design**, not by afterthought. The apps hold a local store (SQLDelight-class, shared in `commonMain`) with three data classes: always-cached (my bookings and QR codes, my wallet balance as-of, today's roster, SOPs I must acknowledge —

render with a staleness banner when unsynced), live-only (telemetry HUD, Grid View, queue positions — these show a connectivity state honestly rather than stale positions), and queued writes (clock-in, work-order notes, incident photos — written locally with client-generated UUIDs and replayed with idempotency keys). A marshal in a mesh dead spot can still photograph damage and write an incident; a customer in the car park can still show their booking QR at the gate. What no client ever does is queue a payment or a kill command — those are online-only actions with explicit failure UI, because a replayed stale kill is worse than a refused one.

2.7 Team plan, build phases, budget

The build runs on a compact senior team; the roadmap detail is Chapter 14.

Role	FTE (build)	FTE (run, Year 1+)
Tech lead / architect	1.0	0.5 (becomes head of platform)
KMP/Compose engineers	2.0	1.0
Kotlin backend engineer	1.0	1.0
Web (TypeScript) engineer	1.0	0.5
Firmware engineer (RCW Node, shared with The Works)	0.5	0.25
QA engineer	0.5	0.5
Product designer	0.5	0.25

Phasing: **MVP (v1)** ships before Phase 1 opening — bookings, wallet + payments, telemetry + kill-switch, fleet + maintenance, POS, RBAC core, Toolbox Talk (Chapter 14 defines the sprint plan). **v1.5** (Months 13–20) adds HR/payroll depth, finance close automation, CRM automations. **v2** (Months 21–36) adds franchise multi-tenancy activation, advanced analytics, and the Phase 2 division modules (aviation/marine mission scoring).

Budget, consistent with Volume 10's software line: **v1 development ≈ \$165k** (team-months at blended \$9–11k/month fully loaded in a favourable engineering market, plus \$12k hardware/bench/licences), inside Phase 1's \$1.85 M capex envelope; **run costs ≈ \$45k/year** (Supabase Pro-class tier, container hosting, PSP fixed fees, push/email volume, observability tooling, store accounts) plus the run-team payroll above, which Volume 10 carries under operating headcount. Volume 10 is the source of truth for the consolidated numbers; any change there sweeps back into this section.

3. Identity, Roles & Access Control

This is the chapter the founder mandated in the brief: RC WORLD OS must give **different interfaces to different users based on access level**, across every user class in the business. The design principle is stated once and enforced everywhere: **the role decides three things — what data the database will serve you (RLS), what actions the API will accept from you (permission checks), and what the app renders for you (role-scoped navigation)**. All three derive from the same role catalogue, so they can never disagree.

3.1 Role catalogue

Roles are assigned per user per site (a franchisee's ops manager has no rights at the flagship). One user may hold multiple roles (a division lead who also marshals); the app composes the union of permissions and offers a workspace switcher.

#	Role code	Title	Primary surface	Notes
1	executive	Founder / GM / Executive	Web console + staff app	Cross-module read; strategic KPIs; limited write (approvals)
2	finance_mgr	Finance manager	Web console	Ledger, reconciliation, refund approval, exports; separation-of-duties rules §3.7
3	hr_mgr	HR manager	Web console + staff app	Staff records, contracts, payroll runs, certifications
4	ops_mgr	Operations manager	Staff app + web console	Park-wide day control: rosters, capacity, incidents, weather closures
5	division_lead	Division lead (Motor-sport, Construction, Aviation, Marine)	Staff app	ops_mgr powers scoped to one division
6	artisan	Artisan / technician	Staff app	Work orders, parts, battery lifecycle, asset status transitions
7	marshal	Track marshal	Staff app	Live Grid View, kill-switch, incident reports, retrieval protocol, Shift start/stop
8	foh	Front-of-house / POS	Staff app (kiosk mode)	Check-in, walk-in bookings, wallet top-ups, retail POS
9	fnb	F&B staff	Staff app (kiosk mode)	F&B POS, stock counts; no access to bookings or fleet
10	franchisee	Franchisee owner	Web console + staff app	Full operator rights scoped to own site(s) ; sees franchisor-published doctrine read-only
11	franchise_staff	Franchise site staff	Staff app	Any of roles 4–9, site-scoped; provisioned by the franchisee
12	customer	Customer	Customer app	Own bookings, wallet, telemetry of own bound vehicle, progression

#	Role code	Title	Primary surface	Notes
13	guardian	Guardian (of minor customer)	Customer app	Manages linked minor profiles: consent, spend limits, visibility (\$10.6)
14	service	Machine identities	—	Ingestion service, kiosks, leaderboards; least-privilege API keys, never user JWTs

3.2 Permission matrix per module

Legend: **F** full (read/write/configure) · **W** read/write within scope · **R** read only · **O** own records only · — no access. “Scope” is always site-scoped for roles 4-11 and division-scoped for role 5.

Mod- ule	exec	fin- ance	HR	ops mgr	div lead	artis- an	mar- shal	FOH	F&B	fran- chisee	custom- er/ guardi- an
Book- ings & queue	R	R	—	F	W	R	W (start/ stop)	W	—	F (own site)	O
Pay- ments & wal- let	R	F	—	R	R	—	—	W (take pay- ment)	W (take pay- ment)	F (own site)	O
Re- funds	ap- prove >\$200	ap- prove ≤\$200; ex- ecute	—	request	request	—	re- quest (break- down)	re- quest ≤\$25 auto	—	per own- site policy	request
Fleet & main- ten- ance	R	R (costs)	—	W	W	F	R + status flags	—	—	F (own site)	O (bound vehicle)
Tele- metry & kill- switch	R	—	—	W + kill	W + kill	R + kill (bench test)	F + kill	—	—	W + kill (own site)	O (own session view)
Fin- ance & ac- count- ing	R	F	R (payroll GL)	R (own budget)	R (own budget)	—	—	—	—	F (own books)	—
HR & payroll	R (ag- greg- ates)	R (cost)	F	W (roster)	W (roster)	O + clock- in	O + clock- in	O + clock- in	O + clock- in	F (own staff)	—
CRM & gami- fication	R	—	—	W	W	—	R (cus- tomer license tier)	R (pro- file at desk)	—	F (own site)	O

Mod- ule	exec	fin- ance	HR	ops mgr	div lead	artis- an	mar- shal	FOH	F&B	fran- chisee	custom- er/ guardi- an
Retail/ F&B stock	R	R	—	W	—	—	—	W (re- tail)	W (F&B)	F (own site)	—
Corpor- ate events	R	R (in- voices)	—	F	W	—	—	R	R (ca- tering brief)	F (own site)	O (their event)
Docu- ments & SOPs	F	R	W (HR docs)	W	W	R + ac- know- ledge	R + ac- know- ledge	R + ac- know- ledge	R + ac- know- ledge	R (doc- trine) + W (local)	R (waivers, own)
System config & roles	R	—	grant roles ≤ mgr	—	—	—	—	—	—	grant own- site roles	—
Audit log	R	R (fin- ancial)	R (HR)	R (ops)	—	—	—	—	—	R (own site)	—

Role grants above manager level (`executive`, `finance_mgr`, franchise-network roles) require a second approver (§3.7). The matrix is stored in the `role_permissions` table (Chapter 12) and rendered in the admin console — the table above is the specification; the database is the implementation.

3.3 One platform, many interfaces: screen-by-screen

The founder's requirement is that each role sees a different application. In practice all staff roles run the same binary; at sign-in the app reads the JWT role claims and assembles a role-scoped navigation graph. What each role experiences:

Executive (web console + app). Home = the Volume 10 KPI wall: today's revenue vs. same-weekday average, park utilization by division (RevPASH-style, §8.5), live visitor count, fleet availability %, incident flag. Drill-down is read-only into every module. One write surface: the approvals inbox (refunds >\$200, purchase orders above threshold, role grants). No POS, no work orders — an executive who needs to edit a booking asks ops, by design.

Finance manager (web console). Home = cash position and reconciliation status: PSP settlement vs. ledger (§5.7), wallet liability balance, unposted journals, AP inbox with invoices awaiting three-way match (§8.3). Tabs: Ledger, Reconciliation, Refunds (their approval queue ≤\$200), Exports (Xero/QuickBooks-class journal, payroll GL), Budget vs. actuals. Cannot edit bookings, rosters, or fleet — but sees the revenue consequences of all three.

HR manager (web console + app). Home = compliance dashboard: expiring certifications (Volume 12 Academy feed), contracts awaiting signature, open positions, absence today. Tabs: Staff records, Contracts, Certifications & training, Payroll runs (compute → review → export, §9.4), Performance reviews, Safety-training records. Sees payroll cost; cannot see customer PII beyond aggregate counts.

Operations manager (staff app, tablet). Home = the day board: weather + minima flags, roster gaps, queue depth per division, fleet-readiness (assets green/amber/red), incident feed. Actions: open/close zones, adjust capacity rules, reassign marshals, trigger weather

closure (auto-credits interrupted Shifts per Marine/Aviation doctrine), approve comp Shifts. The kill-switch is available but two-tap-guarded — marshals are the primary users.

Division lead. The ops manager’s board filtered to one division, plus division KPIs (e.g. Motorsport parity spread from Volume 3, Chapter 14) and division roster editing.

Artisan (staff app, phone — one-hand operable, glove-friendly). Home = my work queue: open work orders sorted by priority, assets awaiting turnaround, PM tasks due today. Core loop: scan asset QR → work order card (history from `maintenance_logs`, predicted issues, parts suggestions) → log parts used (decrements stock) → photos → close with sign-off → asset status transition (§6.3). Battery screens: scan pack QR, log cycle, view internal-resistance trend. No visibility of prices beyond parts cost, no customer data beyond the booking name on a turnaround.

Marshal (staff app, phone, sunlight-readable dark theme). Home = my track now: Grid View of live vehicles on their assigned track with voltage/geofence state per vehicle, Shift clocks, queue next-up list. Actions: start/stop Shift clock, **kill-switch** (single vehicle, guarded swipe), throttle-limit command, launch Tow-Truck Retrieval Protocol (pauses the Shift clock automatically), incident report with photo. The kill-switch control renders only for marshals, division leads, and ops managers whose site+division scope matches the vehicle — enforced by RLS on the command table, not just UI.

Front-of-house / POS (staff app, kiosk mode on counter tablet). Home = check-in scanner. Flows: scan booking QR → verify Toolbox Talk complete → hand off to marshal queue; walk-in sale (§4.3); wallet top-up (card via PSP terminal); retail POS with barcode scan. Cash drawer reconciliation at close. Cannot browse customer records — a scanned QR or a searched booking reference opens exactly one customer card.

F&B staff. F&B POS catalogue, order queue, stock count screen. Nothing else — the role exists so a seasonal barista’s credentials are worthless if phished.

Franchisee owner. The full operator console — but every query is fenced to `site_id ∈ own_sites`. Additionally sees the doctrine layer read-only: franchisor-published SOPs, PM matrices, pricing floors (Volume 12), and their own site’s compliance score. The franchisor’s executive role sees cross-site dashboards (§11.6); the franchisee never sees another site’s data.

Customer (customer app). Home = next booking + wallet balance + license progress ring. Tabs: Book (Shift inventory per division), Wallet & Gears, Garage (live view of their bound vehicle only during a Shift: voltage bar, position dot, lap times), Progression (XP, badges, leaderboards), Profile (waivers, minors they guard). During an active Shift the home surface becomes the live HUD, with the T-5 Top-Up button appearing per doctrine when `queue_roster` shows no waiting customer for that class.

Guardian. The customer app plus a family tab: linked minor profiles, per-minor spend caps, consent toggles, session notifications (“Ava’s Shift started on Track C”), and the minor’s progression view. Minors under the digital-consent age have no independent login (§10.6).

3.4 Identity architecture and JWT claims

Supabase Auth issues JWTs. A `user_site_roles` table (user, site, role, granted_by, granted_at, revoked_at) is the single source of role truth; a Postgres function stamps custom claims into the JWT at token mint:

```
{
  "sub": "d3f6...", "email": "j.moyo@rcworld.example",
  "app_roles": [{ "site": 1, "role": "marshal" }, { "site": 1, "role": "artisan" }],
  "mfa_verified": true, "amr": ["password", "totp"]
}
```

Claims are convenience; the tables are truth. Every RLS policy joins live to `user_site_roles` (revocation is immediate, not token-expiry-delayed) and uses the claim only as a fast-path hint. Customer accounts self-register (email/phone/OAuth); staff accounts are provisioned by HR through an onboarding flow that creates the auth user, assigns roles, and issues a first-login enrolment link — staff never self-register.

3.5 RLS policy patterns

Four patterns cover the whole schema. Helper functions keep policies readable and testable:

```
-- Helper: does the current user hold a role at a site?
create or replace function auth_has_role(p_site bigint, p_roles text[])
returns boolean language sql stable security definer as $$
  select exists (
    select 1 from user_site_roles r
    where r.user_id = auth.uid() and r.site_id = p_site
    and r.role = any(p_roles) and r.revoked_at is null);
$$;
```

Pattern 1 — own-records (customer):

```
alter table bookings enable row level security;

create policy customer_own_bookings on bookings
  for select using (customer_id = auth.uid()
    or exists (select 1 from guardian_links g
      where g.guardian_id = auth.uid() and g.minor_id = bookings.customer_id));
```

Pattern 2 — site-scoped staff read, role-gated write:

```
create policy staff_read_fleet on fleet_inventory
  for select using (auth_has_role(site_id,
    array['executive', 'ops_mgr', 'division_lead', 'artisan', 'marshal', 'finance_mgr', 'franchisee']));

create policy artisan_update_fleet on fleet_inventory
  for update using (auth_has_role(site_id, array['artisan', 'ops_mgr', 'franchisee']))
  with check (status in ('available', 'bound', 'on_track', 'pit', 'maintenance', 'charging', 'retired'));
```

Pattern 3 — command tables (kill-switch): insert-only, tightly scoped, never updatable:

```
create policy marshal_kill_command on vehicle_commands
  for insert with check (
    command in ('KILL', 'LIMIT_20', 'RETURN_TO_PIT')
    and auth_has_role(site_id, array['marshal', 'division_lead', 'ops_mgr', 'franchisee'])
    and exists (select 1 from fleet_inventory f
      where f.id = vehicle_commands.vehicle_id and f.site_id = vehicle_commands.site_id));
-- No update/delete policies exist: commands are append-only and audited.
```

Pattern 4 — column-level privacy (HR): payroll-sensitive columns live in a separate `staff_compensation` table with policies granting only `hr_mgr`, `finance_mgr` (read), and the record's owner (own row read). Joining tables never expose pay data to rostering screens.

Every policy ships with a pgTAP-class test asserting both the allow and the deny case; a policy without a deny test does not merge.

3.6 Audit logging

A single `audit_log` table (append-only; insert via trigger and service code, no update/delete grants to anyone including `service` roles) records: actor, role used, site, action, entity type/id, before/after JSONB diff for sensitive tables (roles, refunds, payroll, price config, firmware rollouts), device id, IP. Kill-switch commands, refunds, role grants, payroll approvals, and firmware promotions are always audited. Retention: 7 years for financial actions, 3 years otherwise (§13.6). The executive and franchisee consoles render scoped audit views; nobody edits history.

3.7 Separation of duties for financial actions

Action	Initiator	Approver	Rule enforced in
Refund $\leq$ \$25 (breakdown protocol auto-credit)	System / FOH / marshal	Automatic, logged	Ledger service
Refund $\leq$ \$200	Any operational role	<code>finance_mgr</code>	Ledger service state machine
Refund $>$ \$200	<code>finance_mgr</code>	<code>executive</code>	Ledger service state machine
Supplier invoice payment	<code>finance_mgr</code> (three-way match, §8.3)	<code>executive</code> above PO threshold	AP workflow
Payroll run release	<code>hr_mgr</code> computes	<code>finance_mgr</code> releases export	Payroll module
Price/config change (Shift prices, membership)	<code>ops_mgr</code> proposes	<code>executive</code> publishes	Config versioning
Role grant $\geq$ manager level	<code>hr_mgr</code> / <code>franchisee</code>	<code>executive</code> (or franchisor for franchise sites)	Role-grant workflow

The invariant: **no single account can create a payee and pay it, or grant itself a role, or approve its own refund.** The approval workflows check `initiator` $\neq$ `approver` at the database level.

3.8 Session and device security

- **MFA (TOTP)** mandatory for `executive`, `finance_mgr`, `hr_mgr`, `ops_mgr`, `franchisee`, and any role holding kill-switch rights; enforced at Supabase Auth level (`amr` checked in RLS for privileged writes).
- **Session lifetimes:** customer 30 days refresh; staff app 12 hours (a shift); web console 8 hours with 15-minute idle lock on finance/HR screens.
- **Kiosk mode:** FOH/F&B tablets run pinned single-app mode, device-enrolled (MDM-class), auto-logout on 90 seconds idle, no credential storage beyond the device-bound session.
- **Device registry:** staff app installs register the device; ops can remote-revoke a lost phone's sessions in one action.
- **Shared-terminal rule:** POS operators re-authenticate with a 6-digit PIN per transaction batch; the till knows who rang every sale.

Field Note. Resist the temptation to give the founder a super-admin bypass “just for launch week”. The first role the RBAC system must constrain is the most senior one — if the founder’s account can silently edit a ledger row, the audit log is theatre and the franchise pitch (Volume 12) loses its best slide: the platform itself enforces the doctrine.

3.9 Worked example: one breakdown, six interfaces

To make “different interfaces for different users” concrete, trace a single physical event — a buggy dies mid-Shift on Track B — through every role that touches it:

1. **Customer.** Their live HUD shows the voltage bar collapse, then a friendly breakdown card: “Your buggy’s down — time to drive the tow truck.” The Shift clock visibly pauses (doctrine: retrieval is never billed). They see only their own vehicle and their own session — RLS pattern 1.
2. **Marshal.** Their Grid View flashes the vehicle amber (under-voltage kill fired node-locally); the vehicle card offers Launch Retrieval Protocol. One tap pauses the Shift clock, sounds nothing (the 85 dB buzzer already fired on the vehicle per doctrine), assigns a recovery crawler, and opens an incident stub. The marshal never sees the customer’s wallet, spend, or contact details — only name, license tier, and session.
3. **Artisan.** The buggy lands in their work queue as it enters `pit → maintenance`: a work order pre-populated with the telemetry trace (voltage curve for the last 3 minutes), the model’s likely-failure list, and suggested parts. They see cost of parts, not prices of Shifts.
4. **Operations manager.** The day board ticks one fleet-availability point down and shows the incident in the feed. If a fresh vehicle wasn’t issued within 5 minutes, they also see the auto-credit event. They can reassign the venue unit’s capacity if the class is now short.
5. **Finance manager.** Nothing, today — the $\leq \$25$ auto-credit posted automatically and appears only in tomorrow’s reconciliation pack as one line among many. That absence is the design: finance sees money movements, not track drama.
6. **Executive.** Nothing individually — unless this is the third breakdown of that model this week, in which case the maintenance-cost-per-vehicle-hour KPI trend line moves and the weekly review asks The Works why.

Same event, same rows in the same database — six different renderings, each scoped by RLS to exactly what that role needs to act on. No role saw more; every role saw enough. This walkthrough is also the acceptance test for the RBAC epic in Sprint 1 (Chapter 14): the demo script executes precisely this scenario on the staging bench with six logged-in devices.

4. Bookings & Scheduling Module

4.1 Session inventory model

Bookings sell **Shift slots**: the intersection of a venue unit (track, zone pod, airfield cage slot, marina berth-block), a vehicle class, and a 20-minute time block — the Shift, the park’s atomic billing unit per canon. The inventory tree:

```

site → division → venue_unit (Track A / Mining pod 1-4 / cage slot / basin block)
      → vehicle_class (touring, drift, excavator, tractor, tug, ...)
      → session_slot (date, block_start, capacity, price_band)

```

Capacity per slot = simultaneously operable vehicles of that class on that venue unit (e.g. Track A touring: 8; Mining pod: 1 excavator + 3 dump trucks per the 1:3 doctrine — the pod is sold as linked slots). Slots are generated 28 days ahead by the capacity rules engine (§4.5) and priced by the pricing engine (§4.6). An Operator Shift books two consecutive blocks and reserves an Artisan battery-swap task in the pit-lane roster automatically.

4.2 Booking flows

App booking (primary). Browse division → class → day grid of slots with live availability → select → pay from wallet or card → booking QR issued. First-time customers are gated: no booking completes until the **Toolbox Talk** induction module for that division is passed and the waiver signed in-app (doctrine; the waiver PDF lands in Storage against the customer record).

Walk-in kiosk / FOH. FOH sells from the same inventory — there is no “walk-in allocation” that can double-book against the app. Kiosk flow: pick class → next available slot → pay (card/cash/wallet) → print/SMS the QR. Toolbox Talk runs on the kiosk for first-timers.

Corporate & party bookings. A `group_booking` reserves multiple venue units across a window (e.g. 2 h, 20 pax corporate per the \$1,400 anchor), takes a configurable deposit (default 30%) through the payments module, holds inventory in held state until final payment T-7 days, and generates a run-sheet for ops (marshal assignments, catering brief to F&B, per-guest Toolbox Talk links sent ahead). Party packages auto-attach retail/F&B bundles.

4.3 Queue management and the Top-Up flow

`queue_roster` (doctrine table) is the live truth of who is waiting for which class at which venue unit. Rows: customer, class, venue unit, joined_at, state (`waiting` → `called` → `checked_in` → `expired`). Walk-ups without bookings join the queue from the app or kiosk and see a live position + estimated wait computed from Shift clocks.

The T-5 Top-Up (doctrine, implemented exactly): at five minutes before Shift end, the rules engine queries `queue_roster` for the vehicle’s class at that venue unit. If no customer is waiting, the customer’s live HUD renders the **Top-Up Session** button; one tap debits the wallet for another block and extends the Shift clock without the vehicle leaving the track. If a queue exists, the button never renders and the turnaround proceeds. The check re-runs at confirmation time to close the race window (a walk-up joining at T-4:50 wins over the top-up).

4.4 Shift lifecycle

```

booked —check-in→ vehicle_bound —first stick input→ running → completed
  | (QR at FOH)      | (QR on transmitter,      |
  |                  | fleet status → bound)    |
  | cancelled (per policy: free ≥24 h, 50% ≥2 h) |
  | on fresh vehicle |                          |
  | no_show (15 min grace, slot released to queue) |
  |                  | interrupted_breakdown → retrieval → resumed
  |                  | interrupted_weather → auto-credit (doctrine)

```

Shift clocks live server-side (Kotlin rules engine) with the marshal app as start/stop authority; the customer HUD mirrors them via Realtime. Breakdown interruptions pause the clock

automatically when the marshal launches the Tow-Truck Retrieval Protocol — retrieval time is never billed, park-wide doctrine.

4.5 Capacity rules engine

Declarative rules evaluated at slot generation and at every booking attempt: max vehicles per venue unit per class; excavator:truck 1:3 pod linkage; license-tier gates (premium classes require the RC WORLD License tier from Volume 9 — the booking UI hides what the customer cannot yet book and shows the progression path instead); weather minima pre-closures (aviation wind bands, marine 30/30 lightning rule feed from division dashboards); maintenance blackouts (a venue unit under grooming per Volume 4's register generates no slots); staffing constraint (slots require a rostered marshal for that venue unit — the roster module (§9.3) feeds capacity, so an un-staffed track sells zero Shifts rather than creating a morning surprise).

4.6 Pricing engine

Price bands resolve in order: base class price (canon: Casual \$15/\$22 premium; Operator \$26/\$38) → peak/off-peak calendar (weekend/holiday multiplier, school-term matrix maintained by ops, published by executive per §3.7) → membership entitlements (Apprentice/Operator/Foreman monthly Shift allowances and % discounts per Volume 9 — entitlement consumption is a wallet-ledger event, §5.3) → promotions (code or segment-targeted, CRM module). Every booking stores the full resolved price breakdown JSONB for finance (revenue recognition needs the components, §8.2).

4.7 Booking communications

Every booking state change has a defined communication, dispatched through the CRM module's channels with consent respected (§10.5): confirmation (push + email with QR and Toolbox Talk link if outstanding), T-24 h reminder with weather note for outdoor divisions, T-2 h "you're up soon" push with live queue context, weather-closure auto-credit notice (doctrine: no refunds debated at the water's edge — the app credits first and explains immediately), and post-visit summary (lap times, tonnage, XP earned, next-tier progress — the retention hook Volume 9 specifies). Group bookings add organizer-facing run-sheet updates and per-guest induction chasers at T-7/T-2 days; an organizer can see which guests haven't completed the Toolbox Talk, because a corporate event where eight of twenty guests hit the induction wall at the gate is an operational failure the software should have prevented the week before.

Field Note. The T-2 h push carries the single highest-leverage sentence in the product: the live wait estimate. Customers who arrive to a known 15-minute queue rate the experience higher than customers who arrive to an unknown 5-minute one. Never suppress the estimate because it looks bad — suppressing it converts a scheduling problem into a trust problem.

5. Payments & Wallet Module

5.1 Payment surfaces

- **In-app:** card via Stripe-class / Adyen-class PSP mobile SDK; **Apple Pay** (iOS) and **Google Pay** (Android) as first-class sheets — family customers at a kiosk queue convert measurably better with wallet-sheet payments than with card entry.
- **POS terminals:** PSP-provided card readers (Stripe Terminal-class) paired to FOH/F&B kiosk tablets; cash accepted at FOH with drawer reconciliation.
- **Invoice:** corporate events pay deposit + balance by card link or bank transfer; bank transfers reconcile in the finance module (§8.3).

5.2 The closed-loop wallet

Every customer has a **wallet** (stored value, USD) and a **Gears balance** (loyalty points, earn-only from behaviour, spend on defined rewards — Volume 9 owns the earn/burn rules). The wallet is closed-loop: loadable, spendable only inside RC WORLD, refundable to source per policy. This keeps the product outside money-transmitter licensing in most jurisdictions — legal review per site is a franchise-launch checklist item (Volume 12).

5.3 Double-entry ledger design

All value movements — wallet, Gears, and the park's own revenue/liability view — post to one **double-entry ledger**. Two tables: `ledger_transactions` (id, site, occurred_at, type, reference, initiator, metadata JSONB) and `ledger_entries` (transaction_id, account_id, direction debit/credit, amount, currency USD | GEARS). Invariant enforced by the ledger service and a database constraint trigger: entries per transaction per currency sum to zero.

Core accounts:

Account	Type	Meaning
<code>customer_wallet:{id}</code>	Liability	Value the park owes this customer
<code>wallet_liability_control</code>	Liability (control)	Sum of all wallets — the finance module's wallet-liability line
<code>psp_clearing</code>	Asset	Card money captured, not yet settled to bank
<code>cash_drawer:{till}</code>	Asset	Physical cash by till
<code>revenue:{product_type}</code>	Revenue	Shifts, memberships, retail, F&B, events — split per §8.2
<code>gears_liability / gears_expense</code>	Liability / Expense	Points issued vs. redeemed, valued at standard cost
<code>refunds_payable</code>	Liability	Approved, not yet settled to source

Worked example — customer loads \$50 by card, then books a \$15 Casual Shift from wallet:

Txn	Debit	Credit	Amount
Wallet load	<code>psp_clearing</code>	<code>customer_wallet:123</code>	\$50.00
(PSP fee accrual)	<code>expense:psp_fees</code>	<code>psp_clearing</code>	\$1.55

Txn	Debit	Credit	Amount
Shift purchase	customer_wallet:123	de-ferred_revenue:shifts	\$15.00
Shift delivered (clock completes)	de-ferred_revenue:shifts	revenue:shifts	\$15.00
Gears earned (e.g. 15 Gears)	gears_expense	gears_liability	15 GEARS

The **ledger service is the only writer**; every other module requests postings through its API. Balances are materialized views refreshed transactionally — no module ever computes a balance by summing on the fly at POS speed.

5.4 POS integration

Retail and F&B sales post to the same ledger (`revenue:retail` , `revenue:fnb`) with basket lines captured for stock decrement (§11.1). Mixed tenders (part wallet, part card, part cash) are one transaction with multiple asset-side entries. Offline POS behaviour: the edge cache server (§13.2) queues card-present authorizations per PSP offline rules and cash sales unconditionally, replaying to the ledger on reconnect with idempotency keys.

5.5 Refunds and the breakdown protocol

Refund paths honour the separation-of-duties table (§3.7). The **breakdown protocol** path is automatic by doctrine: when a Shift is interrupted by vehicle failure and the Tow-Truck Retrieval Protocol runs, the paused clock means the customer loses nothing — but if a fresh vehicle cannot be issued within 5 minutes, the system auto-credits the remaining block value to the wallet and flags the incident to the division lead. Card refunds always return to the original payment method via the PSP; wallet credits are the default and preferred instrument (faster, retains the value in-park).

5.6 PCI-DSS scope minimization

RC WORLD OS **never sees a PAN**. Card entry happens in PSP-hosted fields/SDK sheets (app), PSP terminals (POS), and PSP payment links (invoices); we store only tokens, last-4, and brand. This keeps the platform at SAQ A / SAQ A-EP-class scope rather than SAQ D. Rules that keep it that way: no card data in logs (log scrubber tested in CI), no “card on file” outside PSP vault tokens, webhook signatures verified in Edge Functions, PSP keys in the secrets manager (§13.5) — never in client binaries.

5.7 Reconciliation to finance

Nightly job produces the reconciliation pack: PSP settlement report vs. `psp_clearing` movements (match on PSP transaction ids); cash drawer counts vs. `cash_drawer:*` per till per operator; wallet liability control vs. sum of wallet balances (must equal to the cent — any drift is a sev-2 incident, because it means the ledger invariant broke); Gears liability movement. Exceptions land in the finance manager’s morning queue. Month-end, the pack feeds the GL export (§8.1).

Field Note. Build the reconciliation before the first real payment, not after the first discrepancy. A wallet product without daily liability reconciliation is an unaudited bank. The nightly zero-check on `wallet_liability_control` is ten lines of SQL and it is the single highest-value alarm in the finance module.

6. Fleet Management & Maintenance Module

6.1 Asset registry

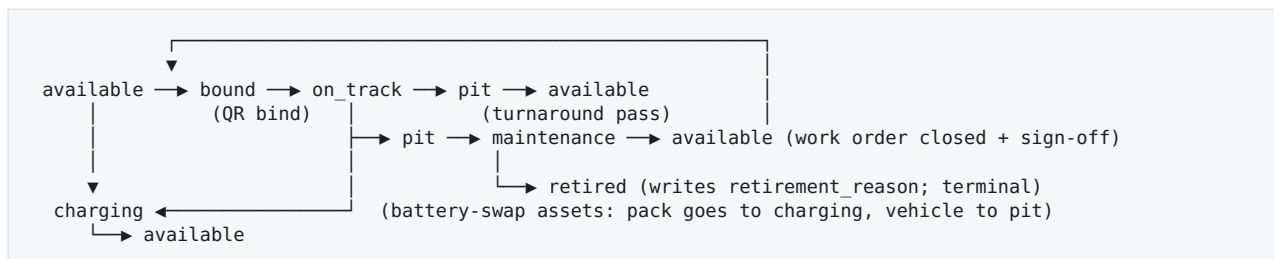
The doctrine table `fleet_inventory` is the registry, extended: `id`, `site_id`, `mac_address` (RCW Node), `asset_tag`, `model_id` (FK to a `vehicle_models` catalogue carrying class, division, scale, wholesale cost, spares profile), `status` (state machine below), `total_hours_run`, `total_shifts`, `purchase_date`, `warranty_until`, `retirement_reason`. Sub-assets (motors, ESCs, servos, the RCW Node itself) register in `asset_components` so a binned motor's history (Volume 3, §4.3 readback discipline) follows it across chassis. Batteries are first-class assets in `battery_packs` (§6.7), never components — they move between vehicles hourly.

6.2 QR/NFC asset tags

Every powered asset, battery pack, and transmitter carries a QR (laser-etched tag; NFC sticker optional where gloves make scanning awkward — Heavy fleet mudguards). The QR encodes only the asset UUID; everything else resolves server-side, so tags never leak data and never go stale. Scans drive the three core loops: customer **vehicle binding** at Shift start (scan transmitter QR → session binds customer→transmitter→vehicle, doctrine), Artisan work orders, and battery cycle logging.

6.3 Status state machine

The original schema's ENUM (active/maintenance/charging) is **extended — explicitly superseding the digest's three-state list** — to the operational reality:



Transition rules enforced in the database (trigger validates legal transitions) and in RLS (`artisan` may move anything; `marshal` may move `on_track` → `pit` and flag `maintenance`; nobody moves `maintenance` → `available` except the closing Artisan via a signed work order — the Volume 3 handshake: no car returns to active without sign-off). Every transition is timestamped; the intervals are the utilization analytics (§8.5).

6.4 Runtime hours and cycle counting

`total_hours_run` accrues from telemetry, not manual logging: the rules engine sums `on_track` time from Shift clocks and adds motion-detected bench time. Cycle counters per class (Shifts, battery swaps, crash-flag events from accelerometer-inferred telemetry anomalies)

feed the PM engine. Doctrine honoured: predictive maintenance from logged runtime hours drives gear-greasing and motor-commutator inspection flags.

6.5 Predictive maintenance and work orders

The PM engine evaluates Volume 7's preventive-maintenance matrices as data, not code: `pm_rules` rows carry (model_id or class, metric ∈ {hours_run, shifts, battery_cycles, days}, threshold, task template, priority). Nightly evaluation + real-time triggers (crash flag, marshal defect flag, three same-symptom repairs in 30 days → engineering review per Volume 6's retirement doctrine) generate **work orders**: asset, type (PM/repair/rebuild/inspection), priority, tasks checklist, parts suggestions from the model's spares profile. The Artisan loop (§3.3) closes them with `parts_used` JSONB (doctrine field), labour minutes, photos, and sign-off. `maintenance_logs` remains the historical record exactly as the digest defines it.

6.6 Parts inventory and procurement link

`parts_stock` per site: SKU (Volume 8 catalogue), bin location in The Works, qty, min/max, unit landed cost, supplier lead time. Work-order `parts_used` decrements stock transactionally; crossing min fires a reorder proposal into the procurement workflow (Volume 8 owns supplier terms; the 4–8 week China-direct lead time is priced into the min levels per Volume 4, §parts doctrine). Three-way match on receipt (§8.3). Stock counts: cycle-count tasks generated weekly for high-turnover bins (pinions, A-arms, tires per the digest's turnover ratings).

6.7 Battery lifecycle tracking

`battery_packs`: id, QR, chemistry/config (2S/3S per canon), `capacity_mah`, connector (XT60 canon), `purchase_date`, `cycle_count`, `last_internal_resistance_mohm` (per cell, JSONB), status (in_service/charging/resting/quarantine/retired). Every swap logs a cycle against pack QR + vehicle asset tag (Volume 4 checklist doctrine). Smart-charger integrations (SkyRC/ISDT-class with logging) or manual IR entry feed the IR trend; retirement triggers: cycle count > model threshold, IR rise > 40% from baseline, physical damage flag (→ quarantine crock per Volume 5 doctrine, then documented disposal). The bunker dashboard shows the 3:1 pool health per class: packs available vs. vehicles active — the morning number that predicts whether the afternoon keeps its Shift schedule.

6.8 Damage and incident reports

Marshal or Artisan files from the app: asset, category (collision/water/electrical/wear/customer-misuse), photos (Storage), Shift/customer linkage if applicable, cost estimate on close. Incident reports link to CCTV bookmarks (§11.4) and feed three consumers: the FMEA failure-data loop (Volume 7), insurance claims (documents module), and the customer-misuse flag on the CRM record (repeated deliberate abuse → license review per Volume 9).

7. Live Telemetry & IoT Platform

The telemetry platform is the system the park cannot open without. It carries the safety doctrine (kill on under-voltage <3.4 V/cell or geofence breach), the revenue doctrine (F1-style power management, lap timing), and the maintenance doctrine (runtime accrual). Design targets: **~150 concurrent nodes, 1 Hz standard payload, kill-command delivery inside a hard latency budget, graceful degradation when the mesh drops.**

7.1 Node payload and transport

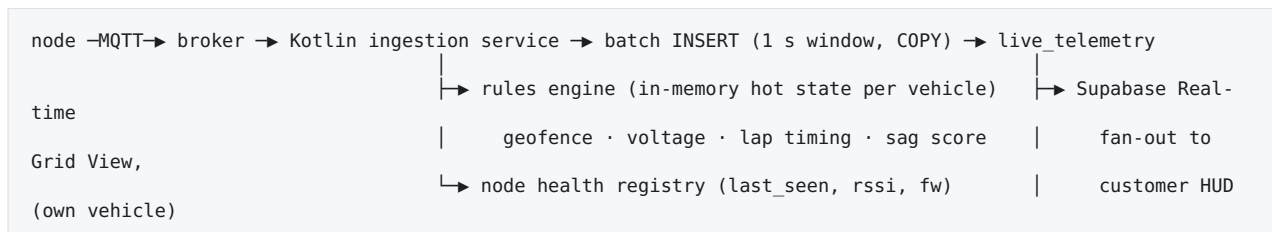
RCW Node payload (per doctrine hardware: ESP32-C3, ATGM336H GPS, voltage divider): `{mac, seq, ts, lat, lng, speed, v_batt_mv, v_cell_mv[], pwm_state, rssi, fw_version}` — ~90 bytes packed. At 1 Hz × 150 nodes this is trivial bandwidth; the engineering problem is connection churn across mesh AP handoffs, not throughput.

MQTT vs HTTP trade-off, decided: the nodes speak **MQTT over TLS** to a broker fronting the Kotlin ingestion service.

Criterion	MQTT	HTTP POST
Persistent session across AP handoff	✓ resumes with session state	✗ new TLS handshake per burst
Downlink (kill command)	✓ broker push on subscribed topic, milliseconds	✗ requires polling — unacceptable for kill latency
Power/CPU on ESP32-C3	✓ lightweight keep-alive	heavier per-message overhead
QoS semantics	✓ QoS 1 for commands (at-least-once + dedupe by seq)	manual retry logic
Ops complexity	broker to run (managed EMQX/HiveMQ-class or self-hosted Mosquitto-class)	none extra

Verdict: MQTT’s downlink is the deciding factor — the kill-switch path needs push, not poll. The doctrine phrase “KILL command over Supabase Realtime” maps as follows: the command originates over Supabase Realtime (marshal app → `vehicle_commands` insert → Realtime notifies the rules engine and Grid View) and is delivered to the node over MQTT; the node also holds autonomous local kill logic (§7.4) so cloud latency is never the last line of defence.

7.2 Ingestion pipeline



The ingestion service buffers one second of payloads and bulk-inserts — 150 rows/s as single-row inserts would be fine for Postgres but wasteful; batching keeps headroom for Phase 2 growth (~200 assets) and burst replays after mesh drops (nodes buffer up to 60 s of payloads in RAM and replay with original timestamps, flagged `replayed=true` so lap timing ignores them).

7.3 Time-series storage, downsampling, retention

`live_telemetry` is **partitioned by day** (native Postgres partitioning; a Timescale-class extension is the v2 upgrade path if Phase 2 volumes demand it). Retention policy:

Tier	Resolution	Retention	Use
Raw	1 Hz	14 days	incident replay, lap-timing audit, engineering diagnosis

Tier	Resolution	Retention	Use
Downsampled	1/60 Hz (per-minute min/avg/max voltage, distance, max speed)	13 months	maintenance analytics, sag-score history, season leaderboard audit
Session aggregates	per Shift (distance, energy proxy, sag score, laps, kills)	7 years	customer progression, finance, legal

Downsampling runs as a nightly Edge Function-scheduled job; raw partitions older than 14 days are dropped (cheap, instant). Session aggregates write at Shift close into `shift_telemetry_summary` — the table every other module actually joins against.

7.4 Geofencing and the kill-switch path, with latency budget

Geofences are polygons per venue unit (geofences table, PostGIS-class geometry) with per-class overlays (crawler course boundary differs from the buggy line on shared terrain). Evaluation is two-layer, per doctrine:

1. **Node-local (authoritative for safety):** each node holds its assigned fence (simplified polygon, ≤ 32 vertices) and the under-voltage threshold, pushed at binding time. Breach or < 3.4 V/cell $\rightarrow$ PWM intercept forces ESC to neutral/brake locally, then reports. **Cloud outage cannot disable the safety net.**
2. **Cloud (authoritative for operations):** the rules engine re-evaluates every payload — catches nodes with stale fences, drives Grid View alarms, and executes marshal-initiated commands.

Marshal kill-command latency budget (tap $\rightarrow$ wheels stop):

Hop	Budget
Marshal app $\rightarrow$ Supabase (insert <code>vehicle_commands</code>)	≤ 150 ms
Realtime $\rightarrow$ rules engine pickup	≤ 100 ms
Rules engine $\rightarrow$ MQTT publish (QoS 1)	≤ 50 ms
Broker $\rightarrow$ node delivery (mesh RTT)	≤ 200 ms
Node PWM intercept actuation	≤ 50 ms
Total (p95)	≤ 550 ms; alarm if p95 > 800 ms over 5 min

The staging bench (§2.5) asserts this budget on every release. Node-local triggers actuate in < 100 ms with no network at all.

7.5 Lap timing from geofence checkpoints

Tracks define ordered **checkpoint lines** (start/finish + 2–3 sector gates). The rules engine runs a per-vehicle state machine: crossing detection by segment intersection between consecutive GPS fixes, with sequence validation (a lap counts only if sectors pass in order — GPS jitter cannot fake a lap) and a minimum-lap-time sanity floor per track. Accuracy: GPS at 1 Hz gives ± 1 –2 s lap resolution — adequate for rental leaderboards and license progression, honest about its limits. League racing (Volume 3, Chapter 13) uses the dedicated transponder timing loop (MYLAPS/EasyLap-class); its API feeds `lap_times` through the same table with `source='transponder'`, so leaderboards blend both with the source badge visible. This honours Volume 5's precedent of external timing feeding RC WORLD OS.

7.6 Voltage-sag scoring

The F1-style energy game (doctrine): the rules engine computes a rolling **sag score** per Shift — the integral of (loaded voltage drop below open-circuit baseline) weighted by throttle-correlated speed changes. Smooth drivers score low sag, keep more usable capacity, and earn the “Smooth Operator” badge family (Volume 9); aggressive drivers see their energy bar drain faster on the HUD and may hit the soft-limit ladder: at sustained sag beyond class threshold → HUD warning → 20% throttle limit command (doctrine) → pit order. Per-class calibration tables live with the parity data (Volume 3) so the game is fair across vehicles.

7.7 Grid View

The operational fleet map (doctrine: real-time grid map): Mapbox/Google Maps SDK-class base with the park’s own tile overlay (Volume 11 site plan), vehicles as class-coloured dots with voltage rings, geofence outlines, breach/kill flashes, queue depth per venue unit, and node-health badges (stale >5 s = amber, >30 s = red). Rendered in the staff app (marshal: own track; ops: whole park) and on the wall display in ops. Customer app renders a single-vehicle version — their bound vehicle only, RLS-enforced.

7.8 OTA firmware management

Nodes check a signed manifest on boot and daily: `firmware_releases` (version, target hardware A/B Micro/Heavy, binary in Storage, SHA-256, signature, rollout ring). Rollout rings: bench (staging nodes) → canary (5 park nodes, ops-selected quiet class) → fleet. A/B partition scheme on the ESP32-C3 with automatic rollback on failed boot health-check. Promotion between rings is an audited executive/lead action. A node never updates while `bound` or `on_track`.

7.9 Offline and degraded modes

Failure	Behaviour
Single AP down	Nodes roam to neighbours (mesh doctrine); brief telemetry gaps back-filled from node RAM buffer
Mesh partition / broker unreachable	Node-local safety active (fence + voltage kill autonomous); nodes buffer 60 s and replay; marshals fall back to line-of-sight control per division SOPs; Shift clocks continue server-side
Internet down, park LAN up	Edge cache server (§13.2) keeps check-in, queue, POS, and Grid View running from local state; telemetry stores locally and syncs on restore
Full power loss	Vehicles’ nodes still kill on voltage locally (powered downstream of main switch, doctrine); park emergency procedures (division volumes) take over

Field Note. The kill-switch’s most likely real-world failure is not the network — it is a node whose fence never updated after a vehicle moved classes. The binding flow therefore pushes and verifies the fence at every QR bind (node echoes fence checksum before status may enter `bound`). A vehicle that cannot confirm its fence does not go on track. This one check eliminates the entire class of “wrong geofence” incidents.

8. Finance & Accounting Module

8.1 General-ledger-lite and export

RC WORLD OS runs a **GL-lite**: the double-entry ledger (\$5.3) plus a chart-of-accounts mapping table, producing trial-balance-grade journals — but statutory books, tax, and audit live in a **Xero/QuickBooks-class** accounting system fed by a nightly journal export (API where available, reviewed CSV otherwise). The boundary is explicit: RC WORLD OS is the operational source of truth (every dollar traceable to a Shift, basket, or work order); the accounting package is the statutory source of truth. The mapping table is finance-manager-maintained and version-controlled; unmapped account activity blocks the export and pages finance rather than silently mis-posting.

8.2 Revenue recognition per product type

Product	Recognition rule	Ledger mechanics
Casual/Operator Shift	On delivery (Shift completed)	Sale posts to <code>deferred_revenue:shifts</code> ; clock completion releases to <code>revenue:shifts</code> (\$5.3 example)
Day Pass / bundles	On visit day, allocated across components	Deferred at sale; released at first check-in of the visit day
Memberships (Apprentice \$29 / Operator \$59 / Foreman \$99)	Ratably over month (MRR)	Monthly charge → <code>deferred_revenue:memberships</code> , daily release job; churn analytics read the same rows
Packages / multi-Shift credits	Per Shift consumed; breakage on expiry	Credit sits in wallet sub-account; expiry sweeps to <code>revenue:breakage</code> per published T&Cs
Wallet loads	Never revenue	Pure liability until spent — the discipline the control account enforces
Corporate events / parties	On event delivery; deposits are liabilities	Deposit → <code>customer_deposits</code> ; delivery releases full contract value
Retail / F&B	At sale	Direct to revenue with COGS posting from stock module

8.3 Accounts payable and procurement link

Supplier invoices (Volume 8 suppliers) enter by upload or email-in; the AP workflow runs **three-way match**: purchase order (from the reorder flow, §6.6) ↔ goods receipt (Works receiving scan) ↔ invoice. Matched invoices queue for payment per §3.7 approvals; mismatches route to procurement with the delta highlighted. Landed-cost allocation (freight, duty on China-direct consolidations) spreads across receipt lines so `parts_stock` unit costs stay honest — Volume 8's landed-cost methodology, executed here.

8.4 Budgeting vs. actuals

Budgets load per site × department × month (finance console, versioned; Volume 10's model is the source). Actuals accrue from the ledger automatically. Division leads see their own budget line (§3.2); variance beyond threshold flags on the executive dashboard. No spreadsheet round-trips: the budget lives where the actuals live.

8.5 KPI dashboard as queries

Volume 10's KPI set is defined as SQL against this schema so every number on the executive wall is reproducible:

KPI	Definition (schema terms)
RevPASH-style utilization	<code>revenue:shifts</code> per venue-unit-hour open: $\text{Shift revenue} \div \Sigma(\text{venue_unit open hours} \times \text{capacity})$, from <code>session_slots</code> × ledger
Maintenance cost per vehicle-hour	$(\text{work-order parts at landed cost} + \text{labour minutes} \times \text{loaded rate}) \div \Sigma \text{total_hours_run}$ delta, per class per month
Membership churn	$\text{memberships lapsed in month} \div \text{active at month start}$, from membership ledger releases; cohort view by join month
Wallet liability & breakage	<code>wallet_liability_control</code> balance; breakage sweep trend
Fleet availability	time-in-status share of <code>available + bound + on_track</code> vs. maintenance per class (from status-transition intervals, §6.3)
90-day second-visit rate (the demand-model canary; targets in Volume 10 §10.4)	$\text{customers with } \geq 2 \text{ visit days within 90 days of first} \div \text{first-visit cohort}$
Battery pool health	$\text{packs in service} \div (\text{vehicles active} \times 3)$ per class — the 3:1 doctrine as a live number

Each KPI ships as a versioned SQL view; the dashboard renders views, never ad-hoc queries — when a definition changes, the view changes in a reviewed migration and every consumer moves together.

9. HR & Payroll Module

9.1 Staff records, contracts, certifications

`staff_profiles` extends the auth user with employment data: position, department/division, start date, contract type, emergency contacts; compensation isolated per the column-privacy pattern (§3.5). Contracts and signed documents live in Storage against the record with e-signature capture. **Certifications** are the compliance backbone: every credential from the Volume 12 Academy — Artisan platform certifications (Volume 7 rebuild sign-offs), marshal safety tickets, LiPo-handling and bunker certification, first aid, FPV spotter rating (Volume 5), waders/water-work sign-off (Volume 6) — is a row with issue/expiry dates and evidence upload. The scheduler will not roster a staff member into a slot whose venue unit requires a certification they lack or that expires before the shift ends; expiring certs alert the staff member at T-30/T-14/T-7 days and the HR dashboard continuously.

9.2 Safety-training records

Toolbox-talk attendance (the staff kind, distinct from the customer induction), incident-response drills, and division safety briefings log against profiles with read-acknowledgement where the content is a document (§11.5). The insurer-facing export — “show me every marshal's current safety record” — is one filtered report, which Volume 5's regulator-evidence table anticipates.

9.3 Rostering against forecast demand

The roster builder plans marshals, Artisans, FOH, and F&B against **forecast demand**: booking curves (advance bookings by day/hour), historical walk-up patterns by weekday/season/weather, and the events calendar. Ops managers see a demand heat-map per venue unit and drag staff into shift slots; the engine validates certifications, rest rules (minimum 11 h between shifts, configurable per jurisdiction), and cost against the labour budget line live. Published rosters push to staff apps; swap requests route peer→lead approval. The capacity feedback loop (§4.5) means an unfilled marshal slot automatically caps that track's sellable Shifts — the roster is not advisory.

9.4 Time & attendance

Clock-in/out happens in the staff app with a **geofence check** (device inside the park polygon; Core Location / FusedLocation per platform) plus the venue-unit check for track roles. No biometric hardware in v1 — the phone, the geofence, and the roster cross-check (clock-in without a rostered shift flags to the lead) are sufficient controls at this scale. Missed punches resolve through a lead-approved correction flow, audited.

9.5 Payroll: compute locally, process externally

The payroll boundary mirrors the accounting boundary: **RC WORLD OS computes; a local payroll processor pays and files**. The payroll calculator (Kotlin service) assembles each period: rostered vs. actual hours from time & attendance, overtime per configured rules (daily/weekly thresholds, penalty rates for late-night race events), allowances (bunker duty, wet-work), leave accruals and deductions. HR reviews the computation sheet per employee with variance-to-last-period highlighting; finance releases (§3.7 separation); the export file goes to the payroll processor in its import format. Processor output (net pay, taxes, employer contributions) posts back as a GL journal. This “process locally, verify locally” framing keeps jurisdiction-specific tax logic — the part that changes with every local election — out of our codebase entirely, which matters double for franchising: each franchise site plugs in its own local processor without touching platform code.

9.6 Performance reviews

Lightweight and evidence-linked: review cycles per role family, self + lead assessment, and auto-attached operational evidence — an Artisan's closed work orders and rework rate, a marshal's incident-response log, FOH transaction accuracy. Reviews inform the Academy development plan (Volume 12) rather than a forced ranking; the schema stores them under HR-only RLS.

10. CRM, Marketing & Gamification Module

10.1 Customer 360

One customer record aggregates: identity and consent state, visit days (from check-ins), life-time and 12-month spend by category (ledger), current wallet/Gears balances, membership status, **RC WORLD License tier**, badge case, bound-vehicle history, incident/misuse flags, marketing segment memberships, and guardian links. FOH sees the thin card (name, tier, today's bookings, flags); marketing sees segments; only the customer sees everything. The

360 is a view over module tables — never a copied “CRM database” that drifts from the truth.

10.2 The progression engine

Volume 9’s design, implemented as data-driven rules evaluated by the rules engine on telemetry and transactional events:

- **XP rules** (`xp_rules` table): event type → XP, with per-day caps and class multipliers. Events: Shift completed, lap improvement vs. personal best, tonnage delivered to the hopper (Construction), mission completed (Marine harbour master), retrieval protocol piloted successfully, low sag score band, race podium.
- **Tier evaluation:** license tiers (Volume 9 catalogue) evaluate on XP thresholds plus gate requirements (e.g. a clean-driving criterion: no geofence kills in last N Shifts; a skills check-off by a marshal for premium classes). Evaluation runs at Shift close; promotions celebrate in-app and unlock booking classes (\$4.5) and transmitter EPA profiles (Volume 3, §12.1).
- **Badge triggers:** declarative predicates over `shift_telemetry_summary` and event streams — “Night Owl” (5 night-race Shifts), “Smooth Operator” (sag score < threshold, 10 Shifts), “Excavator Foreman” (100 t lifetime tonnage), “Tow Master” (10 retrievals). Badges are additive and never revoked; the criteria JSONB is versioned so historical awards stay explicable.

Leaderboards: per track/class daily, weekly, season; per zone (tonnage, missions); global. Computed from `lap_times` and summaries into materialized boards refreshed on event, pushed via Realtime to app and track-side displays. Display names are customer-chosen handles with profanity screening; minors appear only per guardian visibility settings.

10.3 Campaigns and automations

Segmentation over the 360 (visit recency/frequency, spend band, tier, division affinity, distance from park) drives push (FCM/APNs) and email (SES/Postmark-class) campaigns. Automations shipped in v1.5: birthday offer (guardian-mediated for minors), win-back at 45/90 days dormant, post-first-visit sequence, membership renewal risk (usage-decline signal), event announcements by division affinity. Every send is consent-checked at dispatch time, not at segment build time.

10.4 Season passes and entitlements

Season passes and memberships materialize as **entitlement rows** (type, period, remaining allowance, constraints) that the pricing engine (§4.6) consumes. Entitlement usage is a ledger event, so finance sees membership utilization without a reconciliation project, and Volume 9’s economics (allowance breakage, upgrade paths) read straight from the data.

10.5 Consent and privacy (GDPR-class)

- Consent ledger per purpose (transactional, marketing push, marketing email, photo/media, telemetry-derived gamification), timestamped, versioned against the privacy-policy revision; withdrawal is one tap and effective at next dispatch.

- Data subject rights: export (JSON bundle of the customer's rows) and erasure (anonymize identity, retain de-identified transactional/telemetry aggregates required by finance law) as built-in admin workflows, not engineering favours.
- Data minimization: telemetry is vehicle-keyed and joins to customers only through session binding; marketing never receives raw telemetry.

10.6 Minors' data policy

Minors under the local digital-consent age get **no independent account**: a guardian creates a linked minor profile (first name + birth year only), signs the waiver, holds all consents, sets spend caps, and controls leaderboard visibility (default: handle-only, no photos). Progression accrues to the minor profile and migrates to a full account at age of consent with guardian handover. Staff-facing screens show minors with a guardian badge; marketing automations never target minor profiles directly.

11. Additional Modules

11.1 Retail & F&B POS and stock

The POS surfaces (\$5.4) sit on catalogue + stock: retail SKUs (Volume 8's retail range: parts, merch, starter RC kits) and F&B items with recipe-level stock decrement (a flat white decrements beans, milk, cup). Stock features: goods receipt against PO, cycle counts, wastage logging (F&B), min/max reorder into procurement, margin reporting by SKU to finance. Kitchen order screen for the Phase 2 restaurant; the Phase 1 kiosk runs order-and-collect with buzzer numbers.

11.2 Corporate events pipeline

A lightweight pipeline CRM for the events business (from \$1,400/2 h anchor): enquiry → qualified → proposal (templated from package configurator) → deposit paid (→ bookings hold, \$4.2) → confirmed → delivered → invoiced/closed. Stages carry tasks (run-sheet, catering brief, marshal roster block); conversion and pipeline value report to the executive dashboard. Loses record a reason code — the marketing feedback loop Volume 2 asks for.

11.3 Access control integration

Entry gates read the booking/day-pass QR or an NFC wristband (issued at FOH, mapped to the customer session — wristbands suit wet zones and children). Gate controllers (commodity QR/NFC turnstile-class hardware) call a local validation endpoint on the edge server (offline-tolerant, §13.2) which checks entitlement (valid booking today, day pass, member) and logs the movement. Zone-level gates (bunker, The Works, pit lane) run staff-credential NFC with role checks — the physical mirror of RLS.

11.4 CCTV / incident integration

CCTV stays on its own VMS (network video recorder-class, Volume 11 specifies coverage); RC WORLD OS integrates at the bookmark level: an incident report (\$6.8), a kill event, or a marshal flag calls the VMS API to bookmark ± 90 s across the venue unit's cameras, storing the bookmark reference on the incident. Footage never enters our Storage except as expor-

ted evidence clips attached to a closed incident (retention per §13.6). This keeps video privacy surface small and the integration vendor-swappable.

11.5 Document management for SOPs

Every SOP, checklist, and manual section from Volumes 3–8 lives as a versioned document: owner, revision, effective date, distribution list by role/site. **Read-acknowledgement tracking** is the operational point: publishing a revised SOP-MS-002 pushes a task to every affected role; the compliance board shows who has not acknowledged; rostering can require current acknowledgements for safety-critical SOPs the same way it requires certifications (§9.1). Franchise doctrine documents publish read-only to franchisee sites with local-annex support (Volume 12’s mandatory-vs-adaptable split, executed).

11.6 Franchise multi-tenancy

The architecture is site-scoped from day one (§1.4); v2 activates the network features:

- **Partitioning:** every business table carries `site_id`; RLS fences all franchise roles to own sites. Telemetry partitions per site per day. Site-level data export supports a departing franchisee’s wind-down obligations (Volume 12).
- **Franchisor cross-site dashboards:** the executive/franchisor role reads network rollups — same-site revenue growth, utilization, safety incident rates, compliance scores (SOP acknowledgements, cert currency, PM on-time %) — the operational royalty audit Volume 12 defines, computed continuously instead of annually.
- **Per-site configuration:** pricing bands (above franchisor floors), local tax and payroll processor bindings, opening calendars, local vehicle-class mix — all configuration rows, not code branches. Doctrine tables (PM matrices, safety rules, XP rules) publish from the franchisor with site-level read-only enforcement.
- **Tenancy hygiene:** shared platform, shared schema, fenced rows. A single-site franchisee’s traffic cannot see or starve another site’s; per-site rate limits and per-site Realtime channels enforce the isolation operationally as well as logically.

Field Note. The most common multi-tenant mistake is retrofitting `site_id` after launch — a months-long migration through every query and policy. The second most common is forking per site “temporarily”. This plan avoids both by decree: `site_id` is in every business table from migration 0001, and there is exactly one production codebase. Franchisees get configuration, never branches.

12. Data Model

12.1 How to read this chapter

This is the consolidated schema reference: the doctrine tables (`fleet_inventory`, `live_telemetry`, `maintenance_logs`, `queue_roster`) extended into the full entity-relationship design, ~30 core tables grouped by module. Conventions: every table has `id` (UUID or bigint identity), `site_id` (bigint, FK `sites`), `created_at`/`updated_at`; money is `numeric(12,2)` USD; enums are Postgres enums with migration-managed values; soft deletes only where legal retention demands (`revoked_at`/`deleted_at`), never on ledger or audit tables.

ERD in prose. `sites` roots everything. Identity: `user_site_roles` links auth users to roles per site; `staff_profiles` and `customer_profiles` extend users; `guardian_links` connects guardians

to minor profiles. Commercial spine: `venue_units` → `session_slots` → `bookings` → `shifts`; `queue_roster` hangs off `venue_unit` + vehicle class. Money: `ledger_transactions` → `ledger_entries` → `ledger_accounts`; payments and refunds reference PSP objects and post through the ledger. Fleet: `vehicle_models` → `fleet_inventory` → (`asset_components`, `battery_packs` via swap logs, `work_orders` → `maintenance_logs`, `status_transitions`). Telemetry: `fleet_inventory` → `live_telemetry` (partitioned) and `shift_telemetry_summary`; geofences and checkpoints belong to `venue_units`; `vehicle_commands` and `node_registry` complete the IoT loop. A `shift` is the junction where everything meets: customer, booking, vehicle, telemetry summary, ledger postings, XP events.

12.2 Core table listing

Identity & access

Table	Key columns	Relationships / notes
<code>sites</code>	<code>name</code> , <code>timezone</code> , <code>status</code> , <code>franchisee_org_id</code>	Root of tenancy
<code>user_site_roles</code>	<code>user_id</code> , <code>site_id</code> , <code>role</code> , <code>granted_by</code> , <code>revoked_at</code>	RBAC truth (\$3.4)
<code>customer_profiles</code>	<code>user_id</code> , <code>handle</code> , <code>birth_year</code> , <code>license_tier</code> , <code>consent JSONB</code>	1:1 auth user
<code>guardian_links</code>	<code>guardian_id</code> , <code>minor_id</code> , <code>spend_cap</code> , <code>visibility</code>	\$10.6
<code>staff_profiles</code>	<code>user_id</code> , <code>position</code> , <code>division</code> , <code>contract refs</code>	Compensation in separate table
<code>audit_log</code>	<code>actor</code> , <code>role</code> , <code>action</code> , <code>entity</code> , <code>diff JSONB</code> , <code>device</code> , <code>ip</code>	Append-only (\$3.6)

Bookings & scheduling

Table	Key columns	Relationships / notes
<code>venue_units</code>	<code>site_id</code> , <code>division</code> , <code>name</code> , <code>geometry ref</code>	Track A, Mining pod 1...
<code>vehicle_classes</code>	<code>division</code> , <code>name</code> , <code>premium flag</code> , <code>license_tier_required</code>	
<code>session_slots</code>	<code>venue_unit_id</code> , <code>class_id</code> , <code>block_start</code> , <code>capacity</code> , <code>price_band</code>	Generated 28 d ahead (\$4.1)
<code>bookings</code>	<code>customer_id</code> , <code>slot_id</code> (or <code>group_booking_id</code>), <code>state</code> , <code>price_breakdown JSONB</code>	Doctrine flows \$4.2
<code>group_bookings</code>	<code>type</code> (corporate/party), <code>window</code> , <code>deposit ref</code> , <code>run_sheet</code>	
<code>queue_roster</code>	<code>customer_id</code> , <code>class_id</code> , <code>venue_unit_id</code> , <code>joined_at</code> , <code>state</code>	Doctrine table — T-5 Top-Up reads it
<code>shifts</code>	<code>booking_id</code> , <code>vehicle_id</code> , <code>clock events</code> , <code>state</code> , <code>top_up_count</code>	The atomic delivered unit

Payments & ledger

Table	Key columns	Relationships / notes
<code>ledger_accounts</code>	<code>code</code> , <code>type</code> , <code>site scope</code>	Chart \$5.3

Table	Key columns	Relationships / notes
ledger_transactions	type, reference, initiator, occurred_at	
ledger_entries	transaction_id, account_id, direction, amount, currency	Zero-sum constraint
payments	psp_token, method, amount, state, psp_ids	Tokens only — no PANs (§5.6)
refunds	origin, amount, state machine, initiator, approver	§3.7 duties
entitlements	customer_id, type, period, remaining, constraints	Memberships/passes (§10.4)

Fleet, maintenance & telemetry

Table	Key columns	Relationships / notes
vehicle_models	class_id, make/model, scale, wholesale_cost, spares_profile	Volume 8 catalogue link
fleet_inventory	mac_address, asset_tag, model_id, status, total_hours_run, total_shifts	Doctrine table , status extended §6.3
status_transitions	vehicle_id, from, to, actor, at	Utilization analytics
battery_packs	qr, config, cycle_count, ir_trend JSONB, status	3:1 pool (§6.7)
work_orders	vehicle_id, type, priority, tasks, state, signoff	PM engine output
maintenance_logs	vehicle_id, mechanic_id, parts_used JSONB, description, date_resolved	Doctrine table , unchanged
parts_stock	sku, bin, qty, min/max, landed_cost, lead_time	Reorder → procurement
pm_rules	model/class, metric, threshold, task_template	Volume 7 matrices as data
live_telemetry	vehicle_id, ts, lat, lng, speed, voltage fields, pwm_state	Doctrine table , partitioned daily
shift_telemetry_summary	shift_id, distance, sag_score, laps, kills, energy proxy	7-year retention
geofences / checkpoints	venue_unit_id, geometry, class overlays / ordered lines	§7.4–7.5
vehicle_commands	vehicle_id, command, issuer, issued_at, acked_at	Append-only, RLS pattern 3
node_registry	mac, hw form factor, fw_version, last_seen, rssi	OTA + health
firmware_releases	version, target, sha256, signature, ring	§7.8
incidents	category, refs (shift/vehicle/customer), photos, cctv_bookmarks	§6.8, §11.4

Finance, HR, CRM, ops

Table	Key columns	Relationships / notes
suppliers / purchase_orders / goods_receipts / supplier_invoices	three-way-match keys	§8.3, Volume 8
budgets	site, department, month, amount, version	§8.4
staff_compensation	staff_id, pay basis, rates	Column-privacy table (§3.5)
certifications	staff_id, type, issued, expires, evidence ref	Roster gate (§9.1)
roster_shifts / time_entries	slot, staff, venue_unit / clock events, geofence result	§9.3–9.4
payroll_runs	period, computation snapshot, reviewed_by, released_by	Export boundary (§9.5)
xp_rules / xp_events / badges / badge_awards	rule predicates JSONB; event stream; award refs	§10.2
lap_times	shift_id, lap, time_ms, source (gps/transponder)	Blended leaderboards (§7.5)
segments / campaigns / consents	definitions; sends; per-purpose ledger	§10.3, §10.5
products / stock_items / pos_baskets	retail/F&B catalogue, recipe decrement, basket lines	§11.1
event_pipeline	stage, value, tasks, reason codes	§11.2
documents / doc_acknowledgements	revision, owner, distribution / user, doc, at	§11.5
entry_scans	credential, gate, result, at	§11.3

12.3 Schema governance

The digest's original `fleet_inventory` status enum (active/maintenance/charging) is formally superseded by the seven-state machine in §6.3 — this volume states so explicitly per the living-document rule; `active` maps to `available|bound|on_track|pit`. All other doctrine columns are preserved verbatim. Schema changes ship only as reviewed migrations with RLS tests; any volume that binds to a table name (Volumes 3–6 reference `maintenance_logs` and `fleet_inventory` extensively) is protected by this governance: those names and their doctrine columns are frozen APIs.

13. Non-Functional Requirements & Operations

13.1 Availability targets

Service	Target	Rationale
Node-local safety (kill on fence/voltage)	Always — no network dependency	Doctrine; life-safety adjacent
Telemetry pipeline + Grid View	99.5% during park hours	Marshals fall back to line-of-sight SOPs below this
Bookings/check-in/POS	99.5% park hours (with edge degraded mode below)	Revenue path

Service	Target	Rationale
Web console / back office	99% business hours	Tolerates maintenance windows
App availability	Store-distributed; offline-tolerant caches for tickets/QRs	A booking QR must render with no signal

13.2 Edge resilience: the park cache server

One small on-prem server (redundant pair by Phase 2) runs at the ops room: local MQTT broker mirror, a read replica of hot operational tables (today's bookings, queue, fleet status, catalogue), and a write-ahead queue for check-ins, POS sales, and clock events. When the WAN drops, FOH keeps checking in, POS keeps selling (PSP offline rules for cards, unconditional for cash/wallet-cached), marshals keep Grid View from the local broker, and telemetry buffers locally. On restore, the queue replays with idempotency keys; conflicts (a slot double-sold across the partition) resolve by policy: honour both, flag to ops, comp the overflow — never argue with a customer at the gate over a sync conflict.

13.3 Backups and disaster recovery

Supabase point-in-time recovery on Postgres; nightly logical dumps to independent object storage in a second region; Storage (documents, photos, firmware) versioned + replicated. Targets: **RPO ≤ 5 minutes** (WAL-based) for transactional data, ≤ 24 h acceptable for raw telemetry (safety does not depend on history); **RTO ≤ 4 hours** to full back-office, ≤ 1 hour to park-operational (bookings/POS/telemetry) using the runbook-scripted restore. DR drill twice yearly against a scratch project, timed and reported to the executive dashboard like any other KPI.

13.4 Observability

Structured logs (correlation ids across app → service → DB), metrics (ingestion rate, command latency p95 vs. the \$7.4 budget, queue depths, RLS-denial counts, wallet-invariant check), and alerting to the on-call rotation (platform run team, \$2.7) via a PagerDuty-class escalation. Golden alarms: kill-command p95 breach, wallet control drift ≠ 0, node fleet stale >10%, PSP webhook failures, backup job failure. Each golden alarm has a written runbook in the documents module — the platform documents itself with its own SOP machinery.

13.5 Security hardening

Secrets in a managed secrets store (cloud KMS-class), never in repos or client binaries; quarterly key rotation. Dependency and container scanning in CI; signed firmware (§7.8); MDM-enrolled kiosk devices; least-privilege service keys per machine identity (§3.1 role 14). **Penetration test cadence:** external test before public launch, then annually and after any auth/payment architecture change; scope always includes the RLS layer (attempt cross-tenant and cross-role reads) and the MQTT/OTA path (attempt command injection and firmware substitution). Findings feed the same risk register as §14.4.

13.6 Data retention schedule

Data	Retention	Basis
Financial ledger, invoices, payroll exports	7 years	Statutory

Data	Retention	Basis
Audit log (financial actions) / (other)	7 / 3 years	\$3.6
Raw telemetry / downsampled / Shift summaries	14 days / 13 months / 7 years	\$7.3
CCTV bookmarks & evidence clips	90 days; incident-attached until case closes + 2 years	Insurance
Customer PII after erasure request	Anonymized immediately; ledger rows retained de-identified	GDPR-class
Marketing consent records	Life of consent + 3 years	Proof of consent
Staff records post-employment	Per jurisdiction (default 6 years)	Local labour law, per-site config

13.7 Change management and release operations

The park runs seven days a week; the platform changes under it. Release rules: schema migrations deploy in the park's closed hours (with the two-phase expand/contract pattern for anything touching a hot table — add column, backfill, switch readers, drop later); app releases ride store review cycles, so every server change stays backward-compatible with the previous two app versions (enforced by contract tests against pinned older specs); feature flags gate anything customer-visible so a launch is a flag flip at 9 a.m. with the team watching dashboards, not a deploy at midnight with the team asleep. Firmware follows its own ring discipline (§7.8). A one-page change calendar in the ops day board tells the duty manager what changed this week — half of “the system is acting weird” reports resolve by looking at that page.

13.8 Scalability path

One site at 150 assets and ~1,500 daily visitors runs comfortably on a mid-tier Supabase instance and two service containers. Scaling levers, in order of need: read replicas for dashboards; Timescale-class extension for telemetry (Phase 2, ~200 assets); per-site MQTT brokers at franchise sites feeding the shared cloud; regional deployment split only if a franchise cluster lands on another continent (Volume 12's international scenario). Nothing in the design requires re-architecture before ~10 sites.

14. Implementation Roadmap

14.1 Release strategy

Three releases mapped to park phasing: **v1 (MVP)** must be live for Phase 1 opening (Month 12 at the latest; target Month 10 for two months of staff rehearsal); **v1.5** lands Months 13–20; **v2** lands Months 21–36 alongside the Phase 3 franchise pilot. MVP scope per the founder's cut: bookings, wallet + payments, telemetry + kill-switch, fleet + maintenance, POS, RBAC core — plus the Toolbox Talk, without which no booking can legally complete.

14.2 Sprint-level breakdown (v1: 11 two-week sprints ≈ 5.5 months, then hardening)

Sprint	Theme	Key deliverables
0	Foundations	Repos, CI/CD, environments, schema migration 0001 (sites, identity, RBAC tables), design system, KMP app shells on both stores' internal tracks
1	RBAC core	Auth flows, role provisioning, RLS pattern library + pgTAP harness, role-scoped navigation skeleton
2	Telemetry ingest	MQTT broker, ingestion service, <code>live_telemetry</code> partitioning, node bench integration, Grid View alpha
3	Kill-switch & geofence	Command path end-to-end, node fence push/verify at bind, latency-budget test in CI, marshal screen
4	Fleet & maintenance	Registry, status machine, QR binding, work orders, <code>maintenance_logs</code> , parts stock
5	Bookings I	Inventory generation, capacity rules, app booking flow, Toolbox Talk module, waivers
6	Payments & wallet	Ledger service + invariants, PSP integration, Apple Pay/Google Pay, wallet load/spend, reconciliation job
7	Bookings II + queue	<code>queue_roster</code> , walk-in kiosk, T-5 Top-Up flow, Shift clocks, breakdown pause + auto-credit
8	POS & check-in	FOH kiosk mode, retail/F&B POS, card terminals, cash drawer, entry QR validation on edge server
9	Lap timing & customer HUD	Checkpoint engine, sag score v1, live HUD, leaderboards v1, basic XP/badges
10	Back office v1	Finance console (ledger views, refund queue, GL export v1), ops day board, roster v1, staff clock-in
11-13	Hardening	Load test at 150 nodes, offline drills, pen test + fixes, DR drill, staff-app rehearsal content, store review buffer

v1.5 (six sprints): payroll computation + processor export, certification compliance + Academy integration, AP three-way match, budgets vs. actuals, CRM automations, season pass entitlements, document read-acknowledgement. v2 (eight sprints): franchise tenancy activation + franchisor dashboards, per-site config packs, advanced analytics/Bi extracts, Timescale-class telemetry migration, Phase 2 division features (aviation weather-minima automation, marine mission scoring), corporate events pipeline v2.

14.3 Team composition through the roadmap

The §2.7 build team (6.5 FTE) covers v1. v1.5 runs at 4 FTE (payroll/finance depth is backend + web heavy). v2 returns to ~5 FTE with a data engineer added for analytics. The firmware engineer's 0.5 FTE is shared with The Works throughout — node hardware and firmware are one budget line and one owner, avoiding the classic split-brain between “the PCB team” and “the app team”.

14.4 Software program risk register

#	Risk	L	I	Mitigation
R1	Kill-command latency misses budget on real mesh	M	H	Node-local safety is primary (design); staging bench asserts budget every release; AP density survey in Volume 11 build
R2	App Store rejection/delay (kill-switch, stored value)	M	H	Reviewer dossier prepared early (§2.1 Field Note); TestFlight external review two sprints before launch need
R3	Wallet ledger defect → liability drift	L	H	Single-writer ledger service, zero-sum constraint, nightly invariant alarm, property-based tests on posting logic
R4	RLS policy gap leaks cross-role/tenant data	M	H	pgTAP deny-tests mandatory per policy; pen-test scope includes RLS; no service-role key in clients
R5	GPS lap-timing accuracy disappoints racers	H	M	Positioned honestly as rental-grade; transponder loop for leagues (§7.5); expectation set in Volume 9 UX
R6	Scope creep from division wish-lists pre-opening	H	M	MVP cut is founder-signed (this chapter); post-MVP requests enter the v1.5 backlog, not the launch plan
R7	Single-vendor dependence (Supabase)	L	M	Plain Postgres + SQL migrations = portable core; Realtime/Auth abstractions isolated behind interfaces

#	Risk	L	I	Mitigation
R8	Key-person loss (tech lead)	M	M	This volume + ADR log + runbooks as institutional memory; code re-view discipline; no unshared credentials
R9	Payroll/tax misconfiguration at a franchise site	M	M	Process-locally boundary (\$9.5): local processor owns tax; launch checklist requires processor sign-off
R10	Mesh coverage gaps discovered post-build	M	H	RF survey milestone in Volume 11 construction schedule before surfacing; node buffer + replay masks brief gaps

Investor Note. The software line is $\approx 10\%$ of Phase 1 capex and it is the multiplier on the other 90%: utilization (bookings + queue), asset life (predictive maintenance), spend per visit (wallet + gamification), and franchisability (a platform, not a park, is what franchises). The MVP cut above is the minimum that opens the gates; every v1.5/v2 item has a named revenue or margin mechanism behind it in Volumes 9, 10 and 12.

15. Volume Summary & Cross-References

RC WORLD OS is the single operating system of the business: one PostgreSQL schema under Row Level Security, Kotlin services for telemetry/rules/ledger/payroll compute, Kotlin Multiplatform + Compose Multiplatform apps on **both Android and iOS**, and a web admin console — with a build-vs-buy line that keeps custom engineering on the differentiators (telemetry, bookings, gamification, wallet) and buys the commodities (card processing, statutory accounting, payroll filing). The founder's central requirement is delivered as architecture: a fourteen-role catalogue with per-module permission matrices, role-scoped interfaces specified screen by screen, RLS policy patterns with tested SQL, separation of duties on every financial action, and an append-only audit trail. The doctrine systems are implemented exactly — the 20-minute Shift as atomic inventory, the T-5 Top-Up against `queue_roster`, PWM-intercept kill on under-voltage/geofence with a ≤ 550 ms p95 command budget and node-local autonomy, the breakdown protocol's paused clocks and auto-credits, predictive maintenance from runtime hours — and the schema chapter extends the doctrine tables into a governed model of ~ 30 core tables (plus supporting tables) whose doctrine names are frozen APIs. Non-functional commitments (edge degraded mode, RPO ≤ 5 min/RTO ≤ 4 h, retention schedule, pen-test cadence) and an 11-sprint MVP roadmap with team, budget ($\approx \$185k$ v1 inside Volume 10's software line) and risk register complete the program.

Decisions this volume establishes for other volumes (living-document notices): client apps are KMP/CMP on Android **and** iOS, extending the digest's Android-only statement; `fleet_inventory.status` is the seven-state machine of §6.3, superseding the three-state enum;

telemetry transport is MQTT with Supabase Realtime as the command origination and fan-out layer; the wallet is a double-entry ledger and wallet loads are never revenue.

Cross-references.

- **Volume 3** — Motorsport: parity data behind sag-score calibration; transponder league timing feeding `lap_times`; transmitter EPA profiles gated by license tier.
- **Volume 4** — Construction: Heavy-Node fleet, tonnage crediting, pod-linked 1:3 slot model, grooming blackouts feeding capacity.
- **Volumes 5-6** — Aviation/Marine: weather-minima feeds into the capacity engine; external timing precedent; incident and retrieval doctrine implemented in Shift lifecycle.
- **Volume 7** — Engineering & Workshop Manual: PM matrices loaded as `pm_rules`; work-order sign-off handshake; RCW Node build and conformal-coating doctrine.
- **Volume 8** — Procurement: supplier catalogue, landed-cost methodology, reorder linkage from `parts_stock`.
- **Volume 9** — Customer Experience & Loyalty: License tiers, XP/badge/Gears rules the progression engine executes; membership economics behind entitlements.
- **Volume 10** — Finance: source of truth for the software budget line, KPI definitions (\$8.5 implements them as SQL), revenue model the recognition rules serve.
- **Volume 11** — Architecture & Park Design: mesh AP placement and RF survey, conduit and timing-loop provisions, ops room and edge-server housing, CCTV coverage.
- **Volume 12** — Franchise Manual: multi-tenancy features (\$11.6), doctrine publication, per-site payroll/tax boundary, platform licensing terms.